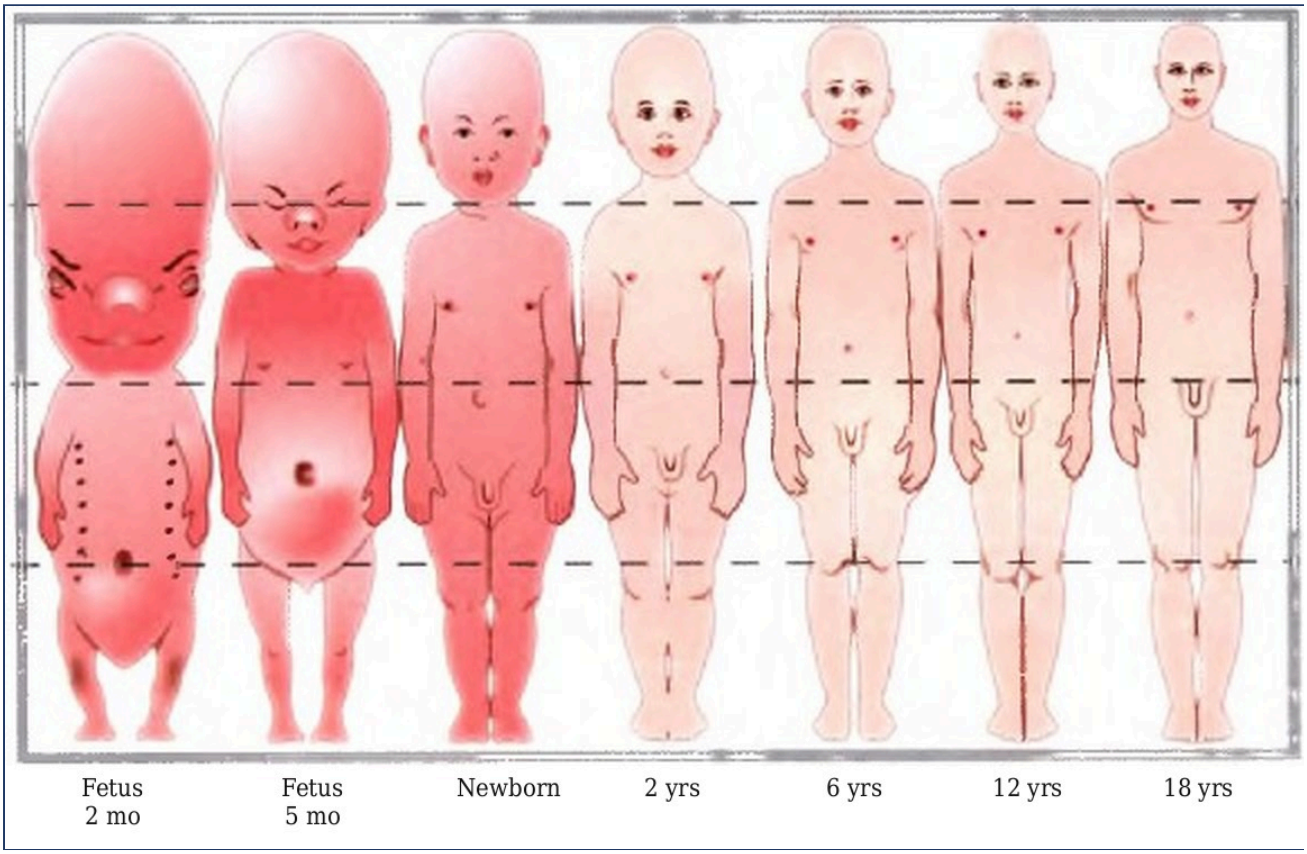
MINISTRY OF HEALTH OF THE REPUBLIC OF UZBEKISTAN
TASHKENT STATE MEDICAL UNIVERSITY
Department of Propaedeutics of Childhood Diseases No. 1

PROPAEDEUTICS OF CHILDHOOD DISEASES

complete study manual

A study manual for practical classes



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This manual is based on the 15 practical class topics of the calendar-thematic plan of the course “Propaedeutics of Childhood Diseases”. Each topic fully covers the anatomical and physiological features of the child's body, examination methods, the semiotics of lesions and their clinical significance. At the end of each chapter there are algorithms for performing practical skills, review questions, test questions and clinical cases with answers. The answers to the tests are given at the end of the book. To make the material easier to memorize, “Remember”, “Clinical significance” and “Memory aid” boxes, tables and figures are used extensively.

The manual is based on the textbooks of Sh.I. Navruzova and N.A. Geppe, R.R. Kildiyarova, A.Sh. Arzikulov, B. Karakhodzhaev, I.M. Lysenko and T.A. Daminov. The figures are taken from these sources, and the source is given under each of them.

The text and structure of the manual were prepared with the help of the artificial intelligence Claude (Anthropic) on the basis of the textbooks listed. The compiler selected the material, defined its structure and checked it. Before applying the clinical information in practice, it is recommended to compare it with the main textbooks.

The manual is intended for 3rd-year students of the faculties of general medicine of medical higher education institutions.

This English edition is a translation of the Uzbek original.

CONTENTS

Preface 7

Introduction to the propaedeutics of childhood diseases. The case history. Taking the history and general inspection of a sick child 8

1.1. The subject, aim and tasks of the discipline 8

1.2. Medical ethics and deontology in pediatrics 10

1.3. Methods of examining the child 10

1.4. The concept of emergency assessment (triage) 11

1.5. Taking the history 12

1.6. Structure of the case history 15

1.7. General rules of objective examination 15

1.8. General inspection 16

1.9. Clinical part 23

Physical development of the child. The newborn and the preterm newborn 32

2.1. The concept of physical development and the factors affecting it 32

2.2. Methods of assessing physical development and the technique of anthropometry 33

2.3. Changes in anthropometric indicators with age and their approximate calculation 35

2.4. Assessing physical development 37

2.5. Semiotics of disorders of physical development 39

2.6. Anatomical and physiological features of the newborn 41

2.7. Borderline (transitional) states of the newborn period 43

2.8. The preterm (premature) newborn 44

2.9. Features of examining the newborn 46

2.10. Clinical part 47

Neuropsychological development of children. Semiotics of nervous system damage. Body temperature, thermoregulation and fever 56

3.1. Anatomical and physiological features of the nervous system 56

3.2. Neuropsychological development 57

3.3. Development of the sense organs 60

3.4. Methods of examining the nervous system 61

3.5. The newborn's innate unconditioned reflexes 62

3.6. Semiotics of nervous system damage 65

3.7. Body temperature and thermoregulation 69

3.8. Fever 69

3.9. Clinical part 71

The skin, subcutaneous fat and lymphatic system: anatomical and physiological features, methods of examination, semiotics 80

4.1. Anatomical and physiological features of the skin 80

4.2. Methods of examining the skin 82

4.3. Skin changes characteristic of the newborn period 86

4.4. Semiotics of skin damage 87

4.5. Subcutaneous fat 93

4.6. Anatomical and physiological features of the lymphatic system 96

4.7. Methods of examining the lymphatic system 98

4.8. Semiotics of lymphatic system involvement 104
4.9. Clinical section 106
The bones and muscles: anatomical and physiological features, methods of examination, semiotics and syndromes 118
5.1. Anatomical and physiological features of bone tissue 118
5.2. Features of the parts of the skeleton 120
5.3. Body proportions by age 124
5.4. Methods of examining the osteoarticular system 124
5.5. Semiotics of osteoarticular system involvement 130
5.6. The muscular system 134
5.7. Clinical section 138
The respiratory system: anatomical and physiological features, the first breath, history taking, inspection, palpation, percussion and auscultation 149
6.1. Anatomical and physiological features of the respiratory organs 149
6.2. Mechanism of the first breath 151
6.3. History taking 153
6.4. Inspection 153
6.5. Palpation of the chest 156
6.6. Percussion of the lungs 157
6.7. Auscultation of the lungs 161
6.8. Clinical section 164
Instrumental and laboratory methods of examining the respiratory system. Semiotics and syndromes of respiratory involvement 175
7.1. Functional methods of examination 175
7.2. Radiological methods of examination 177
7.3. Endoscopic methods 179
7.4. Laboratory methods 180
7.5. Semiotics of the main respiratory symptoms 181
7.6. Syndromes of consolidation (infiltration) of lung tissue and atelectasis 182
7.7. Pleural syndromes 184
7.8. Airway and other lung syndromes 187
7.9. Respiratory failure (RF) syndrome 190
7.10. Clinical section 191
The cardiovascular system: anatomical and physiological features, fetal and neonatal circulation, inspection, palpation, percussion, X-ray and cardiometry 202
8.1. Development of the heart and the fetal circulation 202
8.2. Neonatal circulation: the transitional period 203
8.3. Anatomical and physiological features of the heart and blood vessels 204
8.4. History taking 205
8.5. Inspection 206
8.6. Palpation 207
8.7. Percussion of the heart 209
8.8. Radiological examination and cardiometry 211
8.9. ECG and echocardiography — basic concepts 213
8.10. Clinical section 214

Auscultation of the heart, pulse and arterial pressure. Functional and instrumental examinations of the cardiovascular system, age-related features of the ECG, and semiotics 225
9.1. Auscultation of the heart 225
9.2. The pulse 229
9.3. Arterial pressure 230
9.4. Functional tests 232
9.5. Instrumental investigations 232
9.6. Semiotics 234
9.7. Clinical section 237
Syndromes of cardiovascular involvement: carditis (endocardium, myocardium, pericardium), dystonia, congenital heart defects, heart and vascular failure 249
10.1. Carditis syndrome 249
10.2. Dystonia syndrome 252
10.3. Congenital heart defects 254
10.4. Heart failure syndrome 258
10.5. Acute vascular insufficiency syndrome 260
10.6. Clinical section 261
The hematopoietic system: anatomical and physiological features, age-related features of peripheral blood, the complete blood count, semiotics and syndromes (anemic, hemorrhagic, lymphoproliferative) 273
11.1. Development of hematopoiesis 273
11.2. Hematopoietic factors and iron metabolism 274
11.3. The hemostatic system 275
11.4. Age-related features of peripheral blood 276
11.5. Methods of examining the blood system 278
11.6. Semiotics of blood changes 280
11.7. Main syndromes of blood system involvement 282
11.8. Clinical section 285
The digestive system: anatomical and physiological features, history taking, inspection, examination of the oral cavity, palpation and percussion of the abdomen, liver and spleen, auscultofriction 297
12.1. Development of the digestive system 297
12.2. Anatomical and physiological features of the digestive organs 297
12.3. History taking 300
12.4. Inspection 302
12.5. Palpation of the abdomen 303
12.6. Percussion 308
12.7. Auscultation and auscultofriction 310
12.8. Examination of the rectum and anal area 310
12.9. Clinical section 311
Laboratory, functional, radiological and endoscopic methods of examining the digestive system. Semiotics and main syndromes of digestive system involvement 323
13.1. Laboratory investigations 323
13.2. Functional investigations 325
13.3. Radiological investigations 326

13.4. Endoscopic investigations and biopsy 327
13.5. Semiotics: subjective signs 328
13.6. Main syndromes of digestive system involvement 331
13.7. Clinical section 336
The urinary and immune systems: anatomical and physiological features, methods of examination, urine tests and functional tests, semiotics of involvement 348
14.1. Anatomical and physiological features of the kidneys and urinary tract 348
14.2. Clinical examination of the urinary system 352
14.3. Laboratory investigations 355
14.4. Instrumental investigations 358
14.5. Semiotics of urinary system involvement 359
14.6. The immune system: anatomical and physiological features 363
14.7. Examination of the immune system 366
14.8. Semiotics of immune system involvement 367
14.9. Clinical section 368
Feeding the healthy child: breastfeeding, mixed and artificial feeding, complementary foods. Defense of the case history 383
15.1. Types of feeding 383
15.2. Breastfeeding 383
15.3. Mixed and artificial feeding 387
15.4. Complementary foods 388
15.5. Calculating feeding volume in children under 1 year 390
15.6. Feeding children over 1 year 391
15.7. Feeding history 392
15.8. Writing and defending the case history 392
Answers to the test questions 407
References 409

Preface

Propaedeutics of childhood diseases is the “ABC” of pediatrics. This course teaches the age-specific anatomical and physiological features of the child's body, methods of examining healthy and sick children, and the main signs and syndromes of diseases (their semiotics). The more thoroughly a student masters propaedeutics, the more easily he or she will master faculty and hospital pediatrics, pediatric infectious diseases and pediatric surgery in later years.

The manual follows the order of the 15 practical classes of the calendar-thematic plan. Each chapter corresponds to one practical class and has the same structure:

- **Learning objectives** — what the student should know and be able to do.
- **Theoretical part** — anatomical and physiological features, examination methods and semiotics.
- **Clinical part** — the practical significance of the knowledge acquired, clinical examples.
- **Practical skills** — a step-by-step algorithm for each procedure.
- **Review questions, test questions and clinical cases** — to check independent preparation. The answers to the clinical cases are given within the chapter, and the answers to the tests at the end of the book.
- **Quick review** — key numbers and rules for a revision of a few minutes before an exam.

Four kinds of colored boxes appear in the text:

REMEMBER
Yellow box — a rule, number or definition that must be memorized.

CLINICAL SIGNIFICANCE
Blue box — the significance of the knowledge in clinical practice.

MEMORY AID
Green box — a technique or short rule that makes memorizing easier.

CAUTION
Orange box — common mistakes, dangerous situations and discrepancies between textbooks.

The figures are taken from the textbook “Bolalar kasalliklari propedevtikasi” edited by Sh.I. Navruzova and N.A. Geppe, the textbook “Пропедевтика детских болезней” edited by R.R. Kildiyarova and V.I. Makarova, and the manuals of A.Sh. Arzikulov, B. Karakhodzhaev and I.M. Lysenko. The source of each figure is given under it.

I hope the manual will help students prepare for practical classes, write case histories and prepare for midterm and final examinations.

TOPIC 1

Introduction to the propaedeutics of childhood diseases. The case history. Taking the history and general inspection of a sick child

Learning objectives

The student should know:

- the subject, aim and tasks of the discipline, the periods of childhood;
- the structure of the case history and the content of each of its sections;
- the concept of emergency assessment (triage) and the danger signs;
- the order of taking the history of present illness and the life history, and the features of talking with parents and with children of different ages;
- the clinical criteria for assessing the severity of a sick child's condition;
- the order of general inspection: general condition, consciousness, position, body build, stigmas of dysembryogenesis.

The student should be able to:

- establish contact with the child and the parents and take the history independently;
- take the history of present illness and the life history according to the scheme and draw a conclusion;
- assess the child rapidly (emergency assessment) and identify the danger signs;
- perform a general inspection and assess the severity of the patient's condition;
- record the data obtained correctly in the case history.

1.1. The subject, aim and tasks of the discipline

Propaedeutics (Greek propaideuo — “I teach beforehand”) is an introduction to the study of a discipline, the part that gives its foundations. **The propaedeutics of childhood diseases** is the introductory branch of pediatrics; it teaches the age-specific features of the child's body, the methods of examining healthy and sick children, and the main signs of diseases.

> **REMEMBER**
> A child is not a scaled-down copy of an adult. The child's body is constantly growing and developing, so each age period has its own anatomical and physiological features, its own normal values and its own specific course of disease. A pulse, respiratory rate or blood formula that is normal for an infant is pathological for an older child.

The discipline consists of the following parts:

1. The anatomical and physiological features of the child's body and the laws of physical and neuropsychological development.
2. Methods of questioning the child (taking the history) and of objective examination.
3. The basic semiotics of childhood diseases — symptoms and syndromes.
4. The rules of feeding a healthy child and drawing up a menu.
5. Methods of child care and the basic medical procedures.

The aim of the discipline is to teach the student the features of the child's body according to age and the technique of examining it, and to develop the skill of evaluating the data obtained to identify the affected system and the main syndrome.

Tasks of the discipline:

- to teach the anatomical and physiological features of children of different ages;
- to master the technique of a complete objective examination of the child (inspection, palpation, percussion, auscultation);
- to teach the evaluation of laboratory and instrumental findings according to age;
- to teach the identification of the main symptoms and syndromes and their combination into a syndromal diagnosis;
- to develop the skills of feeding and caring for a healthy child and of writing a case history.

A brief history of pediatrics

The first information on children's medicine is found in Hippocrates' (4th century BC) work “On the Nature of the Child”. In the Middle Ages, scholars of the East made a great contribution to this field. **Abu Bakr al-Razi** (865–925) wrote “A Treatise on Smallpox and Measles” and “Diseases of Children”, and was the first to introduce keeping a daily record of the patient's condition in hospital — the basis of today's case history. **Abu Ali ibn Sina** (980–1037) devoted special chapters of the “Canon of Medicine” to the care of the newborn, breastfeeding, choosing a wet nurse and the treatment of childhood diseases.

Pediatrics took shape as an independent discipline in the 19th century. The founders of the Russian school of pediatrics are N.F. Filatov, N.P. Gundobin, A.A. Kisel, G.N. Speransky, M.S. Maslov and A.F. Tur. N.P. Gundobin created the classification of the periods of childhood that is still used today.

Professors R.S. Gershenovich, K.G. Titov, S.Sh. Shamsiyev, S.N. Yo‘ldosheva, M.A. Mirzamuhamedov, M.R. Nuritdinov, F.X. Nazarmuhamedov, O.S. Mahmudov, X.A. Yunusova and T.A. Daminov made a great contribution to the development of the school of pediatrics in Uzbekistan. Professor **S.Sh. Shamsiyev** headed departments of the propaedeutics of childhood diseases in 1951–1989 and is the author of works such as “Acute Pneumonia in Children” and “Clinical Electrocardiography in Children”. In 1972 the Central Asian Medical Pediatric Institute was opened.

Periods of childhood

Knowing the periods of childhood is necessary for taking the history and evaluating examination findings according to age. In pediatrics a modified classification of N.P. Gundobin is used.

Table 1.1. *Periods of childhood (based on the classification of N.P. Gundobin)*

Period	Duration	Main feature
Intrauterine period	From the beginning of pregnancy until birth (280 days)	Embryonic (0–8 weeks) and fetal (from 9 weeks until birth) stages; teratogenic factors lead to malformations
Intranatal period	From the onset of labor until the umbilical cord is tied	Risk of birth injury and asphyxia
Newborn (neonatal) period	From birth to 28 days	Adaptation to the external environment, borderline (transitional) states
Infancy	From 29 days to 1 year	The fastest growth and development, a tendency to feeding disorders

Period	Duration	Main feature
Early childhood (toddler) period	1–3 years	Speech and movement develop rapidly, infectious diseases become more frequent
Preschool period	3–7 years	Children's groups, airborne infections
Early school age	7–11 (12) years	The milk teeth are replaced by permanent teeth
Senior school (adolescent, pubertal) period	12–17 (18) years	Sexual maturation, endocrine restructuring, autonomic instability

1.2. Medical ethics and deontology in pediatrics

The pediatrician works with two parties at the same time — the child and the parents. That is why the physician's communication skills directly affect the accuracy of the diagnosis.

- Before the examination, **contact should be established** with the child: ask an older child about his name, school and interests; distract a small child with a toy.
- A young child is **not asked for permission** to be examined (he usually says “no”) — the physician acts confidently but gently.
- Words that convey a poor prognosis or that the treatment was wrong are not used in front of the child and the parents.
- The child is addressed by name and the parents respectfully; accusatory phrases such as “Why didn't you bring him earlier?” are not used.
- **Confidentiality** is maintained in conversations with adolescents; it is possible to talk first with the parents and then separately with the adolescent.
- Before an examination with a group of students, the consent of the child (if older) or of the parents is obtained.

The physician's appearance and hands: a clean gown, hair under a cap, hands washed before examining each child, warm, with short nails. The physician should be patient, calm and kind.

1.3. Methods of examining the child

The examination of a child is divided into two groups of methods:

- **Main methods:** subjective — questioning (history); objective (physical) — inspection (inspectio), palpation (palpatio), percussion (percussio), auscultation (auscultatio).
- **Additional methods:** laboratory (blood, urine and stool tests and others) and instrumental (radiography, ultrasound, ECG, endoscopy and others).

MEMORY AID

The order of questioning and examining a child can be remembered with the English acronym **HELP**:

- **H** (history) — the history;
- **E** (examination) — objective examination;
- **L** (logical deduction) — logical conclusion (syndromal diagnosis);
- **P** (plan of management) — the plan of investigation and treatment.

The objective examination is carried out in the following sequence: general inspection → skin and subcutaneous fat → lymph nodes → musculoskeletal system → respiratory system → cardiovascular system → digestive system → urinary system → nervous system and sense organs → endocrine

system. Each system is examined in the order **inspection → palpation → percussion → auscultation**.

REMEMBER

If questioning reveals that an organ is affected, the physician pays more attention to it, but this **does not release him from the obligation to examine all systems fully**.

1.4. The concept of emergency assessment (triage)

Every sick child arriving at a hospital or an admission room is first assessed **rapidly — within 1–2 minutes**. The aim here is not a detailed diagnosis but the detection of a life-threatening condition and the setting of the order of care. This process is called **triage** (French trier — to sort). The World Health Organization (WHO) has developed the **ETAT** system for children (Emergency Triage Assessment and Treatment).

As a result of triage, sick children are divided into three groups:

Table 1.2. *Triage groups according to the WHO ETAT system*

Group	Signs	Action
E — emergency signs	Obstructed or absent breathing; severe respiratory distress; central cyanosis; signs of shock (cold hands, capillary refill time longer than 3 s, rapid and weak pulse); coma or convulsions; severe dehydration in a child with diarrhea (lethargy, sunken eyes, skin pinch goes back very slowly)	Emergency treatment is started immediately
P — priority signs	An infant under 2 months; high fever; trauma; severe pallor; poisoning; severe pain; respiratory distress; restlessness or lethargy; urgent referral; severe wasting; edema of both feet; burns	Seen first, without waiting in the queue
Q — queue (non-urgent)	No emergency or priority signs	Seen in the general queue

MEMORY AID

The emergency signs are easy to check in the **ABCD** order:

- **A** (airway) — patency of the airway;
- **B** (breathing) — breathing (present or absent, severe respiratory distress, cyanosis);
- **C** (circulation; coma; convulsions) — circulation (shock), coma, convulsions;
- **D** (dehydration) — severe dehydration.

Rapid assessment of consciousness — the AVPU scale. In triage, consciousness is assessed with a simple four-level scale:

- **A** (alert) — the child is alert and responds appropriately to the surroundings;
- **V** (voice) — responds only to voice;
- **P** (pain) — responds only to pain;
- **U** (unresponsive) — does not respond to anything.

Responding only to pain (P) or not responding (U) is a sign of coma, that is, an emergency.

General danger signs. In the WHO and UNICEF Integrated Management of Childhood Illness (IMCI) program, the following **general danger signs** are checked in every sick child aged 2 months to 5 years:

1. The child is unable to drink or breastfeed.

2. The child vomits everything.
3. The child has had convulsions during this illness.
4. The child is lethargic or unconscious.
5. The child is convulsing now.

CLINICAL SIGNIFICANCE
A child with at least one general danger sign is considered **severely ill**: he is given immediate care and urgently admitted to hospital. For example, if a 1-year-old child has fever and cough and refuses to breastfeed, this is not a simple cold but may be a sign of severe pneumonia or sepsis.

1.5. Taking the history

The history (anamnesis) (Greek anamnesis — recollection) is the body of information about the patient and the disease obtained by questioning the patient or a person who knows the patient well.

REMEMBER
The specific feature of the history in pediatrics: the information is obtained mainly **from the parents** (or the child's caregiver) and is supplemented by questions addressed to the child only according to the child's age. In a child under 5 years of age the history is taken almost entirely from the parents.

Components of the history:

1. Personal data.
2. Complaints — the main reason for seeking care.
3. History of present illness (anamnesis morbi).
4. Life history (anamnesis vitae): obstetric, neonatal, feeding, developmental, past illnesses, immunizations, allergic, hereditary (family), social and living conditions, and epidemiological history.

Personal data

Surname, first name and patronymic; date of birth (exact age — years, months, and days for a newborn); home address; the parents' names, ages, occupations and places of work; whether the child attends a kindergarten or school; date of admission; the diagnosis of the referring institution. The home address gives information about the epidemiological environment (water supply, sewerage), and the parents' occupation about the family's social level and occupational hazards.

Complaints

Complaints are the subjective signs of the disease. First the **main** and then the **general** complaints are recorded concisely; the complaints on admission and at the time of examination are indicated separately. Each complaint is clarified in detail: site, character, intensity, duration, when it appears, what makes it worse and what relieves it.

Difficulties in obtaining complaints from children:

- a young child cannot express complaints because of age or impaired consciousness, and the parents only report that signs have appeared (for example, “he is crying and pulling his legs up to his tummy”);
- adolescents may exaggerate their condition (aggravation), play it down (dissimulation) or feign a nonexistent disease (simulation).

Questions are asked in language the child and the parents understand; a question should not contain a ready answer (not “Does your tummy hurt on the right?” but “Where does it hurt? Can you show me?”).

History of present illness (anamnesis morbi)

The history of present illness establishes:

1. What the child was like before the illness.
2. When and how the illness began: **acutely** (within hours, 1–2 days) or **gradually**.
3. What the illness is thought to be associated with: chilling, contact with a sick person, a new food, taking a medicine, an injury.
4. The first signs and their subsequent course (dynamics).
5. What investigations were done and their results (the outpatient card and discharge summaries should be reviewed).
6. What treatment was received (including self-treatment) and its effect.
7. The present condition.

> **REMEMBER**
> **A short history** (from a few hours to 1–2 weeks) indicates an acute process, and **a long history** (weeks, months, years) a chronic disease. In a chronic disease the parents often confuse the duration of the disease with the duration of the last exacerbation — that is why the medical records should be reviewed.

After the history of present illness has been taken, a preliminary assumption is made about which organ or system is affected and about the course of the disease (acute or chronic).

Life history (anamnesis vitae)

The life history is taken in two ways depending on the child's age: **in young children (under 3 years)** in detail, starting from the mother's pregnancy and delivery, and in older children more briefly.

1. Obstetric (antenatal and intranatal) history:

- from which pregnancy and which delivery the child was born; how previous pregnancies ended (abortion, miscarriage, stillbirth, preterm delivery);
- how the first and second halves of pregnancy proceeded: toxicosis, gestosis, anemia, threatened miscarriage, the mother's illnesses (especially infections), medicines taken, harmful habits (smoking, alcohol), occupational hazards, nutrition;
- the delivery: at term or preterm, its duration and complications, whether a cesarean section, obstetric forceps or a vacuum extractor was used, the duration of the period after rupture of the membranes.

> **CLINICAL SIGNIFICANCE**
> Spontaneous miscarriages in the mother and threatened abortion may indicate hereditary pathology or intrauterine infection. Gestosis, abnormal labor and a low Apgar score lead to cerebral hypoxia, which later becomes the basis for diseases of the nervous system.

2. Neonatal period: whether the child cried at once and what the cry was like (loud or weak); whether resuscitation was needed; the Apgar score; the birth weight, length and head and chest circumferences; when the baby was put to the breast and how he sucked; when the umbilical stump fell off and how the umbilical wound healed; the degree and duration of physiological jaundice; the

physiological weight loss and its recovery; illnesses in the newborn period; on which day and with what weight the baby was discharged from the maternity hospital.

3. Feeding history: breast, mixed or formula feeding; for how many months the baby was exclusively breastfed; when he was weaned; the type and amount of formula; when and in what order complementary foods were introduced; whether vitamin D was given to prevent rickets; when the child moved to the family table; present diet, appetite.

> **CLINICAL SIGNIFICANCE**
> Early formula feeding, non-adapted formulas and failure to introduce complementary foods on time lead to allergic diseases, digestive diseases, anemia, rickets and hypotrophy. For example, in celiac disease the arrest of weight gain begins a few weeks (usually 1–2 months) after gluten-containing cereals are introduced.

4. Physical and neuropsychological development: when the birth weight doubled; the weight and length at 1 year; when the first teeth erupted and the number of teeth at 1 year; when the child began to hold up his head, smile, babble, roll over, sit, crawl, stand and walk; the first words and phrases; in an older child, school performance and behavior in a group.

5. Past illnesses: what illnesses the child had before and after 1 year of age, their severity and complications; rickets, anemia, diatheses; infectious diseases; operations, injuries, blood transfusions (risk of hepatitis B and C and HIV infection).

6. Immunizations: whether the child has been immunized according to the national immunization schedule; reactions and complications after vaccination; medical contraindications; the results of the Mantoux test.

7. Allergic history: reactions to drugs, foods, household and other allergens (rash, swelling, bronchospasm). This information is necessary for choosing drugs and preventing complications.

8. Hereditary (family) history: the parents' ages and health; chronic, hereditary, allergic, endocrine and mental diseases, tuberculosis, syphilis and alcoholism in the family; consanguineous marriage; the other children in the family and their health, children who have died and the causes of death. The hereditary history is drawn up as **a pedigree (family tree)** covering at least three generations.

9. Social and living conditions: the family's financial situation; housing conditions (area, lighting, dampness, heating, ventilation, number of occupants); who looks after the child; whether the child has a separate bed; daily routine, sleep, walks; bathing; smoking in the family.

10. Epidemiological history: contact with an infectious patient (influenza, ARVI, intestinal infections, tuberculosis, hepatitis) during the last 3 weeks; the quality of food products and water; whether the child has traveled recently. The epidemiological history completes the life history and decides to which ward the patient will be admitted.

Features of talking with the parents, the child and the adolescent

Talking with the parents. Parents describe signs mixed with their own emotions, so the information should be assessed objectively. For example, parents associate a chronic cough with “frequent colds” and do not think of the possibility of bronchial asthma. The conversation takes place in a friendly atmosphere; helping (open) questions are more effective. It is useful to ask at the end of the conversation, “Is there anything else you would like to tell me about your child?” — sometimes the real reason for the visit comes out at this moment.

Talking with the child (5 years and older). A child often describes symptoms more accurately than the parents. The conversation begins with an interesting, non-frightening topic (games, school,

clothes). Then questions the child can understand are asked: “Show me where it hurts. Is the pain strong, or like a needle prick? Does the pain stay in one place or move somewhere else?”

Talking with the adolescent. An adolescent opens up when addressed respectfully, like an adult. The conversation should begin with his interests, friends and school. Confidentiality is guaranteed. Confrontation and long silences make the adolescent close up.

Conclusion from the history

After the history has been taken, a general conclusion is drawn: the presumed pathology, its nature (**hereditary, congenital or acquired**), its course (**acute or chronic**) and the **risk factors** that contributed to the development of the disease are indicated.

1.6. Structure of the case history

The case history is the main medical and legal document reflecting the process of examining, diagnosing and treating the patient. During the course the student writes a training case history according to the following scheme.

Table 1.3. *Scheme of the training case history*

Section	Content
I. Personal data	Full name, age, address, information about the parents, date of admission, diagnosis of the referring institution
II. Complaints	Main and general complaints on admission and at the time of examination
III. History of present illness	Onset, development, investigations, treatment and its effect
IV. Life history	Obstetric, neonatal, feeding, developmental, past illnesses, immunization, allergic, hereditary, social and epidemiological history; conclusion from the history
V. Objective examination (status praesens)	General inspection, physical development, examination by systems (inspection, palpation, percussion, auscultation)
VI. Conclusion from the history and status	Preliminary syndromal diagnosis
VII. Additional investigations	Laboratory and instrumental investigations and their interpretation
VIII. Justification of the syndromal diagnosis	Based on complaints + history + objective + additional data
IX. Calculation of feeding	Daily and single feed volumes, requirements for protein, fat, carbohydrate and energy, a one-day menu
X. Daily notes and temperature chart	Results of observing the patient
XI. References	—

1.7. General rules of objective examination

The following conditions are met for examining a child:

- the room is warm — air temperature **20–22 °C**; if the temperature is below 20 °C, the child is examined partly undressed;

- the lighting is sufficient, preferably **natural daylight**; the child's face is turned toward the light source; in direct light the skin color is seen well, and in side lighting the movements on the body surface (pulsation, peristalsis);
- the child is **fully undressed**: under 3 years — completely; school-age children (especially girls during puberty) are undressed gradually during the examination;
- young children are examined **on a changing table**, older children lying or sitting on a couch or bed; having the parents nearby calms the child;
- the physician stands **on the patient's right side**;
- **a sleeping child is not woken**: during sleep the respiratory rate and pulse are counted and the abdomen is palpated;
- procedures unpleasant for the child (inspection of the throat and ears, examination of a painful area, instrumental investigations) are performed **at the end of the examination**;
- in a preschool child the examination is conducted not according to a strict scheme but according to the child's mood; a restless child is distracted with play.



Figure 1.1. *Preparing a newborn for examination on a changing table (Navruzova, Geppe)*



Figure 1.2. *Preparing an older child for general inspection (Navruzova, Geppe)*



Figure 1.3. *Examining an infant in the mother's arms (Kildiyarova)*



Figure 1.4. *Inspecting the throat in an infant — this procedure is performed at the end of the examination (Kildiyarova)*

1.8. General inspection

The general inspection begins already during the conversation with the child and the mother: the physician notes the child's behavior, movements, voice and response to the surroundings. General inspection is of great diagnostic value — the direction toward the diagnosis is often obtained at first glance.

The general inspection assesses:

1. The general condition (degree of severity).
2. Well-being and mood.
3. The state of consciousness.
4. The patient's position (active, passive, forced).
5. Body build (constitution type), posture, gait.
6. Facial expression.
7. Stigmas of dysembryogenesis and visible malformations.
8. Physical development (see topic 2).

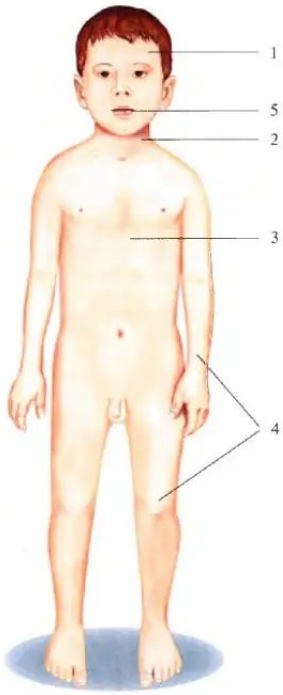


Figure 1.5. *Stages of general inspection in an older child: 1 — head, 2 — neck, 3 — trunk, 4 — limbs, 5 — oral cavity (inspected last) (Navruzova, Geppe)*

The general condition and the criteria for assessing it

The general condition is the physician's **objective conclusion** based on the history and the findings of objective and additional investigations. It determines the scope and order of treatment, the need for additional investigations and where the patient is admitted (intensive care, isolation box, ordinary ward). The condition may change during the day, even within hours.

Grades of the general condition: **satisfactory, moderately severe, severe** and **extremely severe (terminal)**.

REMEMBER

Two main criteria are used to assess the general condition: 1. **The degree of intoxication (toxicosis)** — the main criterion in acute diseases. 2. **The degree of functional impairment of organs and systems** (respiratory, cardiac, renal, hepatic failure) — the main criterion in chronic, long-lasting diseases. In blood diseases the blood count and myelogram, and in kidney diseases the urinalysis and functional tests, are also taken into account.

Table 1.4. *Criteria for assessing the severity of the general condition in acute diseases*

Condition	Clinical description
Satisfactory	The child is active and interested in the surroundings; weakness and restlessness absent or mild; body temperature normal or subfebrile; skin pale pink or slightly flushed; changes in organs and systems absent or mild
Moderately severe	Intoxication is evident: the child is weak, reluctant to communicate, drowsy; temperature 38–39 °C; skin flushed or pale-cyanotic; increased respiratory rate, tachycardia; vomiting, loose stools and, in young children, abdominal distension may occur; the functions of vital organs are subcompensated

Condition	Clinical description
Severe	Impaired consciousness (stupor, sopor, coma); convulsions may occur; temperature 39–40 °C; skin pale, gray or mottled (impaired microcirculation); dyspnea; tachycardia or bradyarrhythmia; repeated vomiting, intestinal paresis (distended abdomen), diarrhea; reduced diuresis; the main physiological systems are decompensated
Extremely severe	Deepening of the above signs and signs directly threatening life; the patient is in the intensive care unit

Well-being

Well-being is the patient's **subjective** sensation. It is good, satisfactory or unsatisfactory and depends on mood. The mood may be even, calm, elevated, excited or unstable; the child's interest in the people around and in toys is noted.

> **REMEMBER**
> The general condition and well-being **are not the same thing**. The general condition is assessed objectively by the physician, and well-being subjectively by the patient. Even with many complaints the general condition may be satisfactory, and vice versa: for example, at the onset of severe poisoning a child may feel well.

State of consciousness

Table 1.5. *Degrees of impaired consciousness*

Degree	Description
Clear	The child responds appropriately to the surroundings, willingly makes contact and answers questions correctly
Somnolence (stupor, torpor)	There is a response to the surroundings, but it is slowed and reduced; the child answers questions correctly but with delay and sluggishly; responds to stimulation by crying
Sopor	Consciousness is clouded, there is no response to the surroundings, the child does not answer questions or gives the same answer, but the response to pain is preserved
Coma	Unconscious, no active movements, no or a very weak response to pain, reflexes reduced or absent; divided into grades I–IV according to its depth

Consciousness may also be impaired in the form of **excitation**: euphoria, negativism, psychomotor agitation, delirium (with hallucinations). In young children intoxication often begins with **lethargy (apathy) and reduced movement (adynamia)**, alternating from time to time with restlessness, after which stupor, sopor, coma and convulsions develop.

> **CLINICAL SIGNIFICANCE**
> The appearance of drowsiness and lethargy in a young child is always a serious sign. The complaint “the child sleeps a lot” may hide the onset of neurotoxicosis, meningitis, severe dehydration or hypoglycemia.

The patient's position

- **Active position** — the child moves freely and readily changes position at the physician's request or according to circumstances.
- **Passive position** — in unconsciousness or extreme weakness: the patient is motionless, the head and limbs hang down, and he cannot change position himself.
- **Forced position** — a specific position the patient adopts to relieve his condition.

Table 1.6. *Examples of forced positions*

Disease	Forced position
Meningitis	Lying on the side with the head thrown back, the legs drawn up to the abdomen and the abdomen drawn in (the “pointer dog” or “cocked gun” position)
Attack of bronchial asthma	Sitting, propped with the hands on the edge of the bed or on the knees (orthopnea)
Heart failure	Sitting or half-sitting position
Exudative pleurisy, lobar pneumonia	Lies on the affected side (the excursion of the healthy lung is preserved)
Acute peritonitis	On the back with the legs bent, does not move — every movement causes pain
Tetralogy of Fallot (cyanotic spell)	Squatting (sitting on the haunches)



Figure 1.6. *Forced position of a child with meningitis: on the side, head thrown back, legs bent (Navruzova, Geppe)*

Body build (constitution), posture and gait

Constitution is the set of relatively stable morphological and functional features of the body formed under the influence of heredity and environment. Three types of body build are distinguished:

Table 1.7. *Types of body build*

Type	Signs
Normosthenic	The body is proportionally developed; the epigastric angle is about 90°
Asthenic	The length of the body predominates over its width; long limbs, a long and narrow chest; acute epigastric angle (less than 90°); poorly developed muscles and subcutaneous fat
Hypersthenic	The width of the body predominates over its length; relatively short limbs, a broad chest, a large abdomen; obtuse epigastric angle (more than 90°)

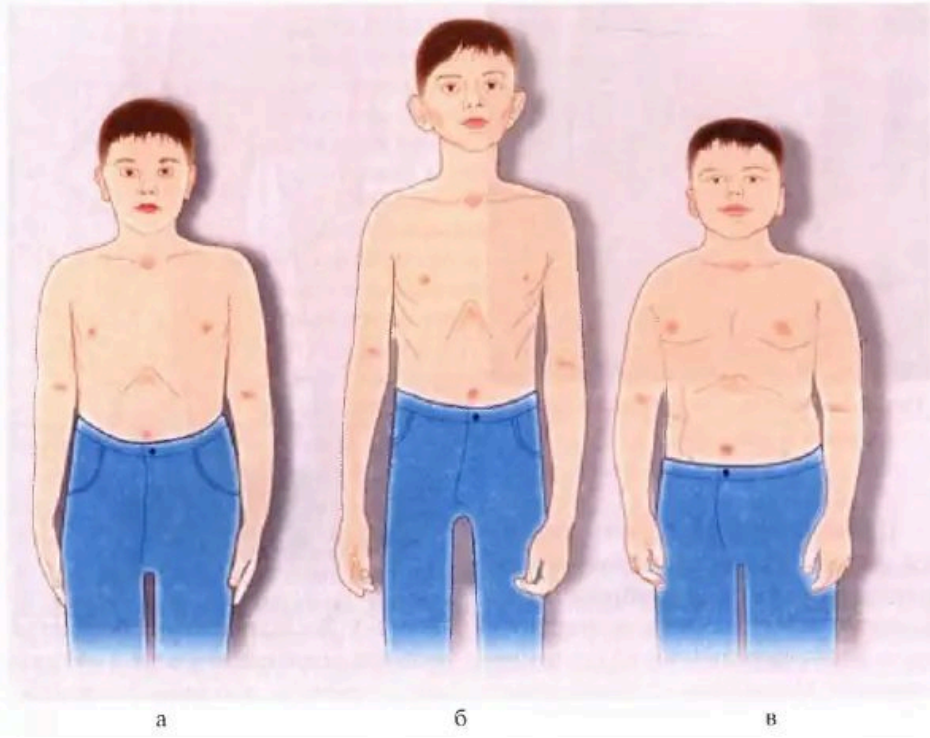


Figure 1.7. *Appearance of children with different body builds (Navruzova, Geppe)*

Posture and gait. A healthy child holds the body upright, walks briskly, and moves freely. Stooping, scoliosis, limping, a “waddling” gait (hip dislocation, muscular dystrophy) and an ataxic gait (cerebellar damage) are noted.

Facial expression

The facial expression gives much information about the child's condition and some diseases. The face of a healthy child is calm, and the eyes are clear and curious.

- **Suffering face** — in pain (otitis, abdominal pain).
- **Hippocratic face** — sunken eyes and temples, a sharpened nose, gray skin, cold sweat — peritonitis, severe dehydration, a terminal state.
- **Adenoid face** — mouth constantly open, puffy face, flattened nasolabial folds, no nasal breathing.
- **Nephritic face** — edema of the face and eyelids, pale skin.
- **Down syndrome** — slanting palpebral fissures (mongoloid slant), epicanthus, flat face, open mouth and a large tongue.
- **Cushingoid (“moon”) face** — round, flushed face (glucocorticoid excess).
- **“Sardonic smile”** — spasm of the facial muscles in tetanus.

Stigmas of dysembryogenesis

Stigmas of dysembryogenesis are minor anomalies in the shape or structure of an organ that do not impair its function. Several stigmas occur as a familial trait even in healthy people.

> **REMEMBER**
> **Up to 5–6** stigmas in one child are considered borderline. If there are more stigmas (**the stigmatization threshold is exceeded**), malformations of the internal organs (heart, genitourinary system) and hereditary syndromes should be sought. The degree of stigmatization is assessed by their total number regardless of location.

Table 1.8. *Common stigmas of dysembryogenesis*

Part of the body	Stigmas
Skull	Microcephaly, hydrocephaly, dolichocephaly, brachycephaly, asymmetry; flat forehead, prominent brow ridges
Face	Broad nasal root, “bird face”, prognathism, micrognathia, progenia, cleft chin
Eyes	Mongoloid or antimongoloid slant, hypertelorism (wide distance between the inner corners of the eyes), epicanthus, distichiasis (double row of eyelashes), coloboma and heterochromia of the iris, coloboma of the eyelid, microphthalmia
Ears	Large, protruding, small or deformed, low-set, asymmetric ears; fused earlobes; accessory auricular tubercle
Mouth	Microstomia, macrostomia, “Gothic” (high-arched) palate, short frenulum of the tongue, macroglossia, furrowed tongue
Teeth	Missing teeth, malposition, overlapping teeth
Neck	Short or long neck, pterygium folds
Trunk	“Pigeon chest”, “cobbler's chest”, widely spaced nipples, accessory nipples, low-set umbilicus, diastasis of the rectus abdominis muscles
Genitals	Cryptorchidism, phimosis, underdevelopment of the penis or labia, enlarged clitoris
Skin	Pigmented spots, hairy congenital nevi, hemangiomas, fistulous tracts
Hands	Transverse palmar crease, polydactyly, brachydactyly (short fingers), arachnodactyly (long thin fingers), syndactyly (fused fingers), clinodactyly (curved finger), short incurved little finger
Feet	Syndactyly, a “sandal gap” between the 1st and 2nd toes, overlapping toes



Figure 1.8. *Coloboma of the upper eyelid (Navruzova, Geppe)*

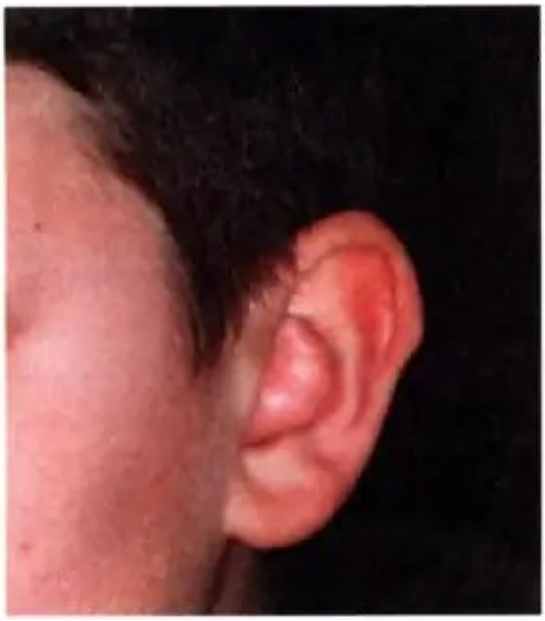


Figure 1.9. *Malformation of the ear: protruding and deformed ear (Navruzova, Geppe)*



Figure 1.10. *Brachydactyly and a transverse palmar crease (Navruzova, Geppe)*



Figure 1.11. *Complete syndactyly of the 3rd and 4th fingers (Navruzova, Geppe)*



Figure 1.12. *Curvature of the little finger — clinodactyly (Navruzova, Geppe)*



Figure 1.13. *A “sandal gap” between the 1st and 2nd toes and syndactyly of the 2nd and 3rd toes (Navruzova, Geppe)*

Assessing the newborn's condition: the Apgar score

The newborn's condition is assessed with the Apgar score **at 1 and 5 minutes** after birth. Each sign is scored 0, 1 or 2 points. The initial letters of the English names of the signs form the word “APGAR”.

Table 1.9. *The Apgar score*

Sign	0 points	1 point	2 points
A — skin color (appearance)	Generalized cyanosis or pallor	Body pink, limbs bluish	Whole body pink
P — heart rate (pulse)	Absent	Less than 100 per minute	More than 100 per minute
G — reflex irritability (grimace)	Absent	Grimaces	Cries, coughs, sneezes
A — muscle tone (activity)	Absent, limbs hanging limp	Limbs slightly flexed	Active movements
R — respiration	Absent	Slow, irregular, weak	Active, loud cry

Interpretation: **8–10 points** — satisfactory condition; **6–7 points** — mild asphyxia; **4–5 points** — moderate asphyxia; **0–3 points** — severe asphyxia. Respiratory distress is assessed with **the Silverman score** in preterm newborns and with **the Downes score** in term newborns (topic 2).

Recording the general inspection in the case history

The result of the general inspection is recorded in the case history in the following order:

1. General condition: satisfactory / moderately severe / severe.
2. Well-being and mood.
3. Consciousness: clear / somnolence / sopor / coma.
4. Response to the surroundings: appropriate / inappropriate.
5. Position: active / passive / forced (which one).
6. Body build: normosthenic / asthenic / hypersthenic.
7. Stigmas of dysembryogenesis: none found, or those found are listed.

Sample: “General condition moderately severe due to intoxication. Well-being unsatisfactory, mood unstable, capricious. Consciousness clear, response to the surroundings appropriate. Position active. Body build normosthenic. Stigmas of dysembryogenesis: Gothic palate, low-set ears (2).”

1.9. Clinical part

The history is half the diagnosis

Experienced pediatricians say that “a correctly taken history gives half the diagnosis”. The following examples show the practical value of the history:

- **Relation to food.** If arrest of weight gain, an enlarging abdomen and large amounts of fatty stools began in a 9-month-old child 1–2 months after semolina porridge was introduced — celiac disease should be considered.
- **Cough during feeding.** Attacks of coughing and cyanosis with every feed in a newborn are a sign of a tracheoesophageal fistula.
- **Family history.** A dry night cough in a child whose parents and elder brother have allergic rhinitis and eczema is characteristic of bronchial asthma.
- **Epidemiological history.** A child who had contact with a child with measles at the kindergarten and has developed fever and conjunctivitis is in the catarrhal period of measles; such a child is admitted to an isolation box ward.
- **Birth history.** Delayed motor development later in a child born with severe asphyxia (Apgar 3 points) is a consequence of perinatal encephalopathy.

The importance of assessing severity correctly

Assessing the general condition is not just for the record — it determines where the patient is admitted, how quickly care is given and how he is monitored.

> **CLINICAL SIGNIFICANCE**
>
> **In an acute disease** the main criterion is intoxication. For example, if a child with ARVI has a fever of 39.5 °C, lethargy, mottled skin and cold hands and feet, the condition is severe (“white hyperthermia”, impaired microcirculation). **In a chronic disease** the main criterion is functional failure.
>
> For example, in a child with a congenital heart defect, even without fever, dyspnea at rest and an enlarged liver (grade II heart failure) make the condition severe.

The forced position — a key to the diagnosis

The patient's position often points to the diagnosis already in the admission room: a febrile child lying with the head thrown back and the legs bent — meningitis; a child sitting propped on the hands and breathing with a wheeze — an asthma attack; a cyanotic child squatting — tetralogy of Fallot.

Practical skills

Skill 1. Taking the history

1. Wash the hands, introduce yourself, establish contact with the parents (and the child) and obtain consent.
2. Establish the personal data.
3. Establish the main and general complaints; explore each complaint in detail (site, character, duration, aggravating and relieving factors).
4. Take the history of present illness: onset, cause, development, investigations, treatment and its effect.
5. Take the life history (appropriate to age): obstetric, neonatal, feeding, developmental, past illnesses, immunizations, allergic, hereditary, social and living conditions, epidemiological history.
6. Draw up a pedigree (at least three generations).
7. At the end of the conversation, give the parents the opportunity to ask further questions.
8. Draw a conclusion from the history: the presumed affected system, the nature and course of the disease, the risk factors.

Skill 2. Emergency assessment (triage)

1. Assess the child immediately on arrival (within 1–2 minutes) by ABCD.
2. **A** — patency of the airway: stridor, signs of a foreign body.
3. **B** — breathing: present or absent, respiratory distress (accessory muscles, nasal flaring, chest indrawing), central cyanosis.
4. **C** — circulation: feel the temperature of the hands, determine the capillary refill time (press on the nail for 3 seconds and release; if the color returns more slowly than in 3 seconds — pathology), pulse; consciousness (AVPU) and convulsions.
5. **D** — dehydration in a child with diarrhea: lethargy, sunken eyes, how fast the skin pinch goes back.
6. If there is an emergency sign — start treatment immediately and call a senior physician; if there is a priority sign — send for examination without waiting; if there are no signs — the general queue.
7. In a child aged 2 months to 5 years, ask about and check the general danger signs.

Skill 3. General inspection and assessment of severity

1. Ensure the conditions for examination: room temperature 20–22 °C, daylight, warm clean hands.
2. During the conversation, observe the child's behavior, movements, voice and response to the surroundings.
3. Undress the child appropriately for age and place him in a position convenient for examination.
4. Assess consciousness (answers to questions, response to voice and pain).

5. Determine the patient's position (active, passive, forced).
6. Assess the signs of intoxication: weakness, temperature, skin color, respiratory rate, pulse, vomiting, diuresis.
7. Identify the signs of functional impairment of organs (respiratory, heart failure and others).
8. Assess the body build, posture, gait and facial expression.
9. Look for and count the stigmas of dysembryogenesis in sequence: head, face, ears, mouth, neck, trunk, limbs.
10. Determine the grade of the general condition and record it in the case history.

Skill 4. Drawing up a pedigree (family tree)

1. The sick child (proband) is marked — he is indicated with an arrow.
2. Males are indicated by a square and females by a circle; the symbol of an affected person is shaded; a line is drawn through the symbol of a deceased person.
3. Members of one generation are placed in one horizontal row in birth order starting from the left; generations are indicated by Roman numerals (I — grandparents, II — parents and their brothers and sisters, III — the proband and his brothers and sisters).
4. Spouses are joined by a horizontal line and their children by a vertical line; a consanguineous marriage is shown by a double line.
5. The disease of each person is recorded and a conclusion is drawn about hereditary predisposition.

Review questions

1. State the subject, aim and tasks of the propaedeutics of childhood diseases.
2. List the periods of childhood according to N.P. Gundobin and state the main feature of each.
3. What are the specific aspects of medical deontology in pediatrics?
4. What is emergency assessment (triage)? Into which groups are patients divided in the ETAT system?
5. List the general danger signs of the IMCI program.
6. What are the features of history taking in pediatrics? What parts does the history consist of?
7. How is the life history taken in a young child?
8. State the scheme of the case history.
9. What are the main criteria for assessing the severity of the patient's condition? Which criterion predominates in acute and in chronic diseases?
10. What is the difference between the general condition and well-being?
11. Describe the degrees of impaired consciousness.
12. Give examples of forced positions of the patient.
13. Describe the types of body build.
14. What are stigmas of dysembryogenesis, and what number of them is clinically significant?

Test questions

1. Not a task of the propaedeutics of childhood diseases:
   A) Studying the anatomical and physiological features of the child's body
   B) Studying the methods of examining the child
   C) Studying the semiotics of diseases
   D) Surgical treatment of childhood diseases
   E) Studying the rules of feeding a healthy child

2. The newborn (neonatal) period lasts:
   A) From birth to 7 days
   B) From birth to 28 days
   C) From birth to 3 months
   D) From birth to 6 months
   E) From birth to 1 year

3. The main feature of history taking in pediatrics:
   A) The history is taken only from the child
   B) The history is taken mainly from the parents and supplemented by questioning the child according to age
   C) The life history is not taken
   D) The epidemiological history is taken only in adults
   E) The hereditary history is not important

4. The main criterion for assessing the severity of the patient's condition in acute diseases:
   A) The number of complaints
   B) The degree of intoxication
   C) Physical development
   D) The type of body build
   E) The number of stigmas of dysembryogenesis

5. The main criterion for assessing severity in chronic diseases:
   A) The degree of intoxication
   B) Body temperature
   C) The degree of functional failure of organs and systems
   D) The patient's well-being
   E) Mood

6. Which of the following is not a grade of the general condition?
   A) Satisfactory
   B) Moderately severe
   C) Severe
   D) Good
   E) Extremely severe

7. Does not answer questions, but the response to pain is preserved. This state of consciousness is:
   A) Clear
   B) Somnolence
   C) Sopor
   D) Coma
   E) Delirium

8. The forced position characteristic of meningitis:
   A) Squatting
   B) Lying on the side with the head thrown back and the legs drawn up to the abdomen
   C) Sitting propped on the hands
   D) On the back with the limbs extended
   E) Lying on the abdomen

9. Characteristic of the asthenic body build:
   A) Obtuse epigastric angle
   B) Short limbs, large abdomen
   C) Body length predominates over width, acute epigastric angle
   D) Epigastric angle 90°
   E) Broad and short chest

10. The borderline number of stigmas of dysembryogenesis:
    A) 1–2
    B) 3
    C) 5–6
    D) 10
    E) 15

11. The room temperature when examining a child should be:
    A) 16–18 °C
    B) 20–22 °C
    C) 24–26 °C
    D) 26–28 °C
    E) It does not matter

12. Procedures unpleasant for the child (inspection of the throat) are performed:
    A) At the beginning of the examination
    B) In the middle of the examination
    C) At the end of the examination
    D) They are not performed at all
    E) Only under anesthesia

13. Which of the following is a general danger sign of the IMCI program?
    A) Nasal congestion
    B) The child is unable to drink or breastfeed
    C) Subfebrile temperature
    D) A single episode of vomiting
    E) Mild cough

14. An Apgar score of 6–7 points is assessed as:
    A) Satisfactory condition
    B) Mild asphyxia
    C) Moderate asphyxia
    D) Severe asphyxia
    E) Clinical death

15. In the AVPU scale the letter “P” means:
    A) The child is alert
    B) Responds only to voice
    C) Responds only to pain
    D) Does not respond to anything
    E) The child is convulsing

16. The scholar who first introduced keeping daily records of the patient's condition in hospital (the basis of the case history):
    A) Abu Bakr al-Razi
    B) S.Sh. Shamsiyev
    C) Hippocrates
    D) N.F. Filatov
    E) Abu Ali ibn Sina

17. The scholar who created the classification of the periods of childhood that is still used today:
    A) N.P. Gundobin
    B) M.S. Maslov
    C) G.N. Speransky
    D) A.F. Tur
    E) A.A. Kisel

18. Which age does early childhood (the toddler period) cover?
    A) 12–17 years
    B) From 29 days to 1 year
    C) 7–11 years
    D) 3–7 years
    E) 1–3 years

19. Which of the following is a sign of shock in triage?
    A) The child is asleep and wakes when roused
    B) A slight decrease in appetite
    C) Cold hands, capillary refill time longer than 3 s, rapid and weak pulse
    D) Mild cough
    E) Temperature 37.5 °C

20. In the WHO triage system, to which group does an infant under 2 months belong?
    A) Q — queue
    B) E — emergency
    C) The general queue
    D) Not assigned to a group
    E) P — priority

21. At what age are the general danger signs of the IMCI program checked in a sick child?
    A) Only over 12 years
    B) 5–10 years
    C) 1–3 years
    D) From 2 months to 5 years
    E) From birth to 28 days

22. Which question in history taking is asked correctly (without a ready answer)?
    A) You had a high fever, didn't you?
    B) The pain gets worse at night, right?
    C) Where does it hurt? Can you show me?
    D) Do you always cough like this?
    E) Does your tummy hurt on the right side?

23. How are males and females indicated in a pedigree (family tree)?
    A) Males by a square, females by a triangle
    B) Males by a square, females by a circle
    C) Males by a triangle, females by a circle
    D) Both by a circle
    E) Males by a circle, females by a square

24. The type of history that completes the life history and decides to which ward the patient will be admitted:
    A) Social and living conditions
    B) Epidemiological
    C) Allergic
    D) Obstetric
    E) Hereditary

25. What is it called when a patient plays down (denies) an existing disease?
    A) Remission
    B) Metastasis
    C) Aggravation
    D) Dissimulation
    E) Simulation

26. In the hereditary (family) history, the pedigree covers at least how many generations?
    A) 3
    B) 5
    C) 4
    D) 2
    E) 1

Clinical cases

Case 1

A child aged 1 year 2 months was brought to the admission room. According to the mother, he has had a fever of 39 °C and a cough for 2 days. Since this morning he has not been breastfeeding and vomits the water he drinks. On examination the child is lethargic, barely opens his eyes and responds weakly to voice. Respiratory rate 58 per minute; the lower part of the chest is drawn in on inspiration.

Questions: 1) What general danger signs does the child have? 2) To which triage group does the child belong? 3) Assess his general condition.

Answer: 1) There are two general danger signs: the child is unable to breastfeed and vomits everything; in addition, lethargy is also a general danger sign. 2) Chest indrawing and tachypnea indicate severe respiratory distress, that is, an emergency (E) sign; lethargy is a priority sign. The child belongs to the emergency group, and treatment is started immediately. 3) The general condition

is severe: obvious intoxication (lethargy, reduced consciousness — weak response to voice, V on AVPU) and signs of respiratory failure. Presumptive diagnosis — severe pneumonia; urgent hospital admission is necessary.

Case 2

A 10-year-old girl came with the complaint of “tummy ache”. Her mother is very anxious: her daughter says that her tummy hurts every morning before going to school. There is no pain at weekends. Objective examination revealed no changes; the condition is satisfactory.

Questions: 1) Which part of the history gives an important direction for the diagnosis? 2) How should the conversation with the child be organized? 3) Explain the difference between the condition and well-being.

Answer: 1) The appearance of pain only on school mornings and its absence at weekends suggest psychogenic (functional) pain; the social history (the school environment, performance, relations with classmates, the family situation) should be taken in detail. At the same time, organic causes (digestive diseases) are excluded. 2) The conversation begins with an interesting, non-frightening topic; it is also useful to talk with the child alone, without the mother; open questions are asked (“How does your day at school go? What upsets you?”); the child is asked to show where it hurts. 3) The condition (the physician's objective assessment) is satisfactory; the well-being (the girl's subjective sensation) may be unsatisfactory because of the pain. It is natural for these two concepts not to coincide.

Case 3

On assessing a newborn at 1 minute: the body is pink and the limbs bluish; heart rate 110 per minute; grimaces when a nasal catheter is inserted; the limbs are slightly flexed; breathing is slow and irregular, and the cry is weak.

Questions: 1) Score by the Apgar scale. 2) Draw a conclusion. 3) When is the reassessment performed?

Answer: 1) Skin color — 1 point, heart rate (more than 100) — 2 points, reflex irritability (grimace) — 1 point, muscle tone (slightly flexed) — 1 point, respiration (slow, irregular) — 1 point. Total: 6 points. 2) Mild asphyxia (6–7 points). 3) The reassessment is performed at 5 minutes; if the score is below 7, the assessment is continued every 5 minutes up to 20 minutes, and resuscitation measures are taken.

Quick review

Table 1.10. *Key facts on Topic 1*

Question	Answer
Meaning of the word propaedeutics	“I teach beforehand” (Greek)
First introduced the case history	Abu Bakr al-Razi
Author of the classification of the periods of childhood	N.P. Gundobin
Neonatal period	0–28 days
Infancy	29 days – 1 year

Question	Answer
Order of examination	HELP: history → objective examination → conclusion → plan
Triage groups	E (emergency), P (priority), Q (queue)
Order of the emergency signs	ABCD: airway, breathing, circulation/coma/convulsions, dehydration
Normal capillary refill time	Less than 3 seconds
Rapid assessment of consciousness	AVPU: alert, responds to voice, responds to pain, unresponsive
General danger signs (IMCI)	Cannot drink, vomits everything, has had convulsions, lethargic or unconscious, convulsing now
Criteria of the condition	In acute disease — intoxication; in chronic disease — functional failure
Grades of the condition	Satisfactory, moderately severe, severe, extremely severe
Well-being	Subjective: good, satisfactory, unsatisfactory
Sopor	No response, reaction to pain preserved
Stigma threshold	5–6
Examination conditions	20–22 °C, daylight, physician on the right
Apgar	At 1 and 5 minutes; 8–10 normal, 6–7 mild, 4–5 moderate, 0–3 severe asphyxia
Aggravation, dissimulation, simulation	Aggravation — exaggerating the signs; dissimulation — playing down or hiding an existing disease; simulation — feigning a nonexistent disease (seen in adolescents)
Abu Bakr al-Razi (865–925)	“A Treatise on Smallpox and Measles”, “Diseases of Children”; introduced daily recording of the patient's condition — the basis of the case history
Abu Ali ibn Sina (980–1037)	“Canon of Medicine”: chapter on newborn care, breastfeeding, choosing a wet nurse and childhood diseases
Components of the history	Personal data → complaints → history of present illness (a. morbi) → life history (a. vitae)
The 10 sections of the life history	Obstetric, neonatal, feeding, physical and neuropsychological development, past illnesses, immunizations, allergic, hereditary, social and living conditions, epidemiological
Epidemiological history	Contact with an infectious patient during the last 3 weeks, quality of food and water, travel; completes the life history
ETAT	Emergency Triage Assessment and Treatment — the WHO system of emergency triage, assessment and treatment; groups: E (emergency), P (priority), Q (queue)
Pedigree symbols	Proband — arrow; male — square, female — circle; affected — filled; deceased — crossed by a line; consanguineous marriage — double line; generations in Roman numerals

TOPIC 2

Physical development of the child. The newborn and the preterm newborn

Learning objectives

The student should know:

- the concept of physical development and the factors affecting it;
- the technique of anthropometric measurements;
- the patterns of change of length (height), weight, and head and chest circumferences with age, and the formulas for approximate calculation;
- the methods of assessing physical development: sigma, centile, WHO Z-score method, indices;
- the semiotics of disorders of physical development;
- the anatomical and physiological features of the newborn and the borderline (transitional) states;
- the degrees and signs of prematurity;
- the features of examining a newborn, the Apgar and Silverman scores.

The student should be able to:

- measure the length (height), weight, and head and chest circumferences of children of different ages;
- evaluate anthropometric indicators by formulas and centile tables and write a conclusion;
- examine a newborn and distinguish transitional states from pathology;
- identify the signs of prematurity;
- assess respiratory distress by the Silverman score.

2.1. The concept of physical development and the factors affecting it

Physical development is the set of morphological and functional features reflecting the process of growth and maturation of the body. In clinical pediatrics it is understood as the child's growth in length, weight gain, development of body parts and biological maturation in the different periods of childhood. Physical development is one of the most important criteria of a child's health.

Four groups of signs of physical development are distinguished:

- **Anthropometric** — length (height), weight, and the circumferences of the head, chest, arm, thigh and calf.
- **Functional** — vital capacity of the lungs, hand grip and back strength, physical working capacity.
- **Somatoscopic** — shape of the spine, degree of fat deposition, muscle development, posture, shape of the legs and feet.
- **Biological** — dental formula, body proportions, ossification centers, secondary sexual characteristics.

Table 2.1. *Factors affecting the child's growth and development*

Factor	Effect
Genetic	More than 100 genes control the synthesis of growth hormones and factors; they determine the rate of growth and the final height

Factor	Effect
Hormonal	Somatotropic hormone (growth hormone) — the main growth hormone; thyroid hormones, insulin; sex hormones at puberty
Nutrition	Protein-energy malnutrition and deficiency of vitamins, amino acids and calcium impair growth
Climatic and geographic	In hot climates and mountainous areas growth slows and puberty is accelerated
Lifestyle	A daily routine, enough sleep and physical activity create good conditions for growth
Diseases	Chronic and severe diseases slow or stop growth
Social factors	The family's financial situation, the quality of care, the parents' harmful habits

Acceleration and retardation

Acceleration (Latin acceleratio — speeding up) is the speeding up of children's growth and biological maturation compared with previous generations. The term was introduced in 1935 by the German scientist E. Koch. In the 20th century the length of newborns increased by 1–2 cm, that of 1-year-old children by 4–5 cm, and that of 15-year-old adolescents by an average of 20 cm over 100 years. Early eruption of permanent teeth, early sexual maturation and early menarche in girls are also signs of acceleration.

- **Harmonious acceleration** — growth in height and biological maturation speed up together.
- **Disharmonious acceleration** — only one indicator speeds up (for example, height increases rapidly, while sexual maturation occurs on time).

Retardation is a slowing of the rate of physical development. Causes: reduced physical activity (sitting for long periods at a computer), the parents' smoking, alcohol and drug use (intrauterine growth restriction), children's harmful habits. The alternation of periods of acceleration and retardation is called **the secular trend**.

2.2. Methods of assessing physical development and the technique of anthropometry

Physical development is assessed by three methods:

1. **Somatoscopic** — visual assessment of body build and proportions.
2. **Somatometric (anthropometric)** — measuring body weight, length (height), and head and chest circumferences.
3. **Physiometric** — measuring functional indicators with instruments (spirometry, dynamometry).

> **REMEMBER**
> All anthropometric measurements are made **in the morning, on an empty stomach**, after voiding urine and stool, with the same instruments. The measurements are repeated regularly: **under 1 year — every month; 1–3 years — every quarter; 3–7 years — twice a year; 7–17 years — once a year**.

Measuring length (height)

The length of children **under 2 years** is measured **lying down with a horizontal length board** (a board 80 cm long and 40 cm wide). The child is placed on the back with the head touching the fixed headboard. The head is held so that the lower margin of the orbit and the upper margin of the tragus are **in one vertical plane**. An assistant or the mother holds the head. With the right hand the

physician straightens the legs by pressing lightly on the knees, and with the left hand moves the sliding footboard against the heels (the feet flexed at a right angle). The distance between the two boards equals the child's length.

The height of children **over 2 years** is measured **standing with a vertical stadiometer**. The child stands with the back to the stadiometer, touching it with **the heels, buttocks, the area between the shoulder blades and the back of the head**; the arms hang along the body, the heels are together and the toes apart. The lower margin of the orbit and the upper margin of the tragus should be **in one horizontal plane**. The sliding headpiece is lowered without pressing on the head.



Figure 2.1. *Measuring the length of an infant on a horizontal length board (Navruzova, Geppe)*



Figure 2.2. *Measuring the height of a child over 2 years with a vertical stadiometer (Navruzova, Geppe)*

Measuring body weight

Children under 3 years are weighed on **electronic baby scales** with a maximum load of up to 15 kg. First a diaper is placed on the platform and the scale is set to zero (or the weight of the diaper is subtracted afterward), then the child is placed lying or sitting. Children over 3 years are weighed undressed on medical scales **to an accuracy of 50 g**; the child stands still in the middle of the platform.

Measuring the head and chest circumferences

- **Head circumference** — the tape measure is passed at the back over **the occipital protuberance** and at the front over **the brow ridges**.
- **Chest circumference** — the tape is passed at the back below **the lower angles of the shoulder blades** and at the front at the level of **the nipples**. In older children the chest circumference is measured three times — during quiet breathing, at maximal inspiration and at maximal expiration; the difference shows the chest excursion.



Figure 2.3. *Measuring the head circumference (Navruzova, Geppe)*



Figure 2.4. *Measuring the chest circumference (Navruzova, Geppe)*



Figure 2.5. *Weighing an infant on electronic scales (Navruzova, Geppe)*

2.3. Changes in anthropometric indicators with age and their approximate calculation

Length (height)

Length is the most stable and leading indicator of physical development. In assessing physical development, length is relied on first. The length of a term newborn is **48–52 cm** (average 50 cm; range 46–56 cm).

Table 2.2. *Growth in length in the first year*

Quarter (months)	Per month	Per quarter
1st quarter (0–3 months)	3 cm	9 cm
2nd quarter (4–6 months)	2.5 cm	7.5 cm
3rd quarter (7–9 months)	1.5–2 cm	4.5–6 cm
4th quarter (10–12 months)	1 cm	3 cm
Total in 1 year		≈ 25 cm

At 1 year the length is **75–77 cm**. In the second year the child grows 12–13 cm, and in the third year 7–8 cm.

Table 2.3. *Formulas for the approximate calculation of height in children over 1 year*

Age	Formula	Reference point
1–4 years	Height = 100 − 8 × (4 − n)	100 cm at 4 years (twice the birth length)
4–8 years	Height = 100 + 6 × (n − 4)	130 cm at 8 years
Over 8 years	Height = 130 + 5 × (n − 8)	≈ 150 cm at 12 years (3 times)

n — the child's age (years).

During puberty a **growth spurt** is seen: 8–10 (12) cm per year. The maximal growth occurs at **10–12 years in girls** and at **13.5–15.5 years in boys**. Growth in height ends at 16–17 years in girls and at 18–19 years in boys.

> **MEMORY AID**
> For height, “**50 – 75 – 100 – 130 – 150**”: 50 cm at birth, 75 cm at 1 year, 100 cm at 4 years (“one meter”), 130 cm at 8 years, 150 cm at 12 years. Height **doubles by 4 years** and **triples by 12 years**.

Body weight

The weight of a term newborn is **2600–4200 g** (average 3500 g in boys and 3200 g in girls). A newborn heavier than 4000 g is considered **large**; a weight below 2500 g is characteristic of prematurity or intrauterine growth restriction.

After birth **physiological weight loss** is seen: the maximal loss occurs on **days 3–4** and amounts to **5–8%** (permissible limit up to 10%); the weight is regained by **days 7–10**.

Weight gain:

- in the first half-year — on average **800 g** per month (25–30 g per day);
- in the second half-year — on average **400 g** per month;
- at 1 year the weight is on average **10.5 kg** — **3 times** the birth weight; the weight **doubles by 4.5–5 months**.

Formula for calculating weight in the first year: the weight of a 6-month-old child ≈ **8200 g**. For a child younger than 6 months, 800 g are subtracted for each missing month; for a child older than 6 months, 400 g are added for each subsequent month. Another method: 800 g for each month of the first half-year and 400 g for each month of the second half-year are added to the birth weight.

Table 2.4. *Calculating weight in children over 1 year*

Age	Formula or rule
2–11 years	Weight (kg) = 10.5 + 2n (n — age)
2–10 years	≈ 2 kg are added per year
10–15 years	3–4 kg are added per year
By height (over 3 years)	A height of 125 cm corresponds to 25 kg; 2 kg are subtracted for each missing 5 cm, and 3 kg (3.5 kg at puberty) are added for each extra 5 cm

MEMORY AID
For weight, the "**3 – 6 – 12**" rule: at 1 year the weight is **3 times** the birth weight; at 6–7 years it is **2 times** the weight at 1 year; at 12 years it is **3 times** the weight at 1 year.

Head circumference

At birth the head circumference is **34–36 cm**. In the first year it grows by 2 cm per month at 0–3 months, by 1 cm per month at 4–6 months and by 0.5 cm per month at 7–12 months. At 1 year it is ≈ **46 cm**, at 5 years ≈ **50 cm**, at 10 years ≈ **55 cm**.

Formulas: in a 6-month-old child the head circumference is **43 cm**; 1.5 cm are subtracted for each missing month, and 0.5 cm are added for each subsequent month. In a child over 1 year it is **50 cm** at 5 years; 1 cm is subtracted for each missing year, and 0.6 cm is added for each subsequent year.

Chest circumference

At birth the chest circumference is **32–34 cm** — **1–2 cm smaller** than the head circumference. **At 3–4 months** the two circumferences become equal, after which the chest circumference grows faster. After 1 year the chest circumference exceeds the head circumference by approximately **n to 2n cm** (n — age).

Formulas: in a 6-month-old child the chest circumference is **45 cm**; 2 cm are subtracted for each missing month, and 0.5 cm are added for each subsequent month. At 10 years it is **63 cm**; 1.5 cm are subtracted for each missing year, and 3 cm are added for each subsequent year.

CLINICAL SIGNIFICANCE

In a newborn, a head circumference **more than 2 cm larger** than the chest circumference may be a sign of hydrocephalus (for example, perinatal damage to the central nervous system). A head circumference small for age is microcephaly. When measuring, the molding of the skull bones during delivery (configuration), the caput succedaneum and edema are taken into account.

Proportionality indices and body proportions

- **The weight–length ratio** is the birth weight (g) / length (cm). In a healthy term newborn it is **60–80** (most often 60–65). **Less than 60** indicates intrauterine growth restriction (hypotrophy).
- **Body mass index (BMI, Quetelet index)** = weight (kg) / height $(m)^2$ — used in school-age children to assess nutritional status and obesity; it is evaluated with centile tables for age and sex.

Body proportions change with age: the height of the head is **1/4** of the body length in a newborn, **1/5** at 2 years, **1/6** at 6 years, **1/7** at 12 years and **1/8** in adults. The midpoint of the body lies above the umbilicus in a preterm newborn, at the umbilicus in a term newborn, between the umbilicus and the pubis at 6 years, and at the pubis in adults.

Philippine test: with the child's head upright, the child's right hand is placed across the top of the head, with the fingers reaching toward the left ear. If the fingertips reach the ear, the test is positive — the first “growth spurt” is complete and the child is biologically ready for school (usually at 5–8 years).

The body surface area of a child is large relative to that of an adult: in a newborn there are 0.06 m^2 per 1 kg of weight, in an adult 0.02 m^2. In a newborn the head accounts for 21% of the body surface, in an adult for 7.5%. That is why a child cools down and overheats quickly; the doses of many drugs are calculated per body surface area.

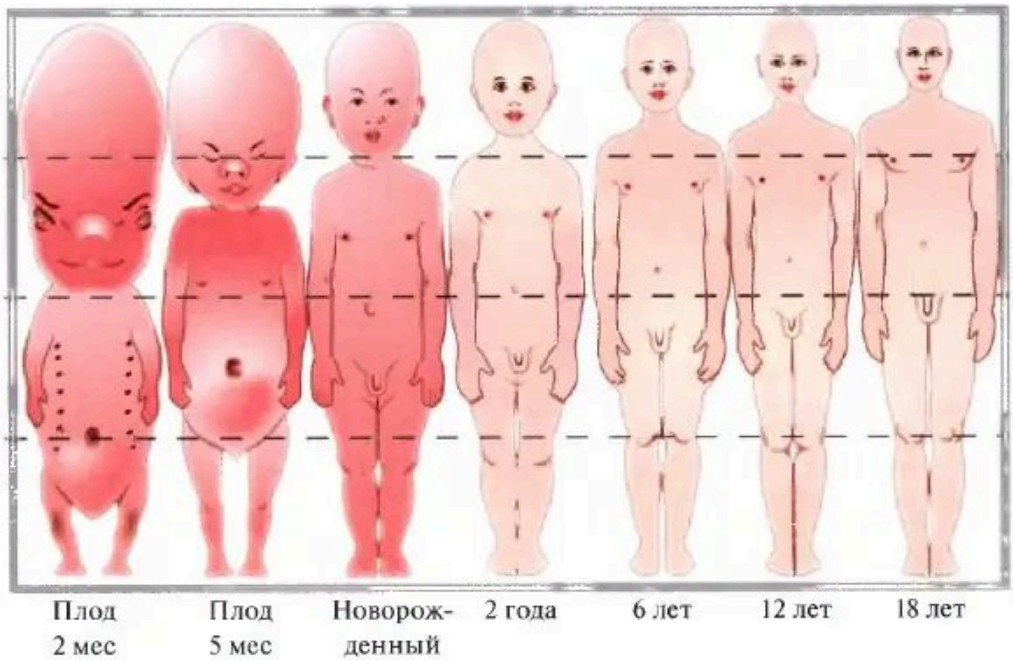


Figure 2.6. *Changes in body proportions in children of different ages: fetus at 2 and 5 months, newborn, 2, 6, 12 and 18 years (Navruzova, Geppe)*

2.4. Assessing physical development

Anthropometric indicators are assessed by the following methods: approximate formulas (above), **the sigma (parametric)** method, **the centile (nonparametric)** method, **the WHO growth standards**

(Z-score) and regression analysis.

The sigma (parametric) method

The child's indicator is compared with the arithmetic mean (M) and the standard deviation (σ) for children of the same age and sex.

Table 2.5. *Assessment by the sigma method*

Range of the indicator	Assessment
M ± 1σ	Average
From M − 1σ to M − 2σ / from M + 1σ to M + 2σ	Below average / above average
From M − 2σ to M − 3σ / from M + 2σ to M + 3σ	Low / high
Below M − 3σ / above M + 3σ	Very low / very high

For development to be harmonious, the difference in sigma deviations between the indicators should not exceed 1σ.

The centile (nonparametric) method

Centile tables and charts show the distribution of an indicator in a population of children in percentages. The table has 7 centile columns (**3, 10, 25, 50, 75, 90, 97**); the intervals between them are called **corridors** (from 1 to 7).

Table 2.6. *Centile corridors and their assessment*

Corridor	Centile interval	Proportion of children	Assessment
1	Below the 3rd centile	3%	Very low
2	3–10	7%	Low
3	10–25	15%	Below average
4	25–75	50%	Average
5	75–90	15%	Above average
6	90–97	7%	High
7	Above the 97th centile	3%	Very high

REMEMBER

Order of centile assessment: 1) determine into which corridor each of height, weight and chest circumference falls; 2) **the level of physical development is determined by the height corridor**; 3) **harmony** is assessed by the difference between the corridors: a difference of **0–1** corridor — harmonious; **2** corridors — disharmonious; **3 or more** — sharply disharmonious development. Corridors 2 and 6 are borderline (require follow-up); in corridors 1 and 7 the probability of pathology is high, and an in-depth examination is needed.

The most important chart is the **weight-for-length** chart. Weight in the **10–25 / 75–90** centile indicates grade I underweight/overweight; in the **3–10 / 90–97** centile — grade II; **below 3 / above 97** — grade III.

The WHO growth standards (Z-score)

In Uzbekistan and many other countries the **WHO growth standards** (2006) are used for children under 5 years. The indicator is assessed by the Z-score (the number of standard deviations from the

median).

Table 2.7. *Interpretation of the Z-score according to the WHO standards*

Z-score	Length/height-for-age	Weight-for-length/height
Above +3	Very tall (endocrine pathology to be excluded)	Obesity
From +2 to +3	Tall (usually normal)	Overweight
From −2 to +2	Normal	Normal
From −2 to −3	Short stature (stunting)	Wasting (underweight)
Below −3	Severe short stature	Severe wasting

Writing the conclusion

The conclusion on physical development states three things: **the level** (by height), **harmony** and, if present, **the degree of underweight or overweight**.

Sample: “Boy aged 8 months. Length 68 cm (corridor 4), weight 9000 g (corridor 5), chest circumference 44 cm (corridor 4). Conclusion: physical development average, harmonious; weight-for-length in the 75–90 centile — grade I overweight.”

2.5. Semiotics of disorders of physical development

Changes in height

Short stature is a height markedly below that of children of the same age and sex (below the 3rd centile or −2 SD); severe short stature (**nanism, dwarfism**) is below −3 SD. Causes:

- endocrine: growth hormone deficiency (pituitary dwarfism — usually detected at 2–4 years: a “doll face”, infantile body build), hypothyroidism, Cushing's disease, severe diabetes mellitus, precocious puberty (early closure of the growth zones);
- chromosomal diseases: Shereshevsky–Turner syndrome and others;
- skeletal dysplasias: achondroplasia, chondrodystrophy, severe rickets;
- chronic somatic diseases (heart, lungs, kidneys, intestine), psychosocial deprivation;
- constitutional (familial) short stature. **Tall stature (gigantism)** — growth hormone excess (pituitary tumor), hyperthyroidism, Marfan syndrome, Klinefelter syndrome (XXY); in families with tall parents, constitutional tall stature is considered normal.

Changes in weight

Intrauterine growth restriction (IUGR) — the newborn's weight and/or length is below the 10th centile for gestational age.

- **Symmetric (hypoplastic)** — both weight and length are equally reduced; the adverse factors acted in the first and second halves of pregnancy, and the changes are more severe.
- **Asymmetric (hypotrophic)** — length is normal and only weight is reduced; the adverse factors acted in the second half of pregnancy.



Figure 2.7. *A newborn with intrauterine growth restriction (left) and a newborn with normal physical indicators (right) (Navruzova, Geppe)*

Acquired chronic nutritional disorders (dystrophies): under 1 year they are called **hypotrophy** and **paratrophy**, and in children over 1 year **protein-energy malnutrition** and **obesity**.

Table 2.8. *Degrees of hypotrophy*

Sign	Grade I	Grade II	Grade III
Weight deficit (relative to length)	11–20%	21–30%	More than 30%
Thinning of the subcutaneous fat	On the abdomen and trunk	On the abdomen, trunk and limbs	Absent everywhere, even on the face
Tissue turgor	Slightly reduced	Clearly reduced, flabby tissues	Sharply reduced, the skin hangs in folds
Length	Not lagging	Lagging by 1–2 cm	Lagging by 4–5 cm
Appetite	Not impaired	Reduced	Reduced to anorexia
Regurgitation and vomiting	Not characteristic	Frequent	Very frequent
Muscle tone	Slight hypotonia	Hypotonia	Marked hypotonia
Psychomotor development	Appropriate for age	Lagging	Acquired skills are lost
Immunity	Slightly reduced	Considerably reduced	Sharply reduced

Hypostature — an equal lag of length and weight with a satisfactory nutritional status. It occurs in congenital heart defects, brain malformations, endocrine and hereditary diseases.

Paratrophy — a chronic nutritional disorder in children under 1 year with excess weight (5–10% or more); the main cause is excessive, one-sided (most often carbohydrate-rich) feeding. The children are flabby and pasty, the skin is pale, tissue turgor and muscle tone are reduced, and they fall ill frequently.

Obesity is excessive fat deposition. Excess weight relative to height: grade I — 15–24%, II — 25–49%, III — 50–99%, IV — more than 100%. **Primary** (alimentary-constitutional) and **secondary**

(hypothyroidism, Itsenko–Cushing syndrome, hypothalamic syndrome, Prader–Willi and Laurence–Moon syndromes) obesity are distinguished.



Figure 2.8. *Boys aged 5 years: normal height (left) and short stature (right) (Navruzova, Geppe)*

2.6. Anatomical and physiological features of the newborn

The neonatal (newborn) period is the period from birth to **28 days**; the period of the body's adaptation to extrauterine life. It is divided into the **early** (from tying of the cord to the end of day 7) and **late** (days 8–28) neonatal periods. **The perinatal period** — from the 22nd week of pregnancy to 7 days after birth.

REMEMBER

Main definitions:

- **Live born** — a baby that has breathed or shows other signs of life (heartbeat, pulsation of the umbilical vessels, voluntary muscle movement) (WHO).
- **Term** — born from **37 completed weeks to 42 weeks** of gestation (37^0–41^6 weeks).
- **Post-term** — born at **42 completed weeks or later**.
- **Preterm (premature)** — born **before 37 completed weeks**.
- **Gestational age** — the number of completed weeks from conception to birth.

A healthy newborn — a baby born to a mother whose pregnancy and delivery were uncomplicated, with an Apgar score of 8–10, an average weight of 3200–3600 g, a weight–length ratio of 60–80, a normal adaptation period, a weight loss not exceeding 6–8%, and fed with breast milk.

Table 2.9. *Features of the newborn by system*

System	Features
Nervous system	Immature, cerebral sulci poorly expressed; sleeps most of the day (≈ 18–20 hours); innate unconditioned reflexes (sucking, swallowing, grasping, blinking) are well expressed

System	Features
Muscles	Physiological hypertonia of the flexor muscles: arms flexed at the elbows, legs drawn up to the abdomen (fetal position); cannot hold up the head; movements uncoordinated, athetoid
Skin	Delicate, velvety, elastic, pink, rich in blood vessels; remnants of lanugo on the shoulders and back; sweat glands poorly developed, sebaceous glands active — overheats and cools down quickly; vascular spots (telangiectasias) on the nape, eyelids and between the eyebrows; a "Mongolian spot" over the sacrum
Subcutaneous fat	Well developed, dense; brown adipose tissue (produces heat)
Bones	Soft, poor in salts; the fontanelles and sutures of the skull are open: the anterior fontanelle is diamond-shaped, from 1.5×1.5 to 3×3 cm; the posterior fontanelle is triangular and often closed at birth; the chest is barrel-shaped, the ribs horizontal; no physiological curves of the spine
Respiration	Narrow nasal passages, trachea and bronchi; breathing shallow, mainly diaphragmatic; RR 40–60 per minute
Cardiovascular	The fetal communications close, and the pulmonary circulation starts to work; HR 140–160 (up to 180 when crying); BP 65–75/36 mm Hg
Digestion	Delicate oral mucosa, a large tongue, Bichat's fat pads in the cheeks; little saliva; weak esophagogastric sphincter — frequent small regurgitations; stomach capacity initially ≈ 10 ml, 90–100 ml by the end of the month; meconium for 1–3 days (dark green, sticky, odorless); the liver protrudes up to 2–3 cm below the costal margin
Urination	Voids 5–6 times a day in the first days, and 20–25 times from the 2nd week
Hematopoiesis	The main site is the red bone marrow; the spleen at the costal margin
Immune system	Maternal IgG has crossed the placenta; protective factors come with colostrum and breast milk; on the whole, immunity is imperfect
Metabolism	The daily fluid requirement increases from 60–80 ml/kg in the first days to 150–165 ml/kg by the end of the 1st month; the water-salt balance is easily disturbed



Figure 2.9. *A newborn: physiological hypertonia of the flexor muscles (Navruzova, Geppe)*

Morphofunctional maturity is the readiness of the newborn's organs and systems to support extrauterine life. A functionally mature newborn has sufficient motor activity, physiological hypertonia of the flexors, active sucking, a loud cry, adequate thermoregulation, a regular breathing rhythm (without attacks of apnea and cyanosis) and a regular heart rhythm.

2.7. Borderline (transitional) states of the newborn period

Borderline (transitional) states are states that arise at the border between intrauterine and extrauterine life and reflect the newborn's adaptation to the new conditions. They are **physiological** and pass on their own, but under unfavorable conditions they **may become pathological**.

Table 2.10. *The main transitional states*

State	Onset and course	Sign of transition to pathology
Physiological weight loss	Maximal on days 3–4, 5–8% (up to 10%); regained by days 7–10, in all babies by the 2nd week	More than 10% or slow recovery
Physiological jaundice	In 60–70% of newborns; appears on day 3, increases until day 6, disappears by days 7–10; bilirubin maximal 130–170 µmol/l; the condition is not affected	Appearance on the 1st day, duration longer than 2 weeks, high bilirubin, worsening of the condition
Simple erythema	Reddening of the skin after the vernix is removed and after bathing; most intense on day 2, disappears by the end of the 1st week	—
Physiological peeling	Large lamellar peeling on days 3–5; abundant in post-term babies	—
Toxic erythema	In 20–30% of newborns, on days 2–5; dense red spots with a papule or vesicle in the center; not found on the palms, soles or mucous membranes; disappears in 2–3 days	Abundant rash, restlessness, eosinophilia
Milia	White-yellow nodules of 1–2 mm (blocked sebaceous glands) on the nasal wings, forehead and chin; disappear in 1–2 weeks	—
Caput succedaneum	Edema of the presenting part; resolves in 1–2 days	—
Sexual (hormonal) crisis	Engorgement of the mammary glands (beginning on days 3–4, maximal on days 7–8, 1.5–2 cm), desquamative vulvovaginitis in girls (60–70%) and vaginal bleeding (on days 4–6, lasting 1–3 days), hyperpigmentation around the nipples and scrotum, edema of the genitals	Squeezing the mammary gland — risk of infection (mastitis); squeezing is forbidden
Transitional disturbance of heat balance	Hypothermia (the first hours after birth) or hyperthermia (on days 3–5, from overheating or fluid deficit)	—
Transitional renal changes	Oliguria (the first 3 days, less than 15 ml/kg), proteinuria, uric acid infarction (uric acid crystals; brick-colored urine; disappears by days 7–10)	—
Transitional blood changes	Polycythemia: Hb 180–220 g/l, erythrocytes $6–8\times10^{12}$/l, leukocytes $10–15\times10^{9}$/l; on day 5 the “first crossover” of neutrophils and lymphocytes	—
Transitional circulation	The arterial duct (ductus Botalli) closes functionally between 10–15 minutes and 10–15 hours after birth; the foramen ovale closes functionally in the first hours; anatomical closure takes weeks and months	Failure of the ducts to close (congenital defect)
Transitional hyperventilation	In the first 2–3 days pulmonary ventilation is 1.5–2 times higher	—

State	Onset and course	Sign of transition to pathology
Transitional immunodeficiency	Physiological immune depression, hypogammaglobulinemia	—

MEMORY AID
Most transitional states follow the "**3–7 rule**": they begin around **day 3** and pass by **days 7–10** (weight loss, jaundice, sexual crisis, uric acid infarction). An earlier onset or longer duration is a sign of pathology.

2.8. The preterm (premature) newborn

A preterm newborn is a baby born **before 37 completed weeks** of pregnancy, usually with a weight **below 2500 g** and a length **below 45 cm**. The main criterion is **the gestational age**; assessment by weight alone is inaccurate (a term newborn with IUGR may also have a low weight).

Table 2.11. *Degrees of prematurity*

Degree	Gestational age	Birth weight
I	35–37 weeks	2001–2500 g
II	32–34 weeks	1501–2000 g
III	29–31 weeks	1001–1500 g
IV	Less than 29 weeks (28 weeks or less)	Less than 1000 g

By weight: **low birth weight** — 1500–2500 g; **very low birth weight** — 1000–1500 g; **extremely low birth weight** — less than 1000 g.

Causes of preterm birth: isthmic-cervical insufficiency, uterine malformations, extragenital diseases and infections of the mother, fetal pathology, pathology of pregnancy and labor, occupational hazards, the mother's age (under 20 and over 35), harmful habits.

Table 2.12. *Morphological signs of term and preterm newborns*

Sign	Term	Preterm
Skin	Pink, elastic	Thin, shiny, translucent, dark red, wrinkled
Lanugo	Only remnants on the shoulders and back	Abundant and thick over the whole body
Subcutaneous fat	Well developed	Almost absent, prone to edema
Body proportions	The head is 1/4 of the body length	The head is large (up to 1/3), the neck and legs short
Position of the umbilicus	In the middle of the body	Displaced downward
Auricle	Elastic, well-defined shape	Soft, pressed against the head, easily folded
Nails	Reach the fingertips	Do not reach the fingertips, soft
Skull	Posterior fontanelle closed	Posterior fontanelle open, sutures open and separated, bones soft
Mammary glands	The areola is distinct, the gland is palpable	The nipple is barely visible

Sign	Term	Preterm
Genitals	In boys the testes are in the scrotum; in girls the labia majora cover the labia minora	The testes have not descended into the scrotum; the labia majora do not cover the labia minora, the genital slit gapes
Plantar creases	Over the whole sole	Only on the anterior part or absent

Signs of functional immaturity:

- **nervous system:** lethargy, a weak cry or squeal, weak or absent sucking and swallowing reflexes, muscle hypotonia; the unconditioned reflexes appear from 1–2 months of age; psychomotor development lags by 1–3 months;
- **thermoregulation:** prone to hypothermia, but also overheats easily (the temperature in the incubator must be strictly controlled);
- **respiration:** RR 36–82, breathing shallow and irregular, attacks of **apnea**; surfactant deficiency — **respiratory distress syndrome** (atelectasis, hyaline membrane disease);
- **cardiovascular:** HR 140–160, BP lower than in term babies (≈ 50–60/30 mm Hg); the fetal communications close anatomically later (at 2–8 weeks);
- **digestion:** small stomach capacity (in the first 10 days 3 ml/kg × days of life), low enzyme activity;
- **jaundice** appears on day 2, is maximal on days 5–8 and lasts up to 2–3 weeks.

Physical development of the preterm child

In the first year, the development of a preterm child is assessed by **postconceptional age** (gestational age + age after birth). For example, a child born at 27 weeks and now 9 weeks old has a postconceptional age of 36 weeks.

- Length increases by 3.5–5 cm per month in the first 3 months; 27–38 cm are added in 1 year.
- At 1–2 months the weight increases by 10–15 g per day for each 1 kg of birth weight. Babies born weighing 1001–1500 g double their weight by 2–3 months, triple it by 3.5–4 months and increase it 6–7 times by 1 year; those born weighing 1501–2000 g double it by 2.5–3 months, triple it by 4–4.5 months and increase it 5–6 times by 1 year.
- The head and chest circumferences become equal (“cross over”) at 3–5 months.
- The indicators catch up with those of term babies at about **2 years**.



Figure 2.10. *An extremely preterm newborn and an incubator (Navruzova, Geppe)*

2.9. Features of examining the newborn

Conditions: ward temperature **22–26 °C**, **30 minutes after** a feed, in daylight, on a heated changing table. The condition is assessed at 1 and 5 minutes by **the Apgar score** (topic 1). The history establishes the obstetric history and risk factors.

Inspection: proportions, position (flexed in a term baby; on the back with the limbs extended in a preterm baby), spontaneous movements, eye movements, facial expressions and cry are assessed. Pathological positions: **the "frog position"** (limbs extended, thighs lying flat to the sides — hypotonia), **opisthotonus** (arching of the body supported on the head and heels).

- **Skin:** a healthy newborn is pink; slight cyanosis of the hands, feet and around the mouth is normal if the rest of the body is pink.
- **Head:** size, shape, fontanelles and sutures. **Caput succedaneum** — doughy, with indistinct borders, lies over several bones and disappears in a few days. **Cephalohematoma** (hemorrhage under the periosteum) — firmer, fluctuant, located **within the border of a single bone** (most often the parietal bone).
- **Neck:** a firm mass in the sternocleidomastoid muscle (hemorrhage — a cause of torticollis); **the clavicle** is palpated (fracture — in breech delivery, with obstetric forceps).
- **Chest:** shape, accessory nipples, respiratory movements, nasal flaring, retraction of the intercostal spaces and sternum.
- **Abdomen:** distension (intestinal obstruction, ascites) or retraction (diaphragmatic hernia); **the umbilical stump falls off on days 4–10**; the state of the umbilical wound (hyperemia, discharge, palpable umbilical vessels).
- **Spine:** a spinal hernia (meningocele, myelomeningocele) is sought.
- **Limbs:** malformations, fractures, symmetry, number of digits; the state of **the hip joint** — **the Ortolani test**: with the newborn on the back, the hip and knee joints are flexed at a right angle and the thighs are abducted; a **"click"** felt as the femoral head slips into place is a sign of congenital dislocation (dysplasia) of the hip. Limited abduction of the thighs also raises suspicion of this.
- **Stigmas of dysembryogenesis:** up to 5–7 stigmas are normal in a newborn.

Palpation and percussion: size, bulging or depression and pulsation of the fontanelles; liver size; pulsation of the femoral and brachial arteries (generally weak pulsation — left ventricular failure; absent femoral pulse — coarctation of the aorta). Because the chest is small, percussion is of limited value.

Auscultation: breathing is mainly weakened vesicular; HR 100–160. A murmur heard immediately after birth is not always a sign of a congenital heart defect — it often disappears during the neonatal period.

The Silverman score

The severity of respiratory distress is assessed with **the Silverman score** in preterm newborns and with **the Downes score** in term newborns. Each sign is scored 0–2 points.

Table 2.13. *The Silverman score*

Sign	0 points	1 point	2 points
Movement of the upper chest	Synchronous with the abdomen	Lags slightly on inspiration	"Seesaw" — the chest sinks while the abdomen rises

Sign	0 points	1 point	2 points
Intercostal retraction	Absent	Slight	Marked
Xiphoid (sternal) retraction	Absent	Slight	Marked
Chin position and nasal flaring	Chin motionless, no nasal flaring	The chin drops on inspiration with the mouth closed; slight nasal flaring	The chin drops with the mouth open; marked nasal flaring
Expiratory noise (grunting)	Absent	Heard with a stethoscope	Heard at a distance

Interpretation: **0 points** — no respiratory distress; **1–3 points** — initial (mild); **4–5 points** — moderate; **6 points or more** — severe respiratory distress syndrome.

REMEMBER

On **the Apgar** score, the higher the score, the better the condition (10 — ideal). On **the Silverman** score it is the opposite: the higher the score, the more severe the respiratory distress (0 — ideal).

2.10. Clinical part

Assessing physical development — a mirror of health

Regular assessment of physical development makes it possible to detect many diseases before their other signs appear.

- **Arrest of weight gain** in infancy may be the first sign of insufficient breast milk (hypogalactia), urinary tract infection, celiac disease, a congenital heart defect or cystic fibrosis.
- **Slowing of growth in height** is an early sign of hypothyroidism, growth hormone deficiency, chronic kidney disease or Crohn's disease.
- **Rapid enlargement of the head circumference** (more than 2 cm per month after 3 months) — hydrocephalus; neurosonography is needed.
- **The dynamics matter, not a single measurement:** if a child has always grown along the 25th centile, this is his individual norm; if he was previously on the 75th centile and has “dropped” to the 25th, this is a sign of disease.

The newborn: the border between normal and pathology

CLINICAL SIGNIFICANCE

Distinguishing physiological jaundice from pathological jaundice is one of the most important tasks in practice. Jaundice **appearing on the first day of life** is never physiological — it is most often a sign of hemolytic disease (Rh or ABO incompatibility) and requires immediate measurement of bilirubin. Jaundice lasting longer than **2 weeks** in a term newborn and longer than **3 weeks** in a preterm newborn also requires investigation (hypothyroidism, biliary atresia, hepatitis).

Similarly, **if the weight falls by more than 10%** or is not regained in 2 weeks, insufficient breast milk, incorrect attachment to the breast or infection is sought.

Practical skills

Skill 1. Measuring the length of an infant

1. Wash the hands, explain the procedure to the mother, spread a diaper on the length board.
2. Lay the child on the back; bring the head into contact with the fixed headboard.
3. Hold the head so that the lower margin of the orbit and the upper margin of the tragus are in one vertical plane (an assistant or the mother).
4. Straighten the legs by pressing lightly on the knees with the right hand.
5. With the left hand, move the sliding footboard against the heels (flexed at a right angle).
6. Read the result, pick up the child, record and evaluate the result.

Skill 2. Measuring the height of a child over 2 years

1. Stand the child without shoes on the platform of the stadiometer with the back to it.
2. Make sure the heels, buttocks, the area between the shoulder blades and the back of the head touch the stadiometer; arms along the body, heels together.
3. Position the head so that the lower margin of the orbit and the upper margin of the tragus are in one horizontal plane.
4. Lower the headpiece without pressing on the head, read and record the result.

Skill 3. Measuring the head and chest circumferences

1. Check and disinfect the tape measure.
2. **Head circumference:** pass the tape at the back over the occipital protuberance and at the front over the brow ridges, tighten slightly and read the result.
3. **Chest circumference:** the child raises the arms to the sides; pass the tape at the back below the lower angles of the shoulder blades; have the child lower the arms and pass the tape at the front at the level of the nipples; measure during quiet breathing (in an older child also at inspiration and expiration).
4. Record the results and compare them with the age norms.

Skill 4. Assessing physical development by the centile method

1. Establish the child's age and sex; measure the height, weight and chest circumference.
2. Find the height corridor in the centile table for age and sex — determine **the level of development**.
3. Find the corridors for weight and chest circumference.
4. Calculate the maximal difference between the corridors — assess **harmony** (0–1 harmonious, 2 disharmonious, 3 or more sharply disharmonious).
5. Determine the degree of underweight or overweight from the weight-for-height chart.
6. Write the conclusion and, if necessary, refer for additional investigation.

Skill 5. Examining a newborn

1. Prepare the conditions: temperature 22–26 °C, a heated changing table, 30 minutes after a feed, wash and warm the hands.
2. Review the obstetric history and the Apgar score.
3. General inspection: position, movements, cry, skin color, proportions.

4. Head: size, shape, fontanelles, sutures, caput succedaneum or cephalohematoma.
5. Face, eyes, mouth (integrity of the palate, frenulum of the tongue), neck, clavicles.
6. Chest: shape, RR (a full 1 minute), participation in breathing, retractions; auscultation of the heart and lungs.
7. Abdomen: shape, palpation of the liver and spleen, umbilical stump and umbilical wound.
8. Genitals, anus, spine.
9. Limbs, hip joints (Ortolani test).
10. Anthropometry, assessment of muscle tone and unconditioned reflexes (topic 3).
11. Identify the transitional states and write the conclusion.

Skill 6. Assessment by the Silverman score

1. Undress the newborn and observe the movements of the chest and abdomen.
2. Assess the five signs in sequence: movement of the upper chest, intercostal retraction, sternal retraction, chin position and nasal flaring, expiratory grunting (with a stethoscope and at a distance).
3. Add up the points and interpret the result (0 — none; 1–3 — mild; 4–5 — moderate; 6 or more — severe).
4. Depending on the result, decide with a senior physician on oxygen therapy or transfer to the intensive care unit.

Review questions

1. What is physical development, and what groups of its signs are there?
2. List the factors affecting physical development.
3. What are acceleration and retardation? What is the difference between harmonious and disharmonious acceleration?
4. How is height measured in children of different ages?
5. How does length increase in the first year? State the formulas for calculating height in a child over 1 year.
6. How does the physiological weight loss proceed?
7. How is weight calculated in the first year and after?
8. How do the head and chest circumferences change with age? When do they become equal?
9. How is physical development assessed by the centile method? How is harmony determined?
10. How is the Z-score interpreted according to the WHO standards?
11. Describe the degrees of hypotrophy.
12. Give the definitions of term, post-term and preterm newborns.
13. List the transitional states of the newborn and the criteria for distinguishing them from pathology.
14. State the degrees of prematurity and its morphological signs.
15. Which signs does the Silverman score assess?

Test questions

1. In a term newborn the physiological weight loss usually:
   A) Is maximal on the 1st day and amounts to 15%
   B) Is maximal on days 3–4 and amounts to 5–8%
   C) Is maximal on day 14
   D) Is not observed
   E) Is regained by 1 month

2. On average, how much does a child grow in length in the first year?
   A) 15 cm
   B) 20 cm
   C) 25 cm
   D) 30 cm
   E) 35 cm

3. The average height of a 4-year-old child:
   A) 75 cm
   B) 85 cm
   C) 100 cm
   D) 115 cm
   E) 130 cm

4. By the “8200 g at 6 months” rule, the approximate normal weight of a 7-month-old child:
   A) 7000 g
   B) 7500 g
   C) 8000 g
   D) 8600 g
   E) 9500 g

5. When do the head and chest circumferences become equal in a term child?
   A) At birth
   B) At 1 month
   C) At 3–4 months
   D) At 1 year
   E) At 3 years

6. By the centile method, development is called harmonious if the difference between the corridors of the indicators is:
   A) 0–1 corridor
   B) 2 corridors
   C) 3 corridors
   D) 4 corridors
   E) 5 corridors

7. In a term newborn, a weight–length ratio below 60 indicates:
   A) Obesity
   B) Intrauterine growth restriction
   C) Hydrocephalus
   D) The norm
   E) Acceleration

8. The weight deficit in grade II hypotrophy:
   A) 5–10%
   B) 11–20%
   C) 21–30%
   D) More than 30%
   E) More than 50%

9. At what gestational age is a term newborn born?
   A) 32–35 weeks
   B) 35–37 weeks
   C) 37–42 weeks
   D) 42–44 weeks
   E) 28–32 weeks

10. Not characteristic of physiological jaundice:
    A) Appearance on day 3
    B) Disappearance by days 7–10
    C) An unaffected condition
    D) Appearance on the 1st day of life
    E) Normal color of stool and urine

11. The correct approach to mammary glands engorged in a sexual crisis:
    A) Squeeze the gland to express the fluid
    B) Prescribe an antibiotic
    C) Do not squeeze the gland, observe
    D) Open it surgically
    E) Prescribe a hormone

12. Corresponds to grade III prematurity:
    A) 35–37 weeks, 2001–2500 g
    B) 32–34 weeks, 1501–2000 g
    C) 29–31 weeks, 1001–1500 g
    D) Less than 28 weeks, less than 1000 g
    E) 37–42 weeks, 2500–4000 g

13. A sign characteristic of prematurity:
    A) The umbilicus is in the middle of the body
    B) The nails do not reach the fingertips
    C) The testes are in the scrotum
    D) Plantar creases over the whole sole
    E) Elastic auricle

14. Characteristic of cephalohematoma:
    A) Located over several bones
    B) Doughy, with indistinct borders
    C) Located only within the border of one bone, fluctuant
    D) Disappears in 1–2 days
    E) Occurs only on the forehead

15. A Silverman score of 6 points is assessed as:
    A) No respiratory distress
    B) Mild respiratory distress
    C) Moderate respiratory distress
    D) Severe respiratory distress syndrome
    E) Normal

16. Among the “biological” signs of physical development are:
    A) Vital capacity of the lungs
    B) Height and weight
    C) Hand grip strength
    D) Shape of the spine
    E) Dental formula, ossification centers, secondary sexual characteristics

17. The scientist who introduced the term “acceleration” in 1935:
    A) E. Koch
    B) G.N. Speransky
    C) S.Sh. Shamsiyev
    D) N.P. Gundobin
    E) A. Quetelet

18. How is the length of a child under 2 years measured?
    A) Lying down with a horizontal length board
    B) Sitting, with a tape measure
    C) With electronic scales
    D) Standing with a vertical stadiometer
    E) Only by an approximate formula

19. A child was 75 cm long at 1 year. What will the height be at 4 years approximately?
    A) 100 cm
    B) 90 cm
    C) 110 cm
    D) 120 cm
    E) 95 cm

20. At what age is the maximal growth (growth spurt) of puberty seen in boys?
    A) 17–19
    B) 13,5–15,5
    C) 10–12
    D) 8–10
    E) 11–12

21. On which days is the body weight regained in a term newborn?
    A) Days 3–4
    B) Days 1–2
    C) Days 7–10
    D) Day 21
    E) Days 14–16

22. What is the average monthly weight gain in the first half-year?
    A) 600 g
    B) 800 g
    C) 1000 g
    D) 400 g
    E) 200 g

23. The normal weight–length ratio (g/cm) in a healthy term newborn:
    A) 110–120
    B) 90–100
    C) 60–80
    D) 40–50
    E) 30–40

24. What proportion of the body length does the height of the head make up in a newborn?
    A) 1/3
    B) 1/8
    C) 1/7
    D) 1/4
    E) 1/6

25. How is the interval between the 25th and 75th centiles (corridor 4) assessed, and what percentage of children does it include?
    A) Above average, 15%
    B) Average, 50%
    C) High, 7%
    D) Below average, 25%
    E) Average, 15%

26. According to the WHO standards, a “length/height-for-age” Z-score below −3 means:
    A) Very tall
    B) Tall
    C) Normal
    D) Short stature
    E) Severe short stature

27. What does it mean if the fingertips reach the ear in the Philippine test (positive test)?
    A) There is scoliosis
    B) Physical development is very low
    C) There is acceleration
    D) The first “growth spurt” is complete; the child is biologically ready for school
    E) The child has rickets

Clinical cases

Case 1

An 8-month-old boy came to the polyclinic for a preventive checkup. At birth he weighed 3200 g and was 50 cm long. Now he weighs 9000 g and is 68 cm long.

Questions: 1) Calculate the normal length and weight. 2) Assess the physical development. 3) What advice will you give the mother?

Answer: 1) Normal length: 50 + 9 (1st quarter) + 7.5 (2nd quarter) + 3.5 (months 7–8, 1.5–2 cm per month) ≈ 70 cm (range 68–70 cm). Normal weight: 8200 + 400 × 2 = 9000 g. 2) His length and weight correspond to his age — physical development is average; a centile table is used for an exact assessment. If the weight-for-length is in the 75–90 centile — grade I overweight, harmonious development. 3) Check the feeding regimen (volume and composition of complementary foods, do not increase carbohydrate foods), physical activity, and continue the monthly control measurements.

Case 2

A newborn on day 3: the mammary glands are engorged up to 1.5 cm on both sides with no changes in the overlying skin. The girl has a grayish-white mucous discharge from the genital slit. The skin is yellow; the jaundice appeared this morning; the baby is active and sucks well. The weight has fallen by 6% from the birth weight.

Questions: 1) What states are being observed? 2) Are they physiological? 3) What is recommended to the mother?

Answer: 1) Sexual (hormonal) crisis (physiological engorgement of the mammary glands, desquamative vulvovaginitis), physiological jaundice, physiological weight loss. 2) Yes, all of them are transitional (borderline) states: their timing (day 3), degree (weight loss 6%) and the unaffected condition of the baby indicate a physiological course. 3) Do not squeeze the mammary glands (risk of mastitis), genital hygiene, continue breastfeeding on demand; if the jaundice does not disappear by days 7–10 or increases — see a physician.

Case 3

A baby was born at 31 weeks of pregnancy weighing 1400 g. The skin is thin and dark red, there is lanugo over the whole body, subcutaneous fat is almost absent, the posterior fontanelle is open, the auricles are soft, and the labia majora do not cover the labia minora. On inspiration there is marked retraction of the intercostal spaces and sternum, the chest moves in a “seesaw” fashion, there is marked nasal flaring, the chin drops with the mouth open, and expiratory grunting is heard at a distance.

Questions: 1) Determine the degree of prematurity. 2) Score by the Silverman scale. 3) What is the cause of the respiratory distress?

Answer: 1) 31 weeks and 1400 g — grade III prematurity (very low birth weight). 2) Chest movement — 2, intercostal retraction — 2, sternal retraction — 2, chin and nasal flaring — 2, grunting — 2; total 10 points — severe respiratory distress syndrome. 3) Immaturity of the surfactant system (hyaline membrane disease, atelectasis); the baby needs respiratory support and surfactant therapy in the intensive care unit.

Quick review

Table 2.14. *Key facts on Topic 2*

Question	Answer
Birth length	48–52 cm (average 50)
Length in 1 year	+25 cm (3–2.5–1.5–1 cm per month by quarters), 75–77 cm at 1 year
Height at 4 / 8 / 12 years	100 / 130 / 150 cm
Height formulas	<4 years: 100 − 8(4 − n); 4–8: 100 + 6(n − 4); >8: 130 + 5(n − 8)
Birth weight	2600–4200 g (average 3200–3500)

Question	Answer
Physiological weight loss	5–8% (up to 10%), maximum on days 3–4, recovery by days 7–10
Weight gain	1st half-year 800 g/month, 2nd half-year 400 g/month; 8200 g at 6 months
Weight at 1 year	10.5 kg (×3); 2–11 years: 10.5 + 2n
Head circumference	34–36 cm; 46 at 1 year; 50 at 5 years; 55 cm at 10 years
Chest circumference	32–34 cm; equals the head at 3–4 months
Weight–length ratio	60–80; <60 — IUGR
Centile corridors	Corridor 4 (25–75) — average; difference 0–1 harmonious, 2 disharmonious, ≥3 sharply disharmonious
WHO Z-score	From −2 to +2 — normal
Hypotrophy	I: 11–20%, II: 21–30%, III: >30%
Term birth	$37–41^{6}$ weeks
Prematurity	I: 35–37 wk (2001–2500 g); II: 32–34 wk; III: 29–31 wk; IV: ≤28 wk (<1000 g)
Newborn RR / HR	40–60 / 140–160
Physiological jaundice	Day 3 → days 7–10, bilirubin ≤170 μmol/l
Umbilical stump	Falls off on days 4–10
Silverman	0 none; 1–3 mild; 4–5 moderate; ≥6 severe
The 4 groups of signs of physical development	Anthropometric, functional, somatoscopic, biological
Methods of assessing physical development (3)	Somatoscopic, somatometric (anthropometric), physiometric (spirometry, dynamometry)
Height formulas (n — age)	1–4 years: 100 − 8×(4−n); 4–8 years: 100 + 6×(n−4); over 8 years: 130 + 5×(n−8)
Weight formula for 2–11 years	Weight (kg) = 10.5 + 2n; ≈ 2 kg per year at 2–10 years, 3–4 kg per year at 10–15 years
Head and chest circumferences (at birth)	Head 34–36 cm, chest 32–34 cm (1–2 cm smaller than the head); equal at 3–4 months; head ≈ 46 cm at 1 year
Ratio of the head to the body length	Newborn 1/4 → 2 years 1/5 → 6 years 1/6 → 12 years 1/7 → adults 1/8
Sigma assessment	M±1σ — average; 1–2σ — below/above average; 2–3σ — low/high; beyond 3σ — very low/very high; harmony: difference ≤ 1σ
Quetelet index (BMI)	Weight (kg) / height $(m)^2$; to assess nutritional status and obesity at school age, with a centile table

TOPIC 3

Neuropsychological development of children. Semiotics of nervous system damage. Body temperature, thermoregulation and fever

Learning objectives

The student should know:

- the anatomical and physiological features of the nervous system in children;
- the stages of development of statics, motor skills, speech, emotions and forms of communication;
- the development of the sense organs (vision, hearing, taste, smell, skin sensation, vestibular apparatus);
- the innate unconditioned reflexes of the newborn and the times of their disappearance;
- the methods of examining the nervous system, the meningeal and encephalitic syndromes, pathological reflexes;
- the features of thermoregulation in children, measuring body temperature, the types of fever.

The student should be able to:

- assess the child's neuropsychological development according to age;
- elicit and assess the unconditioned reflexes in a newborn;
- detect meningeal signs;
- measure body temperature correctly, fill in a temperature chart and determine the type of fever.

3.1. Anatomical and physiological features of the nervous system

The nervous system connects the body with the external environment and maintains the internal balance (homeostasis). At birth the nervous system is not fully formed anatomically and functionally — this determines the specific features of the child's motor, sensory and mental life.

The fetal period. The nervous system begins to form from the ectoderm as the neural tube in the 3rd–4th week of intrauterine development. **The critical period of brain development is weeks 10–18 of pregnancy** (nerve cells divide most intensively). At birth the child has 25% of the cells of the CNS, and at 1 year 90–95%.

Table 3.1. *Main features of the newborn's nervous system*

Feature	Description and significance
Brain weight	350–400 g — ≈ 10% of the body weight (2% in adults); increases 2–2.5 times by 1 year and 3 times by 3 years
Sulci and gyri	The large sulci are present but shallow; the small (tertiary) sulci form by 5–6 years
Cortex	Functionally immature; inhibition predominates in the cortex — the newborn sleeps a lot
Subcortical centers	The thalamo-pallidal system predominates — movements are uncoordinated, athetoid
Myelination	The pyramidal tracts and cranial nerves are not fully myelinated; the pyramidal tracts are myelinated by 2–3 years, and the system as a whole by 5 years — the Babinski reflex is physiological until 2 years

Feature	Description and significance
Medulla oblongata	The most mature part functionally: the centers of respiration, heart activity, sucking, swallowing, coughing and sneezing work
Hypothalamus	Its centers mature unevenly until 14–16 years — thermoregulation and metabolism are imperfect
Blood supply	The brain is rich in blood, and its oxygen demand is 7 times higher than in adults; venous outflow is slow (the diploic veins form after the fontanelles close)
Blood-brain barrier	High permeability — a tendency to neurotoxicosis in infections; bilirubin passes into the brain easily
Spinal cord	In the newborn it reaches vertebra LIII (LI in adults), and is as in adults by 5–6 years — taken into account in lumbar puncture

REMEMBER

Differentiation of nerve cells is completed **by 3 years**; **at 7–8 years** the structure of the cerebral cortex hardly differs from that of adults; the brain is fully formed anatomically **by 20 years**. The younger the nervous system, the faster it develops — especially **in the first 3 months**.

3.2. Neuropsychological development

Neuropsychological development (NPD) is the age-related formation of the child's motor (locomotor) functions, sensation, speech, emotional and mental sphere and skills. Together with physical development, it is the main criterion of a child's health.

NPD is assessed in the following **leading lines**: visual and auditory reactions, emotions, gross movements, hand movements and actions with objects, active speech, understanding of speech, skills; in the second and third years — sensory development and play.

Development of statics and motor skills

Motor functions develop in a **“top-down”** (craniocaudal) direction: first the child holds up the head, then sits, then stands and walks.

Table 3.2. *Main stages of statics and motor skills*

Age	Skill
1 month	Raises the head briefly when placed on the abdomen
2 months	Begins to hold the head in the vertical position
3 months	Holds the head well on the abdomen and in the vertical position; hits a hanging toy with the hands
4 months	Leans on the forearms when on the abdomen, rolls onto the side; grasps a toy and examines it
5 months	Rolls from back to abdomen; stands straight on the feet when supported under the arms
6 months	Rolls from abdomen to back; sits with support; passes a toy from hand to hand
7 months	Crawls; sits independently (7–7.5 months)
8 months	Sits up and lies down by himself; stands holding on
9 months	Stands with support and walks sideways holding on to objects
10 months	Walks when held by the hand; climbs onto and off a sofa

Age	Skill
11 months	Stands independently, takes the first steps
12 months	Walks independently (11.5–12 months); picks up an object with the thumb and another finger (pincer grasp)
1.5 years	Walks well, feeds himself with a spoon
2 years	Runs, steps over an obstacle with alternating feet
3 years	Steps over a 10–15 cm obstacle with alternating feet; rides a tricycle; dresses with the help of adults

MEMORY AID
The main motor milestones can be remembered with the series “**2 – 4 – 6 – 8 – 10 – 12**”: **2 months** — holds up the head; **4 months** — rolls over and grasps a toy; **6 months** — sits; **8 months** — stands; **10 months** — walks when supported; **12 months** — walks independently.

Development of speech

Speech develops in two directions: **sensory speech** (linking a heard word with an object — understanding) and **motor (active) speech** (speaking).

Table 3.3. *Stages of speech development*

Age	Speech
1 month	Laryngeal sounds
2–3 months	Cooing — short vowel sounds
4–5 months	Prolonged, “sing-song” cooing
6–7 months	Babbling — syllables: “ba-ba”, “ma-ma”
7–8 months	Sensory speech: searches for an object with the eyes in response to the question “where is it?”
9–10 months	Repeats the sounds of words; knows his name
11–12 months	The first meaningful words; 8–12 words at 1 year
1.5 years	30–40 words; carries out simple instructions
2 years	Simple sentences; vocabulary of 200–300 words
3 years	Complex sentences; refers to himself as “I”; asks questions; about 1000 words
4–5 years	The “why?” age; tells fairy tales and recites poems

Development of emotions and forms of communication

Table 3.4. *Development of emotions*

Age	Emotional reaction
1–2 weeks	Negative emotion — crying (protest)
3–4 weeks	Positive emotion — concentration of attention, fixed gaze
1.5–2 months	The first smile — in response to adults; the beginning of social life
3 months	“The revival complex” — moves the arms and legs with joy and laughs on seeing loved ones
4–5 months	Laughs out loud; recognizes the mother; distinguishes loved ones from strangers
6–12 months	Emotions become more complex; fear of strangers, fear of separation from the mother

Age	Emotional reaction
1–1.5 years	Emotional life is active, negative emotions weaken; capriciousness appears
2–4 years	Fear, a sense of sympathy
4–6 years	Role-playing games, playing in a group, the beginnings of responsibility

Assessing neuropsychological development

NPD is assessed by observing the child during play, independent activity and carrying out adults' instructions.

> **REMEMBER**
> NPD is considered appropriate for age if the child has acquired the skills **within 1 month (±15 days) in the first year**, **within a quarter in the second year**, **within half a year in the third year**, and **within a year after 3 years**. Lagging behind this means delayed development; being ahead means accelerated development.

In older children, thinking and speech, movement (motor skills), attention and memory, social contacts and mental health are assessed. For example, a **4-year-old** child groups objects (furniture, dishes, clothes), jumps on both feet and buttons up; a **5-year-old** makes up a story from a picture and hops on one foot; a **6-year-old** does a standing long jump of at least 70 cm and counts to ten. **At 7 years** the child can think abstractly and is ready for school.



Figure 3.1. *The child's psychomotor development: speech (2, 4, 10–11 months, 1.5 and 2–3 years), movement (sitting at 7 months, standing at 8–9 months, walking at 1 year) and skills (Navruzova, Geppe)*

Sleep

Table 3.5. *Children's sleep requirements*

Age	Daily sleep
0–2 months	19 hours

Age	Daily sleep
3–5 months	17 hours
6–8 months	15 hours
9–12 months	13 hours
2–3 years	12.5 hours
4–5 years	11.5 hours
6–9 years	10 hours
10–12 years	9.5 hours
13–18 years	8–9 hours

3.3. Development of the sense organs

At birth all the sense organs work, but their reactions are generalized and undifferentiated.

Vision. The newborn's cornea is thick, the lens round and the eyeball short — **all newborns are farsighted (hypermetropic)**. In the first 2–3 weeks there is photophobia and no tears. At 2 weeks tears begin to be secreted and brief monocular fixation appears; at 3 weeks the baby looks binocularly at a stationary object; **at 2 months** he follows a bright toy; **at 3 months** binocular vision forms and he distinguishes red and yellow; at 6 months he distinguishes bright colors; at 3 years color discrimination is developed; visual acuity is low until 4–5 years. Transient strabismus and nystagmus are normal in the first months.

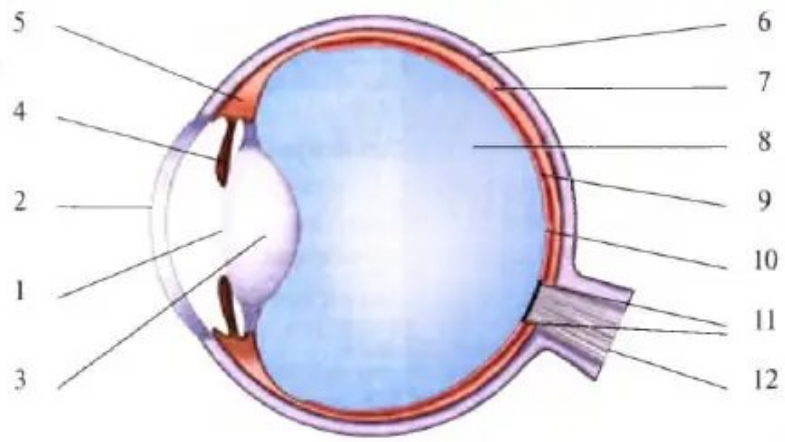


Figure 3.2. *Structure of the eye (diagram): 1 — pupil; 2 — cornea; 3 — lens; 4 — iris; 5 — ciliary body; 6 — sclera; 7 — choroid; 8 — vitreous body; 9 — retina; 10 — fovea; 11 — blind spot; 12 — optic nerve (Navruzova, Geppe)*

Hearing. The auditory analyzer is anatomically and functionally formed at birth — the newborn responds to sound by startling or crying. The external auditory canal is narrow and short, the eardrum thick and horizontal, and **the Eustachian tube short and wide** — a tendency to otitis. From 7–8 weeks the baby turns the head toward a sound, and at 3–4 months recognizes the mother's voice. **Audiological screening** is performed on all newborns in the maternity hospital.

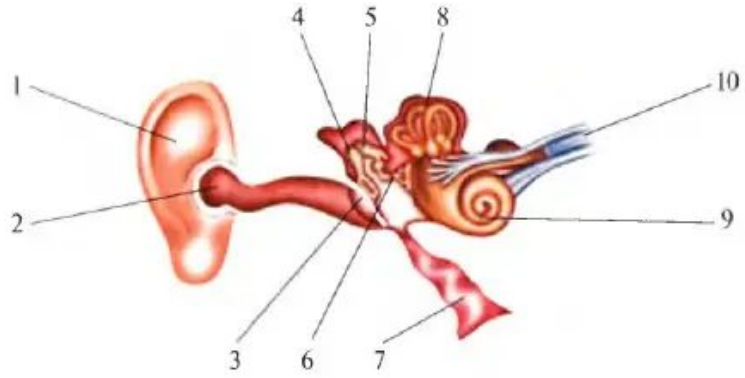


Figure 3.3. *Structure of the ear (diagram): 1 — auricle; 2 — external auditory canal; 3 — eardrum; 4 — malleus; 5 — incus; 6 — stapes; 7 — Eustachian tube; 8 — vestibular apparatus; 9 — cochlea; 10 — auditory nerve (Navruzova, Geppe)*

Taste works from birth: the baby responds positively (sucks) to sweet taste and negatively to bitter, sour and salty tastes; taste is fully differentiated by 4–5 months.

Smell: the baby distinguishes strong odors from the 1st month, pleasant and unpleasant ones at 2–4 months, and faint odors at 7–8 months.

Skin sensation: tactile sensation appears before the other senses. The newborn responds to a touch on the skin with a generalized movement; at 7–8 months he shows precisely the place of the stimulus; he distinguishes pain points at 7–8 years. In the newborn the sensitivity to cold is higher than to heat.

The vestibular apparatus works at birth: when the newborn is rotated, nystagmus appears in the direction opposite to the rotation; rocking calms him.

3.4. Methods of examining the nervous system

History: complaints (headache, dizziness, impaired memory and mood, convulsions, sleep disturbance), pregnancy, delivery and the neonatal period, development of static and mental functions. The parents of a child older than 5–6 years are questioned in the child's absence.

Inspection: the child's response to the surroundings (indifference, drowsiness, excitability), facial expression, mimicry, behavior, consciousness, position, shape and size of the head, fontanelles and sutures, stigmas of dysembryogenesis, coordination of movements (finger-nose and heel-knee tests), gait, speech.

The cranial nerves (12 pairs) are examined by methods appropriate to the child's age: response to odors (I), response to light and following an object (II), eye movements, pupillary reaction, ptosis, strabismus (III, IV, VI), sucking and chewing (V), symmetry of the face at rest and when crying (VII), response to sound and the rotation reflex (VIII), sucking and swallowing (IX, X), turning the head (XI), movements of the tongue (XII).

Muscle tone: the newborn has physiological flexor hypertonia. When the child is pulled to sitting by the hands, the arms extend slightly at the elbows and the head follows the trunk. In older children the tone is assessed by the resistance to passive movements. Reduced tone is **hypotonia**, increased tone **hypertonia**, and alternating tone **dystonia**.

Reflexes: in the newborn and infant, the innate unconditioned reflexes (section 3.5); tendon reflexes (easily elicited and with a wide zone until 2 years), skin (abdominal) reflexes (weak in children).

The autonomic nervous system — dermographism: a line is drawn on the skin with a blunt object. **Red** dermographism — predominance of parasympathetic tone; **white** — predominance of

sympathetic tone; **raised (urticarial)** — excessive vagal tone and increased vascular permeability; **mixed** — autonomic dystonia.

Additional methods: lumbar puncture and analysis of the cerebrospinal fluid (between LIII and LIV), **neurosonography** (ultrasound of the brain through the anterior fontanelle — in newborns and infants), electroencephalography, rheoencephalography, computed and magnetic resonance tomography, Doppler ultrasound of the cerebral vessels, electromyography.

Table 3.6. *Normal values of the cerebrospinal fluid at different ages*

Indicator	Under 14 days	14 days – 3 months	4–6 months	Over 6 months
Color and clarity	Often xanthochromic, clear	Colorless, clear	Colorless, clear	Colorless, clear
Protein, g/l	0,4–0,8	0,2–0,5	0,18–0,36	0,16–0,24
Cytosis, per 1 µl	Up to 10	Up to 8	Up to 7	Up to 3
Cell type	Mainly lymphocytes	Mainly lymphocytes	Lymphocytes	Lymphocytes
Glucose, mmol/l	1,7–3,9	2,2–3,9	2,2–4,4	2,0–4,4

3.5. The newborn's innate unconditioned reflexes

The unconditioned reflexes are divided into **segmental motor automatisms** (controlled through the spinal cord and brainstem) and **suprasegmental postural automatisms** (through the medulla oblongata and midbrain; they regulate tone according to the position of the head and body). The presence or absence of the reflexes, their symmetry and strength, and **their timely appearance and disappearance** are assessed.

Table 3.7. *The main unconditioned reflexes*

Reflex	How it is elicited	Response	Time of disappearance
Sucking	A pacifier or a finger is placed in the mouth	Rhythmic sucking movements	At the end of the 1st year
Search (Kussmaul)	Stroking the corner of the mouth without touching the lips	The corner of the lip drops, the head turns toward the stimulus	3–4 months
Proboscis	Lightly tapping the lips with a finger	The lips are pushed out like a proboscis	2–3 months
Palmomental (Babkin)	Pressing on the thenar eminences of both palms	Opens the mouth, bends the head forward	2–3 months
Grasp (Robinson)	A finger is placed in the baby's palm	Grasps the finger firmly; the baby can be lifted	3–4 months
Moro (embrace)	Striking the table 15–30 cm from the baby's head, or clapping	Phase I: the arms are abducted and the fingers open; phase II: the arms come together as in an embrace	4–5 months
Asymmetric tonic neck (Magnus–de Kleyn)	Turning the head to the side	The arm and leg on the side to which the head is turned extend, and those on the opposite side flex (“fencer's position”)	2–3 months
Protective	Placing the baby face down on the abdomen	Turns the head to the side	1.5–2 months

Reflex	How it is elicited	Response	Time of disappearance
Support	Holding the baby upright under the arms and touching the soles to the table	Flexes and then extends the legs and stands on the whole sole	1–1.5 months
Automatic walking	Holding the baby upright and tilting the body slightly forward	Makes stepping movements	1–1.5 months
Crawling (Bauer)	Placing a palm against the soles of the baby lying on the abdomen	Pushes off with the legs and crawls forward	4 months
Galant	Paravertebral stroking along the spine from top to bottom	The trunk bends in an arc open toward the stimulus	3–4 months
Perez	Running a finger along the spinous processes from the sacrum to the neck of a baby lying prone on the palm	Raises the head and pelvis, flexes the limbs, cries	3–4 months
Babinski	Lightly stroking the outer edge of the sole	The big toe extends, and the others fan out	Physiological until 2 years

Suprasegmental reflexes: the upper Landau reflex (from 4 months) — a baby held in the air by the abdomen raises the head and the upper part of the trunk and holds them in line; **the lower Landau** reflex (from 5–6 months) — also extends and raises the legs; **the labyrinthine reflex**.

CLINICAL SIGNIFICANCE

Absent or weak reflexes (in a newborn) may be a sign of perinatal CNS damage, hypoxia, prematurity or sepsis. **Failure of the reflexes to disappear on time** (for example, persistence of the Moro and grasp reflexes in a 6-month-old child) indicates delayed cortical control and is an early sign of cerebral palsy. **Asymmetry** (a reflex absent on one side) is characteristic of a brachial plexus injury (Erb–Duchenne palsy) or a clavicle fracture.

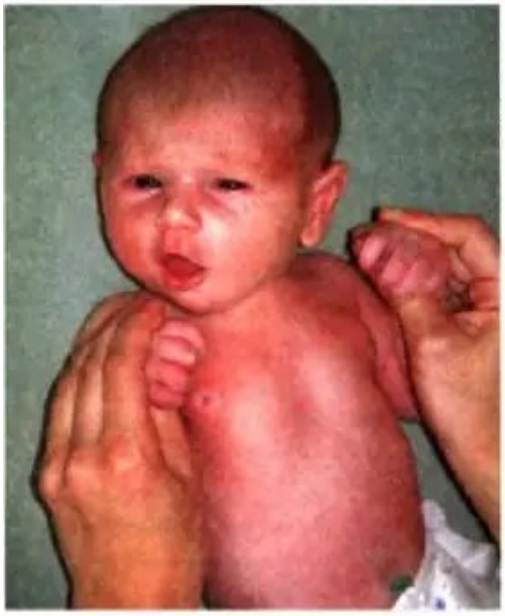


Figure 3.4. *The palmomental (Babkin) reflex (Navruzova, Geppe)*



Figure 3.5. *The search reflex (Navruzova, Geppe)*



Figure 3.6. *The grasp (Robinson) reflex (Navruzova, Geppe)*



Figure 3.7. *The Moro reflex (Navruzova, Geppe)*



Figure 3.8. *The protective reflex (Navruzova, Geppe)*



Figure 3.9. *The Galant reflex (Navruzova, Geppe)*



Figure 3.10. *The Perez reflex (Navruzova, Geppe)*



Figure 3.11. *The support and automatic walking reflexes (Navruzova, Geppe)*



Figure 3.12. *The crawling reflex (Navruzova, Geppe)*



Figure 3.13. *The proboscis reflex (Kildiyarova)*

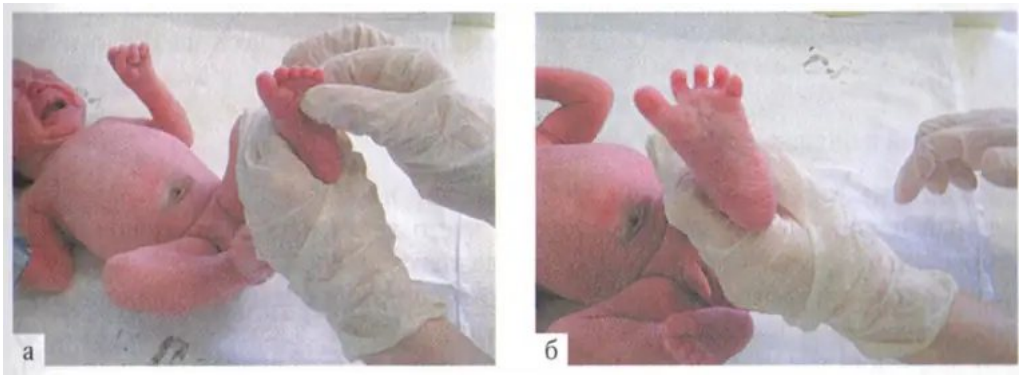


Figure 3.14. *The Babinski reflex: a — stroking the edge of the sole; b — the response (Kildiyarova)*

3.6. Semiotics of nervous system damage

Main complaints and objective signs

Headache, dizziness, vomiting, convulsions, disturbed sleep and behavior, reduced intellect and memory, movement disorders. On inspection: indifference to the surroundings or excitability, a forced position (the “pointer dog” position — meningitis, opisthotonus), the shape and size of the head (hydrocephalus, microcephaly, “tower skull”), convulsions, muscle development and tone, impaired coordination, stigmas of dysembryogenesis (there should be no more than 5).

Focal signs: a difference in the width of the palpebral fissures, ptosis, strabismus (pairs III, IV, VI), drooping of the corner of the mouth and flattening of the nasolabial fold on one side (pair VII — facial nerve paresis), bulbar disorders (pairs IX–X); paresis and paralysis, tremor, hyperkinesis; ataxia, nystagmus; sensory disturbances.



Figure 3.15. *Convergent strabismus (Navruzova, Geppe)*



Figure 3.16. *Left-sided facial nerve paresis: on the affected side there is no mimicry, the palpebral fissure is wide, the nasolabial fold is flattened and the corner of the mouth droops (Navruzova, Geppe)*



Figure 3.17. *Hypertensive-hydrocephalic syndrome (Navruzova, Geppe)*

Main syndromes

Syndrome of reduced excitability (hypoexcitability) — reduced motor and mental activity, a prolonged latent period of the reflexes, hyporeflexia, muscle hypotonia. It occurs in perinatal CNS damage, hereditary diseases (Down syndrome, phenylketonuria), metabolic disorders (hypoglycemia, acidosis) and severe somatic diseases.

Hyperexcitability syndrome — motor restlessness, emotional instability, sleep disturbance, enhanced innate reflexes, tremor, a tendency to convulsions, often muscle hypertonia.

Hydrocephalic and hypertensive-hydrocephalic syndrome — an increase in the volume of cerebrospinal fluid and dilation of the cerebral ventricles: rapid enlargement of the head, separation of the sutures, enlargement and bulging of the anterior fontanelle, dilated veins on the scalp, thinning of the skin of the temples, disproportion between the cranial and facial parts. When intracranial pressure rises, these are joined by a high-pitched, painful cry, vomiting, **Graefe's sign (“setting sun”)** — a strip of sclera visible above the pupil, convergent strabismus, horizontal nystagmus, and a “cracked pot” sound on percussion of the skull.

Convulsive syndrome — paroxysmal involuntary muscle contractions: **clonic** (short, successive contractions), **tonic** (prolonged contraction), **tonic-clonic**; generalized and partial (local). Causes:

- in newborns: brain malformations, perinatal CNS damage (hypoxia, birth injury, hemorrhage), intrauterine infections, metabolic disorders (hypoglycemia, hypocalcemia, hypomagnesemia, hyperbilirubinemia), withdrawal syndrome;
- in infancy: neuroinfections, hyperthermia, spasmophilia;
- in older children: epilepsy, brain tumors, hemorrhage.

Febrile seizures are convulsions, most often generalized tonic-clonic, that occur in children aged 6 months to 5 years against the background of fever (usually above 38 °C) without a neuroinfection. **Breath-holding spells** — cessation of breathing at the peak of a deep inspiration during intense crying or emotional stress, with cyanosis, brief loss of consciousness and sometimes convulsions.

Movement disorder syndrome — hypotonia, hypertonia, dystonia; **cerebral palsy (CP)** — hypotonia and hyporeflexia at birth, dystonia from the 2nd month, then muscle hypertonia (the arms internally rotated and extended, the fingers clenched in a fist, the legs crossed).

Coma syndrome (see topic 1, section 1.8) — grade I: eyes closed, uncoordinated defensive movements, pupillary reflexes preserved; grade II: no defensive reaction to pain and no pupillary reflexes, breathing and cardiac activity preserved; grade III: breathing impaired, pupils dilated and fixed, BP fallen.

Syndrome of autonomic-visceral disorders — mottled and pale skin, acrocyanosis, cold hands and feet, profuse sweating, prolonged subfebrile temperature without infection, emotional instability.

Meningeal syndrome

The meningeal syndrome is irritation of the meninges; it occurs in all forms of acute meningitis. It consists of three groups of signs:

1. **General cerebral signs:** fever, severe headache, vomiting that brings no relief, hyperesthesia (heightened sensitivity of the skin, to light and to sound), impaired consciousness, convulsions.
2. **Meningeal signs** (signs of the meninges).
3. **Changes in the cerebrospinal fluid** (increased pressure, cytosis, protein).

Table 3.8. *Meningeal signs and how to detect them*

Sign	How it is detected	Positive result
Neck stiffness	Passively flexing the head of a child lying on the back toward the chest	The chin does not reach the chest; resistance and pain
Kernig's sign	Flexing the leg of a child lying on the back to 90° at the hip and knee, then extending it at the knee	The leg cannot be extended at the knee
Upper Brudzinski sign	Passive flexion of the head toward the chest	The legs flex reflexively at the hips and knees
Middle (pubic) Brudzinski sign	Pressing on the pubic region	The legs flex at the hips and knees
Lower Brudzinski sign	Strong flexion of one leg at the hip and knee (during the Kernig test)	The other leg also flexes reflexively
Lesage's sign (suspension sign)	Lifting the infant by holding him under the arms	Draws the legs up to the abdomen and holds them flexed
Bulging anterior fontanelle	Palpation of the fontanelle in an infant	The fontanelle is bulging and tense, with increased pulsation

CAUTION

Kernig's sign is physiological in children under 4 months (because of flexor hypertonia) — at this age it is not a sign of meningitis. In infants the most reliable signs of meningitis are **a bulging anterior fontanelle and Lesage's sign**.

Meningism — meningeal signs are present, but the cerebrospinal fluid is unchanged (only the pressure is slightly raised); it arises from a toxic effect in severe infections (influenza, pneumonia, intestinal infection).

Encephalitic syndrome

The encephalitic syndrome is damage to the brain substance (encephalitis, meningoencephalitis, toxic encephalopathy). Signs:

1. **General cerebral signs:** impaired consciousness (from somnolence to coma), psychomotor agitation, convulsions, headache, vomiting, fever.
2. **Focal signs:** paresis and paralysis (hemiparesis), cranial nerve damage (strabismus, ptosis, facial nerve paresis, bulbar disorders), hyperkinesis, ataxia, speech disturbance (aphasia), pathological reflexes.
3. **Cerebrospinal fluid:** pressure increased, protein slightly increased, lymphocytic cytosis slight or normal.

Table 3.9. *Changes in the cerebrospinal fluid in various conditions*

Sign	Serous meningitis	Purulent meningitis	Encephalitis	Cerebral hemorrhage
Pressure	Increased	Sharply increased	Increased	Increased
Clarity	Clear	Turbid	Clear	Slightly turbid, bloody or xanthochromic
Protein	Increased	Sharply increased	Slightly increased	Increased
Cytosis	Increased (hundreds)	Sharply increased (thousands)	Slightly increased	Increased (erythrocytes)
Cells	Lymphocytes	Neutrophils	Lymphocytes	Erythrocytes

Pathological reflexes

In children over 2 years the following reflexes are signs of **damage to the pyramidal tract**:

- **Babinski** — on stroke-like stimulation of the outer edge of the sole, the big toe extends and the others fan out;
- **Oppenheim** — the same response when the fingers are run down the anterior edge of the tibia;
- **Gordon** — on squeezing the calf muscle;
- **Schaeffer** — on squeezing the Achilles tendon;
- **Rossolimo** — flexion of the toes toward the sole when the tips of the toes are struck briskly;
- **ankle clonus** — rhythmic contractions when the foot is sharply dorsiflexed.

Signs of latent spasmophilia (in hypocalcemia; in detail in topic 5): **Chvostek** — contraction of the eyelid and lip muscles when the zygomatic arch is tapped with a hammer; **Trousseau** — the “obstetrician's hand” when the upper arm is compressed; **Lust** — outward turning of the foot when tapped behind the head of the fibula (“ballerina's foot”).

3.7. Body temperature and thermoregulation

Features of thermoregulation in children

Thermoregulation is the balance between heat production (**chemical thermoregulation**) and heat loss (**physical thermoregulation**). Its center is **the hypothalamus**.

- The hypothalamic centers are immature — thermoregulation is **imperfect**, especially in newborns and preterm babies; the child **cools down and overheats quickly**.
- The body surface area is large relative to weight (2–3 times that of adults) — heat loss is high.
- Heat is produced mainly by **brown adipose tissue** (non-shivering thermogenesis); the newborn cannot produce heat by shivering.
- The sweat glands work poorly (full function at 5–7 years) — heat loss is difficult in hot weather, and the newborn overheats easily.
- The skin is rich in blood vessels, and the subcutaneous fat is thin (almost absent in preterm babies).
- Newborns cannot produce a typical febrile reaction — in a severe infection (sepsis) the temperature may not rise or may even fall.

> **REMEMBER**
> The temperature in the delivery room should be **24–26 °C**; the newborn is immediately wrapped in a warm diaper, dried and placed on a radiant warmer table (the “warm chain”). Preterm babies weighing less than 1500 g are cared for **in an incubator** (34.5–35 °C).

Measuring body temperature

Table 3.10. *Normal values of body temperature*

Site of measurement	Normal	Note
Axilla	36.0–37.0 °C (average 36.6)	The main method
Inguinal fold	Close to the axillary value	In newborns and infants
Oral cavity	0.3–0.5 °C higher than in the axilla	In children over 5 years
Rectum	0.5–1.0 °C higher than in the axilla (36.8–37.5)	Core temperature; with care
Ear (tympanic)	Close to the rectal value	Infrared thermometer

The temperature is measured twice a day — **in the morning at 7–8 am and in the evening at 5–7 pm** — and recorded on **the temperature chart** as a curve. In a healthy child the daily fluctuation **does not exceed 1 °C** (lower in the morning, higher in the evening). The axilla should be dry; a mercury thermometer is held for 5–10 minutes, and an electronic thermometer until the signal. The temperature rises slightly when the child moves, cries, and after meals and bathing.

Hypothermia — an axillary temperature below 36.0 °C (below 36.5 °C in newborns): prematurity, chilling, severe infection, hypothyroidism, collapse.

3.8. Fever

Fever (febris) is a rise in body temperature resulting from an **upward shift of the “set point”** of the hypothalamic thermoregulatory center under the action of pyrogens. It is a protective-adaptive reaction of the body: the production of interferon and antibodies increases, and the multiplication of many microbes slows.

Hyperthermia (overheating) differs from fever: the set point does not change, and the temperature rises because of impaired heat loss (a hot room, heavy wrapping, dehydration, heatstroke). Antipyretic drugs are ineffective here — the child must be cooled.

Degrees of fever

Table 3.11. *Types of fever by height (axillary temperature)*

Type	Temperature
Subfebrile	37.1–38.0 °C
Moderate (febrile)	38.1–39.0 °C
High	39.1–41.0 °C
Hyperpyretic	Above 41.0 °C

Stages of fever

1. **Rise of temperature** (stadium incrementi) — heat production exceeds heat loss: the skin is pale and cold, shivering, “goose bumps”, the child feels cold.
2. **Plateau of high temperature** (stadium fastigii) — the skin is hot and flushed, tachycardia, tachypnea, headache.
3. **Fall of temperature** (stadium decrementi) — heat loss predominates, profuse sweating. The fall occurs by **crisis** (sharply within a few hours — risk of collapse) or by **lysis** (gradually over several days).

Types of fever by the temperature curve

Table 3.12. *Types of temperature curve*

Type	Description	When it occurs
Continuous (febris continua)	High temperature, daily fluctuation less than 1 °C	Lobar pneumonia, typhoid fever
Remittent (febris remittens)	Daily fluctuation more than 1 °C, but it does not fall to normal	Pneumonias, viral infections, many bacterial infections
Intermittent (febris intermittens)	High temperature alternates with normal temperature, fluctuation more than 1 °C	Malaria, pyelonephritis, purulent processes
Hectic (wasting) (febris hectica)	Daily fluctuation of 3–5 °C, with profuse sweating	Sepsis, abscess, severe tuberculosis
Relapsing (febris recurrens)	Febrile periods of several days alternate with afebrile periods	Relapsing fever
Undulant (febris undulans)	The temperature rises gradually and falls gradually	Brucellosis, lymphogranulomatosis
Inverse (febris inversa)	The morning temperature is higher than the evening temperature	Tuberculosis, sepsis
Irregular (atypical) (febris irregularis)	Irregular changes	Rheumatism, sepsis, many diseases

MEMORY AID

The types of temperature curve are easy to remember by **the daily fluctuation**: **continuous — less than 1 °C**; **remittent — more than 1 °C, does not fall to normal**; **intermittent — falls to normal**; **hectic — 3–5 °C**.

Clinical types of fever in children

Table 3.13. *"Pink" and "white" fever*

Sign	"Pink" (red) fever	"White" (pale) fever
Skin	Pink, hot, moist	Pale, mottled, dry
Limbs	Warm	Cold, acrocyanosis, bluish nails and lips
Mechanism	Heat production and heat loss are balanced	Spasm of the peripheral vessels — heat loss is impaired
Condition	Relatively satisfactory	Severe: lethargy or agitation, tachycardia, risk of convulsions
Prognosis	Good	Dangerous — emergency care is needed

CLINICAL SIGNIFICANCE

Antipyretic treatment is not necessary in every fever. In previously healthy children over 3 months, the temperature is usually lowered when it is **above 39.0–39.5 °C** or when the child feels unwell. **In the risk group** (children under 3 months, a history of febrile seizures, neurological and heart diseases) and **in "white" fever** the temperature is lowered from **38.0–38.5 °C**. In children only **paracetamol** and **ibuprofen** are recommended; acetylsalicylic acid (risk of Reye syndrome) and metamizole (risk of agranulocytosis) are not used. In "white" fever, warming of the limbs and vasodilator drugs are also needed.

3.9. Clinical part

Delayed development — the first sign of disease

Regular assessment of NPD makes it possible to detect many diseases early:

- If a 3-month-old child **does not hold up the head** and **does not smile** — perinatal CNS damage, hypothyroidism, a visual or hearing defect should be considered.
- If **the grasp and Moro reflexes persist** in a 6-month-old child and the muscle tone is increased — risk of cerebral palsy.
- If a 1-year-old child **does not respond to sound**, does not coo or babble — a hearing defect; an audiological examination is needed.
- If a 2-year-old child does not speak, does not make eye contact and does not respond to his name — autism spectrum disorder, a hearing defect or intellectual delay.
- **Loss of acquired skills (regression)** is always a serious sign: a neurodegenerative disease, severe hypotrophy, encephalitis.

Fever and danger signs

> **CLINICAL SIGNIFICANCE**
> In a febrile child the following **“red flag”** signs require urgent admission: age under 3 months; lethargy, barely opening the eyes; incessant vomiting; convulsions; a bulging fontanelle or neck stiffness; a hemorrhagic rash that does not fade on pressure (meningococcal infection); “white” fever; respiratory distress; signs of dehydration.

Practical skills

Skill 1. Assessing neuropsychological development

1. Establish the child's age (in preterm children — the corrected age).
2. Observe the child during play and independent activity, and ask the mother when the skills appeared.
3. Assess by the leading lines: visual and auditory reactions, emotions, gross movements, hand movements, understanding of speech, active speech, skills.
4. Compare the result with the age norm (±1 month in the 1st year, ±1 quarter in the 2nd year, ±half a year in the 3rd year).
5. Conclusion: NPD appropriate for age / delayed (in which lines) / advanced.

Skill 2. Examining the unconditioned reflexes in a newborn

1. Conditions: a warm room (22–26 °C), 30–60 minutes after a feed, the baby calm and awake.
2. **In the supine position:** the search, sucking, proboscis, Babkin, grasp, Moro, tonic neck and Babinski reflexes.
3. **In the vertical position:** the support and automatic walking reflexes.
4. **In the prone position:** the protective, crawling (Bauer), Galant and Perez reflexes.
5. Assess the presence, strength, symmetry and age-appropriateness of each reflex.
6. Assess the muscle tone (the baby's posture, the pull-to-sit test); record the result.

Skill 3. Detecting meningeal signs

1. Lay the child on the back on a flat surface and remove the pillow.
2. **Neck stiffness:** place the left hand under the child's head and passively flex the head toward the chest.
3. **Upper Brudzinski:** at the same time watch for flexion of the legs.
4. **Kernig:** flex the leg to 90° at the hip and knee and try to extend it at the knee (on both sides).
5. **Lower Brudzinski:** watch the other leg while testing Kernig's sign.
6. **Middle Brudzinski:** press on the pubic region.
7. In infants: palpate **the anterior fontanelle** and test **Lesage's sign**.
8. Record the result (taking into account that Kernig's sign is physiological under 4 months).

Skill 4. Measuring the axillary temperature and filling in the temperature chart

1. Check and disinfect the thermometer; shake a mercury thermometer down below 35 °C.
2. Wipe the axilla dry.

3. Place the tip of the thermometer deep in the axilla; hold the child's arm pressed against the chest (in a young child the mother holds it).
4. Wait 5–10 minutes with a mercury thermometer, or until the signal with an electronic thermometer.
5. Read the result and disinfect the thermometer.
6. Mark the result on the temperature chart with a dot (in the morning “M” or evening “E” column) and join it to the previous dot with a line.
7. Determine the type of fever from the curve.

Review questions

1. State the main anatomical and physiological features of the newborn's nervous system.
2. List the main stages of statics and motor skills.
3. In what stages does speech develop?
4. How do emotions and communication develop? What is the “revival complex”?
5. In which leading lines is NPD assessed, and within what limits is it considered appropriate for age?
6. What are the features of vision and hearing in the newborn?
7. State the newborn's innate unconditioned reflexes, how they are elicited and when they disappear.
8. What are the signs of the hypertensive-hydrocephalic syndrome?
9. What signs make up the meningeal syndrome? How are meningeal signs detected?
10. How does the encephalitic syndrome differ from the meningeal syndrome?
11. Which reflexes are considered pathological?
12. What are the features of thermoregulation in children?
13. How is body temperature measured, and what are its normal values?
14. What is the difference between fever and hyperthermia? State the degrees and stages of fever.
15. State the types of fever by the temperature curve and the difference between “pink” and “white” fever.

Test questions

1. When does the child hold the head well in the vertical position?
   A) At birth
   B) At 2–3 months
   C) At 6 months
   D) At 9 months
   E) At 1 year

2. The child begins to sit independently:
   A) At 3 months
   B) At 4–5 months
   C) At 6–7 months
   D) At 10 months
   E) At 12 months

3. The “revival complex” appears:
   A) At 1 week
   B) At 3 months
   C) At 6 months
   D) At 9 months
   E) At 1 year

4. The vocabulary of a healthy 1-year-old child:
   A) No words
   B) 8–12 words
   C) 30–40 words
   D) 200–300 words
   E) 1000 words

5. When does the Moro reflex disappear?
   A) At 1 month
   B) At 2 months
   C) At 4–5 months
   D) At 1 year
   E) At 2 years

6. Until what age is the Babinski reflex considered physiological?
   A) Until 1 month
   B) Until 3 months
   C) Until 6 months
   D) Until 2 years
   E) Until 5 years

7. How the Galant reflex is elicited:
   A) Pressing on the palm
   B) Stroking the corner of the mouth
   C) Paravertebral stroking along the spine
   D) Touching the soles to the table
   E) Striking the table

8. The most reliable sign of meningitis in an infant:
   A) Kernig's sign
   B) Bulging of the anterior fontanelle and Lesage's sign
   C) The Babinski reflex
   D) The grasp reflex
   E) Red dermographism

9. Until what age may Kernig's sign be physiological?
   A) Until 1 month
   B) Until 4 months
   C) Until 1 year
   D) Until 3 years
   E) Never

10. Graefe's sign (“setting sun”) is characteristic of:
    A) Meningism
    B) The hypertensive-hydrocephalic syndrome
    C) Spasmophilia
    D) Hypothyroidism
    E) Febrile seizures

11. What predominates in the cerebrospinal fluid in purulent meningitis:
    A) Lymphocytes
    B) Neutrophils
    C) Erythrocytes
    D) Eosinophils
    E) No cells

12. The normal axillary temperature:
    A) 35.0–36.0 °C
    B) 36.0–37.0 °C
    C) 37.0–38.0 °C
    D) 37.5–38.5 °C
    E) 38.0–39.0 °C

13. Temperature 38.6 °C. The degree of fever:
    A) Subfebrile
    B) Moderate (febrile)
    C) High
    D) Hyperpyretic
    E) Normal

14. A continuously high temperature with a daily fluctuation of less than 1 °C:
    A) Febris continua
    B) Febris remittens
    C) Febris intermittens
    D) Febris hectica
    E) Febris inversa

15. A sign characteristic of “white” fever:
    A) Pink and moist skin
    B) Warm limbs
    C) Mottled skin, cold limbs
    D) The prognosis is always good
    E) There is no tachycardia

16. At birth, approximately what percentage of the body weight does the brain weight make up?
    A) 20%
    B) 15%
    C) 10%
    D) 5%
    E) 2%

17. Which weeks of pregnancy correspond to the critical period of brain development (the period of the most intensive division of nerve cells)?
    A) 36–40
    B) 10–18
    C) 24–28
    D) 4–8
    E) 1–4

18. To the level of which vertebra does the spinal cord descend in the newborn (taken into account in lumbar puncture)?
    A) L III
    B) L I
    C) S II
    D) L V
    E) T XII

19. The age at which the child begins to sit independently:
    A) 5 months
    B) 10 months
    C) 4 months
    D) 3 months
    E) 7–7.5 months

20. At what age does the child's first smile (in response to adults) appear?
    A) 6 months
    B) 1.5–2 months
    C) 4 months
    D) 9 months
    E) 1–2 weeks

21. At what age does the “revival complex” form?
    A) 2 months
    B) 1 year
    C) 1 month
    D) 6 months
    E) 3 months

22. Approximately how many meaningful words does a child say at 1 year?
    A) 8–12
    B) 100
    C) 30–40
    D) 1–2
    E) 200–300

23. What does red dermographism indicate?
    A) Cerebral edema
    B) Predominance of parasympathetic tone
    C) Reduced vascular permeability
    D) Predominance of sympathetic tone
    E) Absence of the norm

24. What is the refraction of the newborn's eye?
    A) Emmetropia
    B) Nearsighted (myopia)
    C) Astigmatism
    D) Farsighted (hypermetropia)
    E) Color blindness

25. What causes the tendency to otitis in newborns:
    A) A short and wide Eustachian tube
    B) A long and narrow Eustachian tube
    C) Absence of the auditory ossicles
    D) A wide external auditory canal
    E) A thin eardrum

26. The normal protein content of the cerebrospinal fluid in a child over 6 months:
    A) 1.0–1.5 g/l
    B) 0.4–0.8 g/l
    C) 0.18–0.36 g/l
    D) 0.16–0.24 g/l
    E) 0.2–0.5 g/l

27. When does the search (Kussmaul) reflex disappear?
    A) At 1 month
    B) At 2 years
    C) At 3–4 months
    D) At 1 year
    E) At 2–3 months

Clinical cases

Case 1

A mother brought her child for a preventive checkup. When placed on the abdomen, the child raises the head and chest supported on the forearms, rolls onto the side, grasps a hanging toy and examines it, laughs out loud, recognizes the mother and coos in a prolonged sing-song way. The grasp, Babkin, support and automatic walking reflexes cannot be elicited; the upper Landau reflex is present.

Questions: 1) Determine the child's approximate age. 2) Is the NPD appropriate for age? 3) Which reflex is expected to appear in 1–2 months?

Answer: 1) About 4 months: leaning on the forearms on the abdomen, rolling onto the side, grasping a toy, loud laughter, recognizing the mother, sing-song cooing and the upper Landau reflex are characteristic of 4 months. 2) Yes, the NPD is appropriate for age: the grasp, Babkin, support and automatic walking reflexes have disappeared on time. 3) The lower Landau reflex (at 5–6 months) — when held in the air by the abdomen, the child also extends and raises the legs.

Case 2

An 8-month-old child was brought to hospital with fever (39.4 °C) for 2 days and repeated vomiting. On examination the child is lethargic and gives a high-pitched, monotonous cry when stimulated. The anterior fontanelle is bulging and tense, with increased pulsation. When lifted under the arms, he draws the legs up to the abdomen and holds them there. The neck muscles are stiff.

Questions: 1) Which syndrome should be considered? 2) Indicate the symptoms. 3) What investigation is needed to establish the diagnosis, and what result is expected?

Answer: 1) The meningeal syndrome — acute meningitis is suspected. 2) General cerebral signs (fever, repeated vomiting, lethargy, monotonous cry) and meningeal signs: a bulging anterior fontanelle, a positive Lesage's sign, neck stiffness. 3) Lumbar puncture and analysis of the cerebrospinal fluid: the pressure is increased; in purulent meningitis the fluid is turbid, the cytosis is in the thousands with predominance of neutrophils, and the protein is sharply increased; in serous meningitis the fluid is clear with lymphocytic cytosis. A blood count and CSF culture; the child is treated urgently.

Case 3

A 2-year-old child has ARVI. Temperature 39.2 °C. The child is lethargic, the skin is pale and mottled, the hands and feet are cold, the nails bluish, HR 160 per minute. According to the mother, he had a seizure against the background of fever at 1 year of age.

Questions: 1) Determine the degree and clinical type of fever. 2) Why should the child be included in the risk group? 3) What is the physician's management?

Answer: 1) High fever (39.1–41.0 °C); clinical type — “white” (pale) fever: spasm of the peripheral vessels (mottled skin, cold limbs, acrocyanosis, tachycardia). 2) There is a history of a febrile seizure (within the age range of 6 months – 5 years) and “white” fever — the risk of a recurrent seizure is high. 3) Lower the temperature immediately (paracetamol or ibuprofen at the age-appropriate dose) and at the same time warm the limbs and relieve the vascular spasm (vasodilator drugs prescribed by the physician); monitor the condition and decide on hospital admission.

Quick review

Table 3.14. *Key facts on Topic 3*

Question	Answer
Brain weight at birth	350–400 g (10% of the body weight)
Critical period of brain development	Weeks 10–18 of pregnancy
Motor milestones	2 months head, 4 months rolling over, 6 months sitting, 8 months standing, 10 months walking with support, 12 months walking independently
First smile	1.5–2 months
Revival complex	3 months
Cooing / babbling / first words	2–3 months / 6–7 months / 11–12 months
Vocabulary	8–12 at 1 year; 30–40 at 1.5 years; 200–300 at 2 years
NPD appropriate for age	1st year ±1 month; 2nd year ±a quarter; 3rd year ±half a year
Binocular vision	3 months
Reflexes that disappear early	Support and automatic walking 1–1.5 months; protective 1.5–2 months; Babkin and proboscis 2–3 months
Reflexes that disappear at a middle age	Grasp, search, Galant, Perez 3–4 months; Moro 4–5 months; Bauer 4 months
Reflexes that persist long	Sucking until 1 year; Babinski until 2 years

Question	Answer
Landau	Upper from 4 months, lower from 5–6 months
Meningeal signs	Neck stiffness, Kernig, Brudzinski (upper, middle, lower), Lesage, bulging fontanelle
Physiological Kernig	Until 4 months
Graefe's sign	Hypertensive-hydrocephalic syndrome
CSF in purulent / serous meningitis	Neutrophils / lymphocytes
Axillary temperature	36.0–37.0 °C; rectal +0.5–1 °C
Degrees of fever	Subfebrile 37.1–38; moderate 38.1–39; high 39.1–41; hyperpyretic >41
Types of fever	Continuous <1 °C; remittent >1 °C; intermittent falls to normal; hectic 3–5 °C
“White” fever	Pale mottled skin, cold limbs — dangerous
Age of febrile seizures	6 months – 5 years
Features of the nervous system in the newborn	Brain weight 350–400 g (10% of the body weight); cortex immature, inhibition predominates; subcortical structures predominate; myelination incomplete; high permeability of the BBB
Motor development (months)	1 — raises the head; 3 — holds the head; 4 — rolls over; 6 — sits with support; 7 — crawls, sits independently; 9 — stands holding on; 12 — walks
Speech development	2–3 months — cooing; 6–7 months — babbling (ba-ba); 11–12 months — first words (8–12); 2 years — simple sentences; 3 years — complex sentences, ≈1000 words
Development of emotions	Smile at 1.5–2 months; revival complex at 3 months; loud laughter at 4–5 months; fear of strangers at 6–12 months
Daily duration of sleep	0–2 months 19 h; 3–5 months 17 h; 6–8 months 15 h; 9–12 months 13 h; 2–3 years 12.5 h; 6–9 years 10 h; 13–18 years 8–9 h
Types of dermographism	Red — parasympathetic; white — sympathetic; raised (urticarial) — excessive vagal tone; mixed — autonomic dystonia
Development of vision	The newborn is hypermetropic; 2 weeks — tears; 2 months — follows a toy; 3 months — binocular vision; 3 years — color discrimination; visual acuity low until 4–5 years
Normal CSF (over 6 months)	Protein 0.16–0.24 g/l; cytosis ≤ 3 cells/μl; lymphocytes; glucose 2.0–4.4 mmol/l
Unconditioned reflexes (time of disappearance)	Sucking — 1 year; search — 3–4 months; proboscis and palmomental (Babkin) — 2–3 months

TOPIC 4

The skin, subcutaneous fat and lymphatic system: anatomical and physiological features, methods of examination, semiotics

Learning objectives

The student should know:

- the anatomical and physiological features of children's skin, the skin appendages (hair, nails, glands) and the subcutaneous fat;
- the primary and secondary morphological elements of the skin;
- the skin changes characteristic of the newborn period;
- the main signs and syndromes of damage to the skin and subcutaneous fat;
- the anatomical and physiological features of the lymphatic system (lymph nodes, spleen, thymus) and the methods of examining it;
- the semiotics of lymphadenopathy, splenomegaly and lymphatic-hypoplastic diathesis.

The student should be able to:

- inspect the skin and its appendages and describe a rash correctly;
- determine the moisture, temperature and elasticity of the skin by palpation; perform the capillary fragility tests and test dermographism;
- assess the thickness, distribution and consistency of the subcutaneous fat and the tissue turgor; detect edema;
- palpate and describe all groups of peripheral lymph nodes;
- palpate and percuss the spleen, and percuss the thymus;
- write the corresponding sections of the case history.

4.1. Anatomical and physiological features of the skin

The skin is the organ that covers the body from the outside and performs the functions of a barrier between the internal and external environment, metabolism, thermoregulation, respiration and sensation. It consists of three layers: **the epidermis**, **the dermis** (papillary and reticular layers) and **the subcutaneous fat** (hypodermis).

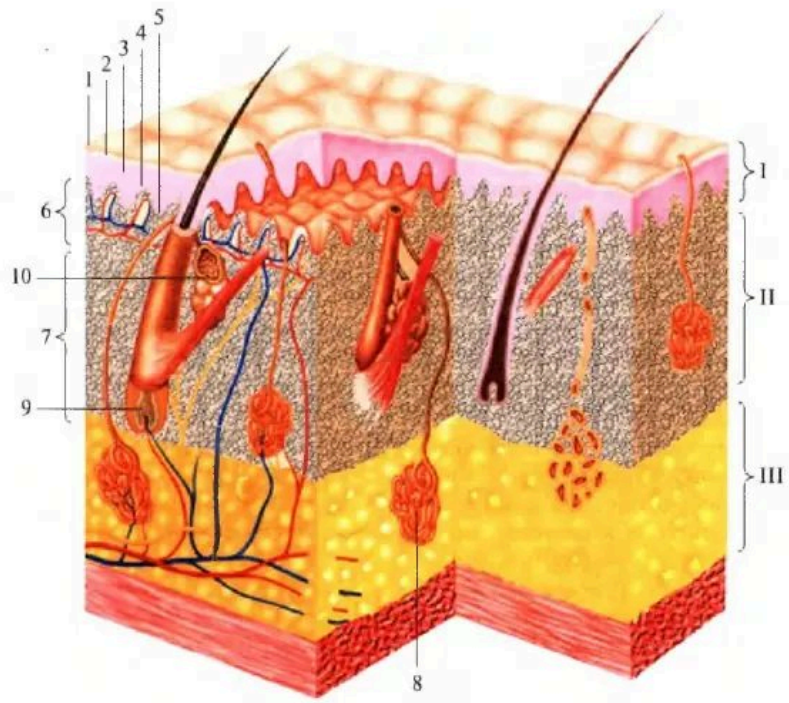


Figure 4.1. *Structure of the skin. I — epidermis: 1 — horny, 2 — lucid, 3 — granular, 4 — spinous, 5 — basal layer; II — dermis: 6 — papillary, 7 — reticular layer; III — subcutaneous fat; 8 — sweat glands, 9 — hair follicle, 10 — sebaceous glands (Navruzova, Geppe)*

Structural features:

- In children under 3 years the skin layers are **1.5–3 times thinner** than in adults; they approach those of adults only **at 7 years**.
- The epidermal cells lie far apart, and the structure is loose; in the newborn the horny layer consists of 2–3 rows and is easily shed.
- The granular layer is poorly developed — the newborn's skin is **translucent and pale pink**.
- **The connection between the epidermis and the dermis is very delicate** (the basement membrane and connective fibers are weak) — in various diseases the epidermis is easily detached and **blisters** form.
- The pH of the skin surface in the newborn is close to neutral and falls by the end of the 1st month; the bactericidal activity of the secretion is weak.
- The capillary network is extensive and well developed — the skin is pink, and absorption and heat loss are high.
- The nerve endings are morphologically immature but functionally mature — pain, tactile and temperature sensation are complete.

Table 4.1. *Functional features of children's skin*

Function	Feature in children
Protective	Weak: the horny layer is thin, keratinization and the connection with the dermis are insufficient; pH close to neutral; local immunity not formed — a tendency to infection and injury
Pigment formation	Immature — weak stimulation of the melanocytes; sensitive to sunlight
Resorptive (absorption)	High — the horny layer is thin and rich in vessels; ointments containing hormones or toxic substances have a systemic effect

Function	Feature in children
Excretory (sweating)	Imperfect
Thermoregulatory	Weak: heat loss exceeds heat production — the child overheats and cools down quickly
Respiration	Several times stronger than in adults
Synthetic	Complete from weeks 3–4: vitamin D_3 is synthesized under the action of UV rays
Sensory	Complete from birth; the newborn first recognizes the mother through tactile sensation

Skin appendages:

- **The sebaceous glands** work already in the fetal period and produce **the cheese-like coating** (vernix caseosa) — it protects the skin from the amniotic fluid and eases passage through the birth canal. The glands work intensively in the 1st year, then their activity decreases and increases again at puberty (acne).
- **The sweat glands** (eccrine) are not fully formed at birth, and their ducts are closed by epithelium. Sweating begins **from weeks 3–4** and is incomplete until 3–4 months; until 3 years sweating occurs only at high temperatures. The sweat glands and the thermoregulatory center mature fully **at 5–7 years**. The apocrine glands begin to work at puberty.
- **Hair.** The newborn has downy hair — **lanugo** — on the shoulders and back (abundant in preterm babies). The hair follicles are poorly developed — furuncles hardly occur in newborns (pseudofurunculosis occurs instead). The eyebrows and eyelashes are poorly developed.
- **The nails** reach the fingertips in a term newborn (a sign of maturity); in the first days of life their growth slows and a “physiological line” forms.

> **REMEMBER**
> A child's skin is **thin, delicate, rich in vessels, highly absorptive and weakly protective**. That is why it needs strict hygiene and a proper temperature regimen, and ointments are used with caution (for example, ointments containing boric acid, hormones or phenol may cause poisoning).

4.2. Methods of examining the skin

The examination is carried out by **questioning, inspection, palpation and special tests**.

Questioning

The time of appearance of changes in color, a rash, itching, sweating or edema; their relation to food, drugs, infections and chemicals; whether similar episodes have occurred before; allergic and skin diseases in the family are established.

Inspection

Inspection is carried out, as far as possible, **in natural daylight** in a warm room. The skin is inspected in sequence **from top to bottom**: the hairy part of the head, face, neck, trunk, **natural folds**, inguinal and femoral regions, palms and soles, the spaces between the fingers.

Inspection assesses:

- the **color** of the skin and its evenness;
- **moisture**;
- **cleanness** (absence of rash, scaling, scratch marks, hemorrhages);

- the state of the **skin vessels** (venous network);
- the **integrity** of the skin;
- the skin **appendages** — hair and nails.

When a rash is found, the following are established: the time of appearance; relation to any factor; the morphological type; size (mm, cm); number (single, few — can be counted, many — cannot be counted); shape (round, oval, star-shaped, annular); color; location (extensor or flexor surfaces, folds) and extent; the skin background (hyperemic or unchanged); the stages and course of the eruption; the secondary elements left (scaling, pigmentation, crusts).

Primary morphological elements of the skin

Primary elements appear on unchanged skin. They are **solid** (macule, papule, tubercle, node, wheal) and **cavitary** — filled with serous, hemorrhagic or purulent fluid (vesicle, bulla, pustule).

Table 4.2. *Primary morphological elements of the skin*

Element	Description
Macule (macula)	A circumscribed change of color that is not raised above the skin surface and has unchanged consistency. An inflammatory macule results from dilation of the dermal vessels and disappears when pressed with glass (diascopy): roseola — up to 5 mm (numerous roseolas of 1–2 mm — a fine punctate rash); macula — 6–20 mm (5–10 mm — a small-spotted rash, >10 mm — a large-spotted rash); erythema — an area of reddening larger than 20–30 mm. Non-inflammatory macules: pigmented (nevus, freckle, chloasma, vitiligo), vascular (hemangioma, telangiectasia), hemorrhagic (petechia — punctate, purpura — small and numerous, ecchymosis — >20 mm) — they do not disappear on pressure
Papule	A solid element raised above the skin surface; located in the epidermis and the superficial dermis; heals without a scar. Miliary — 1–1.5 mm, lenticular — 2–3 mm, plaque — larger than 1 cm
Tubercle (tuberculum)	A circumscribed firm raised element, 5–10 mm; formed by an inflammatory granuloma in the dermis; may leave an ulcer and a scar (tuberculosis, leprosy)
Node (nodus)	A round or oval formation located in the deep dermis or the subcutaneous fat; the result of inflammation or a tumor; may heal without a trace, ulcerate, or leave a scar or atrophy
Wheal (urtica)	A solid raised element formed by edema of the papillary layer, larger than 3–4 mm; itchy; transient — lasts from a few minutes to hours and disappears without a trace (urticaria)
Vesicle (vesicula)	A cavitary element up to 5 mm containing serous or serous-hemorrhagic fluid; dries to form a crust or ruptures to leave an erosion (chickenpox, herpes, eczema)
Bulla	A cavitary element larger than 0.5 cm filled with serous, hemorrhagic or purulent fluid (pemphigus, epidermolysis bullosa, burns)
Pustule (pustula)	An element from 1–2 mm to 10 mm filled with pus, with a ring of hyperemia around it; superficial or deep (vesiculopustulosis, folliculitis)

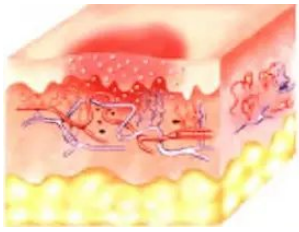


Figure 4.2. *Macule (Navruzova, Geppe)*

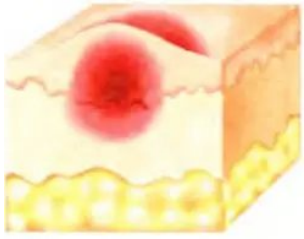


Figure 4.3. *Papule (Navruzova, Geppe)*

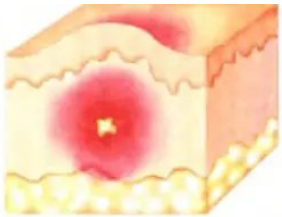


Figure 4.4. *Tubercle (Navruzova, Geppe)*



Figure 4.5. *Node (Navruzova, Geppe)*

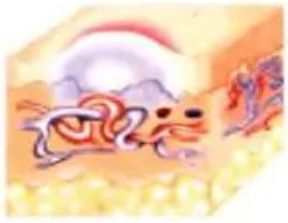


Figure 4.6. *Wheal (Navruzova, Geppe)*

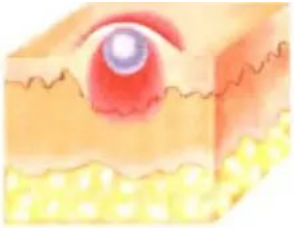


Figure 4.7. *Vesicle (Navruzova, Geppe)*



Figure 4.8. *Bulla (Navruzova, Geppe)*



Figure 4.9. *Pustule (Navruzova, Geppe)*

MEMORY AID

Remember the solid elements in order of depth: **"Macule — Papule — Tubercle — Node"** — from the skin surface downward. The cavitary elements by size and contents: **vesicle < 5 mm < bulla**, **pustule — pus**.

Secondary morphological elements

Secondary elements appear as a result of the evolution of primary elements and help to diagnose the disease retrospectively.

Table 4.3. *Secondary morphological elements*

Element	Description
Scale (squama)	A shed plate of the horny layer of the epidermis; leaf-like (>5 mm), lamellar (1–5 mm), bran-like (<1 mm)
Crust (crusta)	The dried exudate of a vesicle, pustule or weeping surface: serous — clear, purulent — yellow, bloody — dark brown. Crusts on the face of a child with exudative diathesis — milk crust
Erosion	A superficial defect within the epidermis; from rupture of a vesicle or pustule; heals without a trace
Ulcer (ulcus)	A defect involving the epidermis, dermis and deeper tissues; heals with a scar
Fissure (fissura)	A linear defect: superficial — heals without a trace; deep — painful, bleeds, leaves a scar
Excoriation	A scratch mark — a linear abrasion (in itchy dermatoses)
Lichenification	Thickening and hardening of the skin with accentuated skin markings (chronic inflammation, atopic dermatitis)
Vegetation	Rough outgrowths formed by growth of the papillary layer of the epidermis
Scar (cicatrix)	Connective tissue at the site of a deep defect: smooth, atrophic (below the skin level), hypertrophic and keloid (grows beyond the original border)
Secondary macule	Hyper- or hypopigmentation remaining after a rash (pigmentation after measles)



Figure 4.10. *Scale (Navruzova, Geppe)*

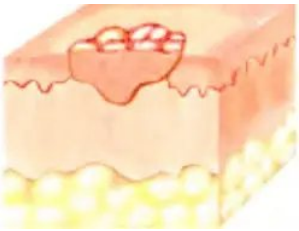


Figure 4.11. *Crust (Navruzova, Geppe)*

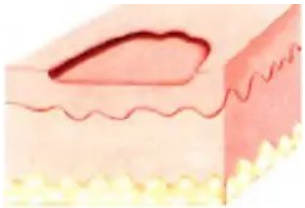


Figure 4.12. *Erosion (Navruzova, Geppe)*

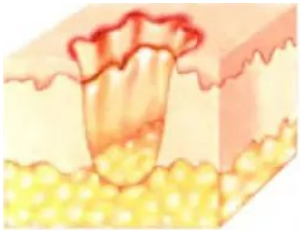


Figure 4.13. *Ulcer (Navruzova, Geppe)*



Figure 4.14. *Fissure (Navruzova, Geppe)*



Figure 4.15. *Lichenification (Navruzova, Geppe)*



Figure 4.16. *Vegetation (Navruzova, Geppe)*



Figure 4.17. *Scar (Navruzova, Geppe)*

CAUTION

An erosion heals without a trace, whereas an ulcer heals with a scar — this is the main criterion for distinguishing them. Small scars remain after chickenpox, irregular deep scars after tuberculosis, and star-shaped scars after a syphilitic gumma.

Palpation

Palpation is carried out from top to bottom, carefully over affected areas.

- **Moisture** is determined by stroking symmetric areas of the body (palms, soles, axillae, groins) with the palm. Normally the skin is slightly shiny and moderately moist.
- **Temperature** is determined by touching symmetric areas with the back of the hand; a local rise in temperature is characteristic of inflammation.
- **Elasticity** — in an area where subcutaneous fat is poorly developed (front of the chest, back of the hand, elbow crease), a fold of **skin only** is picked up with the thumb and index finger. If it smooths out immediately after release, elasticity is preserved; if it smooths out slowly, it is **reduced**.



Figure 4.18. *Determining skin elasticity (Navruzova, Geppe)*

Special tests

The state (fragility) of the vessel wall is assessed by three signs:

1. **The tourniquet sign (Konchalovsky–Rumpel–Leede).** A tourniquet (or a sphygmomanometer cuff) is applied to the middle third of the upper arm so that the venous outflow stops but the radial pulse is preserved. **After 3–5 minutes** the petechiae on the elbow crease and the inner surface of the forearm are counted. **More than 5 petechiae** is pathological (impaired vascular permeability or platelet deficiency).
2. **The pinch sign.** A fold of skin on the front or side of the chest is picked up with the fingers of both hands (2–3 mm apart) and shifted in opposite directions; if a hemorrhage appears in the area of the fold, the sign is positive.

3. **The hammer sign.** The sternum is tapped with a percussion hammer without causing pain; if hemorrhages appear on the skin, the sign is positive.

Dermographism is an assessment of the tone of the skin vessels. Several lines are drawn on the skin of the chest or abdomen with a fingernail or the tip of a spatula:

- normally **after 5–20 seconds** a **white line** appears (white dermographism — sympathetic effect), which is replaced **after 1–10 minutes** by a **red line** (parasympathetic effect);
- a rapidly appearing, broad, long-lasting **red** dermographism indicates parasympathetic predominance (vagotonia); a **white**, long-lasting one — sympathicotonia; a raised one — **urticarial** dermographism (allergic predisposition).



Figure 4.19. *Eliciting dermographism (Navruzova, Geppe)*



Figure 4.20. *Red dermographism (Navruzova, Geppe)*

Other methods: diascopy (pressing a macule with glass — an inflammatory macule disappears, a hemorrhagic one does not); skin biopsy; smears and scrapings (fungi, scabies burrow); the tuberculin (**Mantoux**) test and skin tests with allergens.



Figure 4.21. *The Mantoux reaction: a papule with surrounding hyperemia (Navruzova, Geppe)*

4.3. Skin changes characteristic of the newborn period

Table 4.4. *Skin changes characteristic of the newborn period*

Change	Description
Physiological (simple) erythema	Hyperemia with a cyanotic tinge appearing immediately after birth; maximal on days 1–2, then decreases and ends with fine bran-like physiological peeling (pronounced in preterm babies)
Toxic erythema	Papules and pustules with a pink border appearing on days 3–5 (often 2–5); disappear without a trace in 2–3 days; the general condition is not affected
Mottled skin	Due to imperfect vascular tone; increases in the cold

Change	Description
Harlequin syndrome	When the baby lies on the side, the upper half of the body is white and the lower half red; the border runs along the midline; in immature newborns
Mongolian spot	A bluish-gray spot on the sacrum, buttocks or back; melanocytes lie deep in the dermis; disappears over the years
Milia	Small white-yellowish nodules on the nasal wings and face — blocked sebaceous ducts
Physiological jaundice	Appears on days 2–3 and disappears on days 7–10; immaturity of glucuronyl transferase and increased breakdown of erythrocytes
Miliaria (heat rash)	Small vesicles from blockage of the sweat ducts on overheating (neck, folds)
Diaper dermatitis	Reddening of the skin of the perineum and thighs from wet diapers and contact with urine and stool
Caput succedaneum, cephalohematoma	Associated with birth injuries (see topic 1)



Figure 4.22. *Diaper dermatitis in a newborn (Navruzova, Geppe)*

REMEMBER

Physiological jaundice of the newborn appears **on days 2–3** and disappears **by days 7–10** (up to 2–3 weeks in preterm babies). Jaundice appearing **on the 1st day** or persisting **longer than 2 weeks** is always **pathological** (hemolytic disease, infection, biliary atresia).

4.4. Semiotics of skin damage

Changes in skin color

Pallor. Diffuse pallor of the skin, visible mucous membranes and conjunctivae is a sign of **anemia**. Pallor also occurs in vascular spasm: aortic defects, hypertensive crisis, collapse, “white” fever, shock. Specific tints: **waxy (alabaster)** — chlorosis and kidney diseases; **earthy** — malignant tumors; **“café-au-lait”** — infective endocarditis.

Hyperemia (redness) — dilation of the peripheral vessels (fever, overheating, vasodilators, inflammation, burns, excitement, physical exertion) or erythrocytosis (polycythemia — a dark red-bluish tint). In carbon monoxide poisoning the skin is **bright red (cherry red)**. In Itsenko–Cushing syndrome redness of the cheeks gives a **“matron face”**.



Figure 4.23. *The “butterfly” erythema on the face in systemic lupus erythematosus (Navruzova, Geppe)*



Figure 4.24. *Erythema of the face and eyelids in juvenile dermatomyositis (Navruzova, Geppe)*



Figure 4.25. *Hyperemia of the cheeks and milk crust in exudative-catarrhal diathesis (Navruzova, Geppe)*

Pigmentation disorders:

- **Hyperpigmentation** — congenital pigmented spots (nevus), “café-au-lait” spots (neurofibromatosis — more than 6, >5 mm), a bronze color (adrenal insufficiency — Addison's disease; at friction sites), pigmentation after measles, hemochromatosis, porphyria, heavy metal poisoning.
- **Hypopigmentation** — albinism (complete absence of pigment), **vitiligo** (white spots of various shapes), areas after burns and rashes. In scleroderma, areas of hyper- and hypopigmentation occur together.



Figure 4.26. *A giant congenital pigmented nevus (Navruzova, Geppe)*

Jaundice (icterus) is the accumulation of bilirubin in the tissues. First **the sclerae, the soft palate and the area under the tongue** turn yellow. Causes: hemolysis (hemolytic disease of the newborn, hemolytic anemias), damage to the liver parenchyma (viral hepatitis), obstruction of the bile ducts, hereditary disorders of bilirubin metabolism (Gilbert, Crigler–Najjar syndromes), metabolic diseases

(galactosemia, tyrosinemia, cystic fibrosis). Tints: **lemon-yellow** — hemolytic; **orange (deep yellow)** — parenchymal; **green-yellow** — obstructive jaundice.

CAUTION

Carotene jaundice (from eating large amounts of carrots, pumpkin or tangerines) — only **the palms and soles** turn yellow; **the sclerae and mucous membranes never turn yellow**. This is the main difference from true jaundice.

Cyanosis is a bluish tint of the skin and mucous membranes; it appears when **reduced hemoglobin in capillary blood exceeds 50 g/l**. It is first seen where the epidermis is thin — the lips, nails, earlobes and oral mucosa.

Table 4.5. *Types of cyanosis*

Type	Mechanism and causes	Clinical difference
Central (diffuse)	Oxygenation of arterial blood is impaired: pneumonia, atelectasis, bronchial obstruction, air or fluid in the pleural cavity, damage to the respiratory center and muscles; “blue” congenital heart defects with a right-to-left shunt (tetralogy of Fallot); methemoglobinemia (nitrates, aniline)	The tongue and lips are also blue; the limbs are warm; in heart defects it does not decrease with oxygen
Peripheral (acrocyanosis)	Peripheral blood flow slows and the tissues extract more oxygen: heart failure, shock, cold exposure	The tip of the nose, ears and fingers are blue; the limbs are cold; the tongue is pink
Local	Venous thrombosis or compression (superior vena cava syndrome — upper body, neck, arms), Raynaud syndrome, autonomic dystonia	Only one area

REMEMBER

Physiological acrocyanosis in the newborn (bluish palms and soles) is normal in the first 1–2 days and in the cold. If **cyanosis around the mouth** (perioral) increases with crying, sucking and coughing, cardiac or pulmonary pathology should be excluded.

Skin integrity, blood vessels and peeling

- **Trophic ulcers** — in chronic circulatory failure, diseases of the spinal cord and peripheral nerves, arterial thrombosis, vasculitis. In bedridden patients, **bedsores** on the buttocks, sacrum and shoulder blades.
- **Striae** — broad pink-purple lines: **Itsenko–Cushing syndrome** and long-term glucocorticoid therapy; thin whitish striae may be physiological during rapid growth at puberty.
- **Dilation of the venous network:** on the hairy part of the head — **hydrocephalus**; on the upper chest and between the shoulder blades — enlarged bronchopulmonary lymph nodes; on the anterior abdominal wall the **“caput medusae”** — liver cirrhosis (portal hypertension).
- **Spider angiomas** and **“liver palms”** — chronic liver diseases. **Hemangioma** — a pink or dark red spot, sometimes raised, formed by proliferation of capillaries. **Livedo** — a net-like bluish pattern (vasculitis, antiphospholipid syndrome).



Figure 4.27. *Striae in Itsenko–Cushing syndrome (Navruzova, Geppe)*



Figure 4.28. *Spider angiomas (Navruzova, Geppe)*



Figure 4.29. *Livedo reticularis (Navruzova, Geppe)*

Table 4.6. *Types of peeling and their diagnostic significance*

Type of peeling	In which diseases
Leaf-like (large lamellar) on the palms and soles	Scarlet fever (in the 2nd week), pseudotuberculosis, Kawasaki disease
Bran-like (fine)	Measles (with pigmentation), hypothyroidism
Together with dryness	Hypovitaminosis A and PP, dystrophy, ichthyosis
Coarse, on the palms and soles, after vesicles	Dyshidrosis

Changes in moisture, temperature and elasticity

- **Hyperhidrosis** — overheating, fever (stage of defervescence), physical and mental stress, **hypoglycemia**, **thyrotoxicosis**, autonomic dystonia (palms and axillae — in adolescents), **the initial period of rickets** (sour-smelling sweat on the nape, rubbing against the pillow — bald patch on the nape).
- **Dry skin** — dehydration (hypohidrosis), chronic intoxication, dystrophy, hypovitaminosis A and PP, ichthyosis, **hypothyroidism**.
- **A local rise in temperature** — inflammation of the skin, muscle or joint; a local fall — vascular spasm (Raynaud syndrome, autonomic dystonia).
- **Reduced elasticity** — rapid dehydration (vomiting, diarrhea), severe dystrophy, chronic infections, hypothyroidism.

Features of the rash in certain diseases

Table 4.7. *Features of the rash in childhood infections and other diseases*

Disease	Description of the rash
Scarlet fever	A fine punctate rash on a hyperemic background, dense in the natural folds, on the flexor surfaces of the limbs and in the groin; a pale nasolabial triangle (Filatov's triangle); lamellar peeling of the palms and soles from the 2nd week
Measles	A maculopapular rash on an unchanged background with a tendency to coalesce; appears in stages over 3 days (face → trunk → limbs); preceded by Filatov–Koplik spots; followed by pigmentation and bran-like peeling
Rubella	A pale pink, small-spotted rash appearing all at once; on the extensor surfaces, buttocks and back; the occipital and mastoid lymph nodes are enlarged

Disease	Description of the rash
Chickenpox	Macule → papule → vesicle → crust; polymorphism (elements at different stages at the same time); also on the hairy part of the head and mucous membranes
Meningococcemia	A hemorrhagic, star-shaped rash with central necrosis (legs, buttocks); spreads rapidly; does not disappear on pressure — an emergency
Thrombocytopenic purpura (Werlhof)	Petechiae and ecchymoses of various sizes and colors, asymmetric, spontaneous or after a trivial injury
Hemorrhagic vasculitis (Henoch–Schönlein)	A symmetric, raised (palpable) papular-hemorrhagic rash on the legs, buttocks and around the joints
Urticaria	Itchy, irregularly shaped wheals with a tendency to coalesce; appear quickly and disappear quickly

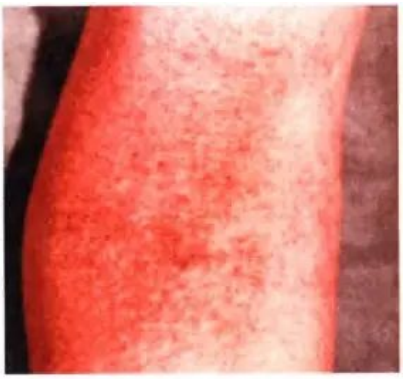


Figure 4.30. *Scarlet fever: fine punctate rash (Navruzova, Geppe)*



Figure 4.31. *Measles: maculopapular rash (Navruzova, Geppe)*

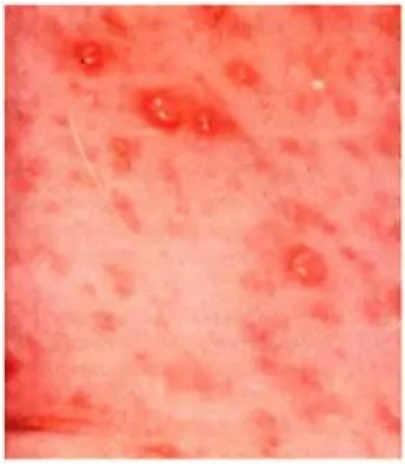


Figure 4.32. *Chickenpox: vesicular rash (Navruzova, Geppe)*



Figure 4.33. *Meningococcemia: hemorrhagic rash with necrosis (Navruzova, Geppe)*



Figure 4.34. *Werlhof disease: ecchymoses (Navruzova, Geppe)*



a

Figure 4.35. *Henoch–Schönlein disease: symmetric petechial-purpuric rash (Navruzova, Geppe)*

CLINICAL SIGNIFICANCE

Fever + a hemorrhagic rash that does not disappear on pressure is regarded as meningococcal infection until another diagnosis is proven. Such a child is given an antibiotic (for example, ceftriaxone) at the prehospital stage and urgently admitted. The rash can be checked with the “glass test”: if the spot does not disappear when a transparent glass is pressed on it, it is hemorrhagic.

Main skin diseases in children (in brief)

- **Pyodermas of the newborn:** vesiculopustulosis, pseudofurunculosis (Figner), epidemic pemphigus of the newborn, exfoliative dermatitis (Ritter) — blisters form easily because of the weak connection between the epidermis and dermis. **Omphalitis** — redness, infiltration and discharge of the umbilical ring.
- **Streptoderma** — impetigo (purulent vesicles, yellow “honey” crusts), erysipelas.
- **Viral:** herpes (grouped vesicles), molluscum contagiosum (pearly papules with a depressed center).
- **Fungal:** trichophytosis, microsporia, candidiasis; **parasitic:** scabies (paired papulovesicles, itching at night, between the fingers), pediculosis.
- **Allergic:** seborrheic dermatitis, **atopic dermatitis** (cheeks, flexor surfaces, itching), strophulus, urticaria, **Quincke's edema** (lips, eyelids, tongue, larynx — risk of respiratory obstruction).
- **Hereditary:** epidermolysis bullosa (blisters from trivial injury), ichthyosis, acrodermatitis enteropathica (impaired zinc metabolism).

Changes in the skin appendages

Table 4.8. *Changes in the nails*

Change	Diagnostic significance
Koilonychia (spoon-shaped nails)	Iron deficiency
“Watch glass” + “drumstick fingers”	Chronic hypoxia: chronic suppurative lung diseases (bronchiectasis, cystic fibrosis), “blue” congenital heart defects, infective endocarditis
Thimble sign (pinpoint pits)	Psoriasis
Leukonychia (white spots)	Injury, dystrophy, chronic diseases
Brittle, breaking nails	Dystrophy, hypovitaminosis, hypothyroidism
Paronychia	Inflammation around the nail
Onychomycosis	Fungal infection: dull, yellow, thick, detaching nail

Table 4.9. *Changes in the hair*

Change	Diagnostic significance
Hair loss on the nape (1st year)	Rickets (sweating and rubbing against the pillow)
Generalized alopecia	Dystrophy, anemia, hypovitaminosis, intoxication, hypothyroidism, SLE, cytostatics
Alopecia areata, patches with “broken-off” hairs	Fungal infection (microsporia, trichophytosis)
Hypertrichosis, hirsutism	Itsenko–Cushing syndrome, long-term glucocorticoid therapy, hereditary
Sparse, dry, brittle hair	Hypothyroidism, dystrophy, trace element deficiency

Sample case-history entry (normal): “The skin is pale pink, clean, moderately moist, warm, with preserved elasticity. No rash, hemorrhages or peeling. The subcutaneous venous network is not visible. Dermographism red, appears after 10 seconds and disappears in 2 minutes. The tourniquet and pinch signs are negative. The hair is shiny and evenly grown; the nails are pink and smooth.”

4.5. Subcutaneous fat

Anatomical and physiological features

Adipose tissue consists mainly of **white fat** and partly of **brown fat**. In brown fat cells oxidation is uncoupled from phosphorylation — energy is converted not into ATP but into **heat** (**non-shivering thermogenesis**).

Table 4.10. *Features of adipose tissue in children*

Indicator	Feature
Mass	≈16% of the body mass in a term newborn; relatively increases by 1 year (rounded body shape, deep folds); low in preterm babies
Course of development	Accumulates intensively in utero from the 7th month; grows rapidly until 9 months (the volume of the fat cell increases 5 times), then decreases; 2 times less at 5 years; increases again at puberty (more in girls)
Distribution	At birth well developed on the face (Bichat's fat pads — make sucking easier), limbs and chest; very scanty in the thoracic and abdominal cavities and the retroperitoneal space (the internal organs are mobile — risk of intussusception and renal ptosis)
Structure	The fat cells are small with large nuclei; with age the cell enlarges and the nucleus becomes smaller
Consistency	Relatively firm with a high melting point — rich in palmitic and stearic (saturated) fatty acids, poor in oleic acid. Tends to harden on cooling (sclerema, scleredema)
Brown fat	1–3% of the body mass in the newborn: back of the neck, axillae, between the shoulder blades, around the thyroid gland and kidneys, along the aorta. Protects the newborn from mild cold for 1–2 days; decreases in the 1st year

MEMORY AID

Fat **is deposited: face → limbs → trunk → abdomen**, and **disappears in the reverse order: abdomen → trunk → limbs → face**. That is why in hypotrophy the face loses weight last (“Bichat's fat pads” persist for a long time).

Methods of examination

The degree of development is assessed by the thickness of the skin fold. A fold of **skin together with the subcutaneous fat** (without muscle) is picked up with the thumb and index finger. Where the fold is measured:

- **on the abdomen** — at the level of the umbilicus, to the left or right of it;
- **on the chest** — at the edge of the sternum;
- **on the back** — below the shoulder blades;
- **on the limbs** — the back of the upper arm (over the triceps), the inner surface of the thigh;
- **on the face** — in the cheek area.



Figure 4.36. *Technique of assessing the thickness of the subcutaneous fat: taking a fold on the abdomen, chest, upper arm and thigh (Navruzova, Geppe)*

Table 4.11. *Average thickness of the abdominal skin fold (A.F. Tur)*

Age	Thickness, cm
Newborn	0,6
6 months	1,3
1 year	1,5
2–3 years	0,8
4–9 years	0,7
10–15 years	0,8

For an accurate objective assessment, the skin fold is measured with **a caliper** at 4 points — biceps, triceps, subscapular and suprailiac — and the sum is compared with centile tables.

Distribution — even or uneven (accumulation on the face, neck, upper body, abdomen). In boys the fat is relatively evenly distributed; in pubertal girls it accumulates on the abdomen, thighs, buttocks and breasts.

Consistency — normally finely granular, uniform and soft; hardening (sclerema), areas of atrophy and nodules are detected.

Tissue turgor — on the inner surface of the upper arm or thigh (upper 1/3), all the soft tissues (skin, fat, muscle) are squeezed with the thumb and index finger and **the resistance** is assessed: tense, with strong resistance — satisfactory turgor; soft, flabby — **reduced**.

Edema — the anterior surface of the shin, the ankle, the sternum, the face (eyelids) and the lumbar region are pressed with a finger for 5–10 seconds: if **a pit** remains, there is edema (pastosity — mild). In young children edema is also seen as deepening of the skin folds and marks left by clothing and socks.

CAUTION

Skin **elasticity** concerns only the skin fold (rapid smoothing); **turgor** is the resistance of all soft tissues; **the thickness of the skin fold** reflects the development of the fat layer. Do not confuse these three concepts.

Semiotics of changes in the subcutaneous fat

Deficiency of the fat layer. In young children — **hypotrophy** (topic 2), at an older age — **dystrophy**, the most severe degree — **cachexia**. Causes: constitutional (asthenic type), insufficient or unbalanced nutrition, digestive diseases (malabsorption), prolonged intoxication, chronic infections (tuberculosis), helminthic invasion, CNS pathology, endocrine diseases (thyrotoxicosis, diabetes mellitus), malignant tumors.

Excessive deposition (obesity). If the body mass exceeds the mass appropriate for height by **more than 15%**, it is obesity (the degrees are in topic 2; it is now also assessed by BMI centiles: **≥95th centile** or, by WHO, **Z-score > +2**). It most often develops **before 4 years and at 7–11 years**.

- **Primary (exogenous-constitutional)** — overeating, low activity; the fat is evenly distributed, the height is appropriate for age or tall.
- **Secondary** — endocrine (hypothyroidism, Itsenko–Cushing, hypogonadism), hypothalamic (tumor, craniopharyngioma, after a neuroinfection), hereditary syndromes (Prader–Willi, Laurence–Moon–Bardet–Biedl, Down).
- **In Itsenko–Cushing syndrome** fat accumulates on **the face** (“moon face”), **the back of the neck** (“buffalo neck”), the upper body and the abdomen, while the limbs are thin; pink-purple striae, hyperemic cheeks.
- **Paratrophy** — with incorrect (carbohydrate-rich) feeding: the mass is excessive, but the turgor is reduced, the tissues are hydrophilic and pale.



Figure 4.37. *Cachexia (Navruzova, Geppe)*



Figure 4.38. *Paratrophy (Navruzova, Geppe)*



Figure 4.39. *Itsenko–Cushing syndrome (Navruzova, Geppe)*

Lipodystrophy (lipoatrophy) — generalized (congenital generalized) or partial absence of the fat layer (Barraquer–Simons disease — in the upper body); local lipoatrophy **at insulin injection sites**. **Lipomatosis** — excessive diffuse or tumor-like accumulation of fat (Madelung and Dercum diseases).

Hardening:

- **Scleredema** — hardening of the skin and subcutaneous fat together with **doughy edema** in preterm and weak newborns on days 3–4; **a pit remains** on pressure; it begins on the legs and

lower abdomen.

- **Sclerema** — **diffuse hardening** of the skin and subcutaneous fat; **no pit remains** on pressure, and the skin cannot be picked up in a fold; the palms, soles and genitals are not affected; the child's condition is severe (cold exposure, sepsis), and movements are restricted.
- **Subcutaneous fat necrosis of the newborn** — circumscribed firm nodules after birth trauma.
- **Scleroderma** — induration of the skin and subcutaneous fat followed by atrophy; areas of hyper- and hypopigmentation.



Figure 4.40. *Juvenile scleroderma: areas of pigmentation and atrophy (Navruzova, Geppe)*

Edema

Edema is the accumulation of fluid in the interstitial spaces. Mechanisms of **generalized edema**: venous congestion (heart failure), activation of the renin–angiotensin–aldosterone system, **hypoproteinemia** (reduced oncotic pressure — nephrotic syndrome, liver diseases, protein deficiency, malabsorption, exudative enteropathy), reduced renal filtration, increased vascular permeability (glomerulonephritis, vasculitis). **Local edema:** Quincke's edema, acute inflammation (phlegmon, erysipelas, osteomyelitis), impaired venous or lymphatic outflow, toxic diphtheria (edema of the cervical subcutaneous fat), mumps (in front of the ear), dermatomyositis (firm edema over the affected muscles).

Table 4.12. *Differences between “cardiac” and “renal” edema*

Sign	Cardiac edema	Renal edema
Location (initially)	Feet, ankles, shins (from below upward)	Face, eyelids (from above downward)
During the day	Increases in the evening, decreases after sleep	Pronounced in the morning
Skin	Cyanotic, cold, firm	Pale, warm, soft, mobile
Accompanying signs	Dyspnea, liver enlargement, tachycardia	Urinary changes (proteinuria), hypertension

Myxedema (hypothyroidism) — a firm edema **that leaves no pit on pressure** (accumulation of mucopolysaccharides); on the face, the anterior surface of the shins, the backs of the hands and the supraclavicular fossae.

4.6. Anatomical and physiological features of the lymphatic system

The lymphatic system includes **the thymus, the spleen, the lymph nodes**, circulating lymphocytes and the lymphoid aggregates of the mucous membranes (tonsils, adenoids, Peyer's patches of the ileum, the appendix). The thymus and bone marrow are **central**, and the rest **peripheral**, lymphoid organs.

The thymus

- It lies in the upper mediastinum behind the sternum; it develops **from the 6th week** of intrauterine life from the 3rd–4th pairs of pharyngeal pouches; it forms **before the other lymphoid organs**.
- In the newborn its mass is **10–15 g** (the largest relative to body mass); the cortex predominates over the medulla.
- Its absolute mass increases **until puberty (9–12 years)** (≈30–40 g), after which **age-related involution** begins — the lymphoid tissue is replaced by fat, but the gland persists throughout life.
- Its function is **the differentiation of T lymphocytes** and the production of thymic hormones (thymosin, thymopoietin).
- Under the influence of adverse factors (infection, stress, antibiotics, glucocorticoids, cytostatics) **accidental (reversible) involution** may occur — at an early age it leads to secondary immunodeficiency.

The spleen

- It lies in the left hypochondrium of the abdominal cavity at the level of ribs IX–XI; its long axis runs along rib X from back to front and downward.
- It is not fully formed at birth: the trabeculae and capsule are weak, the shape is round and lobulated.
- Mass: newborn — **11 g**, 1 year — **26 g**, 12 years — **94 g**, adults — **150 g** (0.25–0.3% of the body mass — constant).
- Functions: immune (formation of T and B lymphocytes and antibodies), destruction of aged erythrocytes and platelets, blood depot; hematopoiesis in the fetal period.

Table 4.13. *Spleen dimensions by age (Kildiyarova)*

Age	Length, cm	Width, cm
Newborn	5	3
1 year	7,8	4,2
8 years	8	5
12 years	11	6
Adults	12	8

Lymph nodes

Lymph nodes are oval formations located in groups where large lymphatic vessels converge; they are the site of tissue fluid drainage, **a barrier** against infection and the multiplication of lymphocytes. The node is covered on the outside by **a capsule** from which **trabeculae** extend inward; under the capsule lies **the marginal sinus**, and in the cortex the **follicles** (B zone), the paracortical (T zone) region and **the medulla**.

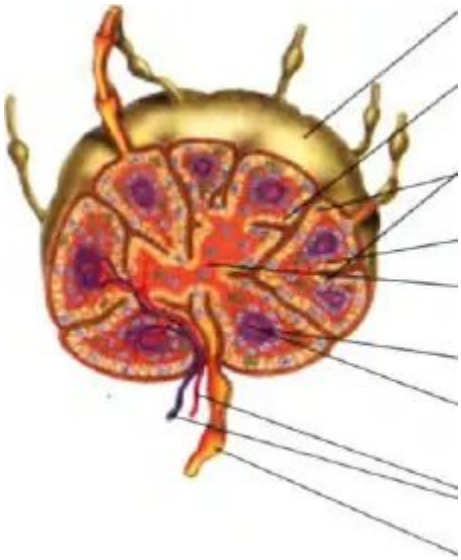


Figure 4.41. *Structure of a lymph node: capsule, trabecula, marginal sinus, cortex and medulla, follicles, afferent and efferent vessels (Navruzova, Geppe)*

Table 4.14. *Age-related features of the lymph nodes*

Age	Features
Fetus	Develop from the mesenchyme from weeks 5–6 of intrauterine life
Newborn and infant	The capsule is very thin, there are almost no trabeculae, the nodes are soft and the sinuses wide — low barrier function, infection tends to generalize; the nodes are difficult to palpate
1–3 years	The connective tissue capsule and trabeculae form; the nodes are easily palpable; inflammatory reaction — lymphadenitis may develop
7–8 years	The barrier function is well developed — infection is retained in the nodes
12–14 years	The structure is complete; growth stops at puberty, partial involution

4.7. Methods of examining the lymphatic system

Questioning and inspection

Questioning: enlargement of lymph nodes, redness and pain over them, the time of appearance, possible causes (recent tonsillitis, skin infection, cat scratch, vaccination, contact with tuberculosis), accompanying signs — fever, sweating (especially at night), weight loss, weakness, itching, bone pain, bleeding.

Inspection: markedly enlarged nodes (change in the configuration of the neck), hyperemia of the skin over them, edema, fistulas and scars; the state of the tonsils.

Palpation of the lymph nodes

Palpation is performed with the tips of fingers II–IV of **both hands**, simultaneously on **symmetric** areas, with sliding circular movements, pressing the node against a firm base (bone). **Order — from top to bottom:**

Table 4.15. *Groups of peripheral lymph nodes: location, drainage area and palpation*

Group	Location and drainage area	Method of palpation; normally
Occipital	Over the occipital protuberance; the hairy part of the head, the back of the neck	The child's head is bent forward; both hands are placed on the nape and the surface of the occipital bone is palpated; often not palpable
Preauricular	Over the parotid gland; the skin of the face, the eye, the ear	In front of the ear, below the zygomatic arch; normally not palpable
Mastoid (retroauricular)	Over the mastoid process; the auricle, the hairy part of the head	The area of the process is palpated; normally not palpable
Submandibular	Below the rami of the mandible; the skin of the face, the gums, the mouth	The head is bent slightly downward; the fingers are placed under the jaw and pressed against the edge of the jaw; single nodes smaller than a pea may be found
Submental	Under the chin in the midline; the lower lip, the floor of the mouth, the teeth	With one hand along the submental midline; normally not palpable
Anterior cervical (and tonsillar)	In front of the sternocleidomastoid muscle (upper cervical triangle); the tonsils, pharynx, nose, mouth	The fingers are run along the anterior edge of the muscle; tonsillar — behind the angle of the jaw; single nodes are palpable
Posterior cervical	Behind the sternocleidomastoid muscle, in front of the trapezius muscle; the skin of the neck, the larynx	Along the posterior edge of the muscle; single nodes may be palpable
Supraclavicular	In the supraclavicular fossa; the upper chest, the pleura, the lung apex	By pressing a finger into the fossa; normally not palpable
Infraclavicular	In the infraclavicular fossa; the skin of the chest, the pleura	In the same way; normally not palpable
Axilla	In the axillary fossa; the skin of the arm (except fingers III–V and the inner surface of the palm)	The child raises the arm to the side, the fingers are inserted deep into the axilla, then the arm is lowered and the fingers slide down along the wall; normally palpable
Thoracic	Below the lower edge of the pectoralis major muscle, medial to the anterior axillary line; the skin of the chest, the parietal pleura, the mammary glands	In this area; normally not palpable
Elbow (cubital)	In the bicipital groove; fingers III–V, the inner surface of the palm	The arm is bent at the elbow at a right angle; the forearm is held with one hand, and the other palpates along the groove; normally not palpable
Inguinal	Along the inguinal ligament; the legs, lower abdomen, buttocks, perineum, genitals, anus	Along the ligament and perpendicular to it; normally palpable
Popliteal	In the popliteal fossa; the foot	With the leg bent at the knee; normally not palpable

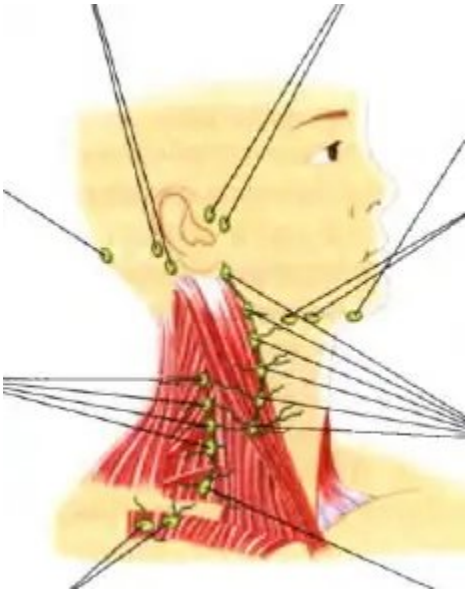


Figure 4.42. *Groups of cervical lymph nodes (Navruzova, Geppe)*



Figure 4.43. *Palpation of the occipital nodes (Navruzova, Geppe)*



Figure 4.44. *Preauricular nodes (Navruzova, Geppe)*



Figure 4.45. *Mastoid nodes (Navruzova, Geppe)*



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Figure 4.46. *Submandibular nodes (Navruzova, Geppe)*



Figure 4.47. *Submental nodes (Navruzova, Geppe)*



Figure 4.48. *Anterior cervical nodes (Navruzova, Geppe)*



Figure 4.49. *Posterior cervical nodes (Navruzova, Geppe)*



Figure 4.50. *Supraclavicular nodes (Navruzova, Geppe)*



Figure 4.51. *Infraclavicular nodes (Navruzova, Geppe)*



Figure 4.52. *Axillary nodes (Navruzova, Geppe)*



Figure 4.53. *Thoracic nodes (Navruzova, Geppe)*



Figure 4.54. *Elbow (cubital) nodes (Navruzova, Geppe)*



Figure 4.55. *Inguinal nodes (Navruzova, Geppe)*



Figure 4.56. *Popliteal nodes (Navruzova, Geppe)*

Criteria for describing lymph nodes

For each palpated group the following are recorded:

1. **Location** (group) and **symmetry**.
2. **Number: single** — up to 3 in a group; **multiple** — more than 3.
3. **Size** — in mm or cm, or by comparison. Normally up to **0.3–0.5 cm** (up to 0.5–1 cm according to Kildiyarova).
4. **Consistency:** normally **soft-elastic**; in pathology dense, hard (“pebble”) or softened (fluctuation — suppuration).
5. **Mobility:** normally mobile; in pathology poorly mobile, immobile.
6. **Relation to the surrounding tissues and to each other:** normally not adherent to each other or to the skin; in pathology they form **a “packet”** (conglomerate).
7. **Pain:** normally painless.
8. **Overlying skin:** normally unchanged; in inflammation hyperemia, warmth, edema, fistula.

Table 4.16. *The 6 degrees of lymph node size*

Degree	Comparison	Approximate size
I	Millet grain	1–2 mm
II	Yellow pea (mung bean)	3–4 mm
III	Pea	5–6 mm
IV	Bean	8–10 mm
V	Hazelnut	15–20 mm
VI	Pigeon egg	25–30 mm

REMEMBER

In a healthy child **no more than 3 groups** are palpable — usually among the **submandibular, cervical, axillary and inguinal** groups: single, no larger than a pea, soft-elastic, mobile, painless. **The submental, supraclavicular and infraclavicular, thoracic, cubital and popliteal** nodes are normally **not palpable**.

MEMORY AID

The groups not palpable normally: **“S-S-I-T-C-P”** — **S**ubmental, **S**upraclavicular, **I**nfraclavicular, **T**horacic, **C**ubital, **P**opliteal.

Examination of the spleen

Palpation — deep sliding palpation according to N.D. Strazhesko:

1. The child lies on the back with the legs slightly bent at the knees and the arms along the body; the doctor sits on the child's right side.
2. The doctor places **the left hand** on the left half of the child's chest in the area of ribs VII–X and squeezes it slightly (restricting respiratory movement).
3. The fingers of **the right hand**, slightly bent, are placed perpendicular to the left costal arch, below it — **starting from below** (from the level of the umbilicus, so as not to miss an enlarged spleen).
4. During the child's expiration a skin fold is formed with the fingers and they are pressed deep into the abdominal cavity; **on deep inspiration** the descending pole of the spleen strikes the fingers.
5. If there is no result, the fingers are moved upward and the maneuver is repeated. If the spleen is not detected in the supine position, it is palpated with the child lying **on the right side** (right leg extended, left leg bent at the knee and hip joints) — the abdominal muscles relax and the spleen moves forward.

In a healthy child the spleen is not palpable (its anterior edge falls 3–4 cm short of the costal arch). If it is palpable, it is **enlarged 1.5–2 times**. When it is detected, the following are assessed: how far it projects below the costal arch (cm), the edge (sharp/blunt, notches), the consistency (soft, dense, hard), the surface, mobility and tenderness.

> **CAUTION**
> In newborns and infants the pole of the spleen may be palpable **up to 1 cm** below the costal margin (the abdominal muscles are lax, and the organ is relatively large). In older children a palpable spleen is always **pathological**.

Percussion — **quiet (soft)** percussion (the spleen borders the intestine and stomach, which give a tympanic sound):

1. The child lies **on the right side** or stands; the left arm is on the head.
2. **Determining the length:** the pleximeter finger is placed along rib X, parallel to it. **Anterior border** — percussion is performed from the costal arch backward along rib X; a mark is made where the dull sound appears (along the edge facing the resonant sound). **Posterior border** — percussion is performed from the posterior axillary line forward along rib X. The distance between the two points is the **length** of the spleen.
3. **Determining the width:** a line perpendicular to rib X is drawn from the middle of the length; with the pleximeter parallel to rib X, percussion is performed toward the center of the spleen **from above** (two intercostal spaces higher) and **from below** (from the costal arch). The borders are marked along the edge on the side of the resonant sound.

Table 4.17. *Percussion dimensions of the spleen (cm)*

Measurement	Preschool age	School age	Adolescents
Length	3–4	5–7	6–8
Width	2–3	3–5	4–6

REMEMBER

The rule of percussion: percussion proceeds **from the resonant (clear) note toward the dull note**, and the border is marked along the edge of the finger **facing the resonant note**. The spleen — **quiet percussion**, because it is surrounded by hollow organs that give tympany.

Percussion of the thymus

It is performed in young children to detect **thymomegaly**:

1. The child lies on the back or sits.
2. Using **quiet direct or finger-to-finger** percussion, the **1st and 2nd intercostal spaces** are percussed to the right and left of the manubrium of the sternum, and also from the midclavicular line toward the sternum (the plexímeter finger parallel to the edge of the sternum).
3. **Normally** there is **no** dullness beyond the borders of the sternum (or only up to the edge of the sternum).
4. Dullness beyond the edge of the sternum on the right and left (widening of the vascular bundle area) suggests **thymic enlargement**. Confirmation — **chest X-ray** and **ultrasound**.

Table 4.18. *Additional methods of examining the lymphatic system*

Method	Purpose
Complete blood count	Leukocytosis/leukopenia, lymphocytosis, atypical mononuclear cells (mononucleosis), blasts (leukemia), cytopenias
Ultrasound (peripheral and abdominal nodes, spleen, thymus)	Size, structure, foci of inflammation, cysts
Chest X-ray	State of the thymus, mediastinal and hilar lymph nodes
CT, MRI	Mediastinal and retroperitoneal nodes (lymphoma staging)
Lymph node aspiration (cytology) and biopsy (histology)	Diagnosis of lymphoma, tuberculosis, metastasis
Serological and PCR tests	EBV, CMV, toxoplasma, HIV, Bartonella
Tuberculin skin test, Diaskintest	Tuberculosis
Rhinoscopy (anterior and posterior)	Adenoids, lymphoid tissue of the pharynx

4.8. Semiotics of lymphatic system involvement

Lymphadenopathy

Lymphadenopathy — enlargement of the lymph nodes (inflammatory and non-inflammatory). **Lymphadenitis** — inflammation of a node: enlarged, **painful**, the overlying skin red, warm and swollen; **fluctuation** when it suppurates. **Micropolyadenia** — small (3–5 mm), dense, mobile, painless nodes in many groups (chronic intoxication, lymphatic diathesis).

Local (regional) lymphadenopathy — a process in the area drained by the node: an infected wound, furuncle, folliculitis, pyoderma, dental caries, stomatitis.

- **Cervical nodes:** tonsillitis (angina), chronic tonsillitis, scarlet fever, diphtheria, adenoiditis; tuberculosis (a dense painless “packet” adherent to the skin, **caseous breakdown and a fistula**, followed by irregular **“stellate” scars**); the initial stage of lymphogranulomatosis (dense-elastic conglomerates fused together — **“a sack of potatoes”**, painless); lymphosarcoma (very hard, painless, enlarges rapidly).

- **Occipital and mastoid: rubella** (before the rash!), infections of the hairy part of the head, pediculosis.
- **Preauricular:** mumps, conjunctivitis.
- **Axillary and cubital: "cat scratch" disease** (bartonellosis — a hand injury), infections of the arm and hand, BCG vaccination complication (left axilla).
- Enlargement of **the supraclavicular** nodes is always a serious sign (lymphoma, mediastinal and abdominal tumors).

Generalized lymphadenopathy — enlargement of 2 or more non-adjacent groups.

Table 4.19. *Causes of generalized lymphadenopathy*

Group	Diseases and features of the nodes
Acute infections	Infectious mononucleosis (all groups, especially cervical — the configuration of the neck changes, "packet"; the spleen is enlarged, atypical mononuclear cells); rubella (occipital, mastoid); measles (cervical, occipital); adenovirus and parainfluenza (cervical, occipital — slight); yersiniosis, pseudotuberculosis
Chronic infections	Toxoplasmosis (marked enlargement, the size of a nut, painless); chronic tuberculous intoxication (many small hard "pebbles" in all groups); brucellosis (1–1.5 cm, with splenic enlargement); HIV/AIDS (an early sign, up to 2–3 cm)
Blood and tumor diseases	Acute leukemia (all groups, hepatosplenomegaly, anemia, hemorrhagic syndrome, bone pain); lymphogranulomatosis (Hodgkin lymphoma); non-Hodgkin lymphomas; neuroblastoma metastases
Autoimmune and other	Diffuse connective tissue diseases (SLE), the systemic form of juvenile rheumatoid arthritis (Still's disease), serum sickness, drug reactions, lymphatic-hypoplastic diathesis

CLINICAL SIGNIFICANCE

"Alarming" signs of lymphadenopathy (a biopsy or an oncohematologist consultation is needed): a node **>2 cm** that does not shrink within 4–6 weeks; **supraclavicular** location; **hard, immobile**, fused conglomerate; **general signs** — prolonged fever, night sweats, weight loss (B symptoms), hepatosplenomegaly, cytopenia or blasts in the blood.

Splenomegaly

Splenomegaly — persistent enlargement of the spleen. **Primary** — diseases of the spleen itself (cyst, tumor, congenital anomalies) — rare. **Secondary** — most often:

- **infections** — infectious mononucleosis, sepsis, bacterial endocarditis, malaria, leishmaniasis, tuberculosis, brucellosis, viral hepatitis, congenital infections (CMV, toxoplasmosis, rubella);
- **hemolytic anemias** ("work" hypertrophy) — microspherocytosis, thalassemia, hemolytic disease of the newborn;
- **liver diseases and portal hypertension** — liver cirrhosis, chronic hepatitis, portal vein thrombosis (after umbilical vein catheterization);
- **tumor and infiltrative processes** — leukemias, lymphomas, histiocytosis, storage diseases (Gaucher, Niemann–Pick);
- **autoimmune** — SLE, the systemic form of juvenile arthritis.

Hypersplenism — enlargement of the spleen together with excessive destruction of blood cells: **splenomegaly + anemia + leukopenia (neutropenia) + thrombocytopenia + compensatory hyperplasia of the bone marrow**. It is seen in all diseases with portal hypertension.

MEMORY AID

Causes of splenomegaly — **"HILTA"**: **H**emolysis, **I**nfection, **L**iver (portal hypertension), **T**umor (leukemia, lymphoma), **A**utoimmune and storage diseases.

Lymphatic-hypoplastic diathesis

Lymphatic-hypoplastic diathesis — a constitutional anomaly with **generalized hyperplasia** of the lymphoid tissue, reduced function of the endocrine system (adrenal glands) and reduced adaptability. Signs may be present from the first months (especially with thymomegaly), but are most often pronounced at **2–5 years**.

Signs:

- **thymomegaly** (widening of the vascular bundle area on percussion, enlarged thymic shadow on X-ray);
- generalized **lymphadenopathy**, hyperplasia of the **adenoids** and tonsils;
- **hepatosplenomegaly**;
- appearance: **pastose**, pale, sluggish, low muscle tone, well-developed fat layer but reduced turgor, long limbs (disproportionate build);
- in the blood — moderate **leukocytosis and lymphocytosis**, neutropenia; arterial hypotension;
- frequent ARVI, reduced adaptation to stress (infection, surgery, anesthesia, vaccination) — **risk of sudden death** due to adrenal insufficiency;
- an enlarged thymus at an early age may cause **respiratory disturbances** (dyspnea, stridor, apnea) and compression of the esophagus.

CLINICAL SIGNIFICANCE

Before anesthesia, surgery or a severe infection in a child with thymomegaly, the pediatrician and anesthesiologist are warned — because of the risk of adrenal insufficiency a glucocorticoid should be kept ready.

4.9. Clinical section

A child with a rash — diagnostic approach

When examining a child with a rash, the answers to three questions are sought:

1. **What is the element?** (macule, papule, vesicle, hemorrhagic) — examined by diascopy.
2. **In what order and where did it appear?** (in stages — measles; simultaneously — rubella, scarlet fever; polymorphic — chickenpox; dense in the folds — scarlet fever; symmetric on the legs — Henoch–Schönlein).
3. **What is the general condition and what are the accompanying signs?** (fever, lymph nodes, mucous membranes, meningeal signs).



Figure 4.57. *Henoch–Schönlein disease: symmetric purpura on the legs (Navruzova, Geppe)*



Figure 4.58. *Meningococcemia (Navruzova, Geppe)*

CLINICAL SIGNIFICANCE

Description of a hemorrhagic rash: in **thrombocytopenia** (Werlhof) — petechiae and ecchymoses are **asymmetric, "leopard skin"**, bleeding from the mucous membranes; in **vasculitis** (Henoch–Schönlein) — a **symmetric, palpable** papular-hemorrhagic rash, often with joint and abdominal pain; platelets are normal; in **meningococcemia** — stellate, necrotic, against a background of fever and intoxication.

Assessment of dehydration by skin signs

According to WHO (IMCI), **the return of the skin pinch** is an important sign of dehydration: the abdominal skin halfway between the umbilicus and the side of the abdomen is pinched in a vertical direction with the thumb and index finger and released. If it goes back **slowly (less than 2 seconds)** — a sign of **some** dehydration; **very slowly (≥2 seconds)** — of **severe** dehydration (assessed together with other signs: condition, eyes, thirst).

A child with lymphadenopathy — algorithm

1. Determine whether it is local or generalized (palpate all groups!).
2. If local — carefully examine the drainage area (tonsils, teeth, skin, scratches).
3. Features of the node: painful, warm, red — **inflammation**; dense, painless, adherent — **tuberculosis, tumor**.
4. Examine the spleen and liver.
5. Complete blood count — the first test; then targeted investigations.

Practical skills

Skill 1. Inspection of the skin and its appendages

1. A warm room, daylight; wash and warm the hands; talk to the child, obtain the mother's consent.
2. Undress the child step by step.
3. Inspect the skin from top to bottom: head, face, neck, trunk, limbs, folds, palms and soles, between the fingers, perineum.

4. Assess the color, moisture, cleanliness, venous network and integrity.
5. If there is a rash — describe it by all criteria; examine it by diascopy.
6. Inspect the hair and nails.
7. Record the result in the case history.

Skill 2. Palpation of the skin and special tests

1. Determine moisture by stroking symmetric areas with the palm.
2. Determine temperature with the back of the hand.
3. Elasticity — pick up a skin fold on the front of the chest or the back of the hand and assess how quickly it smooths out.
4. **Tourniquet sign:** place a sphygmomanometer cuff on the upper arm, raise the pressure to midway between systolic and diastolic (or to a level at which venous outflow stops but the pulse is preserved), hold for 3–5 minutes; count the petechiae in the antecubital area (within a circle 2.5 cm in diameter); **>5 — positive**.
5. **Pinch sign:** pick up a fold in the subclavian area and shift it; assess for hemorrhage.
6. **Dermographism:** draw lines on the skin of the chest with a spatula; record the time of appearance, color, width and time of disappearance.

Skill 3. Assessment of the subcutaneous fat and turgor

1. Pick up a skin-fat fold on the abdomen (at the level of the umbilicus), chest, below the shoulder blade, the back of the upper arm, the inner thigh and the face; measure the thickness (with a caliper or ruler).
2. Compare the result with the age norm; assess the distribution (even/uneven).
3. Consistency — by palpation (soft, finely granular, foci of hardening).
4. Turgor — squeeze all the soft tissues on the inner surface of the thigh and assess the resistance.
5. Detect edema: press on the sternum, the anterior surface of the shin, the ankle and the lumbar region for 5–10 seconds.
6. Conclusion: “The subcutaneous fat is sufficiently (insufficiently / excessively) developed, evenly distributed, soft-elastic in consistency, turgor preserved, no edema.”

Skill 4. Palpation of the peripheral lymph nodes

1. The child sits or stands, the doctor in front of or behind the child.
2. Palpate symmetrically with the fingertips of both hands, with light circular movements, pressing the nodes against the bony base.
3. Order: occipital → preauricular → mastoid → submandibular → submental → anterior cervical and tonsillar → posterior cervical → supraclavicular → infraclavicular → axillary → thoracic → cubital → inguinal → popliteal.
4. For each group: number, size, consistency, mobility, adhesion, tenderness, state of the overlying skin.
5. Conclusion: normal / micropolyadenia / regional or generalized lymphadenopathy / lymphadenitis.

Skill 5. Palpation and percussion of the spleen

1. The child lies on the back with the legs slightly bent; the doctor is on the right side.

2. The left hand fixes the lower left part of the chest; the fingers of the right hand, starting from the level of the umbilicus toward the left costal arch, are pressed deeper during expiration, and the pole of the spleen is sought on deep inspiration.
3. If it is not found — repeat with the child lying on the right side (left leg bent).
4. When it is found: assess the distance below the costal arch, consistency, edge and tenderness.
5. Percussion: on the right side, with quiet percussion along rib X determine the anterior and posterior borders (length), then along a perpendicular line the upper and lower borders (width); compare with the age norm.

Skill 6. Percussion of the thymus

1. The child lies on the back or sits.
2. **Quiet** percussion in the 1st and 2nd intercostal spaces from the midclavicular line toward the sternum, with the pleximeter finger parallel to the edge of the sternum.
3. Mark the border of dullness along the edge on the side of the resonant sound.
4. Normally the dullness does not extend beyond the edge of the sternum; if it does — suspect thymomegaly and confirm radiologically.

Review questions

1. Name the structural features of children's skin. Why are blistering diseases common in newborns?
2. What are the features of the protective, absorptive, thermoregulatory and synthetic functions of children's skin?
3. What are the age-related features of the sebaceous and sweat glands, hair and nails?
4. Name the order of skin inspection and the parameters assessed.
5. List and describe the primary non-cavitary and cavitary morphological elements.
6. What are the secondary morphological elements? How does an erosion differ from an ulcer?
7. How are the tourniquet, pinch and hammer signs tested and what do they show?
8. How is dermographism tested and how is it assessed?
9. Name the skin changes typical of the neonatal period.
10. What are the types of cyanosis and how do they differ? How does carotene jaundice differ from true jaundice?
11. What are the features of the rash in scarlet fever, measles, rubella, chickenpox and meningococcemia?
12. Age-related features of the subcutaneous fat; what is the significance of brown fat?
13. How is the subcutaneous fat examined? What is the difference between turgor, elasticity and skin fold thickness?
14. What is the difference between sclerema and scleredema, and between “cardiac” and “renal” edema?
15. Name the age-related features of the lymph nodes, their groups and the order of palpation. Which groups are not palpable normally?
16. By which criteria are lymph nodes described?
17. Explain the technique of palpation and percussion of the spleen; what are its age-related dimensions?

18. How is percussion of the thymus performed?
19. Name the causes and signs of lymphadenopathy, splenomegaly and hypersplenism.
20. What are the signs and danger of lymphatic-hypoplastic diathesis?

Test questions

1. Which function of children's skin is more developed than in adults?
   A) Protective
   B) Pigment formation
   C) Resorptive (absorption)
   D) Sweating
   E) Thermoregulation

2. The main reason why blisters form easily on the skin of newborns:
   A) High activity of the sebaceous glands
   B) Delicate connection between the epidermis and the dermis
   C) Thickness of the stratum corneum
   D) Intensive work of the sweat glands
   E) Abundance of pigment

3. When does sweating begin in the newborn?
   A) From the day of birth
   B) From the 3rd–4th week
   C) From 6 months
   D) From 1 year
   E) From 3 years

4. Indicate the cavitary primary morphological element:
   A) Papule
   B) Tubercle
   C) Wheal
   D) Vesicle
   E) Roseola

5. Which secondary element heals without a trace?
   A) Ulcer
   B) Erosion
   C) Deep fissure
   D) Scar
   E) Lichenification

6. When is the tourniquet sign considered positive?
   A) If 1 petechia appears
   B) If 2–3 petechiae appear
   C) If more than 5 petechiae appear
   D) If the arm becomes red
   E) If pain appears

7. How soon does white dermographism normally appear?
   A) Immediately
   B) In 5–20 seconds
   C) In 1–2 minutes
   D) In 5–10 minutes
   E) In 30 minutes

8. When does erythema toxicum usually appear in the newborn?
   A) At birth
   B) On days 3–5
   C) In the 2nd week
   D) At 1 month
   E) At 3 months

9. Typical of carotene jaundice:
   A) The sclerae turn yellow
   B) Only the palms and soles turn yellow, the sclerae are unchanged
   C) The stool becomes discolored
   D) The urine darkens
   E) Bilirubin is raised

10. Lamellar peeling on the palms and soles is typical of which disease?
    A) Measles
    B) Rubella
    C) Scarlet fever
    D) Chickenpox
    E) Urticaria

11. What proportion of the body mass of a newborn is brown fat?
    A) 0,1%
    B) 1–3%
    C) 10%
    D) 16%
    E) 25%

12. Thickness of the abdominal skin fold in a 1-year-old child (A.F. Tur):
    A) 0.5 cm
    B) 0.7 cm
    C) 1.5 cm
    D) 3 cm
    E) 4 cm

13. Which lymph nodes are normally not palpable in a healthy 7-year-old child?
    A) Submandibular
    B) Posterior cervical
    C) Axilla
    D) Inguinal
    E) Cubital

14. Along which rib is the pleximeter finger placed for percussion of the spleen?
    A) VI
    B) VIII
    C) X
    D) XII
    E) IV

15. Not typical of lymphatic-hypoplastic diathesis:
    A) Thymomegaly
    B) Generalized lymphadenopathy
    C) Lymphocytosis
    D) Increased adrenal function
    E) Hyperplasia of the adenoids

16. The age at which the skin layers approach those of adults:
    A) 12 years
    B) 5 years
    C) 1 year
    D) 3 years
    E) 7 years

17. The cheese-like coating (vernix caseosa) is the product of which glands?
    A) Sebaceous glands
    B) Eccrine sweat glands
    C) Lacrimal glands
    D) Apocrine glands
    E) Mammary glands

18. At what age do the sweat glands and the thermoregulation center fully mature?
    A) 5–7 years
    B) 1 year
    C) 3–4 months
    D) 14 years
    E) 1 month

19. The reason furuncles are almost never seen in newborns:
    A) The skin pH is acidic
    B) The sebaceous glands do not work
    C) The sweat glands are active
    D) The hair follicles are poorly developed
    E) The skin is thick

20. Which spot disappears on pressure with glass (diascopy)?
    A) Purpura
    B) Petechia
    C) Inflammatory roseola
    D) Hemorrhagic spot
    E) Ecchymosis

21. What is a hemorrhage larger than 20 mm in diameter called?
    A) Roseola
    B) Erythema
    C) Ecchymosis
    D) Petechia
    E) Purpura

22. A non-cavitary, raised, transient, itchy element that disappears without a trace:
    A) Pustule
    B) Papule
    C) Tubercle
    D) Node
    E) Wheal (urtica)

23. A cavitary element up to 5 mm in diameter with serous fluid:
    A) Bulla
    B) Vesicle
    C) Erosion
    D) Papule
    E) Pustule

24. A secondary element that leaves a scar in place of a deep defect:
    A) Erosion
    B) Excoriation
    C) Scale
    D) Superficial fissure
    E) Ulcer

25. Thickening of the skin and accentuation of the skin pattern in chronic inflammation (e.g., atopic dermatitis):
    A) Excoriation
    B) Atrophy
    C) Vegetation
    D) Pigmentation
    E) Lichenification

26. The number of petechiae considered pathological in the tourniquet sign:
    A) More than 3
    B) More than 10
    C) More than 5
    D) More than 1
    E) More than 20

27. How is skin elasticity tested?
    A) By tapping with a percussion hammer
    B) By picking up a skin fold and assessing how quickly it smooths out after release
    C) By applying a tourniquet
    D) By pressing with glass
    E) By stroking with the palm

28. In what order is the skin examined?
    A) From top to bottom
    B) Only at the site of the rash
    C) Only the trunk
    D) At random
    E) From bottom to top

29. What does it mean if red dermographism appears quickly, is wide and persists for a long time?
    A) Vagotonia (parasympathetic predominance)
    B) Normal
    C) Sympathicotonia
    D) Thrombocytopenia
    E) Allergic reaction

30. A condition in which, when lying on the side, the upper half of the body is white and the lower half red, with the border along the midline:
    A) Harlequin syndrome
    B) Mottled skin
    C) Mongolian spot
    D) Toxic erythema
    E) Milia

31. On which days does erythema toxicum appear in the newborn, and what is its course?
    A) On day 1, with peeling
    B) On day 10, leaves a scar
    C) At 1 month, requires antibiotics
    D) On days 3–5, disappears without a trace in 2–3 days, the general condition is not disturbed
    E) Only in term babies, on day 14

32. Cyanosis appears when the amount of reduced hemoglobin in capillary blood exceeds:
    A) 100 g/l
    B) 50 g/l
    C) 10 g/l
    D) 25 g/l
    E) 150 g/l

33. The sign that distinguishes central from peripheral cyanosis:
    A) Only the fingertips are blue
    B) The tongue and lips are also blue, the limbs are warm
    C) The limbs are cold, the tongue is pink
    D) Only one area is blue
    E) Only the ears are blue

34. Leaf-like (large lamellar) peeling on the palms and soles in the 2nd week is typical of which disease?
    A) Measles
    B) Chickenpox
    C) Scarlet fever
    D) Rubella
    E) Urticaria

35. A venous network on the anterior abdominal wall in the form of “caput medusae” is typical of which condition?
    A) Heart failure
    B) Hydrocephalus
    C) Hypothyroidism
    D) Liver cirrhosis (portal hypertension)
    E) Enlargement of the bronchopulmonary lymph nodes

36. A hemorrhagic, stellate rash with central necrosis is typical of which emergency?
    A) Rubella
    B) Measles
    C) Chickenpox
    D) Meningococcemia
    E) Urticaria

Clinical cases

Case 1

A 7-year-old child has had a temperature of 38.7 °C and a sore throat for 2 days. This morning a rash appeared on the skin. On examination: flushed cheeks, a pale nasolabial triangle. A fine punctate rash on a hyperemic background, denser in the axillae, groin and antecubital folds; dark red lines in the folds. The tonsils are red with a purulent coating; the submandibular and anterior cervical lymph nodes are 1.5 cm, painful, mobile.

Questions: 1) Which morphological element is the rash? 2) Which disease should be considered? 3) How do you assess the changes in the lymph nodes? 4) What skin changes are expected after 2 weeks?

Answer: 1) Roseola — inflammatory spots of 1–2 mm (fine punctate rash) on a hyperemic background. 2) Scarlet fever: tonsillitis, a fine punctate rash dense in the folds (Pastia's sign — lines in the folds), a pale nasolabial triangle (Filatov). 3) Regional lymphadenitis — the submandibular and anterior cervical nodes are enlarged (close to degree V), painful — an inflammatory reaction related to lymph drainage from the tonsils. 4) Large lamellar peeling on the palms and soles, branny peeling on the trunk.

Case 2

A 5-year-old girl has been weak, with poor appetite and pallor, for 3 weeks. Her mother noticed “bruises” on her legs and nosebleeds. On examination: pale skin and mucous membranes, petechiae and ecchymoses of various sizes on the legs and trunk, asymmetric. Peripheral lymph nodes of all groups are enlarged (cervical nodes 2–3 cm, dense, painless, poorly mobile). The liver projects 4 cm and the spleen 5 cm below the costal arch, dense. There is bone pain.

Questions: 1) Which syndromes do you identify? 2) What type of lymphadenopathy is it? 3) Which disease should be considered, and what is the first-line investigation?

Answer: 1) Anemic syndrome (pallor, weakness), hemorrhagic syndrome (petechiae, ecchymoses — thrombocytopenic type, nosebleeds), generalized lymphadenopathy, hepatosplenomegaly, ossalgia (bone pain) — i.e., hyperplastic (lymphoproliferative) syndrome. 2) Generalized, “alarming” lymphadenopathy: nodes >2 cm, dense, painless, poorly mobile. 3) Acute leukemia is highly likely. The first investigation is a complete blood count (anemia, thrombocytopenia, blasts, “leukemic hiatus”); confirmation — bone marrow aspiration (myelogram).

Case 3

A preterm (34 weeks) newborn, 4 days old, was kept in a cold ward. The condition is severe, lethargic, the sucking reflex weakened, body temperature 35.2 °C. On the shins, thighs and lower abdomen the skin and subcutaneous fat are swollen and hardened; a pit remains on pressure with a finger; the skin is cold, pale-cyanotic.

Questions: 1) What do you call the change in the subcutaneous fat? 2) How does it differ from sclerema? 3) Which feature of the newborn's subcutaneous fat predisposes to this condition?

Answer: 1) Scleredema of the newborn — hardening of the skin and subcutaneous fat together with edema (a pit remains on pressure) in a weak preterm newborn against a background of cold exposure. 2) In sclerema the hardening is diffuse, no pit remains, the skin cannot be picked up in a fold, the palms, soles and genitals are not affected, and the condition is very severe. 3) Newborn fat is rich in saturated (palmitic, stearic) fatty acids and poor in oleic acid — its melting point is high and it hardens easily in the cold; in addition, thermoregulation is imperfect, and preterm babies have little brown fat.

Quick review

Table 4.20. *Key facts on Topic 4*

Question	Answer
Skin thickness	Up to 3 years 1.5–3 times thinner than in adults; approaches adult values at 7 years
Reason for the tendency to blisters	Delicate epidermis–dermis connection
Strong functions	Absorptive, respiratory
Weak functions	Protective, pigment, sweating, thermoregulation
Onset of sweating	Week 3–4; full maturation at 5–7 years
Primary non-cavitary elements	Macule, papule, tubercle, node, wheal
Primary cavitary	Vesicle (<5 mm), bulla (>5 mm), pustule
Spot sizes	Roseola <5 mm; macula 6–20 mm; erythema >20 mm
Hemorrhagic spots	Petechia, purpura, ecchymosis (>20 mm); do not disappear on pressure
Erosion and ulcer	Erosion — without a trace; ulcer — with a scar
Tourniquet sign	>5 petechiae — positive
Dermographism	White 5–20 s; red in 1–10 minutes
Physiological jaundice	Day 2–3 → day 7–10
Carotene jaundice	Palms and soles; the sclerae do not turn yellow
Central and peripheral cyanosis	Central — limbs warm; peripheral — cold
Scarlet fever	Fine punctate, in the folds; Filatov's triangle; lamellar peeling
Measles	In stages (3 days); pigmentation, branny peeling
Fat mass at birth	16%; brown fat 1–3%
Abdominal skin fold	1 year — 1.5 cm

Question	Answer
Scleredema and sclerema	Scleredema — a pit remains; sclerema — it does not
Cardiac / renal edema	Legs, evening / face, morning
Lymph nodes, normal	≤3 groups, single, 0.3–0.5 cm, soft-elastic, mobile, painless
Not palpable normally	Submental, supra-/infraclavicular, thoracic, cubital, popliteal
Spleen mass	Newborn 11 g; 1 year 26 g; adults 150 g
Percussion of the spleen	Quiet, along rib X; on the right side
Percussion of the thymus	1st–2nd intercostal spaces, dullness beyond the edge of the sternum — thymomegaly
Hypersplenism	Splenomegaly + anemia + leukopenia + thrombocytopenia
Lymphatic-hypoplastic diathesis	Thymomegaly, lymphadenopathy, adenoids, lymphocytosis, adrenal insufficiency
Features of the child's skin	Thin, stratum corneum 2–3 rows; delicate epidermis–dermis connection; pH close to neutral; wide capillary network; high absorption; weak protection and thermoregulation
When sweating begins	From week 3–4; not fully functional until 3–4 months; up to 3 years only at high temperature; full maturation at 5–7 years
Sizes of spot types	Roseola — up to 5 mm; macula — 6–20 mm; erythema — larger than 20–30 mm; ecchymosis — >20 mm (hemorrhage)
Non-cavitary primary elements	Macule, papule, tubercle, node, wheal (urtica)
Cavitary primary elements	Vesicle (<5 mm), bulla (>5 mm), pustule (pustula)
Secondary elements	Scale, crust, erosion, ulcer, fissure, excoriation, lichenification, vegetation, scar, secondary spot
Tests of capillary fragility	Tourniquet sign (Konchalovsky–Rumpel–Leede), pinch sign, hammer sign
Skin changes of the neonatal period	Physiological erythema, erythema toxicum (day 3–5), marbled skin, harlequin syndrome, Mongolian spot, milia, physiological jaundice (day 2–3 → day 7–10)
Shades of jaundice	Lemon-yellow — hemolytic; orange (saffron) — parenchymal; greenish-yellow — obstructive
Types of cyanosis	Central (tongue and lips blue, warm limbs); peripheral/acrocyanosis (cold limbs, pink tongue); local (one area)
Rashes: scarlet fever and measles	Scarlet fever — fine punctate, dense in the folds, Filatov's triangle, lamellar peeling; measles — maculopapular, over 3 days face→trunk→legs, Filatov–Koplik spots, pigmentation

TOPIC 5

The bones and muscles: anatomical and physiological features, methods of examination, semiotics and syndromes

Learning objectives

The student should know:

- anatomical and physiological features of bone tissue, the parts of the skeleton (skull, spine, chest, pelvis, limbs), joints and teeth in children;
- age-related changes in body proportions;
- age-related features of the muscular system, physiological hypertonia;
- methods of examining the bones, joints and muscles and additional investigations;
- the main signs and syndromes of involvement of the osteoarticular and muscular systems (rickets, arthritis, myopathy, hypo- and hypertonia, tetany).

The student should be able to:

- inspect and palpate the head, chest, spine and limbs; measure the anterior fontanelle;
- determine the number of teeth and assess whether it matches the age;
- examine the joints and assess the range of motion; perform the tests for detecting developmental dysplasia of the hip;
- assess the degree of muscle development, tone and strength;
- write the “Musculoskeletal system” section of the case history.

5.1. Anatomical and physiological features of bone tissue

Development. Bones form in two ways: **intramembranous osteogenesis** — directly from connective tissue (the bones of the cranial vault, the facial bones, part of the clavicle) and **endochondral osteogenesis** — first a cartilage model, then bone in its place (tubular bones, vertebrae, pelvis). The skeleton is mainly formed in **weeks 8–9** of intrauterine life.

Growth. Bone grows in length through the **epiphyseal (growth) cartilage** and in width through **the periosteum**.

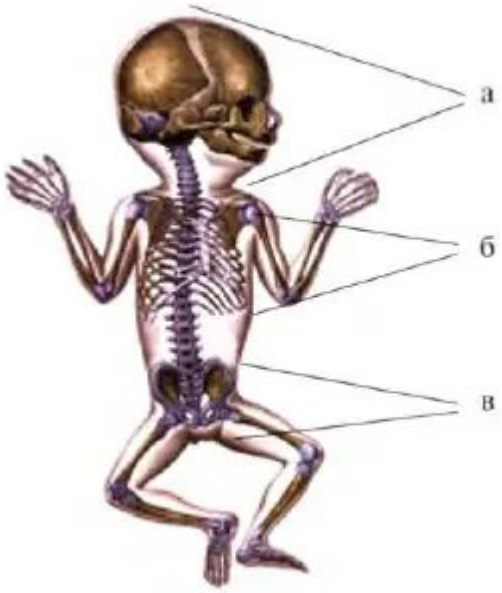


Figure 5.1. *Skeleton of the newborn: a — the skull consists of separate bones; b — the chest is conical, the ribs horizontal; c — the pelvis is narrow, funnel-shaped (Navruzova, Geppe)*

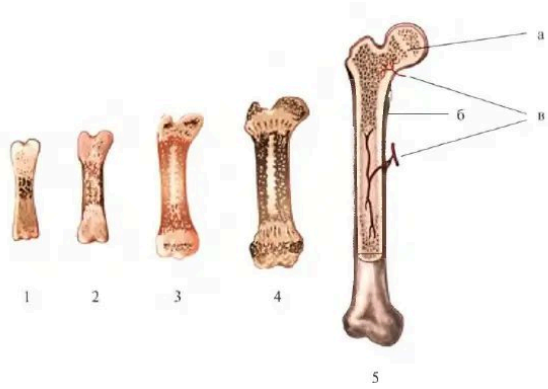


Figure 5.2. *Development of a tubular bone: primary ossification center in the diaphysis and secondary centers in the epiphyses, growth zone, thick periosteum (Navruzova, Geppe)*

Table 5.1. *Features of children's bone tissue and their clinical significance*

Feature	Clinical significance
The structure is coarse-fibered (woven); by 3–4 years it becomes lamellar (Haversian)	The bone is porous and flexible
Few mineral substances, much organic matter and water	The bones are flexible and deform quickly (rickets, incorrect posture), but break rarely
The periosteum is thick, functionally active and well supplied with blood	Subperiosteal fractures of the “greenstick” type; fractures heal quickly
Remodeling is intensive: in the 1st year 50–70% of the bone tissue is renewed (≈5% per year in adults)	High need for calcium, phosphorus and vitamin D; rickets develops quickly when they are deficient
Abundant blood supply; the vessels of the diaphysis and epiphysis are connected	Hematogenous osteomyelitis is most often in the metaphysis; infection easily spreads into the joint
Red bone marrow in all bones (in the fetus and at an early age)	The hematopoietic zone is wide; in adults it remains only in the flat bones and epiphyses
The bone surface is smooth; tuberosities form as the muscles develop	—

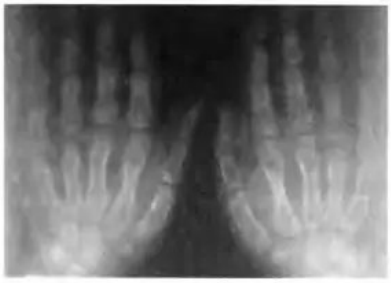
By 12 years the bones approach those of adults in their external and histological structure.

Bone age. The order in which the ossification nuclei appear is an indicator of biological maturity — **bone age**. In the wrist area: **the 1st nucleus at 6 months, the 2nd at 1 year**, then **one each year** (approximately age + 1); by 12–13 years all the nuclei of the wrist bones are formed. In a term newborn there are nuclei in the distal epiphysis of the femur and the proximal epiphysis of the tibia, and in the calcaneus, talus and cuboid bones — **a sign of maturity**.



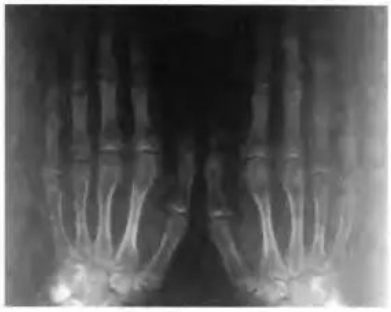
а

Figure 5.3. *Radiological determination of bone age: a — 2 nuclei (1–2 years) (Navruzova, Geppe)*



б

Figure 5.4. *b — 6 nuclei (6–7 years) (Navruzova, Geppe)*



в

Figure 5.5. *c — 11–12 years: all nuclei are formed (Navruzova, Geppe)*

Phosphorus-calcium metabolism is regulated by parathyroid hormone, vitamin D (active form — $1,25(OH)_2D_3$) and calcitonin. Vitamin D_3 (cholecalciferol) is synthesized in the skin under UV light, D_2 (ergocalciferol) comes with food; in the liver it is converted to 25(OH)D and in the kidney to active 1,25(OH)2D. It increases the absorption of calcium and phosphorus in the intestine.

5.2. Features of the parts of the skeleton

The skull

- At birth the skull consists of many separate bones; **the sutures** (coronal, sagittal, lambdoid) are open — the head **molds** in the birth canal. The sutures begin to close at 3–4 months and fully close at **3–5 years**.
- **The anterior fontanelle** — at the junction of the coronal and sagittal sutures, diamond-shaped; its size at birth is **from 1.5×2 to 3×3 cm** (measured between the midpoints of opposite edges); it closes at **1–1.5 years**.
- **The posterior fontanelle** — between the sagittal and lambdoid sutures, triangular; it is closed at birth in **75%** of term babies, and closes at **1–2 months** in the rest.
- **The lateral fontanelles** (sphenoid and mastoid) are closed in the term newborn.
- **The cerebral part of the skull is 8 times larger than the facial part** (2 times in adults). The orbits are relatively wide, the frontal bone has two halves, there are no superciliary arches or frontal sinus, and the jaws are undeveloped.
- In the 1st year the skull grows rapidly and the thickness of the bones increases 3 times; at 2–3 years growth of the facial part accelerates (teething, chewing); the skull base grows actively until 7 years.



Figure 5.6. *Skull of the newborn from the side: 1 — posterior fontanelle, 2 — parietal bone, 3 — anterior fontanelle, 4 — coronal suture, 5 — temporal bone (Navruzova, Geppe)*

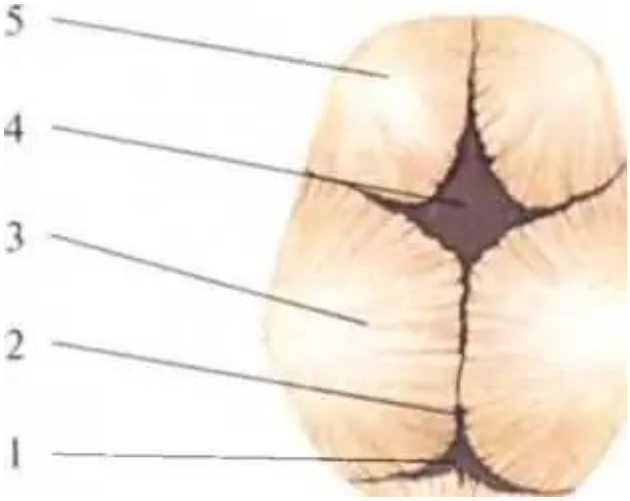


Figure 5.7. *Skull of the newborn from above: 1 — posterior fontanelle, 2 — sagittal suture, 3 — parietal bone, 4 — anterior fontanelle, 5 — frontal bone (Navruzova, Geppe)*

REMEMBER

Early closure of the anterior fontanelle (before 3 months) — microcephaly, craniostenosis, hypervitaminosis D; **late closure** (after 1.5–2 years) — **rickets**, hydrocephalus, hypothyroidism. **Bulging** — raised intracranial pressure (meningitis, hydrocephalus); **sunken — dehydration**.

The spine

In the newborn the spine has **no physiological curves**; it is shaped like an arc concave anteriorly (only the sacral kyphosis is marked). The curves form with the static functions:

Table 5.2. *Formation of the physiological curves of the spine*

Curve	Time of appearance	Related skill
Cervical lordosis	2–3 months	Beginning to hold the head up
Thoracic kyphosis	5–6 months	Beginning to sit
Lumbar lordosis	9–12 months	Standing and walking
Sacral kyphosis	Compensatory, forms together	—

The curves become stable at **6–7 years** and fully fixed **in adolescence** (14–17 years). Because the vertebrae and the muscles supporting them are poorly developed, children easily develop **scoliosis** and **postural disorders** (incorrect sitting, a heavy bag).

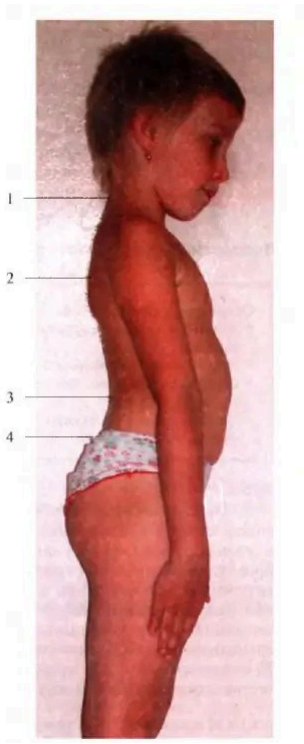


Figure 5.8. *Physiological curves of the spine: 1 — cervical lordosis, 2 — thoracic kyphosis, 3 — lumbar lordosis, 4 — sacral kyphosis (Navruzova, Geppe)*

The chest

- In the newborn it is **conical (barrel-shaped)**; the anteroposterior dimension is greater than or equal to the transverse one; the ribs leave the spine **at a right angle, horizontally — breathing is abdominal (diaphragmatic)**, and chest excursion is small.
- Once the child begins to walk, the sternum descends and the ribs become oblique; **at 3 years** the anteroposterior and transverse dimensions become equal; at school age the transverse dimension predominates and the chest flattens.
- From school age its shape depends on the constitution: **conical** (hypersthenic), **cylindrical** (normosthenic), **flat** (asthenic).

The pelvis and limbs

- At an early age the pelvis is **funnel-shaped** and narrow; sex differences become noticeable **from 9 years** (wider in girls). In the newborn **the acetabulum** is shallow and oval, and a large part of the femoral head lies outside the socket — **predisposition to developmental dysplasia and congenital dislocation of the hip**.
- In the newborn the limbs are relatively **short**; later the legs grow faster than the arms.
- **The foot** is flat until 1 year (fat pad); the arch forms with standing and walking — **flat feet are physiological up to 2 years**.

The joints

At birth the articular-ligamentous apparatus is anatomically formed. The epiphyses are cartilaginous, **the ligaments are extensible but not strong** — the range of motion in the joints is large, and **dislocations and subluxations** occur easily (at 2–4 years a sharp pull on the arm causes **subluxation of the radial head**). Full formation of the articular surfaces, capsule and ligaments is completed at **13–16 years**.

Teeth

There are 20 **primary teeth**; they begin to erupt **from 6–7 months** (5–7 months), and teeth of the same name erupt in pairs.

Table 5.3. *Eruption times of the primary teeth*

Teeth	Eruption time
Lower central incisors (2)	6–7 months
Upper central incisors (2)	8–9 months
Upper lateral incisors (2)	9–10 months
Lower lateral incisors (2)	11–12 months (8 teeth at 1 year)
First primary molars (4)	12–15 months
Canines (4)	18–20 months
Second primary molars (4)	22–24 months (20 teeth at 2 years)

Approximate calculation of the number of primary teeth (up to 2 years): **X = n – 4**, where n is the age in months (e.g., 6 teeth at 10 months).

There are 32 **permanent teeth**; they begin to erupt **from 5–6 years**: 5–7 years — first molars; 7–8 — central incisors; 8–9 — lateral incisors; 10–11 — first premolars; 11–12 — second premolars and canines; 10–14 — second molars; 18–25 — wisdom teeth. **Number of permanent teeth: X = 4n – 20** (n is the age in years).

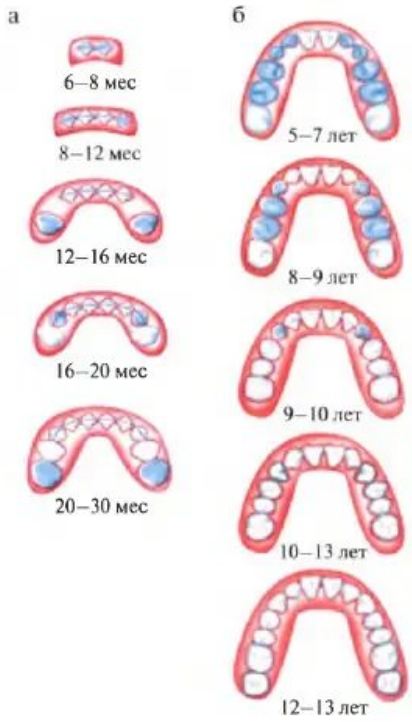


Figure 5.9. *Order of eruption of the primary and permanent teeth (Navruzova, Geppe)*

MEMORY AID

Primary teeth: **“they start at 6 months, 8 at 1 year, 20 at 2 years”**; formulas: primary — **“months minus 4”**, permanent — **“4 × age minus 20”**. Delayed teething (the first tooth after 10 months) may be a sign of **rickets**, hypothyroidism or Down syndrome.

5.3. Body proportions by age

A child's growth is uneven: first the head, then the trunk, and then the limbs grow relatively fast. Therefore the body proportions change with age.

Table 5.4. *Age-related changes in body proportions*

Indicator	Newborn	2 years	6 years	12 years	Adults
Head height / height	1/4	1/5	1/6	1/7	1/8
Midpoint of the body	Umbilicus	Below the umbilicus	Midway between the umbilicus and the pubis	Close to the pubis	Pubic symphysis
Leg length / height	≈1/3	—	—	—	≈1/2
Head circumference and chest circumference	The head is larger (34–35 / 32–34 cm)	They become equal (at about 4 months)	The chest is larger	The chest is larger	The chest is larger

Disturbed body proportions are of diagnostic importance: **short limbs with a normal trunk** (disproportionate short stature) — **chondrodystrophy (achondroplasia)**; **long limbs and long thin fingers** — **Marfan syndrome**; proportionate short stature — pituitary dwarfism; persistence of "infantile" proportions — **hypothyroidism**.

> **REMEMBER**
> In the newborn **the midpoint of the body is at the umbilicus**, in adults — **at the pubic symphysis**; the ratio of the head to the height decreases **from 1/4 to 1/8**. This explains the "big head, short legs" appearance of the newborn.

5.4. Methods of examining the osteoarticular system

Questioning

Complaints: pain in the bones and joints (location, character, time of appearance, morning or evening, relation to movement, provoking factor); deformity; restricted movement and **morning stiffness**; change in gait; joint swelling.

History: osteoarticular, rheumatic and autoimmune diseases in relatives; relation to injury or infection (tonsillitis — rheumatic fever; intestinal and urinary tract infections — reactive arthritis); feeding in the 1st year and **vitamin D prophylaxis**, the age of walking.

Inspection

Inspection is performed **from top to bottom**, with the child **lying, standing and moving**.

Head:

- **shape** — round; **tower-shaped** (craniostenosis), "square" — enlarged **frontal and parietal bosses** (**rickets**), **flattening of the occiput** (rickets), **hydrocephalic** (overhanging forehead, small facial part), **microcephaly**;
- **symmetry**;
- **size** (head circumference — topic 2);
- state of **the upper and lower jaws**, number and condition of the teeth, **the bite (occlusion)**: primary bite — **orthognathic**, permanent — orthognathic; pathological — progenia, prognathia, open bite; "**gothic palate**" (high, narrow).

Chest:

- **shape** — conical, cylindrical, flat, **barrel-shaped** (emphysema, bronchial asthma);
- **symmetry**;
- **the epigastric angle**: **≈90°** — normosthenic; **obtuse** — hypersthenic; **acute** — asthenic;
- **deformities**: **“pigeon chest”** (forward protrusion of the sternum — rickets), **funnel chest (“cobbler's chest”)** — inward depression of the lower sternum (congenital or rickets), **Harrison's groove** (an inward depression at the attachment of the diaphragm, with the costal arches flared forward — rickets), **cardiac hump** (congenital heart defect), **rachitic “rosary”** (thickenings at the costochondral junctions of the ribs).



Figure 5.10. *Funnel chest (“cobbler's chest”) (Navruzova, Geppe)*

Spine: physiological curves and how marked they are (hyperlordosis, hyperkyphosis, **rachitic kyphosis** — in the 1st year, prominent when sitting; **tuberculous spondylitis** — sharp-angled kyphosis — **gibbus**), **scoliosis** (lateral curvature), postural disorders. The child is examined from behind while standing: the symmetry of **the levels of the shoulders and the angles of the shoulder blades, the waist triangles** (the space between the arm and the trunk) and **the iliac crests**. For scoliosis, the line of the spinous processes and **the rib hump** are assessed with the child **bending forward**.

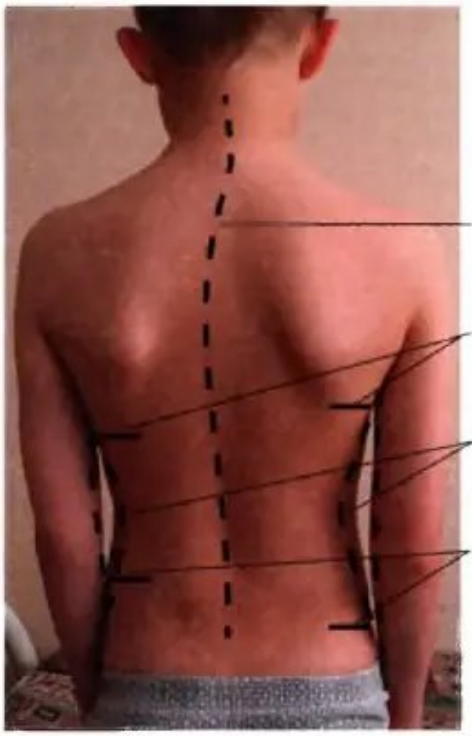


Figure 5.11. *Examination of the spine: 1 — lateral curvature of the spine, 2 — symmetry of the angles of the shoulder blades, 3 — waist triangle, 4 — symmetry of the iliac crests (Navruzova, Geppe)*

Limbs: straightness and symmetry of the contours (fracture, pseudarthrosis), relative length (proportions), shape — **X-shaped (valgus)** and **O-shaped (varus)** deformity of the legs (rickets), **saber-shaped** curvature of the tibia (rickets, congenital syphilis); **flat feet** (pathological after 2 years); **“bracelets”** (thickening of the distal metaphysis of the radius) and **“strings of pearls”** (finger phalanges) — rickets; **“drumstick fingers”** — chronic hypoxia.

Gait:

- **antalgic** — the child quickly shifts from the painful leg to the healthy one (involvement of the hip, knee or foot, back pain);
- **“duck gait”** (waddling) — congenital dislocation of **both hips**, myopathies;
- **limping** — unilateral dislocation, Perthes disease, leg length discrepancy;
- with heel pain (talalgia) — the child walks on tiptoe.

Palpation

- **Head:** the sutures, the anterior fontanelle (size, edges, bulging or depression), **craniotabes** — softening of separate areas of the occipital and parietal bones (they bend like parchment when pressed with a finger) — **an initial sign of rickets** (3–6 months).
- **Ribs:** in a healthy child the thickening at the costochondral junction is hardly noticeable; distinct hemispherical thickenings (ribs V–VIII) — **rachitic “rosary”**.
- **Limbs:** “bracelets”, “strings of pearls”, pain (periostitis, osteomyelitis, tumor, leukemia), pathological fractures.
- **Spine: enthesopathies** (pain at the sites where tendons and ligaments attach to bone), depression or protrusion of the spinous processes, pain.
- **Tapping (percussion): pain** when the flat and tubular bones (sternum, ribs, vertebrae, tibia) are tapped with a finger — blood diseases (**leukemia**), bone metastases.

Examination of the joints

On inspection: shape, size, symmetry, color of the overlying skin; **swelling**, **defiguration** (a reversible change due to intra-articular fluid and thickening of the synovial membrane), **deformity** (a persistent change due to proliferative and destructive processes — subluxation, ankylosis, contracture), hyperemia, pigmentation.

On palpation: local temperature, pain, swelling of the soft tissues, intra-articular fluid (**ballottement** of the patella — on pressure over the patella it "floats" in the fluid), crepitus.

Table 5.5. *Deformities of the finger joints in some diseases*

Deformity	Disease
"Fusiform" — proximal interphalangeal joints	Juvenile rheumatoid arthritis, SLE
"Sausage-shaped" — the whole finger swollen and red (dactylitis)	Reactive arthritis, Reiter disease, psoriatic arthritis
Sclerodactyly — thinning of the fingertips, thickening of the skin	Systemic scleroderma
Arachnodactyly — long, thin "spider" fingers	Marfan syndrome



Figure 5.12. *"Fusiform" deformity (Navruzova, Geppe)*



Figure 5.13. *"Sausage-shaped" deformity (Navruzova, Geppe)*



Figure 5.14. *Sclerodactyly (Navruzova, Geppe)*



Figure 5.15. *Arachnodactyly (Navruzova, Geppe)*

Joint function — the range of active and passive movements (precise measurement — with **a goniometer**):

- **Fingers:** making a full fist (the fingers should touch the palm).
- **Wrist joint:** bringing the palms together ("prayer position") and the backs of the hands together, bent at a right angle to the forearm.

- **Elbow:** fully extending the arm and bending it so that the fingers touch the shoulder.
- **Shoulder:** raising the arms forward and to the side, internal and external rotation; touching the opposite shoulder blade and the opposite ear with the hand.
- **Temporomandibular:** opening the mouth as wide as possible, moving the jaw forward and to the sides.
- **Hip:** flexion (pulling the knee to the chest), **spreading (abduction — normally 40–45°)** the legs to the side, internal and external rotation (≈35–45°).
- **Knee:** touching the buttock with the heel and fully extending the leg (inability to extend — flexion contracture).
- **Ankle:** passive flexion and extension; Achillobursitis (swelling in the area of the Achilles tendon).
- **Spine:** cervical — touching the chest with the chin, throwing the head back, touching the shoulder with the ear, turning the head; **general rigidity** — bending forward without bending the knees and touching the floor with the fingers (normally the distance is **less than 5 cm**); **Ott's test** (thoracic part): a mark on the 7th cervical vertebra and another 30 cm below it — on bending the distance increases by **4–5 cm**; **Schober's test** (lumbar part): a mark on LV and another 10 cm above it — on bending it increases by **4–5 cm**. In rigidity the change does not exceed 1–2 cm.



Figure 5.16. *Making a fist: normal and restricted movement (Navruzova, Geppe)*



а

Figure 5.17. *Function of the wrist joint: a (Navruzova, Geppe)*



б

Figure 5.18. *b (Navruzova, Geppe)*



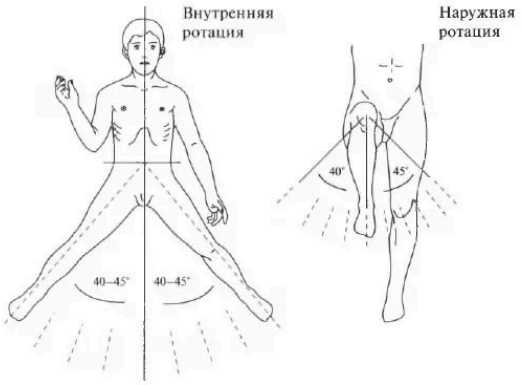
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Figure 5.19. *Rotation of the shoulder joint: touching the opposite shoulder blade and the ear with the hand (Navruzova, Geppe)*



а б

Figure 5.20. *Flexion at the knee joint: normal and restriction (Navruzova, Geppe)*

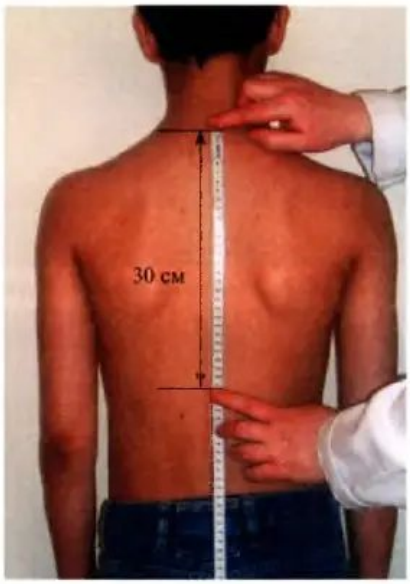


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Figure 5.21. *Function of the hip joint: abduction 40–45°, internal and external rotation (Navruzova, Geppe)*



Figure 5.22. *Testing the general rigidity of the spine (Navruzova, Geppe)*



а

Figure 5.23. *Testing the rigidity of the thoracic spine (Ott's test) (Navruzova, Geppe)*

Detecting developmental dysplasia of the hip (newborn and infant)

All newborns are examined:

1. **Asymmetry of the skin folds** — the thigh, inguinal and gluteal folds are at different levels and of different depth.
2. **Relative shortening of the leg** — in a child lying on the back with the legs bent at the hips and knees, the knees are at different heights.
3. **Restricted spreading (abduction) of the thighs** — when the thighs flexed at 90° are spread apart, they do not reach the table surface (normally 80–90° at 1 month). At an early age physiological hypertonia also causes slight restriction.
4. **"Slipping" (dislocation-reduction) sign — the Marx–Ortolani test**: the child lies on the back with the legs flexed at 90° at the hips and knees; the doctor places the thumbs on the inner surface of the thighs and the other fingers on the outer (greater trochanter) surface and slowly spreads the thighs apart — as the dislocated femoral head enters the socket, **a "click"** is felt.
5. External rotation of the foot.
6. Confirmation — **hip ultrasound** (Graf method, at 1–3 months), and X-ray in older children.

CLINICAL SIGNIFICANCE

Detecting developmental dysplasia of the hip **in the first weeks of life** and treating it with wide swaddling or special splints (the Pavlik harness) ensures full recovery of the joint. If it is detected late (after the child starts walking) — limping, “duck gait”, and surgical treatment is needed.

Additional methods of examination

- **X-ray** — fractures, anomalies, inflammation, tumors, degenerative processes; **bone age** (wrist); in rickets **cup-shaped widening** of the metaphyses, osteoporosis.
- **Ultrasound** (developmental dysplasia of the hip, intra-articular fluid), **CT, MRI**, **densitometry** (bone mineral density), scintigraphy.
- **Biochemistry:** serum **calcium** (total 2.25–2.75 mmol/l; ionized 1.1–1.3), **phosphorus** (1.3–2.3 mmol/l at 1–2 years), **alkaline phosphatase** (raised in rickets), **25(OH)D**, parathyroid hormone; urinary calcium and phosphorus (Sulkowitch test).
- **Immunological:** CRP, rheumatoid factor, ASO, antinuclear antibodies, HLA-B27.
- Analysis of **synovial fluid**, **biopsy** of the synovial membrane and bone.

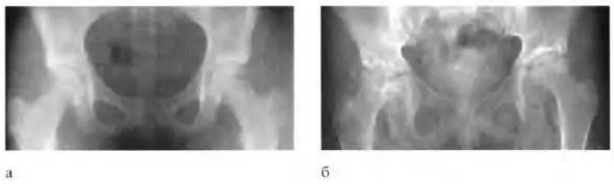


Figure 5.24. *X-ray of the hip joint: a — normal; b — arthritis (narrowing of the joint space, osteoporosis, erosion of the femoral head) (Navruzova, Geppe)*

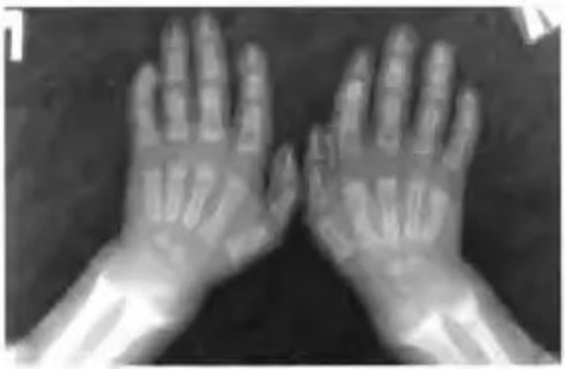


Figure 5.25. *X-ray of the hand in rickets: cup-shaped widening of the metaphyses, osteoporosis (Navruzova, Geppe)*

5.5. Semiotics of osteoarticular system involvement

Skeletal developmental anomalies

- **Skull:** craniostenosis (premature closure of the sutures), “wolf's palate” (non-fusion of the maxillary and palatine bones), micro- and macrocephaly.
- **Vertebrae:** non-fusion of the vertebral arch — **spina bifida** (most often lumbosacral), sacralization, lumbarization.
- **Ribs:** an extra cervical rib, absent or fused ribs, clefts of the sternum.
- **Limbs: congenital dislocation of the hip** (the most common), amelia (absence of a limb), phocomelia, poly- and syndactyly, **congenital clubfoot**, underdevelopment of the clavicle.
- **Systemic skeletal dysplasias: achondroplasia** (short limbs, large head, depressed nasal bridge), **osteogenesis imperfecta** (brittle bones — multiple fractures, **blue sclerae**, deafness).

Bone and joint pain

- **Ossalgia** (bone pain) — fracture, **osteomyelitis**, tumor, **leukemia** (night pain, pain on tapping).
- **“Growing pains”** — at 4–12 years, in the evening and at night in the legs (shins, popliteal area), bilateral; they pass by morning; there are no local changes and no changes in the blood count.

- **Arthralgia** — joint pain without objective changes (infections, onset of rheumatic diseases).
- **Perthes disease** — aseptic necrosis of the femoral head: hip pain increasing with load, limping, the child feels well.
- **Enthesopathies** — Osgood–Schlatter disease (tibial tuberosity), juvenile spondyloarthritis, reactive arthritis.

Rickets

Rickets — a disorder of bone formation and mineralization in the growing body due to **vitamin D deficiency** and disturbed phosphorus-calcium metabolism. Most often **from 2 months to 2 years**. The bones show signs of **softening (osteomalacia)** and **osteoid tissue hyperplasia**.

Table 5.6. *Bone and other signs of rickets*

Area	Signs of osteomalacia (softening)	Signs of osteoid hyperplasia
Head	Softening of the edges of the anterior fontanelle, craniotabes, flattening of the occiput, skull asymmetry	Frontal and parietal bosses (“square head”, “Socrates' forehead”)
The chest	Harrison's groove, lateral compression of the chest, “pigeon chest”, “cobbler's chest”	Rachitic “rosary” (ribs V–VIII)
Limbs	O-shaped and X-shaped deformity, flat pelvis	“Bracelets”, “strings of pearls”
Spine	Rachitic kyphosis, scoliosis	—
Teeth	Late and irregular eruption, enamel defects, caries	—
Nerves and muscles (initial period)	Restlessness, startling, sour-smelling sweat on the nape and a bald nape (hair loss), muscle hypotonia (“frog belly”, “loose joints”, the “pocket-knife” sign — the child easily brings the foot to the face), delayed static functions	



Figure 5.26. *Muscle hypotonia and joint laxity in a child with rickets: the “pocket-knife” sign (Navruzova, Geppe)*

MEMORY AID

Chronology of the signs of rickets: **up to 3 months** — sweating, bald nape, restlessness; **3–6 months** — **craniotabes** (the earliest bone sign); **from 6 months — bosses, “rosary”, “bracelets”** (osteoid hyperplasia); **after 1 year** (when the child begins to stand) — **leg deformities**.

Rickets-like diseases (vitamin D-resistant rickets, de Toni–Debré–Fanconi syndrome, renal tubular acidosis) — tubulopathies developing after 2 years that do not respond to usual doses of vitamin D, with impaired intestinal absorption of calcium or urinary loss of calcium and phosphorus.

Spasmophilia (infantile tetany)

In the initial period of rickets or during recovery (most often in spring, from 6 months to 2 years), **neuromuscular excitability increases** as a result of **a fall in ionized calcium** in the blood.

Signs of **latent tetany**:

- **Chvostek's sign** — contraction of the facial muscles (lip, nasal ala, eyelid) when the area between the zygomatic arch and the corner of the mouth (where the facial nerve emerges) is tapped with a hammer;
- **Trousseau's sign** — when the neurovascular bundle of the upper arm is compressed (with a cuff) for 2–3 minutes, the hand goes into spasm and takes the **"obstetrician's hand"** position;
- **Lust's sign** — outward turning and flexion of the foot when the area below the head of the fibula (n. peroneus) is tapped with a hammer;
- **Maslov's sign** — a brief arrest of breathing when the skin is lightly pricked with a needle.

Manifest tetany — **laryngospasm** (crowing inspiration like a "rooster's crow", cyanosis), **carpopedal spasm** (spasm of the hands and feet), generalized **eclamptic seizures**.

> **CLINICAL SIGNIFICANCE**
> Emergency care for laryngospasm and seizures: lay the child down, splash cold water on the face, fresh air, clear the mouth and nose; **10% calcium gluconate** (or calcium chloride) slowly intravenously; if the seizures continue — diazepam. Vitamin D must not be prescribed without calcium preparations (it worsens the seizures).

Hypermobility

Joint hypermobility — a range of motion greater than normal: **connective tissue dysplasias** (Ehlers–Danlos and Marfan syndromes), muscle hypotonia (rickets, Down syndrome). It is assessed by the Beighton criteria (bending the little finger back more than 90°, touching the forearm with the thumb, hyperextension of the elbow and knee by more than 10°, placing the palms on the floor without bending the knees).

Arthritis

Signs of **arthritis**: **defiguration and/or deformity** of the joint, **swelling** of the soft tissues, local **hyperemia and hyperthermia**, **pain** on palpation, **restriction** of active and passive movements.

Table 5.7. *Main types of arthritis in children*

Arthritis	Features
Acute bacterial (septic)	Most often one large joint; acute pain, all the signs of inflammation, purulent exudate, hectic fever, intoxication
Reactive arthritis (Reiter)	2–4 weeks after an intestinal or genitourinary infection; asymmetric oligoarthritis (2–3 joints), most often of the legs; talalgia, enthesitis, conjunctivitis, urethritis
Juvenile rheumatoid (idiopathic) arthritis	Symmetric polyarthritis with new joints added progressively; morning stiffness, "fusiform" fingers, deformities, contractures, subluxations; lasts more than 6 weeks

Arthritis	Features
Acute rheumatic fever	Migratory (flitting), non-erosive polyarthritis of the large joints; passes quickly and completely; with carditis
SLE, hemorrhagic vasculitis, dermatomyositis	Migratory non-erosive polyarthritis/arthralgia
Juvenile spondyloarthritis, Bechterew disease	Joints of the legs, enthesopathies, spinal rigidity, HLA-B27



Figure 5.27. *Deformity of the right knee joint, flexion contracture (Navruzova, Geppe)*

Osteomyelitis, fractures, tumors

- **Acute hematogenous osteomyelitis** — purulent inflammation of the bone marrow and bone; most often **the metaphyses** of the femur and tibia; acute local pain, the child does not move the leg ("pseudoparalysis"), swelling, high fever, intoxication. X-ray changes appear only **after 10–14 days** (early diagnosis — MRI, scintigraphy).
- **Periostitis** — inflammation of the periosteum: local thickening, pain, unevenness (tuberculosis, syphilis, tumors).
- **Fractures:** pain, swelling, deformity, crepitus, hemorrhage, impaired function; at an early age **"greenstick"** and subperiosteal fractures; fractures through the growth zone may disturb growth. Multiple fractures from trivial trauma — osteogenesis imperfecta; multiple fractures at different stages of healing — possible **child abuse**.
- **Bone tumors** — osteosarcoma, **Ewing sarcoma**, osteochondroma, **osteoid osteoma** (night pain, relieved by aspirin).

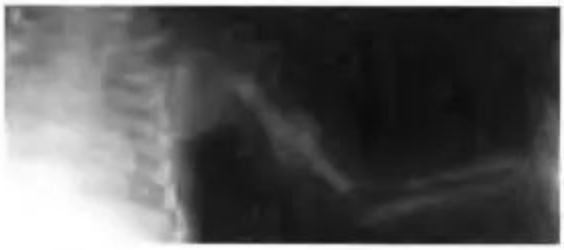


Figure 5.28. *Fracture of the humerus, callus (Navruzova, Geppe)*

Dental pathology

- **Developmental anomalies:** supernumerary teeth, **adentia**, malposition, **Hutchinson's teeth** ("screwdriver-shaped" upper incisors with a crescent-shaped notch — congenital syphilis).
- **Enamel hypoplasia** — disturbed mineral and protein metabolism in the fetus and at an early age (rickets, prematurity).

- **Caries** — demineralization of the enamel (Streptococcus mutans + carbohydrates; a sweet formula and a pacifier at night — “bottle caries”).
- **Fluorosis** — chalky spots on the enamel from excessive fluoride intake.
- **Premature loss** of primary teeth — hypovitaminosis C, hypophosphatasia, diabetes mellitus, leukemia, histiocytosis, mercury poisoning.
- **Late eruption** — rickets, hypothyroidism, Down syndrome, pituitary dwarfism.

5.6. The muscular system

Anatomical and physiological features

- In the newborn the number of muscles is the same as in adults (more than 600), but **the muscle mass** is **≈23–25%** of the body mass (27% at 8 years, **32–33%** at 15 years, **40–44%** in adults). The mass increases not through the number of muscle fibers but through their **thickening**.
- The muscle fibers are thin and multinucleated, with few contractile proteins and preserved **fetal myosin**; they contain much water and little protein, fat and minerals.
- The muscle relief is smooth and becomes defined at 5–7 years; **the fasciae** are delicate and the aponeuroses weak — in the newborn cephalohematoma, **umbilical hernia** (the umbilical ring is not fully closed), forward protrusion of the abdomen (until 3–5 years).
- In the newborn the muscles **do not relax even during sleep** — this produces heat and stimulates the development of muscle tissue.
- **Physiological hypertonia** — in the first 2–4 months **the tone of the flexor muscles** predominates over the extensors: it persists in the arms **until 2–2.5 months** and in the legs **until 3–4 months**. That is why the newborn lies in the “fetal position”.
- In preterm babies (less than 30 weeks) — **generalized hypotonia**: the limbs are extended; at 30–34 weeks the legs flex, and flexion of the arms after 34 weeks — one of the criteria for assessing **gestational age**.
- Muscle development is uneven: first the muscles of **the shoulder and forearm**, then the muscles of the hand (hand movements become well coordinated **at 6–7 years** — readiness for writing). The muscles holding the body upright grow rapidly from 7 years, especially **at 12–16 years**. Precision of movement improves from 10 years and speed at 14 years.
- Muscle strength is higher in boys; relative strength changes little until 6–7 years and increases rapidly **at 13–14 years**.



Figure 5.29. *Physiological hypertonia of the newborn (Navruzova, Geppe)*



Figure 5.30. *Hypotonia in a preterm newborn (Navruzova, Geppe)*

Examination of the muscular system

Complaints: muscle pain (**myalgia**), weakness, rapid fatigue, reduced muscle strength, difficulty walking, frequent falls, inability to climb stairs. **History:** muscle diseases in the family, relation to infection, injury or medication.

1. Degree of development (inspection and palpation — symmetric areas at the same level are compared):

- **good** — at rest the contours of the muscles of the trunk and limbs are clearly visible, the abdomen is drawn in, and the shoulder blades are close to the chest; on tension the relief becomes more pronounced;
- **moderate** — the trunk muscles are moderately and the limb muscles well developed;
- **poor** — at rest the contours are not visible and hardly change on tension; the lower abdomen sags, **the shoulder blades lag behind the chest (“winged scapulae”)**.

To detect asymmetry, the circumference of the limbs is measured with a tape at the same level.

2. Muscle tone — assessed by the resistance to passive movements (the limb should be maximally relaxed). The first impression is obtained from **the child's posture**.

Table 5.8. *Tests for assessing muscle tone in the newborn and infant*

Test	Technique	Assessment
Recoil sign	The legs of a newborn lying on the back are extended, held pressed to the table for 5 seconds, then released	The legs quickly return to the previous flexed position — normal/hypertonia; they do not return — hypotonia
Traction (pull-to-sit) test	A child lying on the back is held by the forearms and pulled to a sitting position	Normally the arms first extend (1st phase), then flex and the child pulls up (2nd phase). In hypertonia the 1st phase is absent, in hypotonia the 2nd phase is absent (the arms extend and the head lags behind)
“Loose shoulders” sign	The child is lifted up, held under the arms	In hypotonia the shoulders rise up to the earlobes and the child “slips out” of the hands
Abduction angle and the “heel-to-ear” test	The thighs are spread apart; the heel is pulled toward the ear	Limited in hypertonia, excessive in hypotonia

Table 5.9. *Types of muscle hypertonia*

Type	Signs	Site of the lesion
Spasticity	Strong resistance at the start of the passive movement, which then suddenly decreases — the “clasp-knife” phenomenon; it increases with rapid movement	Pyramidal tract (central paresis, CP)
Rigidity	The same resistance throughout the movement — the “lead pipe” and “cogwheel” phenomena, plastic tone (“wax doll”)	Extrapyramidal system

Hypotonia — reduced resistance to passive movements, soft muscles, increased range of motion in the joints (excessive flexibility); the child is like a “rag doll”.

3. Muscle strength — assessed with movements in which the child actively resists: in a small child, pulling away a toy held in the hand; in an older child, resistance to extension of the bent arm (leg); sitting, standing, climbing stairs, getting up from the floor (**Gowers' sign** — in myopathy the child

"climbs up" his own legs with his hands). Objectively — **dynamometry** (in children over 5 years, 3 times, the highest result). Strength is also graded on a **0–5 point** scale.

Table 5.10. *Average hand muscle strength in healthy children (kg) (Kildiyarova)*

Age	Boys	Girls
7–9	11–12,5	9–13
10–11	16	14–14,5
12–14	21,5–31	18,5–27

4. Range of movement — active (play, walking, bending, squatting) and passive (flexion-extension of the joints). In the first months the restriction of passive movements is due to physiological hypertonia.

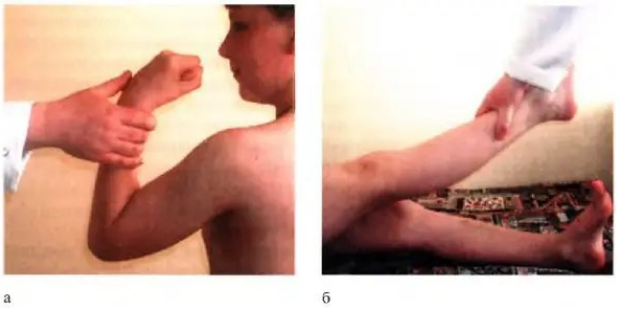


Figure 5.31. *Determining muscle strength: a — in the arm; b — in the leg (Navruzova, Geppe)*



Figure 5.32. *Muscle relief under tension (Navruzova, Geppe)*



Figure 5.33. *Amyotrophy (Navruzova, Geppe)*

Additional investigations: blood **creatine phosphokinase (CPK)**, LDH, transaminases, myoglobin, creatine; autoantibodies; **electromyography** (to distinguish primary muscle from nerve involvement), chronaximetry, ergography; muscle **biopsy**, genetic testing.

Semiotics of muscular system involvement

Hypotrophy and atrophy: a sedentary lifestyle, after surgery, severe somatic disease, dystrophy, poliomyelitis (paralytic form), hereditary muscular dystrophies, chronic infections, long-term glucocorticoid therapy; **local** — joint diseases, nerve damage, immobilization.

Causes of hypotonia:

- in newborns — CNS damage (hypoxia, hemorrhage), prematurity, **hyperbilirubinemia**, sepsis, hypothyroidism, **Down syndrome**, spinal muscular atrophy;
- at an early age — **rickets**, grade II–III hypotrophy;

- at an older age — **rheumatic chorea**, cerebellar damage, myopathies;
- **hypokalemia** (vomiting, diarrhea, diuretics), endocrine diseases.

Causes of hypertonia: damage to the pyramidal tract (**cerebral palsy**, Little disease), brain and spinal cord injuries, meningitis and encephalitis, **tetanus**, hypocalcemia (tetany), Thomsen myotonia.



Figure 5.34. *Hypertonia in cerebral palsy (Navruzova, Geppe)*

Myalgias: infections (influenza, enterovirus — Bornholm disease, leptospirosis, brucellosis), inflammation (**myositis, dermatomyositis**, trichinellosis — pain and induration), ischemia; after physical exertion — metabolic myopathies. Stone-like indurations in the muscles (calcinosis) — dermatomyositis, myositis ossificans.

Muscle weakness: proximal (thigh, shoulder girdle) — **myopathies, dermatomyositis**; **distal** — polyneuropathies, myotonic dystrophy; worsening after exertion — **myasthenia**, metabolic myopathy; endocrine — hypocortisolism, thyrotoxicosis, hypothyroidism, hyperparathyroidism.

Main diseases:

- **Progressive muscular dystrophies** (Duchenne — impaired dystrophin synthesis, boys, from 3–5 years, “duck gait”, **Gowers' sign**, **pseudohypertrophy** of the calf muscles, sharply raised CPK);
- **Inflammatory myopathies** — infectious (bacterial, viral, parasitic), autoimmune (**dermatomyositis** — proximal weakness, violaceous erythema of the eyelids, Gottron's papules);
- **Drug-induced myopathies** (glucocorticoids, penicillamine, etc.);
- **Myotonia** — inability of a muscle to relax for several seconds after contraction (Thomsen disease);
- **Myasthenia** — autoantibodies to acetylcholine receptors; **ptosis, diplopia**, weakness increasing toward the end of the day; neonatal (from the mother) and congenital forms;
- **Paralyses (pareses): peripheral** — reduced tone, atrophy, absent reflexes; **central** — **spastic hypertonia**, hyperreflexia, pathological reflexes;

- **Congenital anomalies — muscular torticollis** (underdevelopment or injury of the sternocleidomastoid muscle), diaphragmatic hernia, absence of the pectoral muscle (Poland syndrome).

5.7. Clinical section

A child suspected of rickets

Rickets is diagnosed on the basis of clinical signs (sweating, bald nape, hypotonia, bone changes) and confirmed by **biochemistry** (reduced calcium and phosphorus, **raised alkaline phosphatase**, reduced 25(OH)D) and **X-ray** (wrist: cup-shaped widening of the metaphyses, a “fringed” border of the growth zone, osteoporosis).

> **CLINICAL SIGNIFICANCE**
> **Prevention of rickets:** during pregnancy — fresh air, proper nutrition and vitamin D for the mother; after birth — **breast milk**, walks, massage and gymnastics; for term babies from 1 month of age **vitamin D 400–500 IU/day** (according to Uzbekistan and WHO recommendations, doses are set by the local protocol), more for preterm babies.

Joint syndrome — diagnostic approach

1. **How many joints?** Mono- (1), oligo- (2–4), polyarthritis (≥5).
2. **Is it symmetric?** Symmetric — JRA, SLE; asymmetric — reactive, septic.
3. **What is the course?** Migratory — rheumatic fever; persistent, progressive — JRA.
4. **Additional signs:** fever, rash (SLE, Still's disease, vasculitis), conjunctivitis (Reiter), heart murmur (rheumatic fever), preceding infection.
5. **Laboratory:** CBC, CRP, ESR, ASO, rheumatoid factor, ANA, HLA-B27, synovial fluid.

A newborn with muscle hypotonia

Hypotonia is “not a disease but a sign”. Its causes are **central** (CNS damage, hypoxia, syndromes) and **peripheral** (spinal muscular atrophy, myopathies, myasthenia). In central hypotonia the reflexes are preserved or increased; in peripheral hypotonia they are **reduced or absent**, and muscle weakness is pronounced.

Practical skills

Skill 1. Measuring the anterior fontanelle and palpating the head

1. Wash and warm the hands; calm the child, on the mother's lap or lying down.
2. Palpate the skull bones and sutures and determine whether they are closed.
3. Find the anterior fontanelle; measure the distance between **the midpoints of opposite sides** in two directions with a tape (not the diagonals!).
4. Assess the edges of the fontanelle (firm/soft), its level (at the level of the skull/bulging/sunken) and its pulsation.
5. Test for craniotabes by pressing on the occipital and parietal bones with a finger.
6. Compare the result with the age norm and record it.

Skill 2. Inspection of the chest and spine

1. The child is undressed and stands (arms along the body).

2. From the front: shape and symmetry of the chest, deformities, the epigastric angle (placing the edges of the palms on the costal arches).
3. From the side: how marked the physiological curves are.
4. From behind: symmetry of the shoulders, the angles of the shoulder blades, the waist triangles and the iliac crests; the line of the spinous processes.
5. The child bends forward — to detect scoliosis and a rib hump.
6. Palpate the ribs for the “rosary”, and the wrists and phalanges for “bracelets” and “strings of pearls”.

Skill 3. Examination of the joints

1. Observe the gait.
2. Inspect each joint symmetrically: shape, swelling, defiguration, deformity, skin color.
3. Palpation: temperature (comparing with the back of the hand), pain, swelling, fluid (ballottement of the patella), crepitus.
4. Assess the range of active, then passive movements with functional tests; record pain and restriction.
5. Mobility of the spine (finger-to-floor distance, Ott's and Schober's tests).
6. Conclusion: the shape of the joints is unchanged, there are no signs of inflammation, movements are in full range and painless (or the changes found).

Skill 4. Testing for developmental dysplasia of the hip (newborn)

1. The child lies on the back on a warm table, calm.
2. Assess the symmetry of the thigh, inguinal and gluteal folds (gluteal folds — with the child prone).
3. Bend the legs at the hips and knees and compare the levels of the knees (relative shortening).
4. Flex the thighs to 90° and slowly spread them apart; assess the abduction angle.
5. Marx–Ortolani test: determine whether a “click” is felt while the thighs are spread.
6. If there is any suspicion — orthopedic consultation and hip ultrasound.

Skill 5. Assessment of the muscular system

1. Inspection: degree of development, relief, symmetry; measure the limb circumference if necessary.
2. Palpation: consistency, pain, indurations.
3. Tone: the child's posture; resistance to passive flexion-extension; in the newborn, the recoil, traction and “loose shoulders” tests.
4. Strength: movements against resistance; dynamometry in an older child.
5. Range of active and passive movements.
6. Conclusion: “The muscles are moderately developed, symmetric, tone normal, strength appropriate for age, painless, movements in full range.”

Skill 6. Testing for the signs of latent tetany

1. **Chvostek:** tap with a hammer or finger below the zygomatic arch, in front of the ear, where the facial nerve emerges; watch for contraction of the corner of the mouth, the nasal ala and the eyelid.

2. **Trousseau:** place a sphygmomanometer cuff on the upper arm, inflate it slightly above systolic pressure and hold for 2–3 minutes; watch for the “obstetrician's hand”.
3. **Lust:** tap with a hammer below the head of the fibula; watch for outward turning and dorsal flexion of the foot.
4. Record the result; if positive — refer for a blood calcium test.

Review questions

1. What are the features of the structure and chemical composition of children's bone tissue, and what is their clinical significance?
2. What is bone age and how is it determined?
3. What are the features of the newborn's skull; what are the sizes and closure times of the fontanelles?
4. When and how do the physiological curves of the spine form?
5. Name the age-related features of the chest and pelvis.
6. What are the eruption times and order of the primary and permanent teeth? Give the formulas for calculating the number of teeth.
7. How do body proportions change with age?
8. What is noted on inspection of the head, chest, spine and limbs?
9. How are the joints examined? What is the difference between defiguration and deformity?
10. Name the signs of developmental dysplasia of the hip and the methods of detecting it.
11. List the bone and other signs of rickets.
12. What are the latent and manifest signs of spasmophilia?
13. What are the main types of arthritis in children and how do they differ?
14. What are the age-related features of the muscular system, and what is physiological hypertonia?
15. How are the degree of muscle development, tone and strength assessed? What is the difference between spasticity and rigidity?
16. Name the causes of muscle hypotonia and hypertonia.

Test questions

1. The main reason children's bones rarely break and deform quickly:
   A) Abundance of mineral substances
   B) Abundance of organic matter and water, scarcity of mineral substances
   C) Thinness of the periosteum
   D) Poor blood supply
   E) Haversian structure
2. When does the anterior fontanelle usually close?
   A) At 3–4 months
   B) At 6 months
   C) At 1–1.5 years
   D) At 2–3 years
   E) At 5 years

3. When does the thoracic kyphosis form?
   A) At birth
   B) When the child begins to hold up the head (2–3 months)
   C) When the child begins to sit (5–6 months)
   D) When the child begins to walk
   E) At 5 years

4. Approximately how many primary teeth should a healthy 10-month-old child have?
   A) 2
   B) 4
   C) 6
   D) 8
   E) 12

5. Where is the midpoint of the body in the newborn?
   A) At the pubic symphysis
   B) At the umbilicus
   C) At the end of the sternum
   D) Midway between the umbilicus and the pubis
   E) At the xiphoid process

6. A wrist X-ray shows 3 ossification nuclei. The child's approximate age:
   A) 6 months
   B) 1 year
   C) 2 years
   D) 4 years
   E) 6 years

7. “Fusiform” fingers are typical of which disease?
   A) Marfan syndrome
   B) Reactive arthritis
   C) Juvenile rheumatoid arthritis
   D) Systemic scleroderma
   E) Rickets

8. The earliest bone sign of rickets:
   A) “Bracelets”
   B) O-shaped deformity of the legs
   C) Craniotabes
   D) Rachitic kyphosis
   E) Harrison's groove

9. A sign typical of osteoid hyperplasia:
   A) Craniotabes
   B) Flattening of the occiput
   C) Rachitic “rosary”
   D) Harrison's groove
   E) X-shaped legs

10. In which condition is Trousseau's sign positive?
    A) Meningitis
    B) Spasmophilia (hypocalcemia)
    C) Hyperthyroidism
    D) Myasthenia
    E) Hypokalemia

11. Gait in bilateral congenital dislocation of the hip:
    A) Antalgic
    B) “Duck gait”
    C) Spastic
    D) Ataxic
    E) Unchanged

12. Until when does physiological hypertonia of the arms persist in the newborn?
    A) Up to 1 week
    B) Up to 2–2.5 months
    C) Until 6 months
    D) Until 1 year
    E) Until 2 years

13. What does the absence of the 2nd phase (flexion of the arms) in the traction test indicate?
    A) Hypertonia
    B) Hypotonia
    C) Normal
    D) Rigidity
    E) Spasticity

14. The “clasp-knife” phenomenon is typical of:
    A) Extrapyramidal rigidity
    B) Pyramidal spasticity
    C) Peripheral paralysis
    D) Myasthenia
    E) Cerebellar damage

15. Not typical of Duchenne muscular dystrophy:
    A) Gowers' sign
    B) Pseudohypertrophy of the calf muscles
    C) Raised CPK
    D) Primary involvement of the distal muscles
    E) “Duck gait”

16. Through which structure do a child's bones grow in length?
    A) Synovial membrane
    B) Joint capsule
    C) Periosteum
    D) Bone marrow
    E) Epiphyseal (growth) cartilage

17. What are “greenstick” fractures in children's bones related to?
    A) Ossification of the epiphyses
    B) Poor blood supply
    C) Abundance of mineral substances
    D) Brittleness of the bones
    E) A thick and active periosteum

18. At what age does the anterior fontanelle close?
    A) 1–1.5 years
    B) 3 years
    C) 2–3 months
    D) 6 months
    E) 5 years

19. Between which sutures is the posterior fontanelle located?
    A) Coronal and sagittal
    B) Sagittal and lambdoid
    C) Coronal and frontal
    D) Sphenoid and coronal
    E) Lambdoid and mastoid

20. At what age does the lumbar lordosis form (related to standing and walking)?
    A) 2–3 months
    B) 9–12 months
    C) 7 years
    D) 5–6 months
    E) 3 years

21. With which skill is the thoracic kyphosis formed?
    A) Holding up the head
    B) Crawling
    C) Walking
    D) Running
    E) Sitting

22. The first primary teeth to erupt and their timing:
    A) Canines, 8 months
    B) Upper central incisors, 4 months
    C) Lower central incisors, 6–7 months
    D) Lower lateral incisors, 5 months
    E) First molars, 10 months

23. At what age does the child have 20 primary teeth?
    A) 5 years
    B) 3.5 years
    C) 1.5 years
    D) 2–2.5 years
    E) 1 year

24. Formula for the approximate number of primary teeth up to 2 years (n — months):
    A) n − 4
    B) n − 2
    C) n + 4
    D) 2n − 10
    E) n / 2

25. Formula for the number of permanent teeth (n — age):
    A) 2n − 4
    B) n − 4
    C) 3n − 10
    D) 4n − 20
    E) n + 4

26. A feature of the newborn's chest:
    A) The transverse dimension is much larger than the anteroposterior
    B) Conical, ribs horizontal, diaphragmatic breathing
    C) Funnel-shaped
    D) Flat, ribs oblique
    E) Cylindrical, thoracic breathing

27. The change in the ribs that occurs in rickets together with forward protrusion of the sternum (“pigeon chest”):
    A) Scoliosis
    B) Funnel chest
    C) Harrison's groove and rachitic rosary
    D) Gibbus
    E) Kyphosis

28. Disproportionate short stature with short limbs and a normal trunk is typical of which disease?
    A) Chondrodystrophy (achondroplasia)
    B) Pituitary dwarfism
    C) Rickets
    D) Marfan syndrome
    E) Hypothyroidism

29. Craniotabes (softening of the occipital and parietal bones) is the initial sign of which disease?
    A) Microcephaly
    B) Rickets
    C) Craniostenosis
    D) Osteomyelitis
    E) Hydrocephalus

30. At the costochondral junctions of which ribs is the rachitic “rosary” palpated?
    A) IX–X
    B) III–IV
    C) V–VIII
    D) I–II
    E) XI–XII

31. What gait is seen in bilateral congenital dislocation of the hip?
    A) “Duck gait”
    B) Ataxic
    C) Steppage
    D) Antalgic
    E) Walking on tiptoe

32. After what age are flat feet considered pathological?
    A) 6 months
    B) 2 years
    C) 1 year
    D) 10 years
    E) 5 years

33. “Fusiform” deformity of the proximal interphalangeal joints is typical of which disease?
    A) Marfan syndrome
    B) Systemic scleroderma
    C) Rickets
    D) Reactive arthritis
    E) Juvenile rheumatoid arthritis

34. Long thin “spider” fingers (arachnodactyly) are typical of which syndrome?
    A) Klinefelter
    B) Chondrodystrophy
    C) Shereshevsky–Turner
    D) Marfan syndrome
    E) Down syndrome

35. A test assessing the mobility of the lumbar spine:
    A) Sulkowitch test
    B) Tourniquet test
    C) Ott's test
    D) Schober's test
    E) Marx–Ortolani test

36. What is felt in the Marx–Ortolani test?
    A) Ballottement of the patella
    B) Crepitus
    C) A “click” as the dislocated femoral head enters the socket
    D) Intra-articular fluid
    E) Tendon pain

37. Which method is used to confirm developmental dysplasia of the hip at 1–3 months?
    A) Densitometry
    B) X-ray
    C) Ultrasound (Graf method)
    D) ECG
    E) CT

38. The normal value of total serum calcium:
    A) 2.25–2.75 mmol/l
    B) 0.5–1.0 mmol/l
    C) 3.5–4.5 mmol/l
    D) 1.1–1.3 mmol/l
    E) 5.0–6.0 mmol/l

Clinical cases

Case 1

A 5-month-old child. Born in winter, mixed-fed, vitamin D prophylaxis was not given. According to the mother, the child is restless, startles, sweats a lot (especially when feeding and during sleep), and has lost hair on the nape. On examination: anterior fontanelle 3×3.5 cm with soft edges; the occiput is flattened, and areas of softening are found on the occipital and parietal bones on pressure with a finger. Muscle tone is reduced, the abdomen is “frog-shaped”. No teeth.

Questions: 1) Which disease do you suspect? 2) Which signs are typical of osteomalacia? 3) Which investigations are needed to confirm the diagnosis? 4) What complication is there a risk of, and how do you test for its latent signs?

Answer: 1) Rickets (onset of the peak period): risk factors — birth in winter, mixed feeding, no prophylaxis; neurovegetative signs (restlessness, sweating, bald nape), muscle hypotonia, bone changes. 2) Craniotabes, flattening of the occiput, softening and enlargement of the edges of the anterior fontanelle. 3) Blood calcium, phosphorus (reduced), alkaline phosphatase (raised), 25(OH)D; wrist X-ray (cup-shaped widening of the metaphyses, osteoporosis). 4) Spasmophilia (especially in spring, when treatment is started) — the Chvostek, Trousseau and Lust signs are tested.

Case 2

A 9-year-old boy has complained of pain and swelling in the left knee and right ankle joints for 3 days. He had an intestinal infection (diarrhea) 3 weeks ago. On examination: the left knee joint is defigured, the skin over it is warm, ballottement of the patella is positive; the right ankle joint is swollen and painful; the 2nd toe of the right foot is swollen along its whole length and red. The conjunctivae of both eyes are red. There is heel pain.

Questions: 1) Describe the joint syndrome. 2) What is the change in the toe called? 3) Presumptive diagnosis? 4) Which investigations are needed?

Answer: 1) Asymmetric oligoarthritis (2 joints) of the legs; in the left knee — defiguration, local hyperthermia, intra-articular fluid (positive ballottement); talalgia (enthesitis). 2) A “sausage” toe (dactylitis). 3) Reactive arthritis (Reiter disease) — after an intestinal infection, arthritis + conjunctivitis (urethritis may also be present). 4) CBC, CRP, ESR, HLA-B27, pathogens (Yersinia, Salmonella, Shigella; Chlamydia) or antibodies in the stool and urine, ultrasound and X-ray of the joints, and synovial fluid analysis if necessary.

Case 3

A 1-month-old girl came for a preventive check-up. The delivery was a breech presentation. On examination: the left thigh and gluteal folds are higher and deeper than on the right; when the legs are flexed at the hips and knees the left knee is lower; abduction of the left thigh is limited to 50° (80° on the right); during abduction a “click” is felt on the left.

Questions: 1) Which pathology do you suspect? 2) Name the signs found. 3) How is it confirmed? 4) What will be the consequence if treatment is not given in time?

Answer: 1) Developmental dysplasia (congenital dislocation) of the left hip; risk factors — breech presentation, female sex. 2) Asymmetry of the skin folds, relative shortening of the left leg, limited abduction, positive “slipping” sign (Marx–Ortolani). 3) Hip ultrasound (Graf method), orthopedic consultation; after 3–4 months — X-ray. 4) Limping when the child begins to walk, leg shortening, lumbar hyperlordosis, later coxarthrosis; need for surgical treatment.

Quick review

Table 5.11. *Key facts on Topic 5*

Question	Answer
Bone structure	Coarse-fibered → lamellar (3–4 years); close to adult at 12 years
Bone properties	Little mineral, much water — flexible, breaks rarely; “greenstick” fracture
Bone age	6 months — 1 nucleus; 1 year — 2; then +1 each year
Anterior fontanelle	1.5×2–3×3 cm; closes at 1–1.5 years
Posterior fontanelle	Closed in 75%; in the rest at 1–2 months
Sutures	Begin to close at 3–4 months, closed at 3–5 years
Curves	Cervical lordosis 2–3 months; thoracic kyphosis 5–6 months; lumbar lordosis 9–12 months
The chest	Conical in the newborn, ribs horizontal; the dimensions become equal at 3 years
Primary teeth	From 6–7 months; 8 at 1 year; 20 at 2 years; X = n – 4
Permanent teeth	From 5–6 years; X = 4n – 20
Head / height	Newborn 1/4 → adults 1/8
Midpoint of the body	Newborn — umbilicus; adults — pubis
Epigastric angle	90° normosthenic; obtuse — hypersthenic; acute — asthenic
Rickets — osteomalacia	Craniotabes, flattening of the occiput, Harrison's groove, leg deformities
Rickets — osteoid hyperplasia	Bosses, “rosary”, “bracelets”, “strings of pearls”
Latent tetany	Chvostek, Trousseau, Lust, Maslov
Fingers	Fusiform — JRA; sausage-shaped — reactive/psoriatic; arachnodactyly — Marfan; sclerodactyly — scleroderma
Hip dysplasia	Fold asymmetry, shortening, limited abduction, Ortolani; ultrasound
Spinal mobility	Ott (30 cm) and Schober (10 cm) — +4–5 cm on bending
Muscle mass	Newborn ≈23–25%; 15 years 32–33%; adults 40–44%
Physiological hypertonia	Arms 2–2.5 months, legs 3–4 months
Traction test	No 1st phase — hypertonia; no 2nd phase — hypotonia
Spasticity / rigidity	“Clasp-knife” — pyramidal; “lead pipe”, “cogwheel” — extrapyramidal
Gowers' sign	Duchenne myopathy

Question	Answer
Features of the child's bone	Coarse-fibered structure (until 3–4 years), little mineral/much organic matter — flexible; thick active periosteum; intensive remodeling; abundant blood supply; red marrow in all bones
Fontanelles	Anterior: diamond-shaped, 1.5×2–3×3 cm, closes at 1–1.5 years; posterior: triangular, closed at birth or at 1–2 months; lateral fontanelles closed at birth
Spinal curves	Cervical lordosis 2–3 months; thoracic kyphosis 5–6 months; lumbar lordosis 9–12 months; stabilization 6–7 years; full fixation 14–17 years
Order of eruption of primary teeth	6–7 months lower central; 8–9 upper central; 9–10 upper lateral; 11–12 lower lateral; 12–15 first molars; 18–20 canines; 22–24 second molars (20 teeth)
Chest deformities	Pigeon chest (sternum forward), funnel chest (cobbler's chest), Harrison's groove, rachitic rosary, cardiac hump
Disturbed body proportions	Short limbs — chondrodystrophy; long thin fingers — Marfan; proportionate short stature — pituitary dwarfism; infantile proportions — hypothyroidism
Signs of hip dysplasia	Asymmetry of the skin folds, leg shortening, limited abduction (normal 80–90°), Marx–Ortolani “click” sign, external rotation of the foot; ultrasound (Graf) at 1–3 months
Spinal mobility tests	Ott (thoracic part, 30 cm below the 7th cervical vertebra, +4–5 cm); Schober (lumbar part, 10 cm above LV, +4–5 cm); ≤1–2 cm in rigidity
Bone signs of rickets	Craniotabes, “square” head, flattening of the occiput, rosary, bracelets, strings of pearls, Harrison's groove, X/O-shaped legs
Types of gait	Antalgic; “duck gait” (bilateral dislocation, myopathy); limping (unilateral dislocation, Perthes); on tiptoe (talalgia)
Joint deformities	Fusiform — JRA; sausage-shaped (dactylitis) — reactive/psoriatic arthritis; sclerodactyly — scleroderma; arachnodactyly — Marfan
Normal calcium-phosphorus metabolism	Total Ca 2.25–2.75 mmol/l; ionized 1.1–1.3; P (1–2 years) 1.3–2.3 mmol/l; alkaline phosphatase is raised in rickets

TOPIC 6

The respiratory system: anatomical and physiological features, the first breath, history taking, inspection, palpation, percussion and auscultation

Learning objectives

The student should know:

- anatomical and physiological features of children's respiratory organs (nose, pharynx, larynx, trachea, bronchi, lungs, pleura, chest) and their clinical significance;
- the mechanism of the first breath, the importance of surfactant, age-related features of the regulation of breathing;
- age norms for respiratory rate, type and rhythm of breathing;
- the rules of comparative and topographic percussion, the lung borders;
- age-specific breath sounds and adventitious breath sounds.

The student should be able to:

- take complaints and history from a child with a respiratory disease;
- inspect the chest, count the respiratory rate and determine the type of dyspnea;
- palpate the chest: determine pain, resistance and vocal fremitus;
- perform comparative and topographic percussion and determine the excursion of the lower lung edge;
- auscultate the lungs and assess the results.

6.1. Anatomical and physiological features of the respiratory organs

Main idea: in the child all the airways are **narrow, soft, rich in blood and lymph vessels and poor in elastic fibers**. For this reason the mucosa swells quickly, the airways become obstructed quickly, and atelectasis and pneumonia are common.

The airways are divided into three parts: **upper** (nose, paranasal sinuses, pharynx), **middle** (larynx, trachea, main, lobar and segmental bronchi), **lower** (bronchioles, alveoli).

Table 6.1. *Age-related features of the respiratory organs and their clinical significance*

Organ	Feature in children	Clinical significance
Nose	Small, narrow nasal passages; no inferior nasal meatus (forms at 4 years); the mucosa is rich in blood vessels; no cavernous tissue (develops at 8–9 years)	With a runny nose, nasal breathing is quickly impaired and feeding becomes difficult; nosebleeds are rare
Paranasal sinuses	The maxillary and ethmoid sinuses are present at birth but small; the frontal and sphenoid sinuses develop from 6–7 years and grow until 15–20 years	Sinusitis is rare at an early age; ethmoiditis may occur
Nasolacrimal duct	Short, wide, valve undeveloped	Infection easily passes from the nose to the eye (conjunctivitis)
Auditory (Eustachian) tube	Short, wide, horizontal	Otitis develops quickly with a runny nose

Organ	Feature in children	Clinical significance
Pharynx	Small, narrow; the tonsils are undeveloped until 1 year and well developed at 4–5 years	Tonsillitis is rare before 1 year; adenoids — mouth breathing
Larynx	Funnel-shaped, located high (the glottis at the level of the 4th cervical vertebra, the 7th in adults); the cartilages are soft, the glottis is narrow, the loose subglottic tissue is prone to edema	At 2–3 years laryngitis → croup (stenosis)
Trachea	Funnel-shaped; ≈4 cm long in the newborn (12 cm in adults); soft cartilage, few elastic fibers	Tracheitis, stenosis, noisy breathing
Bronchi	Short, narrow, soft cartilage, few muscle and elastic fibers, weak mucociliary clearance; the right main bronchus is a continuation of the trachea, short and wide	Foreign bodies and pneumonia more often on the right; bronchiolitis and obstruction before 1 year
Lungs	50–60 g in the newborn; less airy, rich in blood, few elastic fibers, much interstitial tissue; the acinus is not fully mature (grows until 8 years)	Atelectasis, emphysema and pneumonia develop quickly
Pleura	Thin, delicate, the parietal layer loosely attached	When fluid accumulates, the mediastinal organs shift easily
Mediastinum	Relatively large, many lymph nodes	Bronchoadenitis (tuberculosis)
The chest	Barrel-shaped in the newborn, ribs horizontal, epigastric angle obtuse; respiratory muscles weak, the diaphragm high	Breathing is shallow and rapid, mainly diaphragmatic; meteorism makes breathing difficult

REMEMBER

The right main bronchus leaves the trachea at a small angle and is its continuation — this is why **foreign bodies most often enter the right bronchus**. The glottis is at the level of $\mathbf{C_{IV}}$ in children and $\mathbf{C_{VII}}$ in adults.

Lung segments: right lung — **10** segments (3 in the upper lobe, 2 in the middle lobe, 5 in the lower lobe); left lung — **9** segments. In children pneumonia most often occurs in **segments 6, 2 and 10** (poor drainage, the child lies for a long time). The bronchi of **segments 4 and 5** are long and narrow, with lymph nodes nearby — "**middle lobe syndrome**" (chronic atelectasis) may develop.

Stages of lung development (A.I. Strukov):

1. **0–2 years** — the number of alveoli increases rapidly;
2. **2–5 years** — elastic and muscle fibers and peribronchial and perivascular lymphoid tissue grow (pneumonia tends to become chronic);
3. **5–7 years** — formation of the acinar structure is completed;
4. **7–12 years** — lung mass increases mainly through maturation of the lung tissue.

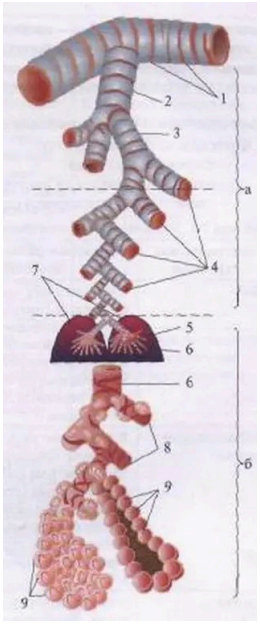


Figure 6.1. *The bronchial and alveolar tree: main, lobar, segmental bronchi → terminal and respiratory bronchioles → alveoli (Arzikulov)*

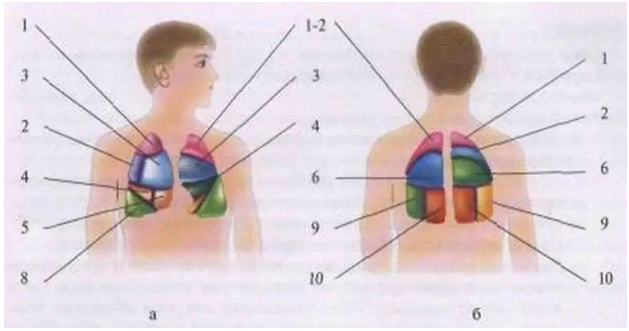


Figure 6.2. *Color diagram of the segments: a — from the front, b — from behind (Arzikulov)*

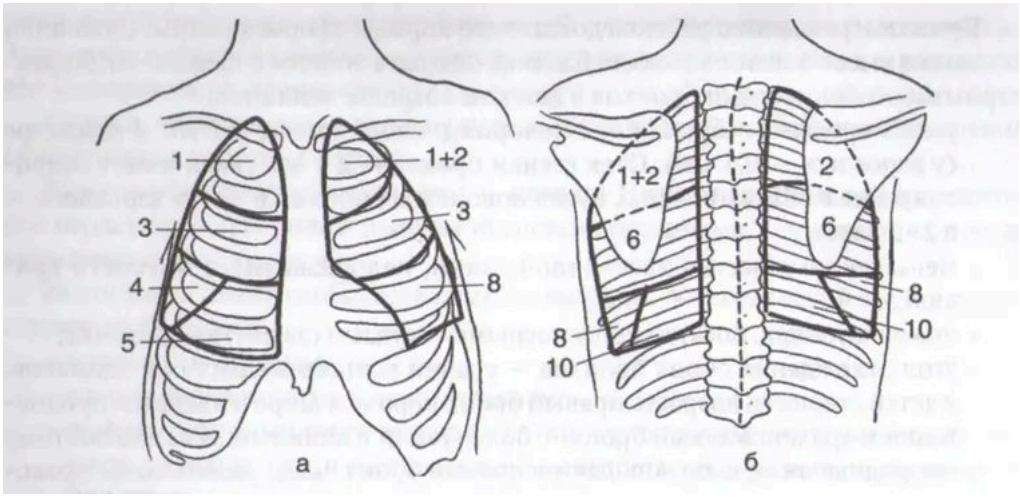


Figure 6.3. *Projection of the segments onto the chest: a — anterior, b — posterior surface. 1 — apical, 2 — posterior, 3 — anterior (upper lobe); 4, 5 — middle lobe (lingular on the left); 6 — superior segment, 7–10 — basal segments (lower lobe) (Kildiyarova, Figure 8.2)*

Functional features:

- Breathing is **shallow and rapid**; the minute volume of breathing relative to body mass is higher than in adults; the oxygen demand is high.
- Gas exchange in the alveoli is intensive (the lungs are rich in blood), but **the respiratory reserve is small** — respiratory failure develops quickly.
- **Local defense** is weak: little secretory IgA, lysozyme and interferon; the ciliated epithelium is immature; the cough reflex is weak (especially in preterm babies).

6.2. Mechanism of the first breath

Main idea: clamping of the cord → **hypoxia + hypercapnia + acidosis** → chemoreceptors → respiratory center → **sharp contraction of the diaphragm** → the lungs expand, and surfactant keeps them open.

1. **The chemical (humoral) factor is the most important.** When the cord is clamped, pO_2 falls from 80 to ≈15 mm Hg, pCO_2 rises from 40 to ≈70 mm, and pH falls below 7.35. The central chemoreceptors are sensitive to pH, the peripheral (carotid, aortic) chemoreceptors to O_2 and CO_2.
2. **Physical (external) stimuli:** cold, light, stimulation of skin receptors, gravity → reticular formation → the excitability of the respiratory center increases.
3. **Removing inhibiting factors:** suctioning mucus and amniotic fluid from the mouth and nose.
4. **Mechanics:** the diaphragm contracts and creates negative intrathoracic pressure (≈80 mm H_2O); the volume of the first breath exceeds 80 ml. Lung fluid is absorbed into the lymphatics and capillaries. A healthy baby takes the first breath with **a loud cry**.

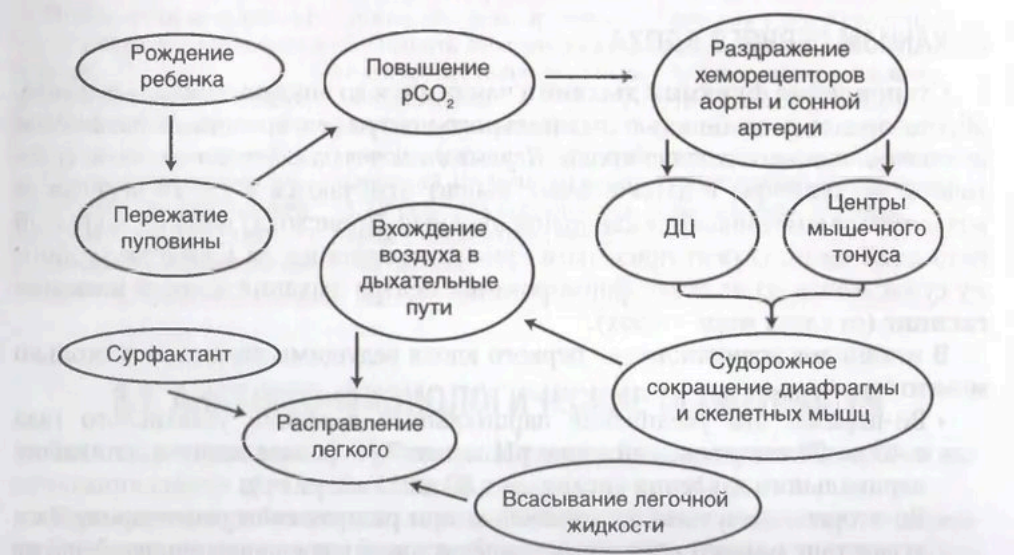


Figure 6.4. *Mechanism of the first breath: birth → clamping of the cord → rise in pCO_2 → stimulation of the aortic and carotid chemoreceptors → respiratory center and muscle tone centers → contraction of the diaphragm and skeletal muscles → entry of air → expansion of the lungs (with the participation of surfactant) → absorption of lung fluid (Kildiyarova, Figure 8.1; figure labels in Russian)*

Surfactant — a surface-active substance covering the alveolar surface (90% lipid, mainly phosphatidylcholine; 10% protein). It is synthesized by **type II alveolocytes**. It prevents the alveoli from collapsing on expiration and has a bactericidal effect. Synthesis enzymes: methyltransferase — active **from week 22–24**, phosphocholine transferase — **from week 35**. Surfactant deficiency causes **respiratory distress syndrome** (hyaline membrane disease), mainly in preterm babies.

> **CLINICAL SIGNIFICANCE**
> If there is a risk of preterm birth (before 34 weeks), the mother is given **glucocorticoids** (dexamethasone or betamethasone) — they accelerate surfactant synthesis in the fetal lungs. Exogenous surfactant is given intratracheally to the preterm newborn.

The respiratory center (medulla oblongata and pons): the medullary part (alternation of inspiration and expiration), the apneustic part (inspiratory spasm), the pneumotaxic part (inhibits the apneustic part). The pneumotaxic part matures **by the end of the 1st year** — at an early age breathing is **arrhythmic**; **apnea** occurs in preterm babies. Up to 3 months, pauses in breathing of **up to 8–10 seconds** are physiological; apnea longer than 20 seconds or accompanied by bradycardia and cyanosis is pathological.

6.3. History taking

Main idea: the leading sign of respiratory disease is **cough**; its character often indicates the level of the lesion.

Main complaints: runny nose (nasal congestion, discharge), cough, sputum, dyspnea and attacks of suffocation, chest pain, hemoptysis, fever, weakness, headache, change of voice; in small children — refusal to feed, restlessness, vomiting.

Table 6.2. *Types of cough and their diagnostic significance*

Type of cough	When it occurs
Harsh, “barking”, with hoarseness or aphonia	Laryngitis, true and false croup
Harsh, “as if coughing into a barrel”	Tracheitis
Dry, exhausting	Pharyngitis, tracheitis, onset of bronchitis, foreign body, bronchial asthma
Moist (productive)	Resolving bronchitis, bronchiectasis, abscess
Short, painful (the child suppresses the cough)	Pleurisy, lobar pneumonia
Paroxysmal, with reprises (a whooping deep inspiration), worse at night, the face turns red, ends with vomiting	Whooping cough
Bitonal (a harsh main tone + a high musical tone)	Enlargement of the bifurcation lymph nodes: tuberculous bronchoadenitis, lymphogranulomatosis, leukemia, mediastinal tumor
Morning, with copious sputum (“a mouthful”)	Bronchiectasis, lung abscess, cavitary tuberculosis
A severe coughing fit during breastfeeding or drinking	Tracheoesophageal fistula, swallowing disorder

Sputum: amount, color (yellowish — purulent, green — stagnant pus, **rusty** — lobar pneumonia), odor (fetid — bronchiectasis, abscess, gangrene), consistency (viscous, glassy — asthma). Small children swallow the sputum. **Hemoptysis** — blood from the lungs is **bright red and frothy**.

Voice: nasal speech — adenoids, cleft palate, paralysis of the soft palate; aphonia — true croup (diphtheria). False croup begins **suddenly** (often at night), true croup **gradually**.

History of the present illness: when and how it began (acutely or gradually), the cause (ARVI, chilling, foreign body — sudden cough during play), the course, the effect of treatment. At an early age pneumonia often develops after ARVI.

Life history: prematurity, whether mechanical ventilation was given; respiratory diseases in the family and contact with **tuberculosis**; past measles or whooping cough; vaccinations; rickets, hypotrophy, allergy (predisposition to asthma); passive smoking; home conditions (dampness, mold, animals).

6.4. Inspection

Main idea: five things are assessed on inspection — **skin color, chest shape, type of breathing, respiratory rate and rhythm, signs of dyspnea**.

General inspection: cyanosis (lips, nasolabial triangle, around the eyes), pallor; **flaring of the nasal alae** (pneumonia in an infant); froth at the corner of the mouth (pneumonia in the newborn); herpes on the lips (ARVI, lobar pneumonia); redness of one cheek (lobar pneumonia on that side);

“**drumstick fingers**” and “**watch-glass**” nails (chronic hypoxia); forced position — **orthopnea** (in an asthma attack the child sits leaning on the hands), in pleurisy the patient lies on the affected side. Nasal discharge, nasal breathing. Inspection of the pharynx and tonsils is performed **last** so as not to upset the child.

Chest shape. In the newborn it is barrel-shaped (the sagittal dimension equals the transverse one); with age the transverse dimension increases. In an older child there are three types: **normosthenic** (conical, epigastric angle ≈90°), **asthenic** (long, narrow, angle <90°, winged scapulae), **hypersthenic** (cylindrical, angle >90°). Pathological shapes: **barrel-shaped (emphysematous)** — asthma, emphysema; **funnel chest** and “**pigeon chest**”, **Harrison's groove** — rickets; **asymmetric** — in exudative pleurisy the affected side bulges, in chronic pneumonia and pneumofibrosis it sinks.

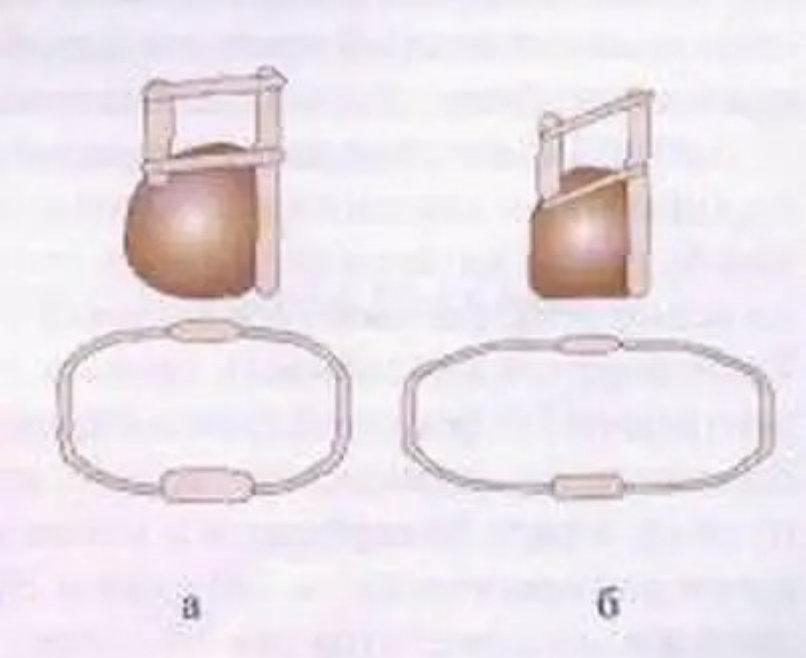


Figure 6.5. *Shape and cross-section of the chest: a — young child (barrel-shaped, ribs horizontal); b — older child (transverse dimension predominates, ribs oblique) (Arzikulov)*

Type of breathing (by age):

Table 6.3. *Age-related changes in the type of breathing*

Age	Type of breathing
Newborn, infant	Diaphragmatic (abdominal)
1–2 years (when the child starts standing)	Mixed (thoracoabdominal)
3–7 years	The thoracic type begins to be added
8–14 years	Depends on sex: boys — abdominal, girls — thoracic type

The **symmetric** participation of both halves of the chest in breathing is checked (the angles of the shoulder blades are watched). Lagging of one side — pleurisy, atelectasis, pneumonia. **Excursion** is measured with a tape (in front over the nipples, behind under the angles of the shoulder blades): at rest, at maximal inspiration and expiration; the difference in older children is 5–8 cm.

The respiratory rate (RR) is counted unnoticed while the child is calm or asleep (with a hand on the chest or abdomen, with a stethoscope held near the nose, or during auscultation). **Up to 3 months — for 1 full minute** (because of arrhythmia and apnea).

Table 6.4. *Respiratory rate and respiration/pulse ratio by age*

Age	RR (per minute)	Respiration : pulse
Newborn	40–60	1 : 2,5–3
Until 1 year	30–35	1 : 3–3,5
1–3 years	25–30	1 : 3,5–4
4–6 years	20–25	1 : 4
7–15 years	18–20	1 : 4–4,5
Over 15 years	16–18	1 : 4–5

CLINICAL SIGNIFICANCE
According to WHO (IMCI), the cut-offs for **tachypnea** (a sign of pneumonia) are: **up to 2 months — ≥60**, **2–12 months — ≥50**, **1–5 years — ≥40** breaths per minute. Lower chest wall indrawing is a sign of **severe pneumonia**.

In respiratory diseases the respiration/pulse ratio narrows to **1:2 or 1:3**. **Tachypnea** — rapid breathing (fever, excitement, lung and heart diseases, anemia, acidosis); **bradypnea** — slow breathing (coma, poisoning with sleeping pills, raised intracranial pressure).

Dyspnea — a disturbance of the rate, depth and rhythm of breathing with increased work of the respiratory muscles.

Table 6.5. *Types of dyspnea*

Type	What is difficult	Signs	When
Inspiratory	Inspiration	Flaring of the nasal alae; retraction of the intercostal spaces and the suprasternal and supraclavicular fossae; noisy breathing (stridor)	Narrowing of the upper airways: croup, foreign body, retropharyngeal abscess
Expiratory	Expiration	The chest is in the “inspiratory” position, expiration is prolonged, the abdominal muscles take part, whistling breathing, wheezing heard at a distance	Bronchial asthma, obstructive bronchitis, bronchiolitis
Mixed	Both	Rapid breathing, the accessory muscles take part	Pneumonia, pleurisy, heart failure

Congenital stridor — a noise on inspiration resembling a cat's purr or a pigeon's “cooing”; it increases with crying and decreases in sleep; it is related to softness of the laryngeal cartilages (laryngomalacia) and disappears **at 2–3 years**. Thymic enlargement and bronchoadenitis must be excluded. Up to 3 months, slight retraction of the intercostal spaces at the attachment of the diaphragm is normal.

Pathological types of breathing (severe disturbance of the respiratory center):

- **Cheyne–Stokes** — the depth of breathing gradually increases, then decreases and ends with a pause (meningitis, encephalitis, severe poisoning, heart failure); it may occur in sleep in preterm babies.
- **Biot** — breaths of the same depth suddenly begin and are suddenly replaced by a long pause (cerebrovascular disorders, meningitis).
- **Grocco** — wave-like breathing without pauses (similar to Cheyne–Stokes, but with shallow breaths instead of the pause).

- **Kussmaul** — deep, noisy, infrequent breathing (the breathing of a “hunted animal”) — diabetic coma, metabolic acidosis, uremia.

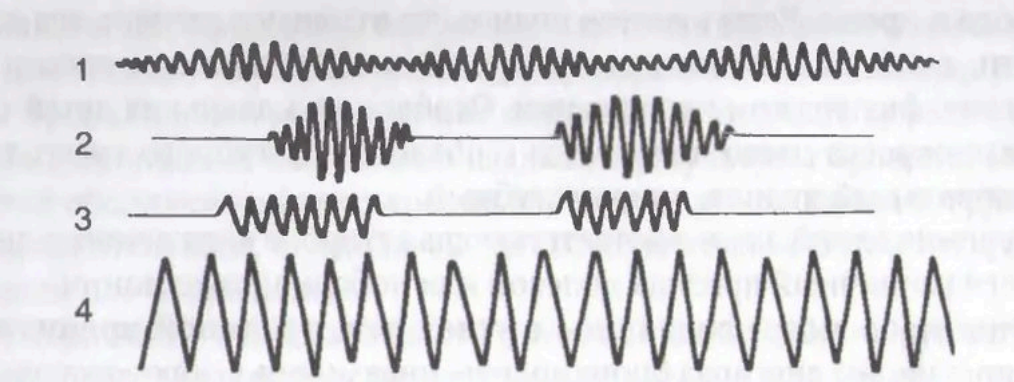


Figure 6.6. *Pathological types of breathing (spirogram): 1 — Grocco; 2 — Cheyne–Stokes; 3 — Biot; 4 — Kussmaul (Kildiyarova, Figure 8.10)*

6.5. Palpation of the chest

Main idea: palpation confirms the findings of inspection; it is performed with both hands over **symmetric** areas.

Palpation determines:

1. **Tenderness** — superficial (muscle, rib, nerve — myositis, neuralgia) or deep pleural. Pleural pain increases with breathing and decreases when the chest is compressed and when bending toward the affected side (in neuralgia it increases instead). **Mussy's sign** — pain on pressure between the heads of the sternocleidomastoid muscle — diaphragmatic pleurisy.
2. **Skin fold thickness** — thickening on the affected side (**Filatov's sign**) — exudative pleurisy, pneumonia.
3. **Resistance (elasticity)** — assessed by compressing the chest with the palms front-to-back and from the sides. If it compresses easily — elastic (normal; increased in rickets); if it does not compress — **rigid** (exudative pleurisy, tumor, ossification of the costal cartilages at an older age).
4. **Vocal fremitus (fremitus vocalis)** — transmission of the vibration of the vocal cords to the chest wall.
5. **In addition:** smoothing and bulging of the intercostal spaces (exudative pleurisy), rachitic “rosary”, subcutaneous **crepitus** (subcutaneous emphysema — crunching like snow), pleural friction felt by hand.

Technique of testing vocal fremitus: both palms are placed on symmetric areas of the chest and moved alternately from the apices to the lower parts. The child says **words with the letter “R”** (“tractor”, “thirty-three”, “ninety-nine”). In an infant who does not speak — during crying or screaming (low accuracy). Normally it is **the same** on both sides.

Table 6.6. *Changes in vocal fremitus*

Vocal fremitus	Cause
Increased	Consolidation of lung tissue (lobar pneumonia, pneumofibrosis), a cavity connected to a bronchus (cavern, abscess)
Weakened or absent	Fluid (exudative pleurisy, hydrothorax) or air (pneumothorax) in the pleural cavity, bronchial obstruction (obstructive atelectasis), emphysema, obesity

MEMORY AID

“Density conducts, fluid and air block”: dense tissue (pneumonia) **increases** vocal fremitus, bronchophony and bronchial breathing; fluid or air in the pleura and an obstructed bronchus **weaken** them.

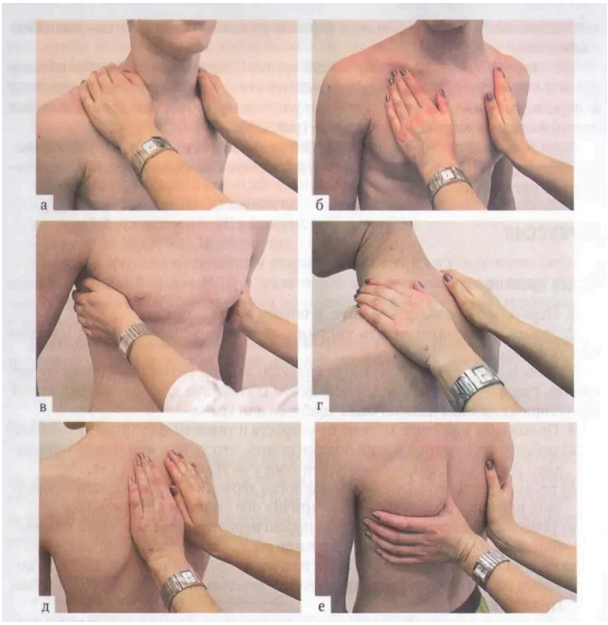


Figure 6.7. *Sequence of testing vocal fremitus at symmetric points (a–f): supra- and infraclavicular, lateral areas, suprascapular, interscapular, subscapular (Kildiyarova, Figure 8.4)*

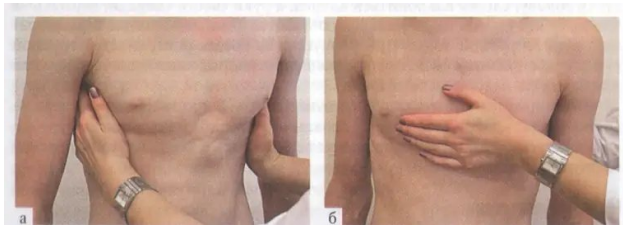


Figure 6.8. *Determining chest resistance: a — front-to-back compression; b — lateral compression (Kildiyarova, Figure 8.5)*

6.6. Percussion of the lungs

Main idea: in the child the stroke is **light (quiet)**, because the chest is thin and elastic — with a strong stroke distant tissues also vibrate and small foci are “lost”. Percussion is always performed **from the clear sound toward the dull sound**.

General rules: a quiet, warm room; the doctor's hands warm and dry, nails short. For anterior percussion the doctor stands on the child's right side, for posterior percussion on the left side. The child's position: over 2 years — standing or sitting; the mother holds an infant upright; up to 2–3 months — lying on the back for the front, and lying with the abdomen on the doctor's palm for the back. For posterior percussion the child bends the head slightly and crosses the arms in front (the shoulder blades move apart); for the lateral areas the hands are placed on the back of the neck. If the child cries — percuss during the deep inspiration between cries. The child's body must be straight and symmetric (otherwise the sound will be asymmetric).

Methods:

- **Indirect (finger-to-finger)** — the 3rd finger of the left hand (pleximeter finger) is placed firmly in the intercostal space **parallel** to the ribs; the 3rd finger of the right hand (plessor finger) gives 2 short strokes on the middle phalanx of the pleximeter; movement only at the wrist joint.
- **Direct** — the fingertip strikes the chest wall directly; used in infants and in hypotrophy. **Obraztsov's method** — the index finger of the right hand slips off the middle finger and strikes the chest.

Comparative percussion

The sound is compared at **symmetric** points of the chest; normally it is the same. A difference is a sign of pathology.

Order:

1. **From the front:** the supraclavicular fossa (finger parallel to the clavicle), directly over the clavicle, infraclavicular; along the midclavicular line in the 2nd–3rd intercostal spaces (lower down on the left is the heart — only on the right).
2. **From the sides:** along the midaxillary lines (hands on the back of the neck).
3. **From behind:** suprascapular (finger horizontal), interscapular (finger **parallel to the spine**), subscapular (finger horizontal along the intercostal spaces).

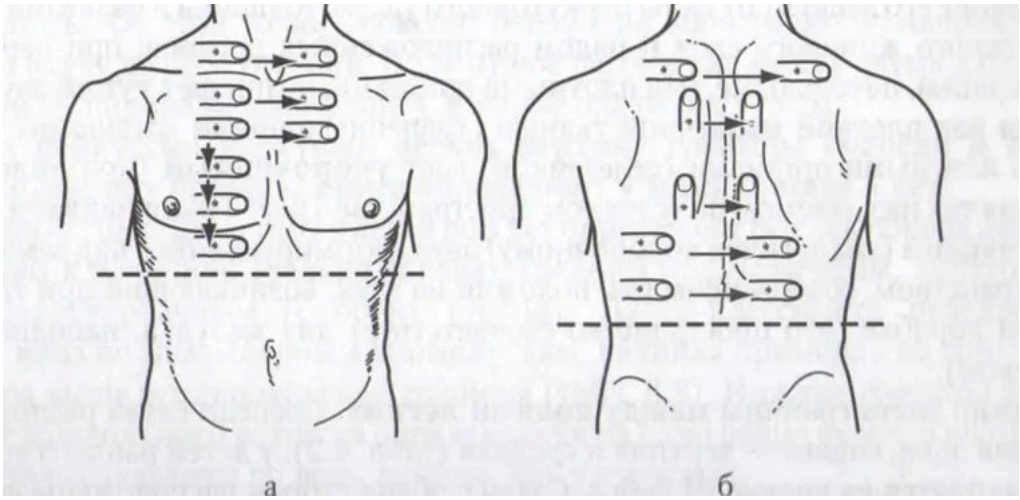


Figure 6.9. *Points of comparative percussion and position of the pleximeter finger: a — anterior, b — posterior surface (Kildiyarova, Figure 8.6)*

Table 6.7. *Types of percussion sound*

Percussion sound	Where normally	In pathology
Clear pulmonary sound	Over healthy lung (slightly tympanic in an infant)	—
Shortened (dulled)	Over the muscles (trapezius, scapular)	Reduced air in the lung: focal pneumonia, atelectasis, pulmonary edema, pneumofibrosis
Dull (“thigh sound”)	Over the heart, liver	Lobar pneumonia, exudative pleurisy, hydrothorax, tumor
Tympanic	Traube's space (gastric air bubble, lower left)	An air-filled cavity: cavern, emptied abscess, large bronchiectasis, pneumothorax
Box-like	—	Increased air in the lung: emphysema, asthma attack, obstructive bronchitis
Metallic / “cracked pot”	—	A large smooth-walled cavity / a superficial cavity connected to a narrow bronchus

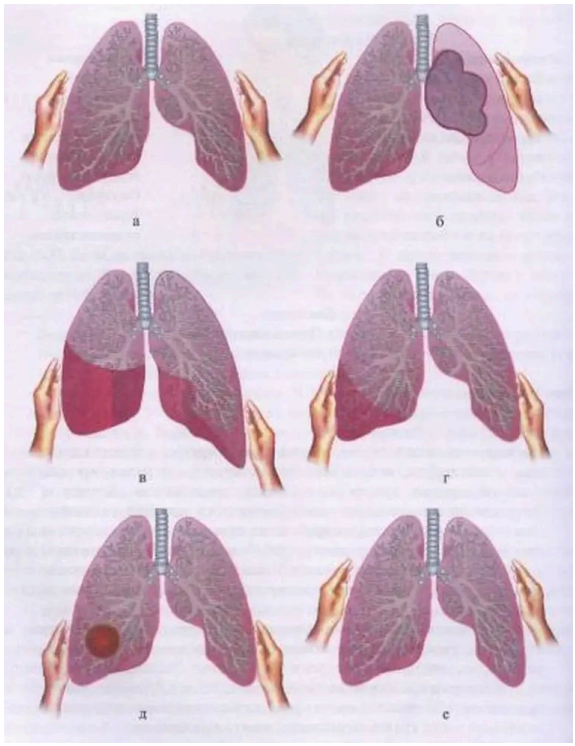


Figure 6.10. *Conditions changing percussion and vocal fremitus: a — normal; b — pneumothorax; c — obstructive atelectasis; d — lobar consolidation; e — focal inflammatory consolidation; f — bronchial narrowing (Arzikulov)*

Topographic percussion

Purpose — to determine the lung borders. **Rules:** the pleximeter finger is placed **parallel** to the border sought; it is moved from the clear sound toward the dull sound; the border is marked along **the edge of the finger facing the clear sound**; quiet percussion is used.

Height of the lung apices (in children over 5–7 years; in younger children the apices lie behind the clavicle and cannot be determined):

- **From the front** — the finger is placed in the supraclavicular fossa parallel to the clavicle and moved upward and medially. Normal: **2–4 cm above the clavicle**.
- **From behind** — from the middle of the spine of the scapula toward the spinous process of **the 7th cervical vertebra**. Normal: **at the level of the C_{VII} process**.
- **Krönig's field** — the strip of clear sound over the apex. The finger is placed at the middle of the edge of the trapezius muscle and percussion is performed alternately toward the neck and toward the shoulder until dullness appears. Normal: **3–5 cm** (4–6 cm according to Arzikulov).

Lower lung borders (at school age; at preschool age approximately **one rib higher**). Up to 3 years they are determined along the midclavicular (right), midaxillary and scapular lines.

Table 6.8. *Lower lung borders (school-age child)*

Line	Right lung	Left lung
Parasternal	5th intercostal space	—
Midclavicular	Rib VI	Forms the cardiac notch
Anterior axillary	Rib VII	Rib VII
Midaxillary	Rib VIII	Rib VIII (IX)
Posterior axillary	Rib IX	Rib IX

Line	Right lung	Left lung
Scapular	Rib X	Rib X
Paravertebral	Spinous process of ThXI	Spinous process of ThXI

MEMORY AID
Lower lung border: **“6-7-8-9-10-11”** — it increases by one rib at each line from the midclavicular line toward the paravertebral line.

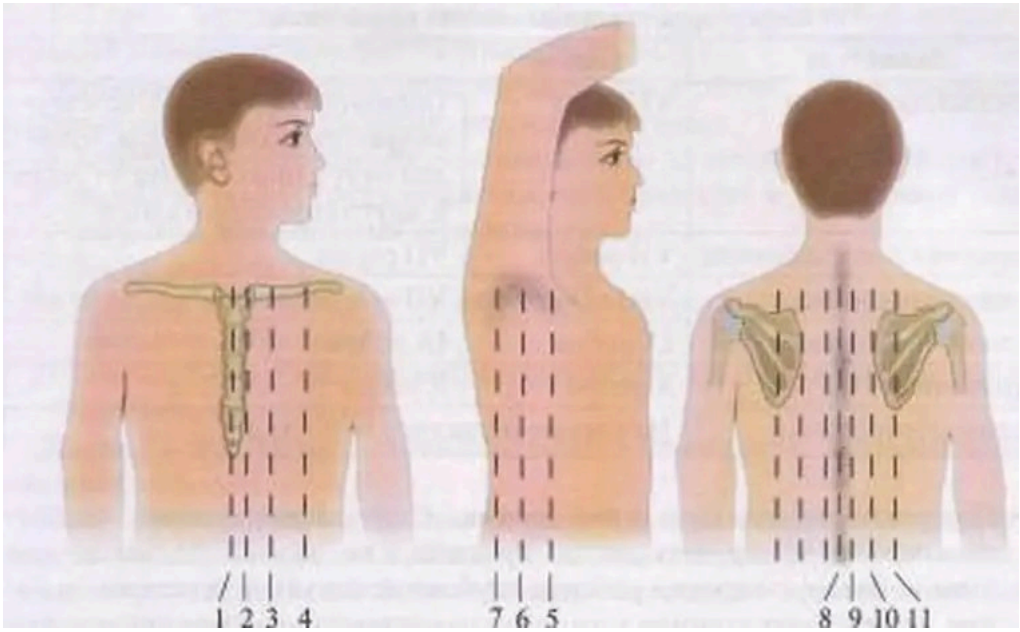


Figure 6.11. *Topographic lines of the chest: 1 — anterior midline, 2 — parasternal, 3 — sternal, 4 — midclavicular; 5–7 — anterior, middle and posterior axillary; 8 — scapular, 9 — paravertebral, 10–11 — posterior midlines (Arzikulov)*

Changes in the borders: downward displacement — emphysema, asthma attack, low position of the diaphragm (visceroptosis). Upward displacement — shrinkage of the lung (pneumofibrosis, atelectasis), fluid or air in the pleura, raised abdominal pressure (hepatosplenomegaly, meteorism, ascites).

Mobility (excursion) of the lower edge — in children who can hold their breath: the border is determined on the middle or posterior axillary line during quiet breathing → the child takes a deep breath and holds it, the border is found → the child breathes out deeply and holds it, it is found again. The distance between the two marks is the excursion, normally **2–6 cm** (depending on age). It is reduced in emphysema, asthma, pneumofibrosis, inflammation, pleural adhesions; it is absent with fluid or air in the pleura, complete fusion of the pleural layers, diaphragmatic paralysis.

Signs for detecting enlargement of the intrathoracic lymph nodes

Table 6.9. *Signs of enlargement of the intrathoracic lymph nodes*

Sign	Technique	If positive
Korányi	Direct percussion along the spinous processes from below (from ThVII–VIII) upward. Normally dullness is at the level of ThII in a small child and ThIV in an older child	Dullness lower down — the bifurcation lymph nodes are enlarged

Sign	Technique	If positive
D'Espine	The stethoscope is moved upward along the same line while the child whispers “chashka chaya” (or “thirty-three”). Normally the sound becomes louder above ThII (small child) / ThIV (older child)	Louder sound lower down — bifurcation nodes
Filosofov's bowl	Percussion in the 1st–2nd intercostal spaces from the right and left toward the sternum (finger parallel to the sternum)	Dullness beyond the edge of the sternum — anterior mediastinal nodes (or thymus)
Arkavin	Percussion along the anterior axillary lines from below toward the axilla	Shortening — bronchopulmonary (hilar) nodes
Franco	Inspection: dilated skin capillaries in the CVII area	The tracheobronchial nodes are enlarged
Smith	A venous hum over the manubrium of the sternum when the infant's head is thrown back	The paratracheal nodes are enlarged

6.7. Auscultation of the lungs

Main idea: auscultation is performed **in the order** of comparative percussion, at symmetric points, listening to **2–3 breaths** at each point; it is essential to know the type of breathing typical of the child's age.

Technique and rules:

- Complete quiet; the clothes are removed (rustling of clothes interferes).
- The child stands or sits; a seriously ill patient lies. In a crying child — during the deep inspiration between cries (crying also helps).
- In small children the diaphragm of the stethoscope **should not be larger than 2 cm in diameter**; it is placed in the intercostal space.
- First listen during breathing through the nose, then during deep breathing through the mouth (adventitious sounds become louder). To induce a deep breath in an infant: close the nose for a few seconds, tickle, press lightly on the trachea.
- Listening points: the apices, the anterior surface, the axillae, suprascapular, interscapular, subscapular and paravertebral areas.

Areas requiring special attention in children: the axillae (in segmental pneumonia bronchial breathing appears early); paravertebral and suprascapular (segments 2, 6, 10 — foci of pneumonia); interscapular (lung root); subscapular (crepitation is heard early); the cardiac area (left lingula).

Basic (physiological) breath sounds

Table 6.10. *Age-specific basic breath sounds*

Type of breathing	Age	Character	Cause
Weakened vesicular	Newborn — 3–6 months	Heard faintly	Few alveoli, shallow breathing, weak muscles
Puerile (increased vesicular)	6 months — 5–7 years	Inspiration and expiration almost equal in duration, loud	The glottis is close, the bronchi narrow, the chest wall thin and elastic, much interstitial tissue

Type of breathing	Age	Character	Cause
Vesicular	Over 7 years	The whole of inspiration and the first 1/3 of expiration; a soft “f” sound	Vibration of the alveolar walls
Laryngotracheal (physiological bronchial)	At any age — over the larynx, the trachea, the manubrium of the sternum, and between the shoulder blades down to ThIII–IV	Expiration longer and louder than inspiration, an “h” sound	Air passing through the glottis

MEMORY AID
“Weakened → puerile → vesicular” = 0–6 months → 6 months–7 years → 7+ years. Hearing puerile breathing in an older child is not a pathology in itself; it is increased vesicular breathing (dyspnea, fever).

Pathological basic breath sounds

Table 6.11. *Pathological types of breathing*

Type	Character	When
Weakened vesicular	Heard faintly	Bronchial obstruction (foreign body, mucus), fluid or air in the pleura, atelectasis, emphysema, pain (dry pleurisy), obesity, the beginning and end of pneumonia
Harsh	Loud, coarse; inspiration and expiration are heard in full	Narrowing of the small bronchi: bronchitis, bronchopneumonia
Bronchial (pathological)	An “h” sound, prolonged expiration — pathological if heard over the lung	Consolidation of lung tissue + preserved bronchial patency: lobar and polysegmental pneumonia, tuberculous infiltrate
Amphoric	A loud, musical type of bronchial breathing (like blowing over an empty jug)	A large smooth-walled cavity connected to a bronchus: cavern, large bronchiectasis

Bronchophony — whispered words (“sixty-six”, “ninety-nine”) heard over the chest wall. Normally the words are not clearly distinguished. **Increased** — over pneumonia and atelectasis (if the bronchus is open), a cavern (metallic); **weakened** — fluid or air in the pleura, obesity. The logic is the same as for vocal fremitus.

Adventitious breath sounds

Main idea: rales — from the bronchi (air + fluid or mucus), **crepitation** — from the alveoli (only at the end of inspiration), **pleural friction rub** — from the pleura (in both phases, superficial).

Dry rales (rhonchi). They arise from vibration of strands of thick mucus on the bronchial wall in the air flow, or from narrowing of the bronchial lumen due to edema and spasm. They are heard in both phases and change after coughing.

Table 6.12. *Dry rales*

Type	Where it arises	Sound	When
Humming (low-pitched, bass)	Large bronchi	Low, bass	Bronchitis, tracheobronchitis
Buzzing	Medium bronchi	Medium pitch	Bronchitis

Type	Where it arises	Sound	When
Whistling (high-pitched)	Small bronchi, bronchioles	Squeaking, mainly on expiration	Bronchospasm: obstructive bronchitis, bronchial asthma

Moist rales. There is liquid secretion (sputum, blood, transudate) in the bronchi, and **bubbles burst** as air passes. Their caliber depends on the caliber of the bronchus.

Table 6.13. *Moist rales*

Type	Where	Feature	When
Fine-bubbling	Small bronchi, bronchioles	Mainly on inspiration; if consonant — the surrounding tissue is inflamed	Bronchopneumonia, bronchiolitis, pulmonary edema
Medium-bubbling	Medium bronchi	In both phases	Bronchitis, bronchiectasis
Coarse-bubbling	Large bronchi, trachea, cavern	May disappear after coughing	Bronchitis, cavern, bronchiectasis, preagonal state

Consonant (loud, distinct) moist rales — the lung around the bronchus is consolidated (pneumonia); **non-consonant** — only the bronchus is affected (bronchitis). In infants rales are hard to hear because chest excursion is small; they are better detected during the deep inspiration while crying.

Crepitation. When there is a small amount of exudate in the alveoli, the stuck-together alveolar walls separate **at the very peak of inspiration**. The sound is a fine uniform crackling, like rubbing hair between the fingers near the ear. **Only at the end of inspiration**, unchanged after coughing. An important sign of pneumonia:

- **crepitatio indux** — days 1–3 of lobar pneumonia (period of exudate accumulation);
- on days 4–6 it is not heard (the alveoli are filled with exudate — bronchial breathing);
- **crepitatio redux** — days 7–10 (resorption of the exudate — a sign of recovery).

Pleural friction rub — friction of the parietal and visceral layers when they become rough and covered with fibrin: **dry pleurisy**, the beginning and end of exudative pleurisy, tuberculosis, tumor, severe dehydration. The sound is like crunching snow or rubbing paper or leather. Most often in **the lower part of the midaxillary line**, together with pain.

Table 6.14. *Distinguishing fine moist rales, crepitation and pleural friction rub*

Sign	Fine moist rales	Crepitus	Pleural friction rub
Phase of breathing	Both phases (more on inspiration)	Only at the end of inspiration	Both phases
After coughing	Change	Unchanged	Unchanged
During “false breathing” with the mouth and nose closed	Absent	Absent	Heard
On pressure with the stethoscope	Unchanged	Unchanged	Increased
Depth of hearing	Deep	Deep	Superficial, under the ear
Pain, felt by hand	Absent	Absent	May be present

CAUTION

In infants with a blocked nose or accumulated mucus, **transmitted (nasopharyngeal) rales** are heard over all areas of the lungs and may lead to a wrong diagnosis. Clear the nose, make the child cough and listen again: transmitted rales disappear or change, while focal rales in the lung remain in place.

6.8. Clinical section

Bringing the results of physical examination together into one rule

Table 6.15. *Physical signs in the main conditions*

State	Percussion	Vocal fremitus and bronchophony	Respiration
Consolidation (lobar pneumonia)	Dull	Increased	Bronchial, crepitation
Fluid in the pleura (exudative pleurisy)	Dull	Weakened / absent	Weakened / absent
Pneumothorax	Tympanic	Weakened / absent	Weakened / absent
Emphysema, asthma	Box-like	Weakened	Weakened, dry rales
Cavern	Tympanic	Increased	Amphoric, coarse-bubbling rales
Obstructive atelectasis	Dull	Weakened	Weakened / absent

Foreign body aspiration

CLINICAL SIGNIFICANCE

Choking and a severe coughing fit that begin **suddenly during play or eating** in a previously healthy child are considered foreign body aspiration until proven otherwise. It most often enters the **right** bronchus. Signs: unilateral weakening of breathing, localized whistling rales, later atelectasis or recurrent pneumonia. Diagnosis and treatment — **bronchoscopy**. Emergency care for a child who cannot cough or speak: under 1 year — 5 back blows between the shoulder blades + 5 chest thrusts; in older children — **the Heimlich maneuver**.

Clinical detection of pneumonia

In small children the first signs of pneumonia are often not physical but **general and respiratory**: fever lasting more than 3 days, **tachypnea**, flaring of the nasal alae, perioral cyanosis, chest indrawing, “grunting” expiration, intoxication. Physical changes (shortened sound, fine-bubbling consonant rales, crepitation) may not be detected if the focus is small — confirmation is by **X-ray**.

Practical skills

Skill 1. Counting the respiratory rate and assessing dyspnea

1. Observe the child unnoticed at rest (if possible, asleep).
2. Place a hand on the child's chest or abdomen (or observe visually).
3. Count the respiratory movements for 1 minute (up to 3 months — always a full minute).
4. Assess the rhythm, depth and type of breathing.
5. Look for signs of dyspnea: flaring of the nasal alae, accessory muscles, retraction of the intercostal spaces and the suprasternal area, stridor, prolonged expiration.

6. Compare the result with the age norm, calculate the ratio to the pulse, and record it.

Skill 2. Determining vocal fremitus

1. The child stands or sits; the doctor's hands are warm.
2. Place the palms on symmetric areas: supra- and infraclavicular, lateral areas, suprascapular, interscapular, subscapular.
3. Ask the child to say words with the letter “R” (in an infant — during crying).
4. Compare the two sides at each pair of points; if necessary, swap the hands and repeat.
5. Conclusion: equal / increased / weakened (in which area).

Skill 3. Comparative percussion

1. The child stands straight and symmetrically; for anterior percussion the doctor stands on the right.
2. Place the pleximeter finger firmly in the intercostal space parallel to the ribs.
3. With a quiet stroke, percuss the supraclavicular, clavicular and infraclavicular areas, then the lateral areas.
4. From behind: the suprascapular, interscapular (finger parallel to the spine) and subscapular areas.
5. Compare the sound at each pair of symmetric points.
6. Conclusion: clear pulmonary sound over the whole lung / site and type of change.

Skill 4. Topographic percussion

1. Height of the apices (from 5–7 years): from the front — relative to the clavicle, from behind — relative to CVII; Krönig's field.
2. Lower borders: on the right starting from the parasternal line, finger parallel to the ribs, from top to bottom, from the clear sound toward the dull sound; along each topographic line.
3. On the left — starting from the anterior axillary line.
4. Mark the border along the edge of the finger facing the clear sound; count the ribs or vertebrae.
5. Determine the excursion of the lower edge on the midaxillary line.
6. Compare the result with the age norm.

Skill 5. Auscultation of the lungs

1. A quiet, warm room; the child undressed; a warm stethoscope.
2. Listen symmetrically at the points of comparative percussion, 2–3 breaths at each point.
3. Determine the basic breath sound and compare it with the age norm.
4. Detect adventitious sounds: type, phase of breathing, location, change after coughing.
5. Test bronchophony.
6. Conclusion: “Puerile (vesicular) breathing in the lungs, no rales” or the changes found.

Review questions

1. What are the anatomical and physiological features of children's upper airways and their clinical significance?
2. Which diseases are children predisposed to by the age-related features of the larynx, trachea and bronchi?

3. List the lung segments. In which segments is pneumonia most common in children and why?
4. Explain the mechanism of the first breath. What is the importance of surfactant?
5. Name the types of cough and their diagnostic significance.
6. What are the age norms for the type of breathing and the respiratory rate? How does the respiration/pulse ratio change?
7. What is the difference between inspiratory, expiratory and mixed dyspnea, and what are their causes?
8. Describe the pathological types of breathing (Cheyne–Stokes, Biot, Grocco, Kussmaul).
9. What is determined by palpation of the chest? When is vocal fremitus increased and when weakened?
10. What are the rules for performing percussion in children? Name the types of percussion sound.
11. What are the normal values for the lung apices, Krönig's field and the lower borders?
12. Name the signs for detecting enlargement of the intrathoracic lymph nodes.
13. What are the age-specific breath sounds? What is puerile breathing and why does it occur?
14. What are the mechanism of formation and the types of dry and moist rales?
15. How can crepitation and pleural friction rub be distinguished from fine moist rales?

Test questions

1. At what age does the inferior nasal meatus form?
   A) At birth
   B) At 6 months
   C) At 1 year
   D) At 4 years
   E) At 10 years

2. The reason foreign bodies most often enter the right bronchus:
   A) The right bronchus is narrow and long
   B) The right bronchus is a continuation of the trachea, short and wide
   C) The left bronchus is short
   D) The right lung is small
   E) The right bronchus has no cartilage

3. The main function of surfactant:
   A) Reducing gas exchange
   B) Preventing the alveoli from collapsing
   C) Narrowing the bronchi
   D) Producing mucus
   E) Triggering the cough reflex

4. The main factor triggering the first breath:
   A) Cold
   B) Light
   C) Hypoxia, hypercapnia and acidosis
   D) Gravity
   E) Sound

5. The normal respiratory rate per minute in a child under 1 year:
   A) 16–18
   B) 20–25
   C) 30–35
   D) 40–60
   E) 60–80

6. Paroxysmal cough with reprises is typical of:
   A) Croup
   B) Whooping cough
   C) Pleurisy
   D) Bronchiectasis
   E) Pharyngitis

7. Inspiratory dyspnea is typical of:
   A) Bronchial asthma
   B) Obstructive bronchitis
   C) False croup
   D) Emphysema
   E) Bronchiolitis

8. In which condition is vocal fremitus increased?
   A) Exudative pleurisy
   B) Pneumothorax
   C) Lobar pneumonia
   D) Obstructive atelectasis
   E) Emphysema

9. A box-like percussion sound is typical of:
   A) Lobar pneumonia
   B) Exudative pleurisy
   C) An attack of bronchial asthma
   D) Atelectasis
   E) Pulmonary edema

10. The lower border of the right lung along the midaxillary line in a school-age child:
    A) Rib VI
    B) Rib VII
    C) Rib VIII
    D) Rib IX
    E) Rib X

11. At what age is puerile breathing heard?
    A) From birth to 3 months
    B) From 6 months to 5–7 years
    C) At 7–14 years
    D) Only in adults
    E) Only in preterm babies

12. In which condition is pathological bronchial breathing heard?
    A) Exudative pleurisy
    B) Emphysema
    C) Lobar pneumonia
    D) Bronchial asthma
    E) Pneumothorax

13. A sound heard only at the end of inspiration that does not change after coughing:
    A) Dry rale
    B) Fine-bubbling moist rale
    C) Crepitus
    D) Pleural friction rub
    E) Coarse-bubbling rale

14. Which process is indicated by whistling dry rales heard on expiration?
    A) Exudate in the alveoli
    B) Spasm of the small bronchi (obstruction)
    C) Inflammation of the pleura
    D) Cavern
    E) Pulmonary edema

15. Enlargement of which lymph nodes does Korányi's sign detect?
    A) Neck
    B) Axilla
    C) Bifurcation
    D) Paratracheal
    E) Inguinal

16. When does the inferior nasal meatus form in children?
    A) At 1 year
    B) Present at birth
    C) At 12 years
    D) At 8 years
    E) At 4 years

17. From what age do the frontal and sphenoid paranasal sinuses develop?
    A) From 12 years
    B) From 3 years
    C) From 1 year
    D) From 6–7 years
    E) At birth

18. The feature of the nasolacrimal duct in children and its consequence:
    A) Closed — no conjunctivitis
    B) Absent — dry eye
    C) Open only in girls
    D) Short, wide, valve undeveloped — infection passes from the nose to the eye
    E) Long and narrow — dacryocystitis

19. The feature of the larynx in children and its complication at 2–3 years:
    A) Funnel-shaped, narrow, subglottic area prone to edema — laryngitis leads to croup
    B) Narrow only in boys
    C) Hard cartilages — foreign body
    D) Located low — aphonia
    E) Wide — no stenosis

20. On which side are foreign bodies and pneumonia more common in children, and why?
    A) On the left, the left bronchus is wide
    B) On the left, the bronchus is narrow
    C) Equally on both sides
    D) Only in the upper lobe
    E) On the right, the right main bronchus is a continuation of the trachea, short and wide

21. How many segments does the right lung have?
    A) 11
    B) 10
    C) 9
    D) 8
    E) 12

22. The most important factor triggering the first breath:
    A) Exposure to cold
    B) Skin receptors
    C) Hypoxia, hypercapnia and acidosis after clamping of the cord (chemical factor)
    D) Gravity
    E) Light

23. Which cells synthesize surfactant?
    A) Type II alveolocytes
    B) Type I alveolocytes
    C) Macrophages
    D) Endothelium
    E) Ciliated epithelium

24. Which syndrome develops in preterm babies with surfactant deficiency?
    A) Bronchiolitis
    B) Whooping cough
    C) Croup
    D) Respiratory distress syndrome (hyaline membrane disease)
    E) Aspiration pneumonia

25. In newborns, pathological apnea is considered to be:
    A) Rapid breathing in sleep
    B) A deep breath after crying
    C) A pause longer than 20 seconds or accompanied by bradycardia and cyanosis
    D) A pause in breathing of up to 8–10 seconds
    E) A 3-second pause

26. A paroxysmal cough with reprises ending in vomiting is typical of which disease?
    A) Pleurisy
    B) Laryngitis
    C) Whooping cough
    D) Pharyngitis
    E) Tracheitis

27. What kind of cough occurs with enlargement of the bifurcation lymph nodes (tuberculous bronchoadenitis, lymphogranulomatosis)?
    A) Moist cough
    B) Short painful cough
    C) Morning cough with copious sputum
    D) Barking cough
    E) Bitonal cough

28. Rusty sputum is typical of which disease?
    A) Bronchiectasis
    B) Abscess
    C) Lobar pneumonia
    D) Asthma
    E) Whooping cough

29. Why is the inspection of the pharynx and tonsils performed last?
    A) Only in adolescents
    B) This is not a rule
    C) Because the room is dark
    D) So as not to upset the child
    E) Because the doctor is tired

30. The normal respiratory rate (RR) in a child under 1 year:
    A) 25–30
    B) 30–35
    C) 20–25
    D) 40–60
    E) 16–18

31. The normal respiration : pulse ratio in the newborn:
    A) 1 : 4
    B) 1 : 5
    C) 1 : 4,5
    D) 1 : 1
    E) 1 : 2,5–3

32. How is the RR counted in a child under 3 months?
    A) For 1 full minute
    B) It is not counted
    C) Only while awake, for 10 seconds
    D) 30 seconds × 2
    E) 15 seconds × 4

33. A sign typical of inspiratory dyspnea:
    A) Retraction of the intercostal spaces and the suprasternal fossae, stridor
    B) Participation of the abdominal muscles only
    C) Whistling breathing, wheezing heard at a distance
    D) The respiration/pulse ratio does not change
    E) Prolonged expiration

34. What kind of dyspnea is seen in an attack of bronchial asthma?
    A) There is no dyspnea
    B) Inspiratory
    C) Not mixed
    D) Expiratory
    E) Only Cheyne–Stokes

35. Deep, noisy, infrequent breathing called the breathing of a “hunted animal”:
    A) Bitonal
    B) Kussmaul
    C) Grocco
    D) Biot
    E) Cheyne–Stokes

36. Breathing whose depth gradually increases, then decreases and ends with a pause:
    A) Cheyne–Stokes
    B) Stridor
    C) Kussmaul
    D) Grocco
    E) Biot

37. What does the child say when vocal fremitus is tested?
    A) Only counts
    B) Words with the letter “S”
    C) The vowel “A”
    D) Says nothing
    E) Words with the letter “R” (“tractor”)

38. Vocal fremitus is weakened or absent:
    A) Over a cavern
    B) In pneumofibrosis
    C) In exudative pleurisy and pneumothorax
    D) In lobar pneumonia
    E) In an abscess

39. With what force is lung percussion performed in a child, and why?
    A) Strong, because the chest is thick
    B) Light (quiet), because the chest is thin and elastic
    C) Strong, because the lungs are small
    D) Only direct
    E) Percussion is not performed

40. Pleural pain increases with breathing, and when bending toward the affected side it:
    A) Increased
    B) Decreases
    C) Always disappears
    D) Moves elsewhere
    E) Unchanged

Clinical cases

Case 1

A 6-year-old child has complained of fever (39.2 °C), cough and right-sided chest pain for 4 days. On examination: the right cheek is flushed, herpes on the lips, flaring of the nasal alae, RR 36/min. The right half of the chest lags behind during breathing. Below the right shoulder blade vocal fremitus is increased; percussion gives a dull sound; on auscultation bronchial breathing and crepitation are heard, and bronchophony is increased.

Questions: 1) Which syndrome do you identify? 2) Why is vocal fremitus increased? 3) Presumptive diagnosis? 4) Is the respiratory rate appropriate for the age?

Answer: 1) The syndrome of consolidation (infiltration) of lung tissue: dull sound, increased vocal fremitus and bronchophony, bronchial breathing, crepitation. 2) Consolidated lung tissue (with an open bronchus) conducts sound vibrations better than air-filled tissue. 3) Right-sided lower lobe lobar pneumonia — flushing of one cheek, herpes and pleural pain are characteristic. 4) No, the norm at 6 years is 20–25/min — tachypnea (a sign of respiratory failure).

Case 2

An 8-month-old child has had a runny nose and cough for 2 days, and last night breathing became difficult. On examination: restless, cyanosis of the nasolabial triangle, RR 64/min. Expiration is prolonged and noisy, the intercostal spaces and the area below the sternum are retracted, and the chest is distended. Percussion gives a box-like sound. On auscultation expiration is prolonged, with diffuse whistling dry rales and many fine-bubbling moist rales on both sides.

Questions: 1) What type of dyspnea is it? 2) Which syndrome? 3) Presumptive diagnosis? 4) What does the box-like sound on percussion indicate?

Answer: 1) Expiratory dyspnea (mixed with expiratory predominance). 2) Bronchial obstruction syndrome and respiratory failure. 3) Acute bronchiolitis (8 months old, after ARVI, diffuse whistling and fine-bubbling rales, marked tachypnea). 4) Hyperinflation of the lungs (acute emphysema) — because of the obstruction, air does not manage to leave on expiration.

Case 3

A 10-year-old child has complained of fever, dry cough and left-sided pain that increases with breathing for 1 week; in the last 2 days the pain has decreased, but the dyspnea has increased. On examination: prefers to lie on the left side, the left half of the chest is enlarged and lags behind during breathing, the intercostal spaces are smoothed. Low on the left, vocal fremitus cannot be detected; percussion gives a dull sound with an oblique upper border; breath sounds are not heard.

Questions: 1) Which syndrome? 2) What is the oblique border found on percussion called? 3) How do you explain the decrease in pain and increase in dyspnea? 4) Which investigations are needed to confirm the diagnosis?

Answer: 1) The syndrome of fluid accumulation in the pleural cavity (left-sided exudative pleurisy). 2) The Damoiseau (Sokolov–Damoiseau) line. 3) It began as dry (fibrinous) pleurisy — the rubbing of the pleural layers caused pain; when fluid accumulated, the layers separated and the pain decreased, but the lung was compressed and respiratory failure increased. 4) Chest X-ray (opacity with a horizontal or oblique level, mediastinal shift toward the healthy side), pleural ultrasound, pleural puncture and fluid analysis (exudate or transudate).

Quick review

Table 6.16. *Key facts on Topic 6*

Question	Answer
Inferior nasal meatus	Forms at 4 years
Glottis	CIV in children, CVII in adults
Length of the trachea	≈4 cm in the newborn
Foreign body	Into the right bronchus (short, wide, continuation of the trachea)
Segments	Right 10, left 9; pneumonia most often — segments 6, 2, 10
Surfactant enzymes	Week 22–24 and week 35
Physiological apnea	Up to 3 months, up to 8–10 s
RR	Newborn 40–60; 1 year 30–35; 5 years 20–25; 10 years 18–20
Respiration : pulse	Newborn 1:2.5–3 → older child 1:4
Tachypnea (IMCI)	<2 months ≥60; 2–12 months ≥50; 1–5 years ≥40
Type of breathing	Newborn — diaphragmatic; 8–14 years: boys — abdominal, girls — thoracic
Inspiratory / expiratory dyspnea	Croup, foreign body / asthma, bronchiolitis
Vocal fremitus ↑ / ↓	Consolidation / fluid, air, obstructed bronchus
Apices	2–4 cm above the clavicle; CVII behind
Krönig's field	3–5 cm
Lower borders	“6-7-8-9-10-11”
Excursion of the lower edge	2–6 cm
Breath sound	0–6 months weakened; 6 months–7 years puerile; 7+ vesicular
Bronchial breathing over the lung	Consolidation (lobar pneumonia)
Crepitus	Only at the end of inspiration; indux days 1–3, redux days 7–10
Pleural friction rub	Both phases, superficial, louder on pressure with the stethoscope
Korányi, d'Espine	Normal: ThII (small) / ThIV (older)
General features of the child's airways	Narrow, soft, rich in blood and lymph vessels, few elastic fibers → predisposition to edema, obstruction, atelectasis, pneumonia
Lung segments	Right — 10 (3+2+5), left — 9; in children pneumonia most often in segments 6, 2, 10; segments 4–5 — “middle lobe syndrome”
Mechanism of the first breath	Clamping of the cord → pO_2↓ (80→15), pCO_2↑ (40→70), pH↓ → chemoreceptors → respiratory center → contraction of the diaphragm; surfactant keeps the lungs open

Question	Answer
Surfactant	90% lipid (phosphatidylcholine) + 10% protein; type II alveolocytes; prevents alveolar collapse; deficiency — RDS
Types of cough	Barking — croup; dry — pharyngitis/foreign body; moist — bronchitis; short painful — pleurisy/lobar pneumonia; with reprises — whooping cough; bitonal — bifurcation lymph nodes
Description of sputum	Yellowish — purulent; green — stagnant pus; rusty — lobar pneumonia; fetid — bronchiectasis/abscess; glassy, viscous — asthma
Type of breathing by age	Newborn — diaphragmatic; 1–2 years — mixed; 8–14 years — abdominal type in boys, thoracic in girls
Normal RR	Newborn 40–60; <1 year 30–35; 1–3 years 25–30; 4–6 years 20–25; 7–15 years 18–20; >15 years 16–18
Respiration : pulse ratio	Newborn 1:2.5–3; 1 year 1:3–3.5; 4–6 years 1:4; adults 1:4–5; in disease it narrows to 1:2–3
Types of dyspnea	Inspiratory — narrowing of the upper airways (croup, foreign body), stridor; expiratory — bronchial asthma, obstructive bronchitis; mixed — pneumonia, pleurisy, heart failure
Pathological types of breathing	Cheyne–Stokes (wave with pause), Biot (sudden pause), Grocco (wave without pause), Kussmaul (deep, noisy — acidosis)
Changes in vocal fremitus	Increased — consolidation (lobar pneumonia), cavern; weakened — fluid/air in the pleura, atelectasis, emphysema
Rules of percussion	Quiet, light stroke; finger-to-finger (indirect), direct/Obraztsov in infants; comparison at opposite symmetric sites; from clear to dull

TOPIC 7

Instrumental and laboratory methods of examining the respiratory system. Semiotics and syndromes of respiratory involvement

Learning objectives

The student should know:

- functional (spirometry, spirography, pneumotachometry, peak flowmetry, breath-holding tests, blood gases, pulse oximetry), radiological (X-ray, tomography, CT, bronchography, fluorography), endoscopic and laboratory methods of examining the respiratory system, their indications and age limitations;
- the semiotics of the main respiratory symptoms;
- the syndromes of upper airway involvement, croup, bronchial obstruction, consolidation (infiltration) of lung tissue, atelectasis, fluid and air in the pleural cavity, a cavity in the lung, respiratory disorders in newborns, pulmonary edema;
- the types and degrees of severity of respiratory failure.

The student should be able to:

- assess spirogram and peak flowmetry results and determine the type of ventilatory disorder;
- “read” a chest X-ray in order;
- assess sputum and pleural fluid analyses;
- identify the leading syndrome on the basis of the physical examination findings;
- determine the degree of respiratory failure.

7.1. Functional methods of examination

Main idea: functional tests require the child's **active participation** — spirography and pneumotachometry are usually performed **after 5 years**; in small children respiratory failure is assessed by clinical signs, pulse oximetry and blood gases.

Breath-holding tests

Performed sitting, with the nose pinched:

- **Stange test** — breath-holding time after a deep inspiration;
- **Genche test** — breath-holding time after expiration (in pediatrics, after 3 deep breaths).

Table 7.1. *Age norms for breath-holding tests (Kildiyarova)*

Age	Genche, s (boys / girls)	Stange, s (boys / girls)
5	12 / 12	24 / 22
7	14 / 15	36 / 30
10	22 / 21	50 / 50
12	22 / 22	60 / 48
15	27 / 26	68 / 60

Shortening of the test times indicates reduced reserve of the cardiorespiratory system (respiratory or heart failure, anemia, detraining).

Spirometry

Spirometry — a method of determining **the vital capacity (VC)** (with a Hutchinson spirometer or a portable spirometer). The child, standing or sitting, takes a maximal breath, pinches the nose and slowly breathes all the air out into the spirometer through the mouthpiece. The measurement is repeated 2–3 times, the highest result is taken and compared with **the predicted value (for age, height and sex)**. A decrease of **more than 15–20%** from the predicted value is pathological.

Spirography

Spirography — graphic recording of respiratory movements, reflecting changes in lung volumes. It is performed in the morning on an empty stomach after 10–15 minutes of rest; the child breathes through a mouthpiece (with a nose clip) or a mask. First 3–5 minutes of quiet breathing and the VC are recorded, then the forced tests.

Main parameters:

- **TV** — tidal volume (the volume of air in one respiratory cycle);
- **MV** — minute volume of breathing = RR × TV; it increases in lung and heart diseases and thyrotoxicosis, and decreases when the respiratory center is depressed;
- **VC** — vital capacity = TV + inspiratory reserve volume + expiratory reserve volume;
- **FVC** — forced vital capacity (a sharp expiration after a maximal deep breath);
- **FEV_1** — forced expiratory volume in the 1st second; **Tiffeneau index** = FEV_1 / VC × 100%, normally **≥70%**;
- **MVV** — maximal voluntary ventilation of the lungs (the maximal volume of air that can be ventilated in 1 minute; calculated from a 10–15-second recording);
- **Breathing reserve** = MVV – MV; it decreases in respiratory failure.

A deviation of **±20%** from the predicted values is considered acceptable.

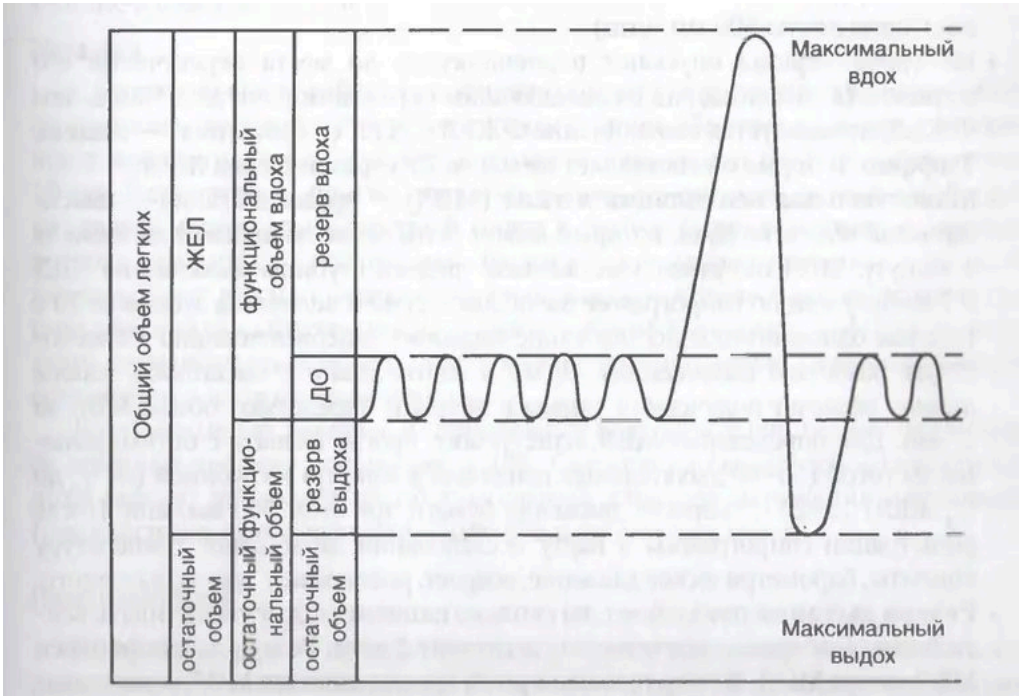


Figure 7.1. *Spirogram diagram: total lung capacity, VC, functional residual capacity, inspiratory and expiratory reserve volumes, tidal volume, residual volume, maximal inspiration and expiration (Kildiyarova, Figure 8.7; figure labels in Russian)*

Table 7.2. *Types of ventilatory disorders*

Type	VC	FEV_1, MVV	Tiffeneau index	Examples
Restrictive	Markedly reduced	Less reduced	Normal	Pneumonia, pleurisy, pneumothorax, pneumofibrosis, chest deformity
Obstructive	Slightly changed	Markedly reduced	Reduced (<70%)	Bronchial asthma, obstructive bronchitis, cystic fibrosis
Mixed	All reduced approximately equally			Severe pneumonia with obstruction, chronic lung diseases

MEMORY AID

Restrictive — "the volume is small" (VC ↓, Tiffeneau normal); **obstructive — "the speed is small"** (FEV_1 and Tiffeneau ↓). An increase in FEV_1 of **more than 12%** in a bronchodilator (salbutamol) test is a sign of reversible obstruction (bronchial asthma).

Pneumotachometry

Pneumotachometry — assessment of bronchial patency: **the peak air flow rate** on inspiration and expiration is measured (from 5–6 years). First a tube 20 mm in diameter is used, then one of 10 mm; the test is repeated 3–4 times and the highest result is taken. A reduced expiratory rate indicates bronchial obstruction (bronchospasm, mucosal edema, secretions).

Peak flowmetry

Peak flowmetry — determining **the peak expiratory flow (PEF)** with a portable device, the peak flow meter. It is simple, cheap and done at home — the main method of **monitoring bronchial asthma** and its control. The child stands, takes a deep breath, seals the lips tightly around the mouthpiece and breathes out **sharply and quickly**; this is repeated 3 times and the highest result is recorded. Measurements are taken **2 times a day** — in the morning (before taking medication) and in the evening.

- **Daily variability >20%** or a **"morning dip"** — bronchial **hyperreactivity** (uncontrolled asthma).
- The result is compared with the norm for the child's height: 107 cm — ≈140, 130 cm — ≈250, 152 cm — ≈360, 175 cm — ≈480 l/min.
- **The "traffic light" system: green zone** — ≥80% of the personal best (good control); **yellow** — 50–80% (step up treatment); **red** — <50% (emergency care).

Blood gases and pulse oximetry

- **Blood gases** (capillary or arterial blood, micro-Astrup apparatus): **PaO_2** (normal in children 80–100 mm Hg), **$PaCO_2$** (35–45 mm), **pH** (7.35–7.45), buffer bases (BE).
- **Pulse oximetry (oxyhemometry)** — continuous, painless transcutaneous measurement of blood oxygen saturation (**SpO_2**); **94–98%** in a healthy child. **SpO_2 <92–94%** is an indication for oxygen therapy (pneumonia, bronchiolitis, asthma attack).

7.2. Radiological methods of examination

- **Chest X-ray** — the most widely used instrumental method; in frontal (anterior) and lateral projections. It detects pneumonia, pleurisy, pneumothorax, atelectasis, foreign bodies (radiopaque), thymomegaly and developmental defects.

- **Fluoroscopy** — real-time observation of diaphragmatic movement and mediastinal shift (high radiation dose — rarely used).
- **Tomography** (layer-by-layer) — the lung root, lymph nodes, cavities.
- **Computed tomography** (spiral, high-resolution CT) — the most accurate method: bronchiectasis, interstitial changes, small foci, tumors; with contrast when necessary.
- **Bronchography** — X-ray after injecting a contrast medium into the bronchi: **bronchiectasis**, bronchial defects, tumors (now mostly replaced by CT).
- **Angiopulmonography**, **radioisotope scanning (scintigraphy)** — pulmonary circulation and ventilation.
- **X-ray of the paranasal sinuses** — sinusitis.
- **Ultrasound of the pleural cavity and lungs** — detecting fluid, marking the puncture site.
- **Fluorography** — mass preventive screening (tuberculosis, tumor); because of the high radiation dose only **after 15 years**.

Order of reading an X-ray:

1. Determine the projection (frontal, lateral).
2. Correct positioning of the child — the clavicles at the same level, the spinous processes in the middle of the sternum.
3. Exposure of the X-ray — if the disc between the 3rd and 4th thoracic vertebrae is visible — normal.
4. The chest skeleton (ribs, spine, clavicles).
5. **Transparency** of the lung fields on both sides.
6. Position of the diaphragmatic domes, the costophrenic sinuses, the gastric air bubble.
7. **Lung roots** — normally from the anterior end of rib II to rib IV, structured.
8. **Lung markings** — normally visible in the inner 2/3 of the lung field; when increased they reach the periphery.
9. Pleura.
10. Mediastinum (cardiac shadow, thymus, position of the trachea).
11. Conclusion. **Reduced transparency (opacity)** — pneumonia, atelectasis, pleurisy, tumor; **increased transparency (lucency)** — emphysema, pneumothorax, bronchial obstruction.

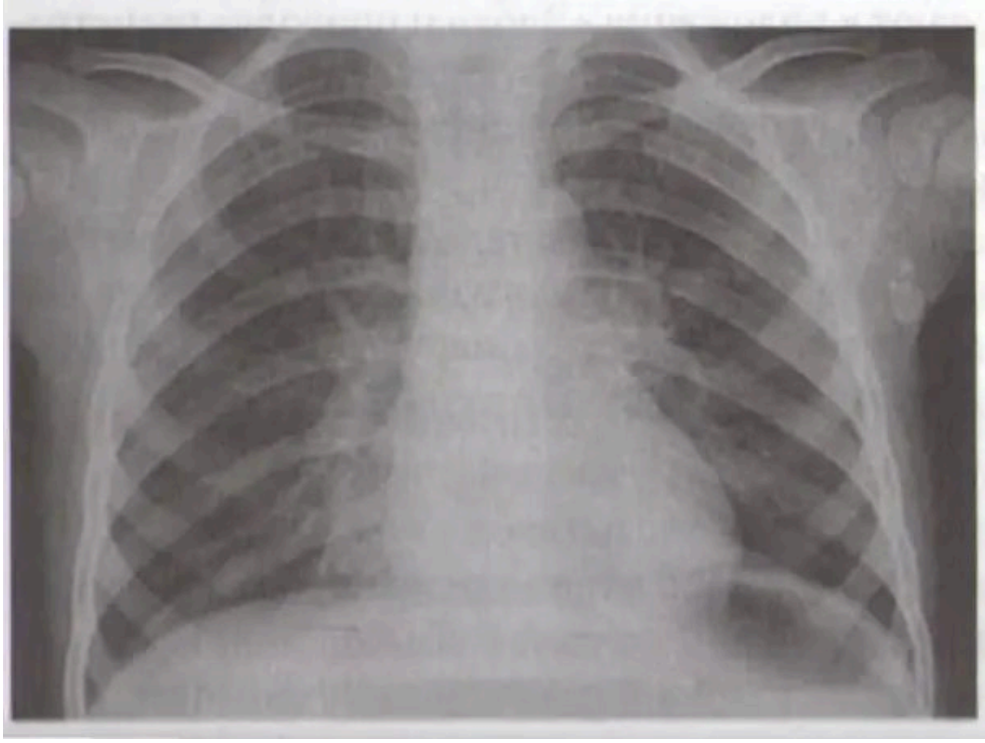


Figure 7.2. *Normal chest X-ray of a child (Kildiyarova, Figure 8.8)*

7.3. Endoscopic methods

- **Rhinoscopy** (anterior and posterior) and **pharyngoscopy** — the nose, adenoids, pharynx.
- **Laryngoscopy** — at an early age direct laryngoscopy (often under anesthesia), in older children indirect (with a mirror); croup, laryngeal anomalies, papillomatosis.
- **Bronchoscopy** — **up to 3–4 years** with a rigid bronchoscope **under anesthesia**, in older children **fiberoptic bronchoscopy** under local anesthesia. Its possibilities: viewing the inner surface of the trachea and bronchi; removing **a foreign body**; suctioning pus, blood and mucus (sanitation); detecting tumors and defects; **biopsy**; obtaining bronchial washings (bronchoalveolar lavage); direct administration of drugs.
- **Thoracoscopy** — viewing the pleural cavity (pleural biopsy, treatment of empyema).

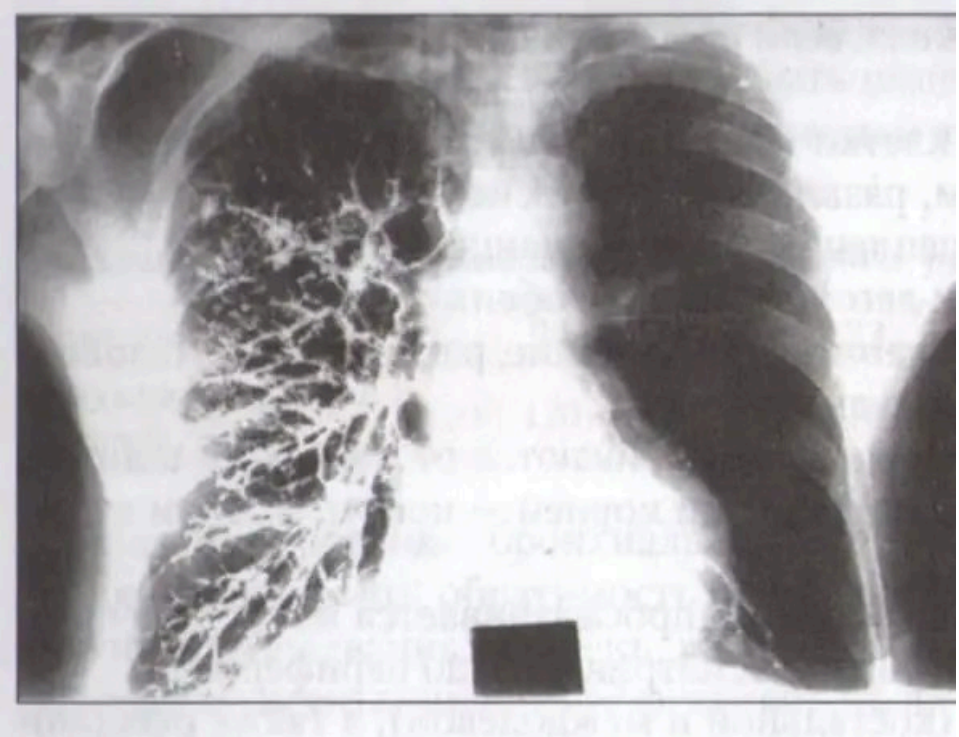


Figure 7.3. *Bronchogram in chronic bronchitis: bronchiectasis (Kildiyarova, Figure 8.9)*

7.4. Laboratory methods

- **Complete blood count:** in bacterial pneumonia **neutrophilic leukocytosis**, a left shift, raised ESR; in viral infection — leukopenia or normal, lymphocytosis; in bronchial asthma and helminthiasis — **eosinophilia**; in whooping cough — marked **leukocytosis and lymphocytosis**.
- **Inflammatory markers:** C-reactive protein, procalcitonin (bacterial infection).
- **Microbiological examination:** culture and bacterioscopy of nasopharyngeal swabs (including for diphtheria — Loeffler's bacillus), sputum, bronchial secretions and pleural fluid; antibiotic sensitivity; PCR and rapid tests for viruses; for tuberculosis — mycobacteria in sputum or gastric washings.
- **Sweat test** — sweat electrolytes (chloride >60 mmol/l) — diagnosis of **cystic fibrosis**.
- **Allergological:** skin tests (scarification, prick test, intradermal), provocation tests; total and specific **IgE** in the blood.
- **Diagnosis of tuberculosis:** Mantoux test, Diaskintest, QuantiFERON test.

Table 7.3. *Diagnostic significance of sputum analysis*

Finding	Significance
Many leukocytes (purulent sputum)	Bacterial inflammation, bronchiectasis, abscess
Elastic fibers	Breakdown of lung tissue (abscess, tuberculous cavern, destructive pneumonia)
Curschmann spirals, Charcot–Leyden crystals, eosinophils	Bronchial asthma
Erythrocytes (hemoptysis)	Destructive process, tuberculosis, tumor, hemosiderosis
Mycobacteria (Ziehl–Neelsen staining)	Tuberculosis
Rusty sputum	Lobar pneumonia

Analysis of **pleural fluid** — distinguishing exudate from transudate (see the table in section 7.7).

7.5. Semiotics of the main respiratory symptoms

Main idea: cough indicates **the level of the lesion** (pharynx, larynx, trachea, bronchus, pleura), while its timing and sputum indicate **the nature of the process**.

Runny nose: the discharge is serous, mucous, mucopurulent or blood-stained. Nasal congestion without discharge — viral and allergic disease, adenoids. **Unilateral purulent, fetid discharge** in a small child — **a foreign body in the nose**. **Nosebleeds** — hemorrhagic diatheses, leukemia, hypoplastic anemia, arterial hypertension, nasal polyps.

Table 7.4. *Character of the cough by level of the lesion*

Site of the lesion	Character of the cough
Pharynx (pharyngitis)	Dry, exhausting, with a scratchy throat
Larynx (laryngitis, croup)	Dry, harsh, barking
Trachea	Harsh, “as if coughing into a barrel”, retrosternal pain
Bronchi, lungs	Dry at first, then moist, productive
Pleura	Painful, short, suppressed
Bifurcation lymph nodes, mediastinal tumor	Bitonal
Whooping cough	Paroxysmal, with reprises, at night, with facial flushing and vomiting
Bronchial asthma	Paroxysmal, dry; viscous sputum after the attack

Timing of the cough:

- **morning** — a cavity in the lung (abscess, bronchiectasis, cavern), chronic bronchitis: sputum accumulated overnight moves when the position changes;
- **evening** — acute bronchitis, pneumonia;
- **night** — tuberculosis, lymphogranulomatosis, tumors (the lymph nodes irritate the bifurcation, and vagal tone rises at night), bronchial asthma (attacks at 4–6 a.m.), adenoiditis (mucus runs down the posterior pharyngeal wall);
- **related to eating** — tracheoesophageal fistula, gastroesophageal reflux, esophageal diverticulum.

If an infrequent cough becomes **frequent and moist** during an illness, this indicates that bronchiolitis or pneumonia has developed. A cough lasting **more than 3–4 weeks** requires exclusion of a chronic disease (asthma, tuberculosis, cystic fibrosis, foreign body, reflux).

Sputum: “a mouthful” of purulent sputum — rupture of an abscess or a suppurating cyst into a bronchus, bronchiectasis. **Hemoptysis** — destructive pneumonia, tuberculous cavern, tumor, hemorrhagic diathesis, pulmonary hemosiderosis.

Voice: a weak squealing cry or no cry in a newborn — asphyxia, CNS damage; hoarse voice and aphonia — acute laryngitis, diphtheria; nasal speech — adenoids; a low “croaky” voice — hypothyroidism.

Dyspnea: inspiratory — narrowing of the upper airways (croup, foreign body, retropharyngeal abscess); expiratory — bronchial obstruction (asthma, obstructive bronchitis, bronchiolitis); mixed —

pneumonia, pleurisy, respiratory failure, heart failure.

Chest pain: increased by deep breathing and coughing — pleurisy, lobar pneumonia (pleuropneumonia); localized, painful on pressure — myositis, intercostal neuralgia, rib injury; in diaphragmatic pleurisy and lower lobe pneumonia the pain **radiates to the abdomen** (risk of mistaking it for acute appendicitis).

Cyanosis: of the nasolabial triangle and around the eyes — mild to moderate respiratory failure; diffuse, gray-marbled — severe respiratory failure. **"Drumstick fingers"** and **"watch-glass" nails** — chronic hypoxia (bronchiectasis, cystic fibrosis).

7.6. Syndromes of consolidation (infiltration) of lung tissue and atelectasis

Main idea: consolidation (open bronchus) → vocal fremitus and bronchophony are **increased**, bronchial breathing; **atelectasis** (closed bronchus) → they are **weakened**, breathing is faint. On percussion the sound is shortened in both.

Infiltration syndrome

It occurs in: **pneumonia**, pulmonary infarction, pulmonary hemorrhage, tuberculous infiltrate, eosinophilic infiltrate.

Table 7.5. *Signs of the infiltration syndrome*

Method	Signs
General	Fever, intoxication (disturbed appetite and sleep, weakness); cough dry at first, then moist; "grunting" expiration; chest pain
Inspection	Pale skin, marbled in a severe condition, cold limbs; perioral cyanosis; flaring of the nasal alae; tachypnea; the affected side lags behind during breathing
Palpation	Vocal fremitus increased over the focus
Percussion	Shortened or dull sound over the focus
Auscultation	Bronchial breathing (large dense focus) or weakened breathing; localized fine-bubbling consonant moist rales; crepitation; increased bronchophony
Special signs	Filatov (skin fold thickening); in right-sided pneumonia Dombrovskaya's sign — the heart sounds are heard as clearly on the right as on the left
Blood	Neutrophilic leukocytosis, left shift, raised ESR
Spirogram	VC, TV, MVV reduced — restrictive type
X-ray	Reduced transparency of the lung field, a focus of opacity (lobar, segmental or focal shadow)

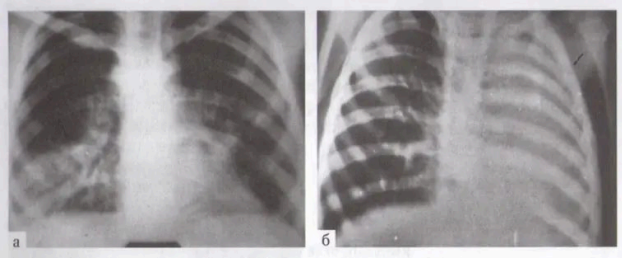


Figure 7.4. *X-ray in pneumonia: a — lower lobe pneumonia; b — polysegmental pneumonia (Kildiyarova, Figure 8.12)*

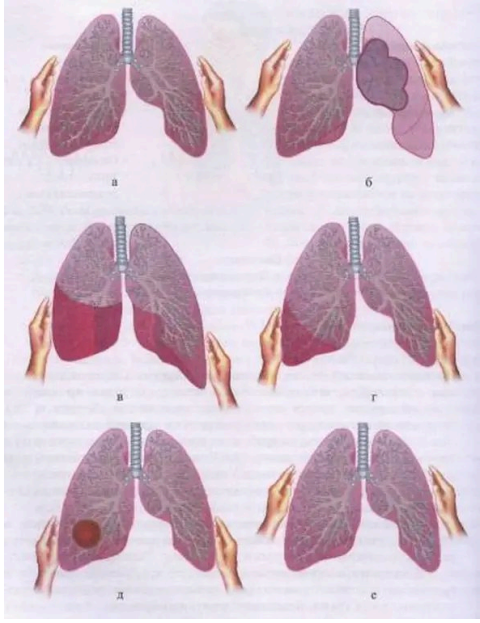


Figure 7.5. *Diagram of the syndromes: a — normal; b — pneumothorax; c — obstructive atelectasis; d — lobar consolidation (lobar pneumonia); e — focal consolidation (bronchopneumonia); f — bronchial narrowing (Arzikulov)*

CLINICAL SIGNIFICANCE

In small children the most reliable clinical signs of pneumonia are **tachypnea**, **lower chest wall indrawing**, fever lasting more than 3 days and **a fall in SpO_2**. Physical signs may not be pronounced with small foci — if in doubt, an X-ray is taken.

Atelectasis syndrome

Atelectasis — loss of air and collapse of part of the lung: obstruction of the airway (thick mucus, foreign body, tumor, an enlarged lymph node compressing from outside) or compression of the lung from outside (fluid, air). Predisposition: narrow bronchi, surfactant deficiency, weak cough reflex; newborns also have **primary (physiological) atelectasis**.

Table 7.6. *Signs of obstructive atelectasis*

Method	Signs
Inspection	With a large atelectasis the condition worsens; tachypnea, tachycardia, cyanosis; the chest on the affected side is flattened (sunken) and lags behind during breathing
Palpation	Vocal fremitus weakened
Percussion	Shortened sound over the atelectasis
Auscultation	Breathing weakened or not heard, no rales
X-ray	Triad: a homogeneous triangular shadow (base at the periphery, apex at the lung root) + mediastinal shift toward the affected side + elevation of the diaphragm on the affected side

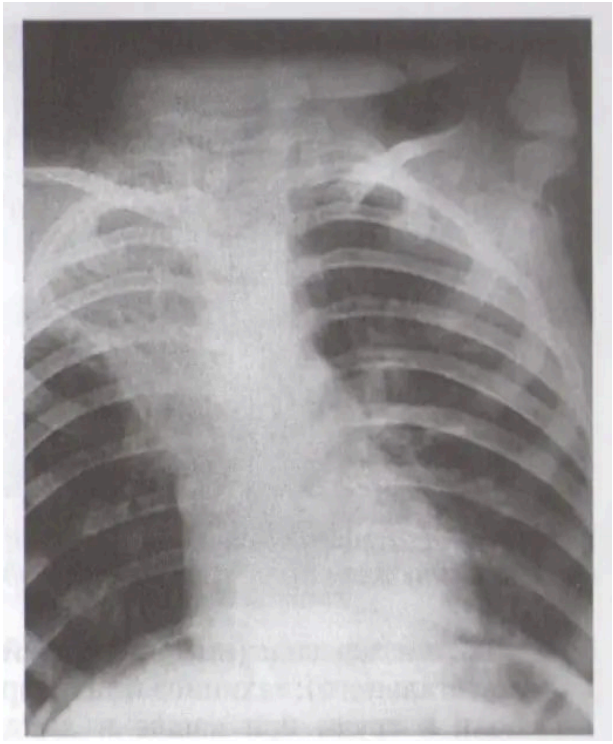


Figure 7.6. *Chest X-ray in atelectasis (Kildiyarova, Figure 8.13)*

CAUTION

The key to distinguishing consolidation from atelectasis is **the state of the bronchus**: an open bronchus (pneumonia) — sound is conducted well (vocal fremitus ↑, bronchial breathing); a closed bronchus (atelectasis) — sound is not conducted (vocal fremitus ↓, breath sounds not heard). On X-ray, in pneumonia the mediastinum stays in place; in atelectasis it is pulled **toward the affected side**.

7.7. Pleural syndromes

Main idea: in the pleura **both fluid and air** block sound — vocal fremitus, bronchophony and breathing are **weakened or absent**; the difference is in percussion: fluid — **dull**, air — **tympanic**. In both, the mediastinum shifts **toward the healthy side**.

Dry (fibrinous) pleurisy

The pleural layers are covered with fibrin. Signs: **stabbing pain** that increases with breathing and coughing; the child lies on the affected side (to restrict movement); the affected side lags behind during breathing; percussion is almost unchanged (mobility of the lower edge is restricted); breathing is weakened because of the pain; **pleural friction rub** (mainly low on the midaxillary line). In diaphragmatic pleurisy — pain radiating to the abdomen and shoulder, hiccups, **Mussy's sign**.

Syndrome of fluid accumulation in the pleural cavity

It occurs in: exudative pleurisy (parapneumonic, tuberculous), **empyema** (purulent), hydrothorax (heart failure, nephrotic syndrome), hemothorax, chylothorax.

Table 7.7. *Signs of the syndrome of fluid in the pleural cavity*

Method	Signs
General	Respiratory failure increases, dyspnea, cyanosis, tachypnea, tachycardia; the pain decreases as fluid accumulates

Method	Signs
Inspection	Forced position — lies on the affected side; the chest is asymmetric, the affected side enlarged and lagging behind during breathing; the intercostal spaces are smoothed or bulging
Palpation	The affected side is rigid; vocal fremitus is not transmitted; the skin fold is thickened (Filatov)
Percussion	Dull (“thigh”) sound; in older children the Damoiseau (Sokolov–Damoiseau) line — the oblique upper border of the fluid (highest on the posterior axillary line); Garland's triangle — on the same side next to the spine, a shortened-tympanic sound over the compressed lung; the Rauchfuss–Grocco triangle — on the opposite side next to the spine, dullness due to the displaced mediastinum
Auscultation	Breathing sharply weakened or not heard; bronchophony weakened; bronchial breathing may be heard at the border of the compressed lung above the fluid
X-ray	Homogeneous opacity with an oblique upper border (or horizontal — if there is fluid and air), the costophrenic sinus obliterated, the mediastinum shifted toward the healthy side; in the lateral decubitus position the shape of the shadow changes

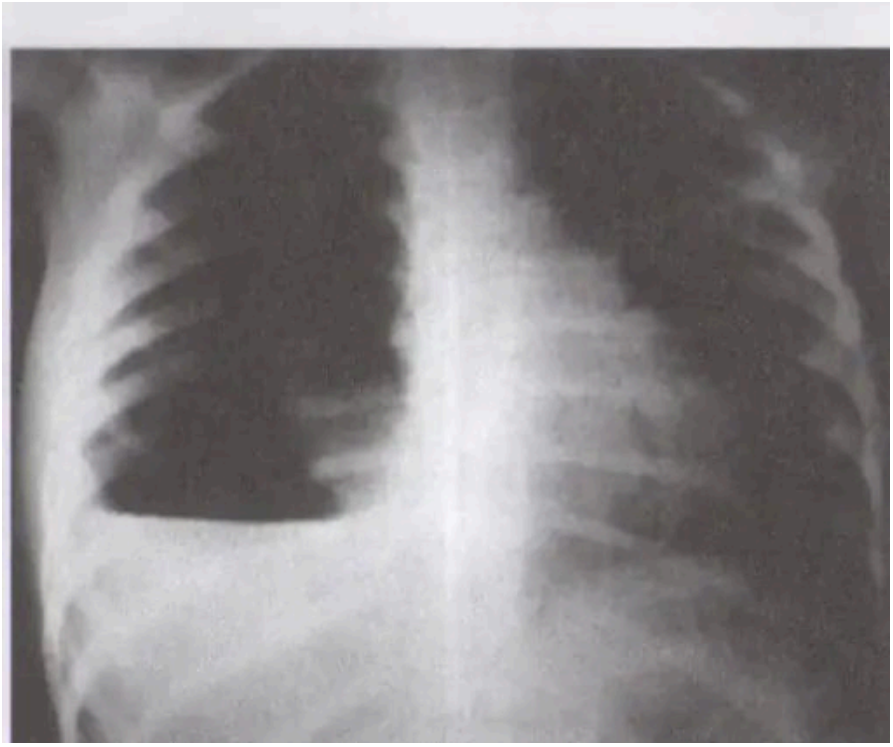


Figure 7.7. *Right-sided pleurisy: fluid level in the upright position (Kildiyarova, Figure 8.14)*

Pleural puncture — for diagnosis and treatment, at the site of the dullest sound (under ultrasound guidance), most often along the posterior axillary or scapular line **in the 7th–8th intercostal space, along the upper edge of the lower rib** (so as not to injure the neurovascular bundle under the rib).

Table 7.8. *Differences between exudate and transudate*

Sign	Exudate (inflammatory)	Transudate (non-inflammatory)
Appearance	Turbid, serofibrinous, purulent, hemorrhagic	Clear, pale yellow
Specific gravity	>1015	<1015
Protein	>30 g/l	<30 g/l
Rivalta test (turbidity in acetic acid solution)	Positive	Negative

Sign	Exudate (inflammatory)	Transudate (non-inflammatory)
Leukocytes	Many: neutrophils in acute infection, lymphocytes in tuberculosis	Few, mononuclear cells
Cause	Pleurisy (bacterial, tuberculous), tumor	Heart failure, nephrotic syndrome, hypoproteinemia

Accumulation of air in the pleural cavity — pneumothorax

Causes: lung destruction (**staphylococcal pneumonia**), rupture of a bulla or cyst, injury, complications of mechanical ventilation, a complication of pleural puncture. **Valvular (tension) pneumothorax** is the most dangerous: with each breath the air increases and compresses the lung and mediastinum; urgent pleural puncture is required. **Pyopneumothorax** — pus and air in the pleura.

Table 7.9. *Signs of pneumothorax*

Method	Signs
General	Sudden chest pain, sharp dyspnea, cyanosis, tachycardia, restlessness
Inspection	The affected side is enlarged, does not take part in breathing, the intercostal spaces are smoothed
Palpation	Vocal fremitus absent
Percussion	Tympanic sound; the cardiac borders are shifted toward the healthy side
Auscultation	Breathing not heard or sharply weakened; bronchophony weakened
X-ray	On the affected side a lucent field without lung markings, the lung collapsed toward the root, the mediastinum shifted toward the healthy side

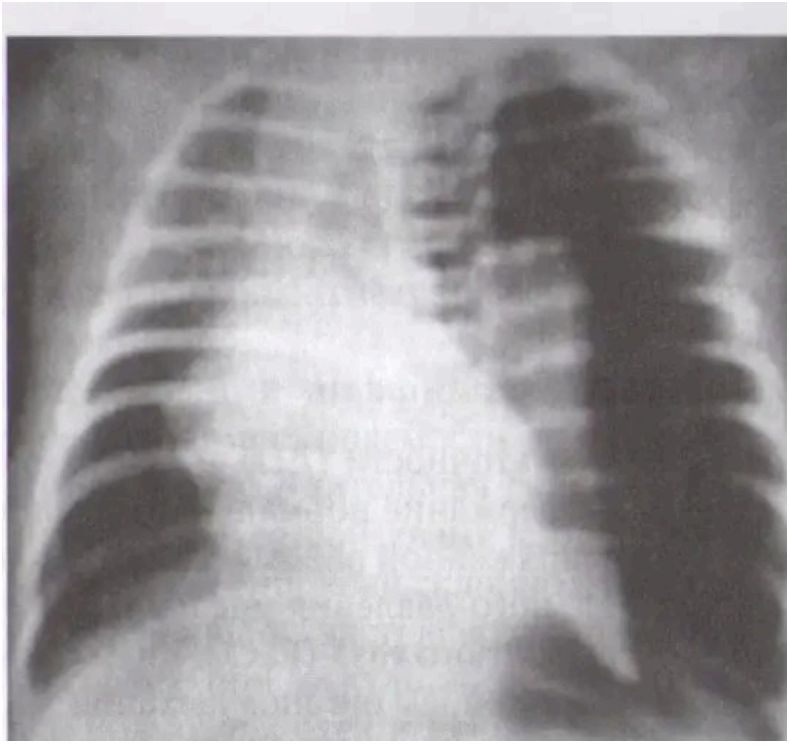


Figure 7.8. *Left-sided pneumothorax (Kildiyarova, Figure 8.15)*

CAUTION

In Kildiyarova's textbook, the signs of pleurisy (dull sound, fluid level) were mistakenly repeated in the pneumothorax section. The correct signs of pneumothorax are given here — **tympanitis** and, on X-ray, **absence of the lung markings** (consistent with Arzikulov's textbook and general clinical data).

7.8. Airway and other lung syndromes

Main idea: narrowing of the upper airways → inspiratory dyspnea (croup); **narrowing of the small bronchi → expiratory** dyspnea, box-like sound, whistling rales (obstruction).

Syndrome of upper airway involvement

In rhinitis, nasopharyngitis, pharyngitis, laryngitis and tracheitis: runny nose (serous-watery at first, then thick), pain on swallowing, a scratchy throat; **dry cough**; hyperemia and granularity of the posterior pharyngeal wall; fever, intoxication. Palpation is unchanged; percussion gives a clear pulmonary sound; breathing is **harsh**, and the rales are **transmitted** in character — **they disappear after coughing**.

Adenoid facies (enlargement of the nasopharyngeal tonsil): a puffy face, the mouth always open, no nasal breathing, snoring and apnea in sleep, malocclusion, a high “gothic” palate, nasal speech, hearing loss (otitis), frequent colds, reduced attention and memory from prolonged hypoxia.

Croup syndrome

Croup — narrowing (stenosis) of the laryngeal lumen. At an early age (6 months – 3 years) it is most often **false croup — stenosing laryngotracheitis** in influenza, parainfluenza and adenovirus infection (edema of the loose tissue below the vocal cords). **The croup triad: barking cough + hoarse voice + inspiratory dyspnea (stridor)**; in addition: restlessness, tachycardia, participation of the accessory muscles, retraction of the suprasternal area and intercostal spaces.

Table 7.10. *Differences between false and true croup*

Sign	False croup (ARVI)	True croup (laryngeal diphtheria)
Onset	Acute, sudden (often at night)	Gradual, in stages
Sound	Dull	Aphonia
Intoxication	Mild	Severe
In the larynx	Subglottic edema	Edema + fibrinous membrane on the vocal cords
Loeffler's bacillus in a nasopharyngeal swab	Not found	Found
Course	Improves quickly	The stenosis progresses, risk of asphyxia; complications (myocarditis, nephritis)

Table 7.11. *Degrees of laryngeal stenosis*

Degree	Signs
I — compensated	Dyspnea and stridor only when the child is upset, absent at rest
II — subcompensated	Stridor even at rest, retraction of the intercostal spaces and the suprasternal area, perioral cyanosis, tachycardia
III — decompensated	Marked dyspnea, generalized cyanosis, severe restlessness or lethargy, paradoxical pulse, weakened breathing
IV — asphyxia	Unconscious, shallow breathing, bradycardia, seizures, risk of cardiac arrest

CLINICAL SIGNIFICANCE

In false croup: calm the child (crying worsens the stenosis), fresh moist air, **inhaled budesonide** or **dexamethasone** (orally or IM), from grade II **adrenaline inhalation**; in grade II–III — urgent hospitalization. Forcible inspection of the pharynx with a spatula **must not** be done (reflex laryngospasm; in epiglottitis — risk of complete obstruction).

Bronchial obstruction syndrome

It occurs in: an attack of bronchial asthma, obstructive bronchitis, **bronchiolitis**, cystic fibrosis, foreign body, bronchopulmonary dysplasia. Mechanism: **spasm of the bronchial smooth muscle + mucosal edema + mucus hypersecretion**; at an early age **edema and hypersecretion** mainly predominate (which is why bronchodilators are less effective).

Table 7.12. *Signs of the bronchial obstruction syndrome*

Method	Signs
General	Dry, paroxysmal cough; restlessness, respiratory failure; tachypnea (up to 60–80/min in bronchiolitis)
Inspection	Expiratory dyspnea, whistling breathing heard at a distance; accessory muscles, flaring of the nasal alae; cyanosis of the nasolabial triangle; distended (barrel-shaped) chest; orthopnea in an asthma attack
Palpation	Vocal fremitus weakened, chest rigid
Percussion	Box-like sound; the lower lung border shifted downward, its excursion reduced; the borders of relative cardiac dullness reduced; the liver and spleen displaced downward (should not be mistaken for enlargement)
Auscultation	Breathing harsh or weakened, prolonged expiration; dry whistling rales on expiration; in bronchiolitis + many fine-bubbling moist rales and crepitation
Spirogram	FEV_1, Tiffeneau index, peak flow and small airway flows reduced — obstructive type; daily PEF variability >20%
X-ray	The lungs are hyperinflated (increased transparency), the ribs horizontal, the diaphragm low and flattened, increased bronchovascular markings
Blood (in asthma)	Eosinophilia, raised IgE

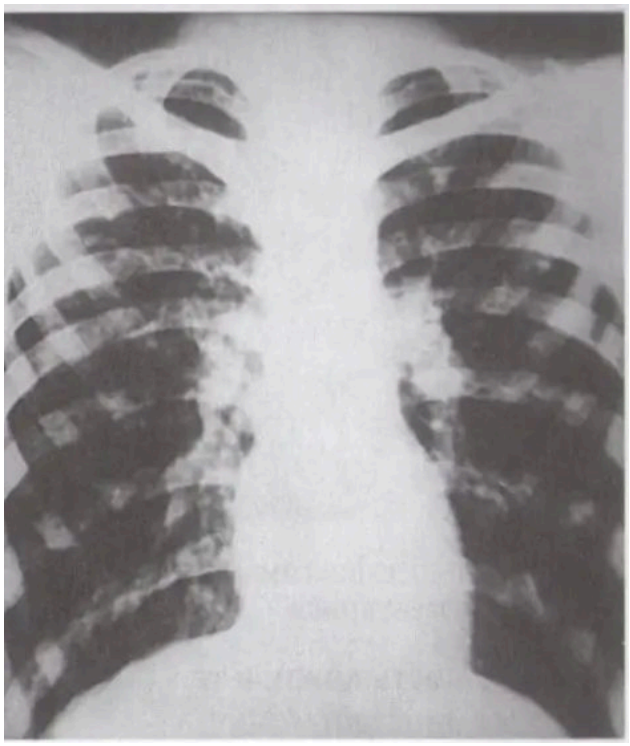


Figure 7.9. *X-ray in the obstructive syndrome: increased transparency of the lung fields, horizontal ribs, increased bronchovascular markings (Kildiyarova, Figure 8.11)*

Emphysema (persistent increase in lung airiness): barrel chest, widened intercostal spaces, reduced excursion; weakened vocal fremitus; **box-like** sound, lowered lower borders; **weakened** breathing; on X-ray increased lung transparency, a flattened and low diaphragm. Local (unilateral) emphysema — foreign body (valve mechanism), congenital lobar emphysema.

Syndrome of a cavity in the lung

An (emptied) abscess, tuberculous cavern, bronchiectasis, suppurated cyst, bullae in destructive (staphylococcal) pneumonia. Signs: restricted respiratory movement on the affected side; vocal fremitus **increased** (if there is a dense inflammatory ridge around the cavity); **tympanic** sound on percussion (shortened if the cavity is small or deep); **amphoric** breathing; **coarse-bubbling consonant** moist rales; increased bronchophony (metallic). X-ray — **a ring-shaped shadow**, possibly with air and a fluid level inside. When an abscess ruptures into a bronchus — “a mouthful” of purulent, fetid sputum.

Syndrome of respiratory disorders in newborns (respiratory distress)

Pneumopathies: atelectasis of the newborn, **hyaline membrane disease**, edematous-hemorrhagic syndrome, meconium aspiration, transient tachypnea of the newborn; mainly **in preterm babies** (surfactant deficiency). Signs: restlessness or lethargy, cyanosis, **frothy discharge on the lips**, retraction of the compliant parts of the chest (sternum, intercostal spaces), **“seesaw” breathing** (thoracoabdominal asynchrony), respiratory arrhythmia and **apnea**, **“grunting” expiration**, weakened breathing, scattered fine-bubbling rales and crepitation, tachycardia, muffled heart sounds, muscle hypotonia, seizures. Severity is assessed with the **Silverman** (preterm) and **Downes** (term) scales (see topic 2).

Pulmonary edema

Escape of fluid from the capillaries into the interstitium and the alveoli: **cardiac** (left ventricular failure, mitral stenosis) and **noncardiac** (drowning, toxic gases, severe pneumonia, sepsis, rapid and

large-volume infusion, nephrotic and nephritic syndromes) causes. Signs: restlessness, fear of death, orthopnea, **cough with frothy, sometimes pink sputum**, rapidly increasing dyspnea and cyanosis, cold sweat, a rapid weak pulse; on percussion a shortened sound in the lower posterior parts; on auscultation **many bilateral fine-bubbling non-consonant** rales, later medium- and coarse-bubbling ("boiling samovar" breathing).

7.9. Respiratory failure (RF) syndrome

Main idea: RF is a condition in which the normal gas composition of the blood is not maintained, or is maintained only **through strain** of the external respiratory apparatus. The cut-offs in children: **PaO_2 <80 mm Hg**, **$PaCO_2$ >40–45 mm Hg**. RF accompanies any respiratory disease and diseases of other organs (CNS, heart, muscles).

Types (by pathogenesis):

- **Ventilatory (hypercapnic)** — lung ventilation is insufficient: PaO_2 low, **$PaCO_2$ high**; the clinical picture is related to **hypercapnia**: headache, sweating, tachycardia, arterial hypertension, drowsiness, tremor, later cerebral edema and coma.
- **Parenchymal (diffusion, hypoxemic)** — the ventilation-perfusion ratio or alveolar-capillary diffusion is disturbed: PaO_2 low, $PaCO_2$ normal or low; the clinical picture is related to **hypoxemia**: cyanosis, restlessness → CNS depression, tachycardia, arrhythmia, muffled heart sounds, arterial hypotension.

Causes of acute RF:

1. **Obstructive (mechanical)** — upper airway obstruction (croup, epiglottitis, foreign body, tongue falling back in unconsciousness), bronchospasm (asthma), viscous secretions (cystic fibrosis), exudate in the bronchi (bronchitis, bronchiolitis, aspiration).
2. **Restrictive** — reduced compliance of the lungs and chest wall: pneumonia, pleurisy, pneumothorax, atelectasis, pulmonary edema, meteorism and a high diaphragm, rib injury.
3. **Neuromuscular** — depression of the respiratory center (CNS injury, neuroinfection, drug poisoning), paralysis of the respiratory muscles (poliomyelitis, myasthenia, botulism, muscular dystrophies).
4. **Diffusion disorder** — pulmonary edema, interstitial pneumonia, fibrosing alveolitis.

Table 7.13. *Degrees of severity of acute respiratory failure*

Sign	Grade I	Grade II	Grade III
Consciousness	Preserved, sometimes restlessness	Agitation, restlessness	Depression, stupor
RR	Up to 30% above normal	30–50% above normal	>50% above normal, or bradypnea, arrhythmic breathing
Accessory muscles	On exertion	Even at rest	Markedly pronounced or paradoxical breathing
Skin color	Pale; cyanosis of the nasolabial triangle on exertion	Pale, cyanosis of the nasolabial area and around the eyes at rest, acrocyanosis	Gray, marbled, generalized cyanosis; does not decrease with oxygen
Pulse	Normal or slightly rapid	Rapid	Markedly rapid or bradycardia, BP reduced
PaO_2, mm Hg	80–71	70–61	<60

Sign	Grade I	Grade II	Grade III
SpO_2	95–90%	90–85%	<85%
$PaCO_2$	Normal or slightly low	Normal or slightly raised	Raised (>50)
pH	7,39–7,36	7,35–7,30	<7,30

Grade IV — hypoxic coma: unconscious; the respiration-pulse ratio is severely disturbed; the skin is gray with purple-blue patches; pathological types of breathing; dilated pupils, seizures; PaO_2 <50 mm, pH <7.2.

REMEMBER

In Arzikulov's textbook the degrees of RF are also given by **blood oxygen saturation**: I — normal or down to 90%; II — 70–80%; III — <70%; IV (hypoxemic coma) — <50%. These are old cut-offs; in practice the SpO_2 values in the table above are used. In a healthy child arterial blood saturation is **94–98%**.

7.10. Clinical section

Distinguishing the syndromes in one table

Table 7.14. *Physical and X-ray signs of the main respiratory syndromes*

Syndrome	The chest	Vocal fremitus	Percussion	Respiration	Adventitious sound	Mediastinum (X-ray)
Infiltration (pneumonia)	The affected side lags behind	Increased	Shortened / dull	Bronchial or weakened	Fine-bubbling consonant, crepitation	Not displaced
Obstructive atelectasis	The affected side is sunken	Weakened	Shortened	Weakened / absent	Absent	Toward the affected side
Fluid (exudative pleurisy)	The affected side bulges	Absent	Dull, Damoiseau line	Not heard	Pleural friction rub at the beginning and end	Toward the healthy side
Air (pneumothorax)	The affected side bulges	Absent	Tympanic	Not heard	Absent	Toward the healthy side
Dry pleurisy	The affected side lags behind	Unchanged	Unchanged	Weakened	Pleural friction rub	Not displaced
Obstruction / emphysema	Barrel-shaped	Weakened	Box-like	Harsh, prolonged expiration	Dry whistling	Lungs transparent, diaphragm low
Cavity (cavern, abscess)	The affected side is restricted	Increased	Tympanic	Amphoric	Coarse-bubbling consonant	Ring-shaped shadow

MEMORY AID

Four rules: 1) **Dense tissue + open bronchus** → sound is conducted well → vocal fremitus and bronchophony ↑, bronchial breathing. 2) **Fluid or air in the pleura, or an obstructed bronchus** → sound is blocked → everything ↓. 3) The mediastinum **moves away from what "pushes"** (fluid, air → toward the healthy side) and **toward what "pulls"** (atelectasis → toward the affected side). 4) **Upper airways → inspiration is difficult** (croup), **small bronchi → expiration is difficult** (asthma, bronchiolitis).

A child in an attack of bronchial asthma

Bronchial asthma is chronic allergic inflammation and hyperreactivity of the bronchi with recurrent attacks of reversible obstruction. Signs of an attack: expiratory dyspnea, whistling breathing heard at a distance, orthopnea, dry cough, box-like sound, dry whistling rales. Diagnosis: history (allergy, heredity, recurrent attacks), reversible obstruction on spirometry (FEV_1 ↑ ≥12% after salbutamol), daily variability >20% on peak flowmetry, eosinophilia, raised IgE, Curschmann spirals and Charcot–Leyden crystals in the sputum.

CLINICAL SIGNIFICANCE

In an asthma attack, **a "silent chest"** (breath sounds and rales are no longer heard), inability to speak, cyanosis, lethargy and SpO_2 <90% are signs of **a life-threatening** attack: urgent oxygen, salbutamol inhalation (via a spacer or nebulizer), systemic glucocorticoids, admission to the intensive care unit.

Practical skills

Skill 1. Performing and assessing peak flowmetry

1. Set the peak flow meter pointer to zero; show the child the technique.
2. The child stands and holds the peak flow meter horizontally (the fingers should not touch the scale).
3. The child takes a deep breath, seals the lips tightly around the mouthpiece and breathes out **sharply and quickly**.
4. Record the result; repeat 3 times and note the highest.
5. Measure in the morning and evening; calculate the daily variability: (evening – morning) / mean × 100%.
6. Compare the result with the norm for height and the personal best (the "traffic light" zones).

Skill 2. Assessing a spirogram

1. Determine the child's data (age, sex, height) and find the predicted values.
2. Compare the actual values of VC, FEV_1, Tiffeneau index and MVV with the predicted values (in %).
3. Determine the type of ventilatory disorder: restrictive (VC ↓, Tiffeneau normal), obstructive (FEV_1, Tiffeneau ↓), mixed.
4. If there is a bronchodilator test — calculate the increase in FEV_1 (≥12% — reversible obstruction).
5. Write the conclusion.

Skill 3. "Reading" a chest X-ray

1. Check the patient's details, the date and the projection.

2. Assess the child's positioning and the exposure of the X-ray.
3. Skeleton, soft tissues.
4. Compare the transparency of the lung fields on both sides; detect foci of opacity or lucency (location, size, shape, homogeneity).
5. Lung markings and roots.
6. Diaphragm, sinuses, pleura.
7. Position and displacement of the mediastinum.
8. Conclusion (e.g., “right lower lobe infiltration”, “left-sided exudative pleurisy”).

Skill 4. Performing pulse oximetry

1. Place the pulse oximeter sensor on the child's finger (in a newborn — on the hand or foot).
2. The child should be calm and the hand warm (a cold hand or movement gives an erroneous result).
3. Once the pulse wave has stabilized (15–30 s), record the SpO_2 and pulse rate.
4. Assessment: ≥95% — normal; 92–94% — borderline; <92% — hypoxemia, oxygen therapy.

Skill 5. Identifying the syndrome on the basis of physical examination findings

1. Inspection data: type of dyspnea, symmetry of the chest, its participation in breathing.
2. Palpation: vocal fremitus increased / weakened.
3. Percussion: dull, shortened, tympanic, box-like sound.
4. Auscultation: type of breathing, rales, crepitation, pleural friction rub, bronchophony.
5. Compare the signs with the table of syndromes and identify the leading syndrome.
6. Plan confirmatory investigations (X-ray, blood tests, puncture, etc.).

Review questions

1. How are breath-holding tests performed and what do they show?
2. What are spirometry and spirography? Name the main spirographic parameters.
3. How can restrictive and obstructive ventilatory disorders be distinguished?
4. How are pneumotachometry and peak flowmetry performed? What is the importance of peak flowmetry in asthma?
5. In what order is a chest X-ray read?
6. What are the indications and age limitations for bronchoscopy, bronchography, CT and fluorography?
7. What is the diagnostic significance of sputum and pleural fluid analysis?
8. Name the diagnostic significance of the character and timing of the cough.
9. What are the signs of the syndrome of consolidation (infiltration) of lung tissue?
10. How does the atelectasis syndrome differ from consolidation?
11. What are the physical and X-ray signs of dry pleurisy, exudative pleurisy and pneumothorax?
12. How does an exudate differ from a transudate?
13. What are the croup syndrome and the degrees of laryngeal stenosis; how do false and true croup differ?

14. What are the mechanism and signs of the bronchial obstruction syndrome? What are its features at an early age?
15. Name the types, causes and degrees of severity of respiratory failure.

Test questions

1. From what age are spirography and pneumotachometry usually performed?
   A) From birth
   B) From 1 year
   C) From 3 years
   D) From 5 years
   E) From 15 years

2. The normal value of the Tiffeneau index:
   A) ≥30%
   B) ≥50%
   C) ≥70%
   D) 100%
   E) 120%

3. Typical of an obstructive ventilatory disorder:
   A) VC sharply reduced, Tiffeneau normal
   B) FEV_1 and Tiffeneau index reduced
   C) All parameters normal
   D) MVV increased
   E) MV reduced

4. Above what level does the daily variability of peak expiratory flow on peak flowmetry indicate bronchial hyperreactivity?
   A) 5%
   B) 10%
   C) 20%
   D) 50%
   E) 80%

5. From what age is fluorography performed?
   A) From 1 year
   B) From 5 years
   C) From 10 years
   D) From 15 years
   E) From 18 years

6. Curschmann spirals and Charcot–Leyden crystals are found in the sputum in:
   A) In lobar pneumonia
   B) Bronchial asthma
   C) Lung abscess
   D) Tuberculosis
   E) Bronchiectasis

7. Shows that the pleural fluid is an exudate:
   A) Specific gravity <1015
   B) Protein <30 g/l
   C) Positive Rivalta test
   D) Clear fluid
   E) Few leukocytes

8. In obstructive atelectasis the mediastinum:
   A) Shifts toward the healthy side
   B) Shifts toward the affected side
   C) Not displaced
   D) Moves down
   E) Widens

9. Percussion sound in pneumothorax:
   A) Dull
   B) Shortened
   C) Tympanic
   D) Clear pulmonary sound
   E) Thigh sound

10. In which condition is the Damoiseau line found?
   A) Lobar pneumonia
   B) Pneumothorax
   C) Exudative pleurisy
   D) Bronchial asthma
   E) Atelectasis

11. Not part of the croup triad:
   A) Barking cough
   B) Hoarse voice
   C) Inspiratory dyspnea
   D) Expiratory dyspnea
   E) All the answers are part of the triad

12. A sign distinguishing true croup from false croup:
   A) Sudden onset
   B) Gradual onset and aphonia
   C) Rapid improvement
   D) Little intoxication
   E) Onset at night

13. On percussion in the bronchial obstruction syndrome:
   A) Dull sound
   B) Box-like sound
   C) Tympanic sound only on one side
   D) Damoiseau line
   E) Elevated lung borders

14. By how much does the RR exceed the norm in grade II respiratory failure?
    A) Up to 10%
    B) Up to 30%
    C) 30–50%
    D) More than 50%
    E) Unchanged

15. The gas change typical of ventilatory respiratory failure:
    A) PaO_2 ↓, $PaCO_2$ ↑
    B) PaO_2 ↓, $PaCO_2$ ↓
    C) PaO_2 ↑, $PaCO_2$ ↑
    D) PaO_2 normal, $PaCO_2$ ↓
    E) No change

16. After what age are spirography and pneumotachometry usually performed?
    A) 5 years
    B) 1 year
    C) 2 years
    D) 10 years
    E) 3 years

17. What does the Stange test measure?
    A) Lung volume
    B) Breath-holding time after expiration
    C) Breath-holding time after a deep inspiration
    D) Respiratory rate
    E) Blood oxygen

18. The formula and normal value of the Tiffeneau index:
    A) FEV_1 / VC × 100%, normal ≥70%
    B) MVV − MV, normal 70%
    C) FEV_1 / MV, normal 50%
    D) RR × TV, normal 70%
    E) VC / FEV_1, normal 2

19. How is the breathing reserve calculated?
    A) VC − FEV_1
    B) FVC / VC
    C) MVV − MV
    D) TV + VC
    E) RR × TV

20. The Tiffeneau index in a restrictive disorder:
    A) Cannot be determined
    B) Increased
    C) Normal
    D) Equal to zero
    E) Reduced

21. What type of disorder is seen in bronchial asthma and obstructive bronchitis?
    A) Mixed restrictive
    B) Only reduced VC
    C) Obstructive (Tiffeneau <70%)
    D) Normal
    E) Restrictive

22. How is the “green zone” assessed on peak flowmetry?
    A) 50–80% — step up treatment
    B) Above 100%
    C) <50% — emergency care
    D) ≥80% of the personal best — asthma well controlled
    E) <20%

23. The daily variability on peak flowmetry indicating bronchial hyperreactivity:
    A) No change
    B) >50%
    C) >10%
    D) >5%
    E) >20%

24. The normal SpO_2 on pulse oximetry in a healthy child and the indication for oxygen:
    A) 100%; <99%
    B) 90%; <70%
    C) 85–90%; <80%
    D) 94–98%; oxygen is given if <92–94%
    E) 70–80%; <60%

25. Increased transparency (lucency) on an X-ray is typical of which condition?
    A) Bronchial tumor
    B) Emphysema, pneumothorax
    C) Atelectasis
    D) Pleurisy
    E) Pneumonia

26. Which disease is indicated by Curschmann spirals and Charcot–Leyden crystals in the sputum?
    A) Lobar pneumonia
    B) Tuberculosis
    C) Abscess
    D) Bronchiectasis
    E) Bronchial asthma

27. Above what chloride level in the sweat test is cystic fibrosis likely?
    A) >60 mmol/l
    B) >40 mmol/l
    C) >100 mmol/l
    D) >20 mmol/l
    E) >10 mmol/l

28. Unilateral purulent, fetid nasal discharge in a small child is typical of which condition?
    A) Influenza
    B) Sinusitis (only in adolescents)
    C) Allergic rhinitis
    D) A foreign body in the nose
    E) Adenoiditis

29. In which diseases does a night cough (when the bifurcation lymph nodes press) occur?
    A) Tuberculosis, lymphogranulomatosis, mediastinal tumors
    B) Allergic rhinitis
    C) Only pharyngitis
    D) Only laryngitis
    E) Influenza

30. If a cough lasts longer than how many weeks is a chronic disease (asthma, tuberculosis, cystic fibrosis, foreign body, reflux) excluded?
    A) 12 weeks
    B) 1 week
    C) 8 weeks
    D) 3–4 weeks
    E) 2 weeks

31. Where does the pain radiate in diaphragmatic pleurisy or lower lobe pneumonia?
    A) To the head
    B) To the abdomen (may resemble appendicitis)
    C) To the arm
    D) To the lower back
    E) To the leg

32. Vocal fremitus and bronchophony in the infiltration (consolidation) syndrome:
    A) Weakened
    B) Uneven
    C) Absent
    D) Increased
    E) Unchanged

33. What is Dombrovskaya's sign?
    A) Stabbing chest pain
    B) In right-sided pneumonia, the heart sounds are heard as clearly on the right as on the left
    C) Pleural friction rub
    D) Tourniquet sign
    E) Thickening of the skin fold

34. The radiological triad of atelectasis:
    A) Fluid level and air
    B) Caverns
    C) A homogeneous triangular shadow, mediastinal shift toward the affected side, elevation of the diaphragm
    D) Unilateral pneumothorax
    E) Lucency, no shift

35. Vocal fremitus and breathing in atelectasis:
    A) Increased, vesicular
    B) Weakened; breathing weakened or not heard
    C) Increased, bronchial
    D) Only rales are present
    E) Normal

36. The percussion sound when there is fluid in the pleural cavity:
    A) Box-like
    B) Metallic
    C) Tympanic
    D) Clear pulmonary sound
    E) Dull (“thigh” sound)

37. The percussion sound when there is air in the pleural cavity (pneumothorax):
    A) Tympanic
    B) Dull
    C) Shortened
    D) Clear pulmonary sound
    E) Non-resonant

38. How does a child with dry (fibrinous) pleurisy lie, and what is heard?
    A) Upright, dry rales
    B) On the healthy side, vesicular breathing
    C) On the back, bronchial breathing
    D) On the abdomen, crepitation
    E) On the affected side, pleural friction rub

39. What does the Damoiseau (Sokolov–Damoiseau) line indicate?
    A) The border of the liver
    B) The oblique upper border of the fluid
    C) The border of the segments
    D) The border of the heart
    E) The border of the pneumothorax

Clinical cases

Case 1

A 2-year-old child suddenly woke at night with a harsh “barking” cough and noisy breathing. In the evening there was a runny nose and a temperature of 37.8 °C. On examination: the child is restless, the voice is hoarse, noisy inspiration (stridor) is heard even at rest, the suprasternal area and intercostal spaces retract on inspiration, perioral cyanosis, RR 42/min, pulse 140/min. In the lungs harsh breathing, transmitted rales.

Questions: 1) Which syndrome? 2) Determine the type of dyspnea and the degree of stenosis. 3) False or true croup? Justify. 4) What are the emergency measures?

Answer: 1) Croup syndrome (acute stenosing laryngotracheitis). 2) Inspiratory dyspnea; stridor at rest, retractions and perioral cyanosis — grade II (subcompensated) stenosis. 3) False croup: against a background of ARVI, sudden onset at night, hoarse voice (not aphonia), little intoxication. 4) Calm

the child, moist fresh air, budesonide inhalation or dexamethasone, adrenaline inhalation, hospitalization; do not forcibly inspect the pharynx.

Case 2

A 12-year-old adolescent complains of cough and attacks of shortness of breath; the attacks worsen in spring, during the flowering season and after playing with a cat. The mother has pollinosis. Spirography: VC — 88% of predicted, FEV_1 — 62%, Tiffeneau index 58%. After salbutamol inhalation FEV_1 increased by 20%. Daily variability on peak flowmetry is 28%. Blood eosinophils 9%.

Questions: 1) Determine the type of ventilatory disorder. 2) Assess the result of the bronchodilator test. 3) Presumptive diagnosis? 4) What may be found in the sputum?

Answer: 1) Obstructive type: FEV_1 and the Tiffeneau index (<70%) markedly reduced, VC almost normal. 2) An increase in FEV_1 of ≥12% is a sign of reversible obstruction (bronchospasm). 3) Bronchial asthma (atopic): allergen-related attacks, hereditary predisposition, reversible obstruction, PEF variability >20%, eosinophilia. 4) Eosinophils, Curschmann spirals, Charcot–Leyden crystals.

Case 3

A 3-year-old child has been treated for pneumonia for 5 days. Today the condition suddenly became much worse: restless, marked dyspnea (RR 60/min), generalized cyanosis, pulse 160/min, SpO_2 84%. The right half of the chest is enlarged and does not take part in breathing. On the right vocal fremitus cannot be detected, percussion gives a tympanic sound, breath sounds are not heard. The heart sounds are shifted to the left. X-ray: on the right a lucency without lung markings, the lung collapsed toward the root, the mediastinum shifted to the left; there are destruction cavities in the lung.

Questions: 1) Which complication has developed? 2) What could be its cause? 3) Determine the degree of respiratory failure. 4) Emergency measures?

Answer: 1) Right-sided pneumothorax (probably valvular — the mediastinum is shifted and the condition is very severe). 2) Destructive (most often staphylococcal) pneumonia — rupture of a bulla or abscess into the pleural cavity (it may also be pyopneumothorax). 3) Grade III RF: RR >50% above normal (25–30 at 3 years), generalized cyanosis, SpO_2 <85%, marked tachycardia. 4) Oxygen, urgent pleural puncture (2nd intercostal space, midclavicular line) and drainage of the pleural cavity, transfer to the surgical department.

Quick review

Table 7.15. *Key facts on Topic 7*

Question	Answer
Spirography, pneumotachometry	From 5 years
Fluorography	From 15 years
Bronchoscopy	Up to 3–4 years rigid bronchoscope, under anesthesia
VC	TV + inspiratory reserve + expiratory reserve
MV	RR × TV
Tiffeneau index	FEV_1 / VC ≥70%
Restrictive / obstructive	VC ↓, Tiffeneau normal / FEV_1, Tiffeneau ↓

Question	Answer
Reversible obstruction	FEV_1 ↑ ≥12% after salbutamol
PEF variability	>20% — hyperreactivity
Normal SpO_2	94–98%; <92% — oxygen
Asthma sputum	Curschmann spirals, Charcot–Leyden crystals, eosinophils
Exudate	>1015, protein >30 g/l, Rivalta positive
Infiltration	Vocal fremitus ↑, dull, bronchial breathing, crepitation
Atelectasis	Vocal fremitus ↓, no breath sounds, mediastinum toward the affected side
Fluid	Dull, Damoiseau line, mediastinum toward the healthy side
Pneumothorax	Tympanitis, no breath sounds, mediastinum toward the healthy side
Croup triad	Barking cough + hoarse voice + inspiratory stridor
True croup	Gradual, aphonia, membrane, Loeffler's bacillus
Obstruction	Expiratory dyspnea, box-like sound, whistling rales
RF cut-off	PaO_2 <80, $PaCO_2$ >40–45 mm
Degrees of RF (RR)	I — up to +30%; II — +30–50%; III — >50% or bradypnea
Stange and Genche tests	Stange — holding time after inspiration; Genche — after expiration; shortening — reduced cardiorespiratory reserve
Spirography parameters	TV, MV = RR × TV, VC, FVC, FEV_1, Tiffeneau = FEV_1/VC ×100% (≥70%), MVV, breathing reserve = MVV − MV
Restrictive and obstructive disorder	Restrictive: VC↓↓, Tiffeneau normal (pneumonia, pleurisy, pneumofibrosis); obstructive: FEV_1↓↓, Tiffeneau <70% (asthma, obstructive bronchitis)
Peak flow meter zones	Green ≥80%; yellow 50–80% (step up treatment); red <50% (emergency care); variability >20% — hyperreactivity
Normal blood gases	PaO_2 80–100 mm; $PaCO_2$ 35–45 mm; pH 7.35–7.45; SpO_2 94–98%
Findings of sputum analysis	Elastic fibers — destruction; Curschmann/Charcot–Leyden/eosinophils — asthma; rusty — lobar pneumonia; mycobacteria (Ziehl–Neelsen) — tuberculosis
Possibilities of bronchoscopy	Viewing the inner surface, removing a foreign body, suctioning secretions, biopsy, bronchoalveolar lavage, drug administration
By timing of the cough	Morning — abscess/bronchiectasis; evening — bronchitis/pneumonia; night — tuberculosis, lymphogranulomatosis, asthma (4–6 a.m.), adenoiditis; with eating — reflux, TE fistula
Infiltration syndrome	Vocal fremitus↑, bronchophony↑, shortened sound on percussion, bronchial breathing, fine-bubbling moist rales, crepitation, opacity on X-ray
Atelectasis syndrome	Vocal fremitus↓, weakened breathing, shortened sound; X-ray: triangular shadow + mediastinal shift + elevated diaphragm
Fluid and air in the pleura	In both, vocal fremitus, bronchophony and breathing are weakened; fluid — dull, air — tympanic; the mediastinum shifts toward the healthy side
Pleurisy triangles	Garland — tympanic next to the spine on the same side; Rauchfuss–Grocco — dullness next to the spine on the opposite side; Damoiseau — the upper border of the fluid
Dry pleurisy	Stabbing pain (increases with breathing/coughing), lies on the affected side, pleural friction rub, Mussy's sign

TOPIC 8

The cardiovascular system: anatomical and physiological features, fetal and neonatal circulation, inspection, palpation, percussion, X-ray and cardiometry

Learning objectives

The student should know:

- the intrauterine development of the heart and the features of the fetal circulation;
- the postnatal reorganization of the circulation and the closure times of the fetal communications;
- the age-related anatomical and physiological features of children's heart and blood vessels;
- the methods of history taking, inspection, palpation and percussion of the cardiovascular system;
- the age norms for the borders of relative and absolute cardiac dullness;
- the basic principles of radiological examination, cardiometry (cardiothoracic index), ECG and echocardiography.

The student should be able to:

- take complaints and history from a child with cardiovascular disease;
- inspect the cardiac area and the vessels;
- palpate and describe the apex beat, detect a “cat's purr”, examine the pulse in the peripheral arteries;
- determine the borders of relative cardiac dullness by percussion and compare them with the age norm;
- calculate the cardiothoracic index on a chest X-ray.

8.1. Development of the heart and the fetal circulation

The heart forms from the mesoderm as a tube in **the 3rd week** of intrauterine development and begins to contract **from day 22**. The division of the heart into atria and ventricles and the formation of the valves take place in **weeks 2–8** — this is **the critical period** for the formation of **congenital heart defects** (viruses, especially rubella, drugs, alcohol, ionizing radiation, maternal diabetes mellitus).

Features of the fetal circulation:

- there is **a placental circulation**: gas exchange takes place not in the lungs but in the placenta;
- **the pulmonary circulation hardly works** — only **≈10%** of the cardiac output passes through the lungs;
- blood passes from both ventricles into the systemic circulation through **three shunts**: **the ductus venosus (Arantius)**, **the foramen ovale**, **the ductus arteriosus (Botallo)**;
- the fetal organs are supplied with **mixed blood**; the most oxygenated blood goes to **the liver, brain and arms** (branches close to the superior vena cava), and the lower part of the body receives less oxygenated blood;
- the pressure in the pulmonary artery and the aorta is almost the same and low.

The path of the blood: oxygen-rich blood comes from the placenta to the fetus via **the umbilical vein** → half to the liver (portal vein), the rest via **the ductus venosus** to the inferior vena cava →

right atrium. Most of the blood from the inferior vena cava passes, with the help of the Eustachian valve, through **the foramen ovale** to the left atrium → left ventricle → aorta → head and arms. Blood from the superior vena cava goes to the right atrium → right ventricle → pulmonary artery → via **the ductus arteriosus** to the descending aorta (pulmonary vascular resistance is high). From the aorta blood returns to the placenta via **the two umbilical arteries**.

REMEMBER

The fetus lives **under conditions of hypoxia**, but is adapted to it: the respiratory surface of the placenta is large, the blood flow is rapid, fetal blood contains **much hemoglobin and many erythrocytes**, **HbF** has a greater affinity for oxygen than adult hemoglobin, and the tissues need little oxygen.

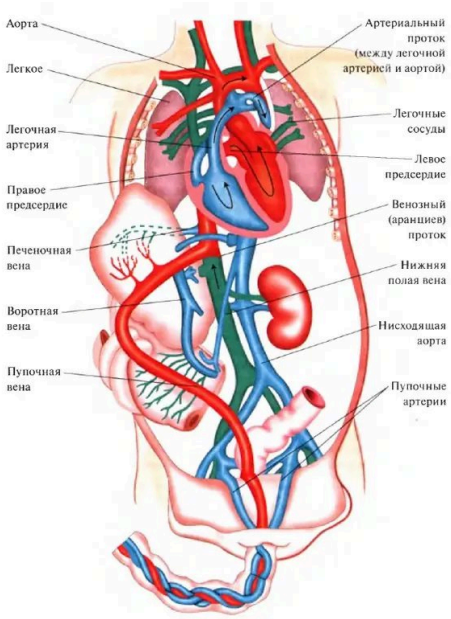


Figure 8.1. *Fetal circulation (Navruzova, Geppe)*

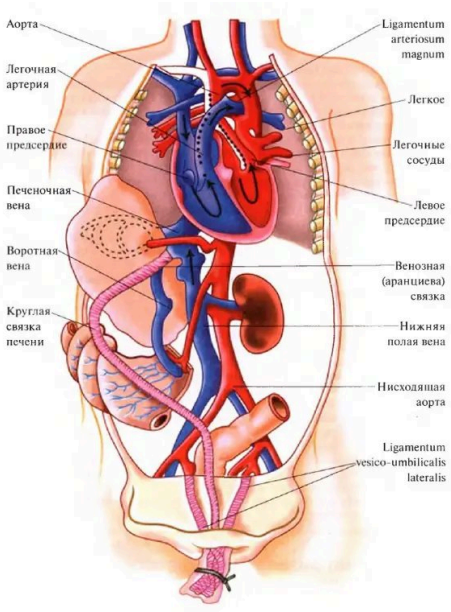


Figure 8.2. *Neonatal circulation (Navruzova, Geppe)*

8.2. Neonatal circulation: the transitional period

After birth the circulatory system is fundamentally reorganized:

- **the placental circulation stops**; the umbilical vessels close;
- with **the first breath** the lungs expand, pulmonary vascular resistance falls sharply and pulmonary blood flow increases **≈5 times — the pulmonary circulation starts working**;
- venous return to the left atrium increases, and the pressure in the left atrium exceeds that in the right atrium — **the valve of the foramen ovale closes**;
- under the influence of increased blood oxygen and decreased prostaglandins, the smooth muscle of **the ductus arteriosus** contracts and it closes;
- the systemic and pulmonary circulations begin to work **in series**; the load on the left ventricle increases.

Table 8.1. *Closure times of the fetal communications (Kildiyarova)*

Communication	Functional closure	Anatomical closure (subsequent fate)
Umbilical vein	After birth (when the clamp is applied)	In 2–3 weeks — the round ligament of the liver
Umbilical arteries	After birth	The lateral umbilical ligaments of the bladder

Communication	Functional closure	Anatomical closure (subsequent fate)
Ductus venosus (Arantius)	In the first minutes	In ≈2 weeks — the ligamentum venosum
Ductus arteriosus (Botallo)	10–15 minutes – 10–15 hours	2–8 weeks – 2–5 months (at 2 months in 90%) — the ligamentum arteriosum
Foramen ovale	With the first breath (hemodynamic), day 7–9	9–12 months (in 50% of children a probe-patent slit may persist until 5 years)

CAUTION

In the neonatal period, with hypoxia, acidosis, cold exposure, pneumonia or RDS, pulmonary vascular resistance may rise again and **the fetal communications may reopen** — the **"persistent fetal circulation" syndrome** (pulmonary hypertension, right-to-left shunt, cyanosis). A hemodynamically insignificant patent foramen ovale is considered **a minor cardiac anomaly**.

8.3. Anatomical and physiological features of the heart and blood vessels

Table 8.2. *Anatomical features of the newborn's heart*

Indicator	Feature
Mass	≈20–25 g, 0.8% of the body mass (0.4–0.5% in adults); it increases 2 times by 1 year, 3 times by 2–3 years and 10 times by 15–16 years
Shape	Spherical (round) — the atria are relatively large, the ventricles insufficiently developed; after 6 years oval as in adults
Ventricles	The walls of the right and left ventricles are almost the same thickness (1.4:1); later the left ventricle develops rapidly (3 times heavier at 16 years)
Position	High (at the level of ThIV–VIII) and transverse (horizontal) — the diaphragm is high, the lungs small, the thymus large; from 2–3 years it becomes oblique
Myocardium	The muscle fibers are thin, with many large nuclei and weak cross-striation; little connective and elastic tissue; rich blood vessels and lymphatic network — predisposition to inflammatory and dystrophic changes
Endocardium, valves	Delicate, elastic
Pericardium	Spherical, closely fitting the heart, mobile, small capacity
Conduction system	Morphologically immature; differentiation is completed at 10–14 years

Growth rate: heart volume increases fastest **in the first 2 years of life, at 5–9 years and in puberty**. Up to 2 years mainly **the atria** grow, at 2–10 years the whole heart, and after 10 years mainly **the ventricles**.

Blood vessels:

- the arteries are relatively **wide**, with thin walls and poorly developed muscle and elastic fibers; their structure becomes similar to that of adults **by 12 years**;
- the ratio of the arterial and venous lumens in the child is **1:1** (1:2 in adults); the veins later grow faster and at 15–16 years are 2 times wider than the arteries;
- in the newborn the pulmonary artery is wider than the aorta (21 and 16 mm); they become equal at 12 years, and in puberty the aorta is wider;

- **the capillaries are wide, short and tortuous**, with higher permeability than in adults — predisposition to **edema** in infections, **exsicosis** in toxicosis, and vascular insufficiency;
- there are **many anastomoses** between the coronary arteries — myocardial infarction is very rare.

Innervation: at an early age the **sympathetic** influence predominates — physiological **tachycardia**. Myelination of the vagus nerve is completed **at 3–4 years**; after that the heart rate (HR) decreases and **respiratory arrhythmia** appears (HR increases on inspiration and slows on expiration); it is maximal **at 4–12 years**.

Functional features:

- **physiological tachycardia** — small heart volume, high oxygen demand, sympathicotonia;
- HR is **labile** — it changes quickly with movement, crying, fever and excitement;
- stroke and minute volume increase with age, but the minute volume relative to body mass is **greater** in the child;
- **arterial pressure is low** — the arteries are wide, the pumping force of the heart small and the peripheral resistance low; it rises gradually with age;
- the circulation rate is high (a full circulation ≈12 s in the newborn, 22 s in adults).

Table 8.3. *Heart rate by age*

Age	HR, per minute
Newborn	140–160
1 year	120–125
5 years	100
10 years	80–85
15 years	70–80

Variants of the adolescent heart (puberty): **the “drop” (small) heart** — with rapid growth in height the heart seems to hang on the vessels; rapid fatigue, dizziness, low BP, a functional systolic murmur, fainting; **the “hypertrophic” heart** — more often in boys, the left ventricle enlarged, BP slightly raised; **“mitral” configuration** — only on X-ray (the left contour is flattened).

8.4. History taking

Complaints:

- **weakness, rapid fatigue** on physical exertion; in the newborn — **tiring during feeding**, stopping feeding to rest, sweating, poor weight gain;
- **dyspnea** — on exertion or at rest;
- **cyanosis** (lips, nails, generalized) — on exertion or constant; **hypercyanotic spells**;
- **edema** (legs, lower back, face);
- **palpitations**, a feeling of “skipped beats”;
- **pain in the heart area** (location, character, duration, radiation, relation to exertion or excitement, relief with medication);
- **fainting**, dizziness, headache;
- pain in the calf muscles on exertion (intermittent claudication) — insufficient arterial circulation in the legs;
- frequent bronchopulmonary diseases (in defects with increased pulmonary blood flow).

History of the present illness: when the complaints appeared (since birth — congenital defect; 1–3 weeks after tonsillitis or ARVI — rheumatic fever, myocarditis), relation to exertion, the course.

Life history: the mother's pregnancy (infections, drugs, alcohol and diabetes in the 1st trimester, occupational hazards); in the family — **congenital heart defects**, cardiomyopathies, rheumatic fever, cases of **early (before 50 years) sudden death**, arterial hypertension; the child's past illnesses (frequent tonsillitis, scarlet fever), previous ECG and echocardiography results.

8.5. Inspection

General inspection

- **Position:** in heart failure the child lies with the head raised, often on the right side; in a severe condition — **orthopnea** (sitting with the legs hanging down); in exudative pericarditis — sits leaning forward or in the **knee-elbow** position; in **tetralogy of Fallot** — **squats** after exertion (the “squatting” position); in acute vascular insufficiency — horizontal, head low.
- **Skin color: pallor** (vascular insufficiency, aortic defects, “white”-type defects, rheumatic fever); **acrocyanosis** (right ventricular failure, slowed peripheral circulation); **generalized (central) cyanosis** — “blue” congenital defects (right-to-left shunt), pulmonary hypertension; **differential cyanosis** — only in the legs or only in the arms (defects of the great vessels); crimson cheeks — mitral stenosis.
- **“Drumstick fingers”** and **“watch-glass” nails** — chronic hypoxia (“blue” defects, infective endocarditis).
- **Edema** — right ventricular failure: first on the feet and ankles, increasing in the evening; then the shins, thighs, lower back and abdomen; in a lying child — the sacrum and lower back; in infants — the face and scrotum.
- **Physical development:** hypotrophy, short stature (congenital defects); **disproportion between the upper and lower body** (an “athletic” shoulder girdle, thin legs) — **coarctation of the aorta**.
- **Precapillary pulse (Quincke)** — a change in the color of the nail bed in time with the pulse when the tip of the nail is lightly pressed — **aortic valve insufficiency**.



Figure 8.3. *“Drumstick fingers” and “watch-glass” nails in a congenital heart defect (Navruzova, Geppe)*

Inspection of the cardiac area

- **The apex beat** — the rhythmic impact of the apex of the heart on the chest wall; it is visible in thin children.
- **The cardiac impulse** — diffuse pulsation in the cardiac area (at the left sternal edge and in the epigastrium); only in pathology (right ventricular hypertrophy, cardiomegaly) or on physical

exertion, in thin children.

- **“Cardiac hump”** — bulging of the chest in the cardiac area; a sign of **cardiomegaly** that developed at an early age (the chest is soft): congenital defects, early carditis.
- Smoothing of the intercostal spaces — pericarditis with a large effusion.
- Epigastric pulsation, pulsation in the 2nd intercostal space to the right and left of the sternum.

Inspection of the blood vessels

- **Strong pulsation of the carotid arteries (“carotid dance”)** and nodding of the head in time with the pulse (**de Musset's sign**) — aortic valve insufficiency.
- **Distension and pulsation of the neck veins** — right ventricular failure, tricuspid valve insufficiency, exudative pericarditis, compression of the superior vena cava (**“Stokes' collar”** — edema of the face and neck).
- Subcutaneous venous collaterals (occlusion of the venae cavae), pulsation of the intercostal arteries (coarctation of the aorta).
- Signs of impaired arterial circulation in the limbs: coldness, pallor, trophic changes.

8.6. Palpation

The apex beat

Technique: the child lies on the back or sits; the doctor stands on the right. The base of the right palm is placed on the sternum and the fingers on the cardiac area toward the anterior axillary line along the intercostal spaces (in girls the breast is moved upward). Once the pulsation is found, its exact location, area, height and force are determined with the tips of the 2nd–3rd fingers. If it cannot be found in the supine position, the child is turned on the left side or leans forward.



Figure 8.4. *Determining the apex beat (Kildiyarova, Figure 9.3)*

Table 8.4. *Characteristics of the apex beat (normal)*

Indicator	Normal
Location	Up to 2 years — in the 4th intercostal space, 1–2 cm outside the left midclavicular line; 2–7 years — in the 5th intercostal space, 1 cm outside the midclavicular line; over 7 years — in the 5th intercostal space, 0.5–1 cm inside the midclavicular line
Area	≈1×1 cm at an early age, 1.5–2 cm² in older children; >2 cm² or in two intercostal spaces — diffuse
Height	The amplitude of the chest wall vibration — moderate
Force (resistance)	The resistance offered to the finger — moderate

Table 8.5. *Changes in the apex beat*

Change	Causes
Displaced to the left and downward	Left ventricular enlargement (aortic defects, mitral insufficiency, myocarditis)
Displaced to the left	Right ventricular enlargement (pushes the heart to the left), mediastinal shift to the left (right-sided pleurisy, pneumothorax; left-sided atelectasis), high diaphragm (meteorism, ascites)
Displaced to the right	Dextrocardia, left-sided pleurisy or pneumothorax
Weakened or not detected	Obesity, emphysema, left-sided pleurisy, exudative pericarditis, severe myocarditis
Increased, high, resistant (“dome-shaped”)	Left ventricular hypertrophy (aortic defects), thyrotoxicosis, fever, physical exertion
Diffuse	Left ventricular dilatation (aortic and mitral insufficiency, myocarditis)
Negative (retraction of the intercostal space)	Adhesive pericarditis

The “cat's purr” and pulsations

The “cat's purr” — a **thrill** felt with the palm, produced by turbulent blood flow at the site of an altered valve apparatus:

- **systolic** — in the 2nd intercostal space to the right of the sternum (aortic stenosis), to the left (pulmonary stenosis), in the 3rd–4th intercostal space at the left sternal edge (**ventricular septal defect**), at the base of the heart (**patent ductus arteriosus** — systolic-diastolic);
- **diastolic** — at the apex of the heart (**mitral stenosis**).

Epigastric pulsation is best detected at the end of a deep inspiration: right ventricular hypertrophy or dilatation (directed from above downward, increases on inspiration); pulsation of the abdominal aorta (from front to back, normal in thin children).

Palpation of the great vessels: in the 2nd intercostal space to the right (aorta) and left (pulmonary artery) of the sternum, in the jugular notch (normally only weak pulsation).



Figure 8.5. *Detecting pulsation: a — in the epigastric area; b — in the jugular notch (Navruzova, Geppe)*

Pulse in the peripheral arteries

The pulse is examined and compared **in both arms and legs**: the radial, brachial, carotid (at the inner edge of the sternocleidomastoid muscle, at the level of the thyroid cartilage), temporal, femoral (at the middle of the inguinal ligament), popliteal, posterior tibial (behind the medial malleolus) and dorsalis pedis arteries. In the newborn **the femoral pulses** must be checked — weak or absent pulses are a sign of **coarctation of the aorta**. The characteristics of the pulse are given in detail in topic 9.

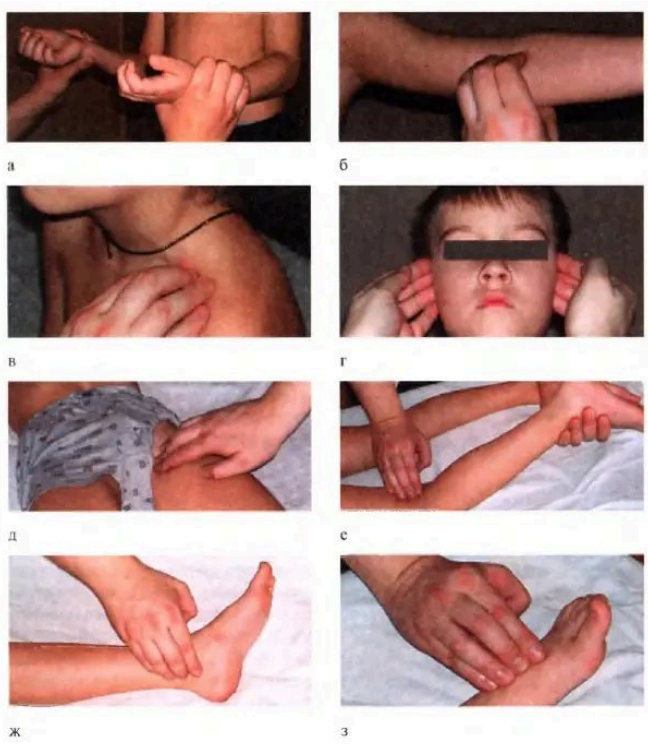


Figure 8.6. *Points for detecting the pulse in the peripheral arteries: radial, brachial, carotid, temporal, femoral, popliteal, posterior tibial and dorsalis pedis arteries (Navruzova, Geppe)*

8.7. Percussion of the heart

Percussion determines: the size of the heart (the borders of relative and absolute dullness), its configuration, position, and the width of the vascular bundle.

Rules:

- percussion should be **quiet** or **very quiet**;
- the pleximeter finger is placed **parallel** to the border sought;
- the pleximeter step is small (not more than the width of a finger);

- the stroke is directed **strictly from front to back**;
- the border is marked along **the outer edge of the finger (facing the clear sound)**;
- in children over 4 years the technique is as in adults; at an early age, direct percussion is done **with one finger only**, and in indirect percussion the plexímeter finger is placed **with the 1st phalanx only**; in infants the most reliable method of determining the left border is **orthopercussion** (strictly in the sagittal direction, with the plexímeter finger placed on its lateral edge).

Order of determining the borders of relative dullness:

1. **Right border.** First, percussing from top to bottom along the right midclavicular line, find **the upper border of liver dullness** (usually rib V–VI). Then move up one intercostal space (the 4th intercostal space), place the finger parallel to the sternum and percuss **from outside inward** — from the clear sound until dullness appears. The right border is formed by **the right atrium**.
2. **Upper border.** The finger is placed along the left parasternal line in the 1st intercostal space parallel to the ribs, and percussion is performed **from top to bottom**; the border is marked along the upper edge of the finger. The upper border is formed by **the pulmonary artery cone and the left atrial appendage**.
3. **Left border.** First find the apex beat; in the same intercostal space place the finger **1–1.5 cm outside** the beat (or from the anterior axillary line), perpendicular to the ribs (parallel to the midclavicular line), and percuss **from outside toward the heart**. The left border is formed by **the left ventricle**.

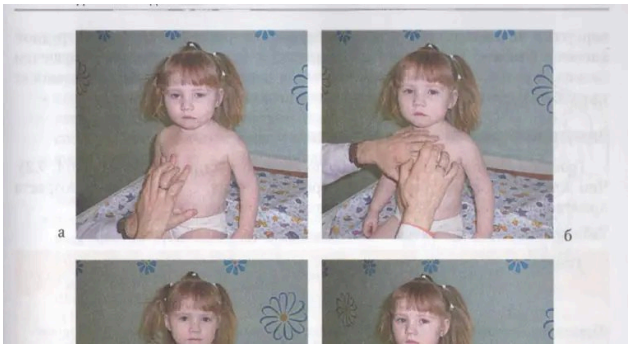


Figure 8.7. *Determining the borders of relative cardiac dullness: a — right; b — upper; c, d — left border (Kildiyarova, Figure 9.5)*



Figure 8.8. *Position of the plexímeter finger when determining the cardiac borders (Navruzova, Geppe)*

Table 8.6. *Borders of cardiac dullness and transverse diameter by age (Kildiyarova)*

Border	0–2 years	2–7 years	7–12 years
Relative dullness			
Right	Right parasternal line	Inside the right parasternal line (midway between the sternal edge and the parasternal line)	Right sternal edge (or 0.5–1 cm outside it)
High	Rib II	2nd intercostal space	Rib III
Left	1.5–2 cm outside the left midclavicular line	0.5–1 cm outside the midclavicular line	Along the midclavicular line or 0.5–1 cm inside it

Border	0–2 years	2–7 years	7–12 years
Transverse diameter, cm	6–9	8–12	9–14
Absolute dullness			
Right	Left sternal edge	Left sternal edge	Left sternal edge
High	Rib III	3rd intercostal space	Rib IV
Left	Midway between the left midclavicular and parasternal lines	The same	The same or close to the midclavicular line

MEMORY AID

With age the heart **"becomes smaller and descends"**: **the upper border rib II → 2nd intercostal space → rib III**; **the right border from the parasternal line → to the edge of the sternum**; **the left border from 2 cm outside the midclavicular line → to the midclavicular line**. The numbers: **"2 – 1 – 0"** cm outside (0–2, 2–7, 7+ years).

Absolute dullness — the part of the heart not covered by the lungs (mainly **the right ventricle**); it is determined by very quiet percussion and is rarely determined in children.

Cardiac configuration — by the borders of relative dullness; in addition, the border is determined in the 3rd intercostal space on the right and the 4th on the left. Normally there is an obtuse angle between the vascular bundle and the left ventricular contour — **the "cardiac waist"**.

Width of the vascular bundle — very quiet percussion in the 1st (2nd at school age) intercostal space from the right and left parasternal lines toward the sternum; normally equal to the width of the manubrium (does not extend beyond the sternum).

Table 8.7. *Changes in the borders of cardiac dullness*

Change	Causes
Enlargement in all directions	Exudative pericarditis, combined defects, severe myocarditis, cardiomyopathy
To the right	Enlargement of the right atrium and ventricle: mitral stenosis, pulmonary hypertension, tricuspid valve insufficiency, atrial septal defect
To the left	Left ventricular hypertrophy or dilatation: aortic defects, mitral insufficiency, arterial hypertension, myocarditis; high diaphragm ("lying heart")
Upward	Left atrial enlargement: mitral stenosis, mitral insufficiency
Reduction	Pulmonary emphysema, asthma attack, low diaphragm; "drop" heart
Widening of the vascular bundle	Thymic enlargement, mediastinal tumor and lymph nodes, dilatation of the aorta or pulmonary artery

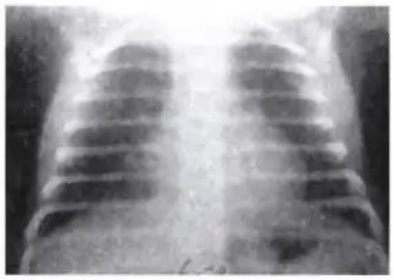
8.8. Radiological examination and cardiometry

Chest X-ray (in frontal and lateral projections, and if necessary in oblique projections with esophageal contrast) gives information on the size, shape (configuration) and position of the heart, the state of the great vessels and **the pulmonary blood flow** (increased or reduced lung markings).

The normal cardiac shadow in the frontal projection:

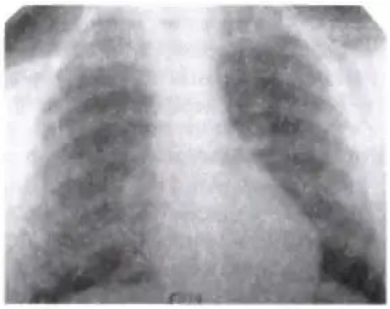
- **right contour** — 2 arcs: above, the superior vena cava (and the ascending aorta), below, the right atrium;
- **left contour** — 4 arcs: the aortic arch, the pulmonary artery, the left atrial appendage, the left ventricle.

In the newborn the cardiac shadow is relatively large, spherical and horizontally positioned, with a wide vascular bundle (thymus); with age it becomes elongated and oblique.



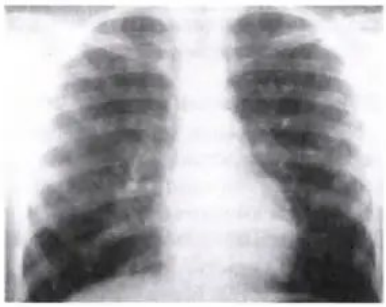
а

Figure 8.9. *Chest X-ray: newborn (Navruzova, Geppe)*



б

Figure 8.10. *3-year-old child (Navruzova, Geppe)*



в

Figure 8.11. *11-year-old child (Navruzova, Geppe)*

Cardiometry — measuring the dimensions of the heart on an X-ray. The most widely used parameter is **the cardiothoracic index (CTI)**:

CTI = (CE + ED) / AB, where CE is the distance from the midline to the outermost point of the right cardiac contour, ED — from the midline to the outermost point of the left contour, AB — the inner transverse diameter of the chest at the level of the diaphragmatic dome.

Normal: ≤0.55 in young children, **≤0.5** in older children. An increased index means **cardiomegaly**.

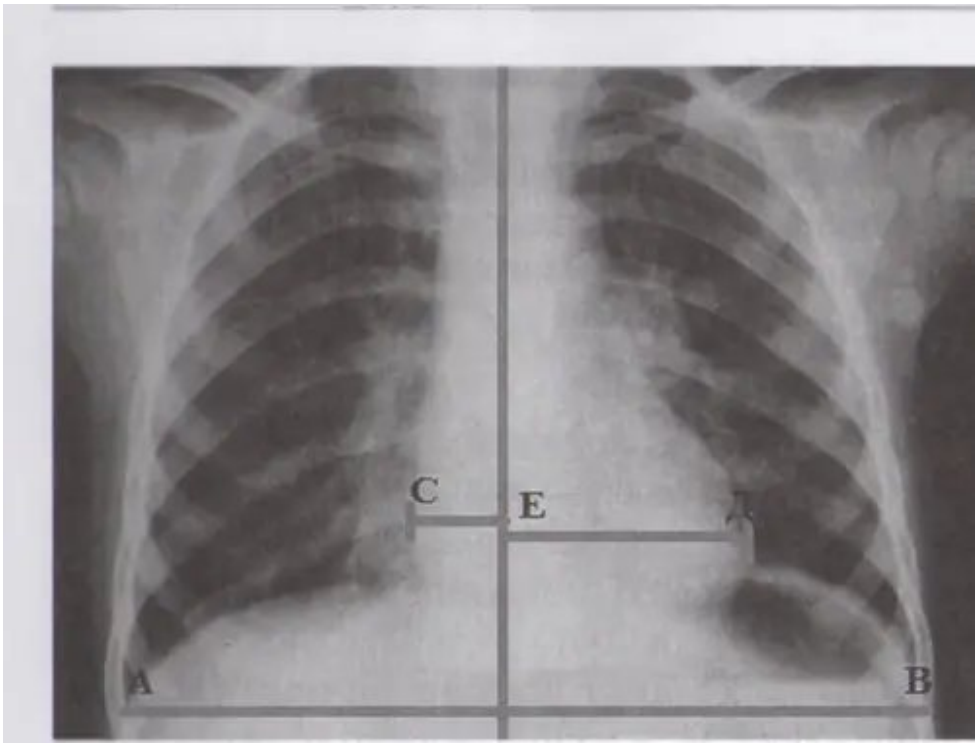


Figure 8.12. *Determining the cardiothoracic index: CTI = (CE + ED)/AB (Kildiyarova, Figure 9.12)*

Pathological configurations:

- **mitral** — the “cardiac waist” is flattened (the arcs of the pulmonary artery and the left atrial appendage bulge): mitral stenosis and insufficiency;
- **aortic** — the “cardiac waist” is deepened, the left ventricle enlarged, the apex displaced downward and to the left (“sitting duck”): aortic defects, arterial hypertension;
- **spherical or triangular (“trapezoid”)** — exudative pericarditis;
- **“wooden shoe” (“Dutch clog”)** — **tetralogy of Fallot** (raised apex, deep “cardiac waist”, reduced lung markings);
- **“egg-shaped”** — transposition of the great arteries; **“snowman” (“figure 8”)** — total anomalous pulmonary venous drainage.



Figure 8.13. *Exudative pericarditis: the cardiac arcs are smoothed, the shadow is round (Navruzova, Geppe)*



Figure 8.14. *Aortic configuration of the heart (Navruzova, Geppe)*

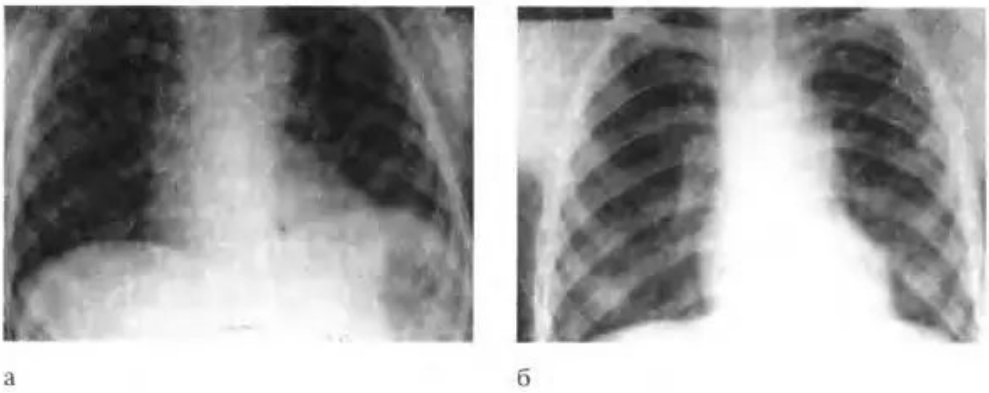


Figure 8.15. *Changes in the vascular bundle: a — bulging of the aortic arch; b — bulging of the pulmonary artery arc (Navruzova, Geppe)*

8.9. ECG and echocardiography — basic concepts

Electrocardiography (ECG) — recording of the electrophysiological processes in the heart; it gives information on the rhythm, HR, conduction, hypertrophy of the heart chambers and the state of the myocardium. 12 standard leads: 3 standard (I, II, III), 3 augmented (aVR, aVL, aVF) and 6 chest (V_1–V_6) leads.

- **P wave** — atrial excitation;
- **PQ interval** — conduction of the impulse from the atria to the ventricles;
- **QRS complex** — ventricular excitation (depolarization);
- **ST segment and T wave** — ventricular repolarization;
- **QT interval** — electrical systole of the ventricles.

HR = 60 / RR (in seconds). The age-related features of the ECG in children (right axis deviation, short waves and intervals, negative T waves) are given in detail in topic 9.

Echocardiography — ultrasound examination of the structure and function of the heart (2.5–7.5 MHz probe); **the main method in pediatric cardiology**. Its possibilities:

- the size of the heart chambers, wall thickness, **contractility** (ejection fraction, normally ≥55–60%);
- the structure and movement of the valves, **regurgitation** and **stenosis** (Doppler echocardiography);
- **congenital defects** (septal defects, patent ductus arteriosus, tetralogy of Fallot, etc.) — the direction and volume of the shunt;
- pulmonary artery pressure;
- the state of the pericardium and **fluid in the pericardial cavity**;
- intracardiac formations (thrombus, tumor, vegetations — infective endocarditis, false chords).

CLINICAL SIGNIFICANCE

For early detection of congenital heart defects in newborns, **pulse oximetry screening** is carried out: SpO_2 is measured on the right hand and a foot at 24–48 hours of life. If **SpO_2 <95%** or the difference between the hand and the foot is **>3%**, the baby is referred for echocardiography.

8.10. Clinical section

Suspicion of a congenital heart defect in a newborn

Parents' complaints and “alarming” signs on examination: rapid tiring and sweating during feeding, poor weight gain, tachypnea at rest, cyanosis (especially increasing with crying), cold limbs, a strong cardiac impulse, a “cardiac hump”, liver enlargement, weak or absent femoral pulses, a BP difference between the arms and legs.

CAUTION

In the newborn, **weak pulses in the femoral arteries** are a sign of **coarctation of the aorta**; when the ductus arteriosus closes (in the 1st–2nd weeks of life) the child's condition may deteriorate sharply (“duct-dependent” defects). Checking the femoral pulses is mandatory at every examination of a newborn.

Physical detection of cardiomegaly

Signs of cardiomegaly: “cardiac hump”, a diffuse and displaced apex beat, a cardiac impulse, widening of the borders of relative dullness on percussion. Confirmation — an increased **CTI** on X-ray and **echocardiography**.

Practical skills

Skill 1. Palpation of the apex beat

1. Wash and warm the hands; the child lies on the back or sits, the doctor on the right.
2. Place the right palm on the sternum and the fingers along the left intercostal spaces toward the anterior axillary line.
3. Where the pulsation is felt, find the outermost and lowest point with the tips of the 2nd–3rd fingers.

4. Determine its location (relative to the intercostal space and the midclavicular line), area, height and force.
5. Check the cardiac area for a “cat's purr” with the palm and determine whether it coincides with systole or diastole (comparing with the carotid pulse).
6. Compare the result with the age norm and record it.

Skill 2. Determining the borders of relative cardiac dullness

1. The child stands or lies, the doctor on the right.
2. Find the upper border of liver dullness along the right midclavicular line.
3. One intercostal space higher, place the finger parallel to the sternum and determine the right border from outside inward.
4. Determine the upper border along the left parasternal line from the 1st intercostal space downward.
5. Determine the left border in the intercostal space of the apex beat from outside inward.
6. Mark the borders along the edge of the finger facing the clear sound; measure the transverse diameter.
7. Compare the result with the age norm.

Skill 3. Examining the pulse in the peripheral arteries

1. Check the synchrony and equality of the pulse in the radial arteries of both arms simultaneously.
2. Examine the pulse symmetrically in the carotid, brachial, femoral, popliteal, posterior tibial and dorsalis pedis arteries.
3. Compare the pulse in the arms and legs (weak femoral pulse — coarctation).
4. Record the result.

Skill 4. Calculating the cardiothoracic index

1. On a frontal X-ray draw the midline (along the spinous processes).
2. Measure the distance from the midline to the outermost point of the right contour (CE) and to the outermost point of the left contour (ED).
3. Measure the inner transverse diameter of the chest at the level of the diaphragmatic dome (AB).
4. Calculate CTI = (CE + ED) / AB.
5. Assessment: ≤0.55 at an early age, ≤0.5 in older children — normal; if increased — cardiomegaly.

Review questions

1. When is the heart formed in utero? What is the critical period for congenital defects?
2. Name the features of the fetal circulation. By what route does the blood circulate?
3. What changes occur in the circulation after birth? When do the fetal communications close?
4. What are the anatomical features of the newborn's heart (mass, shape, position, structure of the myocardium)?
5. What are the features of children's blood vessels and their clinical significance?
6. What are the age norms for heart rate? What is respiratory arrhythmia?
7. What complaints occur in cardiovascular diseases?

8. Which signs on general inspection indicate cardiac pathology?
9. How is the apex beat determined? What are its age norms and changes?
10. What is the “cat's purr” and in which defects is it detected?
11. What are the rules of cardiac percussion and the order of determining the borders of relative dullness?
12. Name the age norms for the borders of relative and absolute cardiac dullness.
13. In which conditions do the cardiac borders change?
14. Name the radiological structure of the cardiac shadow and the pathological configurations.
15. How is the cardiothoracic index calculated and what is its norm? What are the possibilities of ECG and echocardiography?

Test questions

1. From when does the heart begin to contract in utero?
   A) From day 7
   B) From day 22
   C) From week 8
   D) From week 12
   E) From week 20

2. What proportion of the cardiac output passes through the lungs in the fetal circulation?
   A) 100%
   B) 50%
   C) ≈10%
   D) 75%
   E) None at all

3. When does anatomical closure of the ductus arteriosus (Botallo) occur in most children?
   A) Immediately after birth
   B) At 1 week
   C) At 2 months
   D) At 1 year
   E) At 5 years

4. Anatomical closure of the foramen ovale:
   A) At birth
   B) On days 7–9
   C) At 2 months
   D) At 9–12 months
   E) At 7 years

5. The mass of the newborn's heart relative to body mass:
   A) 0,1%
   B) 0,4%
   C) 0,8%
   D) 2%
   E) 5%

6. The normal HR per minute in a 1-year-old child:
   A) 70–80
   B) 80–85
   C) 100
   D) 120–125
   E) 160–180

7. Where is the apex beat found in a child under 2 years?
   A) In the 5th intercostal space, inside the midclavicular line
   B) In the 4th intercostal space, 1–2 cm outside the midclavicular line
   C) In the 3rd intercostal space
   D) In the 6th intercostal space
   E) At the sternal edge

8. A diastolic “cat's purr” at the apex of the heart is typical of:
   A) Aortic stenosis
   B) Mitral stenosis
   C) Ventricular septal defect
   D) Patent ductus arteriosus
   E) Mitral insufficiency

9. Which part forms the right border of relative cardiac dullness?
   A) Left ventricle
   B) Right ventricle
   C) Right atrium
   D) Aorta
   E) Left atrium

10. The upper border of relative cardiac dullness in a 7–12-year-old child:
    A) Rib I
    B) Rib II
    C) 2nd intercostal space
    D) Rib III
    E) Rib IV

11. The left border of relative cardiac dullness in a 0–2-year-old child:
    A) 1.5–2 cm outside the midclavicular line
    B) Along the midclavicular line
    C) 1 cm inside the midclavicular line
    D) On the anterior axillary line
    E) On the parasternal line

12. Enlargement of the borders of cardiac dullness in all directions is typical of:
    A) Emphysema
    B) Exudative pericarditis
    C) Mitral stenosis
    D) “Drop” heart
    E) Arterial hypotension

13. The normal cardiothoracic index in an older child:
    A) ≤0,3
    B) ≤0,4
    C) ≤0,5
    D) ≤0,65
    E) ≤0,75

14. A “wooden shoe”-shaped cardiac shadow on X-ray is typical of:
    A) Patent ductus arteriosus
    B) Tetralogy of Fallot
    C) Exudative pericarditis
    D) Mitral stenosis
    E) Coarctation of the aorta

15. Weak femoral pulses in a newborn are typical of which defect?
    A) Atrial septal defect
    B) Tetralogy of Fallot
    C) Coarctation of the aorta
    D) Mitral stenosis
    E) Pulmonary stenosis

16. In which weeks of fetal development does the division of the heart into atria and ventricles and the formation of the valves take place?
    A) 1–2
    B) 20–24
    C) 30–36
    D) 12–16
    E) 2–8

17. The three shunts of the fetal circulation:
    A) Ductus venosus (Arantius), foramen ovale, ductus arteriosus (Botallo)
    B) Umbilical vein, aorta, pulmonary artery
    C) Coronary, bronchial, portal
    D) Ductus venosus, umbilical artery, aorta
    E) Only the foramen ovale

18. The approximate amount of blood passing through the pulmonary circulation in the fetus:
    A) 70%
    B) 30%
    C) 10%
    D) 100%
    E) 50%

19. Which organs receive the most oxygenated blood in the fetus?
    A) Intestine
    B) Liver, brain, arms
    C) Skin
    D) Legs, kidneys
    E) Lungs

20. When does functional closure of the ductus arteriosus (Botallo) occur?
    A) 2 months
    B) 5 years
    C) 1 year
    D) 10–15 minutes – 10–15 hours
    E) It does not close

21. Approximately when does anatomical closure of the foramen ovale occur?
    A) At 1 week
    B) At 1 month
    C) At 5 years
    D) At 9–12 months
    E) At 2 weeks

22. What percentage of the body mass is the heart in a newborn?
    A) 0,2%
    B) 0,4–0,5%
    C) 0,8%
    D) 1,5%
    E) 3%

23. The shape and position of the heart in children:
    A) Spherical, high and transverse
    B) Oval, oblique, low
    C) Elongated, vertical
    D) Triangular, low
    E) As in adults

24. What appears in the child after myelination of the vagus nerve is completed?
    A) Physiological tachycardia increases
    B) Systolic murmur
    C) Respiratory arrhythmia (HR increases on inspiration)
    D) Constant bradycardia
    E) BP rises

25. Which disease is very rare in children because of the many anastomoses between the coronary arteries?
    A) Pericarditis
    B) Arrhythmia
    C) Endocarditis
    D) Myocardial infarction
    E) Myocarditis

26. The characteristic position of a child with tetralogy of Fallot after exertion:
    A) Stands
    B) Leans forward
    C) Squats (“squatting”)
    D) Lies on the back
    E) Lies with the head low

27. Crimson cheeks are typical of which defect?
    A) Mitral stenosis
    B) Patent ductus arteriosus
    C) Tetralogy of Fallot
    D) Aortic stenosis
    E) Coarctation of the aorta

28. A disproportion between an “athletic” shoulder girdle and thin legs is typical of which defect?
    A) Mitral insufficiency
    B) Patent foramen ovale
    C) Ventricular septal defect
    D) Coarctation of the aorta
    E) Tetralogy of Fallot

29. Where is the apex beat found in a child under 2 years?
    A) 4th intercostal space, 1–2 cm outside the midclavicular line
    B) 5th intercostal space, inside the midclavicular line
    C) 3rd intercostal space
    D) 6th intercostal space, midaxillary line
    E) 2nd intercostal space

30. The normal apex beat in a child over 7 years:
    A) Cannot be determined
    B) In the 5th intercostal space, 0.5–1 cm inside the midclavicular line
    C) In the 4th intercostal space, 2 cm outside the midclavicular line
    D) In the 3rd intercostal space
    E) In the 6th intercostal space

31. Displacement of the apex beat to the left and downward is typical of which condition?
    A) Left-sided pleurisy
    B) Left ventricular enlargement
    C) Dextrocardia
    D) Meteorism
    E) Right ventricular enlargement

32. Retraction of the apex beat (negative apex beat) is typical of which disease?
    A) Myocarditis
    B) Thyrotoxicosis
    C) Exudative pericarditis
    D) Aortic defect
    E) Adhesive pericarditis

33. In which defect is a diastolic “cat's purr” palpated at the apex of the heart?
    A) Tetralogy of Fallot
    B) Pulmonary stenosis
    C) Aortic stenosis
    D) Ventricular septal defect
    E) Mitral stenosis

34. Nodding of the head in time with the pulse (de Musset's sign) is typical of which defect?
    A) Mitral stenosis
    B) Aortic valve insufficiency
    C) Patent foramen ovale
    D) Tricuspid stenosis
    E) Coarctation of the aorta

35. What are weak or absent femoral pulses in a newborn a sign of?
    A) Mitral stenosis
    B) Anemia
    C) Pericarditis
    D) Coarctation of the aorta
    E) Tetralogy of Fallot

36. Which part forms the right border of relative cardiac dullness?
    A) Left atrial appendage
    B) Aorta
    C) Right atrium
    D) Pulmonary artery
    E) Left ventricle

37. Which part forms the left border of relative cardiac dullness?
    A) Left ventricle
    B) Pericardium
    C) Right ventricle
    D) Right atrium
    E) Aorta

38. One of the rules of cardiac percussion:
    A) Only in the supine position
    B) The pleximeter is placed perpendicular to the border
    C) The step is large
    D) A strong stroke is given
    E) The border is marked along the edge of the finger facing the clear sound

39. The most reliable method of determining the left border in infancy:
    A) Palpation only
    B) Orthopercussion
    C) Spirography
    D) Auscultation
    E) X-ray

Clinical cases

Case 1

A 3-month-old child. The mother complains that the baby tires quickly during feeding, stops feeding to rest, sweats a lot and is not gaining weight well. On examination: pale skin, RR 56/min, HR 165/min. There is a bulge over the cardiac area on the left half of the chest. The apex beat is in the 5th intercostal space, 3 cm outside the midclavicular line, diffuse. A systolic thrill is detected at the left

sternal edge in the 3rd–4th intercostal space. On percussion the cardiac borders are enlarged in all directions. The liver projects 3 cm below the costal arch.

Questions: 1) What is the bulge on the chest called and what does it indicate? 2) Assess the apex beat and percussion findings. 3) Which defect is the systolic thrill typical of? 4) Which investigations are needed?

Answer: 1) “Cardiac hump” — cardiomegaly developed at an early age. 2) The apex beat is displaced downward and to the left and diffuse (the norm in a 3-month-old is the 4th intercostal space, 1–2 cm outside the midclavicular line); the cardiac borders are enlarged — cardiomegaly. 3) A systolic thrill in the 3rd–4th intercostal space at the left sternal edge — ventricular septal defect. There are also signs of heart failure: tachycardia, tachypnea, sweating, liver enlargement. 4) Echocardiography (size of the defect, shunt, pulmonary pressure), ECG, chest X-ray (CTI, lung markings).

Case 2

When examining an 8-year-old girl, the doctor determined the cardiac borders: right border — 0.5 cm outside the right sternal edge, upper border — rib III, left border — 0.5 cm inside the left midclavicular line. The apex beat is in the 5th intercostal space, 1 cm inside the midclavicular line, area 1.5 cm², of moderate force. HR 84/min, increasing on inspiration and slowing on expiration.

Questions: 1) Are the cardiac borders appropriate for the age? 2) Assess the apex beat. 3) How do you assess the HR variability?

Answer: 1) Yes, they correspond to the norm for 7–12 years (right — sternal edge, upper — rib III, left — along the midclavicular line or 0.5–1 cm inside it). 2) The location, area and force of the apex beat are normal. 3) The HR is normal for 8 years (80–85); the change related to breathing is physiological respiratory (sinus) arrhythmia, most pronounced at 4–12 years; it disappears when the breath is held.

Case 3

A 12-year-old boy complains of rapid fatigue, headache and leg pain when running. On examination: the shoulder girdle is well developed, the legs are thin. The neck vessels pulsate strongly. BP in the arms is 150/90 mm Hg, in the legs 100/70. The femoral pulses are weak and delayed. The apex beat is displaced to the left and increased. On X-ray “usuration” (notching) of the lower edges of the ribs is found.

Questions: 1) Presumptive diagnosis? 2) Which signs confirm it? 3) Explain the change in the apex beat. 4) What is the normal ratio of BP in the arms and legs?

Answer: 1) Coarctation of the aorta. 2) Disproportion between the upper and lower body, arterial hypertension in the arms, hypotension in the legs, weak femoral pulses, intermittent leg pain (ischemia), rib notching from the pressure of the intercostal collateral arteries. 3) Left ventricular hypertrophy (working against the narrowed aorta) — an increased apex beat displaced to the left. 4) Normally BP in the legs is 10–20 mm Hg higher than in the arms.

Quick review

Table 8.8. *Key facts on Topic 8*

Question	Answer
Formation / contraction of the heart	Week 3 / day 22
Critical period for defects	Weeks 2–8 of pregnancy
Fetal shunts	Ductus venosus, foramen ovale, ductus arteriosus
Blood passing through the lungs (fetus)	≈10%
Closure of the ductus arteriosus	Functional 10–15 minutes–hours; anatomical 2–8 weeks (90% at 2 months)
Closure of the foramen ovale	Functional — with the first breath; anatomical — 9–12 months
Newborn's heart	20–25 g, 0.8% of the body mass; spherical, horizontal, high
HR	Newborn 140–160; 1 year 120; 5 years 100; 10 years 80–85; 15 years 70–80
Respiratory arrhythmia	Maximal at 4–12 years; disappears when the breath is held
The apex beat	<2 years — 4th space, 1–2 cm outside; 2–7 — 5th space, 1 cm outside; >7 — 5th space, 0.5–1 cm inside
Relative dullness (upper)	Rib II → 2nd space → rib III
Relative dullness (right)	Parasternal → inside → sternal edge
Relative dullness (left)	1.5–2 cm → 0.5–1 cm outside the midclavicular line → along the line
Forming the borders	Right — right atrium; upper — pulmonary artery and left atrial appendage; left — left ventricle
“Cat's purr”	Systolic — stenoses, VSD, PDA; diastolic — mitral stenosis
CTI	≤0.55 (early age); ≤0.5 (older age)
Configurations	Mitral, aortic, spherical (pericarditis), “wooden shoe” (Fallot)
Coarctation	BP ↑ in the arms, ↓ in the legs, weak femoral pulse
Shunts of the fetal circulation	Ductus venosus (Arantius), foramen ovale, ductus arteriosus (Botallo); gas exchange in the placenta; pulmonary circulation ≈10%
Postnatal closure	Ductus arteriosus: functional 10–15 hours, anatomical 2–8 weeks (at 2 months in 90%); foramen ovale: 9–12 months; ductus venosus ≈2 weeks
Features of the child's heart	Mass 0.8% (0.4–0.5% in adults); spherical; high and transverse; large atria; the left ventricle later grows rapidly; tight pericardium
Normal HR	Newborn 140–160; 1 year 120–125; 5 years 100; 10 years 80–85; 15 years 70–80
Critical period for congenital heart defects	Weeks 2–8 of pregnancy: viruses (rubella), drugs, alcohol, radiation, maternal diabetes mellitus
Variants of the adolescent heart	“Drop” heart (low BP, fatigue, fainting), “hypertrophic” heart (in boys), “mitral” configuration (only on X-ray)
Cyanosis and defect	Generalized cyanosis — “blue” defects; acrocyanosis — right ventricular failure; differential — great vessels; crimson cheeks — mitral stenosis
Location of the apex beat	<2 years: 4th intercostal space, 1–2 cm outside the midclavicular line; 2–7 years: 5th space, 1 cm outside; >7 years: 5th space, 0.5–1 cm inside

Question	Answer
Displacement of the apex beat	Left and down — left ventricle ↑; left — right ventricle ↑, mediastinal shift; right — dextrocardia; negative — adhesive pericarditis
Borders of relative dullness (7–12 years)	Right — right sternal edge; upper — rib III; left — midclavicular line; transverse 9–14 cm
What forms the borders	Right — right atrium; upper — pulmonary artery cone + left atrial appendage; left — left ventricle
Signs on inspection of the vessels	Carotid dance and de Musset — aortic insufficiency; distended neck veins — right ventricular failure; Stokes' collar — compression of the superior vena cava

TOPIC 9

Auscultation of the heart, pulse and arterial pressure. Functional and instrumental examinations of the cardiovascular system, age-related features of the ECG, and semiotics

Learning objectives

The student should know:

- the rules of cardiac auscultation, the listening points and their order;
- the formation of the heart sounds and their age-related features;
- the classification of heart murmurs, the difference between functional and organic murmurs;
- the characteristics of the pulse and the age norms for arterial pressure;
- the possibilities of functional tests (Martinet, Shalkov, clinoorthostatic), ECG, Holter monitoring and echocardiography;
- the age-related features of the ECG, the main arrhythmias and the signs of hypertrophy;
- the semiotics of changes in heart sounds, murmurs, pulse and arterial pressure.

The student should be able to:

- auscultate the heart at all points and describe the sounds and murmurs;
- distinguish a functional murmur from an organic one;
- examine the pulse for all its characteristics;
- measure arterial pressure correctly in the arms and legs and assess it against the age norm (formula, centiles);
- perform and assess a functional test;
- “read” a child's ECG by the main parameters.

9.1. Auscultation of the heart

Rules:

- a quiet, warm room; the doctor stands on **the child's right side**;
- auscultation is performed **in different positions**: lying on the back, on the left side, standing; if necessary after physical exertion (at school age);
- the results are compared during normal breathing and while holding the breath (after expiration);
- a small-diameter stethoscope is used in small children; when the child cries, listen between breaths;
- auscultation is performed **in a strict order**, then the whole cardiac area, the axilla, the infraclavicular area, the epigastrium and the interscapular area are listened to;
- first the rhythm and HR are assessed, then the sounds, then the murmurs.

Listening points and their order:

1. **Apex of the heart — mitral valve**.
2. **2nd intercostal space, right of the sternum — aortic valve**.
3. **2nd intercostal space, left of the sternum — pulmonary valve**.

4. **The lower end of the sternum, the base of the xiphoid process** (slightly to the right) — **tricuspid valve**.
5. **Botkin–Erb point** — the attachment of ribs III–IV to the left sternal edge — **aortic (and mitral) valve**.

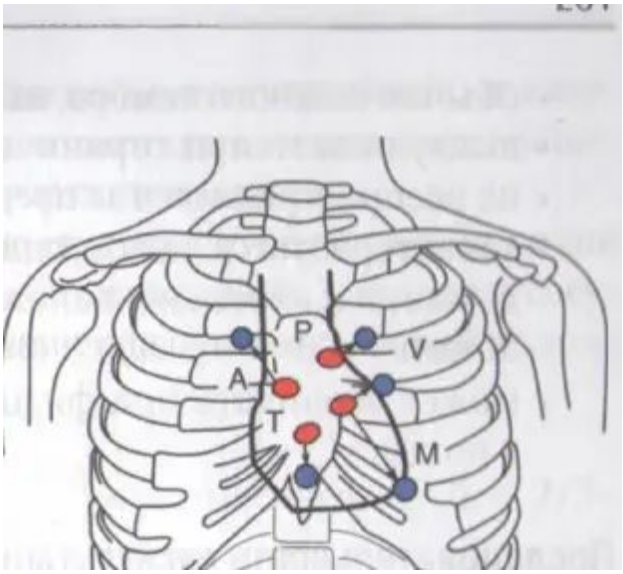


Figure 9.1. *Points of cardiac auscultation (Kildiyarova, Figure 9.7)*

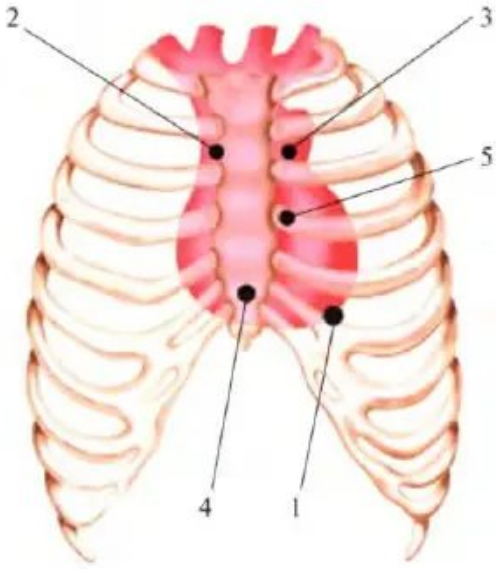


Figure 9.2. *Classic points for listening to the heart sounds: 1 — mitral, 2 — aortic, 3 — pulmonary, 4 — tricuspid valve, 5 — Botkin–Erb point (Navruzova, Geppe)*

MEMORY AID

The order of auscultation — **“Mitral – Aortic – Pulmonary – Tricuspid – Botkin”** = **“MAPTB”**: it starts with the valve most often affected by heart disease.

Heart sounds

Table 9.1. *Description of the heart sounds*

Sound	Formation	Time and place of hearing
S1 (systolic)	At the start of ventricular systole: closure of the atrioventricular (mitral and tricuspid) valves (valvular component), contraction of the ventricular myocardium (muscular), vibration of the initial parts of the aorta and pulmonary artery (vascular), atrial contraction (atrial component)	After the long pause; simultaneous with the apex beat and the carotid pulse; best heard at the apex
S2 (diastolic)	At the start of diastole: closure of the semilunar valves of the aorta and pulmonary artery and vibration of the vessel walls	After the short pause; best heard at the base of the heart (in the 2nd intercostal space)
S3	At the start of diastole: rapid filling of the ventricles and vibration of their walls	After S2; may be heard in thin children and adolescents lying down (physiological)
S4	Atrial contraction at the end of diastole	Before S1; may be present in athletic adolescents

At the apex the sounds are heard as **“LUB-dup”**, at the base as **“lub-DUP”** (Lang). The pause between S1 and S2 (systole) is shorter than the pause between S2 and S1 (diastole).

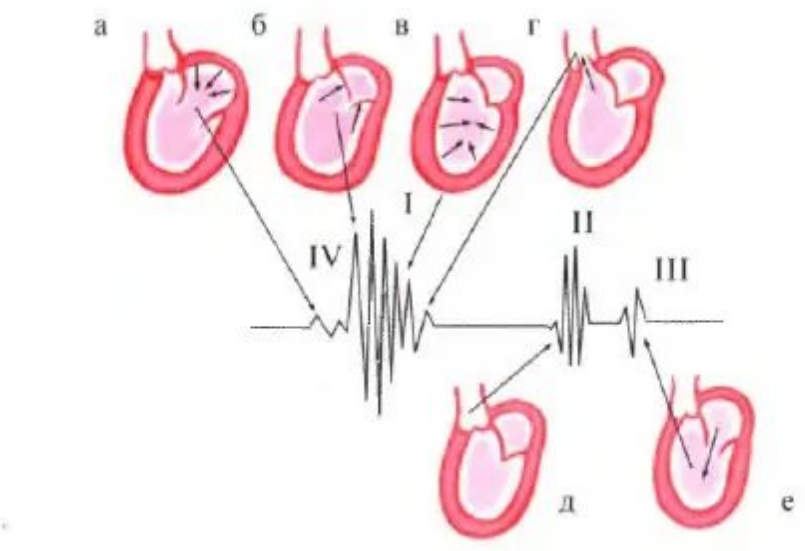


Figure 9.3. *Formation of the heart sounds (diagram): components of S1, S2, S3 and S4 (Navruzova, Geppe)*

Age-related features of the heart sounds:

- **Embryocardia** — a pendulum rhythm: S1 and S2 are identical and the pauses equal. In the newborn it is considered normal **in the first 2 weeks** of life (because of tachycardia); in an older child — **pathological** (acute heart failure, paroxysmal tachycardia, high fever).
- In the neonatal period the sounds may be slightly muffled; later in children the sounds are **louder than in adults** (thin chest wall).
- In **the first 2–3 days** of life S2 is louder than S1 at the apex and at Botkin's point, then they become equal; **from 2–3 months** S1 predominates at the apex.
- In the 1st year S1 is heard better at the base of the heart, then they become equal; **from 2 years** S2 is stronger.
- **From 2 to 12 years** S2 is stronger **over the pulmonary artery** than over the aorta — **a physiological accentuation of S2 over the pulmonary artery**; at 12 years they become equal, and later it is stronger over the aorta.
- In children **physiological splitting of S1 and S2** (at the end of inspiration, changing with body position) and **S3** are frequently heard.

Splitting and reduplication: if there is an imperceptible pause between the two components of a sound — **splitting** (“TRA-ta”); if the pause is distinct — **reduplication**. Physiological splitting is variable and depends on body position and the phase of breathing.

Heart murmurs

A murmur is a longer sound, heard in addition to the heart sounds, resulting from turbulent blood flow. Turbulence depends on changes in **the diameter of the valve orifice or vessel**, **the blood flow velocity** and **the blood viscosity**.

Classification:

- **intracardiac** and **extracardiac** murmurs (pericardial friction rub, pleuropericardial murmur);
- **organic** (anatomical changes in the valves, septa and vessels — congenital and acquired defects, endocarditis) and **inorganic**;
- according to P.S. Moshchich's classification: **organic**, **organic-functional** (impaired cardiac function in myocarditis and arrhythmias — relative valvular insufficiency), **functional** (in

extracardiac diseases: anemia, fever, thyrotoxicosis, autonomic dysfunction) and **physiological (innocent)** — in a healthy child.

Describing a murmur: 1) phase of the cardiac cycle (systolic, diastolic; early, mid, late, pansystolic); 2) duration; 3) loudness (intensity, grade 1–6); 4) timbre (soft, blowing, harsh, scraping, musical); 5) the place where it is best heard (epicenter, punctum maximum); 6) radiation (axilla, neck, interscapular area); 7) relation to the heart sounds; 8) change with body position, breathing and exertion.

Table 9.2. *Differences between functional and organic murmurs*

Sign	Functional (physiological) murmur	Organic murmur
Phase	Systolic only	Systolic or diastolic (diastolic — always organic)
Duration	Short, occupies part of systole	Long, often the whole of systole (pansystolic)
Relation to the heart sounds	Not related to the sounds, the sounds are unchanged	Often related to S1; the sounds are changed
Timbre	Soft, musical, blowing	Harsh, scraping, blowing
Height	Low (grade 1–2)	Often loud (grade ≥3), with a “cat's purr”
Location	In a limited area (often at the apex or over the pulmonary artery)	A distinct epicenter at a particular point
Radiation	Does not radiate	Radiates to the axilla, neck, interscapular area
Dependence on position	Variable: increases lying down, decreases or disappears when standing and after exertion	Constant; increases after exertion
Other signs	Cardiac borders, ECG, echocardiography normal; no circulatory failure	Cardiomegaly, ECG and echocardiographic changes, signs of failure

REMEMBER

A diastolic murmur is never functional. A functional murmur is **“soft, short, systolic, variable, non-radiating”**. A functional systolic murmur is heard at various ages in 50–70% of healthy children.

Extracardiac murmurs:

- **pericardial friction rub** — “creaking”, “scraping”, at the base of the heart or in the area of absolute dullness, **does not radiate anywhere**, variable, does not fully coincide with systole and diastole, **increases on leaning forward and on pressure with the stethoscope**, **persists** when the breath is held;
- **pleuropericardial murmur** — at the left border of the heart, increases at the peak of inspiration, **disappears when the breath is held**.

Auscultation of the blood vessels

In a healthy child two sounds are heard over **the carotid and subclavian arteries**, and sometimes S1 over **the femoral artery**; no sounds are heard over other arteries. Over the neck veins (especially on the right, with the head turned) a **“humming top”** is heard — a normal venous hum (increased in anemia). Murmurs are heard over vessels when arteries are narrowed or dilated and in aortic and

valvular defects; a murmur over the abdominal aorta and renal arteries — renal artery stenosis (a cause of arterial hypertension).

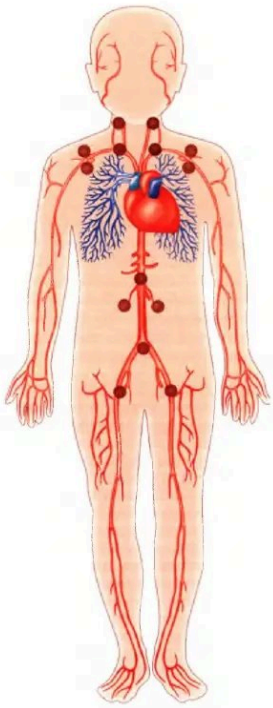


Figure 9.4. *Points for listening to the vessels (Navruzova, Geppe)*

9.2. The pulse

In children **over 2 years** the pulse is examined mainly at **the radial artery** (in younger children — at the temporal, carotid, brachial or femoral arteries). The most accurate information is obtained in the morning, after the child wakes up, at rest (**basal pulse**). First the pulse is examined **in both arms simultaneously** (synchrony and equality), then in one arm for 1 minute.

Table 9.3. *Characteristics of the pulse*

Feature	Description and changes
Rate	Corresponds to the HR; faster than the age norm — tachycardia, slower — bradycardia; pulse deficit — pulse lower than the HR (atrial fibrillation, extrasystole)
Rhythm	Regular or irregular; in children respiratory arrhythmia is normal (faster on inspiration, slower on expiration, disappears when the breath is held) — maximal at 4–12 years, may increase in adolescents (“adolescent arrhythmia”)
Tension	The pressure needed to stop the pulse — related to BP and vascular tone: hard (tense) — arterial hypertension; soft — hypotension
Filling	Depends on the stroke volume and the amount of blood: full, satisfactory, empty (blood loss, collapse)
Size	Depends on tension and filling: large (aortic insufficiency), small; thready pulse — shock, collapse
Shape (speed)	The speed of rise and fall of the pulse wave: rapid, bounding (aortic insufficiency, patent ductus arteriosus, thyrotoxicosis), slow, low (aortic stenosis)
Symmetry	The same in both arms; a difference — vascular defect or aortoarteritis

In a healthy child with fever, the pulse increases by **8–10 beats** for each **1 °C** rise in temperature. At an early age there are **3–3.5** heartbeats per breath, in adults **4**.

Table 9.4. *Changes in the pulse and their diagnostic significance*

Type of pulse	When it occurs
Rapid (tachycardia)	Physiological — newborn and early age, physical and mental stress; pathological — fever, anemia, thyrotoxicosis, pain, myocarditis, heart failure, hypovolemia
Slow (bradycardia)	Physiological — in sleep, in athletes; pathological — AV blocks, sick sinus, raised intracranial pressure, hypothyroidism, obstructive jaundice, dystrophy, some infections (typhoid fever)
Irregular	Respiratory arrhythmia (normal); extrasystole, fibrillation, blocks
Weak or absent in both legs	Coarctation of the aorta
Weak or absent in one arm or leg	Nonspecific aortoarteritis, thrombosis, vascular anomaly
Alternating (alternans)	Severe heart failure, severe myocardial damage — a poor prognostic sign
Rapid and high	Aortic valve insufficiency
Slow and low	Aortic stenosis
Paradoxical (weakens on inspiration)	Exudative pericarditis (tamponade), severe asthma attack, grade III croup

9.3. Arterial pressure

Systolic BP — the maximal pressure in the arteries during left ventricular systole (depends on the stroke volume and the elasticity of the aorta); **diastolic BP** — the minimal pressure in diastole (depends on the tone of the peripheral arterioles); **pulse pressure** — the difference between them (≈40 mm in the newborn, ≈50 mm at 14–15 years).

Technique of measurement (Korotkov method)

1. The child sits quietly for **5 minutes** (small children lying down); there should be no food or active play for 30 minutes before the measurement.
2. **The cuff size must be appropriate for age**: the width of the rubber bladder should cover **≈40%** of the arm circumference and its length **80–100%** (newborn — 3–5 cm, infant — 5–8 cm, preschool — 9–10 cm, school age — 12–13 cm, adolescents — adult cuff). A narrow cuff **overestimates** BP, a wide cuff **underestimates** it.
3. The arm on the table, palm up, muscles relaxed; the heart, the cuff and the zero point of the manometer **at the same level**.
4. The deflated cuff is placed on the upper arm **2 cm above** the antecubital fossa so that 1–2 fingers can pass under it.
5. The brachial artery is found by palpation in the antecubital fossa, and the stethoscope is placed over it.
6. Air is pumped into the cuff to **30–40 mm Hg** above the level at which the radial pulse disappears.
7. The pressure is released slowly (**2–3 mm** per second): **the first clear sound** — systolic BP (Korotkov phase I), **disappearance of the sounds** — diastolic BP (phase V). In small children the sounds may be heard down to zero — then the point of sharp muffling of the sounds (phase IV) is taken.

8. The measurement is repeated **2–3 times** at 2–3-minute intervals and the average is taken; the first time it is measured **in both arms** and **in the legs** (in the leg — with a wide cuff on the thigh, listening over the popliteal artery).



Figure 9.5. *Measuring arterial pressure (Kildiyarova, Figure 9.8)*

Age norms

Table 9.5. *Average arterial pressure by age*

Age	Systolic BP, mm Hg	Diastolic BP
Newborn	60–70 (66–72)	1/2–2/3 of systolic
1 year	80–90	40–50
5 years	100	60–65
10 years	110	70
15 years	115–120	70–75

Approximate formulas:

- **Up to 1 year:** systolic BP = **76 + 2n** (n — number of months); diastolic = **1/2–2/3** of systolic;
- **Over 1 year** (average for boys): systolic = **90 + 2n**, diastolic = **60 + n** (n — number of years);
- upper limit: systolic = 105 + 2n, diastolic = 75 + n; lower limit: systolic = 75 + 2n, diastolic = 45 + n;
- **in girls 5 mm** is subtracted from the values obtained.

In the legs BP is **10–20 mm Hg higher** than in the arms; higher BP in the arms is a sign of coarctation of the aorta.

Centile assessment (tables by sex, age and height): **<10th centile** — low; **10–90** — normal; **90–95** — high normal (risk of hypertension); **≥95th centile** (on 3 measurements) — **arterial hypertension**; **<5th centile** — **arterial hypotension**.

> **CLINICAL SIGNIFICANCE**
> Arterial hypertension is diagnosed only on the basis of BP measured **at least 3 times, on different days**. To exclude “white coat hypertension”, **24-hour ambulatory BP monitoring (ABPM)** is performed.

9.4. Functional tests

Functional tests are performed to determine the adaptation of the cardiovascular system to exertion, the possibility of extending the regimen, and the physical education group.

Martinet test (orthostatic). The child lies for 5–10 minutes and the pulse and BP are measured; then they are measured again standing or after **20 squats** (within 30 seconds), and recovery is observed every minute. **Favorable reaction:** the pulse rises by **≤25% (–50%)**, systolic BP rises by **5–10 mm** (up to 20 mm), diastolic is unchanged or falls slightly, the parameters recover **in 3 minutes**, no complaints.

Graded exercise test according to N.A. Shalkov. HR, RR and BP are measured at rest; the child is given a load (loads 1–3 — sitting up and lying down in bed for patients on bed rest; 4–6 — 5, 10, 20 deep squats or climbing 10–30 steps; 7–8 — exercise and sports loads), and the reaction is assessed.

Table 9.6. *Assessment of the reaction to exertion*

Indicator	Favorable reaction	Unfavorable reaction
Complaints	Absent	Dyspnea, heart pain, weakness, dizziness
Skin	Unchanged	Pallor or cyanosis
Pulse	Rises by ≤25%	Rises by ≥50%
Systolic BP	Rises moderately	Falls or rises sharply
Diastolic BP	Unchanged or falls slightly	Rises or falls sharply
Recovery	In 3 minutes	5–10 minutes or longer

Clinoorthostatic test (COT). The child lies for 10–15 minutes and HR and BP are measured; then the child stands up and stands without leaning for 10 minutes — measured every minute; then lies down again and is observed for 4 minutes. **Normal reaction:** no complaints, HR rises by **20–40%**, systolic and diastolic BP rise slightly (+5…+20 mm), the decrease in pulse pressure does not exceed 50%. The test is used to determine **autonomic support** (sympathicotonic, hyperdiastolic, asympathicotonic, asthenosympathetic and other variants).

Pharmacological tests: atropine (sinus node function, vagal influence), potassium and Obsidan (propranolol) tests (to determine the origin of ST-T changes).

Bicycle ergometry, treadmill test — in older children, to assess physical working capacity and exercise-related arrhythmias and ischemia.

9.5. Instrumental investigations

Electrocardiography

ECG protocol: 1) voltage (1 mV = 10 mm); 2) rhythm source (sinus — a positive P before each QRS); 3) HR (= 60/RR); 4) electrical axis; 5) amplitude and width of the waves; 6) intervals (PQ, QRS, QT); 7) **systolic index** (QT/RR × 100%); 8) ST segment and T wave; 9) age-related features; 10) conclusion.

Table 9.7. *Age-related features of children's ECG (Kildiyarova)*

Indicator	Newborn	Until 2 years	7–15 years
HR	120–140 (–160)	110–120	75–95

Indicator	Newborn	Until 2 years	7–15 years
Electrical axis	Right axis deviation (right-axis pattern)	Right-axis pattern or normal axis	Normal axis, often vertical position
P wave	Tall and pointed in I, II, III	Tall	Positive in II, may be negative or biphasic in III
Q wave	Deep in III	Deep in III	Not constant
QRS	May be notched in lead III; tall R and deep S in the right chest leads	—	May be M- or W-shaped in III; R low in $V_{1–2}$, tall in $V_{5–6}$
T wave	Low, biphasic or negative in V_1–V_4	Negative in V_1–V_3	May be negative in V_1–V_3 (III)
Width of the waves and QRS	≤0.07 s	≤0.07 s	≤0.09 s
Systolic index	45–74%	45–65%	40–58%

Main features of children's ECG:

- the waves and intervals are **short** (the impulse is conducted quickly, HR is high);
- **sinus respiratory arrhythmia** is frequent;
- at an early age **right axis deviation** (relative predominance of the right ventricle), which changes to a normal axis with age;
- **negative T waves** in the right chest leads (V_1–V_3, sometimes V_4) may be normal;
- a deep Q in standard lead III;
- in some children **incomplete right bundle branch block** (a normal variant);
- **early repolarization syndrome** — slight ST elevation in adolescents (often a normal variant).

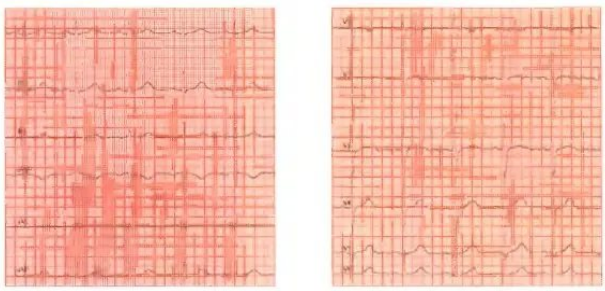


Figure 9.6. *Normal ECG of a 4-year-old child (Navruzova, Geppe)*

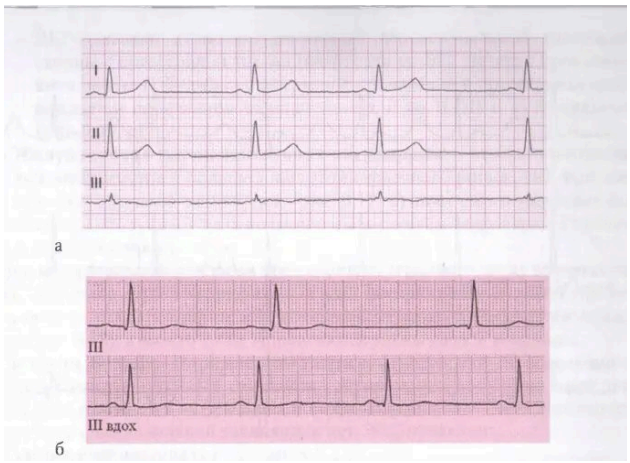


Figure 9.7. *ECG of sinus (a) and respiratory (b) arrhythmia: RR shortens on inspiration (Kildiyarova, Figure 9.15)*

Other electrophysiological methods:

- **Holter monitoring** — continuous ECG recording for 24 hours (or longer): HR dynamics, arrhythmias and their associations, ischemic changes. Indications: fainting, life-threatening arrhythmias, any arrhythmia at an early age, sudden death at a young age in relatives.
- **24-hour ambulatory BP monitoring (ABPM)** — with a portable monitor under natural conditions.
- **Cardiointervalography (CIG)** — mathematical analysis of 100 cardiac cycles; assessment of autonomic tone (sympathetic or parasympathetic predominance) and reactivity.

- **Phonocardiography (PCG)** — graphic recording of heart sounds and murmurs; objective determination of the phase, shape and frequency of a murmur.
- **Echocardiography** (see topic 8) and **Doppler echocardiography** — the direction and velocity of blood flow, regurgitation, gradients.
- **Rheovasography, Doppler ultrasound** — the state of the peripheral vessels.
- **CT, MRI, angiocardiography, cardiac catheterization** — complex defects and preoperative examination.

Laboratory methods: markers of myocardial damage (CK-MB, troponins, LDH), natriuretic peptide (NT-proBNP — heart failure), CRP, ASO (streptococcal infection), blood culture (infective endocarditis), lipid profile.

9.6. Semiotics

Changes in the heart sounds

Table 9.8. *Changes in the heart sounds and their causes*

Change	Causes
Weakening (muffling) of both sounds	Extracardiac: obesity, chest wall edema, emphysema, left-sided pleurisy; cardiac causes: myocarditis, exudative pericarditis, cardiomyopathy, heart failure, collapse
Accentuation of both sounds	Thin children, physical and mental stress, fever, thyrotoxicosis, anemia
Weakening of S1 at the apex	Mitral valve insufficiency, myocarditis (reduced contractility), aortic insufficiency
Accentuation of S1 at the apex (“flapping”)	Mitral stenosis, tachycardia
Accentuation of S2 over the aorta	Arterial hypertension, aortic atherosclerosis (rare in children)
Accentuation of S2 over the pulmonary artery (after 12 years or marked)	Pulmonary hypertension: mitral stenosis and insufficiency, defects with a left-to-right shunt (VSD, ASD, PDA), chronic lung diseases
Weakening of S2 over the aorta	Aortic stenosis, aortic valve insufficiency
Weakening of S2 over the pulmonary artery	Pulmonary stenosis, tetralogy of Fallot
Pathological splitting of S2	Atrial septal defect (fixed splitting), right bundle branch block, pulmonary hypertension
“Gallop” rhythm — a three-part rhythm against a background of tachycardia	Reduced myocardial tone: myocarditis, heart failure, cardiomyopathy — an important sign
“Quail rhythm” — flapping S1 + S2 + mitral opening snap	Mitral stenosis
Embryocardia (in a child over 2 weeks)	Acute heart failure, paroxysmal tachycardia, high fever, severe myocarditis
“Cannon” (Strazhesko) sound — a sharply accentuated S1	Complete AV block (when the atria and ventricles contract simultaneously)

Organic murmurs

Table 9.9. *Main organic murmurs*

Murmur	In which pathology, features
Ejection systolic murmur (midsystolic, diamond-shaped)	Aortic stenosis — 2nd intercostal space right of the sternum (and at Botkin's point), harsh, radiates to the carotid arteries, systolic thrill; pulmonary stenosis — 2nd intercostal space on the left, radiates to the left clavicle and the interscapular area; atrial septal defect — 2nd–3rd intercostal space on the left, low-medium intensity (relative pulmonary stenosis), S2 splitting; hypertrophic cardiomyopathy
Pansystolic (regurgitant) murmur	Mitral insufficiency — at the apex, blowing, merged with S1, radiates to the left axilla, increases lying on the left side, S1 weakened; ventricular septal defect — 3rd–4th intercostal space at the left sternal edge, harsh, loud, widely radiating (also to the back), systolic thrill; tricuspid insufficiency — at the base of the xiphoid process, increases on inspiration
Early diastolic (protodiastolic) murmur	Aortic valve insufficiency — a “pouring” blowing murmur beginning with S2, at the Botkin–Erb point and along the left sternal edge, best heard with the child leaning forward; pulmonary valve insufficiency (in pulmonary hypertension — the Graham Steell murmur)
Mid- and late diastolic (presystolic) murmur	Mitral stenosis — at the apex, low-frequency “rumbling”, with presystolic accentuation, best heard on the left side and on expiration, does not radiate; relative mitral stenosis (Austin Flint murmur — in aortic insufficiency)
Continuous systolic-diastolic (“machinery”) murmur	Patent ductus arteriosus — 2nd intercostal space on the left and below the clavicle; before 1 year only the systolic component may be heard

MEMORY AID

By location of the murmur: **apex** — mitral (systolic — insufficiency, diastolic — stenosis); **2nd space on the right** — aortic stenosis; **Botkin's point** — aortic insufficiency (diastolic); **2nd space on the left** — pulmonary stenosis, ASD, PDA (“machinery”); **3rd–4th space at the left sternal edge** — VSD.

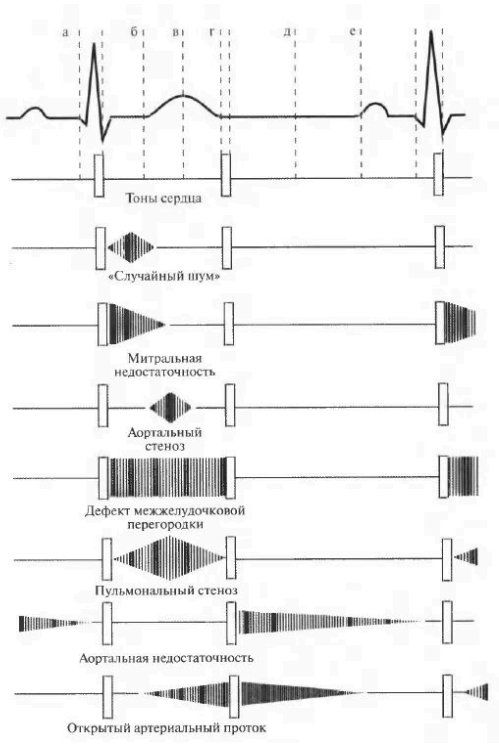


Figure 9.8. *Position of murmurs in the cardiac cycle: a — presystolic; b — protosystolic; c — midsystolic; d — late systolic; e — protodiastolic; f — mid-diastolic (Navruzova, Geppe)*

Causes of functional murmurs

- **physiological (innocent)** — uneven growth of the parts of the heart and the valve rings, false chords (minor cardiac anomalies), a “vibratory” murmur over the pulmonary artery;
- **autonomic dysfunction** — changes in the tone of the papillary muscles (in adolescents);
- **dystrophic changes in the myocardium**, relative valvular insufficiency (organic-functional);
- **increased blood flow velocity** — fever, thyrotoxicosis, physical and mental stress;
- **reduced blood viscosity** — **anemia**.

ECG semiotics

Arrhythmias:

- **Sinus tachycardia** — HR above the age norm, the rhythm source is the sinus node; causes: exertion, fever, anemia, hypoxia, thyrotoxicosis, myocarditis, heart failure, “drop” heart.
- **Sinus bradycardia** — HR below the age norm; vagotonia, athletes, hypothyroidism, raised intracranial pressure, sick sinus syndrome.
- **Sinus (respiratory) arrhythmia** — the RR intervals change with the phase of breathing; it disappears when the breath is held — normal.
- **Extrasystole** — premature contractions: **atrial** (deformed P, narrow QRS, incomplete compensatory pause), **AV nodal**, **ventricular** (wide, deformed QRS without P, discordant ST and T, **complete compensatory pause**). In children it is often functional (autonomic), but also occurs in myocarditis and cardiomyopathy.
- **Paroxysmal tachycardia** — a sharp increase in HR that begins and ends suddenly (160–200 in older children, up to 200–300 at an early age); **supraventricular** (narrow QRS) and **ventricular** (wide QRS — dangerous) forms; the child is restless and pale, and heart failure may develop quickly.
- **Atrial flutter and fibrillation** — rare in children (heart defects, after surgery).

- **WPW syndrome** (Wolff–Parkinson–White) — an accessory pathway: short PQ (<0.1 s), **delta wave** at the start of QRS, widened QRS; predisposition to paroxysmal tachycardia.
- **Blocks:** sinoatrial; **AV blocks** — 1st degree (prolonged PQ), 2nd degree (some QRS complexes are dropped — Mobitz I/II), 3rd degree (complete — the atria and ventricles contract independently, marked bradycardia, Morgagni–Adams–Stokes attacks; congenital — associated with maternal SLE); **bundle branch blocks**.
- **Long QT syndrome** — congenital or drug-induced; risk of fainting and sudden death.

Signs of hypertrophy:

- **left ventricle** — left axis deviation (relative to the age norm), tall R in $V_{5–6}$, deep S in $V_{1–2}$, ST depression and negative T in the left chest leads;
- **right ventricle** — marked right axis deviation, tall R (qR) in $V_{1–2}$, deep S in $V_{5–6}$;
- **left atrium** — wide, double-humped P ("**P mitrale**") in I, II, aVL;
- **right atrium** — tall, pointed P ("**P pulmonale**") in II, III, aVF.

Signs of myocardial damage: reduced voltage of the waves, QRS deformation, arrhythmias and blocks, ST depression of more than 1 mm below the isoline, flattening or inversion of the T wave.

Changes in arterial pressure

- **Arterial hypertension** — **primary (essential)** — most often in adolescents, with obesity and hereditary predisposition; **secondary (symptomatic)** — kidney diseases (glomerulonephritis, pyelonephritis, polycystic disease, renal artery stenosis — the most common cause in children), **coarctation of the aorta**, endocrine diseases (Itsenko–Cushing, pheochromocytoma, thyrotoxicosis, hyperaldosteronism), CNS damage (tumor, injury, encephalitis), drugs (glucocorticoids).
- **Arterial hypotension** — primary (constitutional, autonomic dysfunction — vagotonia) and secondary: adrenal insufficiency, hypothyroidism, pituitary insufficiency, aortic stenosis, shock, collapse, blood loss, dehydration, anaphylaxis.

In more detail — in topic 10 (dystonia syndrome).

9.7. Clinical section

A child with a murmur — diagnostic approach

1. **Determine the phase of the murmur:** diastolic or continuous — **organic**; systolic — continue.
2. **Assess its features:** soft, short, non-radiating, position-dependent — most likely functional; harsh, long, radiating, with a thrill — organic.
3. **Additional signs:** changes in the heart sounds, cardiomegaly, cyanosis, signs of failure, changes in the peripheral pulses, an arm–leg BP difference — organic.
4. **Exclude extracardiac causes**: anemia (CBC), fever, thyrotoxicosis.
5. **Investigations:** ECG, **echocardiography** (the main one), X-ray if necessary.

> **CLINICAL SIGNIFICANCE**
>
> In an infant, **any murmur**, and also a murmur accompanied by **dyspnea, cyanosis, poor weight gain or sweating**, requires a cardiology consultation and echocardiography. In large defects (e.g., a large VSD) there may be **no** murmur in the first weeks of life, because pulmonary vascular resistance is still high.

A child who has fainted

Types of fainting (syncope): **vasovagal** (the most common — prolonged standing, pain, fear, a stuffy room; preceded by nausea, sweating, pallor), **orthostatic** (on standing up abruptly), **sinocarotid**, **post-tussive**, **cardiogenic** (aortic stenosis, hypertrophic cardiomyopathy, tetralogy of Fallot, arrhythmias — complete AV block, long QT, paroxysmal tachycardia). **Dangerous signs of cardiogenic syncope:** fainting during physical exertion, preceding palpitations, fainting while lying down, sudden death at a young age in the family — ECG, Holter and echocardiography are needed.

Practical skills

Skill 1. Auscultation of the heart

1. A warm, quiet room; warm the stethoscope; the child lies on the back, the doctor on the right.
2. Assess the rhythm and HR (if necessary, palpate the pulse simultaneously to detect a pulse deficit).
3. Listen in order: apex → 2nd space right → 2nd space left → base of the xiphoid process → Botkin–Erb point.
4. At each point identify S1 and S2 (S1 coincides with the carotid pulse) and assess their loudness and ratio, splitting and extra sounds.
5. If there is a murmur — describe it by all criteria.
6. Repeat auscultation lying on the left side, standing, and after exertion.
7. Listen over the great vessels (carotid and subclavian arteries, neck veins).
8. Write the conclusion.

Skill 2. Examining the pulse

1. Palpate both radial arteries simultaneously with the 2nd–4th fingers and assess synchrony and equality.
2. Count the rate in one arm for 1 minute.
3. Assess the rhythm; in arrhythmia repeat while the breath is held.
4. Determine the tension, filling, size and shape.
5. Compare the result with the age norm and record it.

Skill 3. Measuring arterial pressure

1. Choose an age-appropriate cuff; the child has been sitting quietly for 5 minutes.
2. Place the cuff 2 cm above the antecubital fossa; the arm at heart level.
3. Find the brachial artery and place the stethoscope.
4. Inflate to 30–40 mm above the point at which the radial pulse disappears; deflate at 2–3 mm per second.
5. Record the systolic and diastolic BP; repeat 2–3 times and take the average.
6. Measure in the other arm and in the legs.
7. Assess by the formula and centile tables.

Skill 4. Performing the Martinet (or Shalkov) functional test

1. Measure the pulse (over 10 seconds) and BP at rest.
2. Ask the child to do 20 deep squats in 30 seconds (or the age-appropriate Shalkov load).

3. Measure the pulse and BP immediately after the exercise, then repeat every minute for 3–5 minutes.
4. Calculate the percentage increase in pulse, the change in BP and the recovery time.
5. Assess the reaction as favorable or unfavorable and write the conclusion.

Skill 5. Analyzing an ECG

1. Check the voltage and paper speed.
2. Determine the rhythm source (is there a P before each QRS).
3. Calculate the HR from the RR interval.
4. Determine the electrical axis (from the QRS in standard leads I and III).
5. Measure the duration of P, PQ, QRS and QT; assess ST and T.
6. Compare the results with the age norm and write the conclusion (e.g., “Sinus rhythm, HR 92/min, respiratory arrhythmia, normal electrical axis — age norm”).

Review questions

1. What are the rules, points and order of cardiac auscultation?
2. How are S1 and S2 formed and how are they distinguished?
3. Name the age-related features of the heart sounds in children. What is embryocardia?
4. How are heart murmurs classified? How is a murmur described?
5. How can a functional murmur be distinguished from an organic one?
6. What are the features of the pericardial friction rub?
7. Which characteristics of the pulse are examined? What is the diagnostic significance of changes in the pulse?
8. How is arterial pressure measured? What is the importance of choosing the cuff?
9. What are the age norms for BP and the formulas for calculating it? What is the ratio of BP in the arms and legs?
10. How are the Martinet, Shalkov and clinoorthostatic tests performed and assessed?
11. Name the age-related features of children's ECG.
12. What are the possibilities and indications of Holter monitoring, ABPM and echocardiography?
13. What changes in the heart sounds and their causes do you know?
14. What are the features of the main organic murmurs (mitral and aortic defects, VSD, PDA)?
15. Name the main arrhythmias seen in children and the ECG signs of hypertrophy.

Test questions

1. Where is the Botkin–Erb point located?
   A) At the apex of the heart
   B) In the 2nd intercostal space right of the sternum
   C) At the attachment of ribs III–IV to the left sternal edge
   D) At the base of the xiphoid process
   E) In the 2nd intercostal space left of the sternum

2. S1 is formed mainly by the closure of which valves?
   A) Aortic and pulmonary
   B) Mitral and tricuspid
   C) Aortic only
   D) Pulmonary only
   E) Venous valves

3. In which period is embryocardia considered normal?
   A) In the first 2 weeks of life
   B) Until 1 year
   C) Until 3 years
   D) At school age
   E) Never

4. Where is S2 heard louder in children aged 2–12 years?
   A) Over the aorta
   B) Over the pulmonary artery
   C) At the apex
   D) At the base of the xiphoid process
   E) The same everywhere

5. Not typical of a functional murmur:
   A) Systolic
   B) Soft timbre
   C) Decreases on standing
   D) Radiates to the axilla
   E) Short

6. A diastolic murmur is:
   A) Always functional
   B) Always organic
   C) Found in anemia
   D) Found in fever
   E) Physiological

7. A pansystolic murmur at the apex, radiating to the left axilla, with a weakened S1. This is typical of:
   A) Mitral stenosis
   B) Mitral insufficiency
   C) Aortic stenosis
   D) Patent ductus arteriosus
   E) Atrial septal defect

8. A continuous systolic-diastolic (“machinery”) murmur is typical of:
   A) Ventricular septal defect
   B) Patent ductus arteriosus
   C) Tetralogy of Fallot
   D) Mitral stenosis
   E) Aortic insufficiency

9. Which condition is the “gallop” rhythm typical of?
   A) A healthy child
   B) Myocarditis
   C) Functional murmur
   D) Respiratory arrhythmia
   E) Vagotonia

10. The average systolic BP in a 5-year-old boy (by the 90 + 2n formula):
   A) 90
   B) 95
   C) 100
   D) 110
   E) 120

11. Normally, BP in the legs compared with the arms is:
   A) 10–20 mm lower
   B) Equal
   C) 10–20 mm higher
   D) 50 mm higher
   E) 2 times lower

12. A rapid and high pulse is typical of which defect?
   A) Aortic stenosis
   B) Aortic valve insufficiency
   C) Mitral stenosis
   D) Pulmonary stenosis
   E) Coarctation of the aorta

13. By how much does the pulse rise in a favorable reaction to the Martinet test?
   A) Up to 25%
   B) 50–75%
   C) 100%
   D) 150%
   E) Unchanged

14. Typical of the newborn's ECG:
   A) Left axis deviation
   B) Right axis deviation (right-axis pattern)
   C) QRS width 0.12 s
   D) Bradycardia
   E) Long PQ

15. A short PQ interval and a delta wave at the start of QRS are typical of:
   A) Complete AV block
   B) WPW syndrome
   C) Ventricular extrasystole
   D) Long QT syndrome
   E) Sinus bradycardia

16. The classic point for listening to the aortic valve:
    A) 2nd intercostal space left of the sternum
    B) 2nd intercostal space right of the sternum
    C) Apex of the heart
    D) Axilla
    E) Base of the xiphoid process

17. Where is the pulmonary valve heard?
    A) At the apex
    B) Below the xiphoid process
    C) 2nd intercostal space, left of the sternum
    D) Botkin–Erb point
    E) 2nd intercostal space, right of the sternum

18. The place where the tricuspid valve is heard:
    A) The lower end of the sternum, the base of the xiphoid process
    B) 5th intercostal space on the left
    C) Apex of the heart
    D) Jugular notch
    E) 2nd intercostal space on the right

19. Where is the Botkin–Erb point?
    A) 2nd intercostal space on the right
    B) Rib VI, right midclavicular line
    C) Axilla
    D) Apex of the heart
    E) The attachment of ribs III–IV to the left sternal edge

20. At which point is S1 best heard?
    A) In the 2nd intercostal space on the right
    B) Between the shoulder blades
    C) At the apex of the heart
    D) In the 2nd intercostal space on the left
    E) In the epigastrium

21. What is embryocardia and when is it normal?
    A) Bradycardia, normal at 1 year
    B) A split sound, at any age
    C) A murmur, normal in adolescents
    D) Gallop rhythm, normal at 5 years
    E) A pendulum rhythm, normal in the first 2 weeks of life

22. From 2 to 12 years, accentuation of S2 over which vessel is considered physiological?
    A) Over the carotid artery
    B) Over the aorta
    C) At the apex
    D) Over the tricuspid valve
    E) Over the pulmonary artery

23. At what age does S2 become equal over the aorta and the pulmonary artery?
    A) 12 years
    B) 1 year
    C) 5 years
    D) 18 years
    E) Never

24. How is a diastolic murmur always regarded?
    A) Physiological
    B) Organic
    C) Inorganic
    D) Functional
    E) Innocent

25. A feature of a functional murmur:
    A) Diastolic, radiating
    B) Increases after exertion
    C) Always long and loud
    D) Systolic, short, soft, non-radiating, changes with position
    E) Harsh, with a cat's purr

26. What is the feature of the pericardial friction rub?
    A) Increases on inspiration, disappears when the breath is held
    B) Does not radiate anywhere, increases on leaning forward and on pressure with the stethoscope, persists when the breath is held
    C) Disappears when the breath is held
    D) Radiates to the axilla
    E) Only in diastole

27. When the breath is held, the pleuropericardial murmur:
    A) Becomes diastolic
    B) Unchanged
    C) Becomes reduplicated
    D) Increased
    E) Absent

28. What does a pulse deficit mean?
    A) Equality of the pulse
    B) The pulse is higher than the HR
    C) No pulse
    D) The pulse is lower than the HR (fibrillation, extrasystole)
    E) Good pulse filling

29. By approximately how much does the pulse rise for each 1 °C rise in temperature in fever?
    A) 2–3 beats
    B) Unchanged
    C) 30 beats
    D) 8–10 beats
    E) 20 beats

30. A rapid, bounding pulse is typical of which condition?
    A) Shock (thready)
    B) Aortic insufficiency, patent ductus arteriosus, thyrotoxicosis
    C) Mitral stenosis
    D) Hypothyroidism
    E) Aortic stenosis

31. When measuring BP in a child, approximately what proportion of the arm circumference should the width of the cuff bladder cover?
    A) 100%
    B) 25%
    C) 10%
    D) 80%
    E) 40%

32. How does a narrow cuff show the BP?
    A) Only raises the pulse
    B) Only raises the diastolic
    C) Does not change it
    D) Overestimated
    E) Underestimated

33. By which phase is diastolic BP determined in the Korotkov method?
    A) Inflating the cuff
    B) Disappearance of the sounds in phase V
    C) The first clear sound in phase I
    D) A murmur in phase II
    E) Disappearance of the pulse

34. The approximate formula for systolic BP in a child under 1 year (n — months):
    A) 105 + 2n
    B) 90 + 2n
    C) 76 + 2n
    D) 100 + n
    E) 60 + n

35. Average systolic BP in a boy over 1 year (n — years):
    A) 76 + 2n
    B) 75 + 2n
    C) 90 + 2n
    D) 60 + n
    E) 105 + 2n

36. When is arterial hypertension diagnosed in centile assessment?
    A) 90th–95th centile
    B) <5th centile
    C) ≥95th centile (on 3 measurements)
    D) <10th centile
    E) 10th–90th centile

37. BP in the legs compared with the arms is usually:
    A) 10–20 mm higher
    B) Cannot be determined
    C) 10–20 mm lower
    D) The same
    E) 50 mm lower

38. If BP in the arms is higher than in the legs, this is a sign of which condition?
    A) Myocarditis
    B) Coarctation of the aorta
    C) Mitral stenosis
    D) Hypothyroidism
    E) Anemia

39. Criteria of a favorable reaction to the Martinet test:
    A) Recovery takes more than 10 minutes
    B) Diastolic BP rises sharply
    C) The pulse rises by ≥50%
    D) The pulse rises by ≤25%, systolic BP rises by 5–10 mm, recovery in 3 minutes
    E) Cyanosis appears

40. What is the electrical axis usually like on the ECG of newborns?
    A) The axis cannot be determined
    B) Normal axis
    C) Deviated to the right (right-axis pattern)
    D) Deviated to the left (left-axis pattern)
    E) Vertical only at 15 years

41. A change that may be normal on the ECG in children:
    A) Negative T wave in V_1–V_3
    B) Ventricular fibrillation
    C) Deep, widened QRS (>0.12 s)
    D) Complete AV block
    E) Marked ST depression

42. What is Holter monitoring?
    A) Continuous 24-hour ECG recording
    B) Phonocardiography
    C) A single BP measurement
    D) X-ray
    E) Echocardiography

43. How is the HR calculated on the ECG?
    A) 60 / RR (in seconds)
    B) QRS / 60
    C) QT / RR
    D) PQ × 60
    E) RR × 60

Clinical cases

Case 1

A 7-year-old child at a preventive check-up. No complaints, physical development appropriate for age. The cardiac borders are normal. On auscultation a short, soft systolic murmur of musical timbre is heard at the apex and at Botkin's point, not radiating anywhere; it is heard well lying down and becomes much weaker on standing and after 10 squats. The heart sounds are clear. ECG — age norm. CBC — Hb 128 g/l.

Questions: 1) How do you assess the murmur? 2) Which signs confirm this conclusion? 3) With which investigation can it be definitively confirmed?

Answer: 1) A functional (innocent, physiological) systolic murmur. 2) Only systolic, short, soft, of musical timbre, non-radiating, changing with body position and exertion (decreasing on standing and after exertion), the heart sounds and cardiac borders are unchanged, the ECG is normal, no complaints, no anemia. 3) Echocardiography — confirms the absence of organic changes (minor anomalies, e.g., a false chord, may be found).

Case 2

An 11-year-old girl had tonsillitis 2 weeks ago. She now complains of weakness, dyspnea and unpleasant sensations in the heart area. On examination: pale skin, HR 118/min (temperature 37.2 °C), soft pulse. The apex beat is weakened, diffuse and displaced to the left. The cardiac border is enlarged 1.5 cm to the left. The heart sounds are muffled, S1 weakened at the apex, “gallop” rhythm, a soft systolic murmur at the apex. ECG: PQ 0.22 s, ST depression, negative T in $V_{5–6}$.

Questions: 1) Which syndrome? 2) How do you assess the tachycardia? 3) What is the origin of the murmur? 4) Explain the ECG changes.

Answer: 1) The syndrome of myocardial damage (myocarditis, probably rheumatic — after tonsillitis). 2) The tachycardia does not correspond to the temperature (at 37.2 °C the pulse should be only slightly raised) — typical of myocarditis. 3) An organic-functional (muscular) murmur — relative mitral insufficiency due to left ventricular dilatation and reduced tone of the papillary muscles. 4) 1st-degree AV block (PQ longer than normal for 11 years), ST depression and negative T — disturbed repolarization in the myocardium (inflammatory and dystrophic changes).

Case 3

In a 14-year-old adolescent, BP of 142/88 mm Hg was found at a school medical check-up. On repeated measurement 3 times (on different days) the result was 138–145/85–90 mm. Height 168 cm, weight 82 kg. Complains of morning headaches. His father has hypertension. The pulses in the arms and legs are equal, BP in the legs is 155/90. Urinalysis is normal.

Questions: 1) How do you assess the BP? 2) Which investigations are needed? 3) Is the hypertension likely to be primary or secondary? Justify.

Answer: 1) The average BP for 14 years is ≈118/74 mm; the values obtained are above the 95th centile and confirmed 3 times — arterial hypertension. 2) ABPM (to exclude “white coat”), ECG and echocardiography (left ventricular hypertrophy), fundoscopy, ultrasound (Doppler) of the kidneys and renal arteries, blood creatinine, electrolytes, glucose, lipids, thyroid hormones. 3) Primary (essential) is likely: adolescence, obesity (BMI ≈29), hereditary predisposition, BP in the legs higher than in the arms (no coarctation), urine unchanged. But secondary causes (renal, endocrine) must be excluded by investigation.

Quick review

Table 9.10. *Key facts on Topic 9*

Question	Answer
Order of auscultation	Apex (mitral) → 2nd right (aorta) → 2nd left (pulmonary a.) → xiphoid process (tricuspid) → Botkin–Erb (3rd–4th left)
S1	Closure of the AV valves; after the long pause, at the apex, coincides with the pulse
S2	Semilunar valves; after the short pause, at the base
Embryocardia	Normal up to 2 weeks, then pathological
Physiological accentuation of S2 over the pulmonary artery	2–12 years
Functional murmur	Systolic, short, soft, non-radiating, position-dependent
Diastolic murmur	Always organic
Mitral insufficiency	Pansystolic at the apex, to the axilla, S1 ↓
Mitral stenosis	Diastolic (presystolic) at the apex, flapping S1, “quail rhythm”
Aortic stenosis	Harsh systolic at the 2nd right, to the carotid arteries
Aortic insufficiency	Protodiastolic at Botkin's point; rapid high pulse, de Musset
VSD	Harsh pansystolic + thrill at the 3rd–4th left sternal edge
PDA	“Machinery” murmur at the 2nd left
Gallop rhythm	Myocarditis, heart failure
BP formula	<1 year: 76 + 2n (months); >1 year: 90 + 2n / 60 + n; girls –5
BP in the legs	10–20 mm higher than in the arms
Hypertension	≥95th centile, 3 times
Martinet test	Pulse ≤25%, recovery 3 minutes — favorable
Newborn ECG	Right-axis pattern, HR 120–160, negative T in $V_{1–4}$
WPW	Short PQ + delta wave
Physiological increase in pulse	8–10 beats per 1 °C
Cardiac auscultation points	Apex — mitral; 2nd space right — aorta; 2nd space left — pulmonary artery; base of the xiphoid process — tricuspid; Botkin–Erb (ribs III–IV, left sternal edge) — aorta
Heart sounds	S1 — systolic, at the apex, after the long pause; S2 — diastolic, at the base; S3 — start of diastole (physiological in thin children); S4 — end of diastole, before S1
Age-specific features of the heart sounds	Embryocardia in the first 2 weeks; from 2–3 months S1 predominates at the apex; 2–12 years S2 accentuated over the pulmonary artery; equal at 12 years
Classification of murmurs (Moshchich)	Organic, organic-functional, functional, physiological (innocent)
Difference between functional and organic murmurs	Functional: systolic, short, soft, non-radiating, position-dependent. Organic: long, harsh, distinct epicenter, radiating, heart sounds changed; diastolic — always organic
Characteristics of the pulse	Rate, rhythm, tension, filling, size, shape (speed), symmetry; respiratory arrhythmia maximal at 4–12 years

Question	Answer
Pulse and temperature	For a 1 °C rise in temperature the pulse rises by 8–10 beats; pulse deficit — fibrillation, extrasystole
Rules of BP measurement	5 minutes' rest; cuff width ≈40% of the arm circumference; phase I — systolic, phase V — diastolic; repeated 2–3 times; the first time in the arms and legs
Normal BP	Newborn 60–70; 1 year 80–90/40–50; 5 years 100/60–65; 10 years 110/70; 15 years 115–120/70–75
BP formulas	<1 year: 76 + 2n (months); >1 year: systolic 90 + 2n, diastolic 60 + n; −5 mm in girls
BP by centiles	<10 low; 10–90 normal; 90–95 high normal; ≥95 hypertension; <5 hypotension
Functional tests	Martinet (orthostatic), Shalkov graded exercise, clinoorthostatic test (COT), pharmacological, bicycle ergometry
ECG protocol	Voltage → rhythm source → HR → electrical axis → waves → intervals (PQ, QRS, QT) → systolic index → ST/T → age-specific features → conclusion
Features of children's ECG	Short waves/intervals; respiratory arrhythmia; right-axis pattern at an early age; negative T in V_1–V_3; deep Q in III; early repolarization (in adolescents)

TOPIC 10

Syndromes of cardiovascular involvement: carditis (endocardium, myocardium, pericardium), dystonia, congenital heart defects, heart and vascular failure

Learning objectives

The student should know:

- the carditis syndrome and its components — the syndromes of endocardial, myocardial and pericardial involvement;
- the main physical signs of acquired heart defects;
- arterial hypertension and hypotension, the autonomic dystonia syndrome;
- the classification of congenital heart defects, the main and additional diagnostic criteria, the clinical picture of the most common defects;
- the causes, types and stages of heart failure;
- the signs of acute vascular insufficiency (fainting, collapse, shock).

The student should be able to:

- identify the leading cardiological syndrome on the basis of physical and instrumental findings;
- determine the hemodynamic group of a congenital heart defect;
- determine the type and stage of heart failure;
- show first-aid measures in a hypercyanotic spell, fainting and collapse.

10.1. Carditis syndrome

Carditis — inflammatory involvement of the heart; the three layers of the heart — **endocardium, myocardium and pericardium** — may be affected separately or together. Simultaneous involvement of all layers is **pancarditis**.

Causes: rheumatic (acute rheumatic fever — after group A β-hemolytic streptococcal infection), nonspecific (viral — Coxsackie B, ECHO, adenoviruses, influenza; bacterial — diphtheria), infective endocarditis, autoimmune diseases (SLE, juvenile arthritis), toxic and allergic damage.

Syndrome of endocardial involvement

It is seen in infective or rheumatic **endocarditis** and **acquired heart defects**. As a result of inflammation the valves become deformed: **insufficiency** (incomplete closure of the cusps, backward flow of blood — regurgitation) or **stenosis** (narrowing of the orifice) develops. The clinical picture mainly reflects involvement of **the mitral** and **aortic** valves.

Table 10.1. *Physical signs of mitral and aortic valve involvement*

Defect	Hemodynamics	Physical and instrumental signs
Mitral valve insufficiency	In systole blood returns from the left ventricle to the left atrium	S1 weakened at the apex; a pansystolic blowing murmur at the apex and Botkin's point, radiating to the left axilla, increasing lying down and after exertion, constant; S2 accentuated over the pulmonary artery; the cardiac border enlarged to the left and upward. ECG — “P mitrale”; echocardiography — mitral regurgitation, enlarged left atrium

Defect	Hemodynamics	Physical and instrumental signs
Mitral stenosis	In diastole the left ventricle does not fill adequately; congestion in the left atrium and lungs	“Mitral flush” of the cheeks, acrocyanosis; a diastolic “cat's purr” at the apex; a flapping S1 at the apex, “quail rhythm”, a diastolic murmur with presystolic accentuation (best heard on the left side); S2 accentuated over the pulmonary artery; the cardiac border upward and to the right. ECG — “P mitrale”, right ventricular hypertrophy
Aortic valve insufficiency	In diastole blood returns from the aorta to the left ventricle	Pallor, “carotid dance”, de Musset's sign, precapillary pulse (Quincke); the pulse rapid and high, the pulse pressure large (diastolic BP sharply low); the apex beat increased, diffuse, displaced to the left and downward; a protodiastolic “pouring” murmur at Botkin's point and along the left sternal edge; S2 weakened over the aorta; “aortic” configuration
Aortic stenosis	Obstruction to the outflow of blood from the left ventricle	Heart pain, dizziness, fainting (on exertion); pallor; the pulse slow and low, systolic BP low; a systolic thrill in the 2nd intercostal space on the right; a harsh midsystolic murmur radiating to the carotid arteries; S2 weakened over the aorta; left ventricular hypertrophy

Distinguishing endocarditis from an established acquired defect is important: endocarditis is an ongoing process, the murmur is less long-lasting and regresses with treatment, the heart chambers are unchanged, it lasts up to 6–8 weeks; **an acquired defect** forms **8–12 months** after the onset of endocarditis — the murmur is long-lasting and widely radiating, and dilatation and hypertrophy of the heart chambers develop.

Signs of **infective endocarditis**: prolonged fever, chills, sweating, weakness, weight loss, pallor (“café-au-lait” color), a new or changed murmur, splenomegaly, petechiae, **Osler nodes** (painful nodules on the fingertips), **Janeway lesions**, **“drumstick fingers”**, emboli; the pathogen in blood culture, **vegetations** on the valves on echocardiography. Risk group — congenital and acquired defects, prosthetic valves, central venous catheters.

Syndrome of myocardial involvement

In **myocarditis** of various etiologies (rheumatic, viral — especially enteroviral, diphtheria), dystrophic and toxic damage, and cardiomyopathies.

Complaints: pain in the heart area (at an early age — restlessness), palpitations, “skipped beats”, dyspnea, weakness, rapid fatigue; in infants — refusal to feed, sweating, vomiting.

Physical signs:

- **pulse changes** — **tachycardia out of proportion to the temperature**, less often bradycardia, arrhythmia;
- the apex beat **weakened, diffuse**, displaced outward;
- the cardiac borders **enlarged** (cardiomegaly);
- **muffled heart sounds**, especially S1 at the apex; splitting, **“gallop” rhythm**;
- **an organic-functional (muscular) systolic murmur** at the apex — relative mitral insufficiency;
- reduced BP;
- signs of heart failure (dyspnea, liver enlargement, edema).

Instrumental signs: ECG — reduced voltage of the waves, QRS deformation, rhythm and conduction disorders (sinus tachycardia, extrasystole, AV blocks, bundle branch block), **ST depression of more than 1 mm**, deformation and inversion of the T wave; X-ray — **CTI >0.5**

(>0.55 at an early age); echocardiography — left ventricular dilatation, **reduced ejection fraction**; blood — raised **CK-MB and troponins**.

> **CAUTION**
> The most important early signs of myocarditis are **tachycardia out of proportion to the temperature**, **muffled heart sounds (especially S1)** and **a gallop rhythm**. An ECG must be taken in weakness, dyspnea and tachycardia that appear 1–3 weeks after tonsillitis or a viral infection.

Syndrome of pericardial involvement

In pericarditis of various etiologies (viral, bacterial, tuberculous, rheumatic, uremic, SLE), accumulation of fluid (blood, transudate) in the pericardial cavity, and systemic diseases.

Dry (fibrinous) pericarditis:

- **severe stabbing** or sharp pain behind the sternum (often in its middle part), **increased by movement, coughing and deep breathing**, decreased on leaning forward; it radiates to the left shoulder and neck;
- **pericardial friction rub** — “creaking”, “scraping”, at the base of the heart or in the area of absolute dullness, **does not radiate anywhere**, variable, increases on leaning forward and on pressure with the stethoscope, **persists when the breath is held**;
- ECG — **ST elevation** in all leads (concordant), then T inversion.

Exudative pericarditis:

- dyspnea, a feeling of heaviness; **forced position** — sitting leaning forward or the knee-elbow position;
- smoothing of the intercostal spaces, distension of the neck veins;
- the apex beat **weakened or not detected**; the apex beat lies **inside** the percussion border;
- cardiac dullness **enlarged in all directions**, **the difference between relative and absolute dullness disappears**;
- **markedly muffled heart sounds**; the friction rub disappears as the fluid increases;
- the pulse rapid, small, **paradoxical**; BP reduced, venous pressure raised;
- X-ray — the cardiac shadow **spherical or triangular**, its arcs smoothed; ECG — **a sharp decrease in the voltage of all waves**; echocardiography — **fluid** in the pericardial cavity (the main diagnostic method).

Cardiac tamponade — fluid accumulates rapidly and compresses the heart: marked dyspnea, cyanosis, tachycardia, paradoxical pulse, a fall in BP, distended neck veins, muffled heart sounds (**Beck's triad**: hypotension + raised venous pressure + muffled heart sounds) — **urgent pericardiocentesis** is required.

Constrictive (adhesive) pericarditis — the pericardial layers thicken and fuse: the heart cannot expand in diastole — ascites, liver enlargement, edema, distended neck veins; **systolic retraction at the site of the apex beat**.



Figure 10.1. *Exudative pericarditis: the cardiac shadow is spherical, its arcs smoothed (Navruzova, Geppe)*

Acute rheumatic fever and carditis

Acute rheumatic fever is a systemic autoimmune disease that develops **2–3 weeks** after a group A β-hemolytic streptococcal infection (tonsillitis, pharyngitis, scarlet fever) and mainly affects **the heart** and joints; most often **at 7–15 years**.

Table 10.2. *Diagnostic criteria of acute rheumatic fever (Kisel–Jones, revised)*

Major criteria	Minor criteria
Carditis (most often with mitral valve involvement)	Arthralgia
Polyarthritis (migratory, large joints)	Fever
Chorea (Sydenham's chorea)	Raised ESR and CRP
Erythema marginatum	Prolonged PQ on the ECG
Rheumatic nodules (subcutaneous)	

The diagnosis requires **2 major** or **1 major + 2 minor** criteria and **evidence of a preceding streptococcal infection** (raised ASO titer, streptococcus in a throat swab).

10.2. Dystonia syndrome

Dystonia syndrome — a disturbance of the regulation of vascular tone; it may occur with a rise or a fall in arterial pressure.

Arterial hypertension syndrome

Definition: systolic and/or diastolic BP **above the 95th centile** for sex, age and height (measured at least 3 times, on different days).

Clinical picture: morning **headache** (often a feeling of pressure in the occiput), dizziness, tinnitus, rapid fatigue, “flies” before the eyes; it often has no symptoms and is found by chance. Physical — the apex beat increased, **S2 accentuated over the aorta**; ECG — first increased left ventricular potentials, then **left ventricular hypertrophy**; fundus — narrowing of the arteries, dilatation of the veins.

Types:

- **primary (essential)** — in adolescents, with obesity, low activity, high salt intake, stress and hereditary predisposition;
- **secondary (symptomatic)** — **renal** (glomerulonephritis, pyelonephritis, polycystic disease, hydronephrosis, renal artery stenosis — the most common cause in children), **cardiovascular**

(coarctation of the aorta, aortic valve insufficiency), **endocrine** (pheochromocytoma, Itsenko–Cushing syndrome, adrenogenital syndrome, hyperaldosteronism, thyrotoxicosis), **neurogenic** (brain tumors, injury, encephalitis), drugs (glucocorticoids, sympathomimetics).

Hypertensive crisis — a sudden sharp rise in BP: the general condition suddenly worsens, severe headache, visual disturbance (flies, a veil), nausea, vomiting, trembling, pallor or flushing of the face, a feeling of fear; seizures may occur.

> **CLINICAL SIGNIFICANCE**
> In a young child (especially under 10 years) severe arterial hypertension is almost always **secondary** — first of all, kidney diseases and coarctation of the aorta must be excluded (urinalysis, renal ultrasound and Doppler, arm–leg BP difference).

Arterial hypotension syndrome

Definition: BP **below the 5th–10th centile**. **Signs:** weakness, rapid fatigue, sensitivity to cold, dizziness and headache (especially in the morning, on exertion, in a stuffy room), a tendency to faint, pallor, cold and moist palms and soles, tachycardia; on auscultation sometimes a “gallop” rhythm; on ECG ST depression, flattened T.

Causes: constitutional (predominance of parasympathetic tone — vagotonia), cardiac (aortic hypoplasia, aortic stenosis), endocrine (hypoglycemia, **adrenal insufficiency** — Addison disease, hypothyroidism, pituitary insufficiency), blood loss, dehydration, intoxication, anaphylaxis, drugs.

Autonomic dystonia syndrome (autonomic dysfunction)

A disturbance of the balance between the sympathetic and parasympathetic divisions of the autonomic nervous system; most often in adolescence (prepubertal and pubertal periods), more often in girls. Cardiovascular disturbances (palpitations, cardialgia, labile BP, fainting) are often leading.

Table 10.3. *Signs of the two types of autonomic tone*

Sign	Sympathicotonia	Vagotonia (parasympathicotonia)
Skin	Pale, dry, cold; white dermographism	Marbled, moist, acrocyanosis; red, persistent dermographism; sweating
Body temperature	Tends to rise (subfebrile temperature)	Tends to fall
Pupils	Wide	Narrow
HR	Tachycardia	Bradycardia, increased respiratory arrhythmia
BP	Tends to rise	Tends to fall
Mood	Excitable, restless, little sleep	Sluggish, drowsy, depressive, tires quickly
Digestion	Tendency to constipation, little saliva	Nausea, tendency to diarrhea, much saliva
Other	Thinness, good appetite, tolerates cold	Tendency to obesity, dislikes stuffy rooms, tendency to faint, allergic reactions

The diagnosis is made by exclusion (organic heart, endocrine and neurological diseases are excluded); **the clinoorthostatic test** and **cardiointervalography** help to assess autonomic tone, reactivity and support.

10.3. Congenital heart defects

Congenital heart defects (CHD) are anatomical defects of the heart and great vessels formed in utero; they occur in **6–8** of every 1000 live births; more than 90 variants are known.

Diagnostic criteria

Table 10.4. *Main and additional diagnostic criteria of congenital heart defects (Kildiyarova)*

Criteria	Manifestation
Main — subjective	Dyspnea, cyanosis or pallor, weight deficit, frequent bronchopulmonary diseases
Main — objective	Change in skin color, hypercyanotic spells, chest deformity (“cardiac hump”), changes in the characteristics of the pulse, cardiomegaly or displacement of the cardiac borders, organic murmurs
Main — instrumental	Strain or hypertrophy of the chambers on ECG; a shunt on echocardiography; cardiomegaly, an anomaly of cardiac position, altered pulmonary blood flow, pulmonary hypertension on X-ray
Additional	Delayed growth and statomotor development (hypoxia), late teething, sweating on exertion; the shade of cyanosis (crimson — pulmonary hypertension, Ebstein anomaly; bluish — single ventricle, tetralogy of Fallot; violet — Eisenmenger syndrome; dark blue — transposition of the great vessels); the location of cyanosis (only in the legs — preductal coarctation; only in the arms — transposition and aortic hypoplasia); sites of thrill, extracardiac radiation of the murmur
Special	Cardiac catheterization (probing), angiocardiography, electrophysiological study

Classification

The classification of CHD is based on **the hemodynamic disturbance** and **the presence or absence of cyanosis**.

Table 10.5. *Hemodynamic classification of congenital heart defects*

Hemodynamics	Acyanotic (“white”)	Cyanotic (“blue”)
Increased pulmonary blood flow (hypervolemia)	Patent ductus arteriosus (PDA), ventricular septal defect (VSD), atrial septal defect (ASD), atrioventricular canal	Transposition of the great arteries, truncus arteriosus, Eisenmenger complex
Reduced pulmonary blood flow (hypovolemia)	Isolated pulmonary stenosis	Tetralogy of Fallot, tricuspid atresia, transposition with pulmonary stenosis, Ebstein anomaly
Reduced systemic blood flow	Isolated aortic stenosis, coarctation of the aorta	—
Hemodynamics undisturbed	Dextrocardia, aortic arch anomalies, a small VSD in the muscular part (Tolochinov–Roger disease)	—

The five defects most common in clinical practice: **VSD, ASD, PDA, coarctation of the aorta, tetralogy of Fallot**.

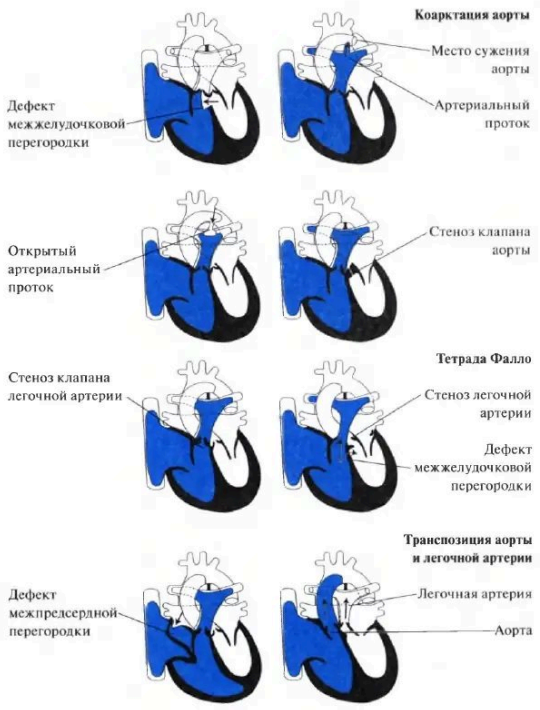


Figure 10.2. *Pathological changes in congenital heart defects (diagram) (Navruzova, Geppe)*

Defects with a left-to-right shunt (VSD, ASD, PDA) — common signs

- frequent and prolonged **bronchopulmonary diseases** at an early age (pulmonary plethora);
- **hypotrophy**, lagging physical development;
- dyspnea, tiring and sweating during feeding;
- crimson acrocyanosis (in pulmonary hypertension);
- **cardiomegaly** (“cardiac hump”, a diffuse cardiac impulse, borders enlarged in all directions), a **systolic thrill** to the left of the sternum;
- **accentuation and splitting of S2 over the pulmonary artery**;
- on X-ray **increased lung markings** (arterial hypervolemia), bulging of the pulmonary artery arc, an enlarged cardiac shadow.

Table 10.6. *Distinguishing signs of the most common acyanotic defects*

Defect	Murmur	ECG	Echocardiography
VSD	A harsh, loud pansystolic murmur in the 3rd–4th intercostal space at the left sternal edge, widely radiating, systolic thrill	Hypertrophy of the left or both ventricles	The defect is visible (size, location), left-to-right shunt
VSD in the muscular part (Tolochinov–Roger)	A loud, scraping systolic murmur in the 4th intercostal space on the left — the only sign (“much ado about nothing”)	Normal	A small defect, often closes spontaneously
ASD	A low- or medium-intensity systolic murmur in the 2nd–3rd intercostal space on the left, radiating to the interscapular area; fixed splitting of S2	Right ventricular and atrial strain, incomplete right bundle branch block	The defect, enlargement of the right chambers

Defect	Murmur	ECG	Echocardiography
PDA	A continuous systolic-diastolic (“machinery”) murmur in the 2nd intercostal space on the left and below the clavicle (only systolic before 1 year); rapid, bounding pulse, low diastolic BP	First right, then left ventricular strain	Patent ductus arteriosus, enlarged left chambers

Tetralogy of Fallot

Tetralogy of Fallot — the most common **“blue”** defect; it consists of four components:

1. **pulmonary stenosis** (narrowing of the right ventricular outflow tract);
2. **a large VSD**;
3. **right ventricular hypertrophy**;
4. **dextroposition of the aorta** (the aorta “overrides” the ventricular septum).

Hemodynamics: reduced pulmonary blood flow, mixed (venous) blood enters the systemic circulation — **central cyanosis**.

Diagnostic criteria:

- **marked cyanosis** (usually increasing from 3–6 months), **hypercyanotic spells**;
- forced position — **squatting** (or lying with the legs drawn up to the abdomen) — systemic resistance rises and pulmonary blood flow increases;
- **“drumstick fingers”** and **“watch-glass” nails**;
- **weakening of S2** over the pulmonary artery;
- **a harsh systolic murmur** over the pulmonary artery (2nd–3rd intercostal space on the left) (stenosis);
- **polycythemia** in the blood (Hb up to 250 g/l, erythrocytes up to 8×10^{12}/l), raised hematocrit;
- ECG — **right ventricular hypertrophy**, right axis deviation;
- X-ray — a heart shaped like **a “wooden shoe” (“Dutch clog”)** (raised apex, deep “cardiac waist”), **reduced lung markings**.

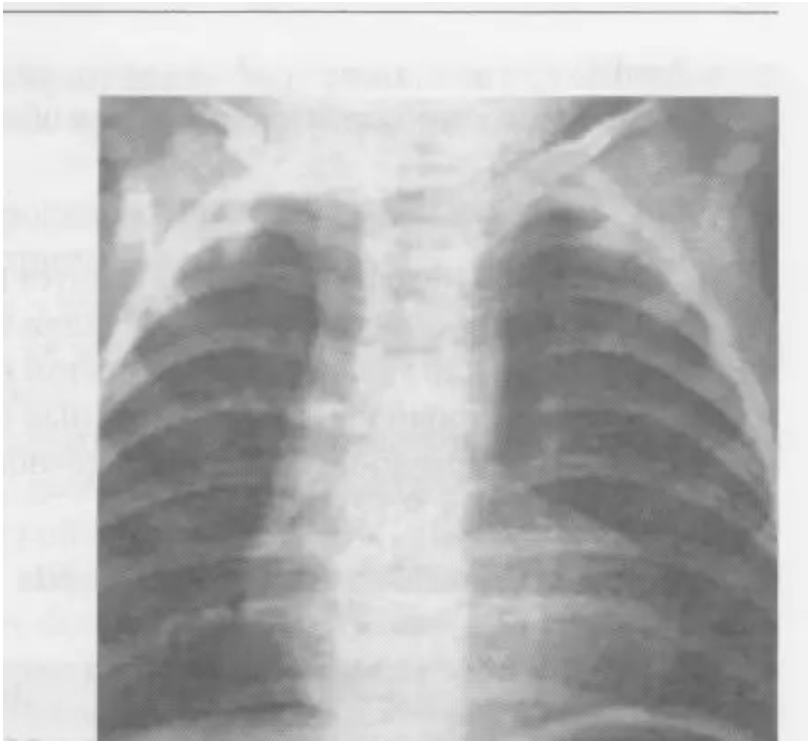


Figure 10.3. *X-ray in tetralogy of Fallot: a "wooden shoe"-shaped heart, reduced lung markings (Kildiyarova, Figure 9.25)*

The hypercyanotic spell is the most dangerous complication of tetralogy of Fallot; most often **at 2–5 years** (from 6 months), provoked by exertion, crying, feeding, defecation, minor surgical procedures, excitement. Mechanism — **spasm** of the right ventricular outflow tract → almost all the venous blood enters the aorta → CNS hypoxia. Signs: restlessness, **marked cyanosis** and dyspnea (deep rapid breathing), tachycardia, **weakening or disappearance** of the murmur (pulmonary blood flow sharply reduced), a fall in BP; in a severe spell — fainting, seizures, coma.

CLINICAL SIGNIFICANCE

First aid in a hypercyanotic spell: calm the child, put the child in **the knee-chest position** (pressing the knees to the chest — peripheral resistance increases), **humidified oxygen**, on the doctor's order **morphine** (or promedol) IM or IV, **sodium bicarbonate** (correction of acidosis), **β-blockers** (propranolol); infusion. Cardiac glycosides **must not be given** (they increase the spasm).

Coarctation of the aorta

Coarctation of the aorta — narrowing in the aortic isthmus (at the transition from the arch to the descending part). There are **two regimes** in the hemodynamics: **arterial hypertension** in the upper part of the body and **hypotension** in the lower part.

Diagnostic criteria: in the newborn poor weight gain, early heart failure (worsening sharply when the ductus arteriosus closes); at an older age — headache, nosebleeds, pain and coldness in the legs; **an "athletic" shoulder girdle, thin legs**; **a BP gradient between the arms and legs** (high in the arms, low in the legs); **weak or absent femoral pulses**; pulsation of the intercostal arteries; a systolic murmur between the shoulder blades and in the 2nd intercostal space on the left; ECG — **left ventricular hypertrophy**; X-ray — a dilated ascending aorta, an enlarged left ventricle, **rib notching** from 8–10 years; echocardiography — the narrowing and turbulent flow.

Transposition of the great arteries

The aorta arises from the right ventricle and the pulmonary artery from the left ventricle — the systemic and pulmonary circulations work **in parallel**; life is possible only thanks to mixing of blood through a patent foramen ovale, PDA or VSD. **Marked generalized cyanosis from birth** (not decreasing with oxygen), dyspnea, rapidly developing heart failure; on X-ray an **"egg on its side"**-shaped heart with a narrow vascular bundle. Urgent surgical treatment (or balloon atrial septostomy) is required.

10.4. Heart failure syndrome

Heart failure — the inability of the heart to supply the organs and tissues with the required amount of blood at a normal or increased venous return; the inability of the heart to convert the venous inflow into an adequate cardiac output.

Table 10.7. *Main diseases with heart failure by age (Kildiyarova)*

Age	Diseases
Up to day 7 of life	CHD (hypoplastic left heart syndrome, coarctation of the aorta, complex defects), persistent fetal circulation, respiratory distress syndrome
7 days – 1 month	CHD (transposition of the great arteries, coarctation of the aorta, defects with a left-to-right shunt), fibroelastosis, nonspecific myocarditis
1 month – 1 year	CHD (VSD, PDA, coarctation of the aorta), nonspecific myocarditis, rhythm and conduction disorders
Over 1 year	Myocarditis (nonspecific, rheumatic), congenital and acquired defects, cardiomyopathies, infective endocarditis, arrhythmias, diseases of other organs (lung, liver, kidney, endocrine)

Types: by rate of development — **acute** and **chronic**; by the chamber mainly affected — **left ventricular** and **right ventricular** failure.

Left ventricular failure

The left atrium and ventricle are affected — **congestion in the pulmonary circulation**: the respiratory surface of the lungs and oxygen diffusion decrease, and hypoxia and acidosis increase.

- nonspecific signs: restlessness, irritability, disturbed sleep, loss of appetite, nausea, vomiting, sometimes abdominal pain;
- **dyspnea** — the earliest, and in infants **the main** sign; it decreases with oxygen and with the head raised; flaring of the nasal alae, participation of the accessory muscles; **cough** (a dry night cough is an early sign), orthopnea in a severe condition;
- **tachycardia** — progressive, **out of proportion to the temperature, persisting in sleep**, resistant to treatment other than cardiac glycosides;
- peripheral or central cyanosis;
- **congestive rales** in the lungs — **fine-bubbling non-consonant moist rales**, symmetric, **in the posterior lower parts**, not disappearing after coughing; the percussion sound is shortened in these areas;
- the cardiac borders enlarged to the left, muffled heart sounds, **"gallop" rhythm**, accentuation or splitting of S2 over the pulmonary artery;
- in a severe condition — **pulmonary edema** (frothy, pink sputum, moist rales over the whole lung);

- X-ray — increased lung markings (venous congestion), reduced transparency in the area of the lung roots.

Right ventricular failure

The right atrium and ventricle are affected — **congestion in the systemic circulation**.

- rapid fatigue, lethargy, weakness, reduced appetite, sweating;
- dyspnea, tachycardia, cyanosis;
- **liver enlargement** — the earliest and most constant sign (first the left lobe, then the right); the liver is soft, elastic, **tender**; **Plesch's sign** (distension of the neck veins increases on pressure over the liver);
- **distension of the neck veins** (increases in the horizontal position);
- **edema** — first in the legs (in a lying child the sacrum and lower back, in infants the face and scrotum), then generalized — **anasarca**, **ascites**, hydrothorax, hydropericardium; diuresis decreases (**oliguria**).

> **CAUTION**
> In an infant the **earliest signs** of heart failure are **tachypnea**, **tachycardia** and **liver enlargement**; peripheral edema appears late and rarely at this age. Early signs: tiring during feeding, sweating (especially of the head), poor weight gain.

Stages of chronic heart failure

Chronic heart failure usually involves disturbances in **both circulations**; its severity is assessed by stages based on the N.D. Strazhesko–V.Kh. Vasilenko classification.

Table 10.8. *Stages of circulatory failure in children (Makarova V.I., Babikova I.V., 2010)*

Signs	I	IIA	IIB	III
Left ventricle				
HR (above normal)	On exertion	15–30%	30–50%	50–60%
RR (above normal)	On exertion	30–50%	50–70%	70–100%
Rhythm disturbance	—	—	+	++
Cough	—	±	+	++
Acrocyanosis	—	±	+	++
Moist rales in the lungs	—	+	++	++
Blood-streaked sputum	—	—	+	++
Pre-edema / pulmonary edema	—	—	—	+
Right ventricle				
Reduced diuresis	—	+	++	+++
Tissue pastosity	—	+	+	+
Peripheral edema	—	—	+	++
Fluid in the cavities	—	—	—	+
Distended neck veins	—	—	+	++
Liver enlargement	—	1–2 cm	3–4 cm	>4 cm

Signs	I	IIA	IIB	III
General				
Acid-base shift	—	—	+	++
Irreversible changes in the internal organs	—	—	—	+

Table 10.9. *Degrees of heart failure (Belozerov Yu.M. et al., 1994)*

Degree	Signs
I	Tachycardia (HR 20–30% above normal) and dyspnea (RR 30–50% above normal) at rest; mucosal cyanosis disappearing with oxygen; muffled heart sounds, enlarged borders
IIA	HR 30–50%, RR 50–70% above normal; mucosal cyanosis, acrocyanosis, periorbital edema; the liver projects 2–3 cm below the costal arch; stroke volume reduced by 20–30%
IIB	The above + pulmonary congestion, oliguria, peripheral edema
III (decompensation)	HR 50–60%, RR 70–100% above normal; incipient pulmonary edema, hepatomegaly, peripheral edema, ascites; in the terminal period — bradypnea, bradycardia, a fall in BP, muscle hypotonia, areflexia

10.5. Acute vascular insufficiency syndrome

Three clinical forms of acute vascular insufficiency: **fainting (syncope), collapse, shock**.

Fainting — a brief loss of consciousness due to acute cerebral hypoxia:

- **prodromal period:** general weakness, darkening before the eyes, ringing in the ears, nausea, sweating — if the child is laid down at this time, fainting can be prevented;
- **sudden brief loss of consciousness** (seconds, minutes), marked pallor, weakening of breathing and circulation;
- dilated pupils, BP sharply reduced, muffled heart sounds, weak pulse; tonic-clonic seizures may occur.

Collapse — an acute fall in vascular tone:

- the condition suddenly worsens, marked weakness, apathy; **consciousness is preserved**, but the child is indifferent;
- pale skin, acrocyanosis, cold clammy sweat; breathing shallow and rapid;
- **the pulse small, rapid, barely detectable**; **BP sharply reduced** (systolic and diastolic); the veins collapsed; the heart sounds loud at first, then muffled.

The main sign distinguishing fainting from collapse — **in fainting consciousness is lost, in collapse it is preserved**.

Shock — an acutely developing life-threatening condition caused by an extremely strong stimulus (injury, infection — septic shock, allergen — anaphylactic shock, blood loss — hypovolemic shock, heart — cardiogenic shock): a severe disturbance of organ perfusion; it has **compensated** (BP normal, tachycardia, cold limbs, capillary refill >2 s, oliguria) and **decompensated** (hypotension, impaired consciousness) stages; its severity is assessed by BP and the character of the pulse.

> **CLINICAL SIGNIFICANCE**
> **First aid in fainting:** lay the child horizontally, **raise the legs by 30–40 cm**, loosen the collar and belt, fresh air, splash cold water on the face, hold ammonia (smelling salts) to the nose. If the child does not regain consciousness or fainting recurs — exclude a cardiogenic cause (ECG, echocardiography, Holter).

10.6. Clinical section

Distinguishing cardiological syndromes

Table 10.10. *Distinguishing signs of the syndromes of endocardial, myocardial and pericardial involvement*

Sign	Endocarditis	Myocarditis	Pericarditis (exudative)
Main complaint	Often few; fever, intoxication in infective endocarditis	Weakness, dyspnea, palpitations	Retrosternal pain (in dry pericarditis), dyspnea
The apex beat	Normal (changes once the defect has formed)	Weakened, diffuse, displaced	Weakened or absent, inside the border
Cardiac borders	Normal at first	Enlarged (mainly to the left)	In all directions, relative = absolute
Heart sounds	Valve-specific changes	Muffled, S1 weakened, gallop	Markedly muffled
Murmur	Organic, constant, at the valve-specific site	Functional-muscular systolic	Pericardial friction rub (in dry pericarditis)
ECG	Strain of the chambers	Reduced voltage, arrhythmia, blocks, ST-T changes	Sharply reduced voltage, ST elevation (in dry pericarditis)
Echocardiography	Valve changes, vegetations, regurgitation	Dilatation, reduced ejection fraction	Fluid in the pericardium

A cyanotic newborn: heart or lung?

Hyperoxia test: if PaO_2 (SpO_2) rises significantly (PaO_2 >150 mm) after breathing 100% oxygen for 10–15 minutes — **a pulmonary cause**; if it hardly rises (PaO_2 <100 mm) — **a "blue" congenital heart defect** (right-to-left shunt). In addition: in lung disease — dyspnea, rales, lung changes on X-ray; in a heart defect — cyanosis with no or little dyspnea, a murmur, changes in the cardiac shadow and lung markings on X-ray.

Practical skills

Skill 1. Examining a cardiac patient and identifying the leading syndrome

1. Take the complaints and history (preceding infection, heredity, signs since birth).
2. General inspection: position, color, edema, "drumstick fingers", physical development.
3. Inspect the cardiac area and vessels, palpate the apex beat, detect a thrill.
4. Percuss the cardiac borders.
5. Auscultation: heart sounds, extra sounds, murmurs, pericardial friction rub.
6. Examine the pulse and BP in the arms and legs.
7. Liver size, neck veins, rales at the lung bases.
8. Group the signs found into syndromes; plan ECG, echocardiography and X-ray.

Skill 2. Determining the stage of heart failure

1. Count the RR and HR and calculate the percentage above the age norm.
2. Detect cyanosis, acrocyanosis, cough, moist rales in the lungs.
3. Determine the liver size (cm below the costal arch); test Plesch's sign.

4. Assess the neck veins, peripheral edema, pastosity, ascites; determine diuresis.
5. Compare the data with the table of stages and determine the stage and type (left, right, total).

Skill 3. First aid in a hypercyanotic spell and fainting

1. Calm the child and eliminate the provoking factor.
2. In a hypercyanotic spell — the knee-chest position; in fainting — horizontal, with the legs raised.
3. Fresh air, humidified oxygen.
4. Monitor the pulse, RR, SpO_2 and BP.
5. Medications on the doctor's order (in a hypercyanotic spell — morphine, sodium bicarbonate, propranolol).
6. Decide on hospitalization.

Skill 4. Determining the BP gradient between the arms and legs

1. Measure BP on the upper arm with an age-appropriate cuff.
2. Measure BP on the thigh with a wide cuff (the child lying prone), listening over the popliteal artery.
3. Palpate the femoral and radial pulses simultaneously to detect delay or weakness.
4. Assessment: normally BP in the legs is 10–20 mm higher; if it is lower in the legs — suspect coarctation of the aorta.

Review questions

1. What is the carditis syndrome? What are its components?
2. What are the physical signs of mitral valve insufficiency and mitral stenosis?
3. What are the signs of aortic valve insufficiency and aortic stenosis?
4. How can endocarditis be distinguished from an established acquired defect? What are the signs of infective endocarditis?
5. Name the clinical and instrumental signs of the syndrome of myocardial involvement.
6. What are the signs of dry and exudative pericarditis? What is cardiac tamponade?
7. Name the diagnostic criteria of acute rheumatic fever.
8. How is the arterial hypertension syndrome determined? What are the causes of secondary hypertension?
9. Name the signs of the arterial hypotension and autonomic dystonia syndromes.
10. What are the main and additional diagnostic criteria of congenital heart defects?
11. Name the hemodynamic classification of congenital heart defects.
12. What are the common and distinguishing signs of VSD, ASD and PDA?
13. What are the components and diagnostic criteria of tetralogy of Fallot? What help is given in a hypercyanotic spell?
14. What are the signs of coarctation of the aorta?
15. Name the signs of left and right ventricular failure, the stages of chronic heart failure and the forms of acute vascular insufficiency.

Test questions

1. Not part of the carditis syndrome:
   A) Endocardial involvement
   B) Myocardial involvement
   C) Pericardial involvement
   D) Coarctation of the aorta
   E) Pancarditis

2. A diastolic murmur with presystolic accentuation at the apex and a flapping S1 are typical of:
   A) Mitral insufficiency
   B) Mitral stenosis
   C) Aortic stenosis
   D) Aortic insufficiency
   E) VSD

3. “Carotid dance”, de Musset's sign, a rapid and high pulse are typical of:
   A) Aortic stenosis
   B) Aortic valve insufficiency
   C) Mitral stenosis
   D) Tetralogy of Fallot
   E) Myocarditis

4. How long after the onset of endocarditis does an acquired heart defect form?
   A) At 1 week
   B) At 1 month
   C) In 8–12 months
   D) In 5 years
   E) Never

5. An important sign of myocarditis:
   A) Tachycardia out of proportion to the temperature
   B) Loud heart sounds
   C) Diastolic murmur
   D) Reduced cardiac borders
   E) Bradycardia and raised BP

6. Not typical of the pericardial friction rub:
   A) Does not radiate anywhere
   B) Increases on pressure with the stethoscope
   C) Disappears when the breath is held
   D) Variable
   E) Increases on leaning forward

7. On the ECG in exudative pericarditis:
   A) A sharp decrease in the voltage of the waves
   B) Tall R waves
   C) Delta wave
   D) Long PQ
   E) No change

8. Not a major criterion of acute rheumatic fever:
   A) Carditis
   B) Polyarthritis
   C) Chorea
   D) Fever
   E) Erythema marginatum

9. The most common cause of secondary arterial hypertension in children:
   A) Kidney diseases
   B) Obesity
   C) Vagotonia
   D) Anemia
   E) Hypothyroidism

10. An acyanotic defect with increased pulmonary blood flow:
    A) Tetralogy of Fallot
    B) Ventricular septal defect
    C) Aortic stenosis
    D) Pulmonary stenosis
    E) Transposition of the great arteries

11. Not part of tetralogy of Fallot:
    A) Pulmonary stenosis
    B) VSD
    C) Right ventricular hypertrophy
    D) Dextroposition of the aorta
    E) Patent ductus arteriosus

12. What position is the child given in a hypercyanotic spell?
    A) Head low, legs high
    B) Knee-chest position
    C) Sitting, legs hanging down
    D) Lying prone
    E) Standing

13. A low-intensity systolic murmur in the 2nd intercostal space on the left and fixed splitting of S2 are typical of:
    A) VSD
    B) ASD
    C) PDA
    D) Tetralogy of Fallot
    E) Coarctation of the aorta

14. The earliest and most constant sign of right ventricular failure:
    A) Moist rales in the lungs
    B) Liver enlargement
    C) Pulmonary edema
    D) Night cough
    E) Bradycardia

15. The main sign distinguishing fainting from collapse:
    A) A fall in BP
    B) Pallor
    C) Loss of consciousness
    D) Tachycardia
    E) Cold sweat

16. What is simultaneous involvement of all three layers of the heart called?
    A) Pericarditis
    B) Myocarditis
    C) Cardiomyopathy
    D) Endocarditis
    E) Pancarditis

17. After which pathogen does acute rheumatic fever develop?
    A) Group A β-hemolytic streptococcus
    B) Pneumococcus
    C) Virus (Coxsackie)
    D) Meningococcus
    E) Staphylococcus

18. The murmur typical of mitral valve insufficiency:
    A) A harsh midsystolic murmur over the aorta
    B) A protodiastolic murmur at Botkin's point
    C) A pansystolic blowing murmur at the apex, radiating to the left axilla
    D) A systolic murmur in the epigastrium
    E) A diastolic murmur at the apex

19. Which defect is typical of a “quail rhythm” and a flapping S1 at the apex?
    A) Aortic insufficiency
    B) Tetralogy of Fallot
    C) Mitral stenosis
    D) Aortic stenosis
    E) Mitral insufficiency

20. What is the pulse pressure in aortic valve insufficiency?
    A) Only the systolic is low
    B) Cannot be determined
    C) Small
    D) Unchanged
    E) Large (diastolic BP sharply low)

21. What is the pulse like in aortic stenosis?
    A) Slow and low
    B) Rapid and high
    C) Paradoxical
    D) Unchanged
    E) Alternating

22. Approximately how long after the onset of endocarditis does an acquired heart defect form?
    A) 5 years
    B) 8–12 months
    C) 1–2 weeks
    D) 3 months
    E) 1 month

23. Signs typical of infective endocarditis:
    A) Hypertension
    B) Sydenham's chorea
    C) Osler nodes, Janeway lesions, splenomegaly, “café-au-lait” pallor
    D) Erythema marginatum
    E) Migratory joint pain

24. The cardiothoracic index (CTI) on X-ray in myocarditis:
    A) CTI >0.5 (>0.55 at an early age)
    B) Never changes
    C) 0,4–0,45
    D) 0,45
    E) <0,4

25. Markers raised in the blood in the diagnosis of myocarditis:
    A) Glucose
    B) CK-MB and troponins
    C) Creatinine
    D) Bilirubin
    E) Cholesterol

26. When does the pain decrease in dry (fibrinous) pericarditis?
    A) On deep inspiration
    B) On movement
    C) On coughing
    D) On leaning forward
    E) On lying down

27. What ECG change occurs in dry pericarditis?
    A) ST depression only in V_1
    B) Complete AV block
    C) Absence of waves
    D) Shortened PQ
    E) ST elevation in all leads (concordant)

28. Beck's triad (cardiac tamponade):
    A) Cyanosis + dyspnea + edema
    B) Hepatomegaly + ascites + edema
    C) Fever + arthralgia + carditis
    D) Chorea + erythema + nodules
    E) Hypotension + raised venous pressure + muffled heart sounds

29. The major criteria of rheumatic fever:
    A) Splenomegaly, petechiae
    B) Carditis, polyarthritis, chorea, erythema marginatum, rheumatic nodules
    C) Only raised ASO
    D) Cyanosis, dyspnea
    E) Arthralgia, fever, raised ESR

30. What is needed for the diagnosis of rheumatic fever?
    A) Only fever
    B) Only an ECG change
    C) Only 1 minor criterion
    D) Only ASO
    E) 2 major or 1 major + 2 minor criteria and evidence of a streptococcal infection

31. The most common cause of secondary arterial hypertension in children:
    A) Kidney diseases (including renal artery stenosis)
    B) Anemia
    C) Hypothyroidism
    D) Obesity
    E) Pheochromocytoma

32. In how many of every 1000 live births do congenital heart defects occur?
    A) 50
    B) 1–2
    C) 6–8
    D) 20–25
    E) 100

33. Which of the following is a “blue” (cyanotic) defect?
    A) Atrial septal defect
    B) Ventricular septal defect
    C) Coarctation of the aorta
    D) Tetralogy of Fallot
    E) Patent ductus arteriosus

34. The shade of cyanosis in transposition of the great arteries:
    A) Violet
    B) Whitish
    C) Yellow
    D) Dark blue
    E) Crimson

35. Defects with a left-to-right shunt include:
    A) Eisenmenger
    B) VSD, ASD, PDA
    C) Transposition
    D) Ebstein
    E) Tetralogy of Fallot

36. A common sign of defects with a left-to-right shunt:
    A) Isolated cyanosis of the legs
    B) Squatting position
    C) Frequent bronchopulmonary diseases, increased lung markings
    D) Constant dark blue cyanosis
    E) Polycythemia

37. What is Tolochinov–Roger disease?
    A) A small VSD in the muscular part
    B) Mitral stenosis
    C) ASD
    D) Tetralogy of Fallot
    E) Patent ductus arteriosus

38. The location of the murmur in VSD:
    A) In the epigastrium
    B) In the 3rd–4th intercostal space at the left sternal edge, loud pansystolic
    C) Diastolic at the apex
    D) Diastolic in the 2nd space on the right
    E) Only diastolic between the shoulder blades

39. The characteristic auscultatory sign of ASD:
    A) Continuous “machinery” murmur
    B) Flapping S1
    C) Fixed splitting of S2
    D) Quail rhythm
    E) Pericardial friction rub

40. The murmur in patent ductus arteriosus:
    A) No murmur
    B) Systolic in the epigastrium
    C) Pansystolic at the apex
    D) A continuous systolic-diastolic (“machinery”) murmur in the 2nd space on the left and below the clavicle
    E) Only diastolic, at the apex

41. The components of tetralogy of Fallot:
    A) Transposition, atresia
    B) Pulmonary stenosis, a large VSD, right ventricular hypertrophy, dextroposition of the aorta
    C) Mitral stenosis, aortic insufficiency
    D) ASD, PDA, VSD, coarctation
    E) Only VSD and ASD

42. The shape of the heart on X-ray in tetralogy of Fallot:
    A) “Mitral” configuration
    B) Spherical
    C) The shape does not change
    D) “Wooden shoe” (Dutch clog)
    E) Triangular

43. What blood change is found in tetralogy of Fallot?
    A) Polycythemia (Hb up to 250 g/l)
    B) Anemia
    C) Thrombocytopenia
    D) Eosinophilia
    E) Leukopenia

44. Why does the squatting position bring relief in tetralogy of Fallot?
    A) Blood pressure falls
    B) The lungs are compressed
    C) Cyanosis increases
    D) The heart becomes smaller
    E) Systemic resistance rises and pulmonary blood flow increases

45. The mechanism of a hypercyanotic spell in tetralogy of Fallot:
    A) Fluid in the pericardium
    B) Left ventricular failure
    C) Aortic valve insufficiency
    D) Spasm of the right ventricular outflow tract, venous blood enters the aorta, CNS hypoxia
    E) Closure of the foramen ovale

46. What happens to the murmur during a hypercyanotic spell?
    A) Becomes diastolic
    B) Unchanged
    C) Only S1 is heard
    D) Weakened or absent
    E) Increased

47. BP in coarctation of the aorta:
    A) Low in the arms, high in the legs
    B) Low everywhere
    C) High in the arms, low in the legs
    D) Only the diastolic is high
    E) The same everywhere

48. Which X-ray sign appears from 8–10 years in coarctation of the aorta?
    A) Pleural fluid
    B) Rib notching
    C) “Wooden shoe”
    D) “Egg on its side”
    E) Caverns

49. Which defect is typical of an “egg on its side”-shaped heart on X-ray?
    A) Transposition of the great arteries
    B) Tetralogy of Fallot
    C) Mitral stenosis
    D) ASD
    E) PDA

50. In transposition of the great arteries, when oxygen is given the cyanosis:
    A) Only half decreases
    B) Increased
    C) Disappears completely
    D) Disappears in the legs
    E) Does not decrease

51. Causes of heart failure up to day 7 of life:
    A) CHD (hypoplastic left heart, coarctation), RDS, persistent fetal circulation
    B) Rheumatic carditis
    C) Infective endocarditis
    D) Cardiomyopathy
    E) Hypertension

Clinical cases

Case 1

A 2.5-year-old child. Since birth the lips have turned blue when crying, and from 6 months the cyanosis has become constant. The child tires quickly when walking and rests by squatting. This morning, while crying, the child suddenly became restless, turned markedly blue, breathed rapidly, then became lethargic. On examination: generalized cyanosis, “drumstick fingers”, “watch-glass” nails. S2 weakened over the pulmonary artery, a harsh systolic murmur in the 2nd–3rd intercostal space on the left (weakened during the spell). CBC: Hb 210 g/l, erythrocytes 7.2×10^{12}/l.

Questions: 1) Presumptive diagnosis and today's condition? 2) Name the four components of the defect. 3) What is the meaning of squatting? 4) What X-ray changes are expected? 5) First aid in the spell?

Answer: 1) Tetralogy of Fallot; today — a hypercyanotic spell. 2) Pulmonary stenosis, a large VSD, right ventricular hypertrophy, dextroposition of the aorta. 3) Squatting compresses the systemic arteries and increases peripheral resistance — the right-to-left shunt decreases, more blood goes to the lungs, and cyanosis decreases. 4) A “wooden shoe”-shaped heart, reduced lung markings. 5) Calming, the knee-chest position, humidified oxygen, on the doctor's order morphine, sodium bicarbonate, propranolol; cardiac glycosides must not be given; urgent hospitalization.

Case 2

A 9-year-old child had tonsillitis 3 weeks ago. For 5 days: fever 38 °C, pain and swelling in the knee and ankle joints (in turn), weakness, dyspnea. On examination: pale skin, HR 124/min. The cardiac border is enlarged 1 cm to the left. At the apex S1 is weakened and there is a blowing systolic murmur radiating to the left axilla. ECG: PQ 0.20 s. ESR 45 mm/h, CRP strongly positive, ASO 800 IU/ml.

Questions: 1) Which disease? Justify the diagnosis with the criteria. 2) Which cardiological syndromes are present? 3) What complication is there a risk of in the future?

Answer: 1) Acute rheumatic fever: major criteria — carditis, migratory polyarthritis; minor criteria — fever, raised ESR and CRP, prolonged PQ; evidence of a streptococcal infection — a history of tonsillitis and a high ASO titer. 2) Endocardial involvement (signs of mitral valve insufficiency — a systolic murmur at the apex radiating to the axilla, weakened S1) and myocardial involvement (tachycardia, enlarged border, prolonged PQ) — rheumatic carditis. 3) Formation of an acquired heart

defect (mitral insufficiency) in 8–12 months, recurrences; therefore long-term bicillin prophylaxis is needed.

Case 3

A 5-month-old child. The mother complains that the baby tires quickly when feeding, sweats a lot, gains weight poorly and often has bronchitis. On examination: pale skin, RR 62/min, HR 168/min. “Cardiac hump”, a diffuse cardiac impulse. A systolic thrill and a harsh loud pansystolic murmur in the 3rd–4th intercostal space at the left sternal edge, widely radiating. S2 accentuated over the pulmonary artery. Fine-bubbling non-consonant moist rales in the lower posterior parts of the lungs. The liver projects 3.5 cm below the costal arch, the neck veins are distended, pastosity of the legs.

Questions: 1) Presumptive diagnosis? 2) Type and stage of heart failure? 3) What X-ray changes are expected? 4) What does the accentuation of S2 over the pulmonary artery indicate?

Answer: 1) Ventricular septal defect (large) — an acyanotic defect with a left-to-right shunt and increased pulmonary blood flow. 2) Total (left and right ventricular) chronic heart failure, stage IIB: HR ≈30% above the norm for 5 months (≈130), RR more than 70% above (≈35), congestive rales in the lungs, liver 3–4 cm, distended neck veins, pastosity. 3) Increased lung markings (arterial hypervolemia), bulging of the pulmonary artery arc, cardiomegaly (CTI >0.55). 4) Pulmonary hypertension.

Quick review

Table 10.11. *Key facts on Topic 10*

Question	Answer
Carditis	Endocardium + myocardium + pericardium; all together — pancarditis
Mitral insufficiency	Pansystolic at the apex, to the axilla; S1 ↓
Mitral stenosis	Presystolic diastolic at the apex; flapping S1; quail rhythm
Aortic insufficiency	Protodiastolic at Botkin's point; carotid dance, de Musset, rapid high pulse
Aortic stenosis	Harsh systolic at the 2nd right → carotid arteries; slow low pulse
Formation of an acquired defect	8–12 months after the onset of endocarditis
Myocarditis	Tachycardia out of proportion to temperature, muffled S1, gallop, ST ↓, CTI ↑
Dry pericarditis	Pain + friction rub; ST ↑
Exudative pericarditis	Borders in all directions, relative = absolute, markedly muffled sounds, voltage ↓
Beck's triad (tamponade)	Hypotension + venous pressure ↑ + muffled heart sounds
ARF criteria	Major: carditis, polyarthritis, chorea, erythema marginatum, nodules; + evidence of streptococcus
Hypertension	≥95th centile; at a young age often secondary (kidney, coarctation)
Sympathicotonia / vagotonia	Tachycardia, BP ↑ / bradycardia, BP ↓
CHD frequency	6–8 / 1000 live births
“White”, pulmonary circulation ↑	VSD, ASD, PDA
“Blue”, pulmonary circulation ↓	Tetralogy of Fallot

Question	Answer
Systemic circulation ↓	Aortic stenosis, coarctation
VSD / ASD / PDA	Harsh pansystolic at the 3rd–4th left / soft at the 2nd–3rd left, fixed splitting of S2 / “machinery”
Fallot	Cyanosis, squatting, S2 ↓ over the pulmonary a., polycythemia, “wooden shoe”
Position in a spell	Knee-chest
Left ventricular failure	Dyspnea, tachycardia, non-consonant moist rales at the lung bases
Right ventricular failure	Liver ↑, neck veins, edema
Fainting / collapse	Unconscious / consciousness preserved
Causes of carditis	Rheumatic (group A streptococcus), viral (Coxsackie B, ECHO, adenovirus), bacterial (diphtheria), infective endocarditis, autoimmune (SLE, JA), toxic-allergic
Mitral and aortic defects	MI: pansystolic murmur at the apex; MS: diastolic thrill, flapping S1, “quail rhythm”; AI: protodiastolic murmur, de Musset, Quincke; AS: harsh midsystolic murmur, slow pulse
Infective endocarditis	Prolonged fever, “café-au-lait”, a new murmur, splenomegaly, petechiae, Osler, Janeway, “drumstick fingers”, vegetations on echocardiography, blood culture
Signs of myocarditis	Tachycardia out of proportion to the temperature, muffled heart sounds, gallop, cardiomegaly, arrhythmia, raised CK-MB and troponin, CTI >0.5
Pericarditis	Dry: stabbing pain, friction rub, ST↑ in all leads; exudative: muffled heart sounds, wide dullness, paradoxical pulse, fluid on echocardiography; tamponade — Beck's triad
Criteria of rheumatic fever	Major: carditis, polyarthritis, chorea, erythema marginatum, nodules; minor: arthralgia, fever, ESR/CRP↑, prolonged PQ; + evidence of streptococcus (ASO)
Definition of arterial hypertension	BP ≥95th centile (at least 3 times, on different days); primary (adolescents) and secondary (renal, cardiovascular, endocrine)
Classification of congenital heart defects	“White” (VSD, ASD, PDA, coarctation, pulmonary stenosis) and “blue” (Fallot, transposition, truncus arteriosus, Ebstein, atresia)
Shade of cyanosis and defect	Crimson — pulmonary hypertension/Ebstein; bluish — Fallot, single ventricle; violet — Eisenmenger; dark blue — transposition
The 5 most common CHD	VSD, ASD, PDA, coarctation of the aorta, tetralogy of Fallot
Murmurs of VSD, ASD, PDA	VSD — pansystolic at the 3rd–4th space on the left; ASD — systolic at the 2nd–3rd space on the left, fixed splitting of S2; PDA — continuous “machinery” murmur at the 2nd space on the left
Tetralogy of Fallot	4 components; cyanosis from 3–6 months, squatting, drumstick fingers, polycythemia, X-ray — “wooden shoe”, reduced lung markings
Autonomic dystonia: sympathicotonia and vagotonia	Sympathicotonic: pale, dry, tachycardia, BP↑, excitable; vagotonic: marbled, moist, bradycardia, BP↓, sluggish, red dermographism
Hypertensive crisis	Severe headache, visual disturbance, vomiting, trembling, fear, seizures; BP rises suddenly and sharply
Coarctation of the aorta	Narrowing of the aortic isthmus; BP↑ in the arms, ↓ in the legs; weak femoral pulses; “athletic” shoulder girdle; rib notching (8–10 years)
Transposition	The aorta from the right ventricle, the pulmonary artery from the left; dark blue cyanosis from birth; X-ray — “egg on its side”; urgent treatment

TOPIC 11

The hematopoietic system: anatomical and physiological features, age-related features of peripheral blood, the complete blood count, semiotics and syndromes (anemic, hemorrhagic, lymphoproliferative)

Learning objectives

The student should know:

- the periods of intrauterine and postnatal hematopoiesis, the types of hemoglobin;
- the basic concepts of hematopoietic factors, iron metabolism and the hemostatic system;
- the age-related features of peripheral blood parameters, the physiological crossovers of the leukocyte differential;
- the methods of examining the blood system, the parameters of the complete blood count and the coagulogram;
- the semiotics of changes in erythrocytes, leukocytes, platelets and hemostasis;
- the anemic, sideropenic, hemorrhagic and lymphoproliferative syndromes and the leukemia syndrome.

The student should be able to:

- take complaints and history from a child with a blood disease, and perform inspection and palpation;
- assess a complete blood count by comparing it with the age norms, and calculate the color index;
- classify anemia by severity, color index and regeneration;
- determine the bleeding time and clotting time, and perform the endothelial tests;
- determine the type of hemorrhagic syndrome on the basis of clinical signs.

11.1. Development of hematopoiesis

Blood is a liquid tissue; it consists of plasma and formed elements — **erythrocytes, leukocytes and platelets**. It supplies the tissues with oxygen and nutrients and performs metabolic and protective functions.

The three periods of intrauterine hematopoiesis:

1. **The embryonic (megaloblastic, yolk sac) period** — from **the 2nd–3rd week** of pregnancy; in the blood islands of the yolk sac, large nucleated cells — **megaloblasts** (primary erythroblasts) with embryonic hemoglobin — are formed.
2. **The hepatic period** — begins **from week 6–7** and reaches its maximum **at 5 months**; erythroblasts and erythrocytes are formed in the liver. **From 3–4 months the spleen** also joins in (erythro-, granulo-, megakaryocytopoiesis); active lymphopoiesis **from week 20**.
3. **The bone marrow (medullary) period** — forms **from 4–5 months** and gradually becomes the main hematopoietic organ.

In line with these periods there are **three types of hemoglobin**: **embryonic (HbP)**, **fetal (HbF)** and **adult (HbA)**. HbP and HbF have **a high affinity for oxygen** — this facilitates the transfer of oxygen

to the fetus. At birth HbF makes up **60–85%** of the total hemoglobin and is gradually replaced by HbA: **≈15% at 1 year**, **≤2% at 3 years**.

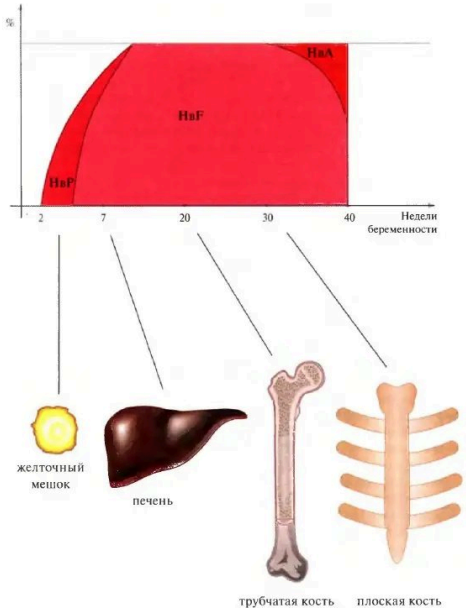


Figure 11.1. *Intrauterine succession of the hematopoietic organs and the types of hemoglobin (Navruzova, Geppe)*

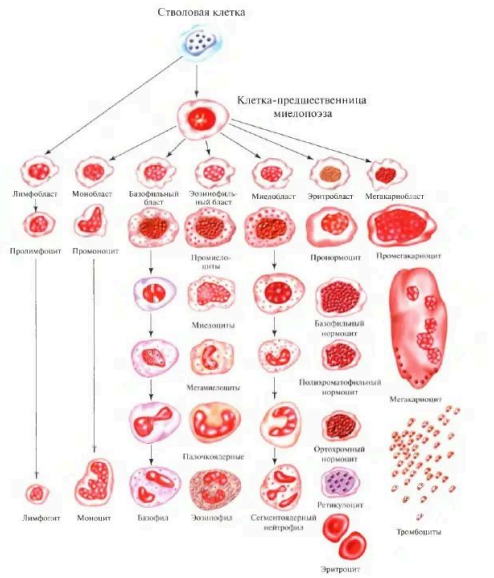


Figure 11.2. *Diagram of hematopoiesis: from the stem cell to mature blood cells (Navruzova, Geppe)*

After birth hematopoiesis in the liver stops; in the spleen only lymphocytes and monocytes are formed, and it destroys aged erythrocytes and platelets. The main organ forming all the formed elements except lymphocytes is **the red bone marrow**.

- In the newborn **all the flat and tubular bones** are filled with red bone marrow; from the first months of life the red marrow is gradually replaced by **yellow (fatty)** marrow; **at 12–15 years** hematopoiesis persists only in **the flat bones** and the epiphyses of the tubular bones.
- In the newborn and young child the bone marrow is **functionally labile**: under adverse influences (severe infection, anemia) it is quickly exhausted or **hematopoiesis may revert to the embryonic type** (extramedullary foci — enlargement of the liver and spleen, young cells in the peripheral blood).

Sites of bone marrow aspiration: in the newborn — **the calcaneus**; up to 1 year — **the (proximal) epiphysis of the tibia**; in older children — **the iliac crest**. Sternal puncture is now rarely used in children. **Trephine biopsy** (from the ilium, under anesthesia) gives more accurate information on the structure of the bone marrow.

Life span of blood cells: erythrocytes — **≈120 days** (in the newborn, erythrocytes with HbF — 60–90 days); platelets — **9–11 days**; neutrophils — a few hours in the blood, ≈14 days in total; eosinophils — 8–12 days; monocytes circulate in the blood for ≈12 hours and become macrophages in the tissues; lymphocytes — from a few days to years.

11.2. Hematopoietic factors and iron metabolism

Hematopoietic factors:

- **erythropoietin** — a glycoprotein hormone, **90% produced in the kidneys** in response to hypoxia; it stimulates the proliferation of erythroid cells and hemoglobin synthesis;
- **thrombopoietin** — accelerates megakaryocytopoiesis; **colony-stimulating factors** and interleukins regulate leukopoiesis;

- **folic acid and vitamin B_{12}** — necessary for DNA synthesis; vitamin B_{12} is absorbed in the small intestine with the help of **the gastric intrinsic factor of Castle**;
- **iron** — part of heme, ensures oxygen transport, is a component of many enzymes, and is necessary for immune function.

The iron store accumulates in the fetus from the mother via the placenta, especially at **weeks 28–32** of pregnancy, unidirectionally against the concentration gradient. For this reason the store is small in **preterm babies**, twins and those born after a complicated pregnancy. After birth the main sources of iron are food (in the 1st half-year — **breast milk**, from which up to **50%** of the iron is absorbed) and endogenous stores. A child's daily requirement is **0.5–1.2 mg** (more in adolescents).

Stages of iron metabolism: Fe^{2+} is absorbed in the intestine → oxidized to Fe^{3+} → bound to **transferrin** and transported to the bone marrow, depots and enzymes → in the bone marrow **Fe + protoporphyrin = heme; heme + globin = hemoglobin**. The store — in the form of **ferritin** and **hemosiderin** in the liver, muscles and bone marrow macrophages.

> **CAUTION**
> In young children with iron deficiency, intestinal iron absorption **does not increase compensatorily** (unlike adults) — this is why they are the group most prone to iron deficiency anemia.

Table 11.1. *Parameters of iron metabolism*

Indicator	Normal	In iron deficiency anemia
Serum iron	Newborn 5.0–19.3 µmol/l; over 1 month 10.6–33.6 µmol/l	Reduced
Total iron-binding capacity of the serum (TIBC)	40.6–62.5 µmol/l	Increased
Latent iron-binding capacity of the serum	TIBC – serum iron	Increased
Transferrin saturation coefficient	25–45%	<17%
Ferritin	15–150 ng/ml	<12–15 ng/ml (the most reliable sign of depleted stores)

11.3. The hemostatic system

Hemostasis — the system that keeps the blood in a liquid state and stops bleeding when a vessel is damaged. Three main components:

1. **The vascular component** — reflex **local angiospasm** on injury; the endothelium normally produces prostacyclin and prevents platelet adhesion.
2. **The platelet component** — platelet attachment to the damaged endothelium (**adhesion**) and to each other (**aggregation**) — the primary platelet plug; platelets contain more than 10 clotting factors (designated by Arabic numerals).
3. **The plasma (coagulation) component** — **13 plasma factors** (designated by Roman numerals) are activated in sequence and a **fibrin** clot forms.

Then **retraction** of the clot (normally 0.3–0.5, i.e., 60–75%) and **fibrinolysis** — dissolution of the clot and restoration of vascular patency — take place. The anticoagulant system — antithrombin III, proteins C and S, heparin.

Features of hemostasis in newborns: in the first days of life **the vitamin K-dependent clotting factors (II, VII, IX, X)**, as well as factors XI and XII, the physiological anticoagulants and the components of fibrinolysis are **physiologically reduced**; the system is balanced, but at a low level. In preterm babies these parameters are even lower — hemorrhagic and thrombotic syndromes, especially **DIC syndrome**, develop quickly. Early cord clamping and late first breastfeeding further reduce the vitamin K-dependent factors.

> **CLINICAL SIGNIFICANCE**
> **Hemorrhagic disease** of the newborn (vitamin K deficiency) presents on days 2–5 of life with melena, blood in vomit, and bleeding from the umbilicus and into the skin. To prevent it, all newborns are given **vitamin K (phytomenadione) intramuscularly** after birth.

11.4. Age-related features of peripheral blood

Blood of the newborn

- **The blood volume** in the newborn is **≈80–90 ml/kg** (≈8–9% of body mass; 90–100 ml/kg in preterm babies), in adults **≈65–70 ml/kg** (≈7%). Some older textbooks give 14–15% for the newborn.
- In **the first days** of life hemoglobin (**180–240 g/l**), erythrocytes (**5–7×10^{12}/l**), the color index (**up to 1.1**) and hematocrit (**≈54%**) are high — the result of intensive erythropoiesis in response to hypoxia in utero and during delivery.
- After birth external respiration is established and hypoxia is replaced by hyperoxia → erythropoietin decreases → erythropoiesis slows; erythrocytes with HbF have a short life span and are easily hemolyzed → **physiological jaundice** (from day 2–3).
- The erythrocytes are polychromatophilic, **anisocytosis** (macrocytes predominate); **normoblasts** may be present in the blood; **reticulocytes** are **2.2–4.2%** in the first days (later 0.6–0.8% — 1%).
- **Leukocytes** are **10–30×10^{9}/l** in the first days, 10–12×10^{9}/l after 2 weeks; the differential shows **neutrophilia** and a left shift (down to myelocytes).
- Platelets **150–400×10^{9}/l**; **ESR low — 1–2 mm/h**; the bleeding time is as in adults; the clotting time is slightly faster or normal.

Physiological crossovers of the leukocyte differential

At birth neutrophils are **60–65%**, lymphocytes **20–30%**. Neutrophils decrease rapidly and lymphocytes increase, and on **days 4–6** they **become equal** (40–45%) — **the first physiological crossover**. Then lymphocytes predominate (up to 60–65% until 1 year, 50–60% until 5 years). From 3–4 years neutrophils gradually increase and **at 4–6 years (on average at 5 years)** they become equal again — **the second physiological crossover**; then neutrophils predominate, and at 12–14 years the differential approaches that of adults (neutrophils 60–65%, lymphocytes 25–30%).

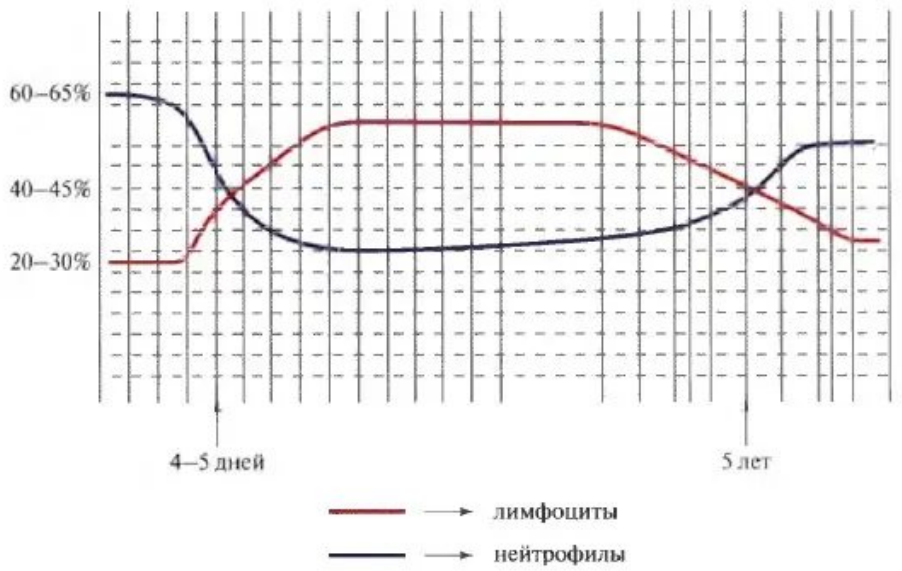


Figure 11.3. *Changes in the numbers of neutrophils and lymphocytes in different periods of childhood: crossovers at 4–5 days and at 5 years (Navruzova, Geppe)*

MEMORY AID

The crossovers: **“5 days — 5 years”**. Between the two crossovers (**from 5 days to 5 years**) **lymphocytes predominate** in the differential — this is normal for children and is not regarded as “lymphocytosis”.

Blood of children under 1 year and older children

- In the 1st year erythrocytes fall to **3.5–4.5×10^{12}/l**, hemoglobin to **110–120 g/l**, and the color index below 1.0; **at 3–6 months** the parameters may fall to their lowest values — **physiological (early) anemia** (rapid growth, increase in blood volume, relative immaturity of erythropoiesis, low iron in the diet).
- **Early anemia in preterm babies** (at 1.5–2 months; low erythropoietin, rapidly increasing blood volume) and **late anemia** (at 4–5 months; iron deficiency — hypochromic) develop; late anemia can be prevented with iron preparations.
- In the 1st year leukocytes are **8–12×10^{9}/l**, with lymphocytes predominating in the differential; ESR **4–7 (5–8) mm/h**.
- **From 2 years to adolescence** the composition of peripheral blood approaches that of adults: hemoglobin **120–140 g/l**, erythrocytes **4.0–4.5×10^{12}/l**.

Table 11.2. *Blood parameters in children of different ages (Geppe)*

Age	Hb, g/l	Erythrocytes, ×10^{12}/l	Reticulocytes, %	Leukocytes, ×10^{9}/l	Platelets, ×10^{9}/l	ESR, mm/h
Day 1	180–240	5–7	1–4	10–30	150–400	2–3
5 days	170–200	4,5–6	0,5–2	9–15	150–400	2–3
6 months	110–160	4,0–5,0	0,5–1,3	8–12	150–400	3–7
1 year	110–130	4,0–4,5	0,5–1	7–11	150–400	4–7
5 years	130–140	4,0–4,5	0,5–1	7–11	150–400	5–8
>10 years	Boys 130–140; girls 120–140	4,0–4,5	0,5–1	5–9	150–400	5–10

Table 11.3. *Erythrocyte indices*

Indicator	Meaning	Normal
MCV — mean corpuscular volume	Micro-, normo-, macrocytosis	Newborn 102–116 fl; 1 month 93–115; 6 months 82–100; 1 year 81–95; from 3 years 80–95 fl
MCH — mean corpuscular hemoglobin	Analogue of the color index	27–31 pg
MCHC — mean corpuscular hemoglobin concentration	Saturation	33–37 g/dl
Color index	(Hb, g/l × 3) / the first 3 digits of the erythrocyte count	0,85–1,05
Hematocrit	Volume of the formed elements	Newborn ≈54%; older children 35–45%

11.5. Methods of examining the blood system

Questioning

Specific complaints: bleeding (from the nose, gums, gastrointestinal tract, menstrual), hemorrhages (“bruises” on the skin, hematomas, hemarthroses), enlarged lymph nodes, pallor of the skin and mucous membranes. **Nonspecific complaints:** weakness, rapid fatigue, dizziness, headache, dyspnea on exertion, reduced appetite, **distorted taste and smell** (eating chalk, soil, ice; liking the smell of gasoline, paint), hair loss, brittle nails, bone and joint pain, fever, sweating, weight loss.

History: the neonatal period (term birth, weight, neonatal jaundice — onset, degree, duration); feeding (breast milk, timing of complementary foods, early feeding with cow's milk, meat intake); past illnesses, medication; **family history** — blood diseases, jaundice, gallstone disease, splenectomy, hemorrhagic syndrome in relatives (hemophilia — in the mother's male relatives), early (before 50 years) infarction and stroke (thrombophilia).

Inspection and palpation

- **General condition:** impaired consciousness (with a rapid fall in Hb or blood volume — anemic coma, hypovolemic shock), ongoing bleeding.
- **Color of the skin and mucous membranes: pallor** (anemia; marked pallor — acute blood loss); pallor + **a lemon-yellow tint** (hemolytic anemia, B_{12} deficiency); waxy pallor (leukemia); **cherry-red** (polycythemia); perioral cyanosis (hypoxia in severe anemia).
- **Hemorrhagic rash:** type (petechiae, purpura, ecchymoses, hematoma), location, symmetry, color (different colors — different ages), circumstances of appearance.
- **Skin appendages and mucous membranes:** dry skin, brittle hair and nails, **koilonychia**, angular stomatitis (“cheilitis”), **a “lacquered” or “geographic” tongue**, glossitis, gum hyperplasia and bleeding (leukemia).
- **Lymph nodes, liver and spleen** (by the methods of topic 4): enlargement, consistency, adhesion; **hepatosplenomegaly** — systemic blood diseases.
- **Joints:** hemarthrosis (hemophilia), joint syndrome (hemorrhagic vasculitis, leukemia).
- **Bones:** pain on tapping (sternum, ribs, spine, tibia — leukemia).
- **Cardiovascular:** tachycardia, a functional systolic murmur at the apex (anemia).

- **Endothelial tests** (tourniquet, pinch, hammer — topic 4) — the state of the vascular wall and platelets.

Laboratory investigations

Complete (clinical) blood count — from a finger (from the heel in the newborn) or a vein; hemoglobin, erythrocytes, color index, reticulocytes, leukocytes and **the leukocyte differential**, platelets, ESR, hematocrit; on an automated analyzer — the erythrocyte indices (MCV, MCH, MCHC, RDW). **ESR** by the Panchenkov method: the column of plasma (mm) above the erythrocytes that have settled in a capillary in 1 hour.



Figure 11.4. *Determining ESR by the Panchenkov method: normal (1, 2) and raised (3) (Navruzova, Geppe)*

Assessment of hemostasis:

- **Bleeding time (Duke)** — the finger or earlobe is pricked with a lancet and the drops of blood are blotted onto filter paper every 15–30 seconds; normally **2–4 minutes**. Prolongation — **reduced platelet count or impaired platelet function** (platelet component).
- **Clotting time** — assesses the plasma component; the norm depends on the method: **Morawitz** — beginning 3, end 5 minutes; **Bürker** — 2.5 and 5.5 minutes; **Lee–White** (venous blood) — 5–10 minutes; **Sitkovsky** — 1.5 and 4 minutes. Prolongation — **deficiency of coagulation factors** (hemophilia).
- **Coagulogram:** activated partial thromboplastin time (**APTT**, 25–55 s), prothrombin index (70–110%) or **INR**, thrombin time (14–16 s), **fibrinogen** (1.7–3.5 g/l), antithrombin III, D-dimers, individual factors (VIII, IX), clot retraction (60–75%), platelet aggregation.

Other investigations: iron metabolism parameters, bilirubin and its fractions, hemoglobin electrophoresis (thalassemia), osmotic resistance of erythrocytes (spherocytosis), Coombs test (immune hemolysis), vitamin B_{12} and folate; **myelogram** (bone marrow aspiration) and trephine biopsy; lymph node biopsy; cytochemical, immunophenotypic, cytogenetic and molecular methods (leukemias).

Table 11.4. *Main parameters of the myelogram (Kildiyarova)*

Indicator	Normal
Leuko-erythroblastic ratio	≈3:1 in infancy
Granulocytes	40–60%
Lymphocytes and reticular cells	10–20%
Monocytes	≤3–5%
Blast cells	≤5% (myeloblasts ≤5–8%)
Megakaryocytes	≈0,5%

11.6. Semiotics of blood changes

Changes in erythrocytes

Anemia — a decrease in the amount of hemoglobin (often also erythrocytes) per unit volume of blood below the age norm. The most common blood pathology in children.

Anemia cut-offs (WHO): in the newborn Hb **<145 g/l**; **6 months – 5 years — <110 g/l**; **5–11 years — <115 g/l**; **over 12 years — <120 g/l** (in local practice <120 g/l is often used for children over 5 years).

Table 11.5. *Classification of anemias*

Criterion	Types
Degree of severity	Mild — Hb 110(120)–90 g/l; moderate — 90–70 g/l; severe — <70 g/l
Color index	Hypochromic — <0.85; normochromic — 0.85–1.05; hyperchromic — >1.05
Erythrocyte size (MCV)	Microcytic — <7 µm, MCV <80 fl; normocytic — 7–7.9 µm, 80–95 fl; macrocytic — >8 µm, MCV >95 fl
Regeneration (reticulocytes)	Hyporegenerative — <0.5–1%; normoregenerative — 1–3%; hyperregenerative — >3%
Origin	After blood loss (posthemorrhagic); from impaired erythropoiesis (deficiency — iron, B_{12}, folate; hypo- and aplastic); from erythrocyte destruction (hemolytic)

Table 11.6. *Features of the main anemias*

Anemia	Color index, MCV	Reticulocytes	Other signs
Iron deficiency	Hypochromic, microcytic (MCV, MCH, MCHC ↓)	Normal or slightly ↑	Anisocytosis, poikilocytosis; serum iron and ferritin ↓, TIBC ↑
B_{12} and folate deficiency	Hyperchromic, macrocytic (MCV, MCH ↑)	Normal or ↓	Megaloblastic hematopoiesis, Howell–Jolly bodies, Cabot rings, hypersegmentation of neutrophils, leuko- and thrombocytopenia
Thalassemia	Hypochromic, microcytic (MCHC normal)	Normal or ↑	Target cells, HbF or HbA_2 ↑, serum iron normal or ↑
Hereditary spherocytosis (Minkowski–Chauffard)	Normochromic (MCHC ↑)	↑	Microspherocytes, osmotic resistance ↓, splenomegaly, jaundice

Anemia	Color index, MCV	Reticulocytes	Other signs
Hemolytic (general)	Normochromic	Sharply ↑	Indirect bilirubin ↑, jaundice, splenomegaly, urobilin in the urine, stercobilin in the stool ↑
Posthemorrhagic (acute)	Normochromic	Normal at first, ↑ from day 3–5	Hypovolemia, shock
Hypo- and aplastic	Normochromic or macrocytic	↓	Pancytopenia; ESR ↑ (40–80 mm/h); in the myelogram all lineages are depressed

Erythrocytosis — an increase in erythrocytes and hemoglobin: secondary — chronic hypoxia ("blue" congenital heart defects, chronic lung diseases), hemoconcentration (dehydration); primary — **polycythemia vera** (rare in children).

Erythrocyte morphology: anisocytosis (different sizes), poikilocytosis (different shapes), micro- and macrocytosis, spherocytes, target cells (thalassemia), sickle erythrocytes; **reticulocytosis and polychromatophilia** — strong regeneration of the bone marrow; **normoblasts** in the peripheral blood — physiological in the neonatal period, later pathological (hemolysis, severe anemia, leukemia).

Changes in leukocytes

Table 11.7. *Diagnostic significance of changes in leukocytes*

Change	Definition	When it occurs
Leukocytosis	Above the age norm; >50×10^9/l — hyperleukocytosis	Bacterial infections, inflammation, leukemias, glucocorticoids
Leukopenia	Below the age norm (<4×10^9/l)	Viral infections (influenza, measles, rubella), hypersplenism, ionizing radiation, cytostatics, sulfonamides, aplastic anemia, some bacterial infections (typhoid fever, brucellosis)
Agranulocytosis	A sharp decrease in or absence of granulocytes (<0.5×10^9/l)	Drugs, autoimmune, bone marrow damage; risk of severe infections
Neutrophilia, left shift (band forms >6%)	Increase in neutrophils	Bacterial infection (pneumonia, sepsis, purulent processes), acute inflammation, autoimmune diseases
Leukemoid reaction	A change resembling leukemia but reversible (young forms present, no "leukemic hiatus")	Severe infections, sepsis, tuberculosis, tumor metastases
Eosinophilia	>4–5%	Helminthiasis, bronchial asthma, allergic diseases (atopic dermatitis, urticaria, drug allergy), serum sickness, lymphogranulomatosis
Eosinopenia	<1% (aneosinophilia)	Severe acute infections, stress, glucocorticoids — a negative prognostic sign
Lymphocytosis	Above the age norm	Viral infections, whooping cough (marked leukocytosis and lymphocytosis), lymphatic-hypoplastic diathesis, tuberculous intoxication, lymphoblastic leukemia
Lymphopenia	Below the age norm	The onset of febrile infections, lymphogranulomatosis, lymphosarcoma, HIV, immunodeficiency, glucocorticoids

Change	Definition	When it occurs
Monocytosis	>8–10%	Infectious mononucleosis (atypical mononuclear cells), tuberculosis, brucellosis, protozoal infections, monocytic leukemia

Changes in platelets

- **Thrombocytosis** (>400×10^9/l) — reactive (inflammation, rheumatic diseases, sepsis, iron deficiency, after blood loss, **after splenectomy**), polycythemia.
- **Thrombocytopenia** (<150×10^9/l) — a separate disease (**immune thrombocytopenia — Werlhof disease**) or a syndrome (leukemia, aplastic anemia, DIC syndrome, hypersplenism, infections, drugs). At **<50×10^9/l** there is a risk of bleeding, at **<20×10^9/l** a risk of spontaneous severe bleeding (including into the brain).

Changes in ESR

- **Increase:** increased fibrinogen and globulins (inflammation, infections, sepsis), anemia, connective tissue diseases, kidney diseases (nephrotic syndrome), leukemias, lymphomas, tumors, after vaccination.
- **Decrease:** the neonatal period, erythrocytosis and hemoconcentration (hematocrit >50%), hyperbilirubinemia, acidosis.

Changes in hemostasis

- **Hypocoagulation** (tendency to bleed) — reduced platelets or impaired platelet function, deficiency of clotting factors (hemophilia, liver diseases, vitamin K deficiency, obstructive jaundice), an excess of anticoagulants.
- **Hypercoagulation** (tendency to thrombosis) — hereditary thrombophilias (deficiency of antithrombin III, proteins C and S), stage 1 of DIC syndrome, antiphospholipid syndrome, burns, major injuries, after surgery, nephrotic syndrome.
- **DIC syndrome** (disseminated intravascular coagulation) — in severe infections (sepsis, meningococcemia), shock, extensive injuries, tumors, hemolysis, exsicosis: first the clotting factors are sharply activated, with **disseminated thromboses** in the small vessels (impaired microcirculation, organ failure), then the factors and platelets are **consumed** (“consumption coagulopathy”) and fibrinolysis is activated, resulting in **bleeding from many sites**.

11.7. Main syndromes of blood system involvement

Anemic syndrome

Signs related to tissue hypoxia due to reduced hemoglobin:

- **pallor** of the skin and mucous membranes (conjunctiva, lips, nail beds, palms);
- weakness, rapid fatigue, lethargy, reduced appetite, dizziness, headache, darkening before the eyes, tinnitus, fainting;
- dyspnea on exertion, **tachycardia**, a **functional systolic murmur** at the apex, a “humming top” over the neck veins, reduced BP;
- in severe anemia — enlarged cardiac borders, heart failure, anemic coma.

Sideropenic syndrome

Signs of tissue iron deficiency in iron deficiency anemia:

- dystrophy of the skin and appendages: **dry skin**, **hair loss and brittleness**, **brittle nails, transverse striations, koilonychia** (spoon-shaped nails);
- atrophy of the mucous membranes: nose, esophagus (difficulty swallowing), stomach (dyspepsia, malabsorption); **angular stomatitis**, glossitis, “lacquered” tongue, gingivitis;
- **distorted taste and smell** — eating chalk, soil, ice, raw meat, dough (**pica**), liking the smell of gasoline, acetone, paint;
- muscle hypotonia and pain (myoglobin deficiency), including of the bladder sphincter — **nocturnal enuresis**, urine leakage on laughing and coughing;
- reduced memory and attention, emotional lability, drowsiness; delay in the child's psychomotor development;
- frequent infections (immunodeficiency).

REMEMBER

The laboratory triad of iron deficiency anemia: **hypochromia** (color index <0.85, MCH ↓) + **microcytosis** (MCV <80 fl) + **reduced ferritin and serum iron** (raised TIBC). In the blood smear — **the “PHA” syndrome**: poikilocytosis, hypochromia, anisocytosis.

Hemolytic syndrome: anemia + **jaundice** (lemon-yellow tint, raised indirect bilirubin) + **splenomegaly** + dark urine (urobilin) + reticulocytosis. In hemolytic crises — fever, weakness, abdominal pain, marked jaundice, deepening anemia.

Hemorrhagic syndrome

Hemorrhagic syndrome — a tendency to bleed: bleeding into cavities and/or hemorrhages into the skin, joints and internal organs. According to which component of hemostasis is disturbed: **coagulopathies** (plasma factors), **thrombocytopenias and thrombocytopathies** (platelet component), **vasopathies** (vascular component).

Table 11.8. *Types of bleeding (according to Z.S. Barkagan)*

Type	Clinical signs	In which diseases
Hematoma type	Painful large hemorrhages into the subcutaneous fat, muscles and joints (hemarthrosis), usually after injury; later deforming arthroses, contractures; prolonged, heavy, delayed (after several hours) bleeding after injury and surgery; small injury — heavy bleeding	Hemophilia (A — factor VIII, B — factor IX deficiency)
Petechial-spotted (microcirculatory) type	Petechiae and ecchymoses on the skin and mucous membranes, spontaneous (often at night) or from trivial injury, asymmetric, of different colors and sizes (“leopard skin”); bleeding from the nose, gums, uterus, kidneys; risk of cerebral hemorrhage	Thrombocytopenias (Werlhof disease), thrombocytopathies, hypo- and dysfibrinogenemias
Mixed (microcirculatory-hematoma) type	The petechial-spotted type predominates, with hemorrhages into the subcutaneous fat; hemarthroses are rare	Von Willebrand disease, DIC syndrome, anticoagulant overdose, deficiency of certain factors

Type	Clinical signs	In which diseases
Vasculitic-purpuric type	Symmetric, slightly raised (palpable) petechiae and purpura on the limbs, around the large joints and on the buttocks; with joint, abdominal (abdominal crises, intestinal bleeding) and renal (hematuria) syndromes	Hemorrhagic vasculitis (Henoch–Schönlein), infectious-toxic vasculitides
Angiomatous type	Recurrent bleeding from one site from angiomatously altered vessels (nose, intestine, lungs); telangiectasias on the skin and mucous membranes	Rendu–Osler disease



Figure 11.5. *Werlhof disease: asymmetric ecchymoses of different sizes and colors (Navruzova, Geppe)*



Figure 11.6. *Henoch–Schönlein disease: symmetric petechial-purpuric rash on the legs (Navruzova, Geppe)*

Table 11.9. *Laboratory differentiation of the hemorrhagic syndrome*

Indicator	Hemophilia	Thrombocytopenia (Werlhof)	Hemorrhagic vasculitis
Platelets	Normal	Sharply reduced	Normal (or raised)
Bleeding time (Duke)	Normal	Prolonged	Normal
Clotting time (Lee–White), APTT	Prolonged	Normal	Normal
Clot retraction	Normal	Impaired	Normal
Endothelial tests (tourniquet, pinch)	Negative	Positive	May be positive
Other	Factor VIII or IX sharply reduced	Antiplatelet antibodies; many megakaryocytes in the myelogram	Neutrophilic leukocytosis, ESR ↑, fibrin degradation products

MEMORY AID

“Hemophilia — hematoma, Werlhof — petechiae, Henoch–Schönlein — symmetric purpura”. In hemophilia **the clotting time** is prolonged, in thrombocytopenia **the bleeding time** is prolonged.

CLINICAL SIGNIFICANCE

Intramuscular injections must not be given to a child with hemophilia (risk of a large hematoma); aspirin and NSAIDs are prohibited. In hemarthrosis — immobilization of the joint, cold, and most importantly urgent **intravenous** administration of the deficient factor concentrate.

Lymphoproliferative (hyperplastic) syndrome

Enlargement of the lymph nodes, liver and spleen (one or several groups of nodes): leukemias, lymphomas, lymphogranulomatosis, histiocytoses, infectious mononucleosis and other infections (see topic 4).

- **In lymphogranulomatosis (Hodgkin lymphoma)** — the cervical and supraclavicular nodes enlarge and form **a conglomerate ("packet")**, mobile at first, then adherent, dense, painless; fever, night sweats, weight loss, itching; on biopsy **Reed–Sternberg (Berezovsky–Sternberg) cells**.
- **In lymphosarcoma** — large tumor masses growing into the surrounding tissues, immobile, with signs of compression (edema, thrombosis, paralysis).

The syndrome of acute leukemia (disturbed proliferation and differentiation of leukocytes)

Mutation of hematopoietic cells and their malignant proliferation. In children most often **acute lymphoblastic leukemia** (2–5 years).

Clinical syndromes:

- **intoxication** — weakness, rapid fatigue, unwillingness to play, loss of appetite, fever, frequent tonsillitis and abdominal pain;
- **anemic** syndrome — pallor (waxy), tachycardia;
- **hemorrhagic** syndrome — petechiae, purpura, bleeding from the nose and gums;
- **hyperplastic (lymphoproliferative)** syndrome — enlargement of the lymph nodes, **liver and spleen**, gum hyperplasia, **skin leukemids** (painless dense bluish infiltrates);
- **osteoarticular** syndrome — pain in the tubular bones and joints (ossalgia), limping;
- **infectious complications** (neutropenia), neuroleukemia (meningeal signs).

Hemogram: normochromic anemia, reticulocytopenia, **thrombocytopenia**, leukocytosis (or leukopenia), **blast cells**, **"leukemic hiatus"** (no intermediate forms between the blasts and the mature granulocytes), raised ESR. **Myelogram:** blasts **≥20–25%** (the diagnostic criterion), the normal hematopoietic lineages depressed.

CAUTION

Acute leukemia must be distinguished from **a leukemoid reaction**: in a leukemoid reaction there is no "leukemic hiatus", intermediate and mature forms are present, there are no specific clinical signs (bone pain, leukemids), and the blood picture recovers after the underlying disease is treated. If in doubt — **bone marrow aspiration**.

11.8. Clinical section

A child with anemia — diagnostic algorithm

1. **Confirm the anemia** and determine its severity (Hb by the age cut-off).
2. **Color index and MCV**: hypochromic-microcytic → iron deficiency (ferritin), thalassemia (Hb electrophoresis); hyperchromic-macrocytic → B_{12} or folate deficiency; normochromic →

continue.

3. **Reticulocytes**: high → hemolysis (bilirubin, Coombs, erythrocyte morphology) or acute blood loss; low → bone marrow failure.
4. **Other lineages**: leukocytes and platelets are also reduced (pancytopenia) or blasts are present → aplastic anemia or **leukemia** — myelogram.
5. Determine **the cause**: feeding, blood loss (gastrointestinal, helminthiasis), malabsorption, prematurity.

CLINICAL SIGNIFICANCE

Prevention of iron deficiency anemia: iron preparations for pregnant and breastfeeding mothers, exclusive breastfeeding up to 6 months, iron-rich complementary foods from 6 months (meat, fortified cereals), no whole cow's milk before 1 year, prophylactic iron preparations for preterm babies from 2–4 weeks.

Practical skills

Skill 1. Assessing a complete blood count

1. Determine the child's age and choose the table of age norms.
2. Assess hemoglobin and erythrocytes; calculate the color index: (Hb × 3) / the first 3 digits of the erythrocyte count.
3. Assess the reticulocytes, MCV, MCH and MCHC.
4. Compare the leukocyte count and the differential with the age features (physiological crossovers); detect a left shift.
5. Assess the platelets and ESR.
6. Conclusion: “Within the age norm” or “Mild hypochromic microcytic anemia, eosinophilia”, etc.

Skill 2. Determining the bleeding time (Duke)

1. Wipe the earlobe or fingertip with alcohol and let it dry.
2. Prick to a depth of 3–4 mm with a sterile lancet; start the stopwatch.
3. Every 15–30 seconds touch the emerging drop of blood with filter paper (without touching the skin).
4. Record the time when no more blood mark remains on the paper.
5. Assessment: normally 2–4 minutes; prolongation — disturbance of the platelet component.

Skill 3. Determining the clotting time (Morawitz or Sitkovsky method)

1. Place a drop of blood from the finger on a dry, clean glass slide; record the time.
2. Every 30 seconds pass a glass rod (or needle) through the drop.
3. The appearance of the first fibrin thread — the beginning of clotting; complete clotting of the drop — the end.
4. Compare with the norm for the method (Morawitz: 3–5 minutes).

Skill 4. Examining a child with a hemorrhagic syndrome

1. History: onset of bleeding, relation to injury, delayed bleeding, bleeding in the family (the mother's male relatives), preceding infection or vaccination.

2. Inspection: type of rash (petechiae, ecchymoses, hematoma), location, symmetry, color; joints (hemarthrosis); mucous membranes.
3. Perform the endothelial tests (tourniquet, pinch).
4. Palpate the lymph nodes, liver and spleen.
5. Determine the type of bleeding (hematoma, petechial-spotted, mixed, vasculitic-purpuric, angiomatous).
6. Draw up an investigation plan: CBC and platelets, bleeding and clotting time, coagulogram, urinalysis (hematuria), stool for occult blood.

Review questions

1. What are the periods of intrauterine hematopoiesis? How do the types of hemoglobin succeed one another?
2. Where does hematopoiesis take place after birth? At which sites is bone marrow aspiration performed?
3. What are the hematopoietic factors and iron metabolism? Why are young children prone to iron deficiency?
4. What components does the hemostatic system consist of? What are the features of hemostasis in newborns?
5. Name the features of the newborn's blood.
6. When do the physiological crossovers of the leukocyte differential occur?
7. What are the norms for blood parameters in children under 1 year and older children? What are physiological anemia and anemia of prematurity?
8. What complaints and objective signs occur in blood diseases?
9. How are the complete blood count, color index and erythrocyte indices assessed?
10. How are the bleeding time and clotting time determined and what do they show?
11. How are anemias classified? What are the laboratory signs of the main anemias?
12. In which conditions do leukocytosis, leukopenia, eosinophilia, lymphocytosis and monocytosis occur?
13. Name the signs of the anemic and sideropenic syndromes.
14. What are the types of hemorrhagic syndrome and their laboratory differences?
15. What are the signs of the lymphoproliferative syndrome and acute leukemia? How does a leukemoid reaction differ from leukemia?

Test questions

1. From when does bone marrow hematopoiesis begin in utero?
   A) From week 2–3
   B) From week 6–7
   C) From month 4–5
   D) From month 8
   E) After birth

2. What proportion of the total hemoglobin is HbF at birth?
   A) 2%
   B) 15%
   C) 30%
   D) 60–85%
   E) 100%

3. From where is bone marrow aspirated in a child under 1 year?
   A) Sternum
   B) Epiphysis of the tibia
   C) Iliac crest
   D) Spine
   E) Rib

4. The hemoglobin level in the newborn in the first days of life:
   A) 100–120 g/l
   B) 120–140 g/l
   C) 140–160 g/l
   D) 180–240 g/l
   E) 250–300 g/l

5. When does the first physiological crossover of the leukocyte differential occur?
   A) At birth
   B) On days 4–6
   C) At 1 month
   D) At 1 year
   E) At 5 years

6. The second physiological crossover:
   A) At 1 year
   B) At 2 years
   C) At 5 years
   D) At 10 years
   E) At 15 years

7. The anemia cut-off (WHO) in a 3-year-old child:
   A) Hb <90 g/l
   B) Hb <100 g/l
   C) Hb <110 g/l
   D) Hb <120 g/l
   E) Hb <130 g/l

8. Typical of iron deficiency anemia:
   A) Hyperchromia, macrocytosis
   B) Hypochromia, microcytosis, reduced ferritin
   C) Reticulocytosis, jaundice
   D) Pancytopenia
   E) Hypersegmentation of neutrophils

9. Not typical of the sideropenic syndrome:
   A) Koilonychia
   B) Distorted taste (eating chalk)
   C) Hair loss
   D) Splenomegaly and jaundice
   E) Angular stomatitis

10. Hyperchromic macrocytic anemia, Howell–Jolly bodies and hypersegmentation of neutrophils are typical of:
   A) Iron deficiency anemia
   B) B_{12} deficiency anemia
   C) Thalassemia
   D) Posthemorrhagic anemia
   E) Spherocytosis

11. In which condition does eosinophilia occur?
   A) Viral infection
   B) Helminthiasis
   C) Sepsis
   D) Glucocorticoid therapy
   E) Stress

12. The type of bleeding typical of hemophilia:
   A) Petechial-spotted
   B) Hematoma
   C) Vasculitic-purpuric
   D) Angiomatous
   E) Mixed

13. The laboratory change in Werlhof disease:
   A) Prolonged clotting time
   B) Thrombocytopenia and prolonged bleeding time
   C) Factor VIII deficiency
   D) Normal platelets
   E) Raised fibrinogen

14. The normal bleeding time (Duke):
   A) 30 seconds
   B) 2–4 minutes
   C) 8–10 minutes
   D) 15 minutes
   E) 30 minutes

15. Typical of the hemogram in acute leukemia:
   A) “Leukemic hiatus”
   B) Reticulocytosis
   C) Thrombocytosis
   D) Reduced ESR
   E) Erythrocytosis

16. In which organ does the first (embryonic) period of intrauterine hematopoiesis take place?
    A) The spleen
    B) Yolk sac
    C) Bone marrow
    D) Liver
    E) Lymph nodes

17. When does hepatic hematopoiesis reach its maximum?
    A) At 3 months
    B) At 2 months
    C) At 5 months
    D) After birth
    E) At 7 months

18. What proportion of the total hemoglobin is fetal hemoglobin (HbF) at birth?
    A) 60–85%
    B) Absent
    C) 100%
    D) 20–30%
    E) 5–10%

19. From where is bone marrow aspirated in a newborn?
    A) Sternum
    B) Spine
    C) Calcaneus
    D) Femoral diaphysis
    E) Iliac crest

20. The average life span of erythrocytes:
    A) 1 year
    B) 9–11 days
    C) 12 hours
    D) 14 days
    E) ≈120 days

21. The life span of platelets:
    A) 1 month
    B) ≈120 days
    C) 9–11 days
    D) 3 months
    E) 12 hours

22. In which organ is erythropoietin mainly produced?
    A) Heart
    B) Liver
    C) The spleen
    D) Kidney (≈90%)
    E) Lungs

23. With the help of which factor is vitamin B_{12} absorbed in the small intestine?
    A) Thrombopoietin
    B) Ferritin
    C) Castle's intrinsic factor
    D) Erythropoietin
    E) Transferrin

24. In which weeks does the iron store accumulate in the fetus especially?
    A) 10–16
    B) 28–32
    C) 36–40
    D) 1–8
    E) After birth

25. Which parameter is the most reliable sign of depleted stores in iron deficiency anemia?
    A) Reduced TIBC
    B) Transferrin saturation 40%
    C) Color index 1.2
    D) Normal serum iron
    E) Reduced ferritin (<12–15 ng/ml)

26. The transferrin saturation coefficient in iron deficiency:
    A) 80%
    B) <17%
    C) 100%
    D) 25–45% (normal)
    E) 60%

27. A feature of hemostasis in newborns:
    A) No platelets
    B) Many anticoagulants
    C) Excess clotting factors
    D) Strong fibrinolysis
    E) The vitamin K-dependent factors (II, VII, IX, X) are physiologically reduced

28. The normal hemoglobin in the newborn in the first days of life:
    A) 100–120 g/l
    B) 60–80 g/l
    C) 180–240 g/l
    D) 300 g/l
    E) 130–140 g/l

29. When is the first physiological crossover of the leukocyte differential?
    A) 5 years
    B) At birth
    C) 12 years
    D) 1 month
    E) Day 4–6

30. When does the second physiological crossover occur?
    A) 4–6 years (on average 5 years)
    B) 14 years
    C) Day 4–6
    D) 1 month
    E) 1 year

31. The anemia cut-off by WHO in a child from 6 months to 5 years:
    A) Hb <110 g/l
    B) Hb <120 g/l
    C) Hb <90 g/l
    D) Hb <145 g/l
    E) Hb <115 g/l

32. The Hb level in mild anemia:
    A) 40 g/l
    B) 60 g/l
    C) 90–70 g/l
    D) 110(120)–90 g/l
    E) <70 g/l

33. If the color index is 0.85–1.05, the anemia is:
    A) Normochromic
    B) Hypochromic
    C) Aplastic
    D) Hyperchromic
    E) Macrocytic

34. MCV and the color index in iron deficiency anemia:
    A) MCV ↑, hypochromic
    B) MCV ↑, hyperchromic
    C) MCV normal, hyperchromic
    D) MCV ↓, hypochromic
    E) Unchanged

35. What is B_{12} and folate deficiency anemia like?
    A) Aplastic
    B) Hyperchromic, macrocytic (megaloblastic)
    C) Hypochromic, microcytic
    D) Normochromic, spherocytic
    E) Hypochromic, thalassemia

36. Osmotic resistance in hereditary spherocytosis (Minkowski–Chauffard):
    A) Unchanged
    B) Cannot be determined
    C) Increased
    D) Reduced
    E) Increased only in the newborn

37. Target cells and increased HbF/HbA_2 are typical of which anemia?
    A) B_{12} deficiency
    B) Posthemorrhagic
    C) Aplastic
    D) Iron deficiency
    E) Thalassemia

38. What are the reticulocytes in hyperregenerative anemia?
    A) 0,5–1%
    B) 0%
    C) >3%
    D) <0,5%
    E) 1–3%

39. Which bilirubin is raised in hemolytic anemia?
    A) Both are reduced
    B) Direct
    C) Only conjugated
    D) Unchanged
    E) Indirect (free)

40. What blood change is seen in aplastic anemia?
    A) Only eosinophilia
    B) Pancytopenia, sharply raised ESR
    C) No change
    D) Only erythrocytosis
    E) Leukocytosis and thrombocytosis

41. The normal bleeding time (Duke):
    A) 2–4 minutes
    B) 1 hour
    C) 10–15 minutes
    D) 30 seconds
    E) 20 minutes

42. Disturbance of which component does a prolonged bleeding time indicate?
    A) Fibrinolysis
    B) Plasma (coagulation) component
    C) Anticoagulant
    D) Platelet component
    E) None

43. Hemarthrosis is typical of which disease?
    A) Hemophilia
    B) B_{12} deficiency
    C) Iron deficiency
    D) Thalassemia
    E) Spherocytosis

44. Waxy pallor and an earthy color are typical of which condition (for leukemia)?
    A) Cyanosis — leukemia
    B) Lemon-yellow — leukemia
    C) Normal
    D) Waxy pallor — leukemia
    E) Cherry-red — leukemia

45. In which condition does pica chlorotica (eating chalk, soil) occur?
    A) Thalassemia
    B) Iron deficiency anemia
    C) Hemophilia
    D) Spherocytosis
    E) Leukemia

Clinical cases

Case 1

A child aged 1 year 3 months. The mother complains of the child's pallor, rapid fatigue, poor appetite and urge to eat lime plaster and soil. Born at 34 weeks. Fed with cow's milk since 4 months, eats meat poorly. On examination: pale skin and mucous membranes, dry skin, sparse and brittle hair, cracks at the corners of the mouth. HR 132/min, a soft systolic murmur at the apex. The liver and spleen are not enlarged. CBC: Hb 82 g/l, erythrocytes 3.6×10^{12}/l, reticulocytes 1.2%, MCV 66 fl, leukocytes 8.0×10^{9}/l, platelets 320×10^{9}/l, ESR 8 mm/h. Smear: anisocytosis, poikilocytosis.

Questions: 1) Calculate the color index. 2) Classify the anemia fully. 3) Which syndromes are present? 4) Presumptive diagnosis and risk factors? 5) Which investigations are needed for confirmation?

Answer: 1) CI = (82 × 3) / 360 = 0.68. 2) Moderate (Hb 90–70 g/l), hypochromic (CI <0.85), microcytic (MCV <80 fl), normoregenerative anemia. 3) Anemic (pallor, weakness, tachycardia, functional murmur) and sideropenic (pica, dry skin, brittle hair, angular stomatitis). 4) Iron deficiency anemia: prematurity (small iron store), early feeding with cow's milk, low meat intake. 5) Serum iron (↓), ferritin (↓), TIBC (↑), transferrin saturation coefficient (↓); stool for occult blood and helminthiasis.

Case 2

A 7-year-old boy. He had ARVI 2 weeks ago. This morning his mother saw many “bruises” and a fine punctate rash on his trunk and legs, and he had a nosebleed. On examination: general condition satisfactory, asymmetric petechiae and ecchymoses of different sizes and colors on the skin and mucous membranes, hemorrhages on the oral mucosa. The tourniquet sign is positive. The lymph nodes, liver and spleen are not enlarged. CBC: Hb 118 g/l, leukocytes 7.2×10^{9}/l, normal differential, platelets 14×10^{9}/l. Bleeding time 12 minutes, clotting time normal.

Questions: 1) Determine the type of bleeding. 2) Presumptive diagnosis? 3) Which laboratory signs confirm it? 4) What must be done to exclude leukemia?

Answer: 1) Petechial-spotted (microcirculatory) type: asymmetric petechiae and ecchymoses of different colors and sizes (“leopard skin”), bleeding from the mucous membranes. 2) Immune thrombocytopenia (Werlhof disease) — 2–3 weeks after a viral infection. 3) Marked thrombocytopenia (<20×10^{9}/l — risk of severe bleeding), prolonged bleeding time, positive endothelial test, normal clotting time, no anemia or changes in other lineages. 4) The absence of

hepatosplenomegaly, lymphadenopathy and bone pain argues against leukemia, but for definitive exclusion — a myelogram (in Werlhof disease many megakaryocytes, no blasts).

Case 3

A 3-year-old child has been weak for 3 weeks, with fever of 37.5–38.5 °C and pallor, and does not want to walk because of leg pain. On examination: waxy pallor of the skin, petechiae on the legs. Lymph nodes of all groups are enlarged (1.5–2 cm), dense, painless. The liver projects 4 cm and the spleen 5 cm below the costal arch. Pain on tapping the sternum and tibias. CBC: Hb 72 g/l, reticulocytes 0.2%, leukocytes 38×10^9/l, blasts 62%, band forms 0%, segmented forms 6%, lymphocytes 32%, platelets 28×10^9/l, ESR 52 mm/h.

Questions: 1) Which syndromes do you identify? 2) Assess the hemogram. 3) Presumptive diagnosis and the method of confirming it?

Answer: 1) Intoxication, anemic (pallor, Hb 72 g/l), hemorrhagic (petechiae, thrombocytopenia), hyperplastic (lymphoproliferative) (lymphadenopathy, hepatosplenomegaly) and osteoarticular (ossalgia) syndromes. 2) Severe normochromic hyporegenerative anemia, thrombocytopenia, leukocytosis, blasts 62%, no intermediate forms (band forms) — “leukemic hiatus”, raised ESR. 3) Acute leukemia (at 3 years most often lymphoblastic); confirmation — bone marrow aspiration (myelogram: blasts ≥20–25%), cytochemical and immunophenotypic studies.

Quick review

Table 11.10. *Key facts on Topic 11*

Question	Answer
Periods of fetal hematopoiesis	Yolk sac (2–3 weeks) → liver (6–7 weeks, max. 5 months) → bone marrow (4–5 months)
HbF	60–85% at birth; 15% at 1 year; ≤2% at 3 years
Aspiration site	Newborn — calcaneus; <1 year — tibia; older — iliac crest
Erythrocyte life span	≈120 days; platelet 9–11 days
Iron accumulation	Weeks 28–32
Blood volume	Newborn 80–90 ml/kg; adults 65–70 ml/kg
Blood of the newborn	Hb 180–240; erythrocytes 5–7; leukocytes 10–30; ESR 1–2
Crossovers	4–6 days and 5 years
Physiological anemia	3–6 months
Anemia cut-off	Newborn <145; 6 months–5 years <110; 5–11 years <115; ≥12 years <120 g/l
Severity	Mild ≥90; moderate 90–70; severe <70 g/l
Color index	(Hb × 3)/erythrocytes; 0.85–1.05
IDA	Hypochromic, microcytic, ferritin ↓, TIBC ↑, “PHA”
B_{12} deficiency	Hyperchromic, macrocytic, megaloblasts, hypersegmentation
Hemolysis	Jaundice + splenomegaly + reticulocytosis
Bleeding time (Duke)	2–4 minutes — platelet component
Clotting time (Morawitz)	3–5 minutes — plasma component

Question	Answer
Hemophilia	Hematoma type, hemarthrosis, clotting time ↑
Werlhof	Petechial-spotted, thrombocytopenia, bleeding time ↑
Henoch–Schönlein	Symmetric purpura + joints, abdomen, kidneys
Acute leukemia	Blasts, “leukemic hiatus”; ≥20–25% blasts in the myelogram
Periods of hematopoiesis	Embryonic (yolk sac, 2–3 weeks) → liver (6–7 weeks, maximum at 5 months; spleen at 3–4 months) → bone marrow (from 4–5 months)
Types of hemoglobin	HbP (embryonic), HbF (fetal; 60–85% at birth, ≈15% at 1 year, ≤2% at 3 years), HbA (adult)
Life span of blood cells	Erythrocyte ≈120 days (60–90 in the newborn); platelet 9–11 days; neutrophil hours in the blood; monocyte ≈12 hours in the blood
Hematopoietic factors	Erythropoietin (kidney), thrombopoietin, B_{12} (+ Castle's factor), folic acid, iron
Parameters of iron metabolism	Ferritin 15–150 ng/ml (<12–15 — deficiency); transferrin saturation 25–45% (<17% — deficiency); TIBC 40.6–62.5 µmol/l (raised in anemia)
Components of hemostasis	Vascular (angiospasm), platelet (adhesion, aggregation), plasma (13 factors, fibrin); then retraction and fibrinolysis
Leukocyte crossovers	I: day 4–6 (neutrophils = lymphocytes ≈40–45%); II: 4–6 years (≈5 years); 12–14 years — adult differential
WHO anemia cut-off	Newborn Hb <145 g/l; 6 months–5 years <110; 5–11 years <115; ≥12 years <120 g/l
Severity of anemia	Mild Hb 110(120)–90; moderate 90–70; severe <70 g/l
Types of anemia (color index/MCV)	Iron deficiency — hypochromic microcytic; B_{12}/folate — hyperchromic macrocytic; thalassemia — hypochromic microcytic; hemolytic/posthemorrhagic/aplastic — normochromic
Erythrocyte indices	MCV 80–95 fl; MCH 27–31 pg; MCHC 33–37 g/dl; color index 0.85–1.05; hematocrit 35–45%
Hemostasis tests	Bleeding time (Duke 2–4 min) — platelet; clotting time — plasma; coagulogram: APTT, PTI, fibrinogen, D-dimer
Clinical signs in anemia	Pallor, weakness, tachycardia, functional murmur, brittle hair/nails, distorted taste (chalk, soil), angular stomatitis
Normal myelogram	Leuko-erythroblastic ratio ≈3:1; blasts ≤5%; granulocytes 40–60%; megakaryocytes ≈0.5%

TOPIC 12

The digestive system: anatomical and physiological features, history taking, inspection, examination of the oral cavity, palpation and percussion of the abdomen, liver and spleen, auscultofriction

Learning objectives

The student should know:

- the embryonic development of the digestive organs and the main periods of formation of congenital defects;
- the age-related anatomical and physiological features of the oral cavity, esophagus, stomach, intestines, liver and pancreas;
- the features of digestion, intestinal microflora and defecation in children;
- the methods of history taking, inspection, palpation, percussion and auscultation in digestive diseases;
- the age norms for liver size and the pain points of the digestive organs.

The student should be able to:

- take complaints and history from a child with a digestive disease;
- inspect the oral cavity, tongue, teeth, pharynx and abdomen;
- perform superficial and deep (Obraztsov–Strazhesko) palpation of the abdomen;
- palpate the liver, gallbladder, pancreas and spleen; test the gallbladder signs;
- percuss the liver according to Kurlov and detect free fluid in the abdominal cavity;
- determine the lower border of the stomach by auscultofriction.

12.1. Development of the digestive system

The digestive organs form at very early stages of embryogenesis: **on day 7–8** the **primitive gut** begins to form from the endoderm, and **on day 12** it divides into an intraembryonic part (the future digestive tube) and an extraembryonic part (the yolk sac).

- **Foregut** — pharynx, esophagus, stomach, part of the duodenum, the pancreatic and hepatic buds;
- **midgut** — duodenum, jejunum and ileum, the right half of the colon;
- **hindgut** — the remaining parts of the colon.

In **weeks 4–10** of pregnancy the primitive gut grows rapidly, leaves the abdominal cavity through the umbilical hernia, **rotates 270°** and returns. Disturbance of these processes results in **intestinal rotation anomalies, embryonic hernias (omphalocele), atresias, stenoses, Meckel's diverticulum**. **From week 16–20** the fetus swallows amniotic fluid (up to 500 ml per day) — the digestive organs “train”; **meconium** forms.

12.2. Anatomical and physiological features of the digestive organs

General features:

- up to 1–1.5 years, especially in the 1st half-year, the digestive system is adapted to **lactotrophic (milk) nutrition**;

- along the whole digestive tube the mucosa is **thin, delicate, drier and easily injured**; the submucosa is **rich in blood vessels** and consists of loose connective tissue; **muscle and elastic fibers are poorly** developed;
- the sucking and swallowing reflexes are well developed;
- the glands have **low secretory activity**: little digestive juice, low enzyme activity;
- conditioned reflex activity is easily disturbed by external influences (overheating, chilling, pain, acute illness, separation from the mother) — predisposition to **functional dyspepsia**.

Oral cavity

- The oral cavity is relatively **small** and filled by the tongue; the tongue is short, wide and thick; the lips and cheeks are thick with well-developed muscles; in the cheeks there are dense **Bichat's fat pads** (rich in solid fatty acids) — the cheeks are not drawn in during sucking; the gums have **ridge-like thickenings** that help sucking, and the lips transverse folds.
- The mucosa is **delicate, dry** and rich in blood vessels — it is easily injured and infected (**candidiasis** — “thrush”, aphthous stomatitis).
- **The salivary glands** are insufficiently developed in the newborn: saliva is scanty and thick, with a neutral or weakly acidic reaction and little amylase. **From 3–4 months** salivation increases sharply and the child cannot swallow it all — **physiological drooling** (3–6 months). Salivary amylase reaches adult levels at 1 year.
- **The act of sucking** consists of three phases: **aspiration** (the lips enclose the nipple and negative pressure is created in the mouth), **compression** (milk is extracted) and **swallowing**. Because the entrance to the larynx lies at the posterior border of the hard palate, the child **can breathe and swallow at the same time**. Absence of the swallowing reflex is a sign of extreme prematurity.

The esophagus and stomach

- **The esophagus** is funnel-shaped, relatively short (10–12 cm in the newborn) and narrow; its mucosa is thin, with few glands, rich in blood vessels; the muscle layer is weak; the physiological narrowings are poorly expressed.
- **The lower esophageal (cardiac) sphincter is poorly** developed, while the muscles of the pyloric part are well developed — predisposition to **regurgitation** and **aerophagia**.
- **The stomach** in infancy lies **horizontally** and varies in shape (round, pear-shaped); once the child begins to walk (9–12 months) it takes a **vertical** position; the fundus and muscle layer are poorly developed.

Table 12.1. *Stomach capacity by age*

Age	Physiological capacity
Newborn (day 1)	7–10 ml (anatomical 30–35 ml)
Day 4	40–50 ml
Day 10	80 ml
From 1 month	Increases by ≈25 ml each month (≈100–150 ml at 3 months)
1 year	250–300 ml
3 years	400–600 ml
10–12 years	1300–1500 ml

- In the gastric mucosa, the pepsin-producing **chief cells** and the hydrochloric acid-secreting **parietal cells** are few and functionally immature; the cells producing protective mucus are few. The acidity of the gastric juice is low (pH ≈5–6 in the newborn, ≈3–4 at 1 year); it reaches adult levels **at 10–12 years**. Proteins are broken down mainly by **chymosin** (rennin) and gastricsin; **gastric lipase** partly breaks down the fats of breast milk.
- Gastric emptying: breast milk — **2–3 hours**, cow's milk and formulas — **3–4 hours** (which is why the intervals are longer with artificial feeding).

The intestines

- The intestines are relatively **long**: in the newborn **6–8 times** the body length (4–5 times in adults); the small intestine is ≈1 m per 1 kg of body mass. The mesentery is long and mobile and the intestinal loops are poorly fixed — predisposition to **intussusception and volvulus**.
- **The duodenum** is ring-shaped, at the level of LI, mobile; it becomes fixed by fatty tissue by 7 years.
- The mucosa is thin, rich in blood vessels, with many villi — **a large absorptive surface**, but **high permeability** (weak barrier function): toxins, allergens and microbes easily pass into the blood — predisposition to toxicosis and allergy.
- **The ileocecal valve (Bauhin's valve)** is relatively weak — cecal flora may pass into the ileum.
- **The appendix** is conical with a wide entrance — at an early age **appendicitis is rare**, but it runs a rapid destructive course and peritonitis develops quickly (the omentum is short).
- **The colon** is not fully mature; formation of the cecum is completed at the end of the 1st year; the sigmoid colon is relatively long and mobile — predisposition to **constipation**.
- **The rectum** in the newborn is cylindrical, without an ampulla or folds, relatively long and with little fatty tissue — predisposition to **prolapse** of the mucosa (straining, whooping cough); it takes up its final position at 2 years.
- **Digestion:** in infancy **membrane (parietal) digestion** predominates over cavity digestion; in the newborn **pinocytosis** (direct absorption of proteins — breast milk immunoglobulins) is preserved. **Lactase** activity is high in the 1st year and then decreases.
- **Motility** is active: food passes through the intestine in **4–18 hours** (≈24 hours in adults) — the bowel empties frequently.

The pancreas and liver

- **The pancreas** is small in the newborn (≈3 g; ≈50 g at 15 years) and insufficiently differentiated; its exocrine activity is low in the 1st year (little amylase and lipase), increases rapidly after complementary foods are introduced and reaches adult levels **at 5–6 years**. The juice contains trypsin, chymotrypsin, carboxypeptidases, elastase (proteins), amylase and lipase.
- **The liver** is **large** relative to body mass: **≈4%** in the newborn (2% in adults), occupying a large part of the abdominal cavity; the left lobe is relatively large (it becomes smaller at 1.5 years); lobulation becomes distinct at the end of the 1st year; its structure approaches that of adults **at 8 years**. It is rich in blood vessels and poor in connective tissue — it **enlarges quickly** in infection and intoxication.
- The **detoxifying** and **conjugating** (glucuronyltransferase) functions of the liver are immature — **physiological jaundice**, sensitivity to drugs. Little bile is secreted and there are few bile acids (poor emulsification and absorption of fats — more fat in the stool); taurine conjugates predominate in bile (antibacterial effect), and there is little cholesterol and salts — **gallstones are rare**.

- **The gallbladder** in the newborn is cylindrical or pear-shaped and lies deep in the liver; at 10–12 years it resembles that of adults.

Intestinal microflora and stool

- The fetal intestine is **sterile**. In the first hours after birth the intestine is **colonized** by microbes (the first "aseptic" phase — 10–20 hours; the second — multiplication; the third — stabilization).
- In a **breastfed** child **bifidobacteria** predominate from the end of the 1st month (90–99%) — the bifidus factor in breast milk (lactulose, oligosaccharides) supports them; they create an acidic environment, protect against pathogenic flora and synthesize vitamins. **With artificial feeding** there are more E. coli, enterococci, lactobacilli and opportunistic flora.
- **Meconium** — the first stool: a dark green, sticky, odorless mass passed in the first **1–3 days** of life (swallowed amniotic fluid, shed epithelium, bile, mucus). **On days 3–5 — transitional stool** (green-yellow, mucous, frequent). From day 5 — permanent stool.

Table 12.2. *Stool of a breastfed and a formula-fed child*

Sign	Breastfed	Formula-fed
Frequency	3–6 times a day in the 1st month (often after each feed), then 1–3	1–2 times
Color	Golden yellow (bilirubin), may turn green in air	Pale yellow or brown
Consistency	Mushy, homogeneous	Firmer, "putty-like"
Odor	Sour	Putrid (unpleasant)
Reaction	Acidic (pH 4.8–5.8)	Neutral or alkaline

REMEMBER
In an infant, **regurgitation 1–2 times a day** (within 1 hour after feeding, without distress, with good weight gain) is physiological; the causes are a weak cardiac sphincter, the horizontal position of the stomach and swallowing air. Normally a child over 1 year should pass stool at least **1 time in 36 hours**.

12.3. History taking

Abdominal pain

Small children cannot localize pain: it shows as **general restlessness**, screaming, crying, refusal of the breast, vomiting, drawing up and kicking the legs. Preschool children often point to the pain **around the umbilicus** (regardless of which organ is affected). In older children the following are determined:

- **location** and **radiation**;
- **character** (dull, stabbing, colicky, cutting, "dagger-like");
- **duration**, **periodicity** and **seasonality** (worsening in autumn and spring — peptic ulcer disease);
- **relation to food:** during eating; **early** (30–60 minutes after eating); **late** (after 1.5–3 hours); **hunger pain** (6–7 hours after eating, relieved by eating); **night pain** (from 23:00 to 3:00);
- relation to defecation;
- aggravating and relieving factors, pain-relieving drugs.

Table 12.3. *Location of pain and its probable causes (Kildiyarova)*

Location of pain	Probable diseases
Right hypochondrium	Liver (hepatitis), biliary tract (cholecystitis, gallstone disease, dyskinesia, giardiasis), head of the pancreas, the right half of the colon; extrapulmonary — right lower lobe pneumonia, pleurisy, kidney diseases
Epigastrium	Cardiac part of the esophagus, stomach and duodenum (functional disorders, gastritis, gastroduodenitis, peptic ulcer disease), hernia of the linea alba
Left hypochondrium	Stomach, body and tail of the pancreas (pancreatitis), the left half of the colon, spleen; left-sided pneumonia, pleurisy, myocarditis
Periumbilical area	Small intestine (enteritis), duodenum, pancreas, mesenteric lymph nodes, giardiasis
Right iliac area	Acute appendicitis, intussusception, Meckel's diverticulum, Crohn disease, mesenteric lymphadenitis; diseases of the kidneys, urinary tract and genitals
Suprapubic	Bladder diseases
Over the whole abdomen	Enterocolitis, intestinal colic, the abdominal form of hemorrhagic vasculitis, intestinal infections, helminthiasis, ascites, peritonitis, renal colic, acute infections (viral hepatitis, meningitis)

Radiation of pain: to the right shoulder and shoulder blade — the hepatobiliary system; to both hypochondria — duodenitis, pancreatitis; to the lower back, the back (“girdle-like”) — **pancreatitis**; to the sacrum, perineum — the anorectal area.

Dyspeptic signs

- **Changes in appetite:** reduced (gastritis, hepatitis, pancreatitis, intoxication), lost (**anorexia**), increased (malabsorption, chronic pancreatitis, recovery period), sharply increased (**bulimia** — diabetes mellitus, CNS pathology), distorted (**pica** — iron deficiency), selective appetite (dislike of fatty food — liver and biliary tract diseases).
- **Nausea and vomiting:** time of vomiting, relation to food, presence or absence of relief after vomiting, character of the vomit (unchanged food, bile, blood, “coffee grounds”, sour smell), volume.
- **Regurgitation** — the return of food to the mouth in infancy without straining of the abdominal muscles.
- **Belching** (air, sour, bitter, putrid) and **heartburn** (burning behind the sternum and in the epigastrium) — gastroesophageal reflux, gastritis.
- **Swallowing disorder (dysphagia)**, painful swallowing (odynophagia).
- **Constipation** — no defecation for more than 36 hours, or difficult, painful defecation, hard stool; **diarrhea** — frequent passage of liquid or mushy stool; **meteorism** — abdominal distension.
- **Unusual taste in the mouth:** sour (gastroduodenitis, peptic ulcer disease), bitter or metallic taste (liver diseases); **thirst** (dehydration, diabetes mellitus); **drooling** (ascariasis, pancreatitis, stomatitis; physiological at 3–6 months).

History

History of the present illness: onset and course of the disease, results of previous investigations and treatment.

Life history: the course of the mother's pregnancy and delivery; the neonatal period (passage of meconium, jaundice); type of feeding and timing of complementary foods, food intolerance (milk, gluten); dyspeptic disorders; allergic history; eating habits (in older children — irregular meals, dry food, fast food); medication; **family history** — peptic ulcer disease, gastritis, gallstone disease, celiac disease, cystic fibrosis, inflammatory bowel diseases.

12.4. Inspection

General inspection

The following are assessed: the child's **general condition**, **position** (with abdominal pain — lying on the side with the legs drawn up to the abdomen; in peritonitis — lying motionless), the degree of **physical development** (hypotrophy — malabsorption, chronic diseases), the color of **the skin and mucous membranes** (pallor, **jaundice**, subicterus), rashes, hemorrhages, **spider angiomas**, **"liver palms"** (palmar erythema), striae, scars, a venous network on the anterior abdominal wall ("**caput medusae**" — portal hypertension).

Examination of the oral cavity

Examination of the oral cavity and pharynx is performed **last** so as not to upset the child. The corners of the mouth are gently pulled with a spatula, and the following are examined in turn:

- **lips** — color, moisture, dryness, cracks, herpes, **angular cheilitis** ("perlèche" — deficiency of iron and B vitamins);
- **oral mucosa** — color (pale, hyperemic, cyanotic, icteric), **aphthae** and ulcers (stomatitis), **candidiasis** (a white cheesy coating), **Filatov–Koplik spots** on the cheeks in measles;
- **gums** — looseness, bleeding (gingivitis, leukemia, scurvy);
- **teeth** — number (appropriate for age), position, bite, **caries**, enamel changes;
- **tongue** — size (macroglossia — hypothyroidism, Down syndrome), moisture, color, state of the papillae, **coating** (white, yellowish, brown), **"geographic" tongue** (exudative diathesis), **"lacquered" tongue** (atrophy of the papillae — iron and B_{12} deficiency, liver diseases), **"raspberry" tongue** (scarlet fever), **short frenulum of the tongue**;
- **hard and soft palate** — clefts, "gothic" palate;
- **the ducts of the salivary glands** (hyperemia, edema — mumps);
- **pharynx** and tonsils;
- **breath odor** — acetone (ketoacidosis, acetonemic vomiting), "liver odor" (liver failure), putrid (caries, gastritis, tonsillitis).

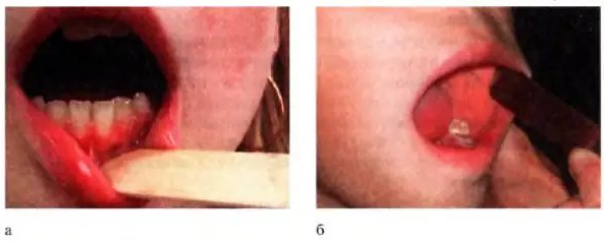


Figure 12.1. *Examination of the oral mucosa: a — gums; b — cheeks (Navruzova, Geppe)*

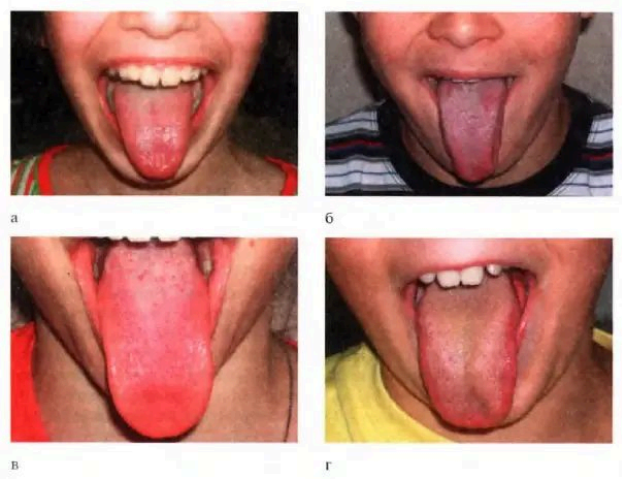


Figure 12.2. *Examination of the tongue: a — normal; b — "geographic" tongue; c — coated at the root; d — brown coating (Navruzova, Geppe)*

Inspection of the abdomen

The abdomen is inspected with the child standing and lying down:

- **shape and size** — normally slightly protuberant in infancy (weak muscles, relatively long intestine); it flattens after 3–5 years;
- **symmetry** — uneven enlargement (hepato- or splenomegaly, tumor, hernia);
- **participation in breathing** — lagging of one half (inflammation), **absence of abdominal breathing** (peritonitis, perforation);
- **visible peristalsis** — obstruction to passage from the intestine or stomach (in pyloric stenosis, the "hourglass" sign in the epigastrium);
- the state of **the umbilicus** (hernia, umbilical wound — in the newborn), **hernia of the linea alba**, separation of the muscles (diastasis);
- **venous network** ("caput medusae").

Table 12.4. *Changes in the shape and size of the abdomen (Kildiyarova)*

State	Shape of the abdomen
Meteorism	Evenly enlarged, skin tense, umbilicus protruding
Enlargement of the liver and/or spleen, tumor, hernia	Unevenly enlarged, asymmetric
Ascites	Markedly enlarged, resembling "an inflated balloon", skin tense and shiny, umbilicus protruding; spreads to the sides when lying ("frog belly")
Rickets	"Frog belly" — widened to the sides, with meteorism and hypotonia of the abdominal wall
Obesity	Evenly enlarged, umbilicus retracted, fat folds
Emaciation; onset of peritonitis; after diarrhea and vomiting	Retracted ("scaphoid")
Malabsorption (celiac disease)	A markedly enlarged abdomen, thin limbs

12.5. Palpation of the abdomen

General rules: a warm room, the doctor's hands warm and nails short; the child lies on the back with the head slightly raised, the arms along the body, the legs slightly bent at the knee and hip joints (the

abdominal press relaxes); the doctor sits on the child's right side, facing the child. Talk to the child to distract attention; a small child may be on the mother's lap. A painful area is palpated **last**. During palpation the doctor watches the child's face (pain reaction).

Abdominal regions: two horizontal lines (through the 10th ribs and the anterior superior iliac spines) and two vertical lines (along the outer edges of the rectus muscles) divide the abdomen into **9 regions**: right and left hypochondrium, epigastrium; right and left lateral regions, umbilical region; right and left iliac regions, suprapubic region.

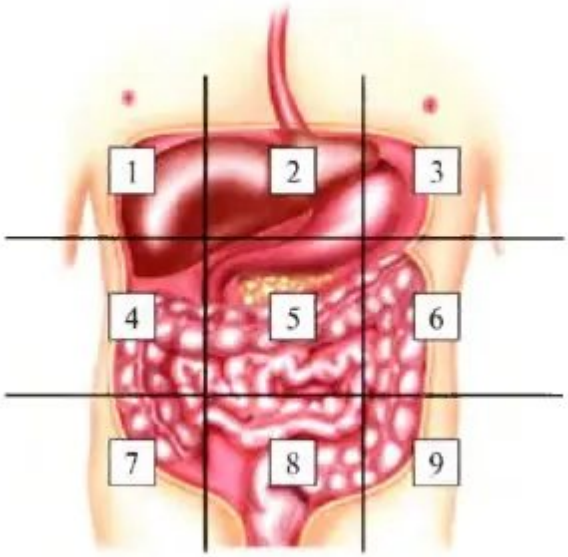


Figure 12.3. *Abdominal regions and the projection of the organs: 1 — right hypochondrium; 2 — epigastrium; 3 — left hypochondrium; 4 — right lateral; 5 — umbilical; 6 — left lateral; 7 — right iliac; 8 — suprapubic; 9 — left iliac region (Navruzova, Geppe)*

Superficial (orienting) palpation

The palm is placed flat on the abdominal wall and gentle pressure is applied with fingers II–V. Order: **starting from the left iliac region, counterclockwise** — left lateral region → left hypochondrium → epigastrium → right hypochondrium → right lateral region → right iliac region → periumbilical and suprapubic regions.

Determined: tenderness of the abdominal wall (localized or diffuse), **tension of the abdominal muscles** (local or diffuse — a “board-like” abdomen — peritonitis), separation of the rectus muscles, hernias (umbilical, inguinal, linea alba), superficial tumors, a markedly enlarged liver and spleen, zones of hyperesthesia (**Zakharyin–Head zones**), asymmetry in the thickness of the subcutaneous fat.

Muscle tension may be **active** (the child is afraid, ticklish — it disappears when attention is distracted) or **passive** (inflammation — it persists). In the newborn the umbilical ring, the periumbilical vessels and the umbilical wound (discharge, hyperemia) are palpated.

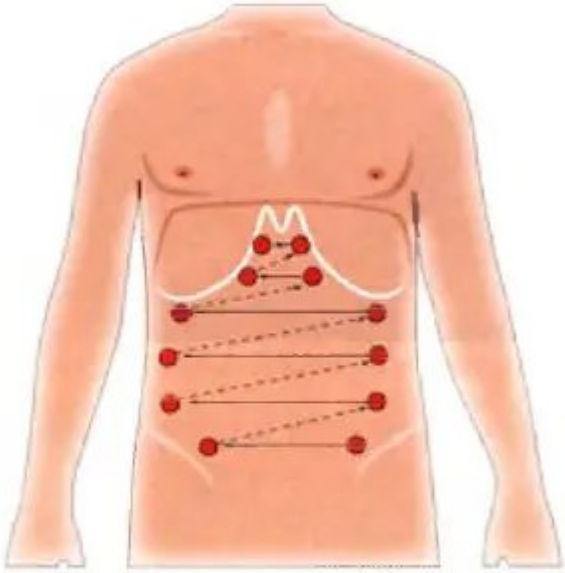


Figure 12.4. *Superficial palpation of the abdomen: directions of hand movement (Navruzova, Geppe)*

Peritoneal signs:

- **Shchetkin–Blumberg sign** — a sharp increase in pain when the hand, slowly pressed into the abdominal wall, is **quickly withdrawn** — irritation of the peritoneum (peritonitis, acute appendicitis);
- **Obraztsov's sign** — increased pain on palpation of the right iliac region while the straight right leg is raised (appendicitis);
- **Rovsing and Sitkovsky** signs (appendicitis) — studied in detail in the surgery course.

Deep (methodical, sliding, topographic) palpation — according to Obraztsov–Strazhesko

Four steps: 1) place the fingers parallel to the organ being palpated; 2) move **a skin fold** perpendicular to the organ (toward the umbilicus); 3) during expiration, **plunge the fingers deep** into the abdominal cavity, to the posterior wall; 4) **slide perpendicular** over the organ.

Order: sigmoid colon → cecum (with the terminal ileum) → ascending colon → transverse colon → descending colon → liver and gallbladder → spleen → stomach and duodenum → pancreas → mesenteric lymph nodes.

- **Sigmoid colon** — in the left iliac region, along the bisector of the left lower quadrant; normally a smooth, dense-elastic, painless, mobile, non-gurgling cylinder 1.5–2 cm in diameter.
- **Cecum** — in the right iliac region, by the same method; soft-elastic, slightly gurgling, painless, mobile.
- **Ascending and descending colon** — in the right and left lateral regions, with the other hand placed under the lower back as “support”.
- **Transverse colon** — with both hands, 2–3 cm above the umbilicus, at the outer edges of the rectus muscles, during expiration.
- **The small intestine** is usually not palpable (deep and mobile).

For each part of the intestine **the location, shape, size, consistency, surface, mobility, gurgling and tenderness** are assessed.



Figure 12.5. *Palpation of the cecum (Navruzova, Geppe)*



Figure 12.6. *Palpation of the descending colon (Navruzova, Geppe)*



Figure 12.7. *Palpation of the transverse colon (Navruzova, Geppe)*

The greater curvature and pyloric part of **the stomach** are difficult to palpate; first its **lower border** is determined by percussion palpation (Obraztsov's "succussion splash") or by **auscultofriction**, then the greater curvature is felt above the umbilicus as a soft thin fold by deep sliding palpation. Epigastric pain — a positive **Chernik's sign**.

Palpation of the liver

Sliding palpation according to Strazhesko — in infants and young children and when the liver is enlarged, only this method: the fingers of the right hand are placed in a line parallel to the lower edge of the liver and slide gently **from top to bottom**; the whole accessible surface of the liver is palpated.

According to Obraztsov (if the liver is not palpable): the right hand is placed flat on the right half of the abdomen at the level of the umbilicus or lower; the left hand encircles the lower right part of the chest (restricting respiratory movement). During expiration the right hand is plunged deep, then the child **takes a deep breath** — the descending edge of the liver strikes the fingers and "slips past" them. Palpation is repeated along the midclavicular, anterior axillary and midline.

Assessed: distance below the costal arch (cm), **edge** (sharp, blunt, rounded), **consistency** (soft-elastic, dense, hard), **surface** (smooth, uneven, bumpy), **tenderness**.



Figure 12.8. *Palpation of the liver (Navruzova, Geppe)*

Table 12.5. *Normal position of the lower edge of the liver (along the right midclavicular line)*

Age	Distance below the costal arch
Up to 6 months (newborn)	Up to 2–3 cm
6 months – 2 years	Up to 1.5–2 cm
3–7 years	Up to 1 cm
Over 7 years	Not palpable (in quiet breathing); along the midline it does not extend below the upper 1/3 of the distance between the umbilicus and the xiphoid process

Normally at an early age the liver is **soft-elastic**, smooth, with a sharp edge, painless; at school age — **dense-elastic** (as in adults).

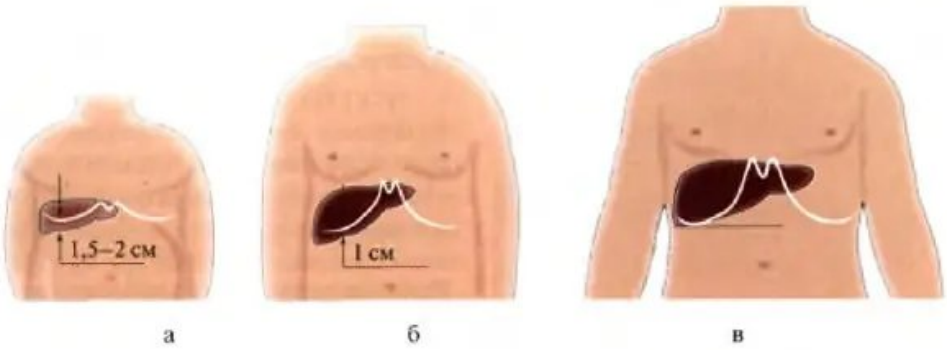


Figure 12.9. *Position of the liver relative to the right costal arch: a — up to 3–4 years; b — from 3–4 to 7 years; c — over 7 years (Navruzova, Geppe)*

The gallbladder

The gallbladder is normally **not palpable** (it does not extend beyond the edge of the liver). Its projection is the point where the outer edge of the right rectus muscle crosses the costal arch (**Kehr's point**). Palpation of an enlarged painless gallbladder — **Courvoisier–Terrier sign**.

Gallbladder ("cystic") signs:

- **Kehr's sign** — pain when a finger is pressed into Kehr's point (on inspiration);
- **Murphy's sign** — pain on deep inspiration while the right hypochondrium is pressed (the child stops breathing in);
- **Ortner–Grekov sign** — pain on tapping the right costal arch with the edge of the palm (compared first on the left, then on the right);
- **Georgievsky–Mussy (phrenicus) sign** — pain on pressure between the heads of the sternocleidomastoid muscle on the right;
- **Razdolsky–Lepine sign** — pain on light percussion over the gallbladder projection;
- **Boas sign** — pain on the right in the area of the transverse processes of ThXI–LI.



Figure 12.10. *Kehr's sign (Navruzova, Geppe)*



Figure 12.11. *Ortner's sign (Navruzova, Geppe)*



Figure 12.12. *Georgievsky–Mussy sign (Navruzova, Geppe)*

The pancreas and mesenteric lymph nodes

Grott's method — the child lies on the back, the clenched right fist placed under the lower back, the legs bent; the fingers are plunged during inspiration between the umbilicus and the hypochondria toward the spine at an angle of 45° (rarely used in practice — pain may occur even in a healthy child).

Pain points and zones:

- **Chauffard's zone** — between the linea alba and the bisector of the right upper quadrant — the head of the pancreas and the duodenum (pyloroduodenal zone);
- **Desjardins' point** — on the bisector of the right upper quadrant, 4–6 cm from the umbilicus (halfway) — the head of the gland;
- **Mayo-Robson point** — on the bisector of the left upper quadrant, in the outer 1/3 of the distance between the umbilicus and the costal arch — the tail of the gland;
- **Gubergrits–Skulsky zone** — on the left, symmetric to Chauffard's zone — the body of the gland;
- **Kach's sign** — pain on the left in the area of the transverse processes of ThX–XI;
- **Georgievsky–Mussy sign on the left** — pancreatitis.

Mesenteric lymph nodes are palpated in **Sternberg's zones** (in the left upper and right lower quadrants, along the outer edge of the rectus muscle); **Porges' point** — the duodenojejunal junction (to the left and above the umbilicus).

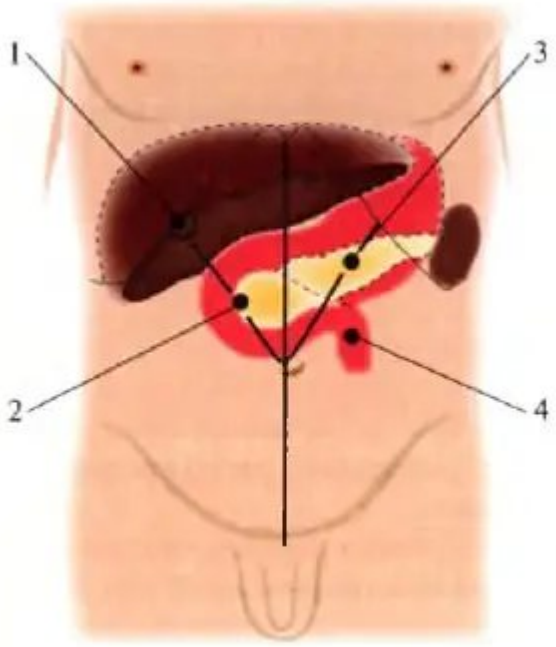


Figure 12.13. *Pain points on the anterior abdominal wall: 1 — Kehr's point (gallbladder); 2 — Desjardins' point; 3 — Mayo-Robson point; 4 — Porges' point; Chauffard's zone (Navruzova, Geppe)*

Spleen — sliding palpation according to Strazhesko, supine and on the right side (see topic 4); normally not palpable.

12.6. Percussion

Over the abdomen there is normally a **tympanic** sound almost everywhere (gas in the intestine); over the liver, the spleen, a stool-filled bowel (sigmoid) and a full bladder — a **dull** sound. Percussion is quiet.

Percussion of the liver

Up to 5 years the liver dimensions are determined along the right anterior axillary, midclavicular and midline. In children **over 5 years — three dimensions according to M.G. Kurlov**:

1. **The first dimension** — along the right midclavicular line: **the upper border** — from top to bottom, from the clear sound to the dull sound (normally rib VI), marked along the upper edge of the finger; **the lower border** — from the level of the umbilicus upward, from tympanitis to dullness (normally the edge of the costal arch), marked along the lower edge of the finger.
2. **The second dimension** — along the midline: the upper point is determined conventionally (because it merges with cardiac dullness) by dropping a perpendicular from the upper point of the first dimension to the midline; the lower border is found by percussion upward from the umbilicus (normally at the junction of the upper and middle thirds of the distance between the xiphoid process and the umbilicus).
3. **The third dimension** — along the left costal arch: the finger is placed perpendicular to the costal arch at the level of the attachment of rib VIII and percussion is performed along the arch toward the sternum; the border is marked along the outer edge of the finger. The distance from the upper point of the second dimension to this point is the third dimension.



Figure 12.14. *Determining the liver borders by percussion (Navruzova, Geppe)*

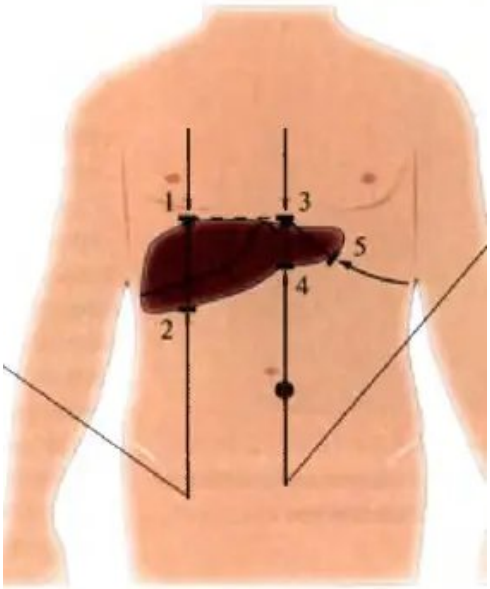


Figure 12.15. *Determining liver dimensions according to Kurlov: 1–2 — first, 3–4 — second, 3–5 — third dimension (Navruzova, Geppe)*

Table 12.6. *Percussion dimensions of the liver (Kildiyarova)*

Dimension	Early age	Preschool age	School age
I (right midclavicular line; up to 5 years — right anterior axillary line), cm	5	6–7	9–10
II (midline; up to 5 years — midclavicular line), cm	4	5	7
III (left costal arch; up to 5 years — midline), cm	5	6	8

MEMORY AID

The Kurlov dimensions in a school-age child — **"9–7–8"** (close to 9–8–7 in adults). Downward displacement of the liver edge with unchanged dimensions — **hepatoptosis** (right-sided pneumothorax, emphysema); upward displacement — ascites, meteorism; in these cases the liver itself is not enlarged.

Disappearance of liver dullness — **perforation** of the stomach or intestine (escape of gas into the abdominal cavity) — a sign requiring urgent surgery.

Detecting free fluid in the abdominal cavity

1. **Fluctuation (fluid wave) sign** — the left palm is placed on the lateral surface of the abdomen and a short tap is given on the opposite side with the fingers of the right hand; the vibration of the fluid strikes the left hand as "a wave". To exclude transmission through the abdominal wall, an assistant presses the edge of the palm on the middle of the abdomen.
2. **Percussion** — with the child lying on the back, percussion is performed from the umbilicus to the sides (finger parallel to the linea alba); **bilateral dullness in the lateral regions** — free fluid.
3. **With a change of position** — the child is turned on the side and percussion is repeated: if the dullness in the upper lateral region **disappears** — free fluid (the fluid has flowed down); if it persists — another cause (a full bowel, a tumor).

Other percussion signs

- **Mendel's sign** — pain on tapping with half-bent fingers over the projection of the greater and lesser curvature of the stomach — peptic ulcer disease, erosive gastritis;
- **Razdolsky's sign** — pain on percussion over the gallbladder projection.

12.7. Auscultation and auscultofriction

Auscultation of the abdomen: normally **faint, periodic sounds of intestinal peristalsis** are heard over the abdominal wall. **Increased peristalsis** — enteritis (loud gurgling over a wide area), the part proximal to a stenosis; **absence of peristalsis ("dead silence")** — intestinal paresis, peritonitis.

Auscultofriction — a method of determining the lower border of the stomach: the stethoscope is placed on the epigastric area (over the stomach projection), and at the same time light "stroke-like" rubbing movements are made with one finger along the abdominal skin **from below upward** (toward the stomach) to the left of the linea alba, radially to the stethoscope. As soon as the finger crosses the border of the stomach the sound becomes sharply **louder** — this is the lower border of the stomach. If light percussion taps are used instead of rubbing — **auscultatory percussion** (auscultopercussion). Normally the lower border of the stomach is **2–4 cm above** the umbilicus.

12.8. Examination of the rectum and anal area

Inspection — the anus and the surrounding skin: fissures, scratch marks (enterobiasis), maceration, **prolapse of the rectal mucosa**, gaping anus, polyps and hemorrhoidal nodes, fistulas, scars, infiltrates.

Digital examination — the child in the knee-elbow position (in a severe condition on the left side, legs drawn up to the abdomen), after defecation; with a gloved little finger (in small children) or index finger lubricated with petroleum jelly. **Determined:** sphincter tone, smoothness of the mucosa, state of the pararectal tissue, pain, tumor, polyp, intussusceptum; after withdrawal of the finger, traces

of **blood, mucus, pus** on the glove. This examination is important for excluding **surgical pathology (intussusception, appendicitis)** in abdominal pain.

12.9. Clinical section

"Acute abdomen" — alarming signs

> **CLINICAL SIGNIFICANCE**
> Signs in abdominal pain that suggest **acute surgical pathology** (appendicitis, intussusception, intestinal obstruction, perforation): refusal of food, change in behavior (intense crying or lying motionless, forced position); waking from pain; vomiting against a background of pain; **no passage of stool or gas**; pallor, cold sweat, **tachycardia out of proportion to the temperature**; **tension of the abdominal muscles**, a positive Shchetkin–Blumberg sign. Such a child is **not given** analgesics, and urgent surgical examination is arranged.

Intussusception

Most often in children **aged 4–12 months** (when complementary foods are introduced); sudden **colicky abdominal pain** (the child cries sharply, draws up the legs, then calms down), vomiting, **a sausage-shaped mass** palpable in the abdomen, later blood-stained mucus like **"raspberry jelly"** from the anus. On digital rectal examination — blood on the glove. Diagnosis — ultrasound; treatment — air reduction or surgery.

Hepatomegaly — is the organ enlarged?

Projection of the liver edge below the costal arch is not always enlargement: **hepatoptosis** (the diaphragm descends in pulmonary emphysema, asthma, right-sided pleurisy, pneumothorax) — the Kurlov dimensions are normal. In true hepatomegaly the dimensions are also increased: hepatitis, heart failure (congestion), storage diseases, hemolytic anemias, leukemias, tumors.

Practical skills

Skill 1. Examination of the oral cavity

1. Wash the hands, put on gloves; good light, a sterile spatula.
2. The child sits (a small child is held by the mother), the head is fixed.
3. Inspect the lips and the corners of the mouth.
4. Retract the cheeks and lips with the spatula and inspect the mucosa, gums and salivary ducts.
5. Assess the number and condition of the teeth and the bite; inspect the tongue (coating, papillae, moisture).
6. Press the middle third of the tongue with the spatula and inspect the pharynx and tonsils (last).
7. Assess the breath odor; record the result.

Skill 2. Superficial palpation of the abdomen

1. The child lies on the back, legs slightly bent, the doctor on the right; warm hands.
2. Place the palm flat on the abdominal wall and press gently with fingers II–V.
3. Palpate all regions symmetrically, starting from the left iliac region counterclockwise; the painful area last.
4. Detect pain, muscle tension, hernias, separation of the muscles, superficial masses.

5. Test the Shchetkin–Blumberg sign.
6. Conclusion: “The abdomen is soft, painless, peritoneal signs negative.”

Skill 3. Deep sliding palpation (Obraztsov–Strazhesko)

1. Sigmoid colon: the right hand in the left iliac region, fingers parallel to the axis of the bowel; move a skin fold toward the umbilicus, plunge deep on expiration, slide perpendicular to the bowel.
2. Palpate the cecum in the right iliac region in the same way.
3. Palpate the ascending and descending colon with the left hand placed under the lower back.
4. Palpate the transverse colon with both hands 2–3 cm above the umbilicus.
5. Record the location, diameter, consistency, surface, mobility, gurgling and tenderness of each part.

Skill 4. Palpation of the liver and percussion according to Kurlov

1. In a small child — sliding palpation from top to bottom according to Strazhesko; in an older child — according to Obraztsov on deep inspiration.
2. Along the midclavicular line determine how far the liver edge projects below the costal arch (cm), its edge, consistency, surface and tenderness.
3. Test the gallbladder signs (Kehr, Murphy, Ortner, Mussy).
4. Determine the three Kurlov dimensions by quiet percussion.
5. Compare the result with the age norm.

Skill 5. Detecting free fluid in the abdominal cavity

1. The child lies on the back; test the fluctuation sign (with an assistant).
2. Percuss from the umbilicus toward both lateral regions; mark the border of dullness.
3. Turn the child on the side and repeat percussion; assess the shift of dullness.
4. Conclusion.

Skill 6. Auscultofriction

1. Place the stethoscope on the epigastric area, to the left of the linea alba.
2. With one finger make light rubbing movements along the abdominal skin from below (from the level of the umbilicus) upward, toward the stethoscope.
3. Mark the place where the sound becomes sharply louder as the lower border of the stomach.
4. Measure the distance from the umbilicus (normally 2–4 cm above).

Review questions

1. How do the digestive organs form in the embryonic period? Which defects are related to disturbed intestinal rotation?
2. Name the general anatomical and physiological features of the digestive system in children.
3. What are the features of the oral cavity adapted to sucking? What is physiological drooling?
4. What are the age-related features of the esophagus and stomach and the stomach capacity? Why are infants prone to regurgitation?
5. What are the age-related features of the intestines and their clinical significance?
6. Name the age-related features of the liver and pancreas.

7. How does the intestinal microflora form? What is the difference between the stools of breastfed and formula-fed children?
8. What complaints occur in digestive diseases? How is abdominal pain described?
9. In what order is the oral cavity examined? What is the diagnostic significance of changes in the tongue?
10. In which conditions does the shape of the abdomen change?
11. How is superficial palpation of the abdomen performed and what is determined?
12. What are the steps and order of deep palpation according to Obraztsov–Strazhesko?
13. How is the liver palpated? What are the age norms for the liver edge?
14. Name the gallbladder signs and the pain points of the pancreas.
15. Explain the methods of liver percussion according to Kurlov, detection of free fluid in the abdominal cavity, and auscultofriction.

Test questions

1. When does the 270° rotation of the intestine take place in utero?
   A) In weeks 1–2
   B) In weeks 4–10
   C) In week 20
   D) In week 30
   E) After birth

2. At what age is physiological drooling seen?
   A) At birth
   B) At 3–6 months
   C) At 1–2 years
   D) At 3 years
   E) At 5 years

3. The main reason for the tendency of infants to regurgitate:
   A) Weakness of the pyloric sphincter
   B) Weakness of the cardiac sphincter and the horizontal position of the stomach
   C) Large stomach capacity
   D) Abundance of hydrochloric acid
   E) Length of the esophagus

4. The stomach capacity in a 1-year-old child:
   A) 30–35 ml
   B) 80–100 ml
   C) 250–300 ml
   D) 600–800 ml
   E) 1000 ml

5. Typical of the stool of a breastfed child:
   A) Putrid odor, alkaline reaction
   B) Golden yellow color, sour odor, acidic reaction
   C) Gray, shiny
   D) “Sheep droppings”
   E) Black color

6. For how many days is meconium passed?
   A) 1–3 days
   B) 7–10 days
   C) 2 weeks
   D) 1 month
   E) It is not passed

7. In a 3–7-year-old child the liver edge along the midclavicular line below the costal arch:
   A) Projects 3–4 cm
   B) Projects 2–3 cm
   C) Projects up to 1 cm
   D) Does not project at all
   E) Projects 5 cm

8. From which region does superficial palpation of the abdomen begin?
   A) Right hypochondrium
   B) Epigastrium
   C) Left iliac region
   D) Periumbilical area
   E) The painful region

9. With which organ does deep sliding palpation begin?
   A) Liver
   B) Stomach
   C) Sigmoid colon
   D) The spleen
   E) Transverse colon

10. Pain on tapping the right costal arch with the edge of the palm is:
    A) Kehr's sign
    B) Murphy's sign
    C) Ortner–Grekov sign
    D) Shchetkin–Blumberg sign
    E) Mendel's sign

11. Pain at the Mayo-Robson point indicates involvement of which organ?
    A) The gallbladder
    B) The tail of the pancreas
    C) Cecum
    D) The spleen
    E) Duodenum

12. The first Kurlov dimension of the liver in a school-age child:
    A) 4–5 cm
    B) 6–7 cm
    C) 9–10 cm
    D) 12–14 cm
    E) 15 cm

13. What is the disappearance of liver dullness a sign of?
    A) Hepatitis
    B) Ascites
    C) Perforation of the stomach or intestine
    D) Hepatoptosis
    E) Heart failure

14. Indicates free fluid in the abdominal cavity:
    A) Dullness in the lateral regions that shifts with a change of position
    B) Tympanitis around the umbilicus
    C) Disappearance of liver dullness
    D) Increased peristalsis
    E) Shchetkin–Blumberg sign

15. Auscultofriction determines:
    A) The lower border of the liver
    B) The lower border of the stomach
    C) The size of the spleen
    D) Intestinal peristalsis
    E) The border of the bladder

16. What does the fetus do from week 16–20 of pregnancy (the digestive organs “train”)?
    A) Drinks the mother's blood
    B) Does not digest
    C) Swallows amniotic fluid
    D) The stomach is empty
    E) Only passes meconium

17. Why are Bichat's fat pads needed?
    A) So that the cheeks are not drawn in during sucking
    B) For breathing
    C) For digestion
    D) For keeping warm
    E) For salivation

18. At what age does salivary amylase reach adult levels?
    A) 5 years
    B) 1 year
    C) 12 years
    D) 6 months
    E) 1 month

19. What is the absence of the swallowing reflex a sign of?
    A) Candidiasis
    B) Pylorospasm
    C) Rickets
    D) Health
    E) Extreme prematurity

20. The lower (cardiac) esophageal sphincter and regurgitation in the newborn:
    A) No sphincter, never vomiting
    B) The esophagus is long
    C) The sphincter is weak, tendency to regurgitation
    D) The sphincter is strong, no regurgitation
    E) The pyloric part is weak

21. The physiological stomach capacity in a 1-year-old child:
    A) 250–300 ml
    B) 80 ml
    C) 600 ml
    D) 1500 ml
    E) 7–10 ml

22. The physiological stomach capacity in the newborn on day 1:
    A) 30–35 ml
    B) 80 ml
    C) 250 ml
    D) 7–10 ml
    E) 150 ml

23. In how much time does breast milk leave the stomach?
    A) 2–3 hours
    B) 6 hours
    C) 8 hours
    D) 3–4 hours
    E) 30 minutes

24. Which enzyme mainly breaks down proteins in the newborn's stomach?
    A) Lipase
    B) Lactase
    C) Amylase
    D) Chymosin (rennin) and gastricsin
    E) Trypsin

25. How many times longer than the body length is the intestine in the newborn?
    A) 4–5
    B) 10–12
    C) 1
    D) 2
    E) 6–8

26. The reason for the predisposition to intussusception and volvulus:
    A) The liver is large
    B) The mucosa is thick
    C) The intestine is short
    D) The mesentery is long and mobile, the intestinal loops are not fixed
    E) The mesentery is short

27. What does the high permeability of the intestinal mucosa in the newborn lead to?
    A) Anemia
    B) Predisposition to toxicosis and allergy
    C) Obesity
    D) Nothing
    E) Constipation

28. The reason for the predisposition to rectal prolapse:
    A) The intestine is short
    B) The rectum is thick
    C) Cylindrical, no ampulla/folds, little fatty tissue
    D) Appendicitis
    E) The sphincter is strong

29. What percentage of the body mass is the liver in the newborn?
    A) ≈0,5%
    B) ≈4%
    C) ≈10%
    D) ≈15%
    E) ≈2%

30. On which days is meconium passed?
    A) On days 1–3 of life
    B) At 3 months
    C) At 1 month
    D) Only in preterm babies
    E) On day 7–10

31. The reaction and odor of the stool in a breastfed newborn:
    A) Alkaline, putrid odor
    B) Acidic (pH 4.8–5.8), sour odor
    C) Neutral, odorless
    D) Acidic, putrid
    E) Alkaline, sour

32. The predominant microorganism of the intestinal flora in breastfeeding:
    A) Klebsiella
    B) E. coli
    C) Enterococci
    D) Candida
    E) Bifidobacteria (90–99%)

33. Where do small children usually point to abdominal pain (regardless of which organ is affected)?
    A) Under the left costal arch
    B) In the right iliac area
    C) In the epigastrium
    D) Suprapubic
    E) Around the umbilicus

34. How many hours after eating does the “hunger” pain typical of peptic ulcer disease appear?
    A) 1.5–3 hours
    B) During eating
    C) 6–7 hours
    D) 30–60 minutes
    E) 12 hours

35. “Girdle-like” pain is typical of disease of which organ?
    A) Liver
    B) Stomach
    C) Bladder
    D) Pancreas (pancreatitis)
    E) Appendix

36. Pain radiating to the right shoulder and shoulder blade:
    A) Pancreatitis
    B) Bladder
    C) Hepatobiliary system
    D) Appendicitis
    E) Colon

37. The most important probable cause of pain in the right iliac region:
    A) Esophageal reflux
    B) Cholecystitis
    C) Pancreatitis
    D) Acute appendicitis
    E) Gastritis

38. What is meant by constipation?
    A) No defecation for more than 36 hours, or difficult, hard stool
    B) Stool 3 times a day
    C) Vomiting
    D) Meteorism
    E) Liquid stool

39. Angular cheilitis (“perlèche”) is typical of which deficiency?
    A) Calcium
    B) Vitamin C
    C) Iodine
    D) Iron and B vitamins
    E) Vitamin K

40. A “raspberry” tongue is typical of which disease?
    A) Diphtheria
    B) Scarlet fever
    C) Influenza
    D) Measles
    E) Chickenpox

41. In which condition does a “lacquered” tongue (atrophy of the papillae) occur?
    A) Iron and B_{12} deficiency, liver diseases
    B) Scarlet fever
    C) Rickets
    D) Influenza
    E) Candidiasis

42. What does an acetone breath odor indicate?
    A) Caries
    B) Liver failure
    C) Gastritis
    D) Tonsillitis
    E) Ketoacidosis, acetonemic vomiting

43. A “frog belly” is typical of which disease?
    A) Peritonitis
    B) Rickets
    C) Pyloric stenosis
    D) Obesity
    E) Appendicitis

44. What is the visible peristalsis in the epigastrium in pyloric stenosis called?
    A) Caput medusae
    B) Ortner
    C) The “hourglass” sign
    D) Filatov
    E) Mussy

45. What does “caput medusae” mean on inspection of the abdomen?
    A) Rickets
    B) Not ascites, meteorism
    C) Abdominal hernia
    D) Hypertension
    E) Portal hypertension

46. When is the painful area palpated during abdominal palpation?
    A) Last
    B) It is not palpated at all
    C) In the middle
    D) Only on percussion
    E) First

47. A “board-like” abdomen is typical of which condition?
    A) Rickets
    B) Peritonitis
    C) Meteorism
    D) Gastritis
    E) Obesity

Clinical cases

Case 1

A 10-year-old girl complains of a dull pain like a feeling of heaviness in the right hypochondrium; the pain increases after fatty and fried food and radiates to the right shoulder; there is a bitter taste in the mouth and nausea. On examination: the tongue is coated with a yellowish coating. The abdomen is soft and tender in the right hypochondrium. The Kehr, Ortner and Murphy signs are positive. The liver is at the edge of the costal arch, 9–7–8 cm according to Kurlov. The spleen is not palpable.

Questions: 1) Involvement of which organ do you suspect? 2) Explain the positive signs. 3) Assess the liver dimensions. 4) Which investigations are needed?

Answer: 1) The gallbladder and biliary tract (biliary dyskinesia or cholecystitis). 2) Kehr — pain on deep palpation over the gallbladder projection; Ortner–Grekov — pain on tapping the right costal arch; Murphy — pain on deep inspiration — signs of gallbladder irritation. 3) The liver dimensions are within the age norm (9–10, 7, 8 cm) — no hepatomegaly. 4) Abdominal ultrasound (shape and wall of the gallbladder, stones) and ultrasound with a cholekinetic breakfast (gallbladder contraction), blood biochemistry (bilirubin, ALT, AST, alkaline phosphatase), stool for Giardia.

Case 2

An 8-month-old child. 4 hours ago the child suddenly began to cry intensely, drawing the legs up to the abdomen, calming down after 10–15 minutes, then the attack recurs. Vomited 2 times. Yesterday meat puree was given for the first time. On examination: the child is pale and lethargic between attacks. The abdomen is slightly distended, a sausage-shaped tender mass is palpable in the right upper quadrant, the right iliac region is “empty”. On digital rectal examination there is blood-stained mucus on the glove.

Questions: 1) Presumptive diagnosis? 2) Which signs confirm the diagnosis? 3) Why are children of this age predisposed to this condition? 4) The doctor's tactics?

Answer: 1) Intestinal intussusception (ileocecal). 2) Age (4–12 months), sudden onset when complementary foods are introduced, colicky pain (attacks of restlessness with quiet intervals), vomiting, a palpable sausage-shaped mass, an empty right iliac region, blood-stained mucus from the rectum (“raspberry jelly”). 3) The long and mobile mesentery, poorly fixed intestinal loops, active motility, and disturbance of the peristaltic rhythm when complementary foods are introduced. 4) Urgent surgical examination, abdominal ultrasound (“target” sign), air or hydrostatic reduction (in the first hours) or surgery; do not give analgesics.

Case 3

A 4-year-old child at a preventive check-up. No complaints. On examination: the abdomen is slightly protuberant, symmetric, takes part in breathing. On superficial palpation the abdomen is soft and painless. The sigmoid colon is palpable in the left iliac region as a smooth, dense-elastic, mobile, painless cylinder 1.5 cm in diameter. The liver projects 1 cm below the costal arch along the midclavicular line, with a sharp, soft-elastic, painless edge. The spleen is not palpable.

Questions: 1) Assess the palpation findings. 2) Is the projection of the liver edge pathological? 3) How are the liver dimensions determined at this age?

Answer: 1) All the findings correspond to the age norm: the abdomen is soft and painless, the sigmoid colon has normal characteristics, the spleen is not palpable. 2) No, at 3–7 years projection of the liver edge up to 1 cm below the costal arch is physiological; a sharp edge, soft-elastic consistency

and absence of pain are normal. 3) Up to 5 years the liver dimensions are determined by percussion along the right anterior axillary, midclavicular and midline (the norm at preschool age is approximately 6–7, 5, 6 cm); the Kurlov method is used from 5 years.

Quick review

Table 12.7. *Key facts on Topic 12*

Question	Answer
Intestinal rotation	Fetal weeks 4–10, 270°
Physiological drooling	3–6 months
Phases of sucking	Aspiration, compression, swallowing
Stomach capacity	Day 1 7–10 ml; day 10 80 ml; 1 year 250–300 ml; 3 years 400–600 ml
Position of the stomach	Horizontal in infancy; vertical from 9–12 months
Emptying	Breast milk 2–3 hours; formula 3–4 hours
Intestinal length	6–8 times the body length in the newborn
Liver mass	4% of the body mass in the newborn
Meconium	1–3 days; transitional stool 3–5 days
Microflora (breast milk)	Bifidobacteria predominate
Liver edge	<6 months 2–3 cm; 6 months–2 years 1.5–2 cm; 3–7 years 1 cm; >7 years does not project
Kurlov (school age)	9–10, 7, 8 cm
Superficial palpation	From the left iliac region counterclockwise
Order of deep palpation	Sigmoid → cecum → ascending → transverse → descending → liver → spleen → stomach → pancreas
Gallbladder signs	Kehr, Murphy, Ortner–Grekov, Mussy, Boas
Pancreas	Chauffard, Desjardins, Mayo-Robson, Kach
Peritoneal sign	Shchetkin–Blumberg
Disappearance of liver dullness	Perforation
Auscultofriction	Lower border of the stomach — 2–4 cm above the umbilicus
Intussusception	4–12 months; colicky pain, sausage-shaped mass, “raspberry jelly”
General features of the digestive organs	Adapted to lactotrophic nutrition; thin mucosa rich in blood; low enzyme activity; predisposition to functional dyspepsia
Gastric juice	pH ≈5–6 in the newborn, 3–4 at 1 year; adult level at 10–12 years; chymosin and gastricsin; emptying: breast milk 2–3 h, cow's milk 3–4 h
Features of the intestine	6–8 times longer than the body length; long mesentery; high permeability; weak Bauhin's valve; long sigmoid; predisposition to rectal prolapse
Liver and pancreas	Liver ≈4% of body mass, immature glucuronyltransferase; few bile acids; pancreas ≈3 g, enzymes at adult level at 5–6 years
Stool dynamics	Meconium 1–3 days; transitional 3–5 days; permanent from day 5; breast milk — golden yellow, mushy, sour odor, 3–6 times

Question	Answer
Microflora	The fetal intestine is sterile; colonization from 10–20 hours; with breast milk bifidobacteria 90–99%; with artificial feeding more E. coli, enterococci, opportunistic pathogens
Location of abdominal pain	Right hypochondrium — liver, biliary tract; epigastrium — stomach, duodenum; periumbilical — small intestine, giardiasis; right iliac — appendicitis; whole abdomen — enterocolitis, peritonitis
Pain and food	Early (30–60 min) — stomach; late (1.5–3 h) — duodenum; hunger (6–7 h) — ulcer; night (23–3 h) — ulcer
Dyspeptic signs	Changes in appetite, nausea, vomiting, regurgitation, belching, heartburn, dysphagia, constipation/diarrhea, meteorism
Changes in the tongue	Raspberry — scarlet fever; lacquered — iron/B_{12} deficiency; geographic — exudative diathesis; macroglossia — hypothyroidism, Down
Breath odor	Acetone — ketoacidosis; “liver odor” — liver failure; putrid — caries, gastritis, tonsillitis
Shape of the abdomen	Meteorism — evenly enlarged; ascites — “inflated balloon”, frog belly; rickets — frog belly; emaciation/peritonitis — scaphoid; celiac disease — large abdomen, thin limbs
Order of abdominal palpation	From the left iliac region counterclockwise; the painful area last; the child on the back, legs bent; watch the face

TOPIC 13

Laboratory, functional, radiological and endoscopic methods of examining the digestive system. Semiotics and main syndromes of digestive system involvement

Learning objectives

The student should know:

- the laboratory (coprology, biochemistry, bacteriology), functional (pH-metry, duodenal intubation), radiological (ultrasound, X-ray, CT, MRI) and endoscopic methods of examining the digestive system, their indications and contraindications;
- the methods of detecting H. pylori infection;
- the main signs of digestive system involvement: abdominal pain, dyspepsia, vomiting, regurgitation, diarrhea, constipation, meteorism, jaundice;
- the coprological, biochemical and instrumental syndromes;
- the main clinical syndromes of digestive system involvement (gastritis, pylorospasm, pyloric stenosis, duodenitis, ulcer, pancreatitis, biliary dysfunction, cytolysis, cholestasis, liver failure, portal hypertension, enteritis, malabsorption, colitis, irritable bowel syndrome).

The student should be able to:

- analyze the coprogram and the biochemical parameters of the liver and pancreas;
- interpret the results of duodenal intubation and ultrasound;
- prepare the patient for instrumental investigations;
- distinguish the types of jaundice by laboratory signs;
- distinguish pylorospasm from pyloric stenosis by clinical and laboratory signs;
- identify the leading syndrome and draw up an investigation plan.

13.1. Laboratory investigations

Stool analysis (coprology)

Coprological examination — macroscopic (amount, color, consistency, odor, visible admixtures — mucus, blood, pus), chemical (reaction, stercobilin, bilirubin, occult blood, soluble protein) and microscopic examination of the stool. The external features of a healthy child's stool depend on the type of feeding and the nature of the food (see topic 12); there are no admixtures; on microscopy **small amounts** of fatty acids, muscle fibers (in children over 1 year), fiber, mucus and **single leukocytes** are found.

The Bristol scale is used to assess **stool form**:

Table 13.1. *The Bristol stool form scale*

Type	Manifestation	Assessment
1	Separate hard lumps (“nuts”), hard to pass	Constipation
2	Sausage-shaped but lumpy	Constipation
3	Sausage-shaped with cracks on the surface	Normal
4	Like a sausage or snake, smooth and soft	Normal (ideal)

Type	Manifestation	Assessment
5	Soft small blobs with clear-cut edges	Tendency to diarrhea
6	Fluffy, with ragged edges, mushy	Diarrhea
7	Watery, no solid pieces	Diarrhea

Coprological signs:

- **Steatorrhea** — fat in the stool: **type 1** — neutral fat (pancreatic lipase deficiency); **type 2** — fatty acids and soaps (insufficient bile secretion, intestinal malabsorption); **type 3** — both. Steatorrhea is **the earliest sign of exocrine pancreatic insufficiency**.
- **Creatorrhea** — muscle fibers "++" or "+++" — deficiency of gastric and pancreatic enzymes, rapid passage of chyme.
- **Amylorrhea** — many starch granules and digestible fiber — impaired carbohydrate breakdown (enteritis).
- **Lienteria** — much fiber and connective tissue — rapid passage of chyme, excess plant food.

The exocrine function of the pancreas is assessed by stool trypsin, quantitative determination of fats and **fecal elastase-1**: normally **>200 µg/g** in children over 1 month; **<200 µg/g** — exocrine insufficiency.

Table 13.2. *Coprological syndromes (Kildiyarova)*

Syndrome	Cause	Inspection and microscopy
Mouth	Inadequate chewing of food, rapid passage	Undigested food residues
Gastrogenic	Secretory insufficiency of the stomach and pancreas	Strongly alkaline reaction, coarse fiber, unchanged muscle fibers, connective tissue, oxalate crystals
Pyloroduodenal	Functional insufficiency of the stomach and duodenum	Unchanged muscle fibers, connective tissue, fiber
Pancreatic insufficiency	Pancreatitis, duodenitis, helminthic invasion	The stool liquid, greasy, yellowish-gray polyfecalia; neutral fat, muscle fibers
Insufficient bile secretion	Cholecystocholangitis, anomalies of the biliary tract	The stool gray, many fatty acids, negative reaction for stercobilin
Liver failure	Acute viral hepatitis, biliary atresia	Acholic (colorless) stool, many fatty acids
Enteral	Enteritis	The stool liquid, homogeneous, yellow; leukocytes, epithelium, fatty acid crystals, soluble protein
Ileocecal	Enterocolitis	Mucous, frothy, with a sour odor; indigestible fiber, starch, iodophilic flora

Bacteriological examination (dysbiosis) — the composition of the intestinal microflora (normal, opportunistic or pathogenic); in dysbiosis the anaerobes (bifidobacteria) decrease and opportunistic flora increases. Stool analysis for **helminth eggs and protozoa** (Giardia) and a perianal scraping (enterobiasis) are also performed.

Biochemical investigations

State of the liver: proteinogram, lipid profile, total, direct and indirect bilirubin, **ALT, AST, LDH** (cytolysis), **alkaline phosphatase, GGT, 5'-nucleotidase** (cholestasis), prothrombin index, albumin, cholinesterase (synthetic function), thymol and sublimate tests.

Pancreas: blood glucose, **amylase** and **lipase**, urinary **diastase**.

Table 13.3. *Biochemical syndromes*

Syndrome	Laboratory signs
Cytolysis	Increased ALT, AST, LDH-5, GGT, aldolase, bilirubin
Cholestasis	Increased direct bilirubin, bile acids, cholesterol, β-lipoproteins, alkaline phosphatase, GGT
Hepatocellular insufficiency	Hypoalbuminemia, reduced prothrombin index (impaired coagulation), reduced cholinesterase
Mesenchymal inflammation	Dysproteinemia (increased γ-globulins), raised thymol test, increased IgM, IgG
Pancreatitis	Increased blood amylase, lipase, trypsin, urinary diastase

13.2. Functional investigations

Contraindications to tube methods: exacerbation of peptic ulcer disease, gastrointestinal bleeding, acute cholecystitis, chronic diseases in the stage of decompensation, diverticula, esophageal anomalies, swallowing disorders, an attack of bronchial asthma, seizure syndrome.

Examination of gastric secretion

Fractional gastric intubation (secretory, acid-forming, enzymatic and motor functions) — now hardly used.

Intragastric pH-metry — the juice is not aspirated; a probe with pH electrodes measures while in contact with the mucosa. In a 5-sensor probe the sensors are located in the duodenum, antrum, body of the stomach, cardia and esophagus; the measurement lasts 30 minutes before stimulation and 1 hour after stimulation (histamine, pentagastrin). The method allows **gastroesophageal and duodenogastric refluxes** to be recorded and the dose of antacid/antisecretory drugs to be chosen. **24-hour pH monitoring** is the “gold standard” for the diagnosis of GERD.

Fractional duodenal intubation

It gives the most complete information on the biliary system; on an empty stomach, the tube is passed into the duodenum and bile is collected. As a choleretic stimulus, **33% (or 25%) magnesium sulfate** or 20% glucose solution is given at **0.3–0.5 ml/kg** (warm).

Table 13.4. *Phases of fractional duodenal intubation (Kildiyarova)*

Phase	Description (normal)	Note
I — common bile duct phase (choledochus)	From insertion of the tube until the stimulus is given; rate 1–1.5 ml/min, 15–20 min, 20–30 ml	If no bile flows — spasm of the sphincter of Oddi or mechanical obstruction; if gallbladder bile flows — hyperkinetic dyskinesia of the gallbladder
II — closed sphincter of Oddi	From the stimulus until new bile appears; 3–5 min	If bile appears within 3 minutes — hypotonia of the sphincter; if prolonged — spasm of the sphincter
III — portion A	From the opening of the sphincter to dark gallbladder bile; 3–5 min, up to 5 ml	If longer than 8 minutes — reduced gallbladder tone or hypertonia of Lutkens' sphincter

Phase	Description (normal)	Note
IV — gallbladder phase, portion B	Thick dark olive bile; 2–3 ml/min, 20–30 min, 30–50 ml	In hypokinetic dyskinesia the time increases to 60 minutes and the volume to 100–150 ml
V — hepatic phase, portion C	Golden yellow bile, 1–1.5 ml/min	—

- **Hyperkinetic (hypermotor)** type — the bile flow rate A >1.5, B >3, C >2 ml/min; the gallbladder reflex time **<5 min**.
- **Hypokinetic (hypomotor)** type — slow bile flow, the volume of portion B and the gallbladder reflex time (**>8 min**) increased.

Microscopy: in portions B and C — biliary epithelium, **cholesterol crystals, calcium bilirubinate**, mucus, **leukocytoids** (inflammation), **vegetative forms of Giardia**. **Biochemistry:** in cholecystitis and cholangitis protein increases in portions B and C; reduced bile acids and lecithin make the bile **lithogenic** (stone formation). The portions are also sent for bacteriological examination — normally the bile is **sterile**.

13.3. Radiological investigations

Ultrasound examination

Radiological examination of the abdomen **begins with ultrasound** — safe, noninvasive, informative, with no contraindications. The size and structure of the organs are assessed. **Ultrasound signs:** in ascites, free fluid (if little — in the lateral canals and small pelvis) and the intestine floating upward; in acute inflammation the ultrasound conductivity of the liver increases; tumors have clear borders; gallbladder anomalies, bile thickening, **stones**; in portal hypertension **the diameter of the portal and splenic veins increases**; thickening of the gastric wall (gastritis); in acute pancreatitis the gland is enlarged with reduced echogenicity, in chronic pancreatitis it is small with indistinct contours; abscess, hematoma, cysts; in pyloric stenosis the pyloric canal is elongated and the muscle layer thickened; in intussusception **the “target” sign**.

Ultrasound with a cholekinetic breakfast — the motor function of the gallbladder: 2 egg yolks or 20 g sorbitol in 100 ml water are given, and the gallbladder volume is measured every 10 minutes for 1 hour. Normally **by 40–50 minutes the gallbladder contracts evenly by 50–60%**; **>60%** — hypermotor, **<50%** and a slow rhythm — hypomotor dyskinesia.

X-ray examination

The main and decisive method for detecting congenital anomalies, developmental defects, acquired organic lesions and complications of peptic ulcer disease:

- **plain abdominal X-ray without contrast** — fluid **levels** and gas **“arches”** in intestinal obstruction; **free gas** under the diaphragm in perforation (a “crescent-shaped” lucency);
- **contrast examination** (barium sulfate suspension) — esophagus, stomach and intestines; in peptic ulcer disease the **“niche” sign**, signs of penetration and stenosis; **irrigography** — colon (Hirschsprung disease, dolichosigma);
- **oral cholecystography**, relaxation duodenography, retrograde cholangiopancreatography — special methods.



Figure 13.1. *Esophageal achalasia — contrast X-ray (Navruzova, Geppe)*



Figure 13.2. *Esophageal stenosis — contrast X-ray (Navruzova, Geppe)*

Computed tomography, MRI and radioisotope investigation

- **CT** — accurately assesses the state of the stomach, liver, gallbladder, pancreas, spleen and intestine without complex preparation; **40–50 times more sensitive** than classic radiography; it is performed after drinking a contrast medium (enhanced CT).
- **MRI** — based on differences in the magnetic properties of tissue molecules; especially effective for studying dynamic processes such as blood flow; **MR cholangiopancreatography** — noninvasive imaging of the biliary and pancreatic ducts.
- **Radioisotope scanning (scintigraphy)** — based on uptake of a radiopharmaceutical by hepatocytes, tumor or inflammatory cells (hepatobiliary scintigraphy, detection of Meckel's diverticulum).

13.4. Endoscopic investigations and biopsy

Endoscopy — viewing the digestive tube with an endoscope; it allows the diagnosis to be clarified, the depth and extent of the process to be assessed, and **a targeted biopsy** to be taken. In children it is performed with special preparation, often under **anesthesia**.

- **Esophagogastroduodenoscopy (EGDS)** — viewing the esophagus, stomach, duodenum (and jejunum) in one examination. **Indications:** pain syndrome, recurrent vomiting, bleeding from the upper parts; **therapeutic endoscopy** — removal of foreign bodies, polypectomy, sclerotherapy of esophageal and gastric varices, dilatation of strictures.
- **Rectosigmoidoscopy** — the rectum and the distal sigmoid colon; **colonoscopy** — the whole colon. The indication is justified by a specialist (gastroenterologist, proctologist, surgeon); at an early age under anesthesia.
- **Endoscopic retrograde cholangiopancreatography (ERCP)** — during duodenoscopy the papilla of Vater is cannulated and the biliary and pancreatic ducts are contrasted; disadvantages — complexity, radiation dose, risk of the complication of **acute pancreatitis**.
- **Video capsule endoscopy** — viewing the small intestine (Crohn disease, occult bleeding).

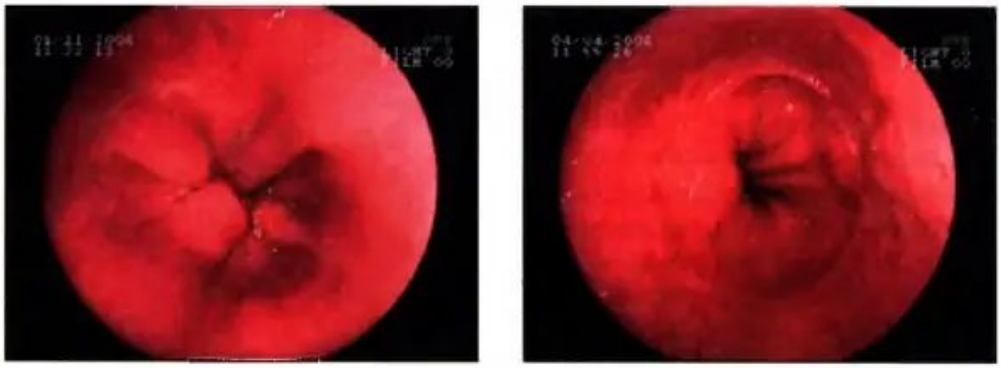


Figure 13.3. *Esophagogastroduodenoscopy: endoscopic appearance of the gastric mucosa (Navruzova, Geppe)*

Biopsy — pathomorphological examination of altered tissue: determining the nature of space-occupying lesions, inflammatory and functional diseases (e.g., villous atrophy of the small intestine in celiac disease, detection of H. pylori). By fine-needle aspiration (cytology) or needle biopsy (a piece of tissue).

Detection of Helicobacter pylori infection

Detection of H. pylori determines the need for **etiotropic (eradication) therapy** in chronic gastric diseases.

- **Bacteriological** — culture of a gastric biopsy on a special medium;
- **morphological** — staining of histological preparations by Giemsa, toluidine blue, Warthin–Starry;
- **cytological** — staining of biopsy smear-imprints by Giemsa, Gram;
- **urea breath test** — labeled (^{13}C or ^{14}C) urea is drunk, and the isotope released by bacterial urease is detected in the exhaled air (noninvasive);
- **rapid urease test** — the biopsy is placed in a medium with a substrate, buffer and indicator (color change);
- **PCR** — the most reliable method;
- **screening methods** — ELISA (antibodies in the blood), H. pylori antigen in the stool, rapid tests.

13.5. Semiotics: subjective signs

Abdominal pain

The location of pain is given in the table in topic 12. **The relation of pain to food** is important:

Table 13.5. *Relation of pain to food intake and defecation*

Type of pain	Time	Probable pathology
Early	Within 30 minutes – 1 hour after eating	Gastritis, gastric ulcer (cardia and body)
Late	1.5–3 hours after eating	Antral gastritis, duodenitis, duodenal ulcer
On an empty stomach (hunger)	6–7 hours after eating; relieved after eating	Duodenal ulcer
Night	From 23:00 to 3:00	Duodenal ulcer

Type of pain	Time	Probable pathology
Decreases after defecation	—	Colitis, irritable bowel syndrome
During defecation	—	Anal fissure, proctitis, distal colitis

The "Moynihan rhythm" is typical of duodenal ulcer: hunger → pain → relief after eating → hunger → pain.

Character of pain: visceral (dull, diffuse, poorly localized — distension or spasm of hollow organs); **parietal (somatic)** — sharp, well localized, increased by movement (peritoneal irritation); **referred (radiating)**. Colicky paroxysmal pain — spasm of a hollow organ (intestinal, biliary, renal colic).

Dyspeptic signs

- **Belching:** of air (aerophagia), **with the smell of rotten eggs** — achylia (hypoacid gastritis); **sour** — increased acidity or fermentation (gastritis, peptic ulcer disease); **bitter** — bile reflux (duodenogastric reflux); **putrid** — decomposition of food residues in stenosis or obstruction of the esophagus, cardia, stomach or duodenum.
- **Heartburn** — a burning, hot or pressing sensation spreading from the epigastrium behind the sternum; in gastroesophageal reflux and diseases with excessive acid production.
- **Hiccups** — spasmodic contractions of the diaphragm; in acute gastric dilatation, meteorism, diseases of the diaphragm.
- **Nausea** — an unpleasant sensation in the epigastrium and pharynx, before or together with vomiting.

Vomiting

Vomiting — an act resulting from stimulation of the vomiting center reflexly via the vagus nerve, directly by toxins, or by CNS processes.

- **Central vomiting** — in CNS pathology; it is not preceded by nausea, is not related to food, is small in volume and **brings no relief**.
- **Peripheral (reflex) vomiting** — in damage to the internal organs; sudden, without nausea, brings no relief.
- **Gastric vomiting** — 0.5–1.5 hours after eating, preceded by nausea, large in volume, with a sour smell, **brings temporary relief**.
- **Acetonemic vomiting** — repeated, acetone breath odor, acetone in the urine.

Table 13.6. *Clinical assessment of blood in vomit (Kildiyarova)*

Appearance of the blood	Level of bleeding	Diseases
Bright red blood	Above the stomach	Bleeding from the nose, gums, pharynx, esophagus, airways
Brown-black, "coffee grounds"	Stomach and/or duodenum	Peptic ulcer disease of the stomach and/or duodenum
Large amount of blood	Massive acute or prolonged capillary bleeding	Esophageal varices in liver cirrhosis and portal hypertension, esophageal burns and ulcers

Regurgitation and rumination

Regurgitation — the **weak return, without straining of the abdominal press,** of swallowed food or secretions to the oral cavity and outside. Up to 6–7 months, if it is **infrequent** and occurs not later than 1 hour after feeding — **physiological**.

Assessed: time of onset (from birth, from 2–3 weeks, from 3–4 months), frequency (2–3 times a day, after each feed, more than 10 times), provoking factors (feeding, crying, swaddling), volume (small, "a mouthful", **"projectile"**), the nature of the contents (fresh or curdled milk, bile, sour or sweetish smell), relation to feeding.

Rumination — for at least 3 months, gastric contents are returned to the mouth by periodic contractions of the abdominal press, diaphragm and tongue muscles, re-chewed and swallowed; in children aged 3–8 months, often with CNS pathology.

Diarrhea, constipation, meteorism

Diarrhea — frequent defecation with altered stool (from mushy to watery). **Acute** — up to 2–3 weeks (intestinal infections, drugs — antibiotics, alimentary causes); **chronic** — more than 3 weeks or recurrent (inflammatory bowel diseases, irritable bowel syndrome, malabsorption — celiac disease, disaccharidase deficiency, cystic fibrosis; pancreatitis, gastrointestinal allergy, parasitoses, endocrine diseases).

Table 13.7. *Types of diarrhea in children (Kildiyarova)*

Type of diarrhea	Infectious causes	Noninfectious causes
Watery (secretory)	Escherichioses, rotavirus, cryptosporidiosis	Overfeeding, cow's milk protein intolerance, carbohydrate malabsorption, irritable bowel syndrome, chloride diarrhea, hyperthyroidism, congenital adrenal hyperplasia
Fatty	Salmonellosis	Cystic fibrosis, Shwachman syndrome, chronic pancreatitis, intestinal lymphangiectasia
Bloody (invasive)	Shigellosis, salmonellosis, campylobacteriosis, yersiniosis, escherichioses	Intestinal polyposis, nonspecific ulcerative colitis, Crohn disease

Constipation — prolongation of the intervals between defecations relative to the individual norm, or regular incomplete emptying of the bowel. A healthy child over 1 year should pass stool at least **1 time in 36 hours**. At an early age, even if stools occur several times a day, stools passed with strong straining, distress and crying, as **"sheep droppings"** or in lumps, with traces of blood and mucus — constipation. **Chronic constipation** — more than 3 months. Causes: alimentary (little fiber, much protein and fat, excess calcium — the most common at an early age), neurogenic, dyskinetic, psychogenic, endocrine (hypothyroidism), mechanical, colonic anomalies (**Hirschsprung disease**, dolichosigma), drugs.

Meteorism — excessive formation, accumulation or impaired passage of gas in the gastrointestinal tract; gurgling in the abdomen, passage of gas; it may also occur in a healthy person after overeating and gas-forming foods.

Jaundice

Jaundice — yellowing of the skin and visible mucous membranes as a result of increased bilirubin in the blood and tissues. Yellowing of the skin and sclera becomes visible when bilirubin is **>68.4–85.5 μmol/l** in the newborn and **>26 μmol/l** in older children.

Physiological (neonatal) jaundice — appears **after 48 hours** of life, usually disappears **by day 7–10** (at the latest by week 2–3), no hepatosplenomegaly, normal stool and urine color, bilirubin increases at the expense of the indirect fraction (up to 252 μmol/l in term babies, up to 175 μmol/l in preterm babies).

False (carotene) jaundice — with a high intake of carrots, oranges, pumpkin and persimmons only **the skin** (palms, soles) turns yellow, **the mucous membranes and sclera are not stained**, bilirubin is normal.

Table 13.8. *Classification of jaundice (Kildiyarova)*

Sign	Prehepatic (hemolytic)	Hepatic (parenchymal)	Posthepatic (obstructive)
Cause	Destruction of erythrocytes	Damage to hepatocytes	Impaired patency of the biliary tract
Diseases	Hemolytic disease of the newborn, hemolytic anemias	Viral hepatitis, liver cirrhosis	Biliary atresia, inspissated bile syndrome, compression of the biliary tract
Blood bilirubin	Indirect ↑↑↑, direct N or ↑	Indirect ↑, direct ↑↑↑	Indirect ↑, direct ↑↑↑, alkaline phosphatase ↑↑↑
Urine	Urobilinogen ↑, bilirubin (–)	Urobilinogen (+), bilirubin (+)	Urobilinogen (–), bilirubin (+) — “beer-colored”
Stool	Normal color (dark)	Normal or slightly discolored	Colorless (acholic)
Skin color	Lemon-yellow	Orange (saffron)	Greenish-yellow
Itching of the skin	Absent	Absent	Present
The spleen	Enlarged	Enlarged	Normal

MEMORY AID
The color of jaundice: **“Lemon — hemolysis, orange — hepatitis, olive — obstructive”**. Itching and acholic stools are a sign of **cholestasis** (obstructive).

13.6. Main syndromes of digestive system involvement

Syndrome of impaired digestion in the oral cavity

Causes: disturbed sucking and swallowing (prematurity, CNS damage, cleft lip and palate, short frenulum of the tongue), stomatitis, dental pathology, salivary gland diseases. Manifestations — difficulty sucking and chewing, drooling, breath odor, refusal of food, the “oral syndrome” on coprology.

Gastritis syndrome

Classification: by course — acute, chronic; by extent — diffuse, focal; by secretion — with increased or reduced acidity; by depth — superficial, atrophic, erosive, hypertrophic.

Clinical picture: dull, distending, pressing or stabbing pain in the epigastrium related to food; night and seasonal pain favor increased acidity, an erosive process and antral location. Dyspepsia: nausea, vomiting after eating, belching of air, food and **with a rotten-egg smell** (hypoacid state), **heartburn** (hyperacid state), a feeling of heaviness, reduced appetite, unstable stool — **diarrhea** (hypoacid) or

constipation (hyperacid). The tongue is coated with a white coating; epigastric tenderness on palpation.

Investigations: coprogram — many unchanged muscle fibers, digestible fiber and connective tissue "+++"; **EGDS** — hyperemia, edema, mucus, fasting fluid, refluxes; H. pylori tests.

Functional disorder of the stomach (functional dyspepsia) — short, inconstant, mild epigastric pains of various character, not related to food; no endoscopic, radiological or morphological changes.

Pylorospasm and pyloric stenosis

Pylorospasm — **a functional spasm of the gastric outlet** in young children. Predisposition: the cylindrical shape of the stomach, weakness of the cardiac muscles and **strong development of the pyloric muscles**, immaturity of the gastric fundus, pressure of the large liver, imperfect neuroregulatory mechanisms. **From the first days after birth**: inconstant, small-volume, frequent regurgitation and vomiting (immediately or after 15–20 minutes); the volume of the vomit is **less than the milk taken**; weight gain is slow but steady; no exsicosis or oliguria, no visible peristalsis; on X-ray passage is not impaired.

Pyloric stenosis — **a developmental defect** of the gastric outlet (hypertrophy of the muscle layer); in 1:2000 births, mainly **in boys**. The pyloric part is thickened in the shape of an olive ("olive") (2–3.5 cm long, 1.5–3 cm wide), almost cartilage-like in density. The clinical picture appears **from week 2–3**: persistent **"projectile" vomiting**, large in volume, **the volume of the vomit is always greater than the volume of the feed**, immediately after feeding; the vomit **never contains bile**; the child loses weight quickly, dehydration, oliguria, stool retention ("**hunger stools**" — scanty stool resembling meconium); **"hourglass"**-shaped peristalsis in the epigastrium; a dense **"olive" of 2–4 cm** is palpable at the edge of the right rectus muscle, midway between the umbilicus and the xiphoid process. On X-ray barium is retained in the stomach for **24 hours or longer**, and emptying does not begin in the first 15–20 minutes. **Ultrasound** is the main diagnostic method (pyloric muscle thickness ≥3–4 mm, canal length ≥15–17 mm). Treatment — **surgical (pyloromyotomy)**.



Figure 13.4. *Pyloric stenosis: on contrast X-ray, emptying of barium from the stomach is delayed even after 20 minutes and 24 hours (Kildiyarova)*

Table 13.9. *Differential diagnosis of pylorospasm, pyloric stenosis and adrenogenital syndrome (Kildiyarova)*

Indicator	Pylorospasm	Pyloric stenosis	Adrenogenital syndrome (salt-wasting)
Onset	From the first days	From week 2–3	From the 1st week
Visible peristalsis	Mild	“Hourglass”	Sometimes
Palpation of the pylorus	Impossible	Possible	Impossible
Blood Na	Normal	Reduced	Sharply reduced
Blood K	Normal	Sharply reduced	Raised (hyperkalemia)
Blood Cl	Normal	Sharply reduced	Sharply reduced
ABB	Unchanged	Metabolic alkalosis	Metabolic acidosis
Urinary Na excretion	Unchanged	Very low	Sharply increased
Excretion of 17-ketosteroids	Unchanged	Reduced	Sharply increased
X-ray	Passage not impaired	Barium retained up to 24 hours	Emptying slowed
Skin pigmentation	Absent	Absent	Present

CAUTION

In Kildiyarova's table, potassium in adrenogenital syndrome is given as “reduced” and the ABB as “metabolic alkalosis”. In salt-wasting adrenogenital syndrome, because of aldosterone deficiency, **hyperkalemia and metabolic acidosis** are seen instead — the table has been corrected according to current data. In pyloric stenosis, by contrast, **hypochloremic hypokalemic metabolic alkalosis** is typical because hydrochloric acid-containing gastric contents are lost.

Involvement of the duodenum (duodenitis)

Pains of various intensity related to food, most often **late** (2–3 hours after eating), sometimes night and early, in the right upper half of the abdomen; nausea, vomiting, unstable stool; **autonomic syndrome** (periodic headache, dizziness, disturbed sleep, rapid fatigue, irritability — related to disturbance of the endocrine function of the duodenum). On examination pallor, weight deficit, coated tongue, tenderness in **the pyloroduodenal (Chauffard) zone**. EGDS confirms the diagnosis.

Peptic ulcer syndrome (peptic ulcer disease)

Predisposing factors: hereditary predisposition, stress (emotional strain, CNS diseases), irregular eating, **H. pylori**, long-term intake of ulcerogenic drugs (NSAIDs).

Duodenal ulcer — hunger, night and late (2–4 hours) pains, **the Moynihan rhythm**, severe, recurrent, to the right of and above the umbilicus; heartburn, sour belching, nausea, vomiting, constipation; autonomic disturbances are more frequent. **Gastric ulcer** — early pain, slightly above the umbilicus along the midline; nausea and vomiting after eating. On examination: forced position, autonomic lability (distal hyperhidrosis, persistent red dermographism, bradycardia, hypotension), tension of the abdominal muscles, zones of skin hyperesthesia, a positive **Mendel's sign**, **Oppenhowski's points** (pain on percussion over the spinous processes of the thoracic vertebrae). Diagnosis — **EGDS** (the ulcer defect), the “niche” sign on contrast X-ray.

Complications: bleeding (“coffee grounds” vomiting, **melena** — black tarry stool), perforation (“dagger-like” pain, “board-like” abdomen, disappearance of liver dullness), penetration, stenosis.

Pancreatitis syndrome

In gross dietary errors, infections (mumps), abdominal trauma, gastrointestinal pathology. Sharp **girdle-like pain** above the umbilicus, nausea, **repeated vomiting** of gastric contents, intoxication, disturbed stool. In the blood — leukocytosis, raised ESR; increased **amylase, lipase, trypsin**; increased urinary diastase. Ultrasound — edema, enlargement and disturbed structure of the gland. In chronic pancreatitis — exocrine insufficiency (steatorrhea, reduced elastase-1).

Biliary tract dysfunction syndrome (biliary dyskinesia)

Causes: long intervals between meals, dry food, abuse of carbohydrate and fatty food, disturbed daily routine; anatomical defects of the gallbladder (kinks, constrictions); reflex, in diseases of other digestive organs. **Gallbladder dysfunction** and **dystonia of the sphincter of Oddi**; by functional state — **hypokinetic (hypomotor)** and **hyperkinetic (hypermotor)** forms.

The main sign is a dull or sharp pain in the right hypochondrium that **increases after eating and after physical exertion**; nausea, vomiting, **a bitter taste in the mouth**, signs of cholestasis, liver enlargement, pain in the right hypochondrium, epigastrium and Chauffard's zone, **positive gallbladder signs**. In the **hyperkinetic** form — short, paroxysmal, colicky pains; in the **hypokinetic** form — constant dull pain with a feeling of heaviness. Diagnosis — ultrasound with a cholekinetic breakfast, duodenal intubation.

Syndromes of liver involvement

Cytolysis syndrome — in viral, infectious and noninfectious, parasitic (echinococcus), toxic (drugs, heavy metal salts, benzene) influences, anomalies of the biliary tract, congenital metabolic disorders. **Clinical picture:** pain and heaviness in the right hypochondrium, dyspepsia (reduced appetite, nausea, bilious vomiting, meteorism, unstable stool), astheno-vegetative signs (weakness, rapid fatigue, sweating, headache), signs of cholestasis (itching, subicterus). The skin is pale, sometimes icteric, dry; scratch marks, ecchymoses, petechiae; **spider angiomas** (telangiectasias), **palmar erythema** (“liver palms”); **hepatosplenomegaly** — the liver dense, tender, smooth. **Dark urine (“beer-colored”)**, discolored stool. Laboratory: increased ALT, AST, LDH, bilirubin mainly at the expense of the direct fraction, dysproteinemia, increased IgM; prolonged detection of HBsAg — a sign of chronicity.

Cholestasis syndrome — disturbed bile flow: **itching of the skin**, pigmentation, **xanthomas**, inconstant jaundice, dyspepsia, astheno-vegetative syndrome; scratch marks, hepatosplenomegaly. Laboratory: high cholesterol, bile acids, β-lipoproteins, total lipids, **alkaline phosphatase**, leucine aminopeptidase; increased direct bilirubin.

Hepatocellular insufficiency syndrome — when only **20–25%** of the liver parenchyma remains functional. The **acute** form (more often at an early age) — viral hepatitis, hepatotropic poisons (death cap, mushrooms, barbiturates) — hepatic necrosis and coma; **chronic** — as a result of cirrhosis, congenital infections, hereditary diseases. **Clinical picture:** change in behavior — lethargy, stupor, adynamia, drowsiness (less often agitation), then confusion and **coma**; intoxication, pale-yellow skin, dystrophy of the muscles and subcutaneous fat, avitaminosis, liver signs (“lacquered” tongue, spider angiomas, palmar erythema), abdominal enlargement (meteorism, hepatosplenomegaly, **ascites**), edema of the legs, **hemorrhagic syndrome** (bleeding from the nose, gums, gastrointestinal tract); laboratory — hypoalbuminemia, reduced prothrombin. **When coma develops the liver begins to shrink** — a poor prognostic sign.

Portal hypertension syndrome — increased pressure in the portal venous system; prehepatic, intrahepatic and posthepatic forms. The main signs — **varicose dilatation of the esophageal and**

gastric veins (X-ray, endoscopy), **splenomegaly**, hepatomegaly, **ascites**, a venous network on the anterior abdominal wall (“caput medusae”).

Syndromes of impaired digestion in the small intestine

Enteritis syndrome — against a background of intestinal infections (coli infection, salmonellosis), food allergy, operations on the small intestine, gross dietary errors, extraintestinal diseases (influenza, pneumonia, otitis). **Diarrhea** (painless, with passage of gas, followed by relief), meteorism, gurgling, pain **around the umbilicus** after eating; signs of intestinal toxicosis and dehydration (gray skin, the fold smooths out slowly). Coprogram: the stool watery, copious, unformed, pale yellow, with a **sour** (fermentation) or **fetid** (putrefaction) odor, mucus mixed with the stool, altered muscle fibers, fatty acids, soaps.

Dyspeptic syndrome of infants (simple dyspepsia) — disturbed feeding routine (overfeeding, too frequent feeding), a diet inappropriate for age, too rapid transfer to artificial feeding. Diarrhea (6–8 times a day, liquid, yellow-green, sour-smelling, with a little mucus and **white flakes** — magnesium and calcium soaps of fatty acids), regurgitation, reduced appetite; the general condition is not disturbed, temperature normal, no dehydration, steady weight gain; no pathogenic flora grows.

Malabsorption syndrome — **impaired absorption** in the small intestine: recurrent diarrhea, **polyfecalia** and progressive dystrophy. **Causes:** congenital and hereditary diseases (**celiac disease**, **cystic fibrosis**, disaccharidase (lactase) deficiency, exudative enteropathy), food allergy (cow's milk proteins), chronic inflammation (enterocolitis, pancreatitis), tumors, intestinal resection (“short bowel” syndrome), long-term antibiotics, massive giardiasis; secondary — against a background of severe diseases.

Clinical picture: the stool watery or frothy, **large in volume**, **gray and clay-like**, fetid (**steatorrhea**), **greasy and shiny**, leaving greasy stains on diapers that are difficult to wash, the potty is hard to clean; vomiting, progressive **anorexia**, abdominal distension, spastic pains, **abdominal enlargement**, emaciation, **lagging physical development** (slowed growth, weight deficit); signs of vitamin and trace element deficiency (anemia, rickets-like changes, hypoproteinemic edema).



Figure 13.5. *Abdominal enlargement and thinning of the limbs in the malabsorption syndrome (celiac disease) (Kildiyarova)*

CLINICAL SIGNIFICANCE

Celiac disease — intolerance of gluten (the protein of wheat, rye, barley); symptoms begin 1–2 months **after cereals (porridge, bread) are introduced**. Diagnosis: **IgA antibodies to tissue transglutaminase** in the blood (with total IgA), antibodies to endomysium, duodenal biopsy (villous atrophy). Treatment — **a lifelong gluten-free diet**. **Lactase deficiency** — frothy, sour-smelling diarrhea and meteorism after milk intake; stool pH <5.5, carbohydrates >0.25%. **Cystic fibrosis** — fatty diarrhea + recurrent bronchopulmonary infections; sweat chloride >60 mmol/l.

Impaired digestion in the colon (colitis syndrome)

Infections (dysentery, food toxicoinfections), parasitoses (amebiasis), helminthiases (enterobiasis, hymenolepiasis, trichuriasis), candidomycosis, noninfectious diseases (**ulcerative colitis, Crohn disease**). Colicky pains in the lower abdomen increasing with defecation, **tenesmus** (false urges), defecation **brings no relief**; frequent stools (10–12 times a day, 20–30 times in severe cases), **mucus, pus, blood** in the stool; in severe cases the stool loses its fecal character (“rectal spit”). Coprogram: much mucus, leukocytes, erythrocytes; pathogenic flora on bacteriological examination.

Irritable bowel syndrome

A functional pathology: abdominal pain or discomfort (usually **relieved by defecation**), dyspepsia (meteorism, gurgling, diarrhea, constipation or their alternation), a feeling of incomplete emptying of the bowel, imperative urges; lasting **at least 3 months**. It is **a diagnosis of exclusion**. **“Red flag” signs** that exclude it: unexplained weight loss, predominance of symptoms **at night**, constant severe pain, fever, hepatosplenomegaly, **blood in the stool**, leukocytosis, anemia, raised ESR, biochemical changes.

13.7. Clinical section

Algorithm for examining a child with abdominal pain

CLINICAL SIGNIFICANCE

1. Exclude **an “acute abdomen”** (see topic 12) — surgical examination. 2) Look for **“red flag”** signs (weight loss, night pain and diarrhea, bloody stools, fever, growth retardation, inflammatory bowel diseases in the family). If they are absent and the pain lasts more than 2 months — most often **functional pain** (functional dyspepsia, irritable bowel syndrome, abdominal migraine). 3) Minimal investigation: CBC and urinalysis, coprogram, helminths and Giardia, **abdominal ultrasound**, biochemistry (ALT, AST, amylase, bilirubin), and if necessary **EGDS** and an H. pylori test.

Approach to a vomiting newborn

CAUTION

In a newborn, **bilious (green) vomiting** is a sign of obstruction below the duodenum (intestinal malrotation and volvulus, atresia) and is regarded as **a surgical emergency**. Non-bilious “projectile” vomiting at 2–3 weeks — pyloric stenosis; vomiting + dehydration + skin pigmentation + hyperkalemia — the salt-wasting form of adrenogenital syndrome.

Cholestatic jaundice in a newborn

> **CLINICAL SIGNIFICANCE**
> Jaundice lasting more than 2 weeks, **colorless stools** and dark urine — suspect **biliary atresia**; direct bilirubin (>20% of total) is determined and the child is referred to a surgeon immediately: the result is much better if the Kasai operation is performed **before 60 days**.

Practical skills

Skill 1. Interpreting a coprogram

1. Macroscopic parameters: amount, form (Bristol scale), color, odor, mucus, blood, pus.
2. Chemical: reaction (pH), stercobilin, bilirubin, occult blood, soluble protein.
3. Microscopy: muscle fibers, connective tissue, neutral fat, fatty acids, soaps, starch, fiber, mucus, leukocytes, erythrocytes, helminth eggs, protozoa.
4. Identify the coprological signs (type of steatorrhea, creatorrhea, amylorrhea).
5. Identify the coprological syndrome and indicate the probably affected part.

Skill 2. Determining the type of jaundice by laboratory signs

1. Assess the ratio of total, direct and indirect bilirubin in the blood.
2. Assess urinary urobilinogen and bilirubin.
3. Assess the stool color and stercobilin.
4. Take into account ALT, AST, alkaline phosphatase, GGT, hemoglobin, reticulocytes.
5. Conclusion: prehepatic, hepatic or posthepatic jaundice.

Skill 3. Preparing a child for instrumental investigations

1. **Abdominal ultrasound:** on an empty stomach (small children 3–4 hours after the last meal, older children 8–12 hours), limit gas-forming foods for 2–3 days beforehand; simethicone in meteorism.
2. **EGDS:** on an empty stomach, 6–8 hours after the last meal (4 hours in infants); informed consent from the parents; remove dentures and foreign bodies.
3. **Contrast X-ray:** on an empty stomach; bowel cleansing before irrigography and colonoscopy (diet, cleansing enema or osmotic laxatives).
4. **Duodenal intubation:** a light supper in the evening, on an empty stomach; insert the tube with the child lying on the right side.
5. Explain the nature of the investigation to the child and the parents.

Skill 4. Clinical examination for pyloric stenosis (“test feed”)

1. Feed the child and observe: time of vomiting, volume (compared with the volume of the feed), bile admixture.
2. Observe the epigastrium from the side in light to detect **“hourglass”**-shaped peristalsis.
3. After vomiting (empty stomach), palpate deeply at the edge of the right rectus muscle, midway between the umbilicus and the xiphoid process, to look for **the “olive”**.
4. Assess the signs of dehydration, diuresis and weight; send for electrolytes and ABB.

Review questions

1. What does coprological examination include? Explain the Bristol scale.
2. What do the types of steatorrhea, creatorrhea and amylorrhea indicate?
3. Name the main coprological syndromes.
4. What are the biochemical parameters assessing the liver and pancreas and the biochemical syndromes?
5. How is intragastric pH-metry performed and what does it determine?
6. Explain the phases of fractional duodenal intubation and the types of dyskinesia.
7. What is the importance of abdominal ultrasound and ultrasound with a cholekinetic breakfast?
8. Name the possibilities of X-ray, CT, MRI and radioisotope investigations.
9. What are the types of endoscopic investigations and their indications? How is H. pylori detected?
10. What are the types of pain by relation to food and their diagnostic significance?
11. What are the types of vomiting and the assessment of blood in the vomit?
12. What are the types of diarrhea, the definition and causes of constipation?
13. How do you distinguish the types of jaundice?
14. How are pylorospasm, pyloric stenosis and adrenogenital syndrome distinguished?
15. What are the main signs of the cytolysis, cholestasis, hepatocellular insufficiency, portal hypertension, malabsorption and colitis syndromes?

Test questions

1. The normal stool types on the Bristol scale:
   A) 1–2
   B) 3–4
   C) 5–6
   D) 6–7
   E) Only 7

2. In which condition is neutral fat in the stool (type 1 steatorrhea) seen?
   A) Insufficient bile secretion
   B) Pancreatic lipase deficiency
   C) Enteritis
   D) Colitis
   E) Gastritis

3. The normal level of fecal elastase-1 (in children over 1 month):
   A) <50 µg/g
   B) 50–100 µg/g
   C) 100–200 µg/g
   D) >200 µg/g
   E) Cannot be determined

4. The enzyme most typical of the cholestasis syndrome:
   A) ALT
   B) Amylase
   C) Alkaline phosphatase
   D) Creatine kinase
   E) LDH

5. The color of gallbladder bile (portion B) in fractional duodenal intubation:
   A) Golden yellow
   B) Dark olive
   C) Colorless
   D) Pale yellow
   E) Red

6. The normal contraction of the gallbladder 40–50 minutes after a cholekinetic breakfast:
   A) 10–20%
   B) 30–40%
   C) 50–60%
   D) 80–90%
   E) 100%

7. A noninvasive method of detecting H. pylori:
   A) Histological
   B) Urea breath test
   C) Rapid urease test on a biopsy
   D) Bacteriological culture
   E) Cytological

8. The pain typical of duodenal ulcer:
   A) During eating
   B) 30 minutes after eating
   C) On an empty stomach and at night
   D) Only on physical exertion
   E) After defecation

9. The source of bleeding in “coffee grounds” vomit:
   A) Nose
   B) Esophageal varices
   C) Stomach or duodenum
   D) Colon
   E) Lungs

10. Not typical of pyloric stenosis:
   A) Onset at 2–3 weeks
   B) Vomit volume greater than the feed volume
   C) Bile in the vomit
   D) “Hourglass” peristalsis
   E) Metabolic alkalosis

11. Typical of obstructive (posthepatic) jaundice:
    A) Lemon-yellow skin, dark stool
    B) Acholic stool, itching of the skin, bilirubin in the urine
    C) An isolated increase in indirect bilirubin
    D) A sharp increase in urinary urobilinogen
    E) Splenomegaly

12. In false (carotene) jaundice:
    A) The sclera turns yellow
    B) Bilirubin is raised
    C) Only the skin turns yellow, the mucous membranes are not stained
    D) The stool becomes discolored
    E) The urine is dark

13. The stool typical of the malabsorption syndrome:
    A) Scanty, hard, “sheep droppings”
    B) Copious, gray-clay, greasy-shiny, fetid
    C) With blood and mucus, small in volume
    D) Black, tarry
    E) Normal

14. Typical of the colitis syndrome:
    A) Relief after defecation, periumbilical pain
    B) Tenesmus, mucus, blood and pus in the stool
    C) Polyfecalia, steatorrhea
    D) Acholic stool
    E) “Projectile” vomiting

15. A “red flag” sign excluding irritable bowel syndrome:
    A) Pain relieved by defecation
    B) Meteorism
    C) Blood in the stool and night symptoms
    D) Alternating diarrhea and constipation
    E) A feeling of incomplete emptying

16. The normal (ideal) stool type on the Bristol scale:
    A) 1
    B) 4
    C) 6
    D) 2
    E) 7

17. What do types 1–2 on the Bristol scale mean?
    A) Malabsorption
    B) Constipation
    C) Meteorism
    D) Normal
    E) Diarrhea

18. What does type 1 (neutral fat) steatorrhea indicate?
    A) Helminthiasis
    B) Enteritis
    C) Insufficient bile secretion
    D) Pancreatic lipase deficiency
    E) Candidiasis

19. What is a large amount of muscle fibers in the stool called?
    A) Melena
    B) Steatorrhea
    C) Lienteria
    D) Creatorrhea
    E) Amylorrhea

20. What does amylorrhea mean?
    A) Blood
    B) Pus
    C) Many starch granules in the stool
    D) Much fat
    E) Muscle fibers

21. The normal fecal elastase-1 (in children over 1 month):
    A) 0
    B) <50 µg/g
    C) <100 µg/g
    D) 10 µg/g
    E) >200 µg/g

22. Acholic (colorless) stool is typical of which condition?
    A) Meteorism
    B) Liver failure, biliary atresia
    C) Constipation
    D) Gastritis
    E) Enteritis

23. Which parameters increase in the cytolysis syndrome?
    A) Only prothrombin
    B) ALT, AST, LDH
    C) Only albumin
    D) Only cholesterol
    E) Only glucose

24. Signs of the cholestasis syndrome:
    A) Only ALT ↑
    B) Bilirubin ↓
    C) Albumin ↑
    D) Prothrombin ↑
    E) Increased direct bilirubin, bile acids, alkaline phosphatase, GGT

25. Signs of hepatocellular insufficiency:
    A) Only bilirubin ↑
    B) γ-globulin ↓
    C) Hypoalbuminemia, reduced prothrombin index and cholinesterase
    D) Only amylase ↑
    E) None

26. Increased in the blood and urine in pancreatitis:
    A) Albumin
    B) Bilirubin
    C) Amylase, lipase; diastase in the urine
    D) Creatinine
    E) Urea

27. The “gold standard” for the diagnosis of GERD:
    A) 24-hour pH monitoring
    B) Ultrasound
    C) Fluoroscopy
    D) Fractional intubation
    E) Coprology

28. What is given as a choleretic stimulus in duodenal intubation?
    A) Atropine
    B) Adrenaline
    C) Saline
    D) Insulin
    E) 33% or 25% magnesium sulfate

29. What kind of bile is portion B in duodenal intubation?
    A) Bloody
    B) Colorless
    C) Purulent
    D) Golden yellow hepatic bile
    E) Thick dark olive gallbladder bile

30. The characteristic ultrasound sign of intussusception:
    A) Stone
    B) Gas arch
    C) Cyst
    D) Free fluid
    E) The “target” sign

31. By how much does the gallbladder normally contract in 40–50 minutes on ultrasound with a cholekinetic breakfast?
    A) 10%
    B) 100%
    C) 50–60%
    D) 0%
    E) 90%

32. The characteristic sign of peptic ulcer disease on contrast X-ray:
    A) The “niche” sign
    B) “Hourglass”
    C) “Wooden shoe”
    D) The “target” sign
    E) “Caput medusae”

33. Which method is used to detect Hirschsprung disease and dolichosigma?
    A) Spirography
    B) Irrigography
    C) Gastroscopy
    D) EGDS
    E) Cardiac ultrasound

34. An indication for EGDS is:
    A) Nasal congestion
    B) A mild cold
    C) Pain syndrome, recurrent vomiting, bleeding from the upper parts
    D) Headache
    E) Not jaundice, only fever

35. Which method is used to view the small intestine?
    A) Video capsule endoscopy
    B) EGDS only
    C) Rectosigmoidoscopy
    D) Ultrasound only
    E) Colonoscopy

36. The most reliable method of detecting H. pylori:
    A) Rapid urease
    B) PCR
    C) None
    D) Screening rapid test
    E) Blood ELISA

37. Early pain (30 min–1 hour after eating) is typical of which pathology?
    A) Anal fissure
    B) Colitis
    C) Duodenal ulcer
    D) Gastritis, gastric ulcer
    E) Cholecystitis only

38. What does the Moynihan rhythm mean?
    A) Hunger → pain → relief after eating → hunger → pain
    B) Constant pain
    C) Pain increases after eating
    D) Pain after defecation
    E) Night vomiting

39. What is pain during defecation typical of?
    A) Anal fissure, proctitis, distal colitis
    B) Appendicitis
    C) Pancreatitis
    D) Cholecystitis
    E) Gastritis

40. Belching with a rotten-egg smell is typical of which condition?
    A) GERD
    B) Rickets
    C) Hyperacid gastritis
    D) Achylia (hypoacid gastritis)
    E) Duodenogastric reflux

41. What does bitter belching indicate?
    A) Anemia
    B) Low acidity
    C) Bile reflux (duodenogastric reflux)
    D) Pancreatitis
    E) Candidiasis

42. A feature of central vomiting:
    A) Large in volume, brings relief
    B) Acetone odor
    C) Always bloody
    D) Not preceded by nausea, brings no relief
    E) 1 hour after eating, sour

43. From where does “coffee grounds” vomit indicate bleeding?
    A) Lungs
    B) Nose
    C) Pharynx
    D) Anus
    E) Stomach and/or duodenum

44. How long does acute diarrhea last?
    A) Up to 2–3 weeks
    B) One day
    C) 1 year
    D) 3 months
    E) 6 months

45. An infection causing bloody (invasive) diarrhea:
    A) Lactase deficiency
    B) Overfeeding
    C) Cryptosporidiosis
    D) Shigellosis
    E) Rotavirus

Clinical cases

Case 1

A 3-week-old boy. For 5 days he has had “projectile” vomiting immediately after each feed; the vomit is larger in volume than the milk taken, sour-smelling, without bile. The baby feeds greedily, but his weight is 250 g less than at birth. The stool is scanty, thick, dark green. Little urine. On examination: dry skin, sunken anterior fontanelle. “Hourglass”-shaped peristalsis is visible in the epigastrium. After vomiting, a dense mobile mass of 2 cm is palpable at the edge of the right rectus muscle. Blood: K 2.9 mmol/l, Cl 85 mmol/l, pH 7.52, BE +9.

Questions: 1) Presumptive diagnosis? 2) Which signs distinguish it from pylorospasm? 3) Explain the electrolyte and ABB changes. 4) Method of confirming the diagnosis and treatment?

Answer: 1) Pyloric stenosis, dehydration, hypochloremic hypokalemic metabolic alkalosis. 2) Onset at 2–3 weeks, vomit volume greater than the feed, “projectile” vomiting, marked weight loss, dehydration, oliguria, “hourglass” peristalsis, palpable “olive”, “hunger stools”, electrolyte disturbances (absent in pylorospasm, where vomiting is from the first days and small in volume). 3) Hydrochloric acid (H^+ and Cl^-) is lost with the vomit — hypochloremia and metabolic alkalosis (pH 7.52, BE +9); the kidney excretes K to retain Na — hypokalemia. 4) Abdominal ultrasound (pyloric muscle thickness ≥3–4 mm, canal length ≥15–17 mm); contrast X-ray (barium retained up to 24 hours). After correction of dehydration and electrolytes — pyloromyotomy (Ramstedt operation).

Case 2

A girl aged 1 year 6 months. Since 10 months (after semolina porridge and bread were introduced) the stools have been copious, 3–4 times a day, gray, shiny, fetid, hard to wash out of diapers. Her appetite is reduced, she is irritable. Weight 8.5 kg (weight deficit), height below the age norm. On examination: the abdomen is markedly enlarged, the limbs thin, the subcutaneous fat layer thinned, the buttocks “bag-like”. Hb 95 g/l. Coprogram: many fatty acids and soaps, starch “++”.

Questions: 1) The leading syndrome? 2) Presumptive diagnosis? 3) Confirmatory investigations? 4) Principle of treatment?

Answer: 1) Malabsorption syndrome (polyfecalia, steatorrhea, dystrophy, growth retardation, anemia). 2) The onset of symptoms after gluten-containing foods (semolina, bread) were introduced — celiac disease. 3) IgA antibodies to tissue transglutaminase in the blood (together with total IgA), antibodies to endomysium; duodenal biopsy for confirmation (villous atrophy, crypt hyperplasia); for differential diagnosis — sweat chloride (cystic fibrosis), fecal elastase-1. 4) A lifelong gluten-free diet (wheat, rye, barley and oat products are excluded; rice, corn and buckwheat are allowed), correction of anemia and vitamin deficiency.

Case 3

A 12-year-old boy has complained for 2 months of pain in the epigastrium and above and to the right of the umbilicus, appearing on an empty stomach and at night; the pain decreases after eating. There is heartburn, sour belching and constipation. His father has peptic ulcer disease. On examination: the tongue is coated with a white coating, the palms are moist, persistent red dermographism. Tenderness on palpation in the pyloroduodenal zone, Mendel's sign positive. Yesterday the stool was black and tarry.

Questions: 1) Presumptive diagnosis? 2) What is the pain rhythm called? 3) What does the black stool mean? 4) Which investigations are needed?

Answer: 1) Duodenal ulcer disease complicated by bleeding. 2) The Moynihan rhythm: hunger → pain → relief after eating; hunger, night and late pains are typical of duodenal ulcer. 3) Melena — a sign of bleeding from the upper parts (stomach, duodenum); breakdown of hemoglobin gives the black color — an emergency. 4) CBC (anemia), stool for occult blood, urgent EGDS (ulcer defect, hemostasis), H. pylori tests (rapid urease on the biopsy, later the urea breath test, stool antigen), abdominal ultrasound; blood group and coagulogram.

Quick review

Table 13.10. *Key facts on Topic 13*

Question	Answer
Normal stool (Bristol)	Types 3–4
Type 1 steatorrhea	Neutral fat — pancreatic lipase deficiency
Type 2 steatorrhea	Fatty acids, soaps — bile insufficiency
Creatorrhea / amylorrhea	Many muscle fibers / much starch
Fecal elastase-1	Normal >200 μg/g
Cytolysis	ALT, AST, LDH ↑
Cholestasis	Alkaline phosphatase, GGT, direct bilirubin, cholesterol ↑
Liver failure	Albumin, prothrombin ↓
Duodenal intubation	A — choledochus, B — gallbladder (dark olive), C — liver (golden yellow)
Gallbladder contraction (ultrasound)	50–60% at 40–50 minutes
H. pylori	Urea breath test, stool antigen (noninvasive); PCR — the most reliable
Intestinal obstruction (X-ray)	Fluid levels, “arches”
Ulcer (X-ray)	“Niche” sign
Perforation	Free gas under the diaphragm, no liver dullness
Duodenal ulcer	Hunger and night pain; Moynihan rhythm
“Coffee grounds”	Bleeding from the stomach / duodenum
Constipation	No stool >36 hours or difficult defecation
Colors of jaundice	Lemon — hemolytic; orange — parenchymal; green — obstructive
Pylorospasm	From the first days, little vomiting, steady weight
Pyloric stenosis	2–3 weeks, “projectile”, non-bilious, “olive”, “hourglass”, metabolic alkalosis
Malabsorption	Polyfecalia, steatorrhea, large abdomen, dystrophy
Celiac disease	Gluten; anti-tTG IgA; gluten-free diet
Colitis	Tenesmus, mucus, blood, pus
Irritable bowel syndrome	≥3 months, relief after defecation, diagnosis of exclusion
Bristol scale	1–2 constipation; 3–4 normal (4 ideal); 5 tendency to diarrhea; 6–7 diarrhea
Coprological signs	Steatorrhea (fat), creatorrhea (muscle fibers), amylorrhea (starch), lienteria (fiber); fecal elastase-1 >200 μg/g

Question	Answer
Laboratory of the liver syndromes	Cytolysis — ALT, AST, LDH; cholestasis — direct bilirubin, ALP, GGT; hepatocellular insufficiency — albumin↓, PI↓; mesenchymal — γ-globulin, thymol
Phases of duodenal intubation	I choledochus (20–30 ml); II closed Oddi; III portion A; IV portion B (30–50 ml, thick); V portion C (hepatic)
Markers of pancreatitis	Blood amylase, lipase, trypsin; urinary diastase; ultrasound: gland enlarged, echogenicity reduced
Gastric pH-metry	Measurement from the mucosa with a probe; GERD and duodenogastric reflux; 24-hour pH monitoring — gold standard
Advantages and signs of ultrasound	Safe, noninvasive; ascites — free fluid; portal hypertension — vein diameter↑; pyloric stenosis — elongated pyloric canal; intussusception — “target”
Endoscopic methods	EGDS, rectosigmoidoscopy, colonoscopy, ERCP, video capsule; biopsy (villous atrophy in celiac disease, H. pylori)
Diagnosis of H. pylori	Urea breath test (noninvasive), rapid urease, bacteriological, histological, cytological, PCR (the most reliable), ELISA, stool antigen
Pain and its relation to food	Early — gastritis, gastric ulcer; late — antral gastritis, duodenitis; hunger/night — duodenal ulcer (Moynihan rhythm)
Types of belching	Air — aerophagia; rotten eggs — achylia; sour — hyperacidity; bitter — bile reflux; putrid — stenosis
Types of vomiting	Central (CNS); peripheral (reflex); gastric (0.5–1.5 h, relief); acetonemic (acetone odor)
Appearance of bleeding	Bright red — esophagus, nose, pharynx; “coffee grounds” — stomach/duodenum; large amount — varices (cirrhosis)
Types of diarrhea	Watery (rotavirus, escherichiosis); fatty (salmonellosis, cystic fibrosis); bloody (shigellosis, ulcerative colitis, Crohn)

TOPIC 14

The urinary and immune systems: anatomical and physiological features, methods of examination, urine tests and functional tests, semiotics of involvement

Learning objectives

The student should know:

- the age-related anatomical and physiological features of the kidneys and urinary tract, the norms of diuresis and voiding frequency;
- the methods of history taking, inspection, palpation and percussion of the urinary system;
- the urinalysis, the Nechiporenko, Hamburger, Addis–Kakovsky and three-glass tests, the Zimnitsky, Rehberg and McClure–Aldrich tests and their norms;
- instrumental investigations (ultrasound, urography, cystography, scintigraphy);
- the main symptoms and syndromes of urinary system involvement;
- the age-related features of the immune system, the “critical periods”, the methods of assessing immunity and the signs of immunodeficiency.

The student should be able to:

- detect edema, palpate the kidneys, test the pain points of the urinary tract, determine Pasternatsky's sign, percuss the border of the bladder;
- calculate the daily diuresis by age;
- interpret the results of urine tests and functional tests;
- identify the leading renal syndrome;
- assess the parameters of the immunogram and identify the warning signs of primary immunodeficiency.

14.1. Anatomical and physiological features of the kidneys and urinary tract

At birth the morphological and functional development of the kidneys is **incomplete**. Defects of the urinary organs make up **≈1/3** of all congenital defects.

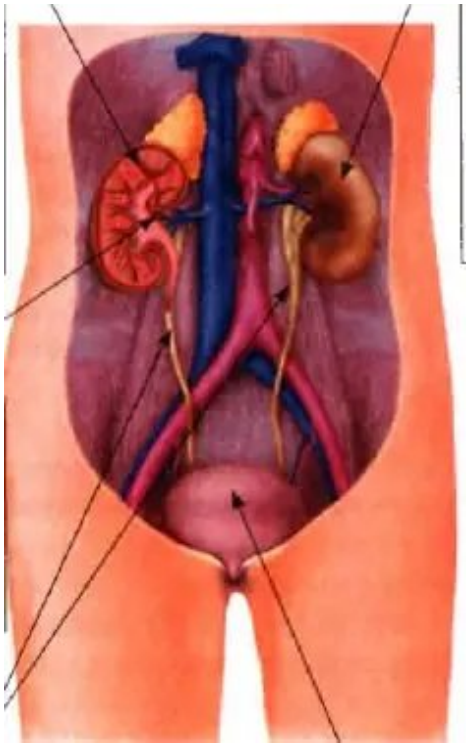


Figure 14.1. *Anatomical and physiological features of the urinary system in young children (Navruzova, Geppe)*

The kidneys

- **Mass and shape:** at an early age the kidneys are relatively **large**: in the newborn the kidney mass to body mass ratio is **1:100**, in adults **1:200**. Up to 1 year the kidney is **round**, then bean-shaped.
- **Position:** because of their relatively large size and the shortness of the lumbar vertebrae, up to 1 year the kidneys lie **lower**: the upper pole at ThXI–XII, the lower pole **below L_{IV}** (below the iliac crest); this feature disappears **by 7 years**. At an early age the kidneys lie almost **parallel**.
- **Mobility:** the perirenal fat and fasciae are poorly developed — the kidneys are **mobile**; the fixation mechanisms form **at 5–8 years**.
- **Structure:** in the 1st year **the lobulated structure** persists (it disappears at 2–5 years); the medulla is thicker than the cortex (1:4; 1:2 in adults). In a term newborn **the number of nephrons is the same as in adults (≈2 million)**, but the glomeruli are small, the nephrons are insufficiently differentiated until 2 years, and the loop of Henle is short. Morphological maturation of the cortex is completed **at 3–5 years**, and full maturation of the kidney **at school age**.

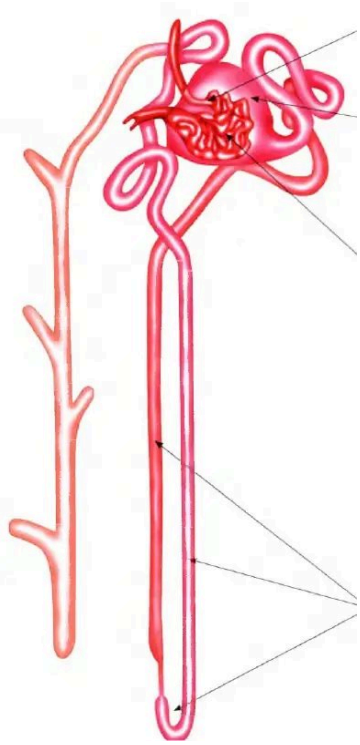


Figure 14.2. *Anatomical structure of the nephron (Navruzova, Geppe)*

Functional features of the kidneys

- **Low glomerular filtration:** the visceral layer of the glomerular capsule has **cuboidal epithelium**; the filtering surface is **5 times smaller** than in adults; renal blood flow is low (**5%** of the cardiac output in the newborn, 25% in adults). Filtration approaches adult levels **by 1 year**.
- **Tubular reabsorption and secretion** develop gradually: amino acids, phosphates and bicarbonates are excreted in large amounts; glucose reabsorption is good; sodium is actively reabsorbed — **predisposition to edema**; water and NaCl loads cannot be excreted quickly; excretion of K, Ca and Mg is reduced.
- **Concentrating function is low** (little ADH, a short loop of Henle, an immature distal tubule) — **low specific gravity** of the newborn's urine; adult concentrating ability develops **at 9–12 months**.
- **Regulation of acid-base balance** is immature (low secretion of hydrogen and ammonium ions) — **acidosis** develops quickly in diseases at an early age.
- The optimal homeostatic function of the kidney is reached **at 10–11 years**. Therefore **edema and dehydration** develop easily in children; the weak excretion of the kidneys must be taken into account when giving drugs (antibiotics) and solutions.

The urinary tract

- **The renal pelvises** are relatively wide and up to 5 years often lie **within the kidney**; **the ureters** are relatively long, wide, hypotonic and tortuous, leaving the pelvis at a right angle; the intravesical part of the ureter is **short** (it lengthens at 10–12 years), and the muscle fibers at its orifice are weak — predisposition to **vesicoureteral reflux**.
- **The bladder** in infancy lies **high** (it is palpable above the pubis) and descends into the small pelvis with age; its mucosa is rich in blood vessels.

Table 14.1. *Physiological capacity of the bladder*

Age	Capacity
Newborn	≈50 ml
1 year	≈100 ml
5–9 years	150–200 ml
12–14 years	300–400 ml

- **The urethra** in girls is **short and wide** at all ages and close to the anus — risk of **ascending infection**; in boys it is long, with marked curvature in infancy (taken into account during catheterization).

REMEMBER

At an early age, the width, hypotonia and tortuosity of the urinary tract and the mobility of the kidney lead to **disturbed urodynamics** and the addition of **a microbial-inflammatory process (pyelonephritis)**.

Diuresis and voiding

- In term newborns the first voiding occurs **on day 1**; in the first **2–3 days** there is little urine — **transient oliguria** (low fluid intake and extrarenal losses). Absence of urine for 1 day is pathological.
- The **formula for daily diuresis** up to 10 years: **600 + 100 × (n – 1)** ml (n — age in years); the ratio of day to night diuresis is **2:1 – 3:1**.

Table 14.2. *Daily diuresis and number of voidings in children (Kildiyarova, Papayan)*

Age	Daily diuresis, ml	Number of voidings per day	Single volume, ml
Until 6 months	300–500	20–25	20–35
6 months – 1 year	300–600	15–16	25–45
1–3 years	760–820	10–12	60–90
3–5 years	900–1070	7–9	70–90
5–7 years	1070–1300	7–9	100–150
7–9 years	1240–1520	7–8	145–190
9–11 years	1520–1670	6–7	220–260
11–13 years	1600–1900	6–7	250–270

- Voiding is **a reflex act**; toilet training can begin **from 5–6 months**; by the end of the 1st year the child asks for the potty while awake, and in the 2nd year this skill becomes stable. In sleep, during excitement, or during an interesting game, involuntary voiding may occur up to 3 years (in sleep — up to 5 years).

MEMORY AID

Diuresis: **“600 + 100 for each year”**. A 7-year-old child: 600 + 100 × 6 = **1200 ml**. Number of voidings: newborn **20–25**, 1 year **15**, 3 years **≈10**, school age **6–7**.

14.2. Clinical examination of the urinary system

Questioning

Main complaints: abdominal and **lumbar pain** (small children cannot localize it); **dysuric disturbances** (frequent or infrequent, painful voiding, incontinence, enuresis); **edema of the face and legs** (the parents notice rapid weight gain); **raised BP**, headache, dizziness; **change in the appearance of the urine** (turbidity, sediment, change of color — “meat washings”); unexplained **fever** or prolonged subfebrile temperature; visual and hearing disturbances; dyspepsia, thirst, itching of the skin; weakness, weight loss.

History: onset of the complaints, results of previous urine tests and treatment; **family history** (glomerulonephritis, pyelonephritis, urinary tract defects, urolithiasis, deafness, renal failure); lagging physical development at an early age; recurrent episodes of fever without signs of infection.

Inspection

Table 14.3. *Signs on inspection in diseases of the urinary organs (Geppe)*

Sign	Note
Pallor	Arteriolar spasm or anemia; waxy pallor — amyloidosis; pallor with an icteric tint, scratch marks, ecchymoses, ammonia breath odor — uremia
Edema (puffiness)	On the face and eyelids, the palpebral fissures narrow (facies nephritica); joint contours are smoothed; marks of clothing and sheets remain; generalized — anasarca, in the cavities — ascites, hydrothorax, hydropericardium
Change in the abdomen	Enlarged in ascites, the umbilicus protrudes when standing, it spreads to the sides when lying (“frog belly”); a bulge above the pubis in acute urinary retention
Bulge in the lumbar region	Paranephritis (on the affected side)
Stigmas of dysembryogenesis	Occur together with hereditary nephropathies and urinary tract defects
Behavior during voiding	An infant cries during or after voiding (cystitis); in the hyporeflexic form of neurogenic bladder dysfunction voiding is prolonged, in several attempts, with pressure on the abdomen



Figure 14.3. *Facies nephritica — edema of the face and eyelids (Navruzova, Geppe)*



Figure 14.4. *Edema of the foot and shin, smoothing of the joint contours (Navruzova, Geppe)*



Figure 14.5. *Marks left by clothing (socks) — a sign of latent edema (Navruzova, Geppe)*



Figure 14.6. *Anasarca in nephrotic syndrome (Navruzova, Geppe)*

Palpation

Detecting edema: the fingers are pressed for 5–10 seconds on the anterior surface of the shin over the tibia — if **a pit** remains, there is edema. To detect **latent edema** the child is weighed daily, the daily diuresis and fluid intake are calculated, and **the McClure–Aldrich test** is performed.

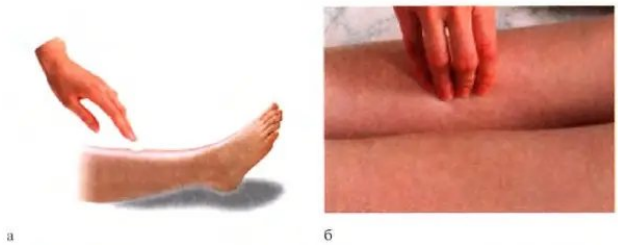


Figure 14.7. *Method of detecting edema of the legs: the fingers are pressed over the tibia (Navruzova, Geppe)*



Figure 14.8. *A pit remaining at the site of pressure (Navruzova, Geppe)*

Palpation of the kidneys — bimanual according to Obraztsov–Strazhesko (lying down): the child lies on the back with the legs slightly bent; the doctor places the left hand under the lower back — in the angle formed by rib XII and the long back muscles; the right hand is placed on the corresponding half of the abdomen, outside the rectus muscle, below the costal arch, plunged deep during expiration, and the hands are made to **"approach"** each other. **According to S.P. Botkin** — standing, bent forward ≈45°. Before palpating the right kidney make sure the liver is not enlarged, and before the left — the spleen; if there is pain, start from the healthy side.

- **Up to 2 years** (especially in thin children) the kidneys (more often **the right**) may be palpable; **in older children they are normally not palpable**.
- Palpability — enlargement (**hydronephrosis, tumor — Wilms tumor, polycystic disease, vicarious hypertrophy of a single kidney**) or displacement (**nephroptosis, dystopia**); in aplasia of the abdominal wall muscles a healthy kidney is also palpable. Painful palpation — **pyelonephritis, paranephritis**.
- Assessed: size, mobility, shape, consistency (a healthy kidney is dense-elastic), surface, tenderness.



Figure 14.9. *Bimanual palpation of the kidney (Obraztsov–Strazhesko) (Navruzova, Geppe)*

Pain points (with the middle fingers, during expiration):

- **upper ureteral points** — where the outer edge of the rectus muscles crosses the horizontal line through the umbilicus;
- **lower ureteral points** — where the inner edge of the rectus muscles crosses the line through the anterior superior iliac spines;
- behind — **the costolumbar points** (where rib XII crosses the long back muscles) and **the costovertebral points** (the angle between rib XII and the spine).



Figure 14.10. *Pain points of the urinary tract: 1 — upper; 2 — lower ureteral points (Navruzova, Geppe)*

Palpation of the bladder — above the pubis; in older children only a **full** bladder is palpable; at an early age an empty bladder may also be palpable. First the time of the last voiding is determined.

Percussion

- **Pasternatsky's sign (tapping the lumbar region):** the left palm is placed on the lumbar region at the level of rib XII, alternately on the right and the left, and **short, weak blows** are given to it with the edge of the right palm (standing or sitting). If pain appears — **positive** (negative, weakly positive, positive, sharply positive): **pyelonephritis, paranephritis, urolithiasis**. The appearance of new erythrocytes in the urine after tapping — **a true Pasternatsky sign**.
- **Upper border of the bladder:** the pleximeter finger is placed parallel to the pubis and quiet percussion is performed along the midline **downward from the umbilicus**; the place where a dull sound appears is the border of the bladder (in acute urinary retention). If dullness is found, it is repeated after voiding.
- Free fluid in the abdominal cavity — see topic 12.



Figure 14.11. *Tapping the lumbar region (Pasternatsky's sign) (Navruzova, Geppe)*

Auscultation: a systolic murmur over the projection of the renal arteries (2–3 cm above the umbilicus, lateral to the midline) — renal artery stenosis (renovascular hypertension).

14.3. Laboratory investigations

Urinalysis

Urine collection: after thorough washing of the external genitals, **the midstream portion of the morning urine** is collected into a clean container; in newborns and infants **urine bags** stuck to the perineum are used. The analysis is performed **no later than 2 hours** after voiding (otherwise it is stored at +4 °C).

Table 14.4. *Norms of urinalysis in children (Geppe, Kildiyarova)*

Indicator	Normal	Features in children
Color	Straw-yellow (urochrome, urobilin)	In the 1st week of life there is much uric acid — reddish-brick color, stains on diapers (uric acid infarction); in infancy pale yellow
Clarity	Completely clear	May become turbid on standing because of salt precipitation
Reaction (pH)	Weakly acidic or neutral (5.5–6.5)	In the newborn 5.4–6.0; with breast milk — weakly alkaline (6.9–7.8); artificial — weakly acidic (5.4–6.9)
Specific gravity	1008–1025 (depends on fluid)	Low in the 1st week (not above 1016–1018); at an early age 1001–1015
Protein	Absent or up to 0.033 g/l	Transient proteinuria in the newborn in the first days (disappears by day 4–10)
Glucose	Absent	Appears when the blood level exceeds 8.9–10 mmol/l (the renal threshold)
Erythrocytes	0–2 per field of view	—
Leukocytes	In boys 0–5(6), in girls up to 5–10	—
Casts	Absent (single hyaline)	—
Epithelium	Single squamous epithelial cells	—
Salts	Absent or up to “+”	Uric acid crystals in the newborn
Bacteria	Absent	—

Quantitative examination of the urinary sediment

Used to detect latent leukocyturia and erythrocyturia.

- **Nechiporenko test** — the number of formed elements **in 1 ml of urine** in the midstream portion of the morning urine; convenient at an early age because the urine is easy to collect.
- **Hamburger test** — after emptying the bladder in the morning, the urine is collected into one container for **3 hours**; the number of elements excreted **in 1 minute** is calculated.
- **Addis–Kakovsky test** — the urine is collected into one container for **24 hours** (from 8–9 a.m. to 8–9 a.m. the next day); for standardization fluid is slightly restricted and protein food is given (the urine is acidic, specific gravity ≈1020 — the cells are well preserved).

Table 14.5. *Norms of the tests for latent leukocyturia and erythrocyturia (Kildiyarova)*

Test	Leukocytes	Erythrocytes	Casts
Nechiporenko (in 1 ml of urine)	Girls — up to 4000, boys — up to 2000	Up to 1000	Absent
Hamburger (in 1 minute)	Girls — up to 2000, boys — up to 1500	Up to 1000	Up to 200
Addis–Kakovsky (per day)	Up to 2,000,000	Up to 1,000,000	Up to 50,000

> **CAUTION**
> Geppe's textbook gives the norm for leukocytes in the Nechiporenko test as **up to 2000** (not divided by sex); Kildiyarova gives **up to 4000** for girls. In practice **leukocytes <2000, erythrocytes <1000 per 1 ml** are often taken as the norm.

Three-glass (two-glass) test — determining **the source** of hematuria and leukocyturia: in the morning, without washing the external genitals, the urine is voided successively into three containers. Changes only in **the 1st portion** — the external genitals or urethra; in **the 3rd portion** — the bladder (neck); **the same in all three portions** — above the bladder (kidney, ureter or bladder).

Bacteriological examination of the urine — the midstream portion collected into a sterile container is cultured; the pathogen, its amount and antibiotic sensitivity are determined. **Diagnostically significant bacteriuria:** in free voiding **≥100,000 (10^5) CFU/ml**; through a catheter **≥10,000 (10^4)/ml**; on suprapubic puncture — **any** amount; in children under 1 year in free voiding, E. coli **≥50,000/ml**. Proteus, Klebsiella and Pseudomonas aeruginosa are significant even in smaller amounts.

Functional tests

Zimnitsky test — assesses the **excretory and concentrating** function of the kidneys and the **circadian rhythm**. During the day the urine is collected into a separate container **every 3 hours** (8 portions; at an early age — at free intervals), **the volume and specific gravity** of each portion are measured, and the fluid intake is taken into account.

- **Excretory function:** normally **70–80%** of the fluid taken is excreted.
- **Concentrating function:** during the day the specific gravity varies within **1008–1025**; in children over 3 years the maximal specific gravity in at least one portion is **1018–1020**; a difference between the maximal and minimal specific gravity of **≥10 units** (optimally 15) — the ability to concentrate and dilute is preserved.
- **Circadian rhythm: daytime diuresis (9:00–21:00) is 2/3–3/4 of the daily diuresis**; predominance of night diuresis — **nocturia** (chronic renal failure, pyelonephritis).

- **Hyposthenuria** — specific gravity constantly **<1010–1012**; **isosthenuria** — monotonous specific gravity around **1008–1010** (the specific gravity of primary urine) — a severe disturbance of the concentrating function; **hypersthenuria** — >1030 (glucosuria, proteinuria, dehydration).

Blood creatinine and urea — easily determined indicators of renal function; an increase indicates impaired filtration function.

Rehberg test — assesses **the glomerular filtration rate (GFR)** by **the endogenous creatinine clearance** and **tubular reabsorption** (creatinine is freely filtered in the glomeruli and is almost neither reabsorbed nor secreted in the tubules). The urine is collected over 24 hours (or 2 hours), and venous blood is taken after the last voiding.

- Clearance: **C = (U × V / P) × (1.73 / S)**, where U and P are the creatinine in the urine and plasma, V is the minute diuresis, and S is the child's body surface area.
- **Tubular reabsorption** = [(clearance – minute diuresis) / clearance] × 100% — normally **97–99%**.
- In practice GFR is calculated by **the Schwartz formula**: **GFR = k × height (cm) / creatinine (μmol/l × 0.0113)**; k — 0.33 in preterm babies; 0.45 in term newborns and up to 2 years; 0.55 at 2–14 years; 0.70 in boys over 14 years.

Table 14.6. *Age norms for the glomerular filtration rate*

Age	GFR, ml/min/1.73 m^2
Newborn	30–50 (≈46)
1 year	Approaches 80–120
1–3 years	≈58 (Kildiyarova)
4–14 years	≈121 ± 18
Adults	≈124 ± 26 (80–120)

McClure–Aldrich test — detects **latent edema**: **0.2 ml of isotonic NaCl is injected strictly intradermally** into the lower third of the inner surface of the forearm, and the time taken for the resulting bleb to resorb is recorded.

Table 14.7. *Norms of the McClure–Aldrich test*

Age	Resorption time of the bleb
Until 1 year	10–15 minutes
1–5 years	20–25 minutes
Over 5 years	40–60 minutes

When tissue hydrophilia is increased (readiness for edema), the bleb resorbs **faster**; with a marked tendency to edema the bleb does not form at all.

Other investigations: blood total protein and fractions, cholesterol, electrolytes (K, Na, Ca, P), ABB; in the urine — daily protein, the protein/creatinine ratio (normal ≤0.2 mg/mmol), microalbuminuria; for the proximal tubules — excretion of amino acids, phosphates, glucose, β_2-microglobulin; for the distal tubules — ammonium and titratable acidity.

14.4. Instrumental investigations

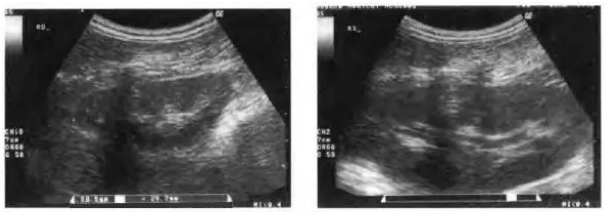
Table 14.8. *Instrumental methods of examining the urinary system (Geppe, Kildiyarova)*

Method	Purpose
Ultrasound (screening method)	Size, shape, position and structure of the kidneys, the pelvicalyceal system, blood flow (Doppler), state of the bladder, stones, developmental defects, fluid in the cavities
Excretory urography	Anatomical and functional state of the kidneys and urinary tract, urodynamics
Voiding cystourethrography	Anatomical and functional state of the bladder and urethra, vesicoureteral reflux
Radioisotope renography, nephroscintigraphy (dynamic, static)	Renal blood flow, tubular secretory function, volume of functioning parenchyma, scars (DMSA)
Renal arteriography	Blood flow in hypoplasia, a shrunken kidney, cysts and tumors; renovascular hypertension
CT, MRI	Tumors, defects, stones
Cystoscopy	State of the bladder mucosa and ureteral orifices (chronic cystitis, ureterocele, polyp, diverticulum, stone, foreign body)
Cystometry, uroflowmetry	Urodynamics of the lower urinary tract (neurogenic bladder dysfunction)
Needle biopsy	Morphological diagnosis, determining treatment and prognosis (nephrotic syndrome, glomerulonephritis)

Table 14.9. *Kidney dimensions on ultrasound (Kildiyarova)*

Age	Length, cm	Width, cm	Thickness, cm
Newborn	4,2	2,2	1,8
1 year	7,0	3,7	2,6
5 years	7,9	4,3	2,8
11 years	9,8	5,2	3,3
15 years	10,7	5,3	3,5

On ultrasound the pelvis measures **≤5 mm** up to 5 years, ≤7 mm at 6–10 years, ≤10 mm at 11–15 years. **Preparation for excretory urography:** gas-forming foods are excluded for 2–3 days, a cleansing enema in the evening and morning; sensitivity to iodine preparations is tested.



а б

Figure 14.12. *Ultrasound: a — the right kidney is reduced (shrunken); b — the left kidney is of normal size (Navruzova, Geppe)*



Figure 14.13. *Ultrasound: a stone (calculus) in the renal pelvis (Navruzova, Geppe)*

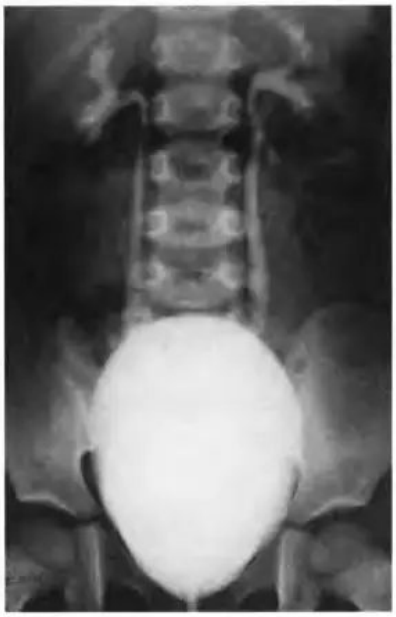


Figure 14.14. *Voiding cystourethrography: bilateral vesicoureteral reflux (Navruzova, Geppe)*



Figure 14.15. *Congenital hydronephrosis — urogram (Navruzova, Geppe)*

14.5. Semiotics of urinary system involvement

Developmental anomalies

- **Kidney: number** — supernumerary kidney, duplication of the kidney, **agenesis**, aplasia; **shape** — **horseshoe** (fused poles), ring-shaped, L- and S-shaped kidney; **position** — **dystopia** (pelvic, iliac), rotation anomaly; **hypoplasia** (simple, dysplastic), **dysplasia**, **polycystic disease**; **congenital hydronephrosis** (most often stenosis of the pelviureteric segment).
- **Ureters:** duplication, stricture, ectopic orifice, diverticulum, **megaureter**, ureterocele.
- **Bladder:** agenesis, **exstrophy**, diverticulum.
- **Urethra:** atresia, stenosis, **posterior urethral valves**, **hypospadias** (the external orifice on the lower surface of the penis), **epispadias** (absence of the anterior wall).

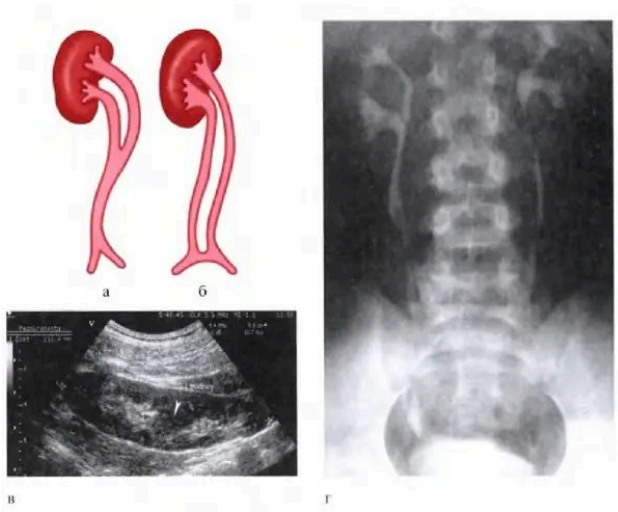


Figure 14.16. *Duplication of the kidney (Navruzova, Geppe)*



Figure 14.17. *Horseshoe kidney — diagram (Navruzova, Geppe)*

Disturbances of urine output and voiding

Table 14.10. *Disturbances of diuresis (Geppe, Kildiyarova)*

Disturbance	Definition	Causes
Polyuria	An increase in daily diuresis to more than 2 times the norm (>1500 ml/m^2 in older children)	Drinking a lot, resolution of edema, diuretics, diabetes mellitus and insipidus, the polyuric phase of renal failure, tubulopathies
Oliguria	A decrease in diuresis to 1/3–1/4 of the age norm	Renal: glomerulonephritis, acute kidney injury, end-stage chronic renal failure; extrarenal: fluid restriction, heavy sweating, profuse diarrhea, persistent vomiting, cardiac edema; transient — in the newborn on day 2–3
Anuria	No urine for 12 hours (check that the bladder is empty by percussion) or ≤50 ml per day (<6–7% of the norm)	True — acute kidney injury (shock, blood loss, poisoning, acute glomerulonephritis); obstructive — obstruction of the upper urinary tract
Ischuria	Acute retention of urine (the bladder is full but does not empty)	Obstruction of the urethra by a stone, trauma, bladder atony, posterior urethral valves
Nocturia	Predominance of night diuresis over day diuresis	Pyelonephritis, resolution of edema, chronic renal failure

Table 14.11. *Disturbances of the act of voiding — the dysuric syndrome (Geppe)*

Disturbance	Description and causes
Pollakiuria (frequent voiding)	With pain — cystitis; in the daytime, increasing with movement — bladder stone; painless — hyperreflexic bladder; in healthy children after chilling or bathing in salt water
Infrequent voiding	Hyporeflexic bladder, heavy sweating, vomiting, diarrhea, increasing edema
Strangury (painful, burning voiding)	Pain at the end of voiding — cystitis (bladder neck); during and after voiding — urethritis
Urge incontinence (inability to hold urine on urging)	Acute cystitis, neurogenic bladder dysfunction, diverticulum or stone
Continuous leakage of urine (without urge)	True — spinal cord injury, spinal hernia; false — ectopic ureteral orifice, bladder exstrophy, fistulas

Disturbance	Description and causes
Enuresis (voiding in sleep)	After 5 years; CNS pathology, neurogenic bladder dysfunction, mental disorders, pathology of the lower urinary tract

Changes in urine color

Table 14.12. *Changes in urine color and their causes (Kildiyarova, Geppe)*

Color	Cause
Colorless, pale	Low specific gravity: diabetes mellitus and insipidus, chronic renal failure, diuretics, infusion
Dark yellow	Increased concentration: fever, vomiting, diarrhea, oliguria; ascorbic acid
Red (“meat washings”)	Erythrocyturia (glomerulonephritis — altered erythrocytes), hemoglobinuria, myoglobinuria, nephrolithiasis, trauma; beets, cherries, some drugs
Orange (saffron)	Uraturia (uric acid infarction in the newborn), rifampicin, nitrofurantoin
Dark brown	Urobilinogenuria (hemolytic anemia)
Black	Hemoglobinuria (hemolysis), alkaptonuria
Greenish-yellow, “beer-colored”	Bilirubinuria — obstructive and parenchymal jaundice
White (“milky”)	Chyluria, pyuria, phosphaturia

Urinary syndrome

Urinary syndrome — pathological changes in the urine: proteinuria, hematuria, leukocyturia, cylindruria, bacteriuria, crystalluria.

- **Proteinuria** — excretion of **>150–200 mg** of protein per day (or more than 0.033 g/l). Degree: **mild** — <1 g/day; **moderate** — 1–2.5 g/day (glomerulonephritis); **marked (nephrotic)** — >2.5–3 g/day or **>50 mg/kg/day**. **Renal (glomerular, tubular)** and **extrarenal** (prerenal — Bence Jones protein; postrenal — inflammation of the urinary tract); **functional** — fever, physical strain, chilling, **orthostatic**. **Selective** (mainly albumin — milder damage) and **nonselective** (large-molecule globulins — deep damage).
- **Hematuria** — **>2–3 erythrocytes** per field of view in the morning urine; **microhematuria** (the color does not change) and **macrohematuria** (>25–30, red urine or “meat washings”). **Glomerular** (dysmorphic erythrocytes, erythrocyte casts, with proteinuria — glomerulonephritis, IgA nephropathy, hereditary nephritis) and **nonglomerular** (isomorphic erythrocytes — urolithiasis, cystitis, tumor, trauma). With pain — renal colic; with dysuria — cystitis.
- **Leukocyturia** — **>5–6 (>10 in girls)** per field of view in the urinalysis, >2000(–4000)/ml by Nechiporenko, >2 million/day by Addis–Kakovsky; **pyuria** — leukocytes cover the whole field of view. **Neutrophilic** type (on the urocytogram) — microbial inflammation (**pyelonephritis, cystitis**, urethritis); **mononuclear/lymphocytic** — glomerulonephritis, interstitial nephritis, SLE.
- **Cylindruria** — **always a sign of renal pathology** (except for single hyaline casts): hyaline — physical strain, fever, nephrotic syndrome; granular and waxy — severe degenerative damage to the tubules; **erythrocyte** — renal hematuria (glomerulonephritis); leukocyte — renal leukocyturia (pyelonephritis).
- **Bacteriuria** — the main sign of urinary infection (significant cut-offs above).

- **Crystalluria:** urates — urate diathesis (neuro-arthritic diathesis), high meat intake; oxalates — oxalate nephropathy; triple phosphates — urinary tract infection.
- **Glucosuria** (diabetes mellitus, **renal glucosuria** — normal blood sugar), **ketonuria** (acetonemic vomiting, diabetes mellitus).

MEMORY AID
Source of hematuria: **“Erythrocyte cast + protein + dysmorphic erythrocyte = glomerulus”**. Leukocyturia + bacteriuria + fever + positive Pasternatsky = **pyelonephritis**; leukocyturia + dysuria, without fever = **cystitis**.

Edema syndrome

Mechanisms of renal edema: **reduced oncotic pressure** of the plasma (hypoalbuminemia); increased capillary permeability; activation of **the renin-angiotensin-aldosterone system** (retention of Na and water); **reduced glomerular filtration**. Renal edema appears **in the morning on the face and eyelids** and descends during the day; it is soft and warm, the skin pale (cardiac edema — in the evening on the legs, with cyanosis, firm).

Pain syndrome

- **Dull, constant lumbar pain** — stretching of the renal capsule (glomerulonephritis, pyelonephritis, hydronephrosis);
- **sharp, paroxysmal pain in the lower back radiating down along the ureter — to the groin and genitals** — **renal colic** (urolithiasis, a blood clot);
- **suprapubic pain** — cystitis, bladder stone, urinary retention;
- pain in the lower back and one side of the abdomen during voiding — **vesicoureteral reflux**.

Main clinical syndromes

Nephritic (acute nephritic) syndrome — **hematuria** (from micro- to macro-, “meat washings”), **arterial hypertension**, moderate **proteinuria** (1–2 g/day), moderate **edema** (face, shins) or latent edema, **oliguria**, cylindruria. Typical of **acute poststreptococcal glomerulonephritis** (**1–3 weeks** after tonsillitis or pyoderma), Henoch–Schönlein nephritis, SLE.

Nephrotic syndrome — **massive edema** (up to anasarca), **marked proteinuria (>50 mg/kg/day or >3 g/day)**, **hypoproteinemia and hypoalbuminemia** (<25 g/l), dysproteinemia (increased α_2-globulins), **hyperlipidemia** (hypercholesterolemia). **Complete** (all signs) and **incomplete**; **pure** (without hematuria and hypertension — most often “minimal change” — in children aged 2–6 years, steroid-sensitive) and **mixed** (with hematuria and/or hypertension). **Primary** (congenital, idiopathic, glomerulonephritides) and **secondary** (SLE, vasculitides, amyloidosis, diabetes mellitus, hepatitis B and C, HIV, heavy metals).

Table 14.13. *Differences between the nephritic and nephrotic syndromes*

Sign	Nephritic syndrome	Nephrotic syndrome
Edema	Moderate (face, eyelids)	Massive, anasarca
Proteinuria	1–2 g/day	>3 g/day (>50 mg/kg)
Hematuria	Marked (“meat washings”)	Absent (in the pure form)
Arterial hypertension	Present	Usually absent
Blood albumin	Normal or slightly low	Sharply low

Sign	Nephritic syndrome	Nephrotic syndrome
Cholesterol	Normal	High
Diuresis	Oliguria	Oliguria (in the edematous phase)
Typical example	Acute poststreptococcal glomerulonephritis	Minimal change nephrotic syndrome

Syndrome of urinary tract involvement (infection) — pyelonephritis, cystitis: pain, dysuria, a positive Pasternatsky sign, **intoxication and fever** (in pyelonephritis); laboratory — **neutrophilic leukocyturia, bacteriuria**, microhematuria; instrumental — dilatation and deformation of the pelvicalyceal system, signs of obstruction. At an early age pyelonephritis often presents with **"unexplained" fever**, vomiting and failure to gain weight.

Tubular dysfunction (tubulopathy) syndrome — polyuria, hypo- and isosthenuria; **proximal** — renal glucosuria, **phosphate diabetes** (vitamin D-resistant rickets: severe rickets-like deformities, low blood P, normal Ca, phosphaturia), **de Toni–Debré–Fanconi syndrome** (glucosuria + aminoaciduria + phosphaturia + rickets-like changes); **distal — nephrogenic diabetes insipidus** (polyuria, polydipsia, hypernatremia), **renal tubular acidosis** (acidosis in the blood, alkaline urine — calcinosis, stones), pseudohypoaldosteronism.

Arterial hypertension syndrome — **renoparenchymal** (glomerulonephritis, pyelonephritis, defects, reflux nephropathy) and **renovascular** (renal artery stenosis, thrombosis, aortoarteritis). The most common cause of secondary hypertension in children is **kidney disease**.

Acute kidney injury (acute renal failure) — a sharp, **reversible** decline in renal function: **oliguria/anuria, hyperazotemia (raised creatinine and urea), hyperkalemia, metabolic acidosis, hyperhydration**. Causes: **prerenal** (shock, dehydration, blood loss), **renal** (hemolytic uremic syndrome — the most common cause at an early age, acute glomerulonephritis, nephrotoxins — aminoglycosides, sepsis), **postrenal** (obstruction). Phases of the course: initial, oligoanuric, polyuric, recovery.

Chronic renal failure (chronic kidney disease) — **irreversible** disturbance of homeostasis as a result of a progressive decrease in the number of functioning nephrons. At the onset — **polyuria, nocturia, isosthenuria**, a moderate rise in creatinine; then intoxication, **anemia**, arterial hypertension, acidosis, **renal osteodystrophy**, growth retardation, hyperkalemia; **the terminal (uremic) stage** — GFR <15–20 ml/min/1.73 m^2, itching of the skin, a grayish-yellow skin color, **ammonia odor**, nausea, vomiting, hemorrhagic syndrome, pericarditis (the "death knell" rub), seizures, uremic coma.

14.6. The immune system: anatomical and physiological features

The immune system — the set of organs, tissues and cells that protects the body from **genetically foreign** substances (antigens) entering from outside or formed in the body.

Table 14.14. *Organs of the immune system (Geppe)*

Central	Peripheral
Red bone marrow, thymus	Lymph nodes, spleen, tonsils (Waldeyer–Pirogov ring), Peyer's patches of the intestine, appendix, skin (Langerhans cells), liver (Kupffer cells), tissue macrophages

(Geppe's table lists the spleen among the central organs; in most textbooks it is considered a peripheral organ.)

Innate (nonspecific) defense

- **Natural barriers:** skin, mucous membranes, ciliated epithelium, the acidic environment of the stomach, the blood-brain barrier.
- **Humoral factors: lysozyme** (saliva, tears, breast milk; high in the newborn), **properdin** (low at birth, increases at 1–3 weeks), **interferons** (low production capacity in the newborn, maximal in puberty), **the complement system** (≈**50%** of adult levels at birth, reaching adult levels by the end of the 1st month).
- **Phagocytosis** (neutrophils, macrophages): in the newborn the engulfing activity is adequate, but **the final stage (digestion) is imperfect** — it reaches adult levels at 6–12 months; phagocytosis of pneumococcus, Klebsiella and Haemophilus is incomplete — these infections are severe and generalized.

Adaptive (specific) immunity

- **Cell-mediated immunity — T lymphocytes** (mature in the thymus): **CD4$^+$ T helpers** (Th1, Th2), **CD8$^+$ cytotoxic T lymphocytes (killers)**, T regulators (suppressors). A T lymphocyte recognizes an antigen together with HLA molecules after it has been processed by **an antigen-presenting cell** (macrophage); it acts against intracellular viruses, tumor cells and intracellular pathogens.
- **Humoral immunity — B lymphocytes** (mature in the bone marrow) become **plasma cells** and produce **immunoglobulins (antibodies)**; they recognize the antigen in its native form; they act against extracellular pathogens.
- **Memory cells** — store information about the antigen and ensure a rapid and strong response on re-encounter (the basis of vaccination).

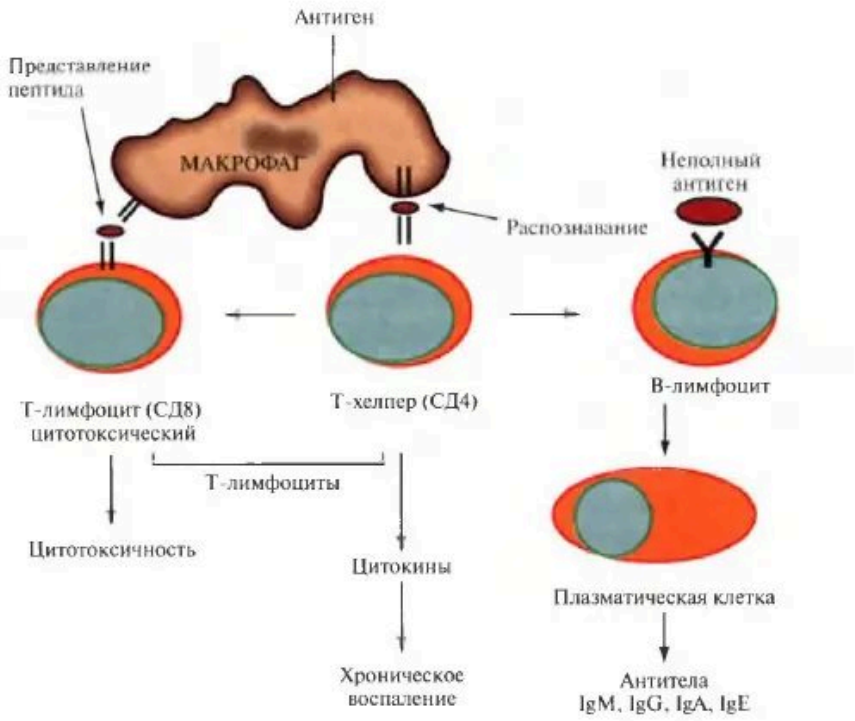


Figure 14.18. *Cells involved in the immune response (Navruzova, Geppe)*

Immunoglobulins and their age-related dynamics

Table 14.15. *Classes of immunoglobulins (Geppe)*

Class	Share of plasma Ig	Main features	Age-related dynamics
IgG	70–80%	Antibodies against viruses, gram-positive bacteria and toxins; immunological memory; crosses the placenta — passive immunity in the newborn	Maternal IgG are broken down by 3–6(9) months; the child's own IgG is 50% of adult levels at 1 year and reaches adult levels at 4–6 years
IgM	5–10%	Antibodies of the primary immune response; gram-negative bacteria, AB0 hemolysins; activates complement by the classical pathway; does not cross the placenta	Increases rapidly after birth, reaches adult levels at 1–2 years; >0.2–0.3 g/l in cord blood — intrauterine infection
IgA (secretory)	10–15%	Local protection of the mucous membranes; abundant in breast milk (colostrum)	Almost absent at birth, increases slowly; adult levels at 10–12 years (secretory IgA — almost at 5–6 years)
IgD	≈0,2%	B-lymphocyte receptor	—
IgE	≈0,01%	Reagins — allergic (type I) reactions, against helminths	Hardly detectable in the newborn, adult levels at 12–13 years; its detection in the fetus — sensitization

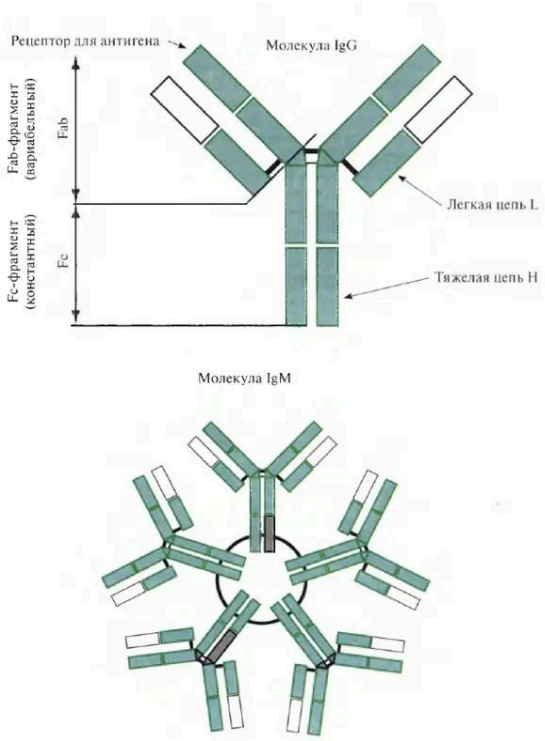


Figure 14.19. *Structure of an immunoglobulin (Navruzova, Geppe)*

The “critical periods” of immune system development

REMEMBER

Critical period 1 — the neonatal period (1st month): physiological depression of the immune system; passive protection by maternal IgG; low phagocytosis, complement and interferon — susceptibility to pyogenic and gram-negative flora and viruses, a tendency of infection to **generalize** (sepsis). The first crossover of the leukocyte differential on day 4–5. **Critical period 2 — 3–6 months:** breakdown of maternal antibodies; the primary immune response (IgM) predominates, **immunological memory is not formed** — whooping cough and measles run an atypical course, reinfection is possible; local immunity is insufficient; allergic reactions and food allergy appear. **Critical period 3 — the 2nd year:** active exploration of the world, increasing contacts; local immunity remains insufficient; the response to the polysaccharide antigens of pneumococcus and Haemophilus is weak; most primary immunodeficiencies become manifest; diseases of the ENT organs and airways recur. **Critical period 4 — 4–6 years:** the second crossover of the leukocyte differential; IgG and IgM reach adult levels, secretory IgA is still low; parasitic invasions, atopic and immune complex diseases, late immunodeficiencies. **Critical period 5 — puberty (12–16 years in boys, 11–15 years in girls):** involution of the thymus, the mass of the lymphoid organs decreases; the influence of sex hormones — autoimmune and lymphoproliferative diseases increase.

Hypersensitivity reactions (according to Gell and Coombs)

Table 14.16. *Types of hypersensitivity reactions*

Type	Mechanism	Time of development	Examples
I — anaphylactic (reaginic)	IgE binds to mast cells, histamine and mediators are released	5–15 minutes	Anaphylaxis, urticaria, bronchial asthma, allergic rhinitis
II — cytotoxic	IgG, IgM against antigens of the body's own cells + complement	Hours	Rh conflict, autoimmune hemolytic anemia, incompatible blood transfusion
III — immune complex	Antigen–IgG/IgM complexes are deposited in the tissues, complement	4–8 hours	Serum sickness, glomerulonephritis, SLE, the Arthus phenomenon, hemorrhagic vasculitis
IV — delayed (cell-mediated)	Sensitized T lymphocytes, cytokines	24–48 hours	Tuberculin (Mantoux) test, contact dermatitis, graft rejection

14.7. Examination of the immune system

History and inspection: frequent and severe infections, recurrences, opportunistic infections, reactions to vaccination (complications after BCG), allergic diseases, family history (death from infection at an early age); examination of the lymph nodes, tonsils and spleen (see topic 4); **absence or hypoplasia** of the tonsils and lymph nodes — a sign of humoral immunodeficiency.

Immunogram (laboratory methods):

- **Cell-mediated immunity:** the number of lymphocytes and T lymphocytes (at 7–14 years T cells make up **≈70%** of lymphocytes, ≈1400/μl), the **CD3, CD4, CD8** subpopulations and **the CD4/CD8 ratio** (normal 1.5–2), the blast transformation reaction with phytohemagglutinin, cytokines; **skin tests** (Mantoux); chest X-ray (thymic hypoplasia or thymomegaly).
- **Humoral immunity:** the number of B lymphocytes (CD19, CD20; ≈25%, ≈500/μl), **serum IgG, IgA, IgM, IgE**, secretory IgA (in saliva), circulating immune complexes, specific post-

vaccination antibodies.

- **Complement system:** total hemolytic activity (CH50), C3, C4, C1 inhibitor.
- **Phagocytosis:** the neutrophil count (**neutropenia** — absolute count <1500/μl), the phagocytic index and number, **the NBT test** (nitroblue tetrazolium — bactericidal activity), chemotaxis, adhesion molecules (CD11/CD18).

14.8. Semiotics of immune system involvement

3 types of immune system involvement: 1) quantitative or functional insufficiency of a component of the immune system (**primary and secondary immunodeficiencies**); 2) impaired recognition of self antigens (**autoimmune diseases**); 3) a hyperreactive immune response (**allergic reactions**).

The 10 warning signs of primary immunodeficiency

1. **6–8 or more** episodes of otitis a year.
2. **4–6** severe sinusitis episodes a year.
3. **More than 2** severe pneumonias a year.
4. Recurrent **deep abscesses** of the skin or internal organs.
5. **More than 2** severe infections (osteomyelitis, sepsis, meningitis).
6. **Persistent oral candidiasis** or fungal infection of the skin in a child over 1 year.
7. The need for **2 months or more** of antibiotic treatment (ineffective).
8. The need for **intravenous antibiotics** to clear an infection.
9. **Primary immunodeficiency** or deaths from infection at an early age in the family history.
10. **Failure to gain weight and growth retardation** in an infant.

CLINICAL SIGNIFICANCE

If there are two or more signs — an immunology consultation and an immunogram are needed. **Live vaccines** (BCG, oral poliomyelitis vaccine, measles) are **contraindicated** in cell-mediated and combined immunodeficiency — risk of generalized vaccine infection.

Table 14.17. *Clinical features of primary immunodeficiencies (Geppe)*

Affected component	Clinical signs	Examples
B lymphocytes (antibodies)	Recurrent purulent otitis, sinusitis, pneumonia and bronchiectasis caused by encapsulated bacteria (pneumococcus, Haemophilus, streptococcus); begins after 6 months (when maternal IgG is exhausted); lymphoid hypoplasia	Bruton agammaglobulinemia (X-linked, in boys, at 6–9 months), common variable immunodeficiency, selective IgA deficiency (the most common), hyper-IgM syndrome, transient hypogammaglobulinemia of infancy
T lymphocytes (cell-mediated)	Viruses, fungi (candidiasis), mycobacteria, Pneumocystis, protozoa; growth retardation, diarrhea, malabsorption; BCG-itis; risk of tumors	DiGeorge syndrome (22q11: thymic aplasia, hypoparathyroidism — hypocalcemic seizures, heart defects, facial anomalies), chronic mucocutaneous candidiasis
Combined (T + B)	Severe bacterial, viral and fungal infections from the 1st–3rd month of life, severe diarrhea, growth arrest	Severe combined immunodeficiency (SCID), adenosine deaminase deficiency, Wiskott–Aldrich syndrome (eczema + thrombocytopenia + infections, in boys), ataxia-telangiectasia (Louis-Bar)

Affected component	Clinical signs	Examples
Phagocytosis	Recurrent staphylococcal and fungal skin infections, subcutaneous abscesses, purulent lymphadenitis, liver and spleen abscesses, osteomyelitis; delayed separation of the umbilical stump (adhesion defect)	Chronic granulomatous disease, Chédiak–Higashi syndrome (partial albinism), leukocyte adhesion defect, hyper-IgE syndrome (Job), Shwachman–Diamond syndrome
Complement	Recurrent bacterial, especially meningococcal and gonococcal (C5–C9), infections; autoimmune diseases (C1–C4 — SLE); C1 inhibitor deficiency — hereditary angioedema	—

Secondary immunodeficiencies

As a result of a pathological process outside the lymphoid tissue: **malnutrition (dystrophy), avitaminoses**; viral infections (measles, rubella, chickenpox, EBV, **HIV — acquired immunodeficiency syndrome**); bacterial and fungal infections; malignant tumors; conditions with **protein loss** (**nephrotic syndrome, extensive burns, exudative enteropathy**); major surgery; **radiation, chemotherapy, glucocorticoid and cytostatic therapy**; splenectomy; severe physical strain and stress.

Allergic and autoimmune conditions

- **Allergic (atopic) diseases** — atopic dermatitis (in infancy), food allergy, allergic rhinitis, bronchial asthma (the “atopic march”); raised IgE in the blood, eosinophilia.
- **Autoimmune diseases** — rheumatic fever, juvenile idiopathic arthritis, SLE, autoimmune thyroiditis, type 1 diabetes mellitus; they increase in puberty.

14.9. Clinical section

Acute poststreptococcal glomerulonephritis

CLINICAL SIGNIFICANCE

1–3 weeks after tonsillitis, scarlet fever or pyoderma — the “**nephritic triad**”: **edema** (face and eyelids in the morning), **arterial hypertension** (headache, vomiting; in a severe case — **eclampsia**: seizures, loss of consciousness), **urine the color of “meat washings”** (macrohematuria) and oliguria. Laboratory: proteinuria 1–2 g/l, erythrocyte casts; in the blood **raised ASO**, **reduced complement C3** (recovers in 6–8 weeks), raised ESR. Treatment: bed rest, restriction of salt and fluid, penicillin, antihypertensive drugs; the main dangers are hypertensive encephalopathy, acute heart failure and acute kidney injury.

Urinary tract infection in a young child

CAUTION

In an infant with **unexplained fever (>38 °C)**, vomiting, poor appetite, failure to gain weight and restlessness, **urinalysis and urine culture** are mandatory (urinary tract infection is the most common bacterial cause of “occult” fever at an early age). After the first febrile pyelonephritis **renal ultrasound** is performed, and if it recurs — **voiding cystourethrography** (vesicoureteral reflux).

Practical skills

Skill 1. Detecting edema

1. Inspect the skin of the face, eyelids, limbs, lower back and abdomen; pay attention to clothing marks.
2. Press with a finger for 5–10 seconds on the anterior surface of the tibia; assess the pit.
3. Check for edema in the lumbosacral area and the abdominal wall (in a lying child).
4. Detect free fluid in the abdominal cavity.
5. If latent edema is suspected — weigh the child, calculate the diuresis and fluid balance, perform the McClure–Aldrich test.

Skill 2. Bimanual palpation of the kidneys (Obraztsov–Strazhesko)

1. The child lies on the back with the legs slightly bent; the doctor on the right; warm hands.
2. Place the left hand under the lower back, in the angle between rib XII and the long muscles.
3. Place the right hand outside the rectus muscle below the costal arch and plunge it deep during expiration.
4. Bring the hands together and try to feel the lower pole of the kidney; on deep inspiration the kidney slides under the fingers.
5. Palpate the left kidney in the same way; in older children repeat standing according to Botkin.
6. Assess size, consistency, surface, mobility, tenderness.

Skill 3. Pain points and Pasternatsky's sign

1. Palpate the upper and lower ureteral points with the child lying on the back, during expiration.
2. Palpate the costolumbar and costovertebral points from behind.
3. With the child sitting or standing, place the left palm on the lumbar region at the level of rib XII.
4. Give short, weak blows with the edge of the right palm; first on the healthy side, then on the affected side.
5. Assess the pain (negative, weakly positive, positive, sharply positive).

Skill 4. Percussion of the border of the bladder

1. The child lies on the back; determine the time of the last voiding.
2. Place the pleximeter finger parallel to the pubis at the level of the umbilicus.
3. Percuss quietly downward along the midline; mark the point where tympanitis changes to dullness.
4. Measure the distance from the pubic symphysis; repeat after voiding if necessary.

Skill 5. Interpreting urine tests and functional tests

1. Compare the urinalysis with the age norms: color, clarity, pH, specific gravity, protein, glucose, sediment.
2. Compare the Nechiporenko, Hamburger or Addis–Kakovsky results with the norm.
3. In the Zimnitsky test assess: the ratio of daily diuresis to fluid intake, day/night diuresis, the maximal specific gravity and the range of specific gravity.
4. Assess the GFR and tubular reabsorption by the Rehberg test or the Schwartz formula.

5. Identify the leading urinary syndrome and the level of involvement (glomerular, tubular, urinary tract).

Skill 6. Performing the McClure–Aldrich test

1. Wipe the lower third of the inner surface of the forearm with alcohol.
2. With a fine needle inject 0.2 ml of isotonic NaCl strictly intradermally (a “lemon peel” bleb).
3. Record the time and observe until the bleb disappears.
4. Compare the result with the age norm.

Skill 7. Assessing an immunogram

1. Determine the absolute numbers of leukocytes, lymphocytes and neutrophils.
2. Compare CD3, CD4, CD8, CD19 and the CD4/CD8 ratio with the age norm.
3. Compare the levels of IgG, IgA, IgM, IgE with the age norm.
4. Identify the affected component by comparing with the clinical signs (type of infections, age at onset).

Review questions

1. What are the anatomical features of the kidneys in children (mass, position, structure)?
2. What are the functional features of the kidneys and their clinical significance?
3. Name the age-related features of the ureters, bladder and urethra.
4. How is daily diuresis calculated? What is the number of voidings at different ages?
5. What complaints and signs on inspection occur in diseases of the urinary system?
6. How are the kidneys palpated? How are the pain points and Pasternatsky's sign tested?
7. What are the norms of the urinalysis and its features in children?
8. How are the Nechiporenko, Hamburger and Addis–Kakovsky tests performed and what are their norms?
9. What does the Zimnitsky test assess? What are hypo- and isosthenuria?
10. What are the essence and norms of the Rehberg test and the McClure–Aldrich test?
11. Name the instrumental methods of examining the urinary system.
12. Describe the disturbances of diuresis and of the act of voiding.
13. What is the diagnostic significance of proteinuria, hematuria, leukocyturia and cylindruria?
14. How are the nephritic and nephrotic syndromes distinguished? What are the signs of acute and chronic renal failure?
15. What are the age-related features of the immune system, its critical periods and the warning signs of primary immunodeficiency?

Test questions

1. The ratio of kidney mass to body mass in the newborn:
   A) 1:50
   B) 1:100
   C) 1:200
   D) 1:300
   E) 1:500

2. The approximate daily diuresis of a 6-year-old child:
   A) 600 ml
   B) 800 ml
   C) 1100 ml
   D) 1500 ml
   E) 2000 ml

3. The number of voidings per day in the newborn:
   A) 5–6
   B) 7–8
   C) 10–12
   D) 20–25
   E) 40–50

4. When does adult concentrating ability develop?
   A) At birth
   B) At 1 month
   C) At 9–12 months
   D) At 5 years
   E) At 15 years

5. The normal number of erythrocytes in 1 ml of urine in the Nechiporenko test:
   A) Up to 100
   B) Up to 1000
   C) Up to 4000
   D) Up to 10,000
   E) Up to 1,000,000

6. For how long is the urine collected in the Addis–Kakovsky test?
   A) 1 hour
   B) 3 hours
   C) 10 hours
   D) 24 hours
   E) A single midstream portion

7. Shows that the concentrating function is preserved in the Zimnitsky test:
   A) Specific gravity 1008–1010 in all portions
   B) Specific gravity 1018–1020 in at least one portion and a range of ≥10 units
   C) Predominance of night diuresis
   D) Specific gravity always below 1005
   E) Diuresis 30% of the fluid intake

8. The normal tubular reabsorption in the Rehberg test:
   A) 50–60%
   B) 70–80%
   C) 85–90%
   D) 97–99%
   E) 100%

9. The normal resorption time of the bleb in the McClure–Aldrich test in a child over 5 years:
   A) 5 minutes
   B) 10–15 minutes
   C) 20–25 minutes
   D) 40–60 minutes
   E) 2 hours

10. Not typical of nephrotic syndrome:
   A) Massive edema
   B) Proteinuria >3 g/day
   C) Hypoalbuminemia
   D) Hypercholesterolemia
   E) Marked macrohematuria and hypertension (in the pure form)

11. Leukocyturia, bacteriuria, fever and a positive Pasternatsky sign are typical of:
   A) Acute glomerulonephritis
   B) Pyelonephritis
   C) Nephrotic syndrome
   D) Renal glucosuria
   E) Diabetes insipidus

12. Isosthenuria is:
   A) Specific gravity >1030
   B) Monotonous specific gravity around 1008–1010
   C) Predominance of night diuresis
   D) Protein in the urine
   E) Absence of urine

13. The only immunoglobulin that crosses the placenta:
   A) IgA
   B) IgM
   C) IgG
   D) IgE
   E) IgD

14. The 2nd critical period of the immune system:
   A) From birth to 28 days
   B) 3–6 months
   C) The 2nd year
   D) 4–6 years
   E) Puberty

15. Recurrent meningococcal infections are typical of which deficiency?
   A) T lymphocytes
   B) IgE
   C) Complement components C5–C9
   D) Neutrophil myeloperoxidase
   E) Lysozyme

16. Approximately what proportion of all congenital defects are defects of the urinary organs?
    A) 1/5
    B) 1/10
    C) 1/2
    D) 1/3
    E) 2/3

17. How are the kidneys positioned in a child under 1 year, and when does this feature disappear?
    A) Only on the left
    B) Higher, at 1 year
    C) Always the same
    D) Lower, disappears by 7 years
    E) Lower, at 14 years

18. When does the lobulated structure of the kidney disappear in the newborn?
    A) At 2–5 years
    B) At 15 years
    C) At 10 years
    D) It does not disappear
    E) At 1 month

19. What are the filtration and tubular function of the newborn's kidney like?
    A) Only secretion is present
    B) Low; low concentrating function, active sodium reabsorption — predisposition to edema
    C) No filtration
    D) High; high concentrating function
    E) As in adults

20. When does the concentrating function of the kidney reach adult levels?
    A) At 7 years
    B) At 12 years
    C) At 9–12 months
    D) At 1 month
    E) At 3 years

21. The reason for the predisposition to vesicoureteral reflux:
    A) The kidney is high
    B) The ureter is short and narrow
    C) The urethra is short
    D) The bladder is small
    E) The intravesical part of the ureter is short, the muscle fibers at its orifice are weak

22. The bladder capacity in the newborn is approximately:
    A) 1000 ml
    B) 150–200 ml
    C) ≈50 ml
    D) ≈100 ml
    E) 300–400 ml

23. The reason for the risk of ascending infection in girls:
    A) The kidney is mobile
    B) The urethra is long
    C) The ureter is narrow
    D) The urethra is short and wide, close to the anus
    E) The bladder is large

24. The formula for daily diuresis up to 10 years (n — age):
    A) 100 × n
    B) 300 + 50n
    C) 500 + n ml
    D) 1000 + 100n
    E) 600 + 100 × (n − 1) ml

25. Complete absence of urine for a day in a newborn:
    A) Physiological
    B) Normal only in preterm babies
    C) Normal up to 2 days
    D) Pathology
    E) Normal

26. The approximate daily diuresis in a child under 6 months:
    A) 2000 ml
    B) 1000–1200 ml
    C) 1520–1670 ml
    D) 760–820 ml
    E) 300–500 ml

27. Where does edema in kidney disease usually begin?
    A) Arms
    B) Face and eyelids (facies nephritica)
    C) Abdomen
    D) Lower back
    E) Feet

28. By whose name is the method of bimanual palpation of the kidney called?
    A) Pasternatsky
    B) Obraztsov–Strazhesko
    C) Ortner
    D) Quincke
    E) Botkin–Erb

29. What is Pasternatsky's sign?
    A) Skin fold
    B) Heart sounds
    C) Abdominal palpation
    D) Pain on tapping the lumbar region or the appearance of blood in the urine
    E) An eye sign

30. What breath odor is there in uremia?
    A) Vinegar
    B) Bitter almonds
    C) Acetone
    D) Putrid
    E) Ammonia

31. The normal pH (reaction) in the urinalysis:
    A) 5,5–6,5
    B) 10–11
    C) 8,5–9,5
    D) 1–2
    E) 2,0–3,0

32. The reason for a reddish-brick color of the urine in the 1st week of life in the newborn:
    A) Abundance of uric acid (uric acid infarction)
    B) Infection
    C) Bilirubin
    D) Hemoglobin
    E) Blood

33. For glucose to appear in the urine, blood glucose must exceed the renal threshold. This threshold is:
    A) 20 mmol/l
    B) 3.3–5.5 mmol/l
    C) 8.9–10 mmol/l
    D) 15 mmol/l
    E) 2 mmol/l

34. The norm for erythrocytes in 1 ml of urine in the Nechiporenko test:
    A) Up to 1000
    B) Up to 100,000
    C) Up to 10,000
    D) 0
    E) Up to 5000

35. For how long is the urine collected in the Hamburger test?
    A) 12 hours
    B) 24 hours
    C) 3 hours
    D) 8 hours
    E) 1 hour

36. If the changes are only in the 1st portion in the three-glass test, the source is:
    A) Ureter
    B) Bladder
    C) Kidney
    D) External genitals or urethra
    E) Prostate only

37. Diagnostically significant bacteriuria (in free voiding):
    A) Any amount
    B) ≥1000 CFU/ml
    C) ≥100,000 CFU/ml
    D) ≥100 CFU/ml
    E) ≥10 CFU/ml

38. What does the Zimnitsky test assess?
    A) Blood flow
    B) Glomerular filtration
    C) Blood pressure
    D) Latent edema
    E) Excretory and concentrating function, circadian rhythm

39. What percentage of the fluid taken should be excreted in the Zimnitsky test?
    A) 100%
    B) 30%
    C) 70–80%
    D) 10–20%
    E) 150%

40. What is isosthenuria?
    A) Daytime diuresis is high
    B) Specific gravity >1030
    C) No voiding at night
    D) Specific gravity 1025
    E) Monotonous urine specific gravity ≈1008–1010

41. What does nocturia mean?
    A) Absence of urine
    B) Bloody urine
    C) High daytime diuresis
    D) Predominance of night diuresis
    E) Urinary incontinence

42. What does the Rehberg test assess?
    A) Urinary sediment
    B) Edema
    C) Concentration
    D) Glomerular filtration (endogenous creatinine clearance) and tubular reabsorption
    E) Bacteriuria

43. Normal tubular reabsorption:
    A) Not 100%, 40%
    B) 70%
    C) 20%
    D) 50%
    E) 97–99%

44. The normal resorption time of the bleb in the McClure–Aldrich test in a child under 1 year:
    A) 10–15 minutes
    B) 20–25 minutes
    C) 1 hour
    D) Immediately
    E) 40–60 minutes

45. The screening method for the urinary tract:
    A) Ultrasound
    B) Biopsy
    C) Angiography
    D) CT
    E) MRI

46. The method detecting vesicoureteral reflux:
    A) Voiding cystourethrography
    B) Spirography
    C) EGDS
    D) PCG
    E) ECG

47. What is hypospadias?
    A) Renal aplasia
    B) The external orifice on the lower surface of the penis
    C) Bladder exstrophy
    D) Urethral atresia
    E) The external orifice on the upper surface

48. What condition is meant by anuria?
    A) ≤50 ml of urine per day or no urine for 12 hours
    B) Urinary incontinence
    C) Voiding at night
    D) Painful voiding
    E) A 2-fold increase in diuresis

49. What is oliguria?
    A) Frequent voiding
    B) A decrease in diuresis to 1/3–1/4 of the age norm
    C) Change in urine color
    D) Blood in the urine
    E) A 2-fold increase in diuresis

50. Causes of polyuria:
    A) Profuse diarrhea
    B) Onset of glomerulonephritis
    C) Cardiac edema
    D) Diabetes mellitus and insipidus, the polyuric phase of renal failure, diuretics
    E) Heavy sweating

51. What is ischuria?
    A) Continuous leakage of urine
    B) Acute retention of urine (the bladder is full but does not empty)
    C) Bloody urine
    D) Absence of urine (from the kidney)
    E) Voiding at night

52. What is strangury?
    A) Urine color
    B) Painful, burning voiding
    C) Frequent voiding
    D) Enuresis
    E) Little urine

53. After what age is enuresis considered pathological?
    A) From 1 year
    B) From 3 years
    C) Only after 10 years
    D) From 2 years
    E) After 5 years

54. What does urine the color of “meat washings” indicate?
    A) Pyuria
    B) Macrohematuria (glomerulonephritis)
    C) Chyluria
    D) Only hemoglobinuria
    E) Bilirubinuria

55. What is “beer-colored” urine typical of?
    A) Phosphaturia
    B) Hematuria
    C) Diabetes
    D) Uraturia
    E) Bilirubinuria (jaundice)

56. The amount of nephrotic proteinuria:
    A) <0.033 g/l
    B) >2.5–3 g/day or >50 mg/kg/day
    C) 1 g/day
    D) 0.2 g/day
    E) <1 g/day

57. A sign typical of glomerular hematuria:
    A) Tumor
    B) Pain (colic)
    C) Dysmorphic erythrocytes and erythrocyte casts
    D) Urinary stone
    E) Isomorphic erythrocytes

58. Which type of cast is always a sign of renal pathology (except for single hyaline casts)?
    A) Only epithelial
    B) Only hyaline
    C) All casts (granular, waxy, erythrocyte, leukocyte)
    D) Casts are not pathological
    E) Only fatty

59. The main signs of the nephritic syndrome:
    A) Massive edema, protein >3 g/day, hyperlipidemia
    B) Hematuria, arterial hypertension, moderate proteinuria and edema, oliguria
    C) Polyuria
    D) Anemia
    E) Dysuria and fever

60. The laboratory signs of the nephrotic syndrome:
    A) Hypoalbuminemia (<25 g/l), hypercholesterolemia, marked proteinuria
    B) Hematuria and hypertension
    C) Glucosuria
    D) Leukocyturia
    E) Bacteriuria

61. How does renal edema differ from cardiac edema?
    A) In the morning on the face, soft, warm, pale; cardiac edema in the evening on the legs, firm, with cyanosis
    B) No difference
    C) Only on the arms
    D) Only on the abdomen
    E) In the evening on the legs, firm

62. The character of renal colic:
    A) Headache
    B) Heart pain
    C) Sharp paroxysmal pain in the lower back, radiating along the ureter to the groin
    D) Dull constant abdominal pain
    E) Leg pain

Clinical cases

Case 1

An 8-year-old boy. He had tonsillitis 2 weeks ago. For 3 days his face and eyelids have been swollen in the morning, with headache, scanty urine the color of “meat washings”. On examination: pale skin, edema of the face and eyelids, pastosity of the shins. BP 140/95 mm Hg. Pasternatsky's sign weakly positive on both sides. Daily diuresis 400 ml. Urinalysis: protein 1.5 g/l, erythrocytes cover the whole field of view (altered), erythrocyte casts 2–3 per field of view, leukocytes 4–5.

Questions: 1) The leading syndrome? 2) Presumptive diagnosis? 3) What is the normal diuresis for this child? 4) Which additional investigations are needed?

Answer: 1) Acute nephritic syndrome: edema, arterial hypertension, macrohematuria (“meat washings”), moderate proteinuria, erythrocyte casts, oliguria. 2) Acute poststreptococcal glomerulonephritis (2 weeks after tonsillitis). 3) 600 + 100 × (8 – 1) = 1300 ml; 400 ml is less than

1/3 of the norm — oliguria. 4) Blood ASO, complement C3 (reduced), creatinine, urea, potassium, total protein, albumin, cholesterol; the Nechiporenko, Zimnitsky and Rehberg (GFR) tests; a throat swab (β-hemolytic streptococcus); renal ultrasound; ECG, fundoscopy.

Case 2

A 4-year-old girl. Over 1 week edema gradually increased: first in the eyelids, then in the legs, and the abdomen enlarged. On examination: anasarca, free fluid in the abdomen, pale “ivory”-colored skin. BP 90/60 mm Hg. Urine: protein 6.6 g/l, erythrocytes 0–1, hyaline casts. Blood: total protein 42 g/l, albumin 18 g/l, cholesterol 11.5 mmol/l. Daily proteinuria 2.8 g (weight 16 kg).

Questions: 1) The leading syndrome? Assess the degree of proteinuria. 2) Determine the form. 3) The mechanism of the edema? 4) How can latent edema be detected?

Answer: 1) Nephrotic syndrome: massive edema, proteinuria 2.8 g / 16 kg = 175 mg/kg/day (>50 mg/kg — nephrotic level), hypoproteinemia, hypoalbuminemia (18 g/l), hypercholesterolemia. 2) Complete, pure nephrotic syndrome (no hematuria or hypertension); by age and clinical picture probably minimal change (steroid-sensitive) idiopathic nephrotic syndrome. 3) Massive proteinuria → hypoalbuminemia → reduced plasma oncotic pressure → fluid moves into the interstitium → hypovolemia → activation of the renin-angiotensin-aldosterone system and ADH → retention of Na and water. 4) Daily weighing, calculating the balance of daily diuresis and fluid intake, the McClure–Aldrich test (the bleb resorbs quickly).

Case 3

A 1-year-old boy. Since 6 months he has had purulent otitis 4 times, pneumonia 2 times and purulent (pneumococcal) meningitis 1 time; the effect of antibiotics is short-lived. The mother's brother died of an infection in childhood. On examination: weight deficit, the tonsils and peripheral lymph nodes are hardly detectable. Blood: IgG 0.5 g/l, IgA and IgM not detected; B lymphocytes (CD19) <1%; T lymphocytes normal.

Questions: 1) Which immune component is affected? 2) Presumptive diagnosis? 3) Why did the disease begin after 6 months? 4) Principle of treatment and vaccination precautions?

Answer: 1) Humoral immunity (B lymphocytes and antibodies) — recurrent purulent infections caused by encapsulated bacteria, lymphoid hypoplasia, a sharp decrease in all Ig classes, absence of B cells, normal T cells. 2) X-linked agammaglobulinemia (Bruton disease) — a boy, a similar condition in the mother's brother (on the maternal side). 3) In the first months of life the child is protected by maternal IgG that crossed the placenta; it is broken down by 3–6 months, and the child's own antibodies are not produced. 4) Lifelong immunoglobulin replacement therapy (intravenous or subcutaneous, keeping IgG at a normal level), active antibiotic treatment of infections; the live oral poliomyelitis vaccine is contraindicated; refer the family for genetic counseling.

Quick review

Table 14.18. *Key facts on Topic 14*

Question	Answer
Kidney mass	1:100 in the newborn, 1:200 in adults
Position of the kidney	Low up to 1 year (lower pole below LIV); as in adults at 7 years
Number of nephrons	≈2 million at birth (as in adults)
Concentrating ability	Close to adult at 9–12 months

Question	Answer
Daily diuresis	600 + 100 × (n – 1)
Number of voidings	Newborn 20–25; 1 year 15; 3 years 10; school age 6–7
Bladder capacity	Newborn 50 ml; 1 year 100 ml
Normal protein	Up to 0.033 g/l
Nechiporenko	L <2000 (<4000 in girls), E <1000 per 1 ml
Addis–Kakovsky	L <2 million, E <1 million, casts <50,000 per day
Hamburger	3 hours; L <2000 (girls), E <1000 per minute
Zimnitsky	8 portions every 3 hours; max. specific gravity ≥1018–1020; range ≥10; daytime diuresis 2/3
Rehberg	Creatinine clearance; reabsorption 97–99%
McClure–Aldrich	0.2 ml NaCl; <1 year 10–15, 1–5 years 20–25, >5 years 40–60 min
Significant bacteriuria	$\geq 10^5$/ml (free voiding)
Oliguria / anuria	<1/3 of the norm / no urine for 12 hours
Nephritic syndrome	Edema + hypertension + hematuria
Nephrotic syndrome	Edema + proteinuria >50 mg/kg + hypoalbuminemia + hyperlipidemia
Pasternatsky's sign	Pyelonephritis, paranephritis, urolithiasis
IgG	Crosses the placenta; maternal IgG is exhausted by 3–6 months
IgM	Raised in cord blood — intrauterine infection
Secretory IgA	In breast milk; increases slowly
Critical periods	Neonatal period, 3–6 months, 2nd year, 4–6 years, puberty
Humoral deficiency	Encapsulated bacteria, otitis, sinusitis, pneumonia (Bruton)
Cell-mediated deficiency	Viruses, fungi, BCG-itis (DiGeorge)
Complement C5–C9	Meningococcal infections
The newborn's kidney	Mass 1:100 (1:200 in adults); lies lower (until 7 years); lobulated (until 2–5 years); nephron differentiation until 2 years; low filtration and concentration
Functional features of the kidney	The filtering surface is 5 times smaller; active Na reabsorption — edema; concentration at 9–12 months; immature acid-base regulation — acidosis; optimal function at 10–11 years
The urinary tract	Ureter long, wide, hypotonic; predisposition to vesicoureteral reflux; in girls the urethra is short and wide; bladder capacity: newborn 50, 1 year 100, 5–9 years 150–200, 12–14 years 300–400 ml
Diuresis	Formula: 600 + 100×(n−1) ml; day:night = 2:1–3:1; up to 6 months 300–500 ml; 1–3 years 760–820; 11–13 years 1600–1900 ml
Kidney palpation and pain points	Obraztsov–Strazhesko (bimanual), Botkin (standing); ureteral points (upper/lower), costolumbar, costovertebral points; Pasternatsky's sign
Signs on inspection in kidney disease	Pallor; facies nephritica (facial edema); anasarca; ascites (frog belly); in uremia ammonia odor, itching, ecchymoses
Toilet skills	Training from 5–6 months; asks for the potty by the end of 1 year; stable at 2 years; voiding in sleep possible up to 5 years

Question	Answer
Normal urinalysis	Straw-yellow, clear, pH 5.5–6.5, specific gravity 1008–1025, protein absent/≤0.033 g/l, no glucose, erythrocytes 0–2, leukocytes 0–5 in boys, 5–10 in girls
Nechiporenko / Hamburger / Addis	Nechiporenko — 1 ml (leuk. 2000–4000, eryth. 1000); Hamburger — 3 hours; Addis–Kakovsky — 24 hours (leuk. ≤2 million, eryth. ≤1 million)
Three-glass test	1st portion — urethra/external organs; 3rd portion — bladder; all portions — kidney/ureter/bladder
Criteria of bacteriuria	In free voiding ≥10^5 CFU/ml; catheter ≥10^4; suprapubic puncture — any; <1 year E. coli ≥5×10^4
Zimnitsky test	8 portions (every 3 hours); excretion 70–80%; specific gravity 1008–1025, range ≥10; daytime diuresis 2/3–3/4; nocturia — renal failure; hypo-/iso-/hypersthenuria
Rehberg test	Endogenous creatinine clearance (GFR); reabsorption 97–99%; Schwartz: GFR = k × height / creatinine; GFR in the newborn 30–50, in children ≈121 ml/min/1.73 m^2
Instrumental methods	Ultrasound (screening), excretory urography, voiding cystourethrography, renography/nephroscintigraphy, angiography, CT/MRI, cystoscopy, needle biopsy
Voiding disturbances	Polyuria, oliguria, anuria (≤50 ml/day), ischuria (acute retention), nocturia, pollakiuria, strangury, enuresis (>5 years)
Urine color	Red — hematuria; orange — uraturia; black — hemoglobinuria; beer-colored — bilirubinuria; white — chyluria, phosphaturia; colorless — diabetes, renal failure
Urinary syndrome	Proteinuria (>150–200 mg/day), hematuria, leukocyturia, cylindruria, bacteriuria, crystalluria, glucosuria, ketonuria
Types of edema	Renal: in the morning on the face, soft, warm, pale. Cardiac: in the evening on the legs, firm, cyanosis. The McClure–Aldrich test detects latent edema
Developmental anomalies	Kidney: agenesis, horseshoe, dystopia, polycystic disease, hydronephrosis; ureters: duplication, ureterocele; urethra: hypospadias, epispadias, posterior urethral valves

TOPIC 15

Feeding the healthy child: breastfeeding, mixed and artificial feeding, complementary foods. Defense of the case history

Learning objectives

The student should know:

- the types of feeding and their definitions;
- the composition of breast milk, the features of colostrum and transitional milk, the advantages of breastfeeding;
- the physiology of lactation, the rules of early breastfeeding and successful breastfeeding, contraindications to breastfeeding;
- the causes and degrees of hypogalactia and measures to overcome it;
- the principles of mixed and artificial feeding, adapted and therapeutic formulas;
- the timing, rules and sequence of introducing complementary foods;
- the methods of calculating food volume in children under 1 year, the principles of feeding children over 1 year;
- the structure of the case history and the procedure for defending it.

The student should be able to:

- assess the breastfeeding technique (position, correct attachment) and explain it to the mother;
- calculate the daily and single feed volume for a child under 1 year;
- draw up an age-appropriate feeding regimen and a plan for introducing complementary foods;
- take a feeding history;
- write and defend a case history.

15.1. Types of feeding

Rational feeding is the basis of the child's physical and mental development, the formation of immune protection and lifelong health. In the 1st year the digestive organs are not yet ready to digest some food ingredients (see topic 12).

Table 15.1. *Types of feeding*

Type	Definition
Natural (breastfeeding)	Breast milk only (breast milk remains the main food even after complementary foods are introduced); exclusive — nothing other than breast milk, not even water, is given
Mixed	When breast milk is insufficient (hypogalactia), supplementary feeding with an adapted formula; breast milk is more than 1/5 of the daily volume (more than 150–200 ml per day)
Artificial	Milk formulas only, or breast milk less than 1/5 of the daily volume

15.2. Breastfeeding

Physiology of lactation

- **The prolactin reflex (milk production reflex):** when the baby sucks, the nerve endings in the nipple are stimulated → **prolactin** is released from the anterior pituitary → milk is produced in

the alveoli. Prolactin is produced more **at night** — **night feeding** maintains lactation; frequent feeding increases milk.

- **The oxytocin reflex (milk ejection reflex):** under the influence of sucking, seeing the baby and hearing its voice, **oxytocin** is released from the posterior pituitary → the myoepithelial cells around the alveoli contract → milk flows; oxytocin also accelerates **involution of the uterus**. Pain, stress and anxiety **inhibit** this reflex.

Composition of breast milk

- **Colostrum** — at the end of pregnancy and in **the first 3–5 days** after delivery; yellowish, thick; rich in **protein** (mainly immunoglobulins — **secretory IgA**), leukocytes, vitamin A, carotene and trace elements; high in calories — even a small amount meets the newborn's needs; **"the first vaccination"**, it speeds up the passage of meconium (mild laxative effect).
- **Transitional milk** — **from day 4–5 to week 2–3**; protein decreases, fat and lactose increase.
- **Mature milk** — **from week 2–3**. Within one feed **"foremilk"** (watery, rich in lactose and water — quenches thirst) and **"hindmilk"** (rich in fat, high in calories — satisfies hunger) are distinguished; therefore the baby should **fully empty** one breast.

Table 15.2. *Composition of colostrum, transitional and mature breast milk and cow's milk (Kildiyarova)*

Type of milk	Proteins, g/l	Fats, g/l	Carbohydrates, g/l	Calories, kcal/l
Colostrum	80–110	28–41	40–53	1500
Transitional milk (from day 4–5)	14–23	29–44	57–66	750–800
Mature breast milk (from week 2–3)	9–13	39–45	68–72	700
Cow's milk	28–32	32–35	35–48	590

MEMORY AID

Breast milk and cow's milk: **"Protein 1:3, lactose 2:1"** — cow's milk contains **3 times more** protein (mainly the poorly digestible **casein**) and half as much lactose; in breast milk **whey proteins (albumin)** predominate, in cow's milk **casein**.

Advantages of breastfeeding

- **Proteins** are finely dispersed (albumin, α-lactalbumin) and easily digested; amino acids (taurine, cystine) are in the optimal ratio for CNS development; there is no antigenic property — **low risk of allergy**.
- **Fats** — many **unsaturated fatty acids** (omega-3, omega-6, long-chain — arachidonic and docosahexaenoic acids) — formation of the brain, retina and cell membranes; **breast milk lipase** breaks down fats already in the stomach.
- **Carbohydrates** — mainly **β-lactose** and **oligosaccharides (bifidus factor)** — promote the growth of **bifidoflora** and improve calcium absorption.
- **Enzymes** (amylase, lipase, proteases) — facilitate digestion.
- **Trace elements** (iron, zinc, calcium) with **high bioavailability** (iron bound to lactoferrin is ≈50% absorbed); an optimal Ca:P ratio (2:1).
- **Protective factors:** secretory IgA, IgG, IgM, **lactoferrin**, **lysozyme**, interferon, complement, macrophages, lymphocytes, neutrophils, bifidus factor — they reduce the risk of intestinal and respiratory infections, otitis, necrotizing enterocolitis, and later of obesity, diabetes mellitus and allergic diseases.

- The milk is **sterile, warm and ready**, and its composition adapts to the baby's needs; it is economically cheap.
- **A psychoemotional bond between mother and baby** forms; in the mother the risk of postpartum bleeding, breast and ovarian cancer and anemia decreases; **lactational amenorrhea** (natural contraception).

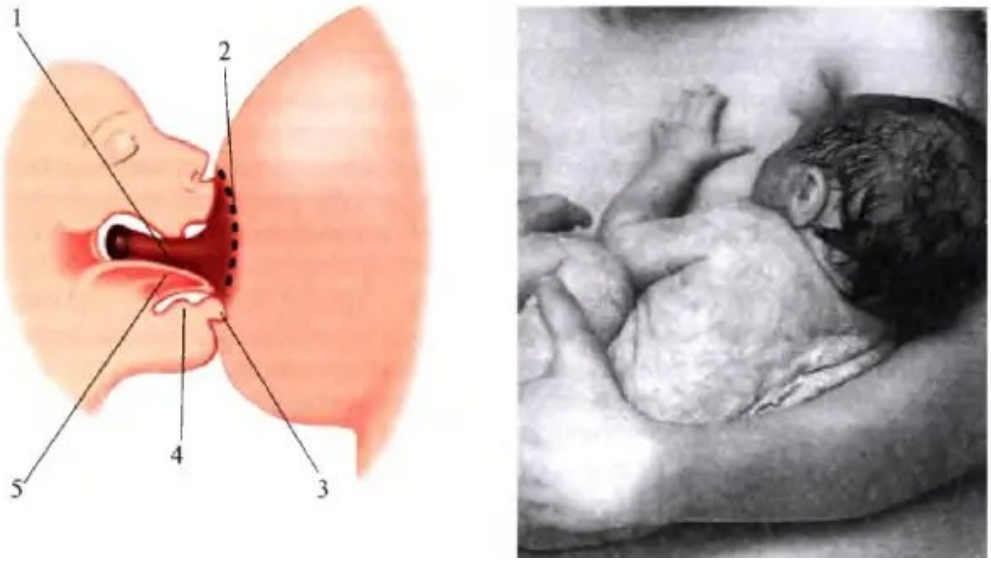


Figure 15.1. *Correct attachment to the breast: 1 — nipple; 2 — areola; 3 — lips; 4 — gums; 5 — tongue (Navruzova, Geppe)*

Rules of early breastfeeding and successful breastfeeding

Putting the baby to the breast **in the delivery room, within the first 30 minutes after delivery** (if there are no contraindications) and **"skin-to-skin" contact**:

- stimulates the rapid onset and long duration of lactation;
- ensures colonization of the skin and intestine with the mother's beneficial microflora; colostrum — immunological protection;
- accelerates **involution of the uterus** and reduces postpartum bleeding;
- prevents purulent-inflammatory diseases in mother and baby;
- quickly establishes the psychoemotional bond.

REMEMBER

The WHO/UNICEF "10 Steps to Successful Breastfeeding" (the main ones): train health workers; inform pregnant women about the benefits of breastfeeding; put the baby to the breast **within the first 30 minutes** after delivery; teach mothers to maintain lactation even when separated from their baby; give the newborn **no food or drink other than breast milk** (unless medically indicated); keep mother and baby **together day and night**; breastfeed **on demand**; give breastfed babies **no teats or pacifiers**; establish breastfeeding support groups.

Breastfeeding technique — signs of a correct position: the baby's head and body are in one line, the face is turned to the breast, the body is close to the mother, and the mother supports the baby's whole body. **Signs of correct attachment:** the mouth is wide open, **the lower lip is turned outward**, **the chin touches the breast**, more of the areola is visible above than below; the baby sucks slowly and deeply, swallowing is heard, and the mother feels no pain.

Feeding regimen

Today **“free feeding” — breastfeeding on demand** is recommended (not chaotic): 8–12 (≈10) times a day in the 1st month, including night feeds; later the baby settles into a regimen.

Table 15.3. *Breastfeeding regimen for children under 1 year (Kildiyarova)*

Age	Number of feeds	Interval, hours	Night break, hours
0–1 month	≈8–12	Free, on demand	—
1–2 months	7	3	6
Before complementary foods	6	3,5	6,5
After complementary foods	5	4	8

A feed lasts **15–20 minutes** (≈50% of the milk is taken in the first 5 minutes). The WHO recommendation: **exclusive breastfeeding up to 6 months**, then continued breastfeeding together with complementary foods **up to 2 years and beyond**.

Signs that the baby is getting enough milk: voiding **more than 6 times a day** (pale urine), regular stools, calm after feeding, **good weight gain** (≥125 g per week, ≥500 g per month in the 1st half-year). “Test weighing” (before and after a feed) is an additional method.

Contraindications and difficulties in breastfeeding

Table 15.4. *Contraindications to breastfeeding (Kildiyarova)*

On the mother's side	On the baby's side
Chronic diseases of the heart, kidneys, liver and lungs in the stage of decompensation	Extreme prematurity (no sucking and swallowing reflexes) — expressed milk is given by tube
Active tuberculosis (bacillary)	Severe cerebral circulatory disturbance, Apgar score <7 (asphyxia, birth trauma, seizures)
Taking cytostatics, some antibiotics (chloramphenicol, tetracycline), radioactive preparations	Defects of the jaws and palate (fed with expressed milk)
Dangerous infections (typhoid fever, cholera, etc.)	Severe developmental defects (gastrointestinal, cardiac)
Acute mental illnesses	Galactosemia, phenylketonuria (special formulas)
HIV infection (when a safe substitute is available)	Hemolytic disease of the newborn (according to some authors, the first 7–10 days)

Difficulties on the mother's side: incorrect nipple shape (flat, inverted), **cracked nipples**, **lactostasis**, **mastitis** (breastfeeding from the healthy breast continues), hypogalactia. **On the baby's side:** rhinitis, stomatitis, candidiasis, cleft palate, short frenulum of the tongue, prematurity.

> **CLINICAL SIGNIFICANCE**
> **Acute respiratory infection or tonsillitis** in the mother is not a contraindication to breastfeeding — the mother wears a mask while feeding (the milk is a source of antibodies for the baby). **Hepatitis B** — breastfeeding is possible if the newborn has been vaccinated. **In mastitis** breastfeeding is not stopped (in purulent mastitis the milk from the affected breast is temporarily expressed and discarded).

Hypogalactia

Hypogalactia — reduced secretory activity of the mammary glands during lactation. **Primary (true)** — hormonal disorders (deficiency of prolactin, oxytocin), endocrine diseases of the mother — rare (≈5%). **Secondary** — the majority: complications of pregnancy and delivery, maternal diseases, **stress**, unwillingness to breastfeed, **violation of the rules of breastfeeding** (late and infrequent breastfeeding, feeding on a strict schedule, a night break, pacifiers and teats, incorrect technique), inadequate nutrition and rest of the mother.

- **Early** (within 10–14 days after delivery) and **late** (after 2–3 weeks).
- **Degrees:** I — milk deficit <25%; II — 26–50%; III — 51–75%; IV — >75%.
- **Lactation crises** — temporary unexplained decreases in milk (at 3–6 weeks, at 3, 4, 7, 8 months), lasting 3–4 days.

Measures: breastfeeding **frequently, on demand**, from **both breasts** at each feed (keeping night feeds); daytime rest and sleep for the mother, complete and varied nutrition, a warm drink and lactogenic teas 15–20 minutes before feeding; a positive emotional climate in the family; breast massage and a warm shower before feeding. Only when all possibilities have been exhausted, in **grade III–IV** hypogalactia, is **supplementary feeding** with an adapted formula (by spoon, cup or syringe — without a teat) prescribed.

15.3. Mixed and artificial feeding

Reasons for transfer to artificial feeding

Grade III–IV hypogalactia or agalactia; contraindications to breastfeeding; absence of the mother; the mother's own choice. **Artificial feeding is a metabolic stress**: protein load, changes in the intestinal microflora, increased risk of infection and allergy.

> **CAUTION**
> **Whole cow's milk, goat's milk and kefir** are not given to a child under 1 year (according to Kildiyarova at least **until 8 months**, preferably until 2 years): much protein and salts — load on the kidneys; little iron and **diapedetic bleeding** from the intestinal mucosa — iron deficiency anemia; no vitamins or protective factors; risk of allergy. In artificial feeding only **adapted formulas** are used.

Breast milk substitutes

Principles of adaptation: reducing the protein content **by 2.5 times**, bringing the whey protein to casein ratio closer to that of breast milk (60:40), adding taurine and nucleotides; optimizing the fat composition with **unsaturated fatty acids** (vegetable oils, ARA, DHA); increasing carbohydrates **1.5–2 times** (lactose, prebiotics — galacto- and fructo-oligosaccharides); fortification with vitamins (12), iron, iodine, zinc, copper and selenium; reducing the content of Ca, K and Na.

- **Starter (formula 1)** — 0–6 months (highly adapted);
- **follow-on (formula 2)** — 6–12 months (more protein and calories);
- **formula 3** — over 1 year;
- **fermented adapted formulas** — from 3–4 weeks, in a 1:1 or 2:1 ratio with a sweet formula (probiotic effect, in dysbiosis, atopic dermatitis, intestinal diseases);
- **therapeutic (special) formulas**.

Table 15.5. *Therapeutic (special) formulas (Geppe)*

State	Formula
Preterm and low-birth-weight newborns	Special formulas for preterm babies (more protein, energy, vitamins and minerals)
Lactase deficiency	Low-lactose or lactose-free formulas
Cow's milk protein allergy	Formulas based on fully (extensively) hydrolyzed protein, amino acid formulas; soy protein-based (after 6 months)
Children at risk of allergy (prevention)	Partially hydrolyzed (“HA”) formulas
Regurgitation and vomiting	Thickened (containing starch or locust bean gum) anti-reflux formulas
Malabsorption, diseases of the pancreas, liver and biliary tract	Formulas enriched with medium-chain triglycerides (do not require emulsification)
Phenylketonuria, galactosemia	Phenylalanine-free, galactose-free formulas

Rules of artificial feeding

- Feed with one adapted formula; **do not change the formula often**; switch gradually to a new formula over **5–7 days**.
- The formula is prepared **fresh before each feed** according to the instructions on the package, at a temperature of **36–37 °C**; the teat hole is small (milk should drip when the bottle is turned over); the utensils are sterile.
- The feeding interval is often **3–3.5 hours** (formula leaves the stomach more slowly), 6–7 times a day, with a 6-hour night break; overfeeding is avoided.
- Complementary foods are also introduced to formula-fed children **at 4–6 months** (they may be introduced earlier than in breastfed children).

Rules of mixed feeding

- First **breastfeed from both breasts**, and only after the breasts are fully emptied give supplementary feeding with formula (**top-up**) — to maintain lactation;
- the top-up is given with **a spoon, cup** or a teat with a small hole (a teat weakens the baby's sucking — “nipple confusion”);
- the volume of the top-up is determined by **test weighing** or by calculating the food volume;
- measures to increase breast milk are continued.

15.4. Complementary foods

Complementary foods — new foods of higher concentration and density introduced into the diet in addition to breast milk (formula): they meet the growing body's increasing needs for protein, energy, iron, zinc and vitamins, develop the chewing apparatus and digestive system, and prepare the child for the family table.

Timing of introduction: according to the recommendations of WHO and the Ministry of Health of Uzbekistan — **from 6 months**; according to the Russian national program (Kildiyarova, Geppe) — individually, **at 4–6 months** (not before 4 months and not later than 6 months). **Signs of readiness:** holds the head independently, sits with support, the tongue-thrust reflex (pushing out the spoon) fades, shows interest in food, the weight has doubled since birth.

Table 15.6. *Consequences of untimely introduction of complementary foods (Kildiyarova)*

Early introduction (before 4 months)	Late introduction (after 6–7 months)
Intestinal dysfunction (regurgitation, vomiting, diarrhea, constipation, dysbiosis) — immaturity of the digestive organs	Delayed skills of eating thick and solid food, refusal to chew and swallow
Food allergy	Delayed development of taste sensations, habituation to uniform liquid food
Risk of intestinal and respiratory infections	Disturbed growth and development, deficiency of iron, zinc, copper, calcium and vitamin D
Decreased lactation (less sucking)	Delayed maturation of the digestive system, constipation
Food aspiration	—

Rules for introducing complementary foods

- Introduced only to a **healthy** child (no new food is introduced during illness or preventive vaccination);
- each new food is started in a **small amount** (½–1 teaspoon) and increased to the required volume over **5–7 days**;
- a new food is given **in the first half of the day** (to monitor tolerance);
- move from **single-component** foods to multi-component ones; only **one** new food at a time;
- food is **homogeneous** at first, then gradually thicker; given **from a spoon**;
- complementary food is given **before breastfeeding**, then the breast is offered;
- a new food is introduced only after the previous one is fully accepted;
- until 1 year, **no salt or sugar is added** to food, and **honey and whole cow's milk** are not given.

Choosing the first complementary food: for children with adequate weight and a tendency to constipation — **vegetable puree** (cauliflower, zucchini, broccoli, pumpkin — soft fiber); for children with low weight and frequent stools — **cereal (porridge)** (first **gluten-free**: rice, buckwheat, corn; gluten-containing — semolina, oats, wheat — later).

Table 15.7. *Timing and volume of complementary food introduction (Kildiyarova, Geppe)*

Food	Age of introduction, months	Volume (4–6 / 7 / 8 / 9–12 months)
Vegetable puree	4–6	10–150 / 170 / 180 / 200 g
Milk porridge (gluten-free first)	4–6	10–150 / 150 / 180 / 200 g
Fruit puree	4–6 (after vegetables and porridge)	5–50 / 5–30 / 80 / 90–100 g
Fruit juice	4–6 (Geppe); in most recommendations — closer to 1 year	5–50 / 5–20 / 80 / 90–100 ml
Meat puree	from 6 (6–7)	5–30 / 30 / 50 / 60–70 g
Cottage cheese	6–7	10–20 / 20–40 / 10–30 / 50 g
Egg yolk (hard-boiled)	7	— / ¼ / ½ / ½ pc
Fish puree	8–9 (1–2 times a week, instead of meat)	— / — / 5–30 / 30–60 g
Kefir, children's yogurt (adapted)	8	— / — / 200 / 200 ml

Food	Age of introduction, months	Volume (4–6 / 7 / 8 / 9–12 months)
Rusks, biscuits	7	— / 3–5 / 5 / 10–15 g
Wheat bread	8	— / — / 5 / 10 g
Vegetable oil	4–6 (added to vegetable puree)	1–3 / 5 / 5 / 6 g
Butter	4–6 (added to porridge)	1–4 / 4 / 5 / 6 g

MEMORY AID
Sequence of complementary foods: **“Vegetables — Porridge — Meat — Cottage cheese — Egg — Fish — Kefir”**. Meat volume: 5–30 g at 6 months, 50 g at 8–9 months (meatballs), 60–70 g at 11–12 months (steamed cutlet). Meat puree should not be introduced **before 6 months**.



Figure 15.2. *Spoon-feeding complementary food (Navruzova, Geppe)*

15.5. Calculating feeding volume in children under 1 year

In the first 10 days of life

Table 15.8. *Methods of calculating feeding volume in the first 10 days (Kildiyarova)*

Method	Formula
Finkelstein	V daily = 70 × n (birth weight <3200 g); V daily = 80 × n (weight >3200 g); n — day of life
Zaitseva	V daily = 2% × M × n; M — birth weight (g), n — day of life
Filatov (single-feed volume)	V single = 10 × n (ml)

From 10 days to 1 year

Table 15.9. *Volumetric (weight-based) method (Kildiyarova)*

Age	Daily volume (fraction of body weight)	Single volume, ml
10 days – 2 months	1/5	100–150
2–4 months	1/6	130–165
4–6 months	1/7	180–200
6–9 months (and up to 1 year)	1/8 (but not more than 1000–1100 ml)	200

Caloric method: in the 1st half-year **115 kcal/kg/day**, in the 2nd half-year **110 kcal/kg/day**; daily volume = (energy requirement × 1000) / caloric content of breast milk (formula) (≈700 kcal/l).

Single-feed volume = daily volume / number of feeds.

Table 15.10. *Daily requirements for basic nutrients and energy in children under 1 year (Kildiyarova)*

Age	Proteins, g/kg	Fats, g/kg	Carbohydrates, g/kg	Energy, kcal/kg
Until 4 months	2,2	6,5	13	115
4–6 months	2,6	6,0	13	115
7–12 months	2,9	5,5	13	110

Protein : fat : carbohydrate ratio — in the 1st half-year ≈1:3:6, in the 2nd half-year ≈1:2:4; after 1 year **1:1:4**.

> **CLINICAL SIGNIFICANCE**
> **Calculation example:** a 3-month-old child weighing 6000 g, fed 6 times a day. Volumetric method: 6000 / 6 = **1000 ml/day**; single-feed volume = 1000 / 6 ≈ **165 ml**. Caloric method: 115 × 6 = 690 kcal; 690 × 1000 / 700 ≈ **985 ml/day**.

15.6. Feeding children over 1 year

Features: stomach capacity increases, saliva and digestive juices increase, the chewing apparatus develops, and the number of teeth grows. **At 1–1.5 years** food is minced (ground, mushy), with a gradual transition to thicker food and pieces; the child is first spoon-fed, then taught **to eat independently**.

- **Number of meals:** up to 1.5 years **4–5 times** a day, then **4 times** (breakfast, lunch, afternoon snack, dinner); intervals of 3.5–4 hours.
- **Dairy products** are the main source of protein and calcium: at 1–3 years **≈600 ml** a day, at older ages **≈500 ml** (milk, kefir, yogurt, cottage cheese, cheese); for children over 1 year, **"growing-up" milk formulas** fortified with vitamins and trace elements are recommended.
- **Meat** (beef, poultry, rabbit), **fish** and **eggs** are complete animal proteins; **sausage products** are not recommended.
- **Cereal products** (buckwheat, rice, oats, corn) — giving buckwheat together with milk improves the amino acid balance.
- **Vegetables, fruits, greens** provide vitamins, minerals and fiber; proteins eaten with vegetables are absorbed better (**85%** instead of 70%).
- **Sugar and sweets** are limited; fast food and carbonated drinks are not recommended.

Table 15.11. *Daily energy requirements of children (Geppe)*

Age	Energy, kcal/day
1–3 years	1500
4–6 years	2000
7–10 years	2300
11–13 years	2700 (boys), 2500 (girls)
14–17 years	3000 (boys), 2600 (girls)

Table 15.12. *Distribution of daily calories among meals (Geppe)*

Age	Breakfast	2nd breakfast	Lunch	Afternoon snack	Dinner
Preschool	25%	—	40%	15%	20%
School	25%	20%	35%	—	20%



Figure 15.3. *A 4-year-old child eating independently (Navruzova, Geppe)*

15.7. Feeding history

In the case history, the following feeding data are recorded (Geppe):

- **type of feeding** (natural, mixed, artificial) and its duration;
- **time of first breastfeeding** (within 30 minutes after birth or later) and the reason;
- how the child took the breast (sucked actively or weakly);
- **feeding regimen** (on demand; every 3–3.5 hours; with a night break);
- **reason, age and type of formula** when switched to mixed or artificial feeding, and its tolerance;
- **complementary foods**: at what age, which foods, in what order, tolerance;
- vitamin D and other supplements;
- age of transition **to the family table**;
- current diet and appetite, food allergy.

15.8. Writing and defending the case history

Structure of the case history

In pediatric propaedeutics, the educational case history reinforces the skill of **complete examination** of the child and **semiotic analysis** of the data obtained.

1. **Title page:** department, patient's full name, age, ward, curator (student) and teacher, period of supervision.
2. **Passport section:** full name, date of birth and age, sex, address, parents' full names, place of work and occupation, child-care institution, date of admission, referral diagnosis.
3. **Complaints** (in the words of the patient and parents): main and additional; review of systems.
4. **History of present illness (anamnesis morbi):** when and how it began, course of development, examinations and treatment received, condition before admission.
5. **Life history (anamnesis vitae):** antenatal period (mother's pregnancy — which one, course, illnesses, risk factors), **intranatal** (delivery — term, course, Apgar score), **postnatal** — birth weight and length, first breastfeeding, separation of the umbilical stump, neonatal period; **physical and neuropsychological development** (head holding, sitting, walking, first words, teething); **feeding history** (15.7); **past illnesses**; **preventive vaccinations** (according to the national schedule, reactions); **allergy history**; **epidemiological history** (contact with infectious patients); **family history** (genealogical tree); **social and living conditions**.
6. **Physical examination (status praesens):** general condition (degree of severity), consciousness, position, **anthropometry and assessment of physical development** (centile tables or z-score, harmony); then **by systems** — nervous system and sense organs, skin, subcutaneous fat, lymph nodes, musculoskeletal system, respiratory, cardiovascular, digestive, urinary and endocrine systems (results of inspection, palpation, percussion and auscultation, with figures).
7. **Semiotic conclusion (identifying pathological changes):** the pathological symptoms found are grouped by systems, combined into **syndromes**, and the **leading syndrome** is identified.
8. **Preliminary (provisional) diagnosis** and its justification (in the propaedeutics course — a syndromic diagnosis).
9. **Examination plan:** laboratory and instrumental tests and interpretation of their results (compared with age norms).
10. **Daily notes (diary)** — dynamics of the child's condition over the days of supervision (temperature, pulse, RR, BP, changes in condition).
11. **Epicrisis** — a brief summary.
12. **References used**, curator's signature.

Procedure for defending the case history

> **REMEMBER**
> At the defense the student: 1) gives brief information about the patient (age, sex, when and with what complaints the patient was admitted); 2) presents **the main complaints and the history of the present illness**; 3) identifies the disease-related **risk factors** from the life history; 4) states **the main pathological findings of the objective examination** and must be able to demonstrate them in practice at the bedside; 5) assesses **physical and neuropsychological development**; 6) groups the symptoms into **syndromes** and justifies **the leading syndrome**; 7) interprets the results of laboratory and instrumental investigations by comparing them with the age norms; 8) answers questions from the teacher and the group (on anatomical and physiological features, methods of examination, semiotics).

Assessment criteria: completeness and orderliness of the case history, correct use of medical terms, literate and precise writing; practical skill in performing the physical examination; correct identification and justification of syndromes; answers to theoretical questions.

CAUTION

The case history is **a medical document**: the **confidentiality** of patient data is preserved, and the rules of deontology are observed in communication with the patient and parents; the examination is carried out with the parents' consent and in their presence.

Practical skills

Skill 1. Assessing breastfeeding technique

1. Help the mother choose a comfortable position (sitting, lying on her side, “football hold”).
2. Assess the child's position: head and body in one line, face toward the breast, body close to the mother, whole body supported.
3. Assess attachment: mouth wide open, lower lip turned outward, chin touching the breast, more of the areola visible above.
4. Assess sucking: slow, deep sucks with pauses, swallowing is heard.
5. Ask the mother about pain and the state of the breasts (cracks, milk stasis); explain mistakes.

Skill 2. Calculating feeding volume and drawing up a regimen

1. Determine the child's age, current weight and type of feeding.
2. In the first 10 days — Finkelstein or Zaitseva formula; after 10 days — calculate the daily volume by the volumetric or caloric method.
3. Divide the daily volume by the age-appropriate number of feeds to determine the single-feed volume.
4. Set the feeding times.
5. If complementary foods are given, replace one feed (then a second) with complementary food and draw up a daily menu.

Skill 3. Preparing an adapted formula

1. Wash hands; sterilize the bottle and nipple.
2. Pour the required volume of boiled water cooled to 40–50 °C into the bottle.
3. Add the indicated amount of dry formula using the scoop from the package (leveled off).
4. Shake until homogeneous; check the temperature by dripping onto the inner surface of the wrist (36–37 °C).
5. Feed the child in a semi-upright position; keep the bottle neck always full of milk (to avoid swallowing air); discard the leftover.

Skill 4. Defending the case history

1. Briefly present the passport data and main complaints.
2. Identify the important factors from the history of present illness and life history.
3. Demonstrate the pathological findings of the physical examination at the bedside.
4. Assess physical and neuropsychological development.
5. Combine the symptoms into syndromes and justify the leading syndrome.
6. Interpret the laboratory and instrumental results; answer questions.

Review questions

1. Define the types of feeding.
2. How is lactation regulated (prolactin and oxytocin reflexes)?
3. What is the composition of colostrum, transitional and mature milk? How does breast milk differ from cow's milk?
4. Name the advantages of breastfeeding for the child and the mother.
5. What is the significance of early breastfeeding, and what are the 10 Steps to Successful Breastfeeding?
6. What are the signs of correct breastfeeding technique?
7. What are the contraindications and difficulties of breastfeeding?
8. What are the causes and degrees of hypogalactia, and how is it managed?
9. What are the reasons and rules for switching to artificial feeding? Why is whole cow's milk not recommended?
10. Name the types of adapted and therapeutic formulas.
11. What are the rules of mixed feeding?
12. When and by what rules are complementary foods introduced? What are the consequences of early and late introduction?
13. How is feeding volume calculated in children under 1 year?
14. What are the principles of feeding children over 1 year?
15. What is the structure of the case history and the procedure for defending it?

Test questions

1. In mixed feeding, breast milk makes up more than what proportion of the daily volume?
   A) 1/10
   B) 1/5
   C) 1/2
   D) 3/4
   E) 100%

2. On which days is colostrum secreted?
   A) In the first 3–5 days after delivery
   B) On day 7–10
   C) In the 2nd–3rd week
   D) After 1 month
   E) Only during pregnancy

3. Protein content of mature breast milk:
   A) 1–3 g/l
   B) 9–13 g/l
   C) 28–32 g/l
   D) 40–50 g/l
   E) 80–110 g/l

4. Hormone responsible for the milk ejection reflex:
   A) Prolactin
   B) Oxytocin
   C) Estrogen
   D) Progesterone
   E) Insulin

5. Not a sign of correct attachment:
   A) Mouth wide open
   B) Lower lip turned outward
   C) Chin touching the breast
   D) Cheeks drawn in, smacking sounds heard
   E) More of the upper areola is visible

6. Absolute contraindication to breastfeeding:
   A) ARVI in the mother
   B) Mastitis in the mother
   C) Galactosemia in the child
   D) Cracked nipples
   E) Tonsillitis in the mother

7. Milk deficit in grade III hypogalactia:
   A) <25%
   B) 26–50%
   C) 51–75%
   D) >75%
   E) No deficit

8. Formula recommended for lactase deficiency:
   A) Anti-reflux
   B) Low-lactose or lactose-free
   C) Medium-chain triglyceride
   D) Phenylalanine-free
   E) Whole cow's milk

9. Formula of choice for cow's milk protein allergy:
   A) Fermented-milk adapted
   B) Based on extensively hydrolyzed protein
   C) Goat's milk
   D) Kefir
   E) Thickened formula

10. Timing of complementary food introduction according to WHO:
   A) 2 months
   B) 3 months
   C) 6 months
   D) 9 months
   E) 12 months

11. First complementary food for a child with adequate weight prone to constipation:
    A) Semolina porridge
    B) Vegetable puree
    C) Meat puree
    D) Cottage cheese
    E) Kefir

12. Before what age is meat puree not introduced?
    A) 4 months
    B) 6 months
    C) 9 months
    D) 12 months
    E) 2 years

13. Daily volume by the Finkelstein formula (birth weight 3500 g, day 5):
    A) 250 ml
    B) 350 ml
    C) 400 ml
    D) 500 ml
    E) 700 ml

14. Daily feeding volume of a 2–4-month-old child by the volumetric method, as a fraction of body weight:
    A) 1/5
    B) 1/6
    C) 1/7
    D) 1/8
    E) 1/10

15. Energy requirement of a child in the 1st half-year:
    A) 70 kcal/kg
    B) 90 kcal/kg
    C) 115 kcal/kg
    D) 150 kcal/kg
    E) 200 kcal/kg

16. What does exclusive breastfeeding mean?
    A) Nothing but breast milk is given, not even water
    B) Breast milk + formula
    C) Breast milk + juice
    D) Formula only
    E) Breast milk + water

17. In mixed feeding, breast milk makes up more than what part of the daily volume?
    A) 1/3
    B) 1/2
    C) 1/10
    D) 1/5
    E) All of the above

18. What does the prolactin reflex lead to?
    A) Uterine contraction
    B) Bleeding
    C) Milk production
    D) Appetite
    E) Milk ejection

19. What does the oxytocin reflex ensure?
    A) Hypogalactia
    B) Mastitis
    C) Milk production
    D) Milk ejection (contraction of myoepithelial cells)
    E) Prolactin

20. When is prolactin produced most?
    A) Only in the evening
    B) No difference
    C) During the day
    D) At noon
    E) At night

21. On which days is colostrum secreted?
    A) Days 10–14
    B) 2 months
    C) 6 months
    D) 1 month
    E) In the first 3–5 days after delivery

22. Caloric content of mature breast milk (kcal/l):
    A) 1500
    B) 590
    C) ≈700
    D) 300
    E) 1000

23. Caloric content of colostrum (kcal/l):
    A) ≈700
    B) ≈1500
    C) ≈2500
    D) ≈300
    E) ≈590

24. Protein content of cow's milk compared with breast milk:
    A) Only lactose
    B) The same
    C) Absent
    D) Higher (28–32 g/l)
    E) Mild

25. What is “hindmilk” like?
    A) No antibodies
    B) Rich in fat, high in calories, satiating
    C) Only lactose
    D) Watery, quenches thirst
    E) Protein-free

26. Absorption of iron from breast milk:
    A) 10%
    B) ≈5%
    C) 100%
    D) 1%
    E) ≈50%

27. Ca:P ratio in breast milk:
    A) 3:1
    B) 2:1
    C) 1:1
    D) 1:2
    E) 1:3

28. When is early breastfeeding carried out?
    A) After 1 week
    B) In the first 30 minutes after delivery
    C) After 3 days
    D) After 6 hours
    E) After 1 day

29. Sign of correct attachment:
    A) Mouth closed
    B) The child takes only the nipple
    C) The mother feels pain
    D) Cheeks drawn in, smacking sound
    E) Mouth wide open, lower lip turned outward, chin touching the breast

30. Sign that the child is getting enough milk:
    A) No stools
    B) Weight loss
    C) Constant crying
    D) Urinating more than 6 times a day
    E) Urinating 2 times a day

31. Adequate weekly weight gain in the 1st half-year:
    A) 1 kg
    B) ≥500 g
    C) ≥10 g
    D) ≥125 g
    E) 50 g

32. According to WHO, how long should exclusive breastfeeding last?
    A) Until 1 month
    B) Up to 12 months
    C) Until 6 months
    D) Up to 9 months
    E) Until 3 months

33. Maternal contraindication to breastfeeding:
    A) Milk stasis
    B) Common cold
    C) Active (bacillary) tuberculosis, taking cytostatics
    D) Cracked nipples
    E) Mild mastitis

34. How is breastfeeding continued in mastitis?
    A) Only from the affected breast
    B) From the healthy breast
    C) Milk is expressed and discarded
    D) Stopped completely
    E) Formula only

35. Grade III hypogalactia (milk deficit):
    A) <25%
    B) 0%
    C) 26–50%
    D) >75%
    E) 51–75%

36. How long do lactation crises usually last?
    A) 2 weeks
    B) 3–4 days
    C) 1 day
    D) 6 months
    E) 1 month

37. Ratio of whey proteins to casein in adapted formulas:
    A) 30:70
    B) 100:0
    C) 50:50
    D) 20:80
    E) 60:40

38. For what age is a starter (Formula 1) adapted formula intended?
    A) 6–12 months
    B) From 2 years
    C) Over 1 year
    D) 0–6 months
    E) Preterm infants only

39. From when are fermented-milk adapted formulas given?
    A) From 3–4 weeks, with sweet formula 1:1 or 2:1
    B) From 1 year
    C) From 3 months
    D) Only from 6 months
    E) From birth

40. Which formula is given for cow's milk protein allergy?
    A) Extensively hydrolyzed protein or amino acid formula
    B) Fermented-milk only
    C) Thickened
    D) Low-lactose
    E) Ordinary cow's milk

41. Which formula is given for lactase deficiency?
    A) Ordinary
    B) Low-lactose or lactose-free
    C) Hydrolyzed
    D) Soy
    E) Anti-reflux

42. In which condition are anti-reflux formulas used?
    A) Diarrhea
    B) Jaundice
    C) Anemia
    D) Allergy
    E) Regurgitation and vomiting

43. Over what period is the switch to a new formula made gradually?
    A) 5–7 days
    B) 1 month
    C) Immediately
    D) 2 hours
    E) 1 day

44. Timing of complementary food introduction (WHO and Uzbekistan Ministry of Health recommendation):
    A) From 2 months
    B) From 12 months
    C) From 6 months
    D) From 9 months
    E) From 3 months

45. Risk of early (before 4 months) introduction of complementary foods:
    A) Not rickets
    B) Delayed chewing
    C) Not obesity, no risk at all
    D) Intestinal dysfunction, food allergy, decreased lactation
    E) Iron deficiency only

46. How is complementary food given?
    A) Before breastfeeding, from a spoon, in the first half of the day
    B) Only at night
    C) 3 new foods at once
    D) From a bottle
    E) After breastfeeding

47. How is the volume of a new food started?
    A) A full portion
    B) Volume does not matter
    C) ½–1 teaspoon, increased over 5–7 days
    D) 100 g
    E) 1 glass

48. First complementary food for a child with low weight and frequent stools:
    A) Gluten-free porridge
    B) Vegetable puree
    C) Sausage
    D) Juice
    E) Cottage cheese

49. From when is meat puree introduced?
    A) From 6–7 months
    B) From birth
    C) From 4 months
    D) From 1 year
    E) From 10 months

50. When is fish puree introduced?
    A) 8–9 months, 1–2 times a week
    B) 6 months
    C) At 3 years
    D) At 1 year
    E) 4 months

51. What is not added to food before 1 year?
    A) Butter
    B) Milk formula
    C) Salt and sugar, honey, whole cow's milk
    D) Vegetable oil
    E) Fruit puree

52. Finkelstein formula (up to day 10 of life, weight >3200 g):
    A) V = 70 × n
    B) V = 2% × M
    C) V = 80 × n
    D) V = 100 × n
    E) V = 10 × n

53. From 10 days to 2 months, what fraction of body weight is the daily feeding volume?
    A) 1/5
    B) 1/6
    C) 1/2
    D) 1/8
    E) 1/7

54. Daily volume in a 6–9-month-old child (fraction of body weight, but not more than):
    A) 1/3
    B) 1/5
    C) 1/6
    D) 1/8, not exceeding 1000–1100 ml
    E) 1500 ml

55. Energy requirement in the 1st half-year by the caloric method:
    A) 80 kcal/kg
    B) 115 kcal/kg/day
    C) 110 kcal/kg
    D) 150 kcal/kg
    E) 60 kcal/kg

56. Protein : fat : carbohydrate ratio in the 1st half-year:
    A) 1:1:4
    B) 1:2:4
    C) 1:1:1
    D) 1:4:8
    E) ≈1:3:6

57. Dairy products for a child aged 1–3 years, approximately per day:
    A) ≈1500 ml
    B) ≈600 ml
    C) ≈100 ml
    D) Not needed
    E) ≈50 ml

Clinical cases

Case 1

A 1.5-month-old child. Birth weight 3300 g, now 4300 g. Breastfed strictly every 3.5 hours with a 6-hour night break; only one breast is offered at each feed. In recent days the child is restless, wakes often between feeds, and urinates 5–6 times a day. The mother says milk “is not staying” in her breasts. She is worried about returning to work and sleeps little.

Questions: 1) Assess the weight gain. 2) What is happening and what are its causes? 3) Doctor's recommendations? 4) Should formula be prescribed immediately?

Answer: 1) Weight has increased by 1000 g in 1.5 months (normally ≈600 g in the 1st month and ≈800 g in the 2nd month — ≈1000–1100 g is expected) — borderline, slightly low. 2) Secondary hypogalactia or a lactation crisis (around week 6); causes: a strict regimen and a long night break (prolactin is produced mostly at night), feeding from only one breast, maternal stress and lack of sleep (the oxytocin reflex weakens). 3) Breastfeed on demand, often, with night feeds and from both

breasts at each feed; check breastfeeding technique; adequate rest, sleep, emotional calm and full nutrition for the mother, a warm drink and lactogenic teas before feeding; support from family members; monitor weight and number of urinations after 3–5 days. 4) No — the degree of hypogalactia has not been established; only if all measures fail and grade III–IV hypogalactia is confirmed (control weighing) is supplementary feeding with an adapted formula prescribed, given by spoon or cup.

Case 2

A 6-month-old child weighing 7500 g, healthy, exclusively breastfed. Psychomotor development is age-appropriate: sits with support, is interested in food. Stools 1 time a day, sometimes 1 time in 2 days, rather hard. The mother asks for advice on complementary foods.

Questions: 1) Is it time to introduce complementary foods? What are the signs of readiness? 2) What should the first complementary food be and why? 3) Rules of introduction? 4) Draw up a plan for the next 2–3 months.

Answer: 1) Yes, according to WHO, 6 months is the time to introduce complementary foods; signs of readiness: sits with support, holds the head, is interested in food, has doubled the birth weight. 2) Because of the tendency to constipation and adequate weight — a single-component vegetable puree (zucchini, cauliflower, broccoli), then vegetable oil is added. 3) Only to a healthy child, in the first half of the day, before breastfeeding, starting with ½–1 teaspoon and increasing to 150 g over 5–7 days; one new food at a time; from a spoon; without salt or sugar; monitor tolerance (rash, stools). 4) 6 months — vegetable puree (single, then mixed); 6.5 months — gluten-free porridge (buckwheat, rice, corn) + butter; 6.5–7 months — meat puree (5–30 g, then 50 g); 7 months — fruit puree, egg yolk (¼), cottage cheese; 8 months — adapted kefir, bread, rusks; 8–9 months — fish puree (1–2 times a week). Breastfeeding continues.

Case 3

A 4-month-old child is artificially fed (the mother stopped breastfeeding at 2 months, saying her milk had “gone”). Weight 6400 g. The child has been fed whole cow's milk diluted with water, 7 times a day. Over the last 2 weeks there has been frequent regurgitation, irregular stools, and redness with itchy rashes on the face and neck. Blood: Hb 98 g/l.

Questions: 1) Feeding errors? 2) Causes of the rash and anemia? 3) Calculate the child's daily and single-feed volumes. 4) Recommendations?

Answer: 1) Whole cow's milk given to a child under 1 year; no adapted formula used; no measures taken to maintain and restore lactation. 2) Cow's milk proteins (casein, β-lactoglobulin) are highly allergenic — rashes typical of atopic dermatitis (cow's milk protein allergy); cow's milk is low in iron, which is poorly absorbed, and diapedetic bleeding from the intestinal mucosa occurs — iron deficiency anemia (Hb 98 g/l). 3) Volumetric method: at 4 months (on the 2–4-month boundary) 1/6 × 6400 ≈ 1070 ml, but it should not exceed 1000 ml — ≈1000 ml/day; with 7 feeds the single-feed volume is ≈140–145 ml (if switched to 6 feeds — ≈165 ml). 4) Stop cow's milk and switch to a therapeutic formula based on extensively hydrolyzed protein; allergist consultation; iron preparations; once formula tolerance improves, at 4–6 months (cautiously, given the allergy) a hypoallergenic single-component complementary food; advise the mother on the possibilities of relactation.

Quick review

Table 15.13. *Key facts on Topic 15*

Question	Answer
Mixed feeding	Breast milk >1/5 of daily volume
Artificial feeding	Breast milk <1/5 or none
First breastfeeding	Within 30 minutes after delivery
Prolactin / oxytocin	Milk production / milk ejection
Colostrum	3–5 days; rich in protein and sIgA; 1500 kcal/l
Mature milk	From week 2–3; protein 9–13 g/l, 700 kcal/l
Cow's milk	Protein 3 times higher (casein), less lactose
Exclusive breastfeeding	Up to 6 months (WHO); breastfeeding up to 2 years
Degrees of hypogalactia	I <25%; II 26–50%; III 51–75%; IV >75%
Whole cow's milk	Not given before 1 year
Lactase deficiency	Lactose-free formula
Cow's milk protein allergy	Extensive hydrolysate
Regurgitation	Anti-reflux (thickened) formula
Complementary foods	WHO — 6 months; Russian Federation program — 4–6 months
First complementary food	Vegetable puree (constipation) or gluten-free porridge (low weight)
Meat	From 6 months; egg yolk — 7 months; fish — 8–9 months
Finkelstein	70 × n (<3200 g); 80 × n (>3200 g)
Zaitseva	2% × M × n
Volumetric method	1/5 – 1/6 – 1/7 – 1/8, maximum 1000–1100 ml
Energy requirement	115 kcal/kg (1st half-year); 110 kcal/kg (2nd half-year)
1–3 years	1500 kcal; dairy products ≈600 ml
Case history	Complaints → history → physical examination → syndromes → leading syndrome → plan
Types of feeding	Natural (breast milk; exclusive — nothing added); mixed (breast milk >1/5); artificial (breast milk <1/5)
Lactation reflexes	Prolactin — milk production (more at night); oxytocin — milk ejection, uterine involution; stress and pain inhibit oxytocin
Types of breast milk and their caloric content	Colostrum 3–5 days, ≈1500 kcal/l; transitional from day 4–5, 750–800; mature from 2–3 weeks, ≈700; cow's milk 590
Advantages of breast milk	Finely dispersed protein, unsaturated fatty acids, β-lactose, bifidus factor, enzymes, secretory IgA, lactoferrin, lysozyme, Ca:P 2:1, sterile
Feeding regimen	In the 1st month on demand 8–12 times; 1–2 months 7 times (interval 3 h, night break 6 h); after complementary foods 5 times; duration 15–20 min
Signs of sufficient milk	Urinating >6 times, regular stools, calm after feeding, weight gain ≥125 g/week
Hypogalactia	Primary ≈5%, secondary common; grades I <25%, II 26–50%, III 51–75%, IV >75%; lactation crises last 3–4 days (at 3–6 weeks, 3, 4, 7, 8 months)

Question	Answer
Contraindications to breastfeeding	Mother: decompensation, active tuberculosis, cytostatics, dangerous infections, mental illness, HIV; child: extreme prematurity, severe asphyxia, galactosemia, phenylketonuria
Adapted formulas	Formula 1 0–6 months; Formula 2 6–12 months; Formula 3 >1 year; fermented-milk from 3–4 weeks; therapeutic (hydrolysate, lactose-free, anti-reflux, MCT, PKU)
Rules of artificial feeding	One formula; switching over 5–7 days; freshly prepared, 36–37 °C; 6–7 times, 3–3.5 h interval, 6 h night break
Timing of complementary foods	WHO/Uzbekistan: from 6 months; Russia: 4–6 months (not before 4 months); signs of readiness: holds the head, sits, extrusion reflex has faded
Rules for complementary foods	To a healthy child; from ½–1 spoon, increased over 5–7 days; in the 1st half of the day; one new food; before breastfeeding; no salt, sugar or honey
Order of complementary foods	Vegetable puree/porridge 4–6 months; fruit 4–6; meat 6–7; cottage cheese 6–7; yolk 7; fish 8–9; kefir 8
Volume calculation (up to 10 days)	Finkelstein: 70×n (<3200 g), 80×n (>3200 g); Zaitseva 2%×M×n; Filatov single-feed 10×n
Volume calculation (10 days–1 year)	10 days–2 months 1/5; 2–4 months 1/6; 4–6 months 1/7; 6–12 months 1/8 (≤1000–1100 ml); caloric: 115 kcal/kg (1st half-year), 110 (2nd half-year)
Protein:fat:carbohydrate ratio	1st half-year ≈1:3:6; 2nd half-year ≈1:2:4; after 1 year 1:1:4

Answers to the test questions

Topic 1: 1-D; 2-B; 3-B; 4-B; 5-C; 6-D; 7-C; 8-B; 9-C; 10-C; 11-B; 12-C; 13-B; 14-B; 15-C; 16-A; 17-A; 18-E; 19-C; 20-E; 21-D; 22-C; 23-B; 24-B; 25-D; 26-A.

Topic 2: 1-B; 2-C; 3-C; 4-D; 5-C; 6-A; 7-B; 8-C; 9-C; 10-D; 11-C; 12-C; 13-B; 14-C; 15-D; 16-E; 17-A; 18-A; 19-A; 20-B; 21-C; 22-B; 23-C; 24-D; 25-B; 26-E; 27-D.

Topic 3: 1-B; 2-C; 3-B; 4-B; 5-C; 6-D; 7-C; 8-B; 9-B; 10-B; 11-B; 12-B; 13-B; 14-A; 15-C; 16-C; 17-B; 18-A; 19-E; 20-B; 21-E; 22-A; 23-B; 24-D; 25-A; 26-D; 27-C.

Topic 4: 1-C; 2-B; 3-B; 4-D; 5-B; 6-C; 7-B; 8-B; 9-B; 10-C; 11-B; 12-C; 13-E; 14-C; 15-D; 16-E; 17-A; 18-A; 19-D; 20-C; 21-C; 22-E; 23-B; 24-E; 25-E; 26-C; 27-B; 28-A; 29-A; 30-A; 31-D; 32-B; 33-B; 34-C; 35-D; 36-D.

Topic 5: 1-B; 2-C; 3-C; 4-C; 5-B; 6-C; 7-C; 8-C; 9-C; 10-B; 11-B; 12-B; 13-B; 14-B; 15-D; 16-E; 17-E; 18-A; 19-B; 20-B; 21-E; 22-C; 23-D; 24-A; 25-D; 26-B; 27-C; 28-A; 29-B; 30-C; 31-A; 32-B; 33-E; 34-D; 35-D; 36-C; 37-C; 38-A.

Topic 6: 1-D; 2-B; 3-B; 4-C; 5-C; 6-B; 7-C; 8-C; 9-C; 10-C; 11-B; 12-C; 13-C; 14-B; 15-C; 16-E; 17-D; 18-D; 19-A; 20-E; 21-B; 22-C; 23-A; 24-D; 25-C; 26-C; 27-E; 28-C; 29-D; 30-B; 31-E; 32-A; 33-A; 34-D; 35-B; 36-A; 37-E; 38-C; 39-B; 40-B.

Topic 7: 1-D; 2-C; 3-B; 4-C; 5-D; 6-B; 7-C; 8-B; 9-C; 10-C; 11-D; 12-B; 13-B; 14-C; 15-A; 16-A; 17-C; 18-A; 19-C; 20-C; 21-C; 22-D; 23-E; 24-D; 25-B; 26-E; 27-A; 28-D; 29-A; 30-D; 31-B; 32-D; 33-B; 34-C; 35-B; 36-E; 37-A; 38-E; 39-B.

Topic 8: 1-B; 2-C; 3-C; 4-D; 5-C; 6-D; 7-B; 8-B; 9-C; 10-D; 11-A; 12-B; 13-C; 14-B; 15-C; 16-E; 17-A; 18-C; 19-B; 20-D; 21-D; 22-C; 23-A; 24-C; 25-D; 26-C; 27-A; 28-D; 29-A; 30-B; 31-B; 32-E; 33-E; 34-B; 35-D; 36-C; 37-A; 38-E; 39-B.

Topic 9: 1-C; 2-B; 3-A; 4-B; 5-D; 6-B; 7-B; 8-B; 9-B; 10-C; 11-C; 12-B; 13-A; 14-B; 15-B; 16-B; 17-C; 18-A; 19-E; 20-C; 21-E; 22-E; 23-A; 24-B; 25-D; 26-B; 27-E; 28-D; 29-D; 30-B; 31-E; 32-D; 33-B; 34-C; 35-C; 36-C; 37-A; 38-B; 39-D; 40-C; 41-A; 42-A; 43-A.

Topic 10: 1-D; 2-B; 3-B; 4-C; 5-A; 6-C; 7-A; 8-D; 9-A; 10-B; 11-E; 12-B; 13-B; 14-B; 15-C; 16-E; 17-A; 18-C; 19-C; 20-E; 21-A; 22-B; 23-C; 24-A; 25-B; 26-D; 27-E; 28-E; 29-B; 30-E; 31-A; 32-C; 33-D; 34-D; 35-B; 36-C; 37-A; 38-B; 39-C; 40-D; 41-B; 42-D; 43-A; 44-E; 45-D; 46-D; 47-C; 48-B; 49-A; 50-E; 51-A.

Topic 11: 1-C; 2-D; 3-B; 4-D; 5-B; 6-C; 7-C; 8-B; 9-D; 10-B; 11-B; 12-B; 13-B; 14-B; 15-A; 16-B; 17-C; 18-A; 19-C; 20-E; 21-C; 22-D; 23-C; 24-B; 25-E; 26-B; 27-E; 28-C; 29-E; 30-A; 31-A; 32-D; 33-A; 34-D; 35-B; 36-D; 37-E; 38-C; 39-E; 40-B; 41-A; 42-D; 43-A; 44-D; 45-B.

Topic 12: 1-B; 2-B; 3-B; 4-C; 5-B; 6-A; 7-C; 8-C; 9-C; 10-C; 11-B; 12-C; 13-C; 14-A; 15-B; 16-C; 17-A; 18-B; 19-E; 20-C; 21-A; 22-D; 23-A; 24-D; 25-E; 26-D; 27-B; 28-C; 29-B; 30-A; 31-B; 32-E; 33-E; 34-C; 35-D; 36-C; 37-D; 38-A; 39-D; 40-B; 41-A; 42-E; 43-B; 44-C; 45-E; 46-A; 47-B.

Topic 13: 1-B; 2-B; 3-D; 4-C; 5-B; 6-C; 7-B; 8-C; 9-C; 10-C; 11-B; 12-C; 13-B; 14-B; 15-C; 16-B; 17-B; 18-D; 19-D; 20-C; 21-E; 22-B; 23-B; 24-E; 25-C; 26-C; 27-A; 28-E; 29-E; 30-E; 31-C; 32-A; 33-B; 34-C; 35-A; 36-B; 37-D; 38-A; 39-A; 40-D; 41-C; 42-D; 43-E; 44-A; 45-D.

Topic 14: 1-B; 2-C; 3-D; 4-C; 5-B; 6-D; 7-B; 8-D; 9-D; 10-E; 11-B; 12-B; 13-C; 14-B; 15-C; 16-D; 17-D; 18-A; 19-B; 20-C; 21-E; 22-C; 23-D; 24-E; 25-D; 26-E; 27-B; 28-B; 29-D; 30-E; 31-A; 32-A;

33-C; 34-A; 35-C; 36-D; 37-C; 38-E; 39-C; 40-E; 41-D; 42-D; 43-E; 44-A; 45-A; 46-A; 47-B; 48-A; 49-B; 50-D; 51-B; 52-B; 53-E; 54-B; 55-E; 56-B; 57-C; 58-C; 59-B; 60-A; 61-A; 62-C.

Topic 15: 1-B; 2-A; 3-B; 4-B; 5-D; 6-C; 7-C; 8-B; 9-B; 10-C; 11-B; 12-B; 13-C; 14-B; 15-C; 16-A; 17-D; 18-C; 19-D; 20-E; 21-E; 22-C; 23-B; 24-D; 25-B; 26-E; 27-B; 28-B; 29-E; 30-D; 31-D; 32-C; 33-C; 34-B; 35-E; 36-B; 37-E; 38-D; 39-A; 40-A; 41-B; 42-E; 43-A; 44-C; 45-D; 46-A; 47-C; 48-A; 49-A; 50-A; 51-C; 52-C; 53-A; 54-D; 55-B; 56-E; 57-B.

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